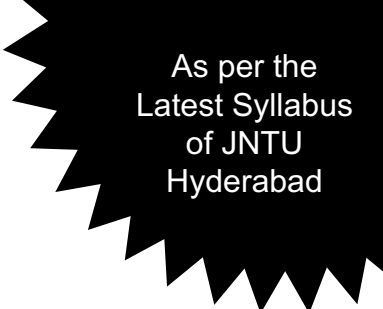




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Hyderabad

Electronic Devices and Circuits

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Preface

This book fully covers the latest syllabus, of the undergraduate engineering course titled *Electronic Devices and Circuits* offered in 2nd year, 1st semester (2—1) by JNTU, Hyderabad.

The book will serve the purpose of a text to the engineering students of degree, diploma, AMIE and graduate IETE courses and also act as a useful reference for those preparing for competitive examinations. Also, it will meet the pressing needs of interested readers who wish to gain sound knowledge and understanding of the principles of electronic devices and circuits. Practicing engineers will find the content of significant relevance in their day-to-day functioning.

The book contains 5 units. Unit 1 deals with *P-N* junction diodes and special purpose electronic devices. Unit 2 explains rectifiers and filters in great detail. Unit 3 deals with bipolar junction transistors and uni-junction transistors. Unit 4 deals with transistor biasing and stabilization. Unit 5 explains field effect transistors and FET amplifiers in great detail.

All the topics have been profusely illustrated with diagrams for easy understanding. Equal emphasis has been laid on mathematical derivations as well as their physical interpretations. Illustrative examples are discussed to emphasise the concepts and typical applications.

A set of questions and exercises has been given at the end of each chapter with a view to help the readers increase their understanding of the subject and to encourage further reading. Questions that have appeared in the university examinations in previous years are included at the appropriate places which will serve to enhance understanding and build the student's confidence.

We are highly indebted to the management of our institutions for encouraging us from time to time and providing all the necessary facilities. Thanks are due to our colleagues, and the students for their valuable suggestions and assistance in the preparation of the manuscript.

We are grateful to McGraw Hill Education (India), for getting the manuscript reviewed by experts, providing feedback and for bringing out this book in a short span of time, and also to Mrs Vibha Mahajan for stimulating interest in this project. We are also thankful to Mr R Gopalakrishnan and Mr S Sankar Kumar for efficiently wordprocessing the major part of the manuscript.

We would like to extend our warmest thanks to our respective wives, Mrs Kalavathy Salivahanan and Mrs Andal Suresh Kumar, for their moral support and constant encouragement and to our children, S Santhosh Kanna and S Subadesh Kanna, S Sree Naga Gowri and S Sree Naga Vani, for their enormous patience and cooperation.

We would also like to thank Mr Vijaya Kishore from Srinivas Institute of Technology and Management, Chittor, and Prof. Sridhar from Lendi College of Engineering, Vishakhapatnam, for their feedback.

Constructive suggestions and corrections for the improvement of the book would be most welcome and highly appreciated.

**S SALIVAHANAN
N SURESH KUMAR**

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ELECTRONIC DEVICES AND CIRCUITS

Objective

This is a fundamental course, basic knowledge of which is required by all the circuit branch engineers. This course focuses:

- To familiarize the student with the principle of operation, analysis and design of junction diode, BJT and FET transistors and amplifier circuits.
- To understand diode as rectifier.
- To study basic principle of filter circuits and various types.

Unit-I

P-N Junction Diode: Qualitative Theory of P-N Junction, P-N Junction as a Diode, Diode Equation, Volt-Ampere Characteristics, Temperature dependence of VI characteristic, Ideal versus Practical – Resistance levels (Static and Dynamic), Transition and Diffusion Capacitances, Diode Equivalent Circuits, Load Line Analysis, Break-down Mechanisms in Semiconductor Diodes, Zener Diode Characteristics.

Special Purpose Electronic Devices: Principle of Operation and Characteristics of Tunnel Diodes (with the help of Energy Band Diagram), Varactor Diode, SCR and Semiconductor Photo Diode.

Unit-II

Rectifiers and Filters: The P-N junction as a Rectifier, Half wave Rectifier, Full wave Rectifier, Bridge Rectifier, Harmonic components in a Rectifier Circuit, Inductor Filters, Capacitor Filters, L-Section Filters, π -Section Filters, Comparison of Filters, Voltage Regulation using Zener Diode.

Unit-III

Bipolar Junction Transistor and UJT: The Junction Transistor, Transistor Current Components, Transistor as an Amplifier, Transistor Construction, BJT Operation, BJT Symbol, Common Base, Common Emitter and Common Collector Configurations, Limits of Operation, BJT Specifications, BJT Hybrid Model, Determination of h-parameters from Transistor Characteristics, Comparison of CB, CE, and CC Amplifier Configurations, UJT and Characteristics.

Unit-IV

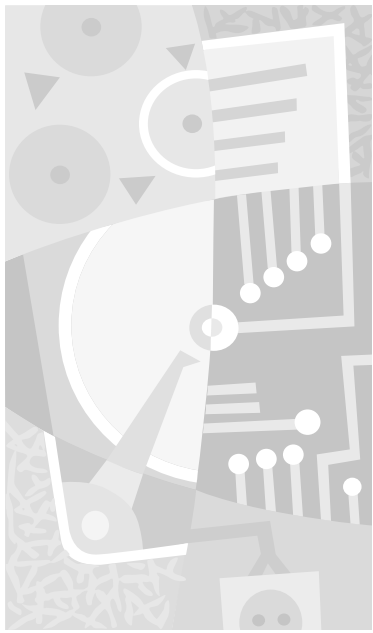
Transistor Biasing and Stabilization: Operating Point, The DC and AC Load lines, Need for Biasing, Fixed Bias, Collector Feedback Bias, Emitter Feedback Bias, Collector – Emitter Feedback Bias, Voltage Divider Bias, Bias Stability, Stabilization Factors, Stabilization against variations in V_{BE} and β , Bias Compensation using Diodes and Transistors, Thermal Runaway, Thermal Stability, Analysis of a Transistor Amplifier Circuit using h-parameters.

Unit-V

Field Effect Transistor and FET Amplifiers

Field Effect Transistor: The junction Field Effect Transistor (Construction, principle of operation, symbol) – Pinch-off Voltage – Volt-Ampere characteristics, The JEFT Small Signal Model, MOSFET (Construction, principle of operation, symbol), MOSFET Characteristics in Enhancement and Depletion modes.

FET Amplifiers: FET Common Source Amplifiers, Common Drain Amplifier, Generalized FET Amplifier, Biasing FET, FET as Voltage Variable Resistor, Comparison of BJT and FET.



1

PN JUNCTION DIODE & SPECIAL-PURPOSE ELECTRONIC DEVICES

1.1 INTRODUCTION

The PN junction diode is one of the semiconductor devices with two semiconductor materials in physical contact, one with excess of holes (P-type) and other with excess of electrons (N-type). A PN junction diode may be found from a single-crystal intrinsic semiconductor by doping part of it with acceptor impurities and the remainder with donors. Such junctions can form the basis of very efficient rectifiers. The most important characteristic of a PN junction is its ability to allow the flow of current in only one direction. In the opposite direction, it offers very high resistance. The high-vacuum diode has largely been replaced by silicon and selenium rectifiers. Semiconductor diodes find wide applications in all phases of electronics, viz. radio and TV, optoelectronics, power supplies, industrial electronics, instrumentation, computers, etc.

In addition to the PN junction diode, other types of diodes are also manufactured for specific applications. These special diodes are two/three terminal devices with their doping levels carefully selected to give the desired characteristics.

1.2 CLASSIFICATION OF SEMICONDUCTORS

Semiconductors are classified as (i) intrinsic (pure) and (ii) extrinsic (impure) types. The extrinsic semiconductors are of N-type and P-type.

1.2.1 Intrinsic Semiconductor

A pure semiconductor is called intrinsic semiconductor. Even at the room temperature, some of the valence electrons may acquire sufficient energy to enter the conduction band to form free electrons. Under the influence of electric field, these electrons constitute electric current. A missing electron in the valence band leaves a vacant space there, which is known as a *hole*, as shown in Fig. 1.1. Holes also contribute to electric current.

In an intrinsic semiconductor, even at room temperature, electron-hole pairs are created. When electric field is applied across an intrinsic semiconductor, the current conduction takes place by two processes, namely, free electrons and holes. Under the influence of electric field, total current through the semiconductor is the sum of currents due to free electrons and holes.

Though the total current inside the semiconductor is due to free electrons and holes, the current in the external wire is fully by electrons. In Fig. 1.2, holes being positively charged move towards the negative terminal of the battery. As the holes reach the negative terminal of the battery, electrons enter the semiconductor near the terminal (X) and combine with the holes. At the same time, the loosely held electrons near the positive terminal (Y) are attracted away from their atoms into the positive terminal. This creates new holes near the positive terminal which again drift towards the negative terminal.

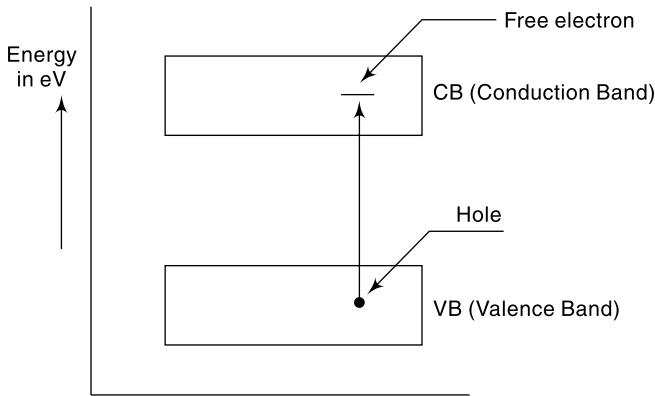


Fig. 1.1 Creation of Electron-hole Pair in a Semiconductor

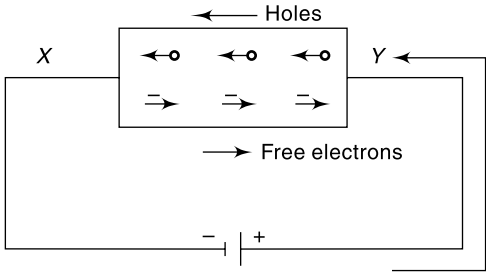


Fig. 1.2 Current Conductor in Semiconductor

1.2.2 Extrinsic Semiconductor

Due to the poor conduction at room temperature, the intrinsic semiconductor as such, is not useful in the electronic devices. Hence, the current conduction capability of the intrinsic semiconductor should be increased. This can be achieved by adding a small amount of impurity to the intrinsic semiconductor, so that it becomes impure or extrinsic semiconductor. This process of adding impurity is known as *doping*.

The amount of impurity added is extremely small, say 1 to 2 atoms of impurity for 10^6 intrinsic atoms.

N-type Semiconductor A small amount of pentavalent impurities such as arsenic, antimony or phosphorus is added to the pure semiconductor (germanium or silicon crystal) to get N-type semiconductor.

Germanium atom has four valence electrons and antimony has five valence electrons. As shown in Fig. 1.3, each antimony atom forms a covalent bond with surrounding four germanium atoms. Thus, four valence electrons of antimony atom form covalent bond with four valence electrons of individual germanium atom and fifth valence electron is left free which is loosely bound to the antimony atom.

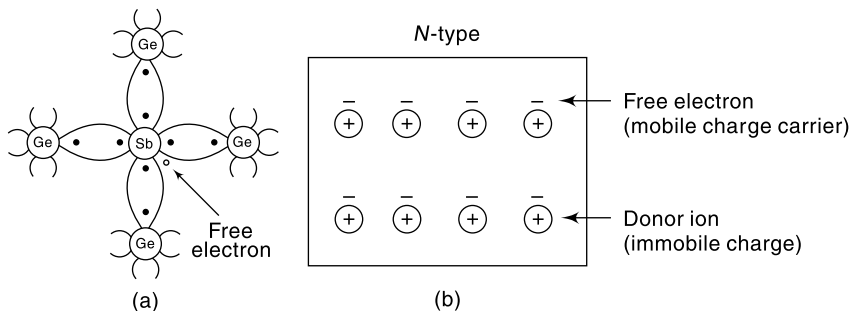


Fig. 1.3 N-type Semiconductor: (a) Formation of Covalent Bonds, and (b) Charged Carriers

This loosely bound electron can be easily excited from the valence band to the conduction band by the application of electric field or increasing the thermal energy. Thus every antimony atom contributes one conduction electron without creating a hole. Such pentavalent impurities are called donor impurities because it donates one electron for conduction. On giving an electron for conduction, the donor atom becomes positively charged ion because it loses one electron. But it cannot take part in conduction because it is firmly fixed in the crystal lattice.

Thus, the addition of pentavalent impurity (antimony) increases the number of electrons in the conduction band thereby increasing the conductivity of N-type semiconductor. As a result of doping, the number of free electrons far exceeds the number of holes in an N-type semiconductor. So electrons are called majority carriers and holes are called minority carriers.

P-type Semiconductor A small amount of trivalent impurities such as aluminium or boron is added to the pure semiconductor to get the P-type semiconductor. Germanium (Ge) atom has four valence electrons and boron has three valence electrons as shown in Fig. 1.4. Three valence electrons in boron form covalent bond with four surrounding atoms of Ge leaving one bond incomplete which gives rise to a hole. Thus trivalent impurity (boron) when added to the intrinsic semiconductor (germanium) introduces a large number of holes in the valence band. These positively charged holes increase the conductivity of P-type semiconductor. Trivalent impurities such as boron is called acceptor

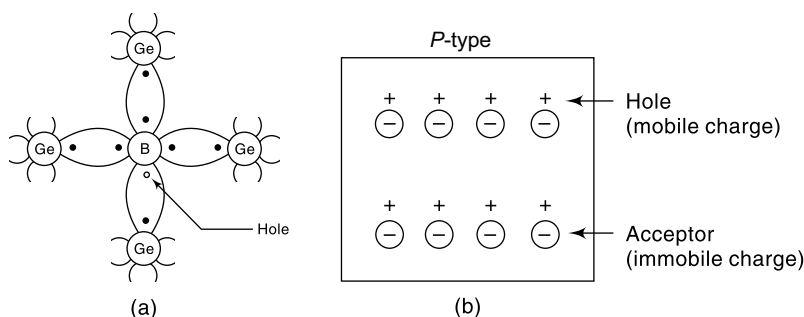


Fig. 1.4 P-type Semiconductor: (a) Formation of Covalent Bonds, and (b) Charged Carriers

impurity because it accepts free electrons in the place of holes. As each boron atom donates a hole for conduction, it becomes a negatively charged ion. As the number of holes is very much greater than the number of free electrons in a P-type material, holes are termed as majority carriers and electrons as minority carriers.

1.3 QUALITATIVE THEORY OF PN JUNCTION DIODE

1.3.1 PN Junction Diode in Equilibrium with no Applied Voltage

In a piece of semiconductor material, if one half is doped by P-type impurity and the other half is doped by N-type impurity, a PN junction is formed. The plane dividing the two halves or zones is called PN junction. As shown in Fig. 1.10, the N-type material has high concentration of free electrons, while P-type material has high concentration of holes. Therefore, at the junction there is a tendency for the free electrons to diffuse over to the P-side and holes to the N-side. This process is called *diffusion*. As the free electrons move across the junction from N-type to P-type, the donor ions become positively charged. Hence a positive charge is built on the N-side for the junction. The free electrons that cross the junction uncover the negative acceptor ions by filling in the holes. Therefore, a net negative charge is established on the P-side of the junction. This net negative charge on the P-side prevents further diffusion of electrons into the P-side. Similarly, the net positive charge on the N-side repels the hole crossing from P-side to N-side. Thus a barrier is set-up near the junction which prevents further movement of charge carriers. i.e. electrons and holes. As a consequence of the induced electric field across the depletion layer, an electrostatic potential difference is established between P-and N-regions, which is called the potential barrier, junction barrier, diffusion potential, or contact potential, V_o . The magnitude of the contact potential V_o varies with doping levels and temperature. V_o is 0.3 V for germanium and 0.72 V for silicon.

The electrostatic field across the junction caused by the positively charged N-type region tends to drive the holes away from the junction and negatively

charged P-type region tends to drive the electrons away from the junction. The majority holes diffusing out of the P-region leave behind negatively charged acceptor atoms bound to the lattice, thus exposing negative space charge in a previously neutral region. Similarly, electrons diffusing from the N-region expose positively ionised donor atoms, and a double space charge layer builds up at the junction as shown in Figs 1.5(a) and (c).

It is noticed that the space-charge layers are of opposite sign to the majority carriers diffusing into them, which tends to reduce the diffusion rate. Thus, the double space of the layer causes an electric field to be set up across the junction directed from N- to P-regions, which is in such a direction to inhibit diffusion of majority electrons and holes, as illustrated in Figs 1.5(a) and (d). The shape of the charge density, ρ , depends upon how the diode is doped. Thus, the junction region is depleted of mobile charge carriers. Hence, it is called the depletion region (layer), the space charge region, or the transition region. The depletion region is of order $0.5 \mu\text{m}$ thick. There are no mobile carriers in this very narrow depletion layer. Hence no current flows across the junction and the system is in equilibrium. To the left of this depletion layer, the carrier concentration is $p \approx N_A$, and to its right it $n \approx N_D$.

Calculation of Depletion Width Let us now consider the width of the depletion region in the junction of Fig. 1.5. The region contains space charge due to the fact that, donors on the N-side and acceptors on the P-side have lost their accompanying electrons and holes. Hence, an electric field is established which, in turn, causes a difference in potential energy, qV_o , between the two parts of the specimen. Thus, a potential is built up across the junction and Fig. 1.5(e) represents the variation in potential. Here, P-side of the junction is at a lower potential than the N-side which means that the electrons on the P-side have a great potential energy.

In this analysis, let us consider an *Alloy Junction* in which there is an abrupt change from acceptor ions on P-side to donor ions on N-side. Assume that the concentration of electrons and holes in the depletion region is negligible and that all of the donors and acceptors are ionised. Hence, the regions of space charge may be described as

$$\rho = -qN_A, 0 > x > X_1$$

$$\rho = +qN_D, X_2 > x > 0$$

$$\rho = 0, \text{ elsewhere}$$

where ρ is the space charge density, as indicated in Fig. 1.5(c) (i). The axes have been chosen in Fig. 1.5(e) in such a way that V_1 and X_1 have negative values. The potential variation in the space charge region can be calculated by using Poission's equation, which is given by

$$\nabla^2 V = - \frac{\rho(x, y, z)}{\epsilon_o \epsilon_r}$$

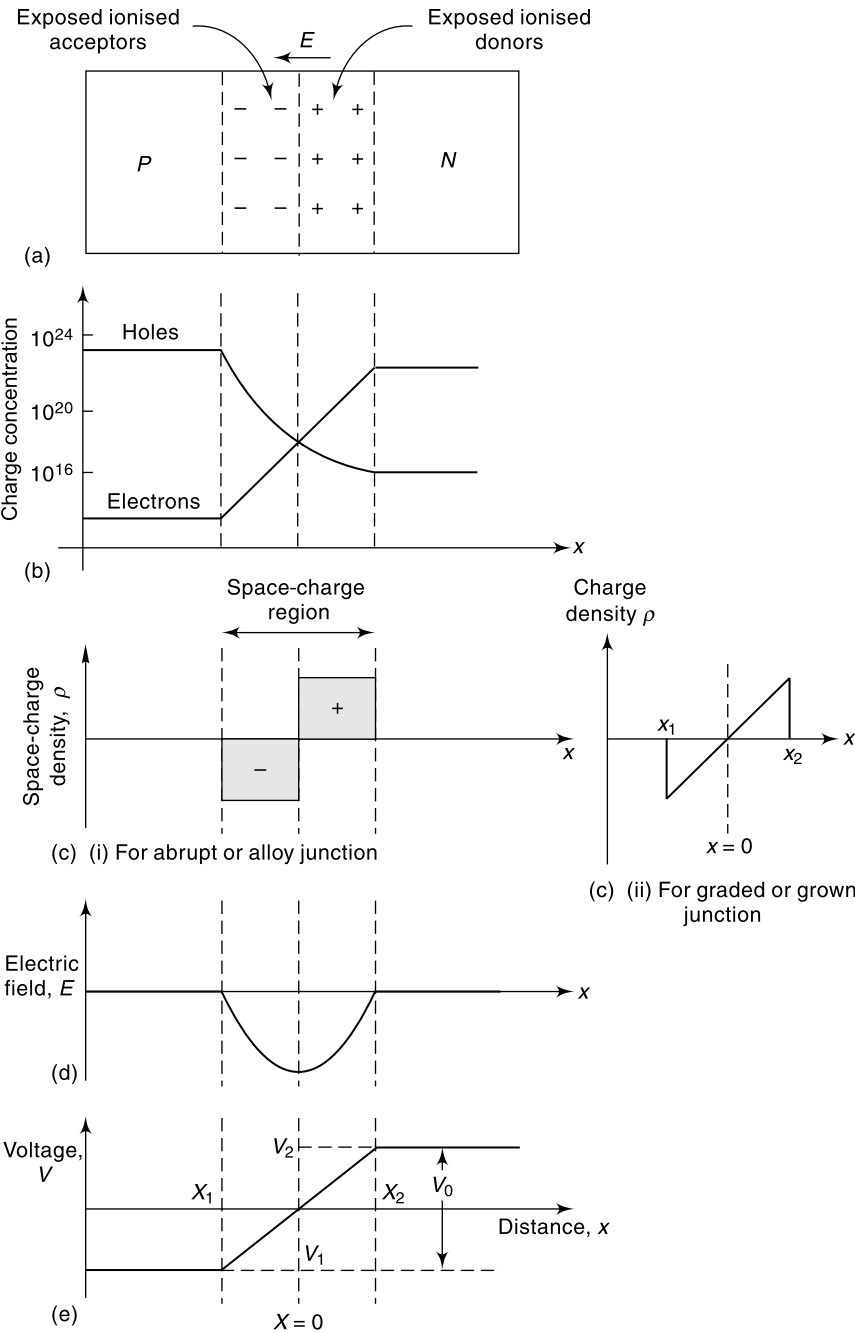


Fig. 1.5 Formation of PN Junction

where ϵ_r is the relative permittivity. The relevant equation for the required one-dimensional problem is

$$\frac{d^2 V}{dx^2} = -\frac{\rho}{\epsilon_o \epsilon_r}$$

Applying the above equation to the P-side of the junction, we get

$$\frac{d^2 V}{dx^2} = \frac{qN_A}{\epsilon_o \epsilon_r}$$

Integrating twice, we get

$$V = \frac{qN_A x^2}{2\epsilon_o \epsilon_r} + Cx + D$$

where C and D are the constants of integration.

From the Fig. 1.5(e), we have $V = 0$ at $x = 0$, and hence $D = 0$. When $x < X_1$ on the P-side, the potential is constant, so that $\frac{dV}{dx} = 0$ at $x = X_1$. Hence,

$$C = -\frac{qN_A}{\epsilon_o \epsilon_r} \cdot X_1$$

Therefore,

$$V = \frac{qN_A x^2}{2\epsilon_o \epsilon_r} - \frac{qN_A}{\epsilon_o \epsilon_r} \cdot X_1 \cdot x$$

i.e.

$$V = \frac{qN_A}{\epsilon_o \epsilon_r} \left(\frac{x^2}{2} - X_1 \cdot x \right)$$

As $V = V_1$ at $x = X_1$, we have

$$V_1 = -\frac{qN_A}{2\epsilon_o \epsilon_r} \cdot X_1^2$$

If we apply the same procedure to the N-side, we get

$$V_2 = \frac{qN_D}{2\epsilon_o \epsilon_r} \cdot X_2^2$$

Therefore, the total built-in potential or the contact potential is V_o , where

$$V_o = V_2 - V_1 = \frac{q}{2\epsilon_o \epsilon_r} (N_A X_1^2 + N_D X_2^2)$$

We know the fact that the positive charge on the N-side must be equal in magnitude to the negative charge on the P-side for the neutral specimen. Hence,

$$N_A X_1 = -N_D X_2$$

Using the fact that X_1 is a negative quantity, we get

$$N_A X_1 = N_D X_2 \text{ i.e. } X_2 = \frac{N_A}{N_D} X_1$$

Substituting in the equation for the contact potential, we get

$$\begin{aligned} V_o &= \frac{qN_A}{2\epsilon_o\epsilon_r} X_1^2 + \frac{qN_D}{2\epsilon_o\epsilon_r} \left[\frac{N_A}{N_D} X_1 \right]^2 \\ &= \frac{qN_A}{2\epsilon_o\epsilon_r} X_1^2 \left[1 + \frac{N_A}{N_D} \right] \end{aligned}$$

Therefore,

$$X_1 = \left[\frac{2\epsilon_o\epsilon_r V_o}{qN_A \left(1 + \frac{N_A}{N_D} \right)} \right]^{1/2}$$

Similarly,

$$X_2 = \left[\frac{2\epsilon_o\epsilon_r V_o}{qN_D \left(1 + \frac{N_D}{N_A} \right)} \right]^{1/2}$$

Let

$$N_{eff} = \frac{N_A N_D}{N_A + N_D}$$

Hence,

$$X_1 = \sqrt{\frac{2\epsilon_o\epsilon_r V_o}{q} N_{eff}} \frac{1}{N_A}$$

$$X_2 = \sqrt{\frac{2\epsilon_o\epsilon_r V_o}{q} N_{eff}} \frac{1}{N_D}$$

We know that,

Therefore, $W = X_1 + X_2$

$$\begin{aligned} &= \sqrt{\frac{2\epsilon_o\epsilon_r V_o}{q} N_{eff}} \left[\frac{1}{N_A} + \frac{1}{N_D} \right] \\ &= \sqrt{\frac{2\epsilon_o\epsilon_r V_o}{q N_{eff}}} = \sqrt{\frac{2\epsilon_o\epsilon_r V_o}{q} \left(\frac{N_A + N_D}{N_A N_D} \right)} \end{aligned}$$

Here, in an *alloy junction*, the depletion width W is proportional to $(V_o)^{1/2}$.

In a *Grown Junction*, the charge density (ρ) varies linearly with distance (x) as shown in Fig.1.5(c) (ii). If a similar analysis is carried for this junction, it is found that W varies as $(V_o)^{1/3}$ instead of $(V_o)^{1/2}$.

1.3.2 Operation and Volt-Ampere Characteristics of a Diode Under Forward Bias Condition

When positive terminal of the battery is connected to the P-type and negative terminal to the N-type of the PN junction diode, the bias applied is known as forward bias.

As shown in Fig. 1.6, the applied potential with external battery acts in opposition to the internal potential barrier and disturbs the equilibrium. As soon as equilibrium is disturbed by the application of an external voltage, the Fermi level is no longer continuous across the junction. Under the forward bias condition, the applied positive potential repels the holes in P-type region so that the holes move towards the junction and the applied negative potential repels the electrons in the N-type region and the electrons move towards the junction. Eventually, when the applied potential is more than the internal barrier potential the depletion region and internal potential barrier disappear.

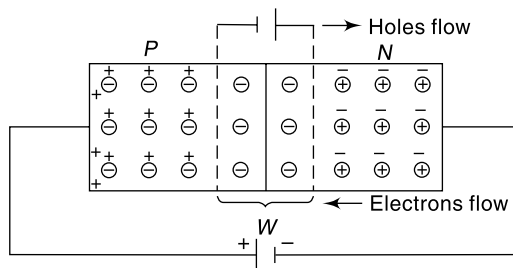


Fig. 1.6 PN Junction Under Forward Bias

Under forward bias condition, the V - I characteristics of a PN junction diode are shown in Fig. 1.7. As the forward voltage (V_F) is increased, for $V_F < V_O$, the forward current I_F is almost zero (region OA) because the potential barrier prevents the holes from P-region and electrons from N-region to flow across the depletion region in the opposite direction.

For $V_F > V_O$, the potential barrier at the junction completely disappears and hence, the holes cross the junction from P-type to N-type and the electrons cross the junction in the opposite direction, resulting in relatively large current flow in the external circuit.

A feature worth to be noted in the forward characteristics shown in Fig. 1.7 is the cut in or threshold voltage (V_r) below which the current is very small. It is 0.3 V and 0.7 V for germanium and silicon, respectively. At the cut in voltage, the potential barrier is overcome and the current through the junction starts to increase rapidly.

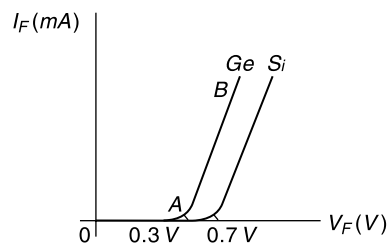


Fig. 1.7 V - I Characteristics of a Diode Under Forward Bias Condition

1.3.3 Operation and Volt-Ampere Characteristics of a diode Under Reverse Bias Condition

When the negative terminal of the battery is connected to the P-type and positive terminal of the battery is connected to the N-type of the PN junction, the bias applied is known as reverse bias.

Under applied reverse bias as shown in Fig. 1.8, holes which form the majority carriers of the P-side move towards the negative terminal of the battery and electrons which form the majority carrier of the N-side are attracted towards

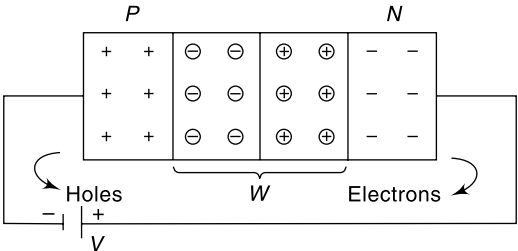


Fig. 1.8 PN Junction Under Reverse Bias

the positive terminal of the battery. Hence, the width of the depletion region which is depleted of mobile charge carriers increases. Thus, the electric field produced by applied reverse bias, is in the same direction as the electric field of the potential barrier. Hence, the resultant potential barrier is increased which prevents the flow of majority carriers in both directions. Therefore, theoretically no current should flow in the external circuit. But in practice, a very small current of the order of a few microamperes flows under reverse bias as shown in Fig. 1.9. Electrons forming covalent bonds of the semiconductor atoms in the P- and N-type regions may absorb sufficient energy from heat and light to cause breaking of some covalent bonds. Hence electron-hole pairs are continually produced in both the regions. Under the reverse bias condition, the thermally generated holes in the P-region are attracted towards the negative terminal of the battery and the electrons in the N-region are attracted towards the positive terminal of the battery. Consequently, the minority carriers, electrons in the P-region and holes in the N-region, wander over to the junction and flow towards their majority carrier side giving rise to a small reverse current. This current is known as *reverse saturation current*, I_o . The magnitude of reverse saturation current mainly depends upon junction temperature because the major source of minority carriers is thermally broken covalent bonds.

For large applied reverse bias, the free electrons from the N-type moving towards the positive terminal of the battery acquire

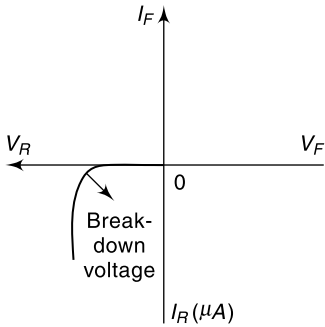


Fig. 1.9 V - I Characteristics Under Reverse Bias

sufficient energy to move with high velocity to dislodge valence electrons from semiconductor atoms in the crystal. These newly liberated electrons, in turn, acquire sufficient energy to dislodge other parent electrons. Thus, a large number of free electrons are formed which is commonly called as an avalanche of free electrons. This leads to the breakdown of the junction leading to very large reverse current. The reverse voltage at which the junction breakdown occurs is known as *Breakdown Voltage*, V_{BD} .

1.3.4 Limiting Values of PN Junction Diode

The PN junction diode will perform satisfactorily only if it is operated within certain limiting values. They are:

(a) Maximum forward current It is the highest instantaneous current under forward bias condition that can flow through the junction.

(b) Peak inverse voltage (PIV) It is the maximum reverse voltage that can be applied to the PN junction. If the voltage across the junction exceeds PIV, under reverse bias condition, the junction gets damaged.

(c) Maximum power rating It is the maximum power that can be dissipated at the junction without damaging the junction. Power dissipation is the product of voltage across the junction and current through the junction.

1.4 PN JUNCTION AS A DIODE

Figure 1.10 shows the current–voltage characteristics of a PN junction. The characteristics of the PN junction vary enormously depending upon the polarity of the applied voltage. For a forward-bias voltage, the current increases exponentially with the increase of voltage. A small change in the forward-bias voltage increases the corresponding forward-bias current by orders of magnitude and hence the forward-bias PN junction will have a very small resistance. The level of current flowing across a forward-biased PN junction largely depends

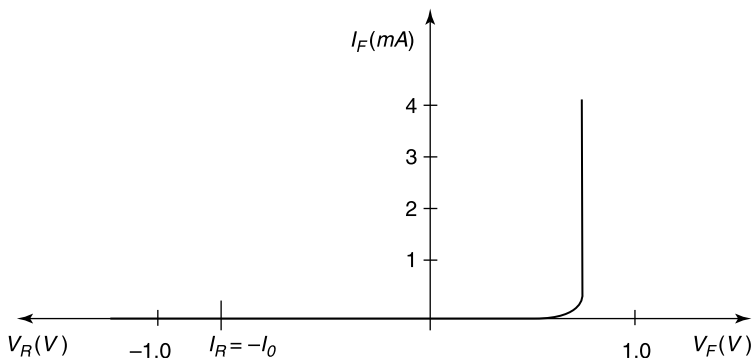


Fig. 1.10 Ideal I - V Characteristics of a PN Junction Diode

upon the junction area. In the reverse-bias direction, the current remains small, i.e., almost zero, irrespective of the magnitude of the applied voltage and hence the reverse-bias PN junction will have a high resistance. The reverse-bias current depends on the junction area, temperature and type of semiconductor material.

The semiconductor device that displays these I - V characteristics is called a PN junction diode. Figure 1.11 shows the PN junction diode with forward-bias and reverse-bias and their circuit symbols. The metal contacts are indicated with which the homogeneous P-type and N-type materials are provided. Thus two metal-semiconductor junctions, one at each end of the diode, are introduced. The contact potential across these junctions is approximately independent of the direction and magnitude of the current. A contact of this type is called an *ohmic contact*, which has low resistance. In the forward-bias, a relatively large current is produced by a fairly small applied voltage. In the reverse-bias, only a very small current, ranging from nanoamps to microamps is produced. The diode can be used as a voltage controlled switch, i.e., OFF for a reverse-bias voltage and ON for a forward-bias voltage.

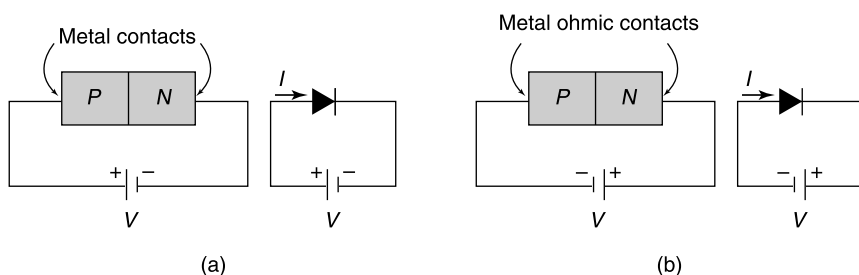


Fig. 1.11 (a) Forward-Biased PN Junction Diode and its Circuit Symbol and
(b) Reverse-Biased PN Junction Diode and its Circuit Symbol

When a diode is reverse-biased by atleast $0.1V$, the diode current is $I_R = -I_o$. As the current is in the reverse direction and is a constant, it is called the diode *reverse saturation current*. Real diodes exhibit reverse-bias currents that are considerably larger than I_o . This additional current is called a *generation current* which is due to electrons and holes being generated within the space-charge region. A typical value of I_o may be 10^{-14} A and a typical value of reverse-bias current may be 10^{-9} A.

1.5

ENERGY BAND STRUCTURE OF OPEN CIRCUITED PN JUNCTION

Consider that a PN junction has P-type and N-type materials in close physical contact at the junction on an atomic scale. Hence, the energy band diagrams of these two regions undergo relative shift to equalise the Fermi level. The Fermi level E_F should be constant throughout the specimen at equilibrium. The distribution of electrons or holes in allowed energy states is dependent on

the position of the Fermi level. If this is not so, electrons on one side of the junction would have an average energy higher than those on the other side, and this causes transfer of electrons and energy until the Fermi levels on the two sides get equalised. However, such a shift does not disturb the relative position of the conduction band, valence band and Fermi level in any region. Equalisation of Fermi levels in the P and N materials of a PN junction is similar to equalisation of levels of water in two containers on being joined together.

The energy band diagram for a PN junction is shown in Fig. 1.12, where the Fermi level E_F is closer to the conduction band edge E_{cn} in the N-type material while it is closer to the valence band edge E_{vp} in the P-type material. It is clear that the conduction band edge E_{cp} in the P-type material is higher than the conduction band edge E_{cn} in the N-type material. Similarly, the valence band edge E_{vp} in the P-type material is higher than the valence band edge E_{vn} in the N-type material. As illustrated in Fig. 1.12, E_1 and E_2 indicate the shifts in the Fermi level from the intrinsic conditions in the P and N materials respectively. Then the total shift in the energy level E_o is given by

$$E_o = E_1 + E_2 = E_{cp} - E_{cn} = E_{vp} - E_{vn}$$

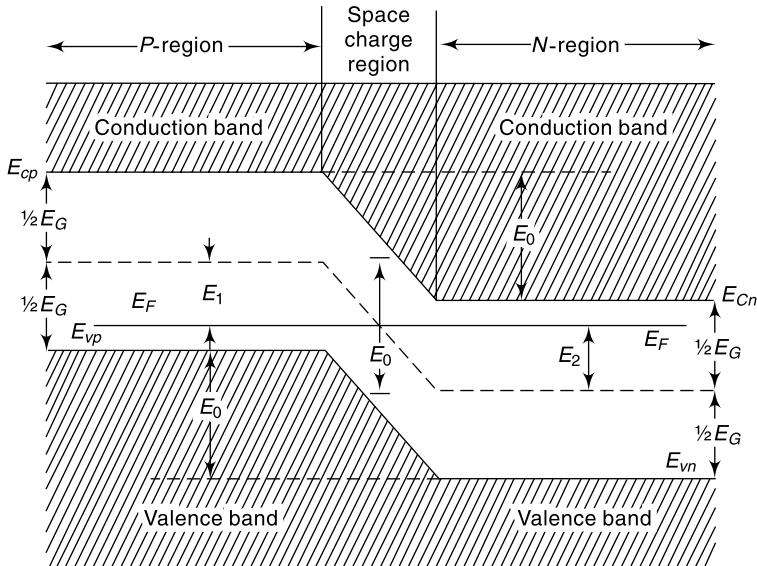


Fig. 1.12 Energy Band Structure

This energy E_o (in eV) is the potential energy of the electrons at the PN junction, and is equal to qV_o , where V_o is the contact potential (in Volts) or contact difference of potential or the barrier potential.

Contact Difference of Potential A contact difference of potential exists across an open circuited PN junction. We now proceed to obtain an expression for E_o . From Fig. 1.12, we find that

$$E_F - E_{vp} = \frac{1}{2} E_G - E_1 \quad (1.1)$$

$$E_{cn} - E_F = \frac{1}{2} E_G - E_2 \quad (1.2)$$

Combining Eqs (1.1) and (1.2), we get

$$E_o = E_1 + E_2 = E_G - (E_{cn} - E_F) - (E_F - E_{vp}) \quad (1.3)$$

We know that

$$np = N_C N_V e^{-E_G/kT} \text{ and}$$

$$np = n_i^2 \text{ (Mass-action law)}$$

From the above equations, we get

$$E_G = kT \ln \frac{N_C N_V}{n_i^2} \quad (1.4)$$

We know that for N-type material $E_F = E_C - kT \ln \frac{N_C}{N_D}$. Therefore, from this equation, we get

$$E_{cn} - E_F = kT \ln \frac{N_C}{n_n} = kT \ln \frac{N_C}{N_D} \quad (1.5)$$

Similarly, for P-type material, $E_F = E_V + kT \ln \frac{N_V}{N_A}$. Therefore, from this equation, we get

$$E_F - E_{vp} = kT \ln \frac{N_V}{p_p} = kT \ln \frac{N_V}{N_A} \quad (1.6)$$

Substituting from Eqs (1.4), (1.5) and (1.6) into Eq. (1.3), we get

$$\begin{aligned} E_o &= kT \left[\ln \frac{N_C N_V}{n_i^2} - \ln \frac{N_C}{N_D} - \ln \frac{N_V}{N_A} \right] \\ &= kT \ln \left[\frac{N_C N_V}{n_i^2} \times \frac{N_D}{N_C} \times \frac{N_A}{N_V} \right] \\ &= kT \ln \frac{N_D N_A}{n_i^2} \end{aligned} \quad (1.7)$$

As $E_o = qV_o$, then the contact difference of potential or barrier voltage is given by

$$V_o = \frac{kT}{q} \ln \frac{N_D N_A}{n_i^2}$$

In the above equations, E is in electron Volts and k is in electron volt per degree Kelvin. The contact difference of potential V_o is expressed in volt and is numerically equal to E_o . From Eq. (1.7), we note that E_o (hence V_o) depends upon the equilibrium concentrations and not on the charge density in the transition region.

An alternative expression for E_o may be obtained by substituting the equations of $n_n \approx N_D$, $P_n = \frac{n_i^2}{N_D}$, $n_p \approx N_A$ and $n_p = \frac{n_i^2}{N_A}$ into Eq. (1.7).

Then we get

$$E_o = kT \ln \frac{P_{po}}{P_{no}} = kT \ln \frac{n_{no}}{n_{po}} \quad (1.8)$$

where subscript o represents the thermal equilibrium condition.

Example 1.1 (a) The resistivities of the P-region and N-region of a germanium diode are 6 W-cm and 4 W-cm, respectively. Calculate the contact potential V_o and potential energy barrier E_o . (b) If the doping densities of both P- and N-regions are doubled, determine V_o and E_o . Given that $q = 1.6 \times 10^{-19} \text{ C}$, $n_i = 2.5 \times 10^{13} / \text{cm}^3$, $m_p = 1800 \text{ cm}^2/\text{V-s}$, $m_n = 3800 \text{ cm}^2/\text{V-s}$ and $V_T = 0.026 \text{ V}$ at 300 K.

Solution (a) Resistivity, $\rho = \frac{1}{\sigma} = \frac{1}{N_A q \mu_p} 6 \Omega\text{-cm}$

$$\text{Therefore, } N_A = \frac{1}{6 q \mu_p} = \frac{1}{6 \times 1.6 \times 10^{-19} \times 1800} = 0.579 \times 10^{15} / \text{cm}^3$$

$$\text{Similarly, } N_D = \frac{1}{4 q \mu_n} = \frac{1}{4 \times 1.6 \times 10^{-19} \times 3800} = 0.579 \times 10^{15} / \text{cm}^3$$

Therefore,

$$V_o = V_T \ln \frac{N_D N_A}{n_i^2} = 0.026 \ln \frac{0.579 \times 0.411 \times 10^{30}}{(2.5 \times 10^{13})^2} = 0.1545 \text{ V}$$

Hence, $E_o = 0.1545 \text{ eV}$.

$$(b) V_o = 0.026 \ln \frac{2 \times 0.579 \times 10^{15} \times 2 \times 0.411 \times 10^{15}}{(2.5 \times 10^{13})^2} = 0.1906 \text{ V}$$

Therefore, $E_o = 0.19006 \text{ eV}$.

1.6 DIODE EQUATION

Let us now derive the expression for the total current as a function of applied voltage assuming that the width of the depletion region is zero. When a forward bias is applied to a diode, holes are injected from the P-side into the N-side. Due to this, the concentration of holes in the N-side (p_n) is increased from its

thermal equilibrium value (p_{no}) and injected hole concentration [$P_n(x)$] decreases exponentially with respect to distance (x)

$$P_n(x) = p_n - p_{no} = P_n(0)e^{-x/L_p}$$

where L_p is the diffusion length for holes in the N-material.

$$p_n(x) = p_{no} + P_n(0)e^{-x/L_p} \quad (1.9)$$

Injected hole concentration at $x = 0$ is

$$P_n(0) = p_n(0) - p_{no} \quad (1.10)$$

These several components of hole concentration in the N-side of a forward biased diode are shown in Fig. 1.13, in which the density $p_n(x)$ decreases exponentially with distance (x).

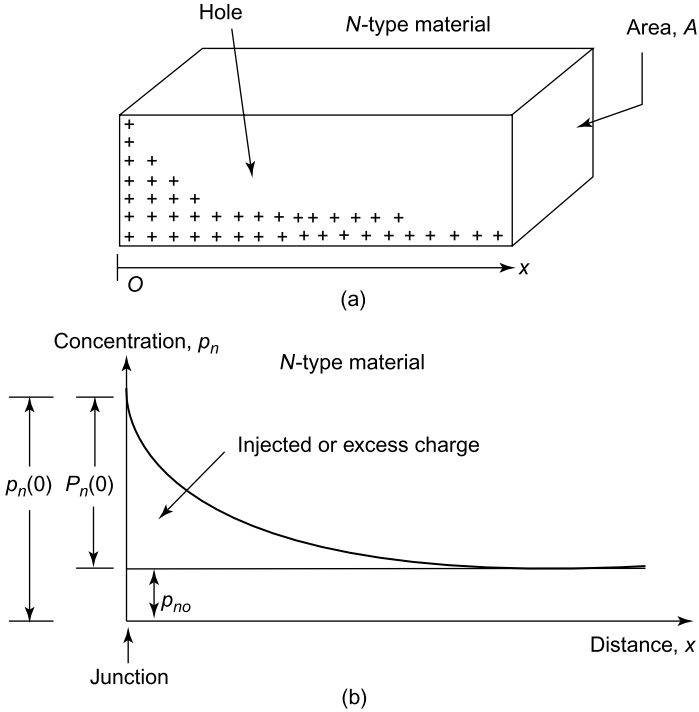


Fig. 1.13 (a) Excess Hole Concentration Varying Along the Axis in an N-type Semiconductor Bar, and (b) The Resulting Diffusion Current

Let p_p and p_n be the hole concentration at the edges of the space charge in the P- and N-sides, respectively. Let $V_B (= V_o - V)$ be the effective barrier potential across the depletion layer. Then

$$p_p = p_n e^{V_B/V_T} \quad (1.11)$$

where V_T is the volt-equivalent of temperature.

This is the Boltzmann's relation of kinetic gas theory. This equation is valid as long as the hole current is small compared with diffusion or drift current. This condition is called Low level injection.

Under open circuit condition (i.e., $V = 0$), $p_p = p_{po}$, $p_n = p_{no}$ and $V_B = V_o$. Equation (1.11) can be changed into

$$p_{po} = p_{no} e^{V_o / V_T} \quad (1.12)$$

Under forward bias condition let V be the applied voltage, then the effective barrier voltage

$$V_B = V_o - V$$

The hole concentration throughout the P-side is constant and equal to the thermal equilibrium value ($p_p = p_{po}$). The hole concentration varies exponentially with distance into the N-side.

$$\text{At } x = 0, p_n = p_n(0)$$

Equation (1.11) can be changed into

$$p_{po} = p_n(0) e^{(V_o - V) / V_T} \quad (1.13)$$

Comparing Eqs (1.12) and (1.13).

$$P_n(0) = P_{no} e^{V / V_T}$$

This boundary condition is called the *law of the junction*. Substituting this into Eq. (1.10), we get

$$P_n(0) = p_{no} (e^{V / V_T} - 1) \quad (1.14)$$

The diffusion hole current in the N-side is

$$\begin{aligned} I_{pn}(x) &= -AeD_p \frac{dp_n(x)}{dx} \\ &= -AeD_p \frac{d}{dx} \left[p_{no} + P_n(0) e^{-x/L_p} \right] \\ &= \frac{AeD_p P_n(0)}{L_p} e^{-x/L_p} \end{aligned}$$

From this equation, it is evident that the injected hole current decreases exponentially with distance.

Forward Currents The hole current crossing the junction into the N-side with $x = 0$ is

$$I_{pn}(0) = \frac{AeD_p P_n(0)}{L_p} = \frac{AeD_p p_{no}}{L_p} (e^{V / V_T} - 1)$$

The electron current crossing the junction into the P-side with $x = 0$ is

$$I_{np}(0) = \frac{AeD_n N_p(0)}{L_n} = \frac{AeD_n n_{po}}{L_n} (e^{V/V_T} - 1)$$

The total diode current,

$$I = I_{pn}(0) + I_{np}(0) = I_o (e^{V/V_T} - 1)$$

where $I_o = \frac{AeD_p p_{no}}{L_p} + \frac{AeD_n n_{po}}{L_n}$ = reverse saturation current.

If we consider carrier generation and recombination in the space-charge region, the general equation of the diode current is approximately given by

$$I = I_o [e^{(V/\eta V_T)} - 1]$$

where V = external voltage applied to the diode, and η = a constant, 1 for germanium and 2 for silicon.

The diode current equation relating the voltage V and current I is given by

$$I = I_o [e^{(V/\eta V_T)} - 1]$$

where I = diode current

I_o = diode reverse saturation current at room temperature

V = external voltage applied to the diode

η = a constant, 1 for germanium and 2 for silicon

$V_T = kT/q = T/11600$, volt-equivalent of temperature. i.e., thermal voltage, where

k = Boltzmann's constant (1.38066×10^{-23} J/K)

q = charge of the electron (1.60219×10^{-19} C)

T = temperature of the diode junction (K) = ($^{\circ}\text{C} + 273^{\circ}$)

At room temperature, ($T = 300$ K), $V_T = 26$ mV. Substituting this value in the current equation, we get

$$I = I_o [e^{40V/\eta} - 1]$$

Therefore, for germanium diode, $I = I_o [e^{40V} - 1]$, since $\eta = 1$ for germanium. For silicon diode, $I = I_o [e^{20V} - 1]$, since $\eta = 2$ for silicon.

If the value of applied voltage is greater than unity, then the equation of diode current for germanium,

$$I = I_o(e^{40V})$$

and for silicon,

$$I = I_o(e^{20V})$$

When the diode is reverse biased, its current equation may be obtained by changing the sign of the applied voltage V . Thus, the diode current with reverse bias is

$$I = I_o [e^{(-V/\eta V_T)} - 1]$$

If $V \gg V_T$, then the term $e^{(-V/\eta V_T)} \ll 1$, therefore $I \approx -I_o$, termed as reverse saturation current, which is valid as long as the external voltage is below the breakdown value.

Reverse Saturation Currents We know that $P_n = \frac{n_i^2}{N_D}$ and $n_p = \frac{n_i^2}{N_A}$. Applying these relationships in the above equation or reverse saturation current, I_o , we get

$$I_o = Ae \left[\frac{D_p}{L_p N_D} + \frac{D_n}{L_n N_A} \right] n_i^2$$

where $n_i^2 = A_o T^3 e^{\frac{-E_{Go}}{kT}} = A_o T^3 e^{\frac{-V_{Go}}{V_T}}$, where V_{Go} is a voltage which is numerically equal to the forbidden gap energy E_{Go} in electron volts.

For a germanium diode, the diffusion constants D_p and D_n vary approximately inversely proportional to T . Hence, the temperature dependence of I_o is

$$I_o = K_1 T^2 e^{\frac{-V_{Go}}{V_T}}$$

where K_1 is a constant independent of temperature.

For a silicon diode, I_o is proportional to n_i instead of n_i^2 . Hence,

$$I_o = K_2 T^{\frac{3}{2}} e^{\frac{-V_{Go}}{2V_T}}$$

where K_2 is a constant independent of temperature.

Example 1.2 When a reverse bias is applied to a germanium PN junction diode, the reverse saturation current at room temperature is $0.3 \mu A$. Determine the current flowing in the diode when $0.15 V$ forward bias is applied at room temperature.

Solution Given $I_o = 0.3 \times 10^{-6} A$ and $V_F = 0.15 V$

The current flowing through the PN diode under forward bias is

$$\begin{aligned} I &= I_o (e^{40 V_F} - 1) \\ &= 0.3 \times 10^{-6} (e^{40 \times 0.15} - 1) \\ &= 120.73 \mu A \end{aligned}$$

Example 1.3 Determine the value of forward current in the case of a PN junction diode, with $I_o = 10$ micro amperes and $V_F = 0.8 V$ at $T = 300 K$. Assume silicon diode. [Dec. 2005, June 2009]

Solution Given $I_o = 10 \mu A$, $V_F = 0.8 V$ and $T = 300 K$

The forward current of a PN junction diode is $I = I_o \left[e^{\frac{V}{\eta V_T}} - 1 \right]$, where $\eta = 2$ for silicon

$$\text{Therefore, } I = 10 \times 10^{-6} \left[e^{\frac{0.8}{2 \times 26 \times 10^{-3}}} - 1 \right] = 48.02 A.$$

Example 1.4 The reverse saturation current of a silicon PN junction diode is $10 \mu\text{A}$. Calculate the diode current for the forward-bias voltage of 0.6 V at 25°C .

Solution Given $V_F = 0.6 \text{ V}$, $T = 273 + 25 = 298 \text{ K}$

$I_o = 10 \mu\text{A} = 1 \times 10^{-5} \text{ A}$ and $\eta = 2$ for silicon,

The volt-equivalent of the temperature (T) is

$$V_T = \frac{T}{11,600} = \frac{298}{11,600} = 25.7 \times 10^{-3} \text{ V}$$

Therefore, the diode current, $I = I_o \left(e^{\frac{V_F}{\eta V_T}} - 1 \right)$

$$I = 10^{-5} \left(\frac{0.6}{e^{2 \times 25.7 \times 10^{-3}} - 1} \right)$$

$$= 1.174 \text{ A}$$

Example 1.5 The diode current is 0.6 mA when the applied voltage is 400 mV , and 20 mA when the applied voltage is 500 mV . Determine η .

Assume $\frac{kT}{q} = 25 \text{ mV}$.

Solution The diode current, $I = I_o \left(e^{\frac{qV}{\eta kT}} - 1 \right)$

$$0.6 \times 10^{-3} = I_o \left(e^{\frac{qV}{\eta kT}} - 1 \right) = I_o e^{\frac{qV}{\eta kT}}$$

$$\text{Therefore,} \quad = I_o \cdot e^{\frac{400}{25\eta}} = I_o \cdot e^{\frac{16}{\eta}} \quad (1.15)$$

$$\text{Also,} \quad 20 \times 10^{-3} = I_o \cdot e^{\frac{500}{25\eta}} = I_o \cdot e^{\frac{20}{\eta}} \quad (1.16)$$

Dividing Eq. (1.16) by (1.15), we get

$$\frac{20 \times 10^{-3}}{0.6 \times 10^{-3}} = \frac{I_o \cdot e^{\frac{20}{\eta}}}{I_o \cdot e^{\frac{16}{\eta}}}$$

$$\text{Therefore,} \quad \frac{100}{3} = e^{\frac{4}{\eta}}$$

Taking natural logarithms on both sides, we get

$$\log_e \frac{100}{3} = \frac{4}{\eta}$$

$$3.507 = \frac{4}{\eta}$$

Therefore, $\eta = \frac{4}{3.507} = 1.14$

1.7

IDEAL VERSUS PRACTICAL—RESISTANCE LEVELS (STATIC AND DYNAMIC)

An ideal diode should offer zero resistance in forward bias and infinite resistance in the reverse bias. But in practice no diode can act as an ideal diode, i.e. an actual diode does not behave as a perfect conductor when forward biased and as a perfect insulator when reverse biased. Let us consider four resistances of the diode (a) dc or static resistance, (b) ac or dynamic resistance, (c) average AC resistance and (d) reverse resistance.

DC or Static Resistance (R_F) It is defined as the ratio of the voltage to the current, V/I , in the forward bias characteristics of the PN junction diode. In the forward bias characteristics of the diode as shown in Fig. 1.14, the DC or static resistance (R_F) at the operating point can be determined by using the corresponding levels of voltage V and current I , i.e. $R_F = \frac{V}{I}$. Here, the DC resistance is independent of the shape of the characteristics in the region surrounding the point of interest. The DC resistance levels at the knee and below will be greater than the resistance levels obtained for the characteristics above the knee. Hence, the DC resistance will be low when the diode current is high. As the static resistance varies widely with V and I , it is not a useful parameter.

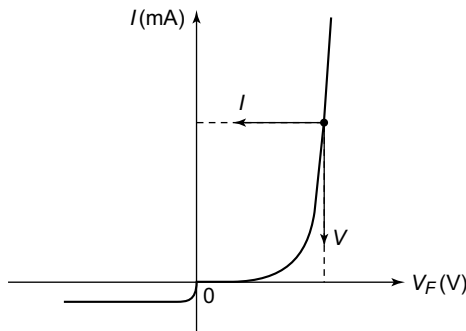


Fig. 1.14 Static Resistance

AC or Dynamic Resistance (r_f) It is defined as the reciprocal of the slope of the volt-ampere characteristics.

$$r_f = \frac{\text{change in voltage}}{\text{resulting change in current}} = \frac{\Delta V}{\Delta I}$$

A straight line drawn tangent to the curve through the Quiescent Point (Q -point) as shown in Fig. 1.15(a) will define a specific change in voltage and current which may be used to determine the AC or dynamic resistance for this region of the diode characteristics. As shown in Fig. 1.15(b), for a small change in voltage, there will be a corresponding change in current, which is equidistant to either side of the Q -point. Hence, the AC or dynamic resistance is determined as $r_f = \frac{\Delta V}{\Delta I}$.

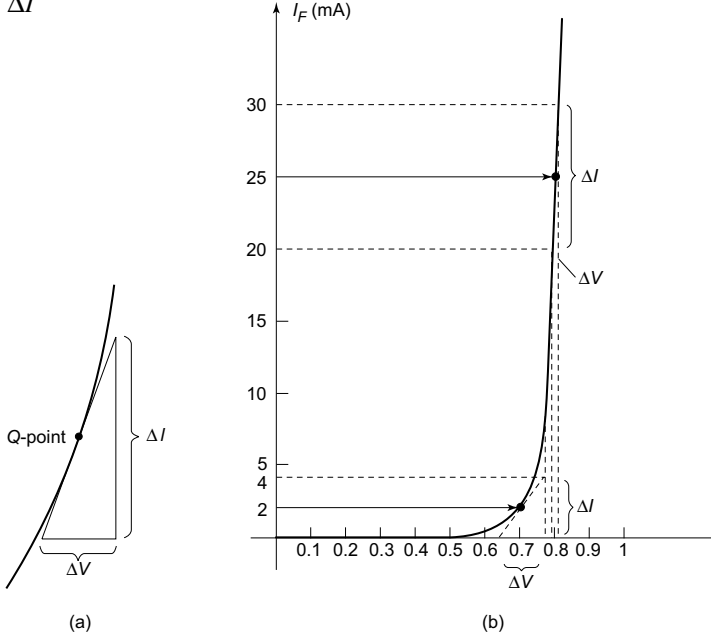


Fig. 1.15 Dynamic Resistance

The derivative of a function at a point is equal to the slope of the tangent line drawn at that point. The Shockley's equation for the forward and reverse bias regions is defined by

$$I = I_o (e^{V/\eta V_T} - 1)$$

Taking the derivative of the above equation w.r.t the applied voltage, V , we get

$$\begin{aligned} \frac{dI}{dV} &= \frac{d}{dV} [I_o (e^{V/\eta V_T} - 1)] \\ &= I_o \left[\frac{1}{\eta V_T} \cdot e^{V/\eta V_T} \right] \\ &= \frac{I_o e^{V/\eta V_T}}{\eta V_T} \end{aligned}$$

$$= \frac{I + I_o}{\eta V_T}$$

Generally, $I \gg I_o$ in the vertical-slope section of the characteristics. Therefore,

$$\frac{dI}{dV} \cong \frac{I}{\eta V_T}$$

Therefore, $\frac{dV}{dI} = r_f = \frac{\eta V_T}{I}$

The dynamic resistance varies inversely with current, i.e. $r_f = \frac{\eta V_T}{I}$, where $V_T = T/1600$, the volt equivalent of temperature (T) of the diode junction ($^{\circ}\text{K}$) and η is a constant whose value is equal to 1 for Germanium and 2 for Silicon diodes. At room temperature, $V_T = 26 \text{ mV}$.

AC resistance of a diode is the sum of bulk resistance r_b and junction resistance r_f . Bulk resistance (r_b) is the sum of ohmic resistances of the P- and N-type semiconductors.

Average AC Resistance It is the resistance associated with the device for the region if the input signal is sufficiently large to produce a wide range of the characteristics as shown in Fig. 1.16. Therefore,

$$r_{av} = \frac{\Delta V}{\Delta I} \Big|_{\text{point to point}}$$

As with the DC and AC resistance levels, the lower the level of currents used to determine the average resistance, the higher is the resistance level.

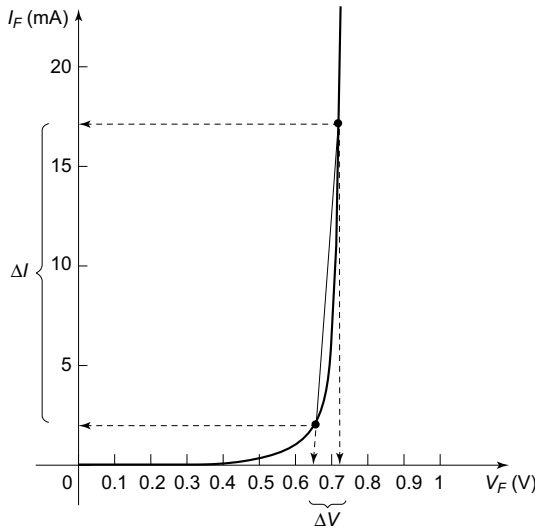


Fig. 1.16 Average AC Resistance

Reverse Resistance It is the resistance offered by the PN junction diode under reverse bias condition. It is very large compared to the forward resistance, which is in the range of several $M\Omega$.

Example 1.6 Determine the forward resistance of a PN junction diode, when the forward current is 5 mA at $T = 300$ K. Assume silicon diode.

Solution Given For a silicon diode, the forward current, $I = 5$ mA, $T = 300$ K
Forward resistance of a PN junction diode, $r_f = \frac{\eta V_T}{I}$, where $V_T = \frac{T}{11,600}$ and $\eta = 2$ for silicon

$$\text{Therefore, } r_f = \frac{2 \times \frac{T}{11,600}}{5 \times 10^{-3}} = \frac{2 \times 300}{11,600 \times 5 \times 10^{-3}} = 10.34 \Omega$$

Example 1.7 Find the value of dc resistance and ac resistance of a Germanium junction diode at 25°C with $I_o = 25 \mu\text{A}$ and at an applied voltage of 0.2 V across the diode. [JNTU Dec. 2004, April/May 2007]

Solution Given $I_o = 25 \mu\text{A}$, $T = 25^\circ\text{C} = 298$ K and $V = 0.2$ Volts

$$I = I_o \left(e^{\frac{V}{\eta V_T}} - 1 \right) = 25 \times 10^{-6} \left(e^{\frac{0.2}{26 \times 10^{-3}}} - 1 \right) = 54.79 \text{ mA}$$

$$\text{DC resistance } R_F = \frac{V}{I} = \frac{0.2}{54.79 \times 10^{-3}} = 3.65 \Omega$$

$$\text{For germanium, } \eta = 1, V_T = \frac{KT}{q} = 25.71 \text{ mV}$$

$$\text{AC resistance } r_F = \frac{\eta V_T}{I} = \frac{25.71 \times 10^{-3}}{54.79 \times 10^{-3}} = 0.47 \Omega.$$

Example 1.8 Calculate the dynamic forward and reverse resistances of a PN junction diode when the applied voltage is 0.25 V at $T = 300$ K given $I_o = 2 \mu\text{A}$. [JNTU Dec. 2004, Aug. 2008]

Solution Given $V = 0.25$ V, $T = 300$ K, $I_o = 2 \mu\text{A}$

$$\text{At } T = 300 \text{ K, } V_T = 26 \text{ mV.}$$

Assuming it to be silicon diode, $\eta = 2$.

$$\text{Therefore, } I = I_o \left(e^{\frac{V}{\eta V_T}} - 1 \right) = 2 \times 10^{-6} \left(e^{\frac{0.25}{2 \times 26 \times 10^{-3}}} - 1 \right) = 0.24 \text{ mA}$$

$$r_F = \frac{\eta V_T}{I} = \frac{2 \times 26 \times 10^{-3}}{0.24 \times 10^{-3}} = 216.67 \Omega$$

For germanium diode, $\eta = 1$.

$$I = I_o \left(e^{\frac{V}{\eta V_T}} - 1 \right) = 2 \times 10^{-6} \left(e^{\frac{0.25}{26 \times 10^{-3}}} - 1 \right) = 0.03 \text{ A}$$

$$r_F = \frac{\eta V_T}{I} = \frac{26 \times 10^{-3}}{0.03} = 0.867 \Omega$$

$$\text{Reverse resistance} = \frac{V}{I_o} = \frac{0.25}{2 \times 10^{-6}} = 125 \text{ k}\Omega$$

Example 1.9 A pn-junction diode has a reverse saturation current of $30 \mu\text{A}$ at a temperature of 125°C . At the same temperature, find the dynamic resistance for 0.2 V bias in forward and reverse directions.

[JNTU Aug. 2006, April/May 2007, Feb 2010]

Solution Given The reverse saturation current, $I_o = 30 \times 10^{-6} \text{ A}$ and $V = 0.2 \text{ V}$

$$\text{We know that the dynamic resistance} = \frac{\eta V_T}{I_o e^{V/\eta V_T}}$$

$$\text{Here, } \eta = 1 \text{ for germanium and } V_T = \frac{T}{11,600} = \frac{125 + 273}{11,600} = 34.3 \text{ mV}$$

$$\text{Therefore, forward dynamic resistance } r_f = \frac{34.3 \times 10^{-3}}{30 \times 10^{-6} (e^{0.2/34.3 \times 10^{-3}})} = 3.356 \Omega$$

$$\begin{aligned} \text{Reverse dynamic resistance, } r_r &= \frac{\eta V_T}{I_o e^{-V/\eta V_T}} = \frac{34.3 \times 10^{-3}}{30 \times 10^{-6} (e^{-0.2/34.3 \times 10^{-3}})} \\ &= 389.5 \text{ k}\Omega \end{aligned}$$

Example 1.10 If two similar Germanium diodes are connected back to back and the voltage V is impressed upon, calculate the voltage across each diode and current through each diode. Assume similar value of $I_o = 1 \mu\text{A}$ for both the diodes and $\eta = 1$.

[JNTU Dec. 2004]

Solution The arrangement is shown in Fig. 1.17.

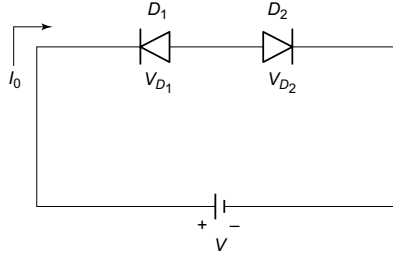


Fig. 1.17

As D_1 is reverse biased, the total current flowing in the circuit is $I_0 = 1 \mu\text{A}$. The diode D_2 is forward biased and its forward current is equal to the reverse current $I_0 = 1 \mu\text{A}$, which can flow as D_1 is reverse biased.

For diode D_2 , $I = I_0 = 1 \mu\text{A}$ and voltage across D_2 is V_{D2} .

$$\text{Therefore, } I = I_0 \left[e^{V/\eta V_T} - 1 \right]$$

$$\text{or } I_0 = I_0 \left[e^{V_{D2}/\eta V_T} - 1 \right]$$

$$\text{Hence, } e^{V_{D2}/\eta V_T} = 1 + 1 = 2$$

$$\frac{V_{D2}}{\eta V_T} = \ln 2$$

$$V_{D2} = \eta V_T \times \ln 2 = 1 \times 26 \times 10^{-3} \times 0.6931 = 0.01802 \text{ V}$$

$$\text{Therefore, } V_{D1} = V - V_{D2} = V - 0.01802 \text{ V}$$

The current through the diodes D_1 and D_2 is $I = I_0 = 1 \mu\text{A}$

1.8

TRANSITION OR SPACE CHARGE (OR DEPLETION REGION) CAPACITANCE (C_T)

Under reverse bias condition, the majority carriers move away from the junction, thereby uncovering more immobile charges. Hence the width of the space-charge layer at the junction increases with reverse voltage. This increase in uncovered charge with applied voltage may be considered a capacitive effect. The parallel layers of oppositely charged immobile ions on the two sides of the junction form the capacitance, C_T , which is expressed as

$$C_T = \left| \frac{dQ}{dV} \right|$$

where dQ is the increase in charge caused by a change in voltage dV . A change in voltage dV in a time dt will result in a current $I = dQ/dt$ given by

$$I = C_T \frac{dV}{dt}$$

Therefore C_T is important while considering a diode or a transistor as a circuit element. The quantity C_T is called the transition, space-charge, barrier or depletion region capacitance.

1.8.1 Step-graded Junction

A PN junction is formed from a single-crystal intrinsic semiconductor by dop ing part of it with acceptor impurities and the remainder with donors. A junction between P-type and N-type materials may be fabricated in a variety of ways.

The change in impurity concentration from P to N-type semiconductor occurs in a very short length, typically much less than $1\ \mu\text{m}$. In an abrupt PN junction, there is a sudden step change from acceptor ions on one side to donor ions on the other side. Such a junction is fabricated by placing trivalent indium against N-type germanium and heating the combination to a high temperature for a short time. Since some of the indium dissolves into the germanium, the N-type germanium is changed into P-type at the junction. Such a step-graded junction is called an *alloy*, or *fusion, junction*. A step-graded junction is also formed between emitter and base of an integrated transistor.

A diffused junction is graded in which case the donor and acceptor concentrations are functions of distance across the junction. Then the acceptor density, N_A , gradually decreases and the donor density, N_D , gradually increases till $N_A = N_D$ is reached. Therefore, N_D increases and N_A decreases to zero.

It is not necessary for the abrupt junction to be symmetrical, that is, the doping concentrations at either side of the junction are dissimilar.

As shown in Fig. 1.18, consider a PN diode which is asymmetrically doped at the junction. Since the net charge is zero, then $eN_AW_p = eN_DW_n$. If $N_A \ll N_D$, Then $W_p \gg W_n$.

The relationship between potential and charge density is given by the Poisson's equation, $\frac{d^2V}{dx^2} = \frac{eN_A}{\epsilon}$, where ϵ is the permittivity of the semiconductor.

Integrating the above equation twice,

$$\iint d^2V = \iint \frac{eN_A}{\epsilon} dx^2$$

Therefore,
$$V = \frac{eN_Ax^2}{2\epsilon}$$

At $x = W_p \approx W$, $V = V_B$, the barrier potential that appears across the uncovered acceptor ions. Thus

$$V_B = \frac{eN_AW^2}{2\epsilon} \quad (1.17)$$

Here $V_B = V_o - V$, where V is a negative number for an applied reverse bias and V_o is the contact potential. Hence, the width of the depletion layer increases with applied reverse voltage, i.e. $V_b \propto W^2$. Therefore, $W \propto \sqrt{V_B}$.

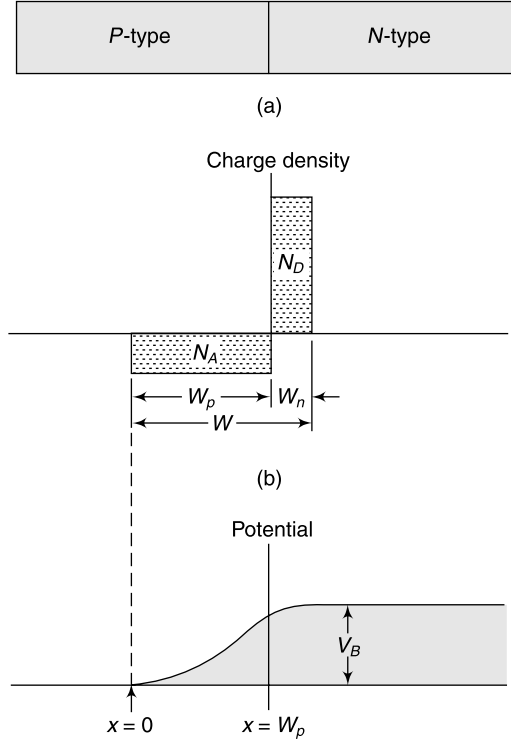


Fig. 1.18 Charge Density and Potential Variation at an Alloy PN Junction

The total charge density of a P-type material with area of the junction A is given by

$$Q = eN_A W A$$

Differentiating the above equation w.r.t. V , we get

$$C_T = \left| \frac{dQ}{dV} \right| = A e N_A \left| \frac{dW}{dV} \right| \quad (1.18)$$

Differentiating Eq. (1.17) w.r.t. V , we get

$$1 = \frac{e N_A 2W}{2\epsilon} \left| \frac{dW}{dV} \right|$$

Therefore,
$$\left| \frac{dW}{dV} \right| = \frac{\epsilon}{e N_A W} \quad (1.19)$$

Substituting Eq. (1.19) in Eq. (1.18), we have

$$C_T = \left| \frac{dQ}{dV} \right| = A e N_A \frac{\epsilon}{e N_A W}$$

Therefore, $C_T = \frac{\epsilon A}{W}$

Here ϵ is the permittivity of the material, A is the cross-sectional area of the junction and W is the width of the depletion layer over which the ions are uncovered. The depletion width, W , is given by

$$W = \left[\frac{2 \epsilon_o \epsilon_r (V_o - V)}{q} \left(\frac{N_A + N_D}{N_A N_D} \right) \right]^{1/2}$$

where V is the applied voltage and V_o is the barrier potential, or the contact potential.

When no external voltage is applied, i.e. $V = 0$, the width of the depletion region of a PN junction diode is of the order of 0.5 microns. The movement of majority carriers across the junction causes opposite charges to be stored at this distance W apart. This depletion region acts as a dielectric between the two conducting P- and N-regions. Therefore, these regions act as a parallel plate capacitor whose transition capacitance C_T is approximately 20 pF with no external bias.

When forward bias $+V$ is applied, the effective barrier potential, $V_B = [V_o - (+V)]$, is lowered and hence the width of the depletion region W decreases and C_T increases. Under reverse bias condition, the majority carriers move away from the junction, thereby uncovering more immobile charges. Now the effective barrier potential, $V_B = [V_o - (-V)]$, is increased and hence, W increases with reverse voltage and C_T decreases correspondingly. The values of C_T range from 5 to 200 pF, the larger values being for the high power diodes. This property of voltage variable capacitance with the reverse bias appears in varactors, vari-caps or volta-caps.

1.9 DIFFUSION (OR STORAGE) CAPACITANCE (C_D)

The capacitance that exists in a forward biased junction is called a diffusion or storage capacitance (C_D), whose value is usually much larger than C_T , which exists in a reverse-biased junction. This is also defined as the rate of change of injected charge with applied voltage, i.e. $C_D = \frac{dQ}{dV}$, where dQ represents the change in the number of minority carriers stored outside the depletion region when a change in voltage across the diode, dV , is applied.

Calculation of C_D Let us assume that the P material in one side of the diode is heavily doped in comparison with the N-side. Since the holes move from the P to the N-side, the hole current $I \approx I_{pn}(0)$.

The excess minority charge Q existing on the N-side is given by

$$Q = \int_0^{\infty} A e P_n(0) e^{-x/L_p} dx = \left[\frac{A e P_n(0) e^{-x/L_p}}{1/L_p} \right]_0^{\infty} = L_p A e P_n(0)$$

Differentiating the above equation, we get

$$C_D = \frac{dQ}{dV} = AeL_p \frac{d[P_n(0)]}{dV} \quad (1.20)$$

We know that the diffusion hole current in the N-side is $I_{pn}(x) = \frac{AeD_p P_n(0)}{L_p} e^{-x/L_p}$. The hole current crossing the junction into the N-side with $x = 0$ is $I_{pn}(0) = \frac{AeD_p P_n(0)}{L_p}$.

Therefore,

$$I = \frac{AeD_p P_n(0)}{L_p}$$

$$P_n(0) = \frac{IL_p}{AeD_p}$$

Differentiating the above equation w.r.t. “ V ”, we get

$$\frac{d[P_n(0)]}{dV} = \frac{dI}{dV} \frac{L_p}{AeD_p}$$

Upon substituting in Eq. (1.20), we have

$$C_D = \frac{dQ}{dV} = \frac{dI}{dV} \frac{L_p^2}{D_p}$$

Therefore, $C_D = g\tau$, where $g = \frac{dI}{dV}$ is the diode conductance and $\tau = \frac{L_p^2}{D_p}$ is the mean life time of holes in the N-region.

From diode current equation, $g = \frac{I}{\eta V_T}$.

Therefore,

$$C_D = \frac{\tau I}{\eta V_T}$$

where τ is the mean life time for holes and electrons.

Diffusion capacitance C_D increases exponentially with forward bias or, alternatively, that it is proportional to diode forward current. The values of C_D range from 10 to 1000 pF, the larger values being associated with the diode carrying a larger anode current, I .

The effect of C_D is negligible for a reverse-biased PN junction. As the value of C_D is inversely proportional to frequency, it is high at low frequencies and it decreases with the increase in frequency.

1.10 TEMPERATURE DEPENDENCE OF V-I CHARACTERISTICS OF DIODES

The reverse saturation current I_o is temperature dependent while voltage equivalent of temperature V_T is also temperature dependent. Hence, the diode current involving I_o and V_T is temperature dependent. The overall diode characteristics depends on the temperature.

The dependence of I_o on temperature T is given by

$$I_o = K T^m e^{-V_{Go}/\eta V_T} \quad (1.21)$$

where K = constant independent of temperature (not the Boltzmann's constant)
 $m = 2$ for Ge and 1.5 for Si

and V_{Go} = Forbidden energy gap = 0.785V for Ge and 1.21V for Si.

As temperature increases, the value of I_o increases and hence the diode current increases. To keep diode current constant, it is necessary to reduce the applied voltage V of the diode.

Let us calculate, the rate of change of the applied voltage to keep the diode current constant. For a constant diode current, $\frac{dI}{dT} = 0$. Then a change in voltage has to be calculated.

A diode current equation is given by

$$I = I_o (e^{V/\eta V_T} - 1)$$

Since $I \gg I_o$ for a forward characteristics, we have

$$I = I_o e^{V/\eta V_T} \quad (1.22)$$

Substituting Eq. (1.21) into Eq. (1.22), we get

$$\begin{aligned} I &= K T^m e^{-V_{Go}/\eta V_T} \cdot e^{V/\eta V_T} \\ &= K T^m e^{(V-V_{Go})/\eta V_T} \end{aligned} \quad (1.23)$$

Since $V_T = kT$, where k is Boltzmann's constant,

$$I = K T^m e^{(V-V_{Go})/\eta k T} \quad (1.24)$$

For a constant diode current, $\frac{dI}{dT} = 0$. Hence, differentiating the above equation with respect to T , we get

$$\frac{dI}{dT} = K \left[m T^{m-1} e^{(V-V_{Go})/\eta k T} + T^m e^{(V-V_{Go})/\eta k T} \cdot \frac{d}{dT} \left(\frac{V-V_{Go}}{\eta k T} \right) \right]$$

$$\frac{dI}{dT} = K e^{(V-V_{Go})/\eta k T} \left[m T^{m-1} + \frac{T^m}{\eta k} \left(\frac{T \frac{dV}{dT} - (V-V_{Go}) \times 1}{T^2} \right) \right]$$

$$\frac{dI}{dT} = K e^{(V-V_{Go})/\eta k T} \left[\frac{m T^m}{T} + \frac{T^m}{\eta k T^2} \left(T \frac{dV}{dT} - (V - V_{Go}) \right) \right]$$

Note that V_{Go} is forbidden energy gap at $0^\circ K$ and hence it is a constant from differentiation point of view.

Taking T^m outside and rearranging the above equation, we get

$$\begin{aligned} \frac{dI}{dT} &= K e^{(V-V_{Go})/\eta k T} \times T^m \left[\frac{m \eta k T + \left(T \frac{dV}{dT} - (V - V_{Go}) \right)}{\eta k T^2} \right] \\ &= K e^{(V-V_{Go})/\eta k T} \times \frac{T^m}{\eta k T^2} \left[m \eta k T + \left(T \frac{dV}{dT} - (V - V_{Go}) \right) \right] \end{aligned}$$

Replacing kT with V_T , we get

$$\frac{dI}{dT} = K e^{(V-V_{Go})/\eta k T} \times \frac{T^{m-1}}{\eta V_T} \left[m \eta V_T + \left(T \frac{dV}{dT} - (V - V_{Go}) \right) \right]$$

Now $\frac{dI}{dT} = 0$ for constant diode current. Hence, equating the above equation to zero, we get

$$m \eta V_T + T \frac{dV}{dT} - (V - V_{Go}) = 0$$

$$T \frac{dV}{dT} = V - V_{Go} - m \eta V_T$$

$$\frac{dV}{dT} = \frac{V - (V_{Go} + m \eta V_T)}{T}$$

This is the required change in voltage necessary to keep diode current constant.

Hence for germanium, at cut-in voltage $V = V_\gamma = 0.2 \text{ V}$ and with $m = 2, \eta = 1$, $T = 30^\circ \text{ K}$ and $V_{Go} = 0.785 \text{ V}$ in the above equation, we get

$$\frac{dV}{dT} = \frac{0.2 - (0.785 + 2 \times 1 \times 26 \times 10^{-3})}{300} = -2.12 \text{ mV}/^\circ \text{ C for Ge}$$

The negative sign indicates that the voltage must be reduced at a rate of 2.12 mV per degree change in temperature to keep diode current constant.

Similarly, $\frac{dV}{dT} = -2.3 \text{ mV}/^\circ \text{ C}$, for Si

Practically, the value of $\frac{dV}{dT}$ is assumed to be $-2.5 \text{ mV}/^\circ\text{C}$ for either Ge or Si at room temperature.

$$\text{Thus, } \frac{dV}{dT} = -2.5 \text{ mV}/^\circ\text{C}$$

The negative sign indicates that dV/dT decreases with increase in temperature.

1.10.1 Effect of Temperature on Reverse Saturation Current

To study by what rate I_o changes with respect to temperature, consider Eq. (1.21) again. That is,

$$I_o = KT^m e^{-V_{Go}/\eta V_T}$$

Taking logarithm on both sides, we get

$$\begin{aligned} \ln(I_o) &= \ln(KT^m e^{-V_{Go}/\eta V_T}) \\ &= \ln K + \ln T^m - \frac{V_{Go}}{\eta V_T} \\ &= \ln K + m \ln T - \frac{V_{Go}}{\eta V_T} \end{aligned}$$

Substituting $V_T = kT$, we get

$$\ln(I_o) = \ln K + m \ln T - \frac{V_{Go}}{\eta k T}$$

Differentiating this equation with respect to T, we get

$$\frac{d \ln(I_o)}{dT} = 0 + \frac{m}{T} - \frac{V_{Go}}{\eta k} \cdot \left(-\frac{1}{T^2}\right) = \frac{m}{T} + \frac{V_{Go}}{\eta k T^2}$$

Replacing kT with V_T we have

$$\frac{d[\ln I_o]}{dT} = \frac{m}{T} + \frac{V_{Go}}{\eta T V_T}$$

For germanium, substituting the values of various terms at room temperature, we get

$$\frac{d[\ln I_o]}{dT} = \frac{2}{300} + \frac{0.785}{1 \times 300 \times 26 \times 10^{-3}} = 0.11 \text{ per } ^\circ\text{C}$$

This indicates that I_o increases by 11 % per degree rise in temperature. For silicon, we get

$$\frac{d[\ln I_o]}{dT} = 0.08 \text{ per } ^\circ\text{C}$$

This indicates that I_o increases by 8 % per degree rise in temperature.

Practically it is found that the reverse saturation current I_o increases by 7 % per $^\circ\text{C}$ change in temperature for both silicon and germanium diodes. If at $T^\circ\text{C}$ is 1 μA , then at $(T+1)^\circ\text{C}$, it becomes 1.07 μA and so on. From this, it can be concluded that reverse saturation current approximately doubles i.e 1.07^{10} for every 10°C rise in temperature.

The above result can be mathematically represented as,

$$I_{o2} = \left(2^{\frac{T_2 - T_1}{10}} \right) I_{o1} = \left(2^{\frac{\Delta T}{10}} \right) I_{o1}$$

where I_{o2} is the reverse saturation current at T_2 and I_{o1} is the reverse saturation current at T_1 .

1.10.2 Temperature Dependence of V-I Characteristics

The rise in temperature increases the generation of electron-hole pairs in semiconductors and increases their conductivity. As a result, the current through the PN junction diode increases with temperature as given by the diode current equation,

$$I = I_o \left[e^{(V/\eta V_T)} - 1 \right]$$

The reverse saturation current I_o of diode increases approximately 7 percent/ $^\circ\text{C}$ for both germanium and silicon. Since $(1.07)^{10} \approx 2$, reverse saturation current approximately doubles for every 10°C rise in temperature. Hence, if the temperature is increased at fixed voltage, the current I increases. To bring the current I to its original value, the voltage V has to be reduced. It is found that at room temperature for either germanium or silicon, $\frac{dV}{dT} \approx -2.5 \text{ mV}/^\circ\text{C}$ in order to maintain the current I to a constant value.

At room temperature, i.e at 300 K, the value of barrier voltage or cut-in voltage is about 0.3 V for germanium and 0.7 V for silicon. The barrier voltage is temperature dependent and it decreases by 2 mV/ $^\circ\text{C}$ for both germanium and silicon. This fact may be expressed in mathematical form, which is given by

$$I_{o2} = I_{o1} \times 2^{(T_2 - T_1)/10}$$

where I_{o1} = saturation current of the diode at temperature (T_1), and I_{o2} = saturation current of the diode at temperature (T_2).

Figure 1.19 shows the effect of increased temperature on the characteristic curve of a PN junction diode. A germanium diode can be used up to a maximum of 75°C and a silicon diode to a maximum of 175°C .

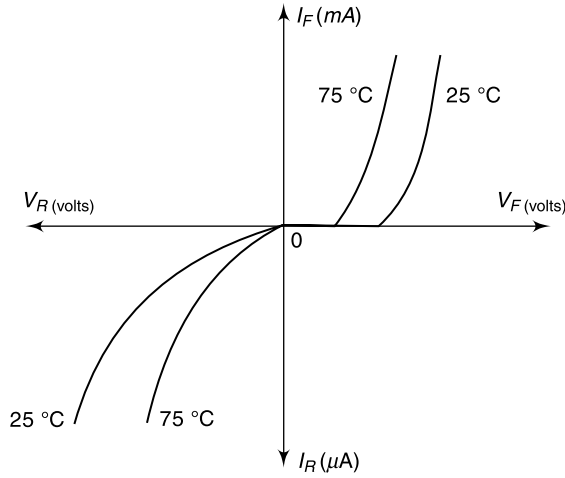


Fig. 1.19 Effect of Temperature on the Diode Characteristics

Example 1.11 The voltage across a silicon diode at room temperature (300 K) is 0.7 Volts when 2 mA current flows through it. If the voltage increases to 0.75 V, calculate the diode current (assume $V_T = 26$ mV).

Solution Given data

Room temperature = 300 K

Voltage across a silicon diode, $V_{D1} = 0.7$ V

Current through the diode $I_{D1} = 2$ mA

When the voltage increases to 0.75 V, V_{D2} , then

$$\frac{I_{D2}}{I_{D1}} = \frac{I_o (e^{V_{D2}/V_T \eta} - 1)}{I_o (e^{V_{D1}/V_T \eta} - 1)} = \frac{e^{0.75/26 \times 10^{-3}} \times 2 - 1}{e^{0.7/26 \times 10^{-3}} \times 2 - 1} = 2.615$$

Therefore,
$$I_{D2} = 2.615 \times I_{D1}$$

$$= 2.615 \times 2 \times 10^{-3} = 5.23 \text{ mA}$$

Example 1.12 A silicon diode has a saturation current of 7.5 μ A at room temperature 300 K. Calculate the saturation current at 400 K.

Solution Given $I_{o1} = 7.5 \times 10^{-6}$ A at $T_1 = 300$ K = 27 °C and $T_2 = 400$ K = 127 °C

Therefore, the saturation current at 400 K is

$$I_{o2} = I_{o1} \times 2^{(T_2 - T_1)/10}$$

$$= 7.5 \times 10^{-6} \times 2^{(127 - 27)/10}$$

$$= 7.5 \times 10^{-6} \times 2^{10} = 7.68 \text{ mA}$$

Example 1.13 The reverse saturation current of the Ge transistor is $2 \mu A$ at room temperature of $25^\circ C$ and increases by a factor of 2 for each temperature increase of $10^\circ C$. Find the reverse saturation current of the transistor at a temperature of $75^\circ C$. [JNTU Jan 2010]

Solution Given $I_{o1} = 2 \mu A$ at $T_1 = 25^\circ C$, $T_2 = 75^\circ C$
Therefore, the reverse saturation current of the transistor at $T_2 = 75^\circ C$ is

$$\begin{aligned} I_{o2} &= I_{o1} \times 2^{(T_2 - T_1)/10} = 2 \times 10^{-6} \times 2^{\left(\frac{75-25}{10}\right)} \\ &= 2 \times 10^{-6} \times 2^5 = 64 \mu A \end{aligned}$$

1.11 DIODE EQUIVALENT CIRCUITS

The PN junction diode is considered as a circuit element. The basic diode circuit shown in Fig. 1.19 consists of a dc voltage V_S which is supplied across a resistor and a diode. In order to find the instantaneous diode voltage V and current I , the circuit can be analysed when the instantaneous source voltage is V_S . From Kirchhoff's voltage law (KVL), the instantaneous diode voltage is

$$V_S = IR + V \quad (1.25)$$

which can be expressed as

$$I = \frac{V_S}{R} - \frac{V}{R} \quad (1.26)$$

The ideal diode current equation relating the diode voltage V_D and the current I_D is

$$I = I_o [e^{(V/\eta V_T)} - 1] \quad (1.27)$$

where I_o is the diode reverse saturation current at room temperature, η is a constant (1 for Ge and 2 for Si) and V_T is the thermal voltage, i.e., $V_T = \frac{T}{11,600}$

in which T is the temperature of the diode junction (300 K) or $V_T = 26 \text{ mV}$ at room temperature.

Substituting Eq. (1.27) into Eq. (1.25), we get

$$V_S = I_o R [e^{(V/\eta V_T)} - 1] + V \quad (1.28)$$

which has only one unknown variable V . Since Eq. (1.28) is a transcendental equation, this equation cannot be solved directly.

As these equations contain both linear and exponential terms, it is difficult to solve by hand. The iteration (trial and error) technique can be used to find a solution to this equation. The graphical analysis technique involves plotting two simultaneous equations and locating their point of intersection, which is called the quiescent point, or the Q -point.

1.11.1 Load Line Analysis

The use of the load line construction allows the graphical analysis of many circuits including devices which are much more complicated than the PN diode. Two plots are needed to determine the operating point of the diode. One plot is drawn using Kirchhoff's current and voltage laws and other is by plotting the volt–ampere characteristic of the diode. From KCL, $I = I_R$ and from KVL, $V + V_R = V_S$ as shown in Fig. 1.20. The relationship between diode current I and voltage V is obtained from the diode characteristics curve as shown in Fig. 1.21(a). From Ohm's law, the current through the resistor and the voltage across the resistor are linearly related as shown in Fig. 1.21(b).

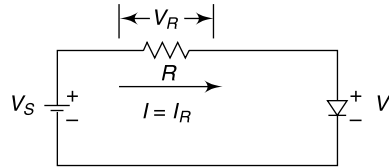


Fig. 1.20 Simple Diode Circuit

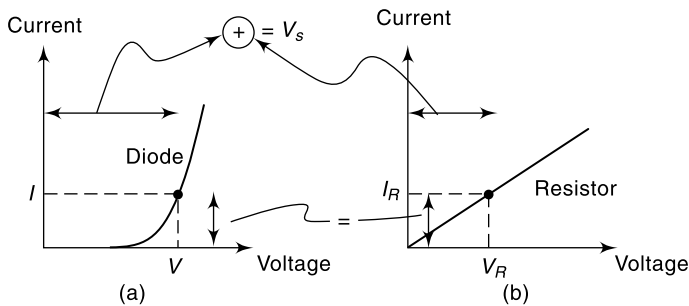


Fig. 1.21 (a) Diode Characteristics for the Circuit Shown in Fig. 1.20 and (b) Output Across Resistor

To draw the load line, flip the resistor curve horizontally such that the slope of the curve is $-1/R$ and push the two curves, as shown in Figs 1.22(a) and 1.22(b), together horizontally until the y -axes are V_S apart.

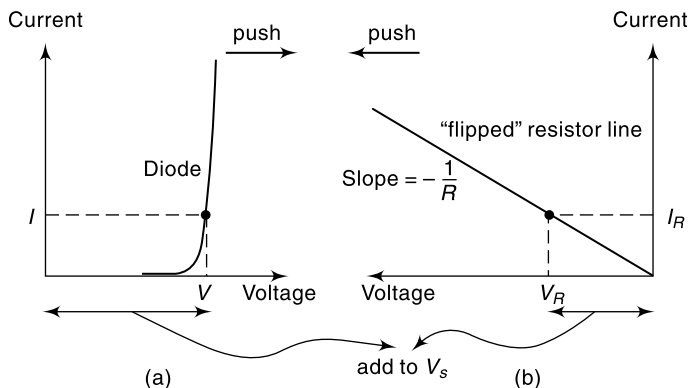


Fig. 1.22 (a) Forward Voltage–Current Characteristic of a Diode and (b) Flipped Resistor Line with Slope $= -\frac{1}{R}$

The intersection point of the flipped resistor line called load line and the diode static characteristics curve is the operating point of the device, as shown in Fig. 1.23.

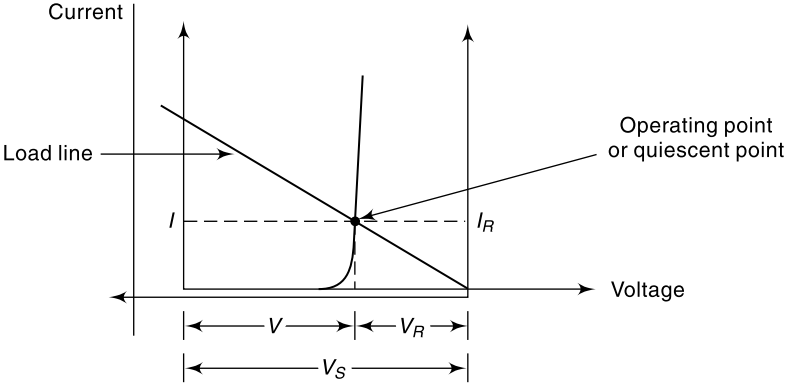


Fig. 1.23 Intersection Point of the Load Line with the Voltage–Current Characteristic of the Diode

In the second method to draw the load line, I is determined when the device is short-circuited and V is determined when the device is open-circuited, so that the point B ($V = 0$, $I = V_S/R$) and A ($V = V_S$, $I = 0$) lies somewhere on the y -axis and x -axis respectively of the diode characteristics curve as shown in Fig. 1.24. Thus a straight line drawn connecting the points A and B is called the load line. This load line intersects the diode characteristics at some point which is chosen as operating point for the device. The operating point provides the diode voltage V , appearing across the diode and the current I , flowing through the diode.

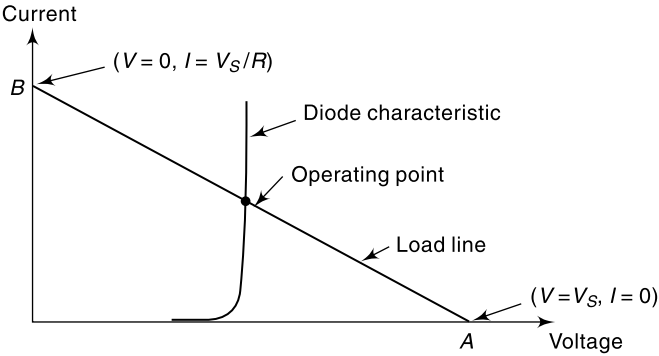


Fig. 1.24 Load Line Curve for Fixing the Operating Point

1.12

PIECEWISE LINEAR DIODE MODEL/DIODE EQUIVALENT CIRCUITS

As the volt–ampere relationship of the diode is non-linear, the analysis of circuits containing diodes is difficult. With the help of piecewise linear approximation model of the diode, the results can easily be obtained. The particular regions of operation of the volt–ampere characteristics of the diode are broken into linear segments and the concept of a diode cut-in voltage is also used in this piecewise linear model.

If the reverse resistance R_r is included in the diode characteristics, then the piecewise linear and continuous volt–ampere characteristic is obtained. Piecewise linear model is used when a more accurate model than ideal-diode model is needed but not resorted to nonlinear equation or graphical technique. The piecewise linear model for diode is obtained by using following steps:

1. Approximate the actual V - I characteristics by straight-line segments.
2. Model each section (forward and reverse characteristics) with a resistance in series with a constant voltage source.

The simple piecewise linear equivalent for the diode is shown in Fig. 1.25.

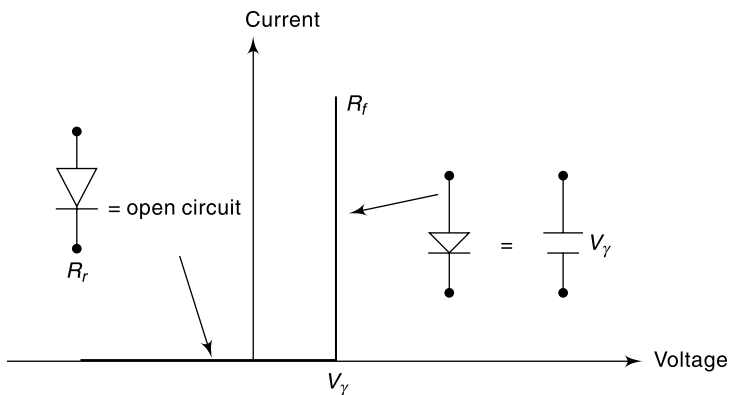


Fig. 1.25 Characteristics Piecewise Linear Diode

Since the diode is a binary device, it can exist in only one of two possible states, i.e., the diode is either ON or OFF state at a given time. If the voltage applied across the diode exceeds the cut-in voltage, V_γ , the diode is forward biased and is said to be in ON state with diode forward resistance R_f . For a reverse bias, the diode is open circuited and is said to be in OFF state with infinitely large reverse resistance R_r .

The piecewise linear model is used for analysis of the diode circuits. Consider a circuit containing several diodes, resistors and power supplies. This type of circuit is analyzed by assuming the state of the diode. For ON state, the diode is replaced with R_f value and for the OFF state, the diode is replaced with R_r value.

After replacing the diode with this piecewise model, the entire circuit is linear and now by using Kirchhoff's voltage and current laws, the entire current and voltage in the circuit can be calculated. The assumption that a diode is ON can be verified by observing the sign of the current through it. If the current is in forward direction, then the assumption of the diode is justified. However, if the current is in the reverse direction, then the diode assumption is incorrect. Under this circumstance, the analysis must begin again with the diode assumed to be OFF.

Similar to the above trial-and-error method, the diode OFF condition is tested by finding the voltage across it. If the voltage is either in the reverse direction or in the forward direction with a voltage less than V_γ , the turn-on or cut-in voltage of the diode, then the diode assumption is correct. However, if the diode voltage is in the forward direction with voltage greater than V_γ , the condition of the diode is ON and the original assumption is not correct. Under this circumstance, the analysis must begin again with the diode assumed to be ON.

1.13

BREAKDOWN MECHANISMS IN SEMICONDUCTOR DIODES

The diode equation predicts that, under reverse bias conditions, a small constant current, the saturation current, I_o , flows due to minority carriers, which is independent of the magnitude of the bias voltage. But this prediction is not entirely true in practical diodes. There is a gradual increase of reverse current with increasing bias due to the ohmic leakage currents around the surface of the junction. Also, there is a sudden increase in reverse current due to some sort of *breakdown*, when the reverse bias voltage approaches a particular value called breakdown voltage, V_{BD} , as shown in Fig. 1.26. Once breakdown occurs, the diode is no longer blocking current and the diode current can be controlled only by the resistance of the external circuit.

The breakdown occurs due to avalanche effect in which thermally generated minority carriers cross the depletion region and acquire sufficient kinetic energy from the applied potential to produce new carriers by removing valence electrons from their bonds. These new carriers will in turn collide with other atoms and will increase the number of electrons and holes available for conduction. This multiplication effect of free carriers may be represented by the following equation:

$$M = \frac{1}{1 - \left(\frac{V}{V_{BD}} \right)^n}$$

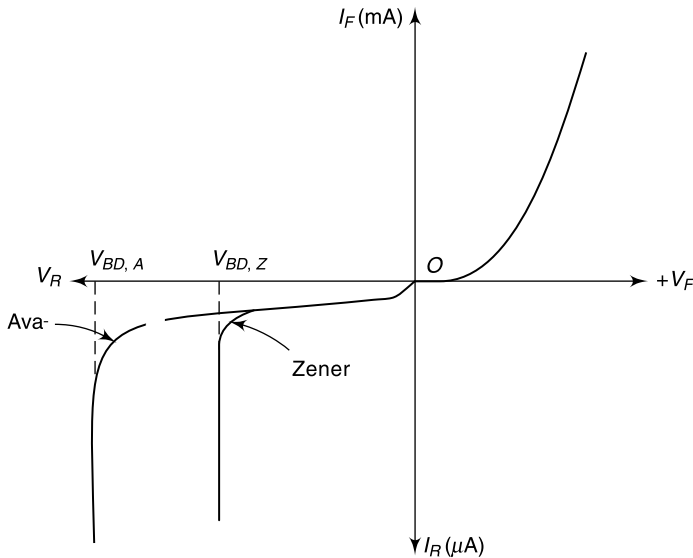


Fig. 1.26 Breakdown in PN Junction Diode

where M = carrier multiplication factor, which is the ratio of the total number of electrons leaving the depletion region to the number entering the region

V = applied reverse voltage

V_{BD} = reverse breakdown voltage

n = empirical constant, which depends on the lattice material and the carrier type, for N-type silicon, $n \approx 4$ and for P-type, $n \approx 2$

It is evident from the above equation that M is being very small for $V = 0.9 V_{BD}$. But when $V > 0.9 V_{BD}$, M is very large and the resulting reverse current is given by $I_R = MI_o$, where I_o is the reverse saturation current before breakdown.

As V approaches the breakdown voltage V_{BD} , the value of M will become infinite and there is a rapid increase in carrier density and a corresponding increase in current. Because of the cumulative increase in carrier density after each collision, the process is known as *avalanche breakdown*.

Even if the initially available carriers do not gain enough energy to disrupt bonds, it is possible to initiate breakdown through a direct rupture of the bonds because of the existence of strong electric field. Under these circumstances the breakdown is referred to as *Zener breakdown*.

Voltage reference diodes that utilise the almost constant voltage characteristics in the breakdown region are also called variously *avalanche diodes* or sometimes *Zener diodes*. The Zener effect is in diodes with breakdown voltages below about 6 V. The operating voltage in avalanche breakdown are from several volts to several hundred volts with power ratings up to 50 W.

True Zener diode action displays a negative temperature coefficient, i.e., breakdown voltage decreases with increasing temperature. True avalanche diode

action exhibits a positive temperature coefficient, i.e. breakdown voltage increases with increasing temperature.

It is clear that the breakdown voltage for a particular diode can be controlled during manufacture by altering the doping levels in the junction. The breakdown voltage for silicon diodes can be made to occur at a voltage as low as 5 V with 10^{17} impurity atoms per cubic cm or as high as 1000 V when doped to a level of only 10^{14} impurity atoms per cubic cm.

1.14 PN DIODE APPLICATIONS

An ideal PN junction diode is a two terminal polarity sensitive device that has zero resistance (diode conducts) when it is forward biased and infinite resistance (diode does not conduct) when reverse biased. Due to this characteristic the diode finds a number of applications as follows.

- (a) rectifiers in dc power supplies
- (b) switch in digital logic circuits used in computers
- (c) clamping network used as dc restorer in TV receivers and voltage multipliers
- (d) clipping circuits used as wave shaping circuits used in computers, radars, radio and TV receivers
- (e) demodulation (detector) circuits.

The same PN junction with different doping concentration finds special applications as follows:

- (a) detectors (APD, PIN photo diode) in optical communication circuits
- (b) Zener diodes in voltage regulators
- (c) varactor diodes in tuning sections of radio and TV receivers
- (d) light emitting diodes in digital displays
- (e) LASER diodes in optical communications
- (f) Tunnel diodes as a relaxation oscillator at microwave frequencies.

1.15 ZENER DIODE CHARACTERISTICS

When the reverse voltage reaches breakdown voltage in normal PN junction diode, the current through the junction and the power dissipated at the junction will be high. Such an operation is destructive and the diode gets damaged. Whereas diodes can be designed with adequate power dissipation capabilities to operate in the breakdown region. One such a diode is known as Zener diode. Zener diode is heavily doped than the ordinary diode.

From the V - I characteristics of the Zener diode, shown in Fig. 1.27, it is found that

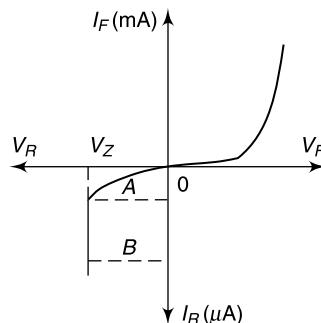


Fig. 1.27 V - I Characteristics of a Zener Diode

the operation of Zener diode is same as that of ordinary PN diode under forward-biased condition. Whereas under reverse-biased condition, breakdown of the junction occurs. The breakdown voltage depends upon the amount of doping. If the diode is heavily doped, depletion layer will be thin and, consequently, breakdown occurs at lower reverse voltage and further, the breakdown voltage is sharp. Whereas a lightly doped diode has a higher breakdown voltage. Thus breakdown voltage can be selected with the amount of doping.

The sharp increasing current under breakdown conditions are due to the following two mechanisms.

- (1) Avalanche breakdown
- (2) Zener breakdown

1.15.1 Avalanche Breakdown

As the applied reverse bias increases, the field across the junction increases correspondingly. Thermally generated carriers while traversing the junction acquire a large amount of kinetic energy from this field. As a result the velocity of these carriers increases. These electrons disrupt covalent bonds by colliding with immobile ions and create new electron–hole pairs. These new carriers again acquire sufficient energy from the field and collide with other immobile ions thereby generating further electron–hole pairs. This process is cumulative in nature and results in generation of avalanche of charge carriers within a short time. This mechanism of carrier generation is known as Avalanche multiplication. This process results in flow of large amount of current at the same value of reverse bias.

1.15.2 Zener Breakdown

When the P and N regions are heavily doped, direct rupture of covalent bonds takes place because of the strong electric fields, at the junction of PN diode. The new electron–hole pairs so created increase the reverse current in a reverse biased PN diode. The increase in current takes place at a constant value of reverse bias typically below 6 V for heavily doped diodes. As a result of heavy doping of P and N regions, the depletion region width becomes very small and for an applied voltage of 6 V or less, the field across the depletion region becomes very high, of the order of 10^7 V/m, making conditions suitable for Zener breakdown. For lightly doped diodes, Zener breakdown voltage becomes high and breakdown is then predominantly by Avalanche multiplication. Though Zener breakdown occurs for lower breakdown voltage and Avalanche breakdown occurs for higher breakdown voltage, such diodes are normally called Zener diodes.

1.15.3 Effect of Temperature on Zener Diode

The Zener voltage V_Z changes with the temperature. The percentage change in the Zener voltage V_Z for every $^{\circ}\text{C}$ change in temperature is called temperature coefficient (TC) of a Zener diode. It is denoted as TC and expressed as $\% / ^{\circ}\text{C}$.

Mathematically it can be defined as

$$TC = \frac{\Delta V_Z}{V_Z (T_1 - T_0)} \times [100] \text{ \% / } ^\circ\text{C}$$

where T_1 is the final temperature of junction while T_0 is generally 25°C at which nominal Zener voltage V_Z is specified. ΔV_Z is the resulting change in the Zener voltage due to the temperature variation.

A positive value of TC indicates that there is an increase in V_Z due to increase in temperature or decrease in V_Z due to decrease in temperature. The negative value of TC indicates that there is an increase in V_Z due to decrease in temperature or decrease in V_Z due to increase in temperature.

From Eqn. (1), we have

$$\Delta V_Z = \frac{V_Z TC (T_1 - T_0)}{100} = \frac{V_Z TC \Delta T}{100}$$

where ΔT is the change in temperature .

Sometimes TC for a Zener diode is expressed in $\text{mV}/^\circ\text{C}$ and in such a case, the corresponding change in V_Z can be obtained as

$$\Delta V_Z = TC (T_1 - T_0) = TC \times \Delta T$$

For a Zener diode with V_Z less than 6V , the temperature coefficient is negative and hence V_Z decreases as temperature increases. While for a Zener diode with V_Z greater than 6V , the temperature coefficient is positive and hence V_Z increases as temperature increases.

1.15.4 Applications

From the Zener characteristics shown in Fig. 1.27, under the reverse bias condition, the voltage across the diode remains almost constant although the current through the diode increases as shown in region AB . Thus, the voltage across the Zener diode serves as a reference voltage. Hence, the diode can be used as a voltage regulator.

In Fig. 1.28 it is required to provide constant voltage across load resistance R_L , whereas the input voltage may be varying over a range. As shown, Zener

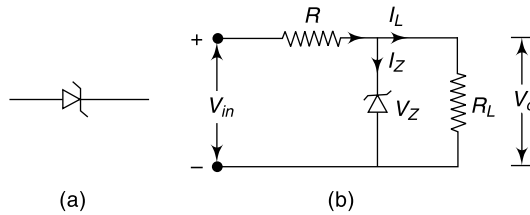


Fig. 1.28 Zener Diode (a) Circuit Symbol and (b) as a Voltage Regulator

diode is reverse biased and as long as the input voltage does not fall below V_Z (Zener breakdown voltage), the voltage across the diode will be constant and hence the load voltage will also be constant.

1.16

PRINCIPLE OF OPERATION AND CHARACTERISTICS OF TUNNEL DIODE

The Tunnel or Esaki diode is a thin-junction diode which exhibits negative resistance under low forward bias conditions.

An ordinary PN junction diode has an impurity concentration of about 1 part in 10^8 . With this amount of doping the width of the depletion layer is of the order of 5 microns. This potential barrier restrains the flow of carriers from the majority carrier side to the minority carrier side. If the concentration of impurity atoms is greatly increased to the level of 1 part in 10^3 , the device characteristics are completely changed. The width of the junction barrier varies inversely as the square root of the impurity concentration and therefore, is reduced from 5 microns to less than $100 \text{ \AA}$ (10^{-8} m). This thickness is only about 1/50th of the wavelength of visible light. For such thin potential energy barriers, the electrons will penetrate through the junction rather than surmounting them. This quantum mechanical behavior is referred to as tunneling and hence, these high-impurity-density PN junction devices are called tunnel diodes.

The V - I characteristic for a typical germanium tunnel diode is shown in Fig. 1.29. It is seen that at first forward current rises sharply as applied voltage is increased, where it would have risen slowly for an ordinary PN junction diode (which is shown as dashed line for comparison). Also, reverse current is much larger for comparable back bias than in other diodes due to the thinness of the

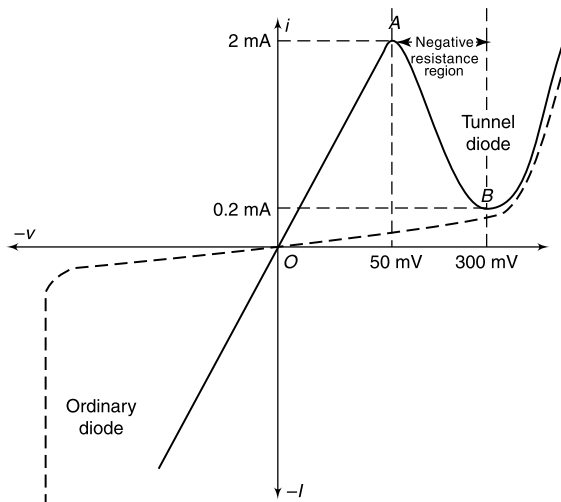


Fig. 1.29 V - I Characteristic of Tunnel Diode

junction. The interesting portion of the characteristic starts at the point *A* on the curve, i.e. the peak voltage. As the forward bias is increased beyond this point, the forward current drops and continues to drop until point *B* is reached. This is the valley voltage. At *B*, the current starts to increase once again and does so very rapidly as bias is increased further. Beyond this point, characteristic resembles that of an ordinary diode. Apart from the peak voltage and valley voltage, the other two parameters normally used to specify the diode behaviour are the peak current and the peak-to-valley current ratio, which are 2 mA and 10 respectively, as shown.

The V - I characteristic of the tunnel diode illustrates that it exhibits dynamic resistance between *A* and *B*. Figure 1.30 shows energy level diagrams of the tunnel diode for three interesting bias levels. The shaded areas show the energy states occupied by electrons in the valence band, whereas the cross hatched regions represent energy states in the conduction band occupied by the electrons. The levels to which the energy states are occupied by electrons on either side of the junctions are shown by dotted lines. When the bias is zero, these lines are at the same height. Unless energy is imparted to the electrons from some external source, the energy possessed by the electrons on the N-side of the junction is insufficient to permit them to climb over the junction barrier to reach the P-side. However, quantum mechanics show that there is a finite probability for the electrons to tunnel through the junction to reach the other side, provided there are allowed empty energy states in the P-side of the junction at the same energy level. Hence, the forward current is zero.

When a small forward bias is applied to the junction, the energy level of the P-side is lower as compared with the N-side. As shown in Fig. 1.30(b), electrons in the conduction band of the N-side see empty energy level on the P-side.

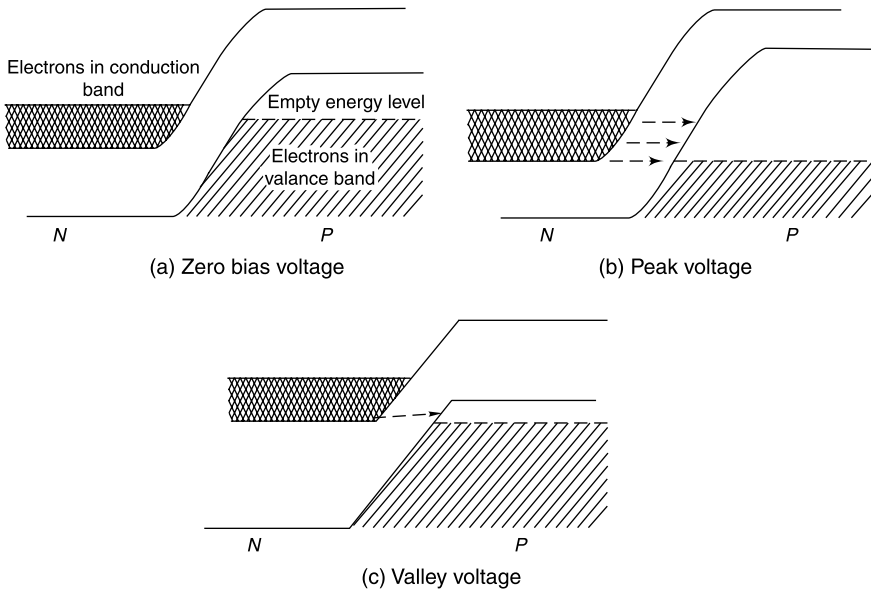


Fig. 1.30 Energy Level Diagrams of Tunnel Diode

Hence, tunneling from N-side to P-side takes place. Tunneling in other directions is not possible because the valence band electrons on the P-side are now opposite to the forbidden energy gap on the N-side. The energy band diagram shown in Fig. 1.30(b), is for the peak of the diode characteristic.

When the forward bias is raised beyond this point, tunneling will decrease as shown in Fig. 1.30(c). The energy of the P-side is now depressed further, with the result that fewer conduction band electrons on the N-side are opposite to the unoccupied P-side energy levels. As the bias is raised, forward current drops. This corresponds to the negative resistance region of the diode characteristic. As forward bias is raised still further, tunneling stops altogether and it behaves as a normal PN junction diode.

Equivalent Circuit The equivalent circuit of the tunnel diode when biased in the negative resistance region is as shown in Fig. 1.31(a) and Fig. 1.31(b) shown the symbol of tunnel diode. In the circuit, R_s is the series resistance and L_s is the series inductance which may be ignored except at highest frequencies. The resulting diode equivalent circuit is thus reduced to parallel combination of the junction capacitance C_j and the negative resistance $-R_n$. Typical values of the circuit components are $R_s = 6 \Omega$, $L_s = 0.1 \text{ nH}$, $C_j = 0.6 \text{ pF}$ and $R_n = 75 \Omega$.

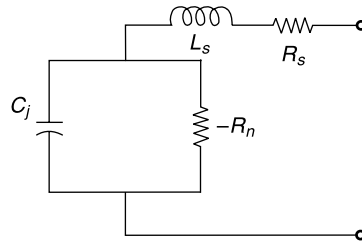


Fig. 1.13(a) Equivalent Circuit of Tunnel Diode

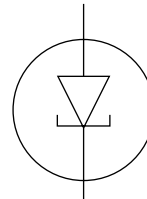


Fig. 1.31(b) Circuit Symbol of Tunnel Diode

Applications

1. Tunnel diode is used as an ultra-high speed switch with switching speed of the order of ns or ps
2. As logic memory storage device
3. As microwave oscillator
4. In relaxation oscillator circuit
5. As an amplifier

Advantages

1. Low noise
2. Ease of operation
3. High speed
4. Low power

Disadvantages

1. Voltage range over which it can be operated is 1 V or less
2. Being a two terminal device, there is no isolation between the input and output circuit

1.17

PRINCIPLE OF OPERATION AND CHARACTERISTICS OF VARACTOR DIODE

The varactor, also called a varicap, tuning or voltage variable capacitor diode, is also a junction diode with a small impurity dose at its junction, which has the useful property that its junction or transition capacitance is easily varied electronically.

When any diode is reverse biased, a depletion region is formed, as seen in Fig. 1.32(a). The larger the reverse bias applied across the diode, the width of the depletion layer “ W ” becomes wider. Conversely, by decreasing the reverse bias voltage, the depletion region width “ W ” becomes narrower. This depletion region is devoid of majority carriers and acts like an insulator preventing conduction between the N and P regions of the diode, just like a dielectric, which separates the two plates of a capacitor. The varactor diode with its symbol is shown in Fig. 1.32(b).

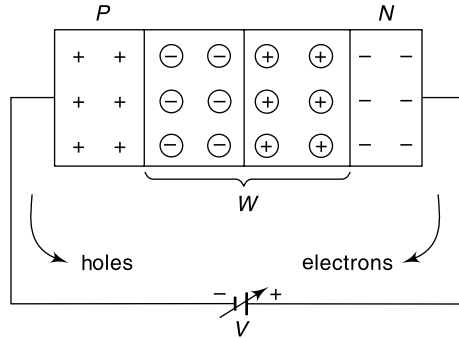


Fig. 1.32(a) Depletion Region in a Reverse Biased PN Junction



Fig. 1.32(b) Circuit Symbol of Varactor Diode

As the capacitance is inversely proportional to the distance between the plates $\left(C_T \propto \frac{1}{W}\right)$, the transition capacitance C_T varies inversely with the reverse voltage as shown in Fig. 1.33. Consequently, an increase in reverse bias

voltage will result in an increase in the depletion region width and a subsequent decrease in transition capacitance C_T . At zero volt, the varactor depletion region W is small and the capacitance is large at approximately 600 pF. When the reverse bias voltage across the varactor is 15 V, the capacitance is 30 pF.

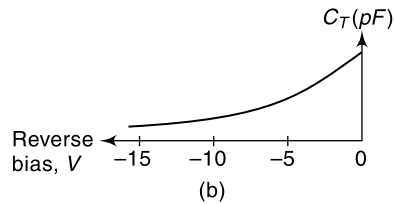


Fig. 1.33 Characteristics of Varactor Diode

The varactor diodes are used in FM radio and TV receivers, AFC circuits, self adjusting bridge circuits and adjustable bandpass filters. With improvement in the type of materials used and construction, varactor diodes find application in tuning of LC resonant circuit in microwave frequency multipliers and in very low noise microwave parametric amplifiers.

1.18

PRINCIPLE OF OPERATION OF SILICON CONTROLLED RECTIFIER (SCR)

Thyristor, in general, is a semiconductor device having three or more junctions. Such a device acts as a switch without any bias and can be fabricated to have voltage ratings of several hundred volts and current ratings from a few amperes to almost thousand amperes. The family of thyristors consists of PNPN diode (Shockley diode), SCR, LASCR, TRIAC, DIAC, GTO etc.

1.18.1 PNPN Diode (Shockley Diode)

As shown in Fig. 1.34, it is a four layer PNPN silicon device with two terminals. When an external voltage is applied to the device in such a way that anode is positive with respect to cathode, junctions J_1 and J_3 are forward biased and J_2 is reverse biased. Then the applied voltage appears across the reverse biased junction J_2 . Now the current flowing through the device is only reverse saturation current.

However, as this applied voltage is increased, the current increases slowly until the so called firing or breakover voltage (V_{BO}) is reached. Once firing takes place, the current increases abruptly and the voltage drop across the device decreases sharply. At this point, the diode switches over from 'OFF' to 'ON' state. Once the device is fired into conduction, a minimum amount of current known as *holding current*, I_H , is required to flow to keep the device in ON state. To turn the device OFF from ON state, the current has to be reduced below I_H by reducing the applied voltage close to zero, i.e. below *holding voltage*, V_H . Thus the diode acts as a switch during forward bias condition. The characteristic curve of a PNPN diode is shown in Fig. 1.35.

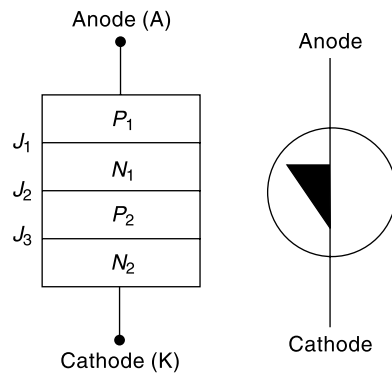


Fig. 1.34 PNPN Diode: (a) Basic Structure and (b) Circuit Symbol

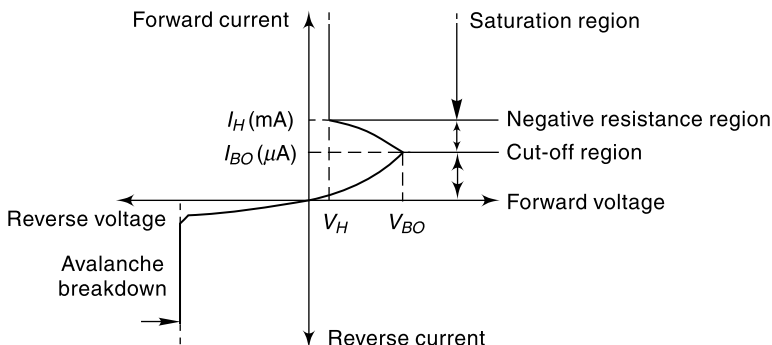


Fig. 1.35 Characteristic Curve of PNPN Diode

1.18.2 SCR (Silicon Controlled Rectifier)

The basic structure and circuit symbol of SCR is shown in Fig. 1.36. It is a four layer three terminal device in which the end P-layer acts as anode, the end N-layer acts as cathode and P-layer nearer to cathode acts as gate. As leakage current in silicon is very small compared to germanium, SCRs are made of silicon and not germanium.

Characteristics of SCR The characteristics of SCR are shown in Fig. 1.37. SCR acts as a switch when it is forward biased. When the gate is kept open, i.e. gate current $I_G = 0$, operation of SCR is similar to PNPN diode. When $I_G < 0$, the amount of reverse bias applied to J_2 is increased. So the breakover voltage V_{BO} is increased. When $I_G > 0$, the amount of reverse bias applied to J_2 is

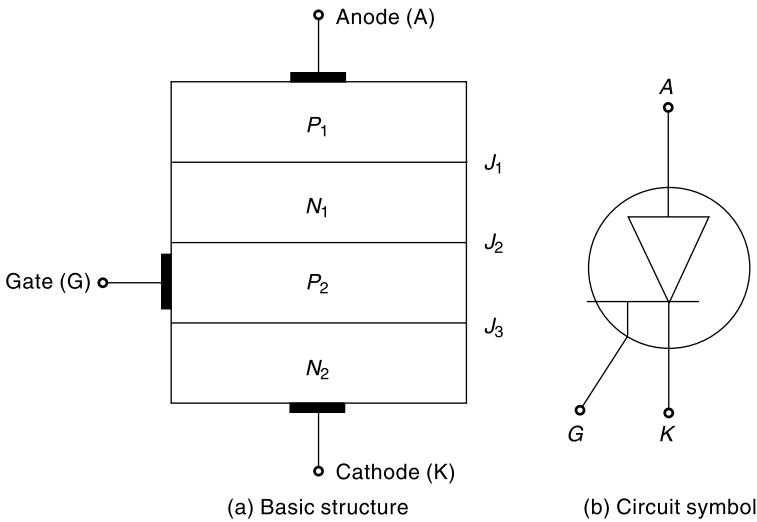


Fig. 1.36 Basic Structure and Circuit Symbol of SCR

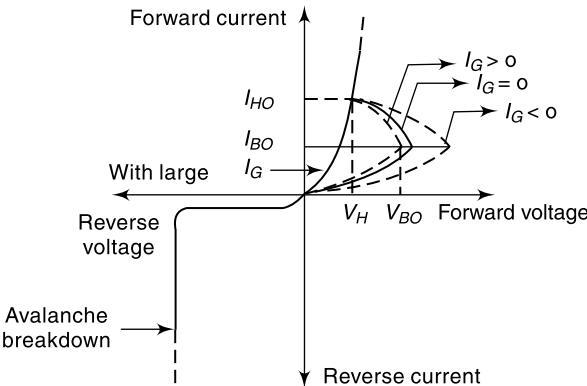


Fig. 1.37 Characteristics of SCR

decreased thereby decreasing the breakover voltage. With very large positive gate current breakover may occur at a very low voltage such that the characteristics of SCR is similar to that of ordinary PN diode. As the voltage at which SCR is switched 'ON' can be controlled by varying the gate current I_G , it is commonly called as controlled switch. Once SCR is turned ON, the gate loses control, i.e. the gate cannot be used to switch the device OFF. One way to turn the device OFF is by lowering the anode current below the holding current I_H by reducing the supply voltage below holding voltage V_H , keeping the gate open.

SCR is used in relay control, motor control, phase control, heater control, battery chargers, inverters, regulated power supplies and as static switches.

Two-transistor Version of SCR The operation of SCR can be explained in a very simple way by considering it in terms of two transistors, called as the two transistor version of SCR. As shown in Fig. 1.38, an SCR can be split into two parts and displaced mechanically from one another but connected electrically. Thus the device may be considered to be constituted by two transistors T_1 (PNP) and T_2 (NPN) connected back to back.

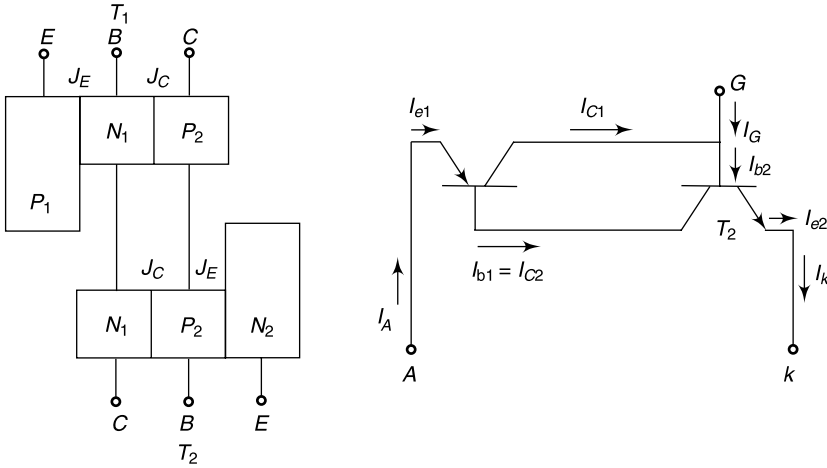


Fig. 1.38 Two Transistor Version of SCR

Assuming the leakage current of T_1 to be negligibly small, we obtain

$$I_{b1} = I_A - I_{e1} = I_A - \alpha_1 I_A = (1 - \alpha_1) I_A \quad (1.29)$$

Also, from the Fig. 1.38, it is clear that

$$I_{b1} = I_{C2} \quad (1.30)$$

and

$$I_{C2} = \alpha_2 I_K \quad (1.31)$$

Substituting the values given in Eqs (1.30) and (1.31) in Eq. (1.29), we get

$$(1 - \alpha_1) I_A = \alpha_2 I_K \quad (1.32)$$

We know that

$$I_K = I_A + I_G \quad (1.33)$$

Substituting Eq. (1.33) in Eq. (1.32), we obtain

$$(1 - \alpha_1) I_A = \alpha_2 (I_A + I_G)$$

$$\text{i.e.} \quad (1 - \alpha_1 - \alpha_2) I_A = \alpha_2 I_G$$

$$\text{i.e.} \quad I_A = \left[\frac{\alpha_2 I_G}{1 - (\alpha_1 + \alpha_2)} \right] \quad (1.34)$$

Equation (1.34) indicates that if $(\alpha_1 + \alpha_2) = 1$, then $I_A = \infty$, i.e. the anode current I_A suddenly reaches a very high value approaching infinity. Therefore, the device suddenly triggers into ON state from the original OFF state. This characteristic of the device is known as its *regenerative action*.

The value of $(\alpha_1 + \alpha_2)$ can be made almost equal to unity by giving a proper value of positive current I_G for a short duration. This signal I_G applied at the gate which is the base of T_2 will cause a flow of collector current I_{C2} by transferring T_2 to its ON state. As $I_{C2} = I_{b1}$, the transistor T_1 will also be switched ON. Now, the action is regenerative since each of the transistors would supply base current to the other. At this point even if the gate signal is removed, the device keeps on conducting, till the current level is maintained to a minimum value of holding current.

1.18.3 Thyristor Ratings

Latching current (I_L) Latching current is the minimum current required to latch or trigger the device from its OFF-state to its ON-state.

Holding current (I_H) Holding current is the minimum value of current to hold the device in ON-state. For turning the device OFF, the anode current should be lowered below I_H by increasing the external circuit resistance.

Gate current (I_G) Gate current is the current applied to the gate of the device for control purposes. The minimum gate current is the minimum value of current required at the gate for triggering the device. The maximum gate current is the maximum value of current applied to the device without damaging the gate. More the gate current, earlier is the triggering of the device and vice-versa.

Voltage safety factor (V_f) Voltage safety factor V_f is a ratio which is related to the PIV, the RMS value of the normal operating voltage as,

$$V_f = \frac{\text{peak inverse voltage (PIV)}}{\sqrt{2} \times \text{RMS value of the operating voltage}}$$

The value of V_f normally lies between 2 and 2.7. For a safe operation, the normal working voltage of the device is much below its PIV.

1.18.4 Rectifier Circuits Using SCR

SCRs are much superior in performance than ordinary diode rectifiers. They find their main applications as rectifiers. Some of the rectifier circuits have been explained in the following sections.

SCR Half Wave Rectifier Though the SCR is basically a switch, it can be used in linear applications like rectification. Fig. 1.39 shows the circuit of an SCR half wave rectifier.

During the negative halfcycle, the SCR does not conduct irrespective of the gate current, as the anode is negative with respect to cathode and also PIV is less than the reverse breakdown voltage.

During the positive half cycle of ac voltage appearing across secondary, the SCR will conduct provided proper gate current is made to flow. The greater the gate current, the lesser the supply voltage at which the SCR is triggered ON. Referring to Fig. 1.39(b), the gate current is adjusted to such a value that SCR is turned ON at a positive voltage V_1 of ac secondary voltage which is less than the peak voltage V_m . Beyond this, the SCR will be conducting till the applied voltage becomes zero. The angle at which the SCR starts conducting during the positive half cycle is called firing angle θ . Therefore, the conduction angle is $(180^\circ - \theta)$.

The SCR will block not only the negative part of the applied sinusoidal voltage, but will also block the part of the positive waveform up to a point SCR is triggered ON. If the angle θ is zero, this will be an ordinary half wave rectification. Therefore by proper adjustment of gate current, the SCR can be made to conduct full or part of a positive half cycle, thereby controlling the power fed to the load.

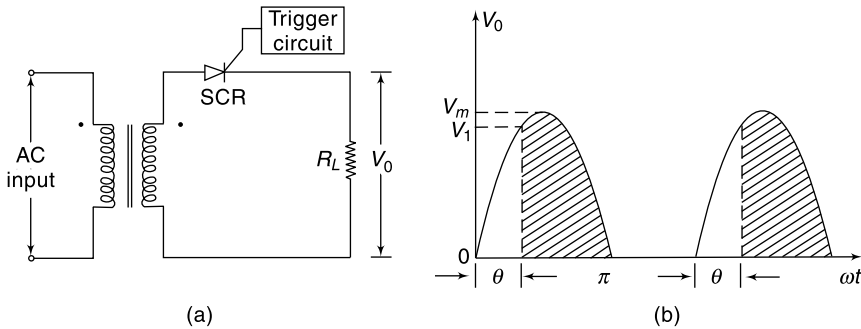


Fig. 1.39 SCR Half Wave Rectifier

Analysis Let $V = V_m \sin \omega t$ be alternating voltage that appear across the secondary of the transformer. In SCR halfwave rectifier, θ is the firing angle and the rectifier conducts from θ to 180° (π radians) during the positive half cycle.

$$\begin{aligned}
 \text{Therefore, average or dc output, } V_{av} &= \frac{1}{2\pi} \int_{\theta}^{\pi} V_m \sin \omega t \, d\omega t \\
 &= \frac{1}{2\pi} [-V_m \cos \omega t]_{\theta}^{\pi} \\
 &= \frac{V_m}{2\pi} (1 + \cos \theta)
 \end{aligned}$$

For $\theta = 0^\circ$, $V_{av} = \frac{V_m}{\pi}$. Here the full positive half cycle will appear across the load. This is the value of average voltage for ordinary halfwave rectifier.

When $\theta = 90^\circ$, $V_{av} = \frac{V_m}{2\pi}$. This shows that greater the firing angle θ , the smaller is the average voltage and vice-versa.

$$\begin{aligned}
 \text{Similarly, } V_{rms} &= \sqrt{\frac{1}{2\pi} \int_{\theta}^{\pi} (V_m \sin \omega t)^2 d\omega t} \\
 &= \sqrt{\frac{V_m^2}{4\pi} \int_{\theta}^{\pi} (1 - \cos 2\omega t) d\omega t} \\
 &= \sqrt{\frac{V_m^2}{4\pi} \left[\omega t - \frac{\sin 2\omega t}{2} \right]_{\theta}^{\pi}} \\
 &= \frac{V_m}{2} \left[\frac{1}{\pi} (\pi - \theta + \sin 2\theta) \right]^{\frac{1}{2}}
 \end{aligned}$$

$$\text{If } \theta = 0, \text{ then } V_{rms} = \frac{V_m}{2}$$

Example 1.14 In an SCR half wave rectifier, the forward breakdown voltage of SCR is 110 V for a gate current of 1 mA. If a 50 Hz sinusoidal voltage of 220 V peak is applied, find firing angle, conduction angle, average voltage, average current, power output and the time during which SCR remains OFF. Assume load resistance is 100 Ω and the holding current to be zero.

Solution We know that $V_1 = V_m \sin \theta$
 $110 = 220 \sin \theta$ i.e. $\sin \theta = 0.5$
Therefore, firing angle, $\theta = \sin^{-1}(0.5) = 30^\circ$
Average voltage, $V_{av} = \frac{V_m}{2\pi} (1 + \cos \theta)$
 $= \frac{220}{2\pi} (1 + \cos 30^\circ) = 65.37 \text{ V}$
Average current, $I_{av} = V_{av}/R_L = 65.37/100 = 0.6537 \text{ A}$
Power output $= V_{av} I_{av} = (65.37)(0.6537) = 42.7326 \text{ W}$
As $V_1 = V_m \sin \theta = V_m \sin \omega t$,
 $\omega t = \theta = 30^\circ = \frac{\pi}{6}$
 $2\pi \times 50 t = \frac{\pi}{6}$

Therefore, the time during which the SCR remains OFF is

$$t = 1/(2 \times 6 \times 50) = 1/600 = 1.667 \text{ ms}$$

SCR Full Wave Rectifier The SCR full wave rectifier is shown in Fig. 1.40. It is exactly similar to an ordinary full wave rectifier except that the two diodes have been replaced by two SCRs. The angle of conduction can be changed by adjusting the gate currents.

During the positive half cycle of the input signal, anode of SCR1 becomes positive and at the same time the anode of SCR2 becomes negative. When the input voltage reaches V_1 as shown in Fig. 1.40(b), SCR1 starts conducting and therefore only the shaded portion of positive half cycle will pass through the load.

During the negative half cycle of the input, the anode of SCR1 becomes negative and the anode of SCR2 becomes positive. Hence, SCR1 does not conduct and SCR2 conducts when the input voltage becomes V_1 .

The main advantage of this circuit over ordinary full wave rectifier circuit is that any voltage can be made available at the output by simply changing the firing angle of the SCRs.

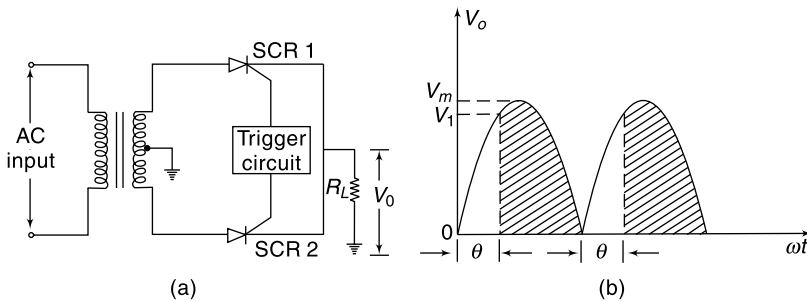


Fig. 1.40 SCR Full Wave Rectifier

Analysis Referring to Fig. 1.40, let $V = V_m \sin \omega t$ be the alternating voltage that appears between center tap and either end of secondary and θ be the firing angle.

$$\begin{aligned} V_{av} &= \frac{1}{\pi} \int_{\theta}^{\pi} V_m \sin \omega t \, d\omega t \\ &= \frac{V_m}{\pi} [-\cos \omega t]_{\theta}^{\pi} \\ &= \frac{V_m}{\pi} [1 + \cos \theta] \end{aligned}$$

This is double that of a half wave rectifier, as negative half cycle is also rectified.

Example 1.15 A full wave controlled rectifier employs 2 SCRs and 2 diodes in bridge configuration to rectify 230 V, 50 Hz ac mains and give an output of 150 V to a resistive load of $10 \, \Omega$. Find the firing angle, the time during which the SCR remains OFF and the load current.

Solution For an SCR full wave rectifier,

$$V_{dc} = \frac{V_m}{\pi}(1 + \cos\theta)$$

$$150 = \frac{230 \times \sqrt{2}}{\pi}(1 + \cos\theta)$$

Therefore,

$$\theta = 63.33^\circ$$

For 50 Hz,

$$T = 20 \text{ ms for } 360^\circ$$

Therefore

$$t = \frac{20}{360^\circ} \times 63.33^\circ = 3.52 \text{ ms}$$

Load current,

$$I_{av} = \frac{V_{av}}{R_L} = \frac{150}{10} = 15 \text{ A}$$

Example 1.16 When an SCR full wave rectifier is connected across a sinusoidal voltage of $400 \sin 314t$, the RMS value of the current flowing through the device is 20 A. Find the power rating of the SCR.

Solution As the supply voltage is $400 \sin 314t$, $V_m = 400 \text{ V}$

Peak inverse voltage (PIV) = $\sqrt{3} V_m = \sqrt{3} \times 400 = 692.8 \text{ V}$

RMS value of current = 20 A

Average value of current, $I_{av} = \text{RMS value/form factor} = 20/1.11 = 18 \text{ A}$

Power rating of the SCR = $\text{PIV} \times I_{av} = 692.8 \times 18 = 12.47 \text{ kW}$

SCR Bridge Rectifier The SCR bridge rectifier is shown in Fig. 1.41. During the positive half cycle of the input ac voltage, SCR1 and diode D1 conduct whereas SCR 2 and diode D2 do not conduct. During the negative half cycle, SCR2 and diode D2 conduct. As shown in Fig. 1.41(b), the conduction angle and hence the output voltage can be changed by adjusting the gate currents of SCR1 and SCR2. Here, the dc output voltage, $V_{av} = \frac{V_m}{\pi}[1 + \cos\theta]$, which is equal to that of SCR full wave rectifier.

If the current is lowered below I_H by increasing the external circuit resistance, the SCR will switch OFF.

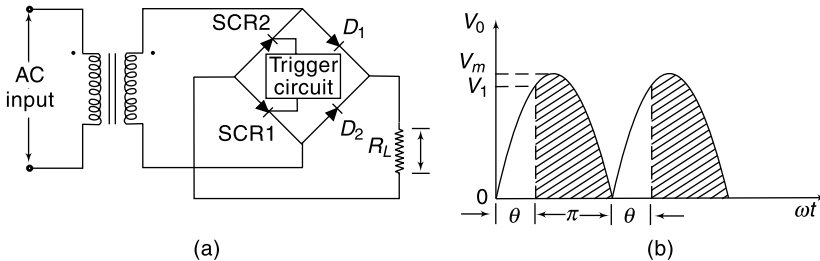


Fig. 1.41 SCR Bridge Rectifier

1.18.5 LASCR (Light Activated SCR)

The LASCR shown in Fig. 1.42 is triggered by irradiating with light. The arrows represent incoming light that passes through a window and falls on the depletion layer closer to the middle junction J_2 of SCR. The incident light generates electron-hole pairs in the device thus increasing the number of charge carriers. This leads to the instantaneous flow of current within the device and the device turns ON. For light triggering to occur, the device must have high value of rate of change of voltage with time, dV/dt .

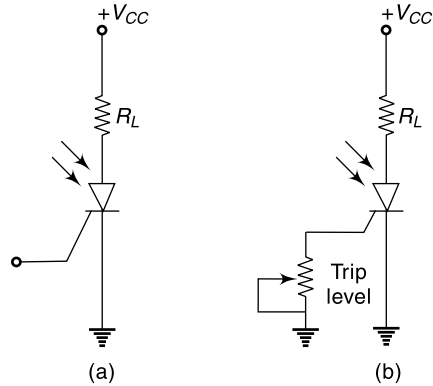


Fig. 1.42 Light Activated SCR

1.19

PRINCIPLE OF OPERATION OF SEMICONDUCTOR PHOTODIODE

Silicon photodiode is a light sensitive device, also called *photodetector*, which converts light signals into electrical signals. The construction and symbol of a photodiode are shown in Fig. 1.43. The diode is made of semiconductor PN junction kept in a sealed plastic or glass casing. The cover is so designed that the light rays are allowed to fall on one surface across the junction. The remaining sides of the casing are painted to restrict the penetration of light rays. A lens permits light to fall on the junction. When light falls on reverse biased PN photodiode junction, hole-electron pairs are created. The movement of these hole-electron pairs in a properly connected circuit results in current flow. The magnitude of the photocurrent depends on the number of charge carriers generated and hence, on the illumination of the diode element.

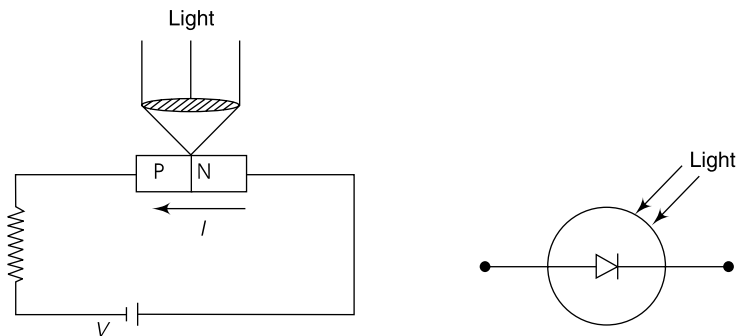


Fig. 1.43 Photodiode (a) Construction, and (b) Symbol

This current is also affected by the frequency of the light falling on the junction of the photodiode. The magnitude of the current under large reverse bias is given by

$$I = I_S + I_o (1 - e^{V/\eta V_T})$$

where I_o = reverse saturation current

I_S = short-circuit current which is proportional to the light intensity

V = voltage across the diode

V_T = volt equivalent of temperature

η = parameter, 1 for Ge and 2 for Si.

The characteristics of a photodiode are shown in Fig. 1.44. The reverse current increases in direct proportion to the level of illumination. Even when no light is applied, there is a minimum reverse leakage current called *dark current*, flowing through the device. Germanium has a higher dark current than silicon, but it also has a higher level of reverse current.

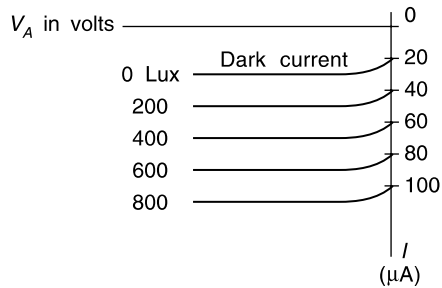


Fig. 1.44 Characteristics of Photodiode

Photodiodes are used as light detectors, demodulators and encoders. They are also used in optical communication system, high speed counting and switching circuits. Further, they are used in computer card punching and tapes, light operated switches, sound track films and electronic control circuits.

1.20

SPECIFICATIONS OF SEMICONDUCTOR DIODES AND SPECIAL PURPOSE ELECTRONIC DEVICES

The specifications of some of the commonly used PN junction diode and Zener diode are given in Table A.1 and Table A.2 respectively in Appendix A.

The specifications of some of the special purpose electronic devices such as varactor diode, tunnel diode, photodiode and SCR are given in Tables A.3, A.4, A.5 and A.10 respectively in Appendix A.

REVIEW QUESTIONS

1. What is meant by intrinsic semiconductor?
2. Explain the differences between intrinsic and extrinsic semiconductors.
3. Explain what a hole is. How do they move in intrinsic semiconductor?
4. What is meant by doping in a semiconductor?
5. Discuss the following with respect to semiconductor: (i) doping (ii) dopant (iii) donor and (iv) acceptor.

6. Explain “majority and minority carriers” in a semiconductor.
7. What is meant by N-type impurity in a semiconductor?
8. What is meant by P-type impurity in a semiconductor?
9. What is a PN junction? How is it formed?
10. Explain the formation of depletion region in a PN junction.
11. Draw the energy band diagram of a PN junction and explain the working of a diode.
12. (a) The resistivities of the P-region and N-region of a silicon diode are $6\ \Omega\text{-cm}$ and $4\ \Omega\text{-cm}$ respectively. Calculate the contact potential V_o and potential energy barrier E_o ; (b) if the doping densities of both P- and N-regions are tripled, determine V_o and E_o . Given that $q = 1.6 \times 10^{-19}\ \text{C}$, $n_i = 1.5 \times 10^{10}/\text{cm}^3$, $\mu_p = 500\ \text{cm}^2/\text{V-s}$, $\mu_n = 1300\ \text{cm}^2/\text{V-s}$ and $V_T = 0.026\ \text{V}$ at 300 K.
 [Ans: (a) $V_o = 0.6\ \text{V}$, $E_o = 0.6\ \text{eV}$ (b) $V_o = 0.66\ \text{V}$, $E_o = 0.66\ \text{eV}$]
13. Show that in an intrinsic semiconductor, the Fermi level is located at the middle of the unallowable energy gap.
14. Sketch the conduction and valence bands before and after diffusion of carriers in a PN junction.
15. Explain how a barrier pointial is developed at the PN junction.
16. Describe the action of PN junction diode under forward bias and reverse bias.
17. Explain how unidirectional current flow is possible through a PN junction diode.
18. Explain V - I characteristics of a PN junction diode.
19. Indicate the differences between the characteristics of silicon and germanium diodes and state approximately their cut-in voltages.
20. Explain the following terms in a PN junction diode:
 - (a) Maximum forward current
 - (b) Peak inverse voltage, and
 - (c) Maximum power rating
21. Explain the terms: (i) Static resistance, (ii) Dynamic resistance, (iii) Junction resistance, and (iv) Reverse resistance of a diode.
22. Write the volt-ampere equation for a PN diode. Give the meaning of each symbol.
23. What are the factors governing the reverse saturation current in a PN junction diode?
24. Determine the forward bias voltage applied to a silicon diode to cause a forward current of 10 mA and reverse saturation current, $I_o = 25 \times 10^{-7}\ \text{A}$ at room temperature.
 [Ans: 0.4 V]
25. The reverse saturation current I_o in a germanium diode is $6\ \mu\text{A}$. Calculate the current flowing through the diode when the applied forward bias voltages are 0.2, 0.3 and 0.4 V at room temperature. [Ans: 13.15 mA, 21.5 mA, 28.8 mA]
26. Define the term transition capacitance C_T of a PN diode.

27. Explain the term diffusion capacitance C_D of a forward biased diode.
28. Explain the effect of temperature of a diode.
29. Give the ideal diode current-voltage relationship. Describe the meaning of I_o and V_T .
30. What is iteration method of analysis?
31. Define a load line in a simple diode circuit.
32. Describe the piecewise linear approximation model of a diode.
33. Distinguish between avalanche and Zener mechanisms.
34. Can an ordinary rectifier diode be used as a Zener diode? Explain.
35. Mention the applications of PN junction diode.
36. Explain Avalanche breakdown and Zener breakdown.
37. Draw the V - I characteristic of Zener diode and explain its operation.
38. Show that the Zener diode can be used as a voltage regulator.
39. What is "tunneling?"
40. From the energy band diagram explain the V - I characteristic of a tunnel diode.
41. Draw the equivalent circuit of a tunnel diode and explain it.
42. List out the applications of tunnel diode and mention its advantages and disadvantages.
43. Explain the principle behind the varactor diode and list out its applications.
44. What is a thyristor? Mention some of them.
45. Describe the operation of Shockley diode.
46. Describe the working principle of an SCR with V - I characteristics.
47. Draw the two transistor model of an SCR and explain its breakdown operation.
48. Explain why an SCR is operated only in the forward biased condition.
49. Explain how triggering of an SCR can be controlled by the gate signal supplied.
50. Explain the terms (a) firing angle, and (b) conduction angle of an SCR.
51. Once the SCR is triggered, the gate loses its control. Explain.
52. Explain the two transistor analogy of an SCR.
53. A half wave rectifier circuit employing an SCR is adjusted to have a gate current of 1 mA and its forward breakdown voltage is 150 V. If a sinusoidal voltage of 400 V peak is applied, determine (i) firing angle, (ii) average output voltage, (iii) average current for a load resistance of 200 Ω , and (iv) power output. [Ans: (i) 22° (ii) 122.6 V (iii) 0.613 A (iv) 75.15 W]
54. A sinusoidal voltage $V_1 = 200 \sin 314t$ is applied to an SCR whose forward break-down voltage is 150 V. Determine the time during which SCR remains OFF. [Ans: 2.7 ms]
55. The brightness of a 100 W, 110 V lamp is to be varied by controlling firing angle of SCR full wave circuit. The RMS value of ac voltage appearing across each SCR is 110 V. Find the RMS voltage and current in the lamp and firing angle of 60°. [Ans: 98.9 V, 0.82 A]

56. Explain the principle and working of photodiode.
57. Write the equation for the volt-ampere characteristics of a photodiode. Define each term in the equation.
58. Explain the volt-ampere characteristics of a semiconductor photodiode.
59. List the applications of photodiode.

OBJECTIVE TYPE QUESTIONS

1. Valence electrons are the
 - (a) loosely packed electrons
 - (b) mobile electrons
 - (c) electrons present in the outermost orbit
 - (d) all the above
2. The element that does not have three valence electrons is
 - (a) boron
 - (b) aluminium
 - (c) germanium
 - (d) gallium
3. The element having four valence electrons is
 - (a) silicon
 - (b) germanium
 - (c) both (a) and (b) above
 - (d) none of the above
4. The forbidden energy gap for silicon is
 - (a) 1.12 eV
 - (b) 0.32 eV
 - (c) 0.72 eV
 - (d) 7.2 eV
5. The forbidden energy gap for germanium is
 - (a) 0.12 eV
 - (b) 0.32 eV
 - (c) 0.72 eV
 - (d) 0.92 eV
6. The resistivity of a semiconductor
 - (a) increases as the temperature increases
 - (b) decreases as the temperature increases
 - (c) remains constant even when temperature varies
 - (d) none of the above
7. Semiconductor has a
 - (a) zero temperature coefficient of resistance
 - (b) positive temperature coefficient of resistance
 - (c) negative temperature coefficient of resistance
 - (d) none of the above
8. The donor impurity element is
 - (a) aluminium
 - (b) boron
 - (c) phosphorous
 - (d) indium
9. The acceptor impurity element is
 - (a) antimony
 - (b) gallium
 - (c) arsenic
 - (d) phosphorous
10. The element which does not have five valence electrons is
 - (a) antimony
 - (b) arsenic
 - (c) gallium
 - (d) phosphorous
11. The forbidden bandgap of the semiconductor material
 - (a) increases with increase in temperature
 - (b) decreases with increase in temperature
 - (c) does not vary with temperature
 - (d) can increase or decrease with increase in temperature depending upon the semiconductor material
12. One of the following is not a semiconductor
 - (a) Gallium arsenide
 - (b) Indium
 - (c) Germanium
 - (d) Silicon

13. Which one of the following has the ability to act as an open circuit for dc and a short circuit for ac of high frequency?
(a) Inductor (b) Capacitor
(c) Resistor (d) None of the above
14. The static resistance of a diode is
(a) its opposition to the DC current flow
(b) its opposition to AC current flow
(c) resistance of diode when forward-biased
(d) none of these
15. When the reverse bias is applied to a junction diode, it
(a) lowers the potential barrier
(b) raises the potential barrier
(c) greatly decreases the minority-carrier current
(d) greatly increases the majority-carrier current
16. Doping of semiconductor is
(a) the process of purifying semiconductor materials
(b) the process of adding certain impurities to the semiconductor material in controlled amounts
(c) the process of converting semiconductor material into some form of active device such as FET, UJT etc.
(d) one of the steps used in fabrication of ICs
17. Referring to energy level diagram of semiconductor materials, the width of forbidden band-gap is about
(a) 10 eV (b) 100 eV (c) 1 eV (d) 0.1 eV
18. A PN-junction diode
(a) has high resistance in both forward and reverse directions
(b) has low resistance in the forward direction
(c) has high resistance in the forward direction
(d) has low resistance in the reverse direction
19. If a PN-junction diode is not biased, the junction current at equilibrium is
(a) zero as no charges cross the junction
(b) zero as equal number of carriers cross the barrier
(c) mainly due to diffusion of majority carriers
(d) mainly due to diffusion of minority carriers
20. In a PN-junction, the potential barrier is due to the charges on either side of the junction, which consists of
(a) fixed donor and acceptor ions (b) majority carriers only
(c) minority carriers only (d) both majority and minority carriers
21. In a PN-junction, the region containing the uncompensated acceptor and donor ions is called
(a) transition zone (b) depletion region
(c) neutral region (d) active region
22. In a forward-biased PN-junction diode, the
(a) positive terminal of the battery is connected to the P-side and the negative side to the N-side
(b) positive terminal of the battery is connected to the N-side and the negative side to the P-side

- (c) N-side is connected directly to the N-side
 - (d) junction is earthed
23. When a PN junction is forward biased
- (a) electrons in the N region are injected into the P region
 - (b) holes in the P region are injected into the N region
 - (c) both (a) and (b)
 - (d) None of the above
24. When we apply reverse bias to a junction diode, it
- (a) lowers the potential barrier
 - (b) raises the potential barrier
 - (c) greatly decreases the minority-carrier current
 - (d) greatly increases the majority-carrier current
25. Under normal operating voltage, the reverse current in a silicon diode is about
- (a) 10 mA
 - (b) 1 μ A
 - (c) 1000 μ A
 - (d) None
26. The depletion region in a PN diode is due to
- (a) reverse biasing
 - (b) forward biasing
 - (c) an area created by crystal doping
 - (d) an area void of current carriers
27. When a diode is forward biased,
- (a) barrier potential increases
 - (b) barrier potential decreases
 - (c) majority current decreases
 - (d) minority current decreases
28. For a Germanium PN junction, the maximum value of barrier potential is
- (a) 0.3 V
 - (b) 0.7 V
 - (c) 1.3 V
 - (d) 1.7 V
29. For a Silicon PN junction, the maximum value of barrier potential is
- (a) 0.3 V
 - (b) 0.7 V
 - (c) 1.3 V
 - (d) 1.7 V
30. The resistivity of a semiconductor depends on the
- (a) shape of the semiconductor
 - (b) atomic nature of the semiconductor
 - (c) width of the semiconductor
 - (d) length of the semiconductor
31. When holes leave the *p*-material to fill electrons in the *n*-material, the process is called
- (a) mixing
 - (b) depletion
 - (c) diffusion
 - (d) none of the above
32. The depletion region in a PN diode is due to
- (a) reverse biasing
 - (b) forward biasing
 - (c) an area created by crystal doping
 - (d) an area void of current carriers
33. When a diode is forward biased, the
- (a) barrier potential increases
 - (b) barrier potential decreases
 - (c) majority current decreases
 - (d) minority current decreases
34. In a semiconductor diode, V - I relationship is such that
- (a) current varies linearly with voltage
 - (b) current increases exponentially with voltage
 - (c) current varies inversely with voltage
 - (d) none of these

35. The capacitance appearing across a reverse-biased semiconductor junction
 - (a) increases with increase in bias voltage
 - (b) decreases with increase in bias voltage
 - (c) is independent of bias voltage
 - (d) none of these
36. Zener breakdown occurs
 - (a) due to normally generated minority carriers
 - (b) in lightly doped junctions
 - (c) due to rupture of covalent bonds
 - (d) mostly in germanium junctions
37. A breakdown which is caused by cumulative multiplication of carriers through field-induced impact ionization occurs in
 - (a) Zener diode
 - (b) tunnel diode
 - (c) varactor diode
 - (d) avalanche diode
38. Zener diode is usually operated
 - (a) in forward-bias mode
 - (b) in reverse-bias mode
 - (c) near cut-in voltage
 - (d) in forward-linear region
39. For a highly doped diode
 - (a) Zener breakdown is like to take place
 - (b) avalanche breakdown is likely to take place
 - (c) either (a) or (b) will take place
 - (d) neither (a) or (b) will take place
40. Which one of the following diodes is used for voltage stabilization?
 - (a) PN-junction
 - (b) Tunnel
 - (c) Varactor
 - (d) Zener
41. In Zener and avalanche breakdown diodes, the current flow is due to
 - (a) majority carriers
 - (b) minority carriers
 - (c) majority and minority carriers
 - (d) none of these
42. The breakdown that occurs in the reverse biased condition in a narrow junction diode is
 - (a) Zener breakdown
 - (b) avalanche breakdown
 - (c) both (a) and (b)
 - (d) none of these
43. The breakdown that occurs in the reverse biased condition in a wider junction diode is
 - (a) Zener breakdown
 - (b) avalanche breakdown
 - (c) both (a) and (b)
 - (d) none of these
44. A Zener breakdown diode has
 - (a) a positive temperature coefficient
 - (b) a negative temperature coefficient
 - (c) a breakdown voltage that is independent of temperature
 - (d) none of these
45. An avalanche breakdown diode has
 - (a) a positive temperature coefficient
 - (b) a negative temperature coefficient
 - (c) a breakdown voltage that is independent of temperature
 - (d) none of these

46. In avalanche multiplication, pickup the correct answer
(a) Disruption of covalent bond occur by collision
(b) Direct rupture
(c) (a) and (b) both
(d) None of the above
47. Which of the following statement is best situated for a Zener diode?
(a) It is rectifier diode (b) It works in the forward bias region
(c) It is a constant voltage device (d) It is mostly used in clipping circuit
48. Which of the following statement is best suited for a Zener diode?
(a) It is rectifier diode (b) It works in the forward bias region
(c) It is a constant voltage device (d) It is mostly used in clipping circuit
49. There are two semiconductor diodes A and B. One of them is Zener whereas other is avalanche. Their ratings are 5.6V and 24V, respectively, then
(a) A is Zener, B is avalanche (b) A is avalanche, B is Zener
(c) both of them are Zener diodes (d) both of them are avalanche diodes
50. The diode used in voltage regulator is
(a) PN junction diode (b) Varactor diode
(c) Zener diode (d) GUNN diode
51. A tunnel diode
(a) is a reverse recovery diode (b) has heavy doping
(c) is a power diode (d) has light doping
52. Which one of the following diodes shows the negative resistance region?
(a) PN-junction (b) Tunnel
(c) Zener (d) Varactor
53. The most important application of a tunnel diode is as a
(a) rectifier (b) switching device
(c) voltage controlled device (d) none of these
54. The V - I characteristics of a tunnel diode exhibit a
(a) multivalued function of voltage (b) multivalued function of current
(c) single valued function of current (d) none of these
55. An SCR conducts appreciable current when its with respect to cathode.
(a) anode and gate are both negative
(b) anode and gate are both positive
(c) anode is negative and gate is positive
(d) gate is negative and anode is positive
56. An SCR may be turned OFF by
(a) interrupting its anode current
(b) reversing polarity of its anode-cathode voltage
(c) low-current dropout
(d) all of the above
57. The $\frac{dv}{dt}$ effect in an SCR can result in
(a) a high rate of rise of anode voltage
(b) an increased junction capacitance

- (c) a false triggering
 - (d) a low capacitive charging current
58. The $\frac{di}{dt}$ effect in an SCR leads to the formation of
- (a) local hot spots (b) conduction zone
 - (c) charge-spreading zone (d) none of the above
59. An SCR turns off from conducting state to blocking state on
- (a) reducing gate current
 - (b) reversing gate voltage
 - (c) reducing anode current below holding current value
 - (d) applying ac to the gate
60. When a thyristor is negatively biased,
- (a) all the three junctions are negatively biased
 - (b) outer junctions are positively biased and the inner junction is negatively biased
 - (c) outer junctions are negatively biased and the inner junctions is positively biased
 - (d) the junction near the anode is negatively biased and the one near the cathode is positively biased.
61. The minimum value of current required to maintain conduction in an SCR is called its
- (a) commutation (b) holding
 - (c) gate trigger (d) breakover
62. A photodiode is used in reverse bias because the
- (a) majority swept are reverse across the junction
 - (b) only one side is illuminated
 - (c) reverse current is small as compared to photocurrent
 - (d) reverse current is large as compared to photocurrent

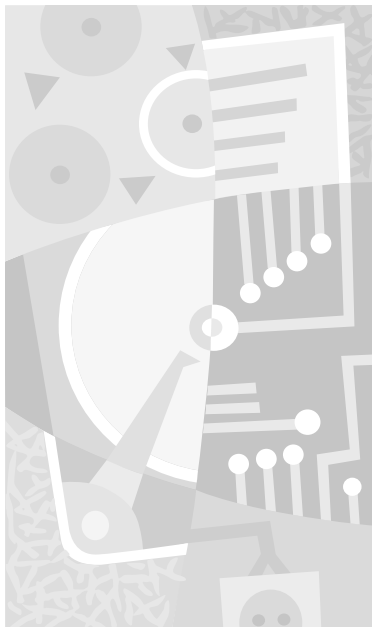
State whether the following statements are true (T) or false (F)

- 63. Silicon and germanium are semiconductors with a valence of 4.
- 64. Hole current is the movement of negative charges in the opposite direction from the electron flow.
- 65. Doping with phosphorous makes silicon P-type, with majority electron charges and minority hole charge.
- 66. Hole current into an electrode is considered as the positive direction of current.
- 67. The internal barrier potential at a PN-junction for silicon is approximately 9 V.
- 68. Atoms of the impurity element in a doped semiconductor provide fixed ion charges.
- 69. A depletion layer consists of free charge carriers.
- 70. An N-type semiconductor has free electrons, while a P-type has hole charges.
- 71. A hole has the same amount of positive charge as a proton, equal to an electron but with opposite polarity.
- 72. The proton is a stable charge in the nucleus that is not free to move.
- 73. A hole is a positive charge outside the nucleus present only in semiconductors because of unfilled covalent bond.

74. The forward current of a PN-junction can be controlled to provide amplification.
75. Gallium is generally used for germanium as donor impurity element.
76. Phosphorous in its purest form acts as an intrinsic semiconductor.
77. Gallium is generally used as an acceptor impurity for germanium.
78. A Zener diode operates in the reverse biased region.
79. A Zener diode is used as a rectifier.
80. The diode current equation is $I_0 = I \left[e^{(V/nV_T)} - 1 \right]$.
81. To obtain N type semiconductor, the impurity added to a pure semiconductor is pentavalent.
82. A semiconductor has a negative temperature coefficient of resistance.
83. In the reverse bias region, the reverse saturation current of a silicon diode rises for every 10°C in temperature.
84. When PN junction is biased in the forward direction majority carriers in each region are injected into the other region.
85. The tunnel diode exhibits positive resistance under low forward bias conditions.
86. The tunnel diode can be used as a logic memory storage device.
87. A varactor diode is also called a voltage variable capacitor.
88. The SCR is a silicon rectifier with a gate electrode to control when the current flows from cathode to anode.
89. The SCR is a silicon diode rectifier with a gate-control electrode.
90. Avalanche photodiodes internally do not multiply the primary signal photocurrent before it enters the input circuitry of the following amplifier.

ANSWERS

- | | | | | |
|---------|---------|---------|---------|---------|
| 1. (c) | 2. (d) | 3. (c) | 4. (a) | 5. (c) |
| 6. (b) | 7. (c) | 8. (c) | 9. (b) | 10. (c) |
| 11. (b) | 12. (b) | 13. (b) | 14. (a) | 15. (b) |
| 16. (b) | 17. (c) | 18. (b) | 19. (b) | 20. (a) |
| 21. (b) | 22. (a) | 23. (c) | 24. (b) | 25. (b) |
| 26. (a) | 27. (b) | 28. (a) | 29. (b) | 30. (d) |
| 31. (c) | 32. (a) | 33. (b) | 34. (b) | 35. (b) |
| 36. (c) | 37. (d) | 38. (b) | 39. (a) | 40. (d) |
| 41. (b) | 42. (a) | 43. (b) | 44. (a) | 45. (b) |
| 46. (a) | 47. (c) | 48. (c) | 49. (a) | 50. (c) |
| 51. (b) | 52. (b) | 53. (c) | 54. (b) | 55. (b) |
| 56. (d) | 57. (c) | 58. (a) | 59. (c) | 60. (c) |
| 61. (b) | 62. (c) | 63. (T) | 64. (F) | 65. (F) |
| 66. (T) | 67. (F) | 68. (T) | 69. (F) | 70. (T) |
| 71. (T) | 72. (T) | 73. (T) | 74. (T) | 75. (F) |
| 76. (F) | 77. (T) | 78. (T) | 79. (F) | 80. (F) |
| 81. (T) | 82. (T) | 83. (T) | 84. (T) | 85. (F) |
| 86. (T) | 87. (T) | 88. (T) | 89. (T) | 90. (F) |



2

RECTIFIERS AND FILTERS

2.1 INTRODUCTION

All electronic circuits need dc power supply either from battery or power pack units. It may not be economical and convenient to depend upon battery power supply. Hence, many electronic equipment contain circuits which convert the ac supply voltage into dc voltage at the required level. The unit containing these circuits is called the *Linear Mode Power Supply* (LMPS). In the absence of ac mains supply, the dc supply from battery can be converted into required ac voltage which may be used by computer and other electronic systems for their operation. Also, in certain applications, dc to dc conversion is required. Such a power supply unit that converts dc into ac or dc is called *Switched Mode Power Supply* (SMPS).

1. Linear mode power supply (LMPS): ac/dc power supply—Converter
2. Switched mode power supply (SMPS): (i) dc/dc power supply—converter
(ii) dc/ac power supply—Inverter

An ac/dc power supply converts ac mains (230 V, 50 Hz) into required dc voltages and is found in all mains operable systems.

DC/DC power supplies or dc/dc converters are used in portable systems, DC/AC power supplies or inverter are used in portable main operable system and as a supplement to ac mains in non-portable mains operable system, where a disruption in the power supply can affect the job being done by the system. An inverter is a form of UPS (Uninterrupted Power Supply) or SPS (Standby Power Supply) and is very popular in computer system.

Based on the regular concept, power supplies are classified as either *linear* or *switched mode power supply*. The main difference between LMPS and SMPS is seen from their block diagrams given in Figs 2.1(a) and (b).

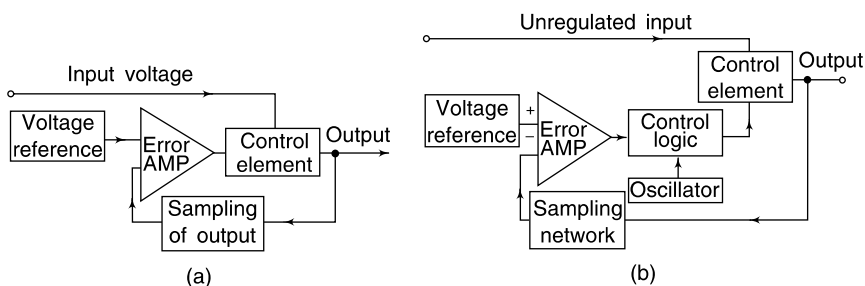


Fig. 2.1 (a) Principle of a Normal Linear Mode Power Supply,
(b) Block Diagram of a Switched Mode Power Supply

2.2 LINEAR MODE POWER SUPPLY

The basic building blocks of the linear power supply are shown in Fig. 2.2. A transformer supplies ac voltage at the required level. This bidirectional ac voltage is converted into a unidirectional pulsating dc using a rectifier. The unwanted ripple contents of this pulsating dc are removed by a filter to get pure dc voltage. The output of the filter is fed to a regulator which gives a steady dc output independent of load variations and input supply fluctuations.

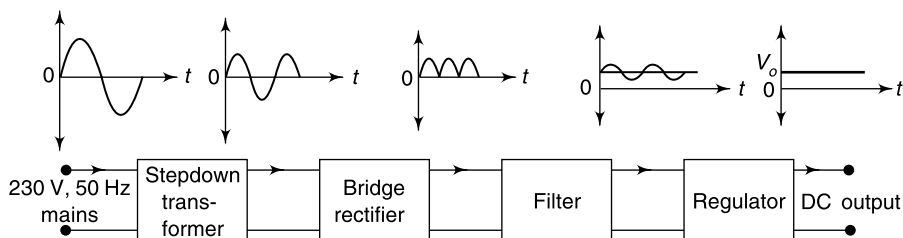


Fig. 2.2 Basic Building Block of Linear Mode Power Supply

2.2.1 Requirements of Linear Mode Power Supply

1. The most important consideration in designing a power supply is the dc voltage at the output. It should be able to give minimum operable dc voltage at the rated current.
2. It should be able to furnish the maximum current needed for the unit, maintaining the voltage constant. In other words, the regulation of the power supply should be good.
3. The ac ripple should be low.
4. The power supply should be protected in the event of short-circuit on the load side.
5. Over voltage (spike and surges) protection must be incorporated.
6. The response of the power supply to temperature changes should be minimum.

2.3 P-N JUNCTION DIODE AS A RECTIFIER

A PN junction diode is a two terminal device that is polarity sensitive. When the diode is forward biased, the diode conducts and allows current to flow through it without any resistance, i.e. the diode is ON. When the diode is reverse biased, the diode does not conduct and no current flows through it, i.e. the diode is OFF, or providing a blocking function. Thus an ideal diode acts as a switch, either open or closed, depending upon the polarity of the voltage placed across it. The ideal diode has zero resistance under forward bias and infinite resistance under reverse bias.

2.4 RECTIFIERS

Rectifier is defined as an electronic device used for converting ac voltage into unidirectional voltage. A rectifier utilizes unidirectional conduction device like a vacuum diode or PN junction diode. Rectifiers are classified depending upon the period of conduction as Half-wave rectifier and Full-wave rectifier.

2.4.1 Half-wave Rectifier

It converts an ac voltage into a pulsating dc voltage using only one half of the applied ac voltage. The rectifying diode conducts during one half of the ac cycle only. Figure 2.3 shows the basic circuit and waveforms of a half wave rectifier.

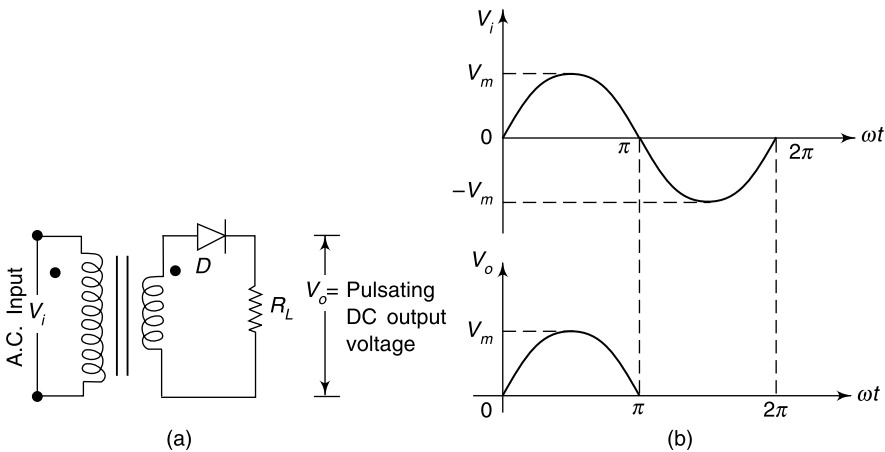


Fig. 2.3 (a) Basic Structure of a Half-wave Rectifier and
(b) Input and Output Waveforms of Half-wave Rectifier

Let V_i be the voltage to the primary of the transformer and given by the equation

$$V_i = V_m \sin \omega t; V_m \gg V_r$$

where V_r is the cut-in voltage of the diode. During the positive half-cycle of the input signal, the anode of the diode becomes more positive with respect to the cathode and hence, diode D conducts. For an ideal diode, the forward voltage drop is zero. So the whole input voltage will appear across the load resistance, R_L .

During negative half-cycle of the input signal, the anode of the diode becomes negative with respect to the cathode and hence, diode D does not conduct. For an ideal diode, the impedance offered by the diode is infinity. So the whole input voltage appears across diode D . Hence, the voltage drop across R_L is zero.

Ripple factor (Γ) The ratio of rms value of ac component to the dc component in the output is known as ripple factor (Γ)

$$\Gamma = \frac{\text{rms value of ac component}}{\text{dc value of component}} = \frac{V_{r, rms}}{V_{dc}}$$

where

$$V_{r, rms} = \sqrt{V_{rms}^2 - V_{dc}^2}$$

$$\Gamma = \sqrt{\left(\frac{V_{rms}}{V_{dc}}\right)^2 - 1}$$

V_{av} is the average or the dc content of the voltage across the load and is given by

$$V_{av} = V_{dc} = \frac{1}{2\pi} \left[\int_0^{\pi} V_m \sin \omega t d(\omega t) + \int_{\pi}^{2\pi} 0 d(\omega t) \right]$$

$$= \frac{V_m}{2\pi} [-\cos \omega t]_0^{\pi} = \frac{V_m}{\pi}$$

Therefore,

$$I_{dc} = \frac{V_{dc}}{R_L} = \frac{V_m}{\pi R_L} = \frac{I_m}{\pi}$$

If the values of diode forward resistance (r_f) and the transformer secondary winding resistance (r_s) are also taken into account, then

$$V_{dc} = \frac{V_m}{\pi} - I_{dc} (r_s + r_f)$$

$$I_{dc} = \frac{V_{dc}}{(r_s + r_f) + R_L} = \frac{V_m}{\pi (r_s + r_f + R_L)}$$

RMS voltage at the load resistance can be calculated as

$$\begin{aligned} V_{rms} &= \left[\frac{1}{2\pi} \int_0^{\pi} V_m^2 \sin^2 \omega t d(\omega t) \right]^{\frac{1}{2}} \\ &= V_m \left[\frac{1}{4\pi} \int_0^{\pi} (1 - \cos 2\omega t) d(\omega t) \right]^{\frac{1}{2}} = \frac{V_m}{2} \end{aligned}$$

Therefore
$$\Gamma = \sqrt{\left[\frac{V_m / 2}{V_m / \pi} \right]^2} - 1 = \sqrt{\left(\frac{\pi}{2} \right)^2} - 1 = 1.21$$

From this expression it is clear that the amount of ac present in the output is 121% of the dc voltage. So the half-wave rectifier is not practically useful in converting ac into dc.

Efficiency (η) The ratio of dc output power to ac input power is known as rectifier efficiency (η).

$$\begin{aligned} \eta &= \frac{\text{dc output power}}{\text{ac input power}} = \frac{P_{dc}}{P_{ac}} \\ &= \frac{\frac{(V_{dc})^2}{R_L}}{\frac{(V_{rms})^2}{R_L}} = \frac{\left(\frac{V_m}{\pi} \right)^2}{\left(\frac{V_m}{2} \right)^2} = \frac{4}{\pi^2} = 0.406 = 40.6\% \end{aligned}$$

The maximum efficiency of a half-wave rectifier is 40.6%.

Peak Inverse Voltage (PIV) It is defined as the maximum reverse voltage that a diode can withstand without destroying the junction. The peak inverse voltage across a diode is the peak of the negative half cycle. For half-wave rectifier, PIV is V_m .

Transformer Utilisation Factor (TUF) In the design of any power supply, the rating of the transformer should be determined. This can be done with a knowledge of the dc power delivered to the load and the type of rectifying circuit used.

$$\begin{aligned} \text{TUF} &= \frac{\text{dc power delivered to the load}}{\text{ac rating of the transformer secondary}} \\ &= \frac{P_{dc}}{P_{ac \text{ rated}}} \end{aligned}$$

In the half-wave rectifying circuit, the rated voltage of the transformer secondary is $V_m / \sqrt{2}$, but the actual rms current flowing through the winding is only $\frac{I_m}{2}$, not $I_m / \sqrt{2}$.

$$\begin{aligned} \text{TUF} &= \frac{\frac{I_m^2}{\pi^2} R_L}{\frac{V_m}{\sqrt{2}} \times \frac{I_m}{2}} = \frac{\frac{V_m^2}{\pi^2} \frac{I}{R_L}}{\frac{V_m}{\sqrt{2}} \frac{V_m}{2 R_L}} = \frac{2\sqrt{2}}{\pi^2} = 0.287 \end{aligned}$$

The TUF for a halfwave rectifier is 0.287.

Form Factor

$$\begin{aligned}\text{Form factor} &= \frac{\text{rms value}}{\text{average value}} \\ &= \frac{V_m / 2}{V_m / \pi} = \frac{\pi}{2} = 1.57\end{aligned}$$

Peak Factor

$$\begin{aligned}\text{Peak factor} &= \frac{\text{peak value}}{\text{rms value}} \\ &= \frac{V_m}{V_m / 2} = 2\end{aligned}$$

Example 2.1 A half-wave rectifier, having a resistive load of 1000Ω , rectifies an alternating voltage of 325 V peak value and the diode has a forward resistance of 100Ω . Calculate (a) peak, average and rms value of current (b) dc power output (c) ac input power, and (d) efficiency of the rectifier.

Solution (a) Peak value of current, $I_m = \frac{V_m}{r_f + R_L} = \frac{325}{100 + 1000} = 295.45 \text{ mA}$

Average current, $I_{dc} = \frac{I_m}{\pi} = \frac{295.45}{\pi} \text{ mA} = 94.046 \text{ mA}$

RMS value of current, $I_{rms} = \frac{I_m}{2} = \frac{295.45}{2} = 147.725 \text{ mA}$

(b) DC power output, $P_{dc} = I_{dc}^2 \times R_L$
 $= (94.046 \times 10^{-3})^2 \times 1000 = 8.845 \text{ W}$

(c) AC input power, $P_{ac} = (I_{rms})^2 \times (r_f + R_L)$
 $= (147.725 \times 10^{-3})^2 (1100) = 24 \text{ W}$

(d) Efficiency of rectification, $\eta = \frac{P_{dc}}{P_{ac}} = \frac{8.845}{24} = 36.85\%$

Example 2.2 A HWR has a load of $3.5 \text{ k}\Omega$. If the diode resistance and secondary coil resistance together have a resistance of 800Ω and the input voltage has a signal voltage of peak value 240 V . Calculate

(a) Peak, average and rms value of current flowing.

(b) dc power output

(c) ac power input

(d) efficiency of the rectifier

[JNTU May 2003, Dec 2003]

Solution Load resistance in a HWR, $R_L = 3.5 \text{ k}\Omega$
 Diode and secondary coil resistance, $r_f + r_s = 800 \Omega$
 Peak value of input voltage = 240 V

(a) Peak value of current,
$$I_m = \frac{V_m}{r_s + r_f + R_L} = \frac{240}{4300} = 55.81 \text{ mA}$$

Average value of current,
$$I_{dc} = \frac{I_m}{\pi} = \frac{55.81 \times 10^{-3}}{\pi} = 17.77 \text{ mA}$$

RMS value of current,
$$I_{rms} = \frac{I_m}{2} = \frac{55.81 \times 10^{-3}}{2} = 27.905 \text{ mA}$$

(b) DC power output is

$$P_{dc} = (I_{dc})^2 R_L = (17.77 \times 10^{-3})^2 \times 3500 = 1.105 \text{ W}$$

(c) AC power input is

$$P_{ac} = (I_{rms})^2 \times (r_f + R_L) = (27.905 \times 10^{-3})^2 \times 4300 = 3.348 \text{ W}$$

(d) Efficiency of the rectifier is

$$\eta = \frac{P_{dc}}{P_{ac}} = \frac{1.105}{3.348} \times 100 = 33\%$$

Example 2.3 A half-wave rectifier is used to supply 24 V dc to a resistive load of 500 Ω and the diode has a forward resistance of 50 Ω . Calculate the maximum value of the ac voltage required at the input.

Solution Average value of load current,
$$I_{dc} = \frac{V_{dc}}{R_L} = \frac{24}{500} = 48 \text{ mA}$$

Maximum value of load current,
$$I_m = \pi \times I_{dc} = \pi \times 48 \text{ mA} = 150.8 \text{ mA}$$

Therefore, maximum ac voltage required at the input,

$$\begin{aligned} V_m &= I_m \times (r_f + R_L) \\ &= 150.8 \times 10^{-3} \times 550 = 82.94 \text{ V} \end{aligned}$$

Example 2.4 An ac supply of 230 V is applied to a half-wave rectifier circuit through transformer of turns ratio 5:1. Assume the diode is an ideal one. The load resistance is 300 Ω . Find (a) dc output voltage (b) PIV (c) maximum, and (d) average values of power delivered to the load.

Solution (a) The transformer secondary voltage
$$= \frac{230}{5} = 46 \text{ V}$$

Maximum value of secondary voltage,
$$V_m = \sqrt{2} \times 46 = 65 \text{ V}$$

Therefore, dc output voltage,
$$V_{dc} = \frac{V_m}{\pi} = \frac{65}{\pi} = 20.7 \text{ V}$$

(b) PIV of a diode
$$= V_m = 65 \text{ V}$$

(c) Maximum value of load current, $I_m = \frac{V_m}{R_L} = \frac{65}{300} = 0.217 \text{ A}$

Therefore, maximum value of power delivered to the load,

$$P_m = I_m^2 \times R_L = (0.217)^2 \times 300 = 14.1 \text{ W}$$

(d) The average value of load current, $I_{dc} = \frac{V_{dc}}{R_L} = \frac{20.7}{300} = 0.069 \text{ A}$

Therefore, average value of power delivered to the load,

$$P_{dc} = I_{dc}^2 \times R_L = (0.069)^2 \times 300 = 1.43 \text{ W}$$

Example 2.5 A HWR circuit supplies 100 mA dc to a 250 Ω load. Find the dc output voltage, PIV rating of a diode and the rms voltage for the transformer supplying the rectifier. [JNTU May 2008 and Jan 2010]

Solution Given $I_{dc} = 100 \text{ mA}$, $R_L = 250 \Omega$

(a) The dc output voltage, $V_{dc} = I_{dc} \times R_L = 100 \times 10^{-3} \times 250 = 25 \text{ V}$

(b) The maximum value of secondary voltage,

$$V_m = \pi \times V_{dc} = \pi \times 25 = 78.54 \text{ V}$$

(c) PIV rating of a diode = $V_m = 78.54 \text{ V}$

(d) rms voltage for the transformer supplying the rectifier

$$V_{rms} = \frac{V_m}{2} = \frac{78.54}{2} = 39.27 \text{ V}$$

Example 2.6 A voltage of $200 \cos \omega t$ is applied to HWR with load resistance of 5 k Ω . Find the maximum dc current component, rms current, ripple factor, TUF and rectifier efficiency. [JNTU May 2008 and Jan 2010]

Solution Given Applied Voltage = $200 \cos \omega t$, $V_m = 200 \text{ V}$, $R_L = 5 \text{ k}\Omega$

(a) To find dc current

$$I_m = \frac{V_m}{R_L} = \frac{200}{5 \times 10^3} = 40 \text{ mA}$$

Therefore, $I_{dc} = \frac{I_m}{\pi} = \frac{40 \times 10^{-3}}{\pi} = 12.7 \times 10^{-3} \text{ A} = 12.73 \text{ mA}$

(b) To find rms current

$$I_{rms} = \frac{I_m}{2} = \frac{40 \times 10^{-3}}{2} = 20 \text{ mA}$$

(c) Ripple factor $\Gamma = \sqrt{\left(\frac{I_{rms}}{I_{dc}}\right)^2 - 1} = \sqrt{\left(\frac{20 \times 10^{-3}}{12.73 \times 10^{-3}}\right)^2 - 1} = 1.21$

(d) To determine TUF

$$TUF = \frac{P_{dc}}{P_{ac(rated)}}$$

$$P_{dc} = I_{dc}^2 R_L = (12.73 \times 10^{-3})^2 \times 5 \times 10^3 = 0.81 \text{ W}$$

$$P_{ac(rated)} = \frac{V_m}{\sqrt{2}} \times \frac{I_m}{2} = \frac{200}{\sqrt{2}} \times \frac{40 \times 10^{-3}}{2} = 2.828$$

Therefore, $TUF = \frac{P_{dc}}{P_{ac(rated)}} = \frac{0.81}{2.828} = 0.2863$

(e) Rectifier Efficiency, $\eta = \frac{P_{dc}}{P_{ac}}$

$$P_{dc} = 0.81 \text{ W}$$

$$P_{ac} = I_{rms}^2 R_L = (20 \times 10^{-3})^2 \times 5 \times 10^3 = 2 \text{ W}$$

Therefore, $\eta = \frac{P_{dc}}{P_{ac}} \times 100 = \frac{0.81}{2} \times 100 = 40.5\%$

Example 2.7 A diode has an internal resistance of 20Ω and 1000Ω load from a 110 V rms source of supply. Calculate (i) the efficiency of rectification (ii) the percentage regulation from no load to full load.

[JNTU May/June 2006 and Aug 2008]

Solution Given, $r_f = 20 \Omega$, $R_L = 1000 \Omega$ and $V_{rms} \text{ (secondary)} = 110 \text{ V}$

The half-wave rectifier uses a single diode.

Therefore, $V_m = \sqrt{2} V_{rms} \text{ (secondary)} = 155.56 \text{ V}$

$$I_m = \frac{V_m}{r_f + R_L} = \frac{155.56}{20 + 1000} = 0.1525 \text{ A}$$

$$I_{dc} = \frac{I_m}{\pi} = \frac{0.1525}{\pi} = 0.04854 \text{ A}$$

$$V_{dc} = I_{dc} R_L = 0.04854 \times 1000 = 48.54 \text{ V}$$

$$P_{dc} = V_{dc} I_{dc} = 48.54 \times 0.04854 = 2.36 \text{ W}$$

$$P_{ac} = I_{rms}^2 (r_f + R_L) = \left(\frac{I_m}{2} \right)^2 (r_f + R_L)$$

$$\text{(since } I_{rms} = \frac{I_m}{2} \text{ for half-wave)}$$

$$= \left(\frac{0.1525}{2} \right)^2 (1000 + 20) = 5.93 \text{ W}$$

Efficiency, $\eta = \frac{P_{dc}}{P_{ac}} \times 100 = \frac{2.36}{5.93} \times 100 = 39.7346\%$

$$\begin{aligned}\text{Percentage of line regulation} &= \frac{V_{NL} - V_{FL}}{V_{FL}} \times 100 = \frac{\frac{V_m}{\pi} - V_{dc}}{V_{dc}} \times 100 \\ &= \frac{\frac{155.56}{\pi} - 48.54}{48.54} \times 100 = 2\%\end{aligned}$$

Example 2.8 Show that maximum d.c. output power $P_{dc} = V_{dc} \times I_{dc}$ in a half-wave single phase circuit occur when the load resistance equals diode resistance r_f .
[JNTU Nov/Dec 2004]

Solution For a half wave rectifier,

$$I_m = \frac{V_m}{r_f + R_L}$$

and
$$I_{dc} = \frac{I_m}{\pi} = \frac{V_m}{\pi(r_f + R_L)}$$

and
$$V_{dc} = I_{dc} \times R_L$$

Therefore,
$$P_{dc} = V_{dc} \times I_{dc} = I_{dc}^2 R_L = \frac{V_m^2 R_L}{\pi^2 (r_f + R_L)^2}$$

For this power to be maximum,

$$\frac{dP_{dc}}{dR_L} = 0$$

$$\frac{d}{dR_L} \left[\frac{V_m^2 R_L}{\pi^2 (r_f + R_L)^2} \right] = \frac{V_m^2}{\pi^2} \left[\frac{(r_f + R_L)^2 - R_L \times 2(r_f + R_L)}{(r_f + R_L)^4} \right] = 0$$

$$(r_f + R_L)^2 - 2R_L(r_f + R_L) = 0$$

$$r_f^2 + 2r_f R_L + R_L^2 - 2r_f R_L - 2R_L^2 = 0$$

$$r_f^2 - R_L^2 = 0$$

$$R_L^2 = r_f^2$$

Thus the power output is maximum if $R_L = r_f$.

Example 2.9 The transformer of a half-wave rectifier has a secondary voltage of $30 V_{rms}$ with a winding resistance of 10Ω . The semiconductor diode in the circuit has a forward resistance of 100Ω . Calculate (a) No load dc voltage (b) dc output voltage at $I_L = 25 \text{ mA}$ (c) % regulation at $I_L = 25 \text{ mA}$ (d) ripple voltage across the load (e) ripple frequency (f) ripple factor (g) dc power output and (h) PIV of the semiconductor diode.

[JNTU Aug 2008]

Solution $V_{rms}(\text{secondary}) = 30 \text{ V}$, $r_s = 10 \Omega$, $r_f = 100 \Omega$

$$V_m = \sqrt{2} \times V_{rms} = \sqrt{2} \times 30 = 42.4264 \text{ V}$$

$$(a) \quad V_{dc} = \frac{V_m}{\pi} = \frac{42.4264}{\pi} = 13.5047 \text{ V}$$

$$(b) \quad I_L = I_{dc} = 25 \text{ mA}$$

$$V_{dc} = I_{dc} R_L = \frac{I_m}{\pi} R_L = \frac{V_m}{\pi(r_f + r_s + R_L)} \times R_L$$

$$\text{Here } R_L = \frac{V_{dc}}{I_{dc}}$$

$$\text{Therefore, } V_{dc} = \frac{V_m}{\pi \left(r_f + r_s + \frac{V_{dc}}{I_{dc}} \right)} \times \frac{V_{dc}}{I_{dc}}$$

$$V_{dc} = \frac{42.4264 V_{dc}}{\pi \left(100 + 10 + \frac{V_{dc}}{25 \times 10^{-3}} \right)} \times \frac{1}{25 \times 10^{-3}}$$

$$V_{dc} (110 + 40 V_{dc}) = 540.1897 V_{dc}$$

$$V_{dc} = \frac{540.1897 - 110}{40} = 10.7547 \text{ V}$$

$$(c) \quad \text{Percentage of regulation} = \frac{V_{dc(NL)} - V_{dc(FL)}}{V_{dc(FL)}} \times 100$$

$$= \frac{13.5047 - 10.7547}{10.7547} \times 100 = 25.569\%$$

$$(d) \quad I_m = \frac{V_m}{r_f + r_s + R_L}, \text{ where } R_L = \frac{V_{dc}}{I_{dc}} = \frac{10.7547}{25 \times 10^{-3}} = 430.188 \Omega$$

$$\text{Therefore, } I_m = \frac{42.4264}{100 + 10 + 430.188} = 0.07854 \text{ A}$$

$$I_{rms} = \frac{I_m}{2} = 0.03927 \text{ A}$$

$$r = \sqrt{\left(\frac{I_{rms}}{I_{dc}}\right)^2} - 1 = \sqrt{\left(\frac{0.03927}{25 \times 10^{-3}}\right)^2} - 1 = 1.21$$

Ripple voltage $\Gamma \times V_{dc} = 1.21 \times 10.7547 = 13.02791 \text{ V}$

(e) Ripple frequency $= f = 50 \text{ Hz}$

(f) $\Gamma = \text{ripple factor} = 1.21$

(g) $P_{dc} = V_{dc} I_{dc} = 10.7547 \times 25 \times 10^{-3} = 0.2688 \text{ W}$

(h) $\text{PIV} = V_m = 42.4264 \text{ V}$

2.4.2 Full-wave Rectifier

It converts an ac voltage into a pulsating dc voltage using both half cycles of the applied ac voltage. It uses two diodes of which one conducts during one half-cycle while the other diode conducts during the other half-cycle of the applied ac voltage. There are two types of full-wave rectifiers viz. (i) center tapped transformer full-wave rectifier and (ii) bridge rectifier.

Figure 2.4 shows the basic circuit and waveforms of center tapped transformer full-wave rectifier. During positive half of the input signal, anode of diode D_1 becomes positive and at the same time the anode of diode D_2 becomes negative. Hence, D_1 conducts and D_2 does not conduct. The load current flows through D_1 and the voltage drop across R_L will be equal to the input voltage.

During the negative half-cycle of the input, the anode of D_1 becomes negative and the anode of D_2 becomes positive. Hence, D_1 does not conduct and D_2 conducts. The load current flows through D_2 and the voltage drop across R_L will be equal to the input voltage.

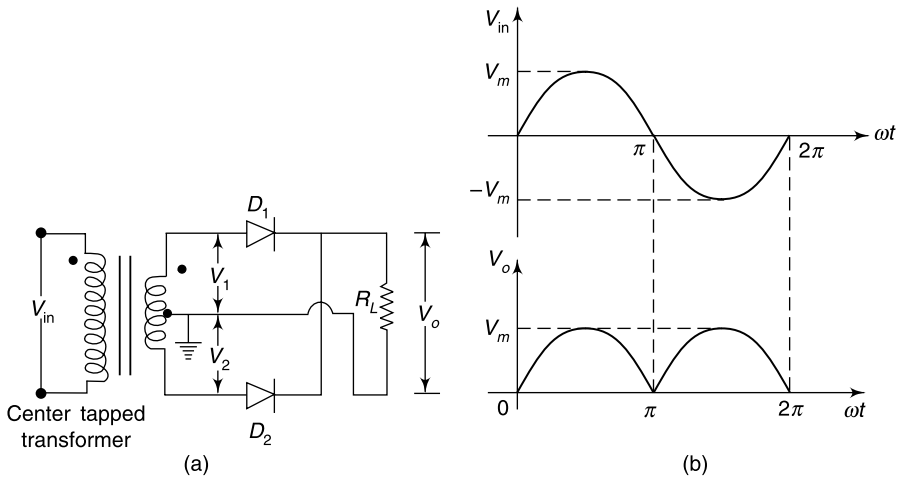


Fig. 2.4 Full-wave Rectifier

Ripple Factor (Γ)

$$\Gamma = \sqrt{\left(\frac{V_{rms}}{V_{dc}}\right)^2} - 1$$

The average voltage or dc voltage available across the load resistance is

$$\begin{aligned}
 V_{dc} &= \frac{1}{\pi} \int_0^{\pi} V_m \sin \omega t \, d(\omega t) \\
 &= \frac{V_m}{\pi} [-\cos \omega t]_0^{\pi} = \frac{2V_m}{\pi} \\
 I_{dc} &= \frac{V_{dc}}{R_L} = \frac{2V_m}{\pi R_L} = \frac{2I_m}{\pi} \text{ and } I_{rms} = \frac{I_m}{\sqrt{2}}
 \end{aligned}$$

If the diode forward resistance (r_f) and the transformer secondary winding resistance (r_s) are included in the analysis, then

$$\begin{aligned}
 V_{dc} &= \frac{2V_m}{\pi} - I_{dc} (r_s + r_f) \\
 I_{dc} &= \frac{V_{dc}}{(r_s + r_f) + R_L} = \frac{2V_m}{\pi (r_s + r_f + R_L)}
 \end{aligned}$$

RMS value of the voltage at the load resistance is

$$V_{rms} = \sqrt{\left[\frac{1}{\pi} \int_0^{\pi} V_m^2 \sin^2 \omega t \, d(\omega t) \right]} = \frac{V_m}{\sqrt{2}}$$

Therefore,

$$\Gamma = \sqrt{\left(\frac{V_m / \sqrt{2}}{2V_m / \pi} \right)^2} - 1 = \sqrt{\frac{\pi^2}{8}} - 1 = 0.482$$

Efficiency (η) The ratio of dc output power to ac input power is known as rectifier efficiency (η).

$$\begin{aligned}
 \eta &= \frac{\text{dc output power}}{\text{ac input power}} = \frac{P_{dc}}{P_{ac}} \\
 &= \frac{(V_{dc})^2 / R_L}{(V_{rms})^2 / R_L} = \frac{\left[\frac{2V_m}{\pi} \right]^2}{\left[\frac{V_m}{\sqrt{2}} \right]^2} = \frac{8}{\pi^2} = 0.812 = 81.2\%
 \end{aligned}$$

The maximum efficiency of a full-wave rectifier is 81.2%

Transformer Utilisation Factor (TUF) The average TUF in a full-wave rectifying circuit is determined by considering the primary and secondary winding separately and it gives a value of 0.693.

Form Factor

$$\begin{aligned}
 \text{Form factor} &= \frac{\text{rms value of the output voltage}}{\text{average value of the output voltage}} \\
 &= \frac{V_m / \sqrt{2}}{2V_m / \pi} = \frac{\pi}{2\sqrt{2}} = 1.11
 \end{aligned}$$

Peak Factor

$$\text{Peak factor} = \frac{\text{peak value of the output voltage}}{\text{rms value of the output voltage}} = \frac{V_m}{V_m / \sqrt{2}} = \sqrt{2}$$

Peak inverse voltage for full-wave rectifier is $2V_m$ because the entire secondary voltage appears across the non-conducting diode.

Example 2.10 A 230 V, 60 Hz voltage is applied to the primary of a 5:1 step-down, center-tap transformer used in a full wave rectifier having a load of 900 Ω . If the diode resistance and secondary coil resistance together has a resistance of 100 Ω , determine (a) dc voltage across the load, (b) dc current flowing through the load, (c) dc power delivered to the load, (d) PIV across each diode, (e) ripple voltage and its frequency and (f) rectification efficiency. [JNTU May 2003, 2004, April/May 2007 and June 2009]

Solution The voltage across the two ends of secondary = $\frac{230}{5} = 46$ V

Voltage from center tapping to one end, $V_{rms} = \frac{46}{2} = 23$ V

$$\begin{aligned} \text{(a) dc voltage across the load, } V_{dc} &= \frac{2V_m}{\pi} = \frac{2 \times 23 \times \sqrt{2}}{\pi} \\ &= 20.7 \text{ V} \end{aligned}$$

$$\begin{aligned} \text{(b) dc current flowing through the load, } I_{dc} &= \frac{V_{dc}}{(r_s + r_f + R_L)} = \frac{20.7}{1000} \\ &= 20.7 \text{ mA} \end{aligned}$$

(c) dc power delivered to the load,

$$P_{dc} = (I_{dc})^2 \times R_L = (20.7 \times 10^{-3})^2 \times 900 = 0.386 \text{ W}$$

$$\text{(d) PIV across each diode} = 2V_m = 2 \times 23 \times \sqrt{2} = 65 \text{ V}$$

$$\begin{aligned} \text{(e) Ripple voltage, } V_{r,rms} &= \sqrt{(V_{rms})^2 - (V_{dc})^2} \\ &= \sqrt{(23)^2 - (20.7)^2} = 10.05 \text{ V} \end{aligned}$$

$$\text{Frequency of ripple voltage} = 2 \times 60 = 120 \text{ Hz}$$

$$\begin{aligned} \text{(f) Rectification efficiency, } \eta &= \frac{P_{dc}}{P_{ac}} = \frac{(V_{dc})^2 / R_L}{(V_{rms})^2 / R_L} = \frac{(V_{dc})^2}{(V_{rms})^2} \\ &= \frac{(20.7)^2}{(23)^2} = \frac{428.49}{529} = 0.81 \end{aligned}$$

Therefore, percentage of efficiency = 81%

Example 2.11 Determine (a) dc output voltage (b) PIV and (c) rectification efficiency for the circuit shown in Fig. 2.5.

[JNTU May 2008, Aug 2008 and June 2009]

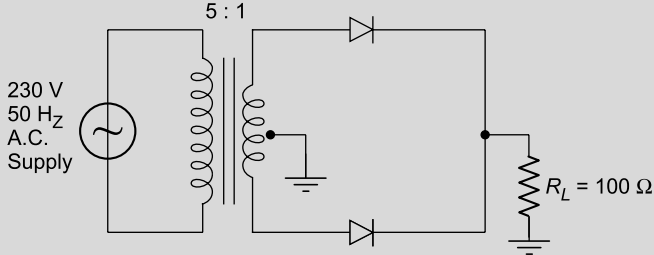


Fig. 2.5

Solution V_{rms} (primary) = 230 V, $N_1 : N_2 = 5 : 1$, $R_L = 100 \Omega$

V_{rms} (primary) = R.M.S. voltage from either end of secondary to center tap

$$\frac{V_{rms}(\text{primary})}{2V_{rms}(\text{secondary})} = \frac{N_1}{N_2}$$

$$V_{rms}(\text{secondary}) = \frac{230}{5 \times 2} = 23 \text{ V}$$

$$V_m = \sqrt{2} V_{rms}(\text{secondary}) = \sqrt{2} \times 23 = 32.53 \text{ V}$$

$$I_m = \frac{V_m}{R_L + r_s + r_f} = \frac{32.53}{100 + 0 + 0} = 0.3253 \text{ A}$$

$$I_{dc} = \frac{2I_m}{\pi} = \frac{2 \times 0.3253}{\pi} = 0.2071 \text{ A}$$

(a) $V_{dc} = I_{dc} R_L = 0.2071 \times 100 = 20.71 \text{ V}$

(b) $PIV = 2V_m = 2 \times 32.53 = 65.06 \text{ V}$

(c) $P_{dc} = I_{dc}^2 R_L = (0.2071)^2 \times 100 = 4.289 \text{ W}$

Therefore, percentage of efficiency, $\eta = \frac{P_{dc}}{P_{ac}} \times 100 = \frac{4.289}{5.29} \times 100 = 81.07\%$

Example 2.12 Draw the circuit diagram of a full-wave rectifier using center tapped transformer to obtain an output d.c. voltage of $V_{dc} = 18 \text{ V}$ at 200 mA and V_{dc} no load equals to 20 V . Assume suitable value of r_f and transformer resistance and also mention transformer rating and sketch the input and output waveforms.

[JNTU Dec 2004]

Solution The circuit diagram of a full-wave rectifier using centre tapped transformer is shown in Fig. 2.6(a).

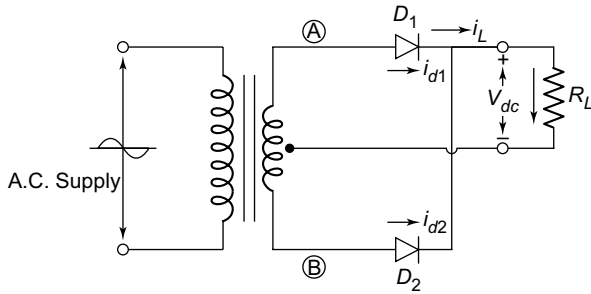


Fig. 2.6(a)

Given $V_{dc} = 18 \text{ V}$, $I_{dc} = 200 \text{ mA}$ and $V_{dc(NL)} = 20 \text{ V}$

$$V_{dc(FL)} = I_{dc} R_L$$

$$18 = 200 \times 10^{-3} R_L$$

Therefore, $R_L = 90 \Omega$

Assume $r_f = 2 \Omega$ and $r_s = 8 \Omega$

$$\text{Now } V_m = \frac{\pi}{2} \times V_{dc(NL)} = \frac{\pi}{2} \times 20 = 31.42 \text{ V}$$

$$\text{Therefore, } V_{rms} = \frac{V_m}{\sqrt{2}} = \frac{31.42}{\sqrt{2}} = 22.21 \text{ V}$$

Hence, the transformer secondary voltage rating = $22.21 - 0 = 22.21 \text{ V}$

$$I_m = \frac{V_m}{r_s + r_f + R_L} = \frac{31.42}{8 + 2 + 90} = 0.3142 \text{ V}$$

$$I_{rms} = \frac{I_m}{\sqrt{2}} = \frac{0.3142}{\sqrt{2}} = 0.2221 \text{ A}$$

VA rating of transformer = $I_{rms} V_{rms} = 22.21 \times 0.2221 = 4.93 \text{ VA}$

The waveforms are as shown in Fig. 2.6(b).

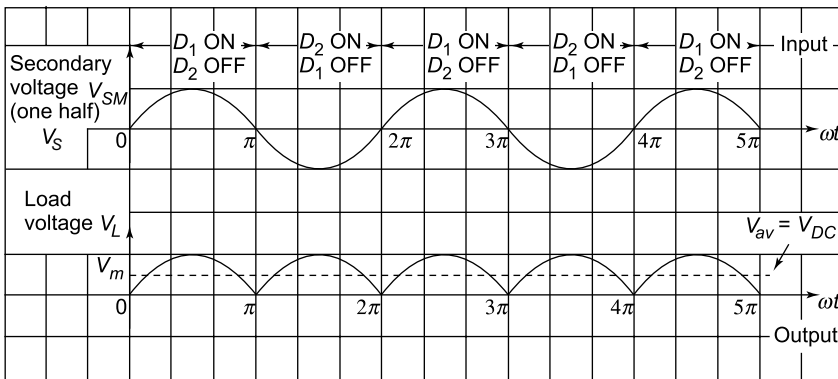


Fig. 2.6(b)

Example 2.13 In a full wave rectifier, the transformer rms secondary voltage from center tap to each end of the secondary is 50 V. The load resistance is 900 Ω . If the diode resistance and transformer secondary winding resistance together has a resistance of 100 Ω , determine the average load current and rms value of load current? [JNTU Jan 2010]

Solution Voltage from center tapping to one end, $V_{rms} = 50$ V

$$\text{Maximum load current, } I_m = \frac{V_m}{r_s + r_f + R_L} = \frac{V_{rms} \times \sqrt{2}}{r_s + r_f + R_L} = \frac{70.7}{1000} = 70.7 \text{ mA}$$

$$\text{Average load current, } I_{dc} = \frac{2I_m}{\pi} = \frac{2 \times 70.7 \times 10^{-3}}{\pi} = 45 \text{ mA}$$

$$\text{RMS value of load current, } I_{rms} = \frac{I_m}{\sqrt{2}} = \frac{70.7 \times 10^{-3}}{\sqrt{2}} = 50 \text{ mA}$$

Example 2.14 A full-wave rectifier has a center-tap transformer of 100–0–100 V and each one of the diodes is rated at $I_{max} = 400$ mA and $I_{av} = 150$ mA. Neglecting the voltage drop across the diodes, determine (a) the value of load resistor that gives the largest dc power output, (b) dc load voltage and current, and (c) PIV of each diode.

Solution (a) We know that the maximum value of current flowing through the diode for normal operation should not exceed 80% of its rated current.

$$\text{Therefore, } I_{max} = 0.8 \times 400 = 320 \text{ mA}$$

The maximum value of the secondary voltage,

$$V_m = \sqrt{2} \times 100 = 141.4 \text{ V}$$

Therefore, the value of load resistor that gives the largest dc power output

$$R_L = \frac{V_m}{I_{max}} = \frac{141.4}{320 \times 10^{-3}} = 442 \text{ } \Omega$$

$$(b) \text{ DC (load) voltage, } V_{dc} = \frac{2V_m}{\pi} = \frac{2 \times 141.4}{\pi} = 90 \text{ V}$$

$$\text{DC load current, } I_{dc} = \frac{V_{dc}}{R_L} = \frac{90}{442} = 0.204 \text{ A}$$

$$(c) \text{ PIV of each diode } = 2 V_m = 2 \times 141.4 = 282.8 \text{ V}$$

Example 2.15 In a full wave rectifier, the required dc voltage is 9 V and the diode drop is 0.8 V. Calculate ac rms input voltage required in case of bridge rectifier circuit and center tapped full wave rectifier circuit.

[JNTU May 2004, Aug 2008]

Solution The dc voltage across the load of the full wave rectifier circuit,

$$V_{dc} = \frac{2V_m}{\pi} - 0.8 = \frac{2\sqrt{2} \times V_{rms}}{\pi} - 0.8$$

where V_{rms} is the rms input voltage from center tapping to one end. That is,

$$9.8 = \frac{2\sqrt{2} V_{rms}}{\pi}$$

Therefore,
$$V_{rms} = \frac{9.8\pi}{2\sqrt{2}} = 10.885 \text{ V}.$$

Hence, the voltage across the two ends of the secondary $= 2 \times 10.885$
 $= 21.77 \text{ V}$

In the bridge rectifier,
$$V_{dc} = 9 = \frac{2\sqrt{2} V_{rms}}{\pi} - 2 \times 0.8$$

Therefore, the voltage across two ends of secondary, $V_{rms} = \frac{10.6\pi}{2\sqrt{2}} = 11.77 \text{ V}.$

Example 2.16 A full-wave rectifier delivers 50 W to a load of 200 Ω . If the ripple factor is 1%, calculate the ac ripple voltage across the load.

Solution DC power delivered to the load,

$$P_{dc} = \frac{V_{dc}^2}{R_L}$$

Therefore,
$$V_{dc} = \sqrt{P_{dc} \times R_L} = \sqrt{50 \times 200} = 100 \text{ V}$$

The ripple factor,
$$\Gamma = \frac{V_{ac}}{V_{dc}}$$

i.e.
$$0.01 = \frac{V_{ac}}{100}$$

Therefore, the ac ripple voltage across the load, $V_{ac} = 1 \text{ V}$

Example 2.17 A full-wave rectifier circuit uses two silicon diodes with a forward resistance of 20 Ω each. A dc voltmeter connected across the load of 1 k Ω reads 55.4 Volts. Calculate

- I_{rms}
- average voltage across each diode
- ripple factor
- transformer secondary voltage rating

[JNTU April/May 2007]

Solution Given $V_{dc} = 55.4 \text{ V}$ and $R_L = 1 \text{ k}\Omega$

(a)
$$I_{dc} = \frac{V_{dc}}{(r_f + R_L)} = \frac{55.4}{20 + 1000} = 54.31 \times 10^{-3} \text{ A}$$

We know that $I_{dc} = \frac{2I_m}{\pi}$ and $I_{rms} = \frac{I_m}{\sqrt{2}}$

$$I_m = I_{dc} \times \frac{\pi}{2} = 54.31 \times 10^{-3} \times \frac{\pi}{2} = 85.31 \text{ mA}$$

$$I_{rms} = \frac{I_m}{\sqrt{2}} = \frac{85.31 \times 10^{-3}}{\sqrt{2}} = 60.32 \text{ mA}$$

(b) The average voltage across each silicon diode will be 0.72 V.

(c) To find ripple factor Γ

$$\begin{aligned} \Gamma &= \sqrt{\left(\frac{I_{rms}}{I_{dc}}\right)^2 - 1} \\ &= \sqrt{\left(\frac{60.32 \times 10^{-3}}{54.31 \times 10^{-3}}\right)^2 - 1} = 0.4833 \end{aligned}$$

(d) To find transformer secondary voltage rating

We know that, $V_{dc} = \frac{2V_m}{\pi} - I_{dc}(r_s + r_f)$

where r_f is the diode forward resistance and r_s is the transformer secondary winding resistance.

$$55.4 = \frac{2V_m}{\pi} - 54.31 \times 10^{-3} \times 20 = \frac{2V_m}{\pi} - 1.086$$

$$56.49 = \frac{2V_m}{\pi}$$

Therefore, $V_m = 56.49 \times \frac{\pi}{2} = 88.73 \text{ V}$

$$V_{rms} = \frac{V_m}{\sqrt{2}} = \frac{88.73}{\sqrt{2}} = 62.74 \text{ V}$$

Hence, transformer secondary voltage rating is $65 \text{ V} - 0 = 65 \text{ V}$

2.4.3 Bridge Rectifier

The need for a center tapped transformer in a full-wave rectifier is eliminated in the bridge rectifier. As shown in Fig. 2.7 the bridge rectifier has four diodes connected to form a bridge. The ac input voltage is applied to diagonally opposite ends of the bridge. The load resistance is connected between the other two ends of the bridge.

For the positive half-cycle of the input ac voltage, diodes D_1 and D_3 conduct, whereas diodes D_2 and D_4 do not conduct. The conducting diodes will be in series through the load resistance R_L . So the load current flows through R_L .

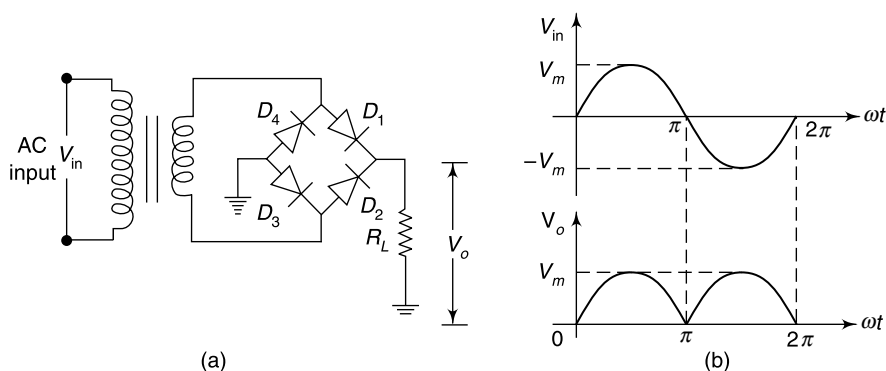


Fig. 2.7 Bridge Rectifier

During the negative half-cycle of the input ac voltage, diodes D_2 and D_4 conduct, whereas diodes D_1 and D_3 do not conduct. The conducting diode D_2 and D_4 will be in series through the load R_L and the current flows through R_L in the same direction as in the previous half-cycle. Thus a bidirectional wave is converted into a unidirectional one.

The average values of output voltage and load current for bridge rectifier are the same as for a center-tapped full wave rectifier. Hence,

$$V_{dc} = \frac{2V_m}{\pi} \quad \text{and} \quad I_{dc} = \frac{V_{dc}}{R_L} = \frac{2V_m}{\pi R_L} = \frac{2I_m}{\pi}$$

If the values of the transformer secondary winding resistance (r_s) and diode forward resistance (r_f) are considered in the analysis, then

$$V_{dc} = \frac{2V_m}{\pi} - I_{dc} (r_s + r_f)$$

$$I_{dc} = \frac{2V_m}{\pi} = \frac{2V_m}{\pi (r_s + r_f + R_L)}$$

The maximum efficiency of a bridge rectifier is 81.2% and the ripple factor is 0.48. The PIV is V_m .

Advantages of the Bridge Rectifier In the bridge rectifier the ripple factor and efficiency of the rectification are the same as for the full-wave rectifier. The PIV across either of the non-conducting diodes is equal to the peak value of the transformer secondary voltage, V_m . The bulky center tapped transformer is not required. Transformer utilisation factor is considerably high. Since the current flowing in the transformer secondary is purely alternating, the TUF increases to 0.812, which is the main reason for the popularity of a bridge rectifier. The bridge rectifiers are used in applications allowing floating output terminals, i.e. no output terminal is grounded.

The bridge rectifier has only one disadvantage that it requires four diodes as compared to two diodes for center-tapped full wave rectifier. But the diodes are readily available at cheaper rate in the market. Apart from this, the PIV rating required for the diodes in a bridge rectifier is only half of that for a center tapped full-wave rectifier. This is a great advantage, which offsets the disadvantage of using extra two diodes in a bridge rectifier.

Example 2.18 A 230 V, 50 Hz voltage is applied to the primary of a 4:1 step-down transformer used in a bridge rectifier having a load resistance of 600 Ω . Assuming the diodes to be ideal, determine (a) dc output voltage, (b) dc power delivered to the load, (c) PIV, and (d) output frequency.

Solution (a) The rms value of the transformer secondary voltage,

$$V_{rms} = \frac{230}{4} = 57.5 \text{ V}$$

The maximum value of the secondary voltage

$$V_m = \sqrt{2} \times 57.5 = 81.3 \text{ V}$$

Therefore, dc output voltage, $V_{dc} = \frac{2V_m}{\pi} = \frac{2 \times 81.3}{\pi} = 52 \text{ V}$

(b) DC power delivered to the load,

$$P_{dc} = \frac{V_{dc}^2}{R_L} = \frac{52^2}{1000} = 2.704 \text{ W}$$

(c) PIV across each diode = $V_m = 81.3 \text{ V}$

(d) Output frequency = $2 \times 50 = 100 \text{ Hz}$

Example 2.19 In a Bridge rectifier, the transformer is connected to 200 V, 60 Hz mains and the turns ratio of the step down transformer is 11:1. Assuming the diode is ideal, find

(a) I_{dc}

(b) Voltage across the load

(c) PIV

[JNTU May 2003]

Solution Given in a bridge rectifier, input voltage = 200 V, 60 Hz and turns ratio = 11:1

(a) To find the voltage across load, V_{dc} :

$$V_{dc} = \frac{2V_m}{\pi}$$

where $V_m = V_{rms} \sqrt{2}$

$$V_{rms} \text{ (secondary)} = \frac{V_{rms} \text{ (primary)}}{\text{Turns Ratio}} = \frac{200}{11} = 18.18 \text{ V}$$

Therefore $V_m = 18.18 \times \sqrt{2} = 25.7$

Hence, $V_{dc} = \frac{2 \times 25.7}{\pi} = 16.36 \text{ V}$

(b) To find I_{dc} :

Assuming that, $R_L = 600 \Omega$, then

$$I_{dc} = \frac{V_{dc}}{R_L} = \frac{16.36}{600} = 27.26 \text{ mA}$$

(c) To find PIV:

$$PIV = V_m = 25.7 \text{ V}$$



Example 2.20 A bridge rectifier uses four identical diodes having forward resistance of $5\ \Omega$ and the secondary voltage is 30 V (rms) . Determine the dc output voltage for $I_{dc} = 200\text{ mA}$ and value of the output ripple voltage.
[JNTU May 2004, June 2009]

Solution Given Transformer secondary resistance = $5\ \Omega$
Secondary voltage $V_{rms} = 30\text{ V}$, $I_{dc} = 200\text{ mA}$
Since only two diodes of the bridge rectifier circuit will conduct during positive or negative half cycle of the input signal, the diode forward resistance $r_f = 2 \times 5\ \Omega = 10\ \Omega$

We know that,

$$V_{dc} = \frac{2V_m}{\pi} - I_{dc} (r_f + r_s) \text{ where } V_m = \sqrt{2} V_{rms} = \sqrt{2} \times 30\text{ V}$$

Therefore,
$$V_{dc} = \frac{2 \times \sqrt{2} \times 30}{\pi} - 200 \times 10^{-3} (10 + 5) = 24\text{ V}$$

$$\text{Ripple factor} = \frac{\text{rms value of ripple at the output}}{\text{Average value of output voltage}}$$

Therefore,
$$0.48 = \frac{\text{rms value of ripple at the output}}{24}$$

Hence, rms value of ripple at the output = $0.48 \times 24 = 11.52\text{ V}$

Comparison of Rectifiers

The comparison of rectifiers is given in Table 2.1

Table 2.1 A Comparison of Rectifiers

Particulars	Type of rectifier		
	Half-wave	Full-wave	Bridge
No. of diodes	1	2	4
Maximum efficiency	40.6%	81.2%	81.2%
V_{dc} (no load)	V_m/π	$2V_m/\pi$	$2V_m/\pi$
Average current/diode	I_{dc}	$I_{dc}/2$	$I_{dc}/2$
Ripple factor	1.21	0.48	0.48
Peak inverse voltage	V_m	$2V_m$	V_m
Output frequency	f	2f	2f
Transformer utilisation factor	0.287	0.693	0.812
Form factor	1.57	1.11	1.11
Peak factor	2	$\sqrt{2}$	$\sqrt{2}$

2.5 HARMONIC COMPONENTS IN A RECTIFIER CIRCUIT

The term *harmonic* is defined as “a sinusoidal component of a periodic waveform or quantity possessing a frequency, which is an integral multiple of the fundamental frequency.” By definition, a perfect sine wave has no harmonics, except fundamental component at one frequency. Harmonics are present in waveforms that are not perfect sine waves due to distortion from nonlinear loads. The French mathematician named Fourier discovered that a distorted waveform can be represented as a series of sine waves, with each being an integer multiple of the fundamental frequency and each with a specific magnitude.

That is, the harmonic frequencies are integer multiples [2, 3, 4...] of the fundamental frequency. For example, the second harmonic on a 50 Hz system is 2×50 or 100 Hz. The sixth harmonic in a 50 Hz system, or the fifth harmonic in a 60 Hz system is 300 Hz. There are a number of different types of equipment that may experience faulty operations or failures due to high harmonic voltage and/or current levels. The amount of the harmonic voltage and current levels that a system can tolerate is dependent on the equipment and the source.

The sum of the fundamental and all the harmonics is called the ***Fourier series***. This series can be viewed as a spectrum analysis where the fundamental frequency and the harmonic component are identified.

The result of such an analysis for the current waveform of a half-wave rectifier circuit using a single diode is given by

$$i = I_m \left[\frac{1}{\pi} + \frac{1}{2} \sin \omega t - \frac{2}{\pi} \sum_{k=2,4,6..} \frac{\cos k \omega t}{(k+1)(k-1)} \right]$$

The angular frequency of the power supply is the lowest angular frequency present in the above expression. All the other terms are the even harmonics of the power frequency.

The full-wave rectifier consists of two half-wave rectifier circuits, arranged in such a way that one circuit conducts during one half cycle and the second circuit operates during the second half cycle. Therefore, the currents are functionally related by the expression $i_1(\alpha) = i_2(\alpha + \pi)$. Thus, the total current of the full-wave rectifier is $i = i_1 + i_2$ as expressed by

$$i = I_m \left[\frac{2}{\pi} - \frac{4}{\pi} \sum_{\substack{k=even \\ k \neq 0}} \frac{\cos k \omega t}{(k+1)(k-1)} \right]$$

From the above equation, it can be seen that the fundamental angular frequency is eliminated and the lowest frequency is the second harmonic term 2ω . This is the advantage that the full-wave rectifier presents in filtering of the output. Additionally, the current pulses in the two halves of the transformer winding are in such directions that the magnetic cycles formed through the iron core is essentially that of the alternating current. This avoids any dc saturation of the transformer core that could give rise to additional harmonics at the output.

2.6 FILTERS

The output of a rectifier contains dc component as well as ac component. Filters are used to minimise the undesirable ac, i.e. ripple leaving only the dc component to appear at the output.

The ripple in the rectified wave being very high, the factor being 48% in the full-wave rectifier; majority of the applications which cannot tolerate this, will need an output which has been further processed.

Figure 2.8 shows the concept of a filter, where the full wave rectified output voltage is applied at its input. The output of a filter is not exactly a constant dc level. But it also contains a small amount of ac component. Some important filters are:

- | | |
|---------------------------------|-------------------------------|
| (a) Inductor filter | (b) Capacitor filter |
| (c) LC or L-section filter, and | (d) CLC or π -type filter |

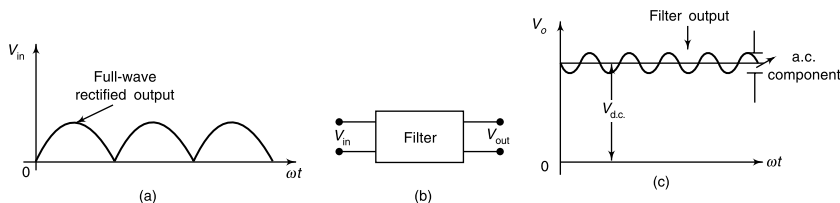


Fig. 2.8 Concept of a Filter

2.6.1 Inductor Filter

Figure 2.9 shows the inductor filter. When the output of the rectifier passes through an inductor, it blocks the ac component and allows only the dc component to reach the load.

The ripple factor of the Inductor filter is given by

$$\Gamma = \frac{R_L}{3\sqrt{2}\omega L}$$

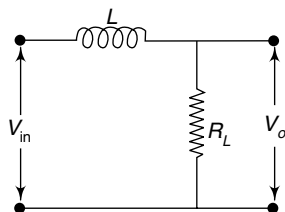


Fig. 2.9 Inductor Filter

It shows that the ripple factor will decrease when L is increased and R_L is decreased. Clearly, the inductor filter is more effective only when the load current is high (small R_L). The larger value of the inductor can reduce the ripple and at the same time the output dc voltage will be lowered as the inductor has a higher dc resistance.

The operation of the inductor filter depends on its well known fundamental property to oppose any change of current passing through it.

To analyse this filter for a full-wave, the Fourier series can be written as

$$V_o = \frac{2V_m}{\pi} - \frac{4V_m}{\pi} \left[\frac{1}{3} \cos 2\omega t + \frac{1}{15} \cos 4\omega t + \frac{1}{35} \cos 6\omega t + \dots \right]$$

The dc component is $\frac{2V_m}{\pi}$.

Assuming the third and higher terms contribute little output, the output voltage is

$$V_o = \frac{2V_m}{\pi} - \frac{4V_m}{3\pi} \cos 2\omega t$$

The diode, choke and transformer resistances can be neglected since they are very small as compared with R_L . Therefore, the dc component of current

$I_m = \frac{V_m}{R_L}$. The impedance of series combination of L and R_L at 2ω is

$$Z = \sqrt{R_L^2 + (2\omega L)^2} = \sqrt{R_L^2 + 4\omega^2 L^2}$$

Therefore, for the ac component,

$$I_m = \frac{V_m}{\sqrt{R_L^2 + 4\omega^2 L^2}}$$

Therefore, the resulting current i is given by,

$$i = \frac{2V_m}{\pi R_L} - \frac{4V_m}{3\pi} \cdot \frac{\cos(2\omega t - \phi)}{\sqrt{R_L^2 + 4\omega^2 L^2}}$$

where

$$\phi = \tan^{-1} \left(\frac{2\omega L}{R_L} \right)$$

The ripple factor, which can be defined as the ratio of the rms value of the ripple to the dc value of the wave, is

$$\Gamma = \frac{\frac{4V_m}{3\pi\sqrt{2}\sqrt{R_L^2 + 4\omega^2 L^2}}}{\frac{2V_m}{\pi R_L}} = \frac{2}{3\sqrt{2}} \times \frac{1}{\sqrt{1 + \frac{4\omega^2 L^2}{R_L^2}}}$$

If $\frac{4\omega^2 L^2}{R_L^2} \gg 1$, then a simplified expression for Γ is

$$\Gamma = \frac{R_L}{3\sqrt{2}\omega L}$$

In case, the load resistance is infinity, i.e. the output is an open circuit, then the ripple factor is

$$\Gamma = \frac{2}{3\sqrt{2}} = 0.471$$

This is slightly less than the value of 0.482. The difference being attributable to the omission of higher harmonics as mentioned. It is clear that the inductor filter should only be used where R_L is consistently small.

Example 2.21 Calculate the value of inductance to use in the inductor filter connected to a full-wave rectifier operating at 60 Hz to provide a dc output with 4% ripple for a $100\ \Omega$ load.

Solution We know that the ripple factor for inductor filter is $\Gamma = \frac{R_L}{3\sqrt{2}\omega L}$

$$\text{Therefore, } 0.04 = \frac{100}{3\sqrt{2}(2\pi \times 60 \times L)} = \frac{0.0625}{L}$$

$$L = \frac{0.0625}{0.04} = 1.5625\text{ H}$$

2.6.2 Capacitor Filter

An inexpensive filter for light loads is found in the capacitor filter which is connected directly across the load, as shown in Fig. 2.10(a). The property of a capacitor is that it allows ac component and blocks the dc component. The operation of a capacitor filter is to short the ripple to ground and leave the dc to appear at the output when it is connected across a pulsating dc voltage.

During the positive half-cycle, the capacitor charges up to the peak value of the transformer secondary voltage, V_m , and will try to maintain this value as the full-wave input drops to zero. The capacitor will discharge through R_L slowly until the transformer secondary voltage again increases to a value greater than the capacitor voltage. The diode conducts for a period which depends on the capacitor voltage (equal to the load voltage). The diode will conduct when the transformer secondary voltage becomes more than the 'cut-in' voltage of the diode. The diode stops

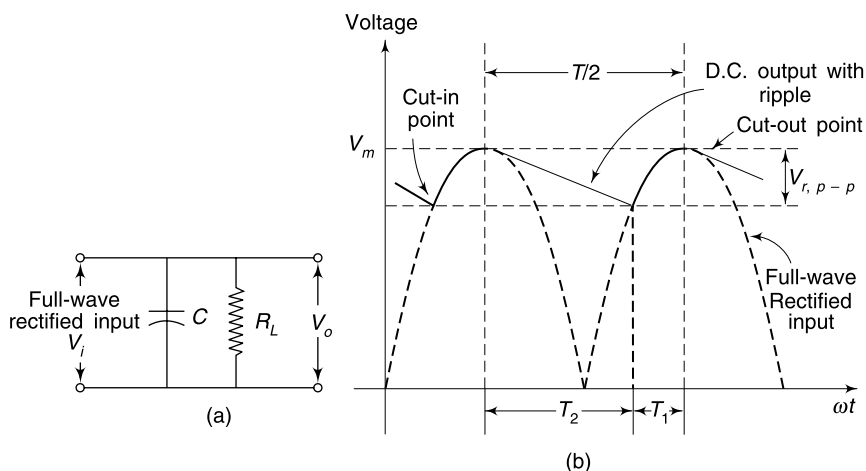


Fig. 2.10 (a) Capacitor Filter, (b) Ripple Voltage Triangular Waveform

conducting when the transformer voltage becomes less than the diode voltage. This is called cut-out voltage.

Referring to Fig. 2.10(b) with slight approximation, the ripple voltage waveform can be assumed as triangular. From the cut-in point to the cut-out point, whatever charge the capacitor acquires is equal to the charge the capacitor has lost during the period of non-conduction, i.e. from cut-out point to the next cut-in point.

$$\text{The charge it has acquired} = V_{r,p-p} \times C$$

$$\text{The charge it has lost} = I_{dc} \times T_2$$

$$\text{Therefore, } V_{r,p-p} \times C = I_{dc} \times T_2$$

If the value of the capacitor is fairly large, or the value of the load resistance is very large, then it can be assumed that the time T_2 is equal to half the periodic time of the waveform.

$$\text{i.e. } T_2 = \frac{T}{2} = \frac{1}{2f}, \text{ then } V_{r,p-p} = \frac{I_{dc}}{2fC}$$

With the assumptions made above, the ripple waveform will be triangular in nature and the rms value of the ripple is given by

$$V_{r,rms} = \frac{V_{r,p-p}}{2\sqrt{3}}$$

Therefore from the above equation, we have

$$\begin{aligned} V_{r,rms} &= \frac{I_{dc}}{4\sqrt{3}fC} \\ &= \frac{V_{dc}}{4\sqrt{3}fCR_L}, \text{ since } I_{dc} = \frac{V_{dc}}{R_L} \end{aligned}$$

$$\text{Therefore, ripple factor } \Gamma = \frac{V_{r,rms}}{V_{dc}} = \frac{1}{4\sqrt{3}fCR_L}$$

The ripple may be decreased by increasing C or R_L (or both) with a resulting increase in dc output voltage.

$$\text{If } f = 50 \text{ Hz, } C \text{ in } \mu\text{F and } R_L \text{ in } \Omega, \Gamma = \frac{2890}{CR_L}.$$



Example 2.22 Calculate the value of capacitance to use in a capacitor filter connected to a full-wave rectifier operating at a standard aircraft power frequency of 400 Hz, if the ripple factor is 10% for a load of 500 Ω .

Solution We know that the ripple factor for capacitor filter is

$$\Gamma = \frac{1}{4\sqrt{3}fCR_L}$$

$$\text{Therefore, } 0.01 = \frac{1}{4\sqrt{3} \times 400 \times C \times 500} = \frac{0.722 \times 10^{-6}}{C}$$

$$C = \frac{0.722 \times 10^{-6}}{0.01} = 72.2 \mu\text{F}$$

Example 2.23 A 15-0-15 Volts (rms) ideal transformer is used with a full wave rectifier circuit with diodes having forward drop of 1 Volt. The load is a resistance of 100 Ohm and a capacitor of 10,000 μF is used as a filter across the load resistance calculate the dc load current and voltage. [JNTU Dec 2004, June 2009]

Solution Given transformer secondary voltage = 15-0-15 V (rms); Diode forward drop = 1 V; $R_L = 100 \Omega$; $C = 10,000 \mu\text{F}$

$$\text{We know that, } V_{dc} = V_m - \frac{V_{r,pp}}{2} = V_m - \frac{I_{dc}}{4fC}$$

$$\text{Therefore, } V_{dc} = V_m - \frac{V_{dc}}{R_L 4fC}, \left[\text{since } I_{dc} = \frac{V_{dc}}{R_L} \right]$$

Simplifying, we get

$$V_{dc} = \left[\frac{4fR_L C}{4fR_L C + 1} \right] V_m$$

$$\text{We know that } V_m = V_{rms} \times \sqrt{2} = 15 \times \sqrt{2}$$

$$\text{Therefore, } V_{dc} = \left[\frac{4 \times 50 \times 100 \times 10000 \times 10^{-6}}{4 \times 50 \times 100 \times 10000 \times 10^{-6} + 1} \right] \times 15 \times \sqrt{2} = 21.105 \text{ V} = 21.105 \text{ V}$$

Considering the given voltage drop of 1 Volt due to diodes,

$$V_{dc} = 21.105 - 1 = 20.105 \text{ V}$$

$$I_{dc} = \frac{V_{dc}}{R_L} = \frac{20.105}{100} = 0.20105 \text{ A}$$

Example 2.24 A full-wave rectified voltage of 18 V peak is applied across a 500 μF filter capacitor. Calculate the ripple and dc voltages if the load takes a current of 100 mA. [JNTU April/May 2007]

Solution Given $V_m = 18 \text{ V}$, $C = 500 \mu\text{F}$ and $I_{dc} = 100 \text{ mA}$

$$V_{dc} = V_m - \frac{I_{dc}}{4fC} = 18 - \frac{100 \times 10^{-3}}{4 \times 50 \times 500 \times 10^{-6}} = 17 \text{ V}$$

$$V_{\gamma, rms} = \frac{I_{dc}}{4\sqrt{3}fC} = \frac{100 \times 10^{-3}}{4\sqrt{3} \times 50 \times 500 \times 10^{-6}} = 0.577 \text{ V}$$

$$\text{Therefore, ripple } \Gamma = \frac{V_{rms}}{V_{dc}} = \frac{0.577}{17} \times 100 = 3.39\%$$

Example 2.25 A bridge rectifier with capacitor filter is fed from 220V to 40V step down transformer. If average d.c. current in load is 1A and capacitor filter of 800 μF , calculate the load regulation and ripple factor. Assume power line frequency of 50Hz. Neglect diode forward resistance and d.c. resistance of secondary of transformer. [JNTU August 2008]

Solution

$$V_{rms} = 40 \text{ V}, I_{dc} = 1 \text{ A}, C = 800 \mu\text{F} \text{ and } f = 50 \text{ Hz}$$

$$V_m = \sqrt{2}V_{rms} = \sqrt{2} \times 40 = 56.5685 \text{ V}$$

$$V_{dc(FL)} = V_m - \frac{I_{dc}}{4fC} = 56.5685 - \frac{1}{4 \times 50 \times 800 \times 10^{-6}} = 50.3185 \text{ V}$$

$$\text{On no load, } I_{dc} = 0.$$

$$\text{Hence, } V_{dc(NL)} = V_m = 56.5685 \text{ V}$$

$$\begin{aligned} \text{Therefore, percentage of regulation} &= \frac{V_{dc(NL)} - V_{dc(FL)}}{I_{dc(FL)}} \times 100 \\ &= \frac{56.5685 - 50.3185}{50.1385} \times 100 = 12.42\% \end{aligned}$$

$$R_L = \frac{V_{dc}}{I_{dc}} = \frac{50.1385}{1} = 50.1385 \Omega$$

$$\gamma = \frac{1}{4\sqrt{3}fCR_L} = \frac{1}{4\sqrt{3} \times 50 \times 800 \times 10^{-6} \times 50.1385} = 0.0717 \text{ i.e. } 7.17\%$$

2.6.3 L-Section or LC Filter

We know that the ripple factor is directly proportional to the load resistance R_L in the inductor filter and inversely proportional to R_L in the capacitor filter. Therefore, if these two filters are combined as LC filter or L-section filter as shown in Fig. 2.11, the ripple factor will be independent of R_L .

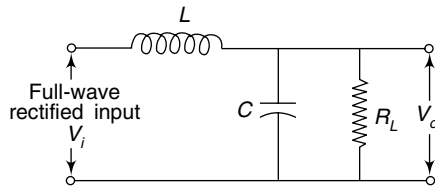


Fig. 2.11 LC Filter

If the value of the inductance is increased, it will increase the time of conduction. At some critical value of inductance, one diode, either D_1 or D_2 in full-wave rectifier, will always be conducting.

From Fourier series, the output voltage can be expressed as

$$V_o = \frac{2V_m}{\pi} - \frac{4V_m}{3\pi} \cos 2\omega t$$

$$\text{The dc output voltage, } V_{dc} = \frac{2V_m}{\pi}$$

Therefore,
$$I_{rms} = \frac{4V_m}{3\pi\sqrt{2}} \cdot \frac{1}{X_L} = \frac{\sqrt{2}}{3} \cdot \frac{V_{dc}}{X_L}$$

This current flowing through X_c creates the ripple voltage in the output.

Therefore,
$$V_{r, rms} = I_{rms} \cdot X_C = \frac{\sqrt{2}}{3} \cdot V_{dc} \cdot \frac{X_C}{X_L}$$

The ripple factor,
$$\Gamma = \frac{V_{r, rms}}{V_{dc}} = \frac{\sqrt{2}}{3} \cdot \frac{X_C}{X_L}$$

$$= \frac{\sqrt{2}}{3} \cdot \frac{1}{4\omega^2 CL}, \text{ since } X_C = \frac{1}{2\omega C} \text{ and } X_L = 2\omega L$$

If $f = 50$ Hz, C is in μF and L is in Henry, ripple factor $\Gamma = \frac{1.194}{LC}$.

LC Filter with Bleeder Resistor It was assumed in the analysis given above that for a critical value of inductor, either of the diodes is always conducting, i.e. current does not fall to zero. The incoming current consists of two components:

(a) $I_{dc} = \frac{V_{dc}}{R_L}$ and (b) a sinusoidal varying components with peak value of $\frac{4V_m}{3\pi X_L}$. The negative peak of the ac current must always be less than dc, i.e., $\sqrt{2} I_{rms} \leq \frac{V_{dc}}{R_L}$.

We know that for LC filter, $I_{rms} = \frac{\sqrt{2}}{3} \cdot \frac{V_{dc}}{X_L}$

Hence
$$\frac{2V_{dc}}{3X_L} \leq \frac{V_{dc}}{R_L}, \text{ i.e. } X_L \geq \frac{2}{3} R_L$$

i.e., $L_C = \frac{R_L}{3\omega}$, where L_C is the critical inductance.

It should be noted that the condition $X_L \geq 2/3 R_L$ cannot be satisfied for all load requirements. At no load, i.e. when the load resistance is infinity, the value of the inductance will also tend to be infinity. To overcome this problem, a bleeder resistor R_B , is connected in parallel with the load resistance as shown in Fig. 2.12.

Therefore, a minimum current will always be present for optimum operation of the inductor. It improves voltage regulation of the supply by acting as the pre-load on the supply. Also, it provides safety by acting as a discharging path for capacitor.

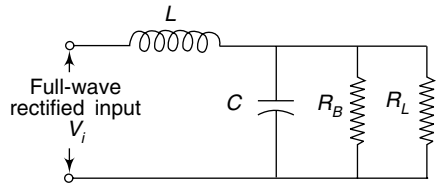


Fig. 2.12 Bleeder Resistor Connected at the LC Filter Output

Example 2.26 Design a filter for full-wave circuit with LC filter to provide an output voltage of 10 V with a load current of 200 mA and the ripple is limited to 2%.

Solution The effective load resistance $R_L = \frac{10}{200 \times 10^{-3}} = 50 \Omega$

We know that the ripple factor, $\Gamma = \frac{1.194}{LC}$

$$\text{i.e.} \quad 0.02 = \frac{1.194}{LC}$$

$$\text{i.e.} \quad LC = \frac{1.194}{0.02} = 59.7$$

$$\text{Critical value of} \quad L = \frac{R_L}{3\omega} = \frac{50}{3 \times 2\pi f} = 53 \text{ mH}$$

Taking $L = 60 \text{ mH}$ (about 20% higher), C will be about $1000 \mu\text{F}$.

Example 2.27 A full wave rectifier (FWR) supplies a load requiring 300 V at 200 mA . Calculate the transformer secondary voltage for
 (a) a capacitor input filter using a capacitor of 10 mF and
 (b) a choke input filter using a choke of 10 H and a capacitance of $10 \mu\text{F}$.
 Neglect the resistance of choke. [JNTU May/June and Sept 2006, Feb 2010]

Solution Given $V_{\text{dc}} = 300 \text{ V}$; $I_{\text{dc}} = 200 \text{ mA}$

(a) For the capacitor filter with $C = 10 \mu\text{F}$,

$$V_{\text{dc}} = V_m - \frac{I_{\text{dc}}}{4fc}$$

$$300 = V_m - \frac{200 \times 10^{-3}}{4(50)(10 \times 10^{-6})} = V_m - 100$$

$$\text{Therefore,} \quad V_m = 400 \text{ V}_{(p-p)}$$

$$V_{\text{rms}} = \frac{V}{\sqrt{2}} = 282.84 \text{ V}$$

(b) For the choke, i.e., LC filter with $L = 10 \text{ H}$; $C = 10 \mu\text{F}$

$$V_{\text{dc}} = \frac{2V_m}{\pi}$$

$$300 = \frac{2V_m}{\pi}$$

$$\text{Therefore,} \quad V_m = 471.23 \text{ V}$$

$$V_{\text{rms}} = \frac{V_m}{\sqrt{2}} = 333.21 \text{ V}$$

Example 2.28 Determine the ripple factor of a L -type choke input filter comprising a $10H$ choke and $8\mu F$ capacitor used with a FWR. Compare with a simple $8\mu F$ capacitor input filter at a load current of $50mA$ and also at $150mA$. Assume the dc voltage of $50V$. [JNTU May 2008 and Jan. 2010]

Solution $V_{dc} = 50V$, $L = 10H$, $C = 8\mu F$

Assume $f = 50Hz$ i.e. $\omega = 2\pi f = 100\pi$ rad/sec.

For LC filter, the ripple factor is

$$\Gamma = \frac{1}{6\sqrt{2}\omega^2 LC} = \frac{1}{6\sqrt{2} \times (100\pi)^2 \times 10 \times 8 \times 10^{-6}} = 0.01492 \text{ i.e. } 1.492\%$$

For simple capacitor filter, $C = 8\mu F$,

(i) $I_L = 50mA$

$$R_L = \frac{V_{dc}}{I_L} = \frac{50}{50 \times 10^{-3}} = 1000\Omega$$

$$\Gamma = \frac{1}{4\sqrt{3}fCR_L} = \frac{1}{4\sqrt{3} \times 50 \times 8 \times 10^{-6} \times 1000} = 0.3608 \text{ i.e. } 36.08\%$$

(ii) $I_L = 150mA$

$$R_L = \frac{V_{dc}}{I_L} = \frac{50}{150 \times 10^{-3}} = 333.33\Omega$$

$$\Gamma = \frac{1}{4\sqrt{3}fCR_L} = \frac{1}{4\sqrt{3} \times 50 \times 8 \times 10^{-6} \times 333.33} = 1.082 \text{ i.e. } 108.2\%$$

Thus it is inferred that LC choke input filter is more effective than capacitor input filter and the ripple factor of LC choke input filter does not depend on the load resistance.

Example 2.29 In a full-wave rectifier using an LC filter $L = 10H$, $C = 100\mu F$ and $R_L = 500\Omega$. Calculate I_{dc} , V_{dc} and ripple factor for an input of $V_i = 30\sin(100\pi t)V$. [JNTU Dec 2004 and Aug 2008]

Solution Comparing the input with $V_i = V_m \sin \omega t$,

$$V_m (\text{secondary}) = V_m = 30V$$

$$V_{dc} = \frac{2V_m}{\pi} = \frac{2 \times 30}{\pi} = 19.0985V$$

$$I_{dc} = \frac{V_{dc}}{R_L} = \frac{19.0985}{500} = 0.03819A = 38.19mA$$

$$\begin{aligned} \text{Ripple factor} &= \frac{1}{6\sqrt{2}\omega^2 LC} \\ &= \frac{1}{6\sqrt{2} \times (100\pi)^2 \times 10 \times 100 \times 10^{-6}} = 1.194 \times 10^{-3} \end{aligned}$$

Multiple LC Filter Better filtering can be achieved using two or more L-section filters as shown in Fig. 2.13.

The ripple factor,
$$\Gamma = \frac{\sqrt{2}}{3} \cdot \frac{X_{C2}}{X_{L2}} \cdot \frac{X_{C1}}{X_{L1}}$$

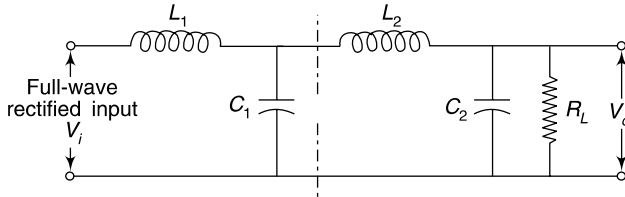


Fig. 2.13 Multiple LC Filter

2.6.4 CLC or π -section Filter

Figure 2.14 shows the CLC or π -type filter which basically consists of a capacitor filter followed by an LC section.

This filter offers a fairly smooth output, and is characterised by a highly peaked diode currents and poor regulation. Proceeding the analysis in the same ways as that for the single L-section filter, we obtain

$$\Gamma = \sqrt{2} \cdot \frac{X_{C1}}{R_L} \cdot \frac{X_{C2}}{X_L}$$

The term ' R_L ' in the above equation should be noted.

If $f = 50$ Hz, C in μF , L in H and R_L in Ω , then
$$\Gamma = \frac{5700}{LC_1 C_2 R_L}$$

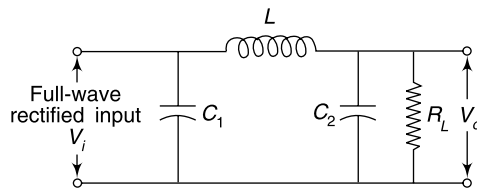


Fig. 2.14 CLC or π -type Filter

Example 2.30 Design a CLC or π -section filter for $V_{dc} = 10$ V, $I_L = 200$ mA and $\Gamma = 2\%$.

Solution

$$R_L = \frac{10}{200 \times 10^{-3}} = 50 \Omega$$

$$0.02 = \frac{5700}{LC_1 C_2 \times 50} = \frac{114}{LC_1 C_2}$$

If we assume $L = 10 \text{ H}$ and $C_1 = C_2 = C$, we have

$$0.02 = \frac{5700}{LC_1 C_2 \times 50} = \frac{114}{LC^2} = \frac{11.4}{C^2}$$

$$C^2 = 570; \text{ therefore, } C = \sqrt{570} \approx 24 \mu\text{F}$$

Example 2.31 A full-wave single phase rectifier employs a π -section filter consisting of two $4 \mu\text{F}$ capacitances and a 20 H choke. The transformer voltage to the center tap is 300 V rms . The load current is 500 mA . Calculate the dc output voltage and the ripple voltage. The resistance of the choke is 200Ω .

Solution $C_1 = C_2 = 4 \mu\text{F}$ $L = 20 \text{ H}$

$$I_C = 500 \text{ mA}, R_x = 200 \Omega$$

Maximum value of secondary voltage,

$$V_{sm} = \sqrt{2} \times 300 = 424.2 \text{ V}$$

$$R_L = \frac{V_{dc}}{I_{dc}} = \frac{270.19}{500 \times 10^{-3}} = 540 \Omega$$

$$V_{dc} = V_{s(max)} - \frac{V_r}{2} - I_{dc} R_x$$

$$\text{Ripple voltage, } V_r = \frac{I_{dc}}{2fC}$$

$$= \frac{500 \times 10^{-3}}{2 \times 50 \times 4 \times 10^{-6}} = 1.25 \text{ mV}$$

$$\text{DC output voltage, } V_{dc} = 424.2 - \frac{1.25 \times 10^{-3}}{2} - (500 \times 10^{-3} \times 200) = 324.19 \text{ V}$$

2.6.5 R-C Filters

Consider the CLC filter with the inductor L replaced by a resistor R . This type of filter called R - C filter is shown in Fig. 2.15. The expression for the ripple factor can be obtained by replacing X_L by R giving.

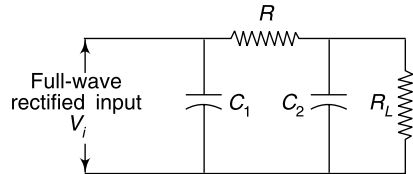


Fig. 2.15 R-C Filter

$$\Gamma = \sqrt{2} \frac{X_{C1}}{R_L} \cdot \frac{X_{C2}}{R}$$

Therefore, if resistor R is chosen equal to the reactance of the inductor which it replaces, the ripple remains unchanged.

The resistance R will increase the voltage drop and hence, the regulation will be poor. This type of filters are often used for economic reasons, as well as the space and weight requirement of the iron-cored choke for the LC filter. Such R - C filters are often used only for low current power supplies.

2.6.6 Comparison of Filters

Table 2.2 shows the comparison of various types of filters, when used with full-wave circuits. In all these filters, the resistances of diodes, transformer and filter elements are considered negligible and a 60 Hz power line is assumed.

Table 2.2 Comparison of Various Types of Filters

	Type of Filter				
	None	L	C	L -Section	π -Section
V_{dc} at no load	$0.636 V_m$	$0.636 V_m$	V_m	V_m	V_m
V_{dc} at load I_{dc}	$0.636 V_m$	$0.636 V_m$	$V_m - \frac{4170 I_{dc}}{C}$	$0.636 V_m$	$V_m - \frac{4170 I_{dc}}{C}$
Ripple factor Γ	0.48	$\frac{R_L}{16000 L}$	$\frac{2410}{CR_L}$	$\frac{0.83}{LC}$	$\frac{3330}{LC_1 C_2 R_L}$
Peak inverse voltage (PIV)	$2V_m$	$2V_m$	$2V_m$	$2V_m$	$2V_m$

2.7 VOLTAGE REGULATION USING ZENER DIODE

In an unregulated power supply, the output voltage changes whenever the input voltage or load changes. An ideal regulated power supply is an electronic circuit designed to provide a predetermined dc voltage V_o which is independent of the load current and variations in the input voltage. A voltage regulator is an electronic circuit that provides a stable dc voltage independent of the load current, temperature and ac line voltage variations.

Factors Determining the Stability The output dc voltage V_o depends on the input unregulated dc voltage V_{in} , load current I_L and temperature T . Hence, the change in output voltage of power supply can be expressed as follows:

$$\Delta V_o = \frac{\partial V_o}{\partial V_{in}} \Delta V_{in} + \frac{\partial V_o}{\partial I_L} \Delta I_L + \frac{\partial V_o}{\partial T} \Delta T$$

or

$$\Delta V_o = S_V \Delta V_{in} + R_o \Delta I_L + S_T \Delta T$$

where the three coefficients are defined as

$$\text{Input regulation factor, } S_V = \frac{\Delta V_o}{\Delta V_{in}} \bigg|_{\Delta I_L = 0; \Delta T = 0}$$

$$\text{Output resistance, } R_o = \frac{\Delta V_o}{\Delta I_L} \bigg|_{\Delta V_{in} = 0; \Delta T = 0}$$

$$\text{Temperature coefficient, } S_T = \frac{\Delta V_o}{\Delta T} \bigg|_{\Delta V_{in} = 0; \Delta I_L = 0}$$

Smaller the value of the three coefficients, better the regulation of the power supply.

Line Regulation Line regulation is defined as the change in output voltage for a change in line supply voltage keeping the load current and temperature constant. Line regulation is given by

$$\text{Line regulation} = \frac{\text{change in output voltage}}{\text{change in input voltage}} = \frac{\Delta V_o}{\Delta V_{in}}$$

Load Regulation Load regulation is expressed as

$$\text{Load regulation} = \frac{(V_{\text{no load}} - V_{\text{full load}})}{V_{\text{no load}}}$$

or

$$\text{Load regulation} = \frac{V_{\text{no load}} - V_{\text{full load}}}{V_{\text{full load}}}$$

where $V_{\text{no load}}$ is the output voltage at zero load current and $V_{\text{full load}}$ is the output voltage at rated load current. This is usually denoted in percentage. The plot of the output voltage V_o versus the load current I_L for a full wave rectifier is given in Fig. 2.16. The drop in the characteristics is a measure of the internal resistance of the power supply.

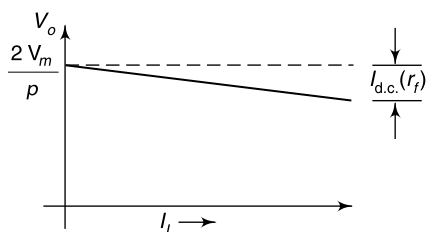


Fig. 2.16 Load Regulation Characteristics

2.7.1 Zener Diode Shunt Regulator

A zener diode, under reverse bias breakdown condition, can be used to regulate the voltage across a load, irrespective of the supply voltage or load current

variations. A simple zener voltage regulator circuit is shown in Fig. 2.17. The zener diode is selected with V_z equal to the voltage desired across the load. The zener diode has a characteristic that under reverse bias condition, the voltage across it practically remains constant, even if the current through it changes by a large extent. Under normal conditions, the input current $I_i = I_L + I_z$ flows through resistor R . The input voltage V_i can be written as $V_i = I_i R + V_z = (I_L + I_z) R + V_z$.

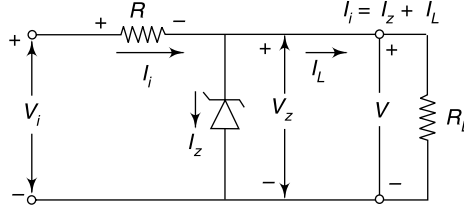


Fig. 2.17 Zener Voltage Regulator

When the input voltage V_i increases (say due to supply voltage variations), as the voltage across zener diode remains constant, the drop across resistor R will increase with a corresponding increase in $I_L + I_z$. As V_z is a constant, the voltage across the load will also remain constant and hence, I_L will be a constant. Therefore, an increase in $I_L + I_z$ will result in an increase in I_z which will not alter the voltage across the load.

It must be ensured that the reverse voltage applied to the zener diode never exceeds PIV of the diode and at the same time, the applied input voltage must be greater than the breakdown voltage of the zener diode for its operation. The zener diodes can be used as 'stand-alone' regulator circuits and also as reference voltage sources.

Example 2.32 Design a Zener shunt-voltage regulator with the following specifications:

$V_o = 10\text{ V}$; $V_{in} = 20\text{--}30\text{ V}$; $I_L = (30\text{--}50)\text{ mA}$; $I_z = (20\text{--}40)\text{ mA}$

Solution Refer to Fig. 2.15.

Selection of Zener diode

$$V_z = V_o = 10\text{ V}$$

$$I_{z\text{ max}} = 40\text{ mA}$$

$$P_z = V_z \times I_{z\text{ max}} = 10 \times 40 \times 10^{-3} = 0.4\text{ W}$$

Hence a 0.5W 10V Zener can be selected

Value of load resistance, R_L

$$R_{L\text{ min}} = \frac{V_o}{I_{L\text{ max}}} = \frac{10}{50 \times 10^{-3}} = 200\ \Omega$$

$$R_{L\text{ max}} = \frac{V_o}{I_{L\text{ min}}} = \frac{10}{30 \times 10^{-3}} = 333\ \Omega$$

Value of load resistance, R

$$R_{\max} = \frac{V_{in(\max)} - V_o}{I_{L\min} + I_{z(\max)}}$$

$$= \frac{30 - 10}{(30 + 40) \times 10^{-3}} = 286 \Omega$$

$$R_{\min} = \frac{V_{in(\min)} - V_o}{I_{L\max} + I_{z(\min)}}$$

$$= \frac{20 - 10}{(50 + 20) \times 10^{-3}} = 143 \Omega$$

Therefore,

$$R = \frac{R_{\max} + R_{\min}}{2} = 215 \Omega$$

Example 2.33 In a Zener regulator, the dc input is $10V \pm 20\%$. The output requirements are 5 V, 20 mA. Assume $I_{z(\min)}$ and $I_{z(\max)}$ as 5 mA and 80 mA. Design the Zener regulator.

Solution The minimum Zener current is $I_{z(\min)} = 5$ mA when the input voltage is minimum. Here the input voltage varies between $10V \pm 20\%$ i.e. 8 V and 12 V.

Therefore, the input voltage $V_{i(\min)} = 8$ V.

Given load current $I_L = 20$ mA and the voltage across the load, $V_o = 5$ V.

Therefore,

$$R_L = \frac{V_o}{I_L} = \frac{5V}{20 \times 10^{-3}} = 250 \Omega$$

Hence, the series resistance $R = \frac{V_{i(\min)} - V_o}{(I_{z(\min)} + I_L)}$

$$= \frac{(8 - 5)}{(5 + 20) \times 10^{-3}} = 120 \Omega$$

The various values are given in the Zener regulator shown in Fig. 2.18.

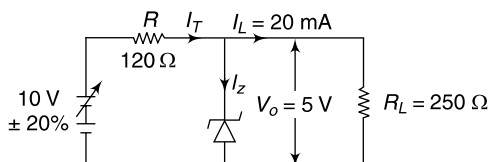


Fig. 2.18

Example 2.34 If dc unregulated input is 20 V, $V_o = 10$ V, load current is 0 – 20 mA, design the regulator. Assume for the Zener, $I_{z(\min)} = 10$ mA, $I_{z(\max)} = 100$ mA.

Solution Given input voltage, $V_i = 20$ V

Output voltage, $V_o = 10$ V

Load current varies from 0 to 20 mA

$$I_{z(\min)} = 10 \text{ mA}, I_{z(\max)} = 100 \text{ mA}.$$

Here, $V_z = V_o = 10$ V (constant)

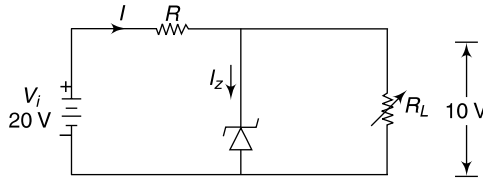


Fig. 2.19

Applying KVL to a closed loop circuit,

$$20 = IR + 10$$

or $IR = 10$

Therefore, $R = \frac{10}{I} \Omega$, where I is the loop current in amperes

(a) Let $I_z = I_{z(\min)}$ and $I_L = 0$

$$\text{The total current } I = I_L + I_z = 10 \text{ mA}$$

$$\text{Therefore, } R = \frac{10 \text{ V}}{10 \text{ mA}} = 1000 \Omega$$

(b) For $I_z = I_{z(\max)} = 100$ mA and $I_L = 20$ mA

$$I = I_L + I_z = 20 + 100 = 120 \text{ mA}$$

$$\text{Therefore, } R = \frac{10 \text{ V}}{120 \text{ mA}} = 83.33 \Omega$$

(c) The range of R varies from 83.33Ω to 1000Ω .

Example 2.35 Design a Zener voltage regulator to meet the following specifications: Output voltage = 5 V, Load current = 10 mA, Zener wattage = 400 mW and Input voltage = $10 \text{ V} \pm 2 \text{ V}$.

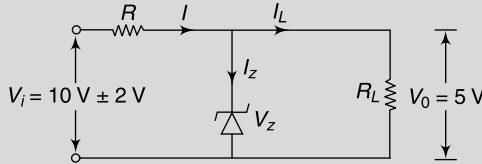


Fig. 2.20

Solution Given $V_o = 5 \text{ V}$, $I_L = 10 \text{ mA}$

Here, load resistance is
$$R_L = \frac{V_o}{I_L} = \frac{5}{10 \times 10^{-3}} = 500 \Omega$$

Maximum Zener Current
$$I_{zm} = \frac{400 \text{ mW}}{5 \text{ V}} = 80 \text{ mA}$$

The minimum input voltage required will be when $I_Z = 0$. Under this condition,

$$I = I_L = 10 \text{ mA}$$

Minimum input voltage
$$V_{i(\min)} = V_o + IR$$

Hence,
$$V_{i(\max)} = 10 - 2 = 8 \text{ V}$$

or
$$8 = 5 + (10 \times 10^{-3})R$$

Therefore,
$$R = \frac{3}{10 \times 10^{-3}} = 300 \Omega$$

Now, maximum input voltage,
$$V_{i(\max)} = 5 + [(80 + 10)10^{-3}]R$$

or
$$12 = 5 + (90 \times 10^{-3})R$$

$$R = \frac{7}{90 \times 10^{-3}} = 77.77 \Omega$$

The value of R is chosen between 77.77Ω and 300Ω .

Example 2.36 A 24 V, 600 mW Zener diode is used for providing a 24 V stabilized supply to a variable load. If the input voltage is 32 V, calculate (a) the value of series resistance required and (b) diode current when the load is 1200Ω .

Solution Given $V_o = 24 \text{ V}$, $V_i = 32 \text{ V}$, $P_Z = 600 \text{ mW}$

Max. Zener current, $I_{zm} = \frac{600 \times 10^{-3}}{24} = 25 \text{ mA}$

$$R_{\max} = \frac{V_i - V_o}{I_{L(\min)} + I_{z(\max)}} = \frac{32 - 24}{(20 + 25) \times 10^{-3}} = \frac{8}{45 \times 10^{-3}} = \frac{1600}{9} = 177.78 \Omega$$

The load current, $I_L = \frac{V_o}{R_L} = \frac{24}{1200} = 20 \text{ mA}$

Example 2.37 A Zener voltage regulator circuit is to maintain constant voltage at 60 V, over a current range from 5 to 50 mA. The input supply voltage is 200 V. Determine the value of resistance R to be connected in the circuit, for voltage regulation from load current $I_L = 0 \text{ mA}$ to $I_L \text{ max}$, the maximum possible value of I_L . What is the value of $I_L \text{ max}$? [JNTU Dec 2005, Aug 2008]

Solution Given $V_z = V_o = 60 \text{ V}$ and $V_{in} = 200 \text{ V}$

(a) To find the value of resistance (R):

$$\begin{aligned} R &= \frac{V_{in} - V_o}{I_{z(\min)} + I_{L(\max)}} \\ &= \frac{200 - 60}{50 \times 10^{-3}} = \frac{140}{50 \times 10^{-3}} = 2.8 \text{ k}\Omega \end{aligned}$$

(b) To find the value of $I_{L(\max)}$:

If $I_{z(\max)} = 50 \text{ mA}$, $I_{L(\min)} = 0 \text{ mA}$

If $I_{z(\min)} = 5 \text{ mA}$, $I_{L(\max)} = 45 \text{ mA}$

Therefore, $I_{L(\max)} = 45 \text{ mA}$

Example 2.38 For the Zener voltage regulation shown, determine the range of R_L and I_L that gives the stabilizer voltage of 10 V.

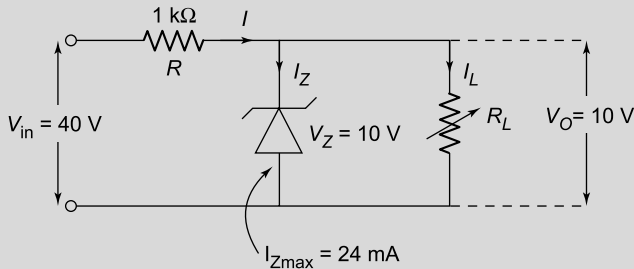


Fig. 2.21

[JNTU April 2003]

Solution From the circuit, $I = I_z + I_L$

But from R , $I = \frac{V_{in} - V_o}{R} = \frac{40 - 10}{1 \times 10^3} = 30 \text{ mA}$

When I_L is minimum, I_Z is maximum and vice versa.

$$I = I_{Z\max} + I_{L\min}$$

$$30\text{mA} = 24\text{mA} + I_{L\min}$$

$$\text{Therefore, } I_{L\min} = 6\text{ mA}$$

$$\text{But } I_{L\min} = \frac{V_o}{R_{L\max}}$$

$$R_{L\max} = \frac{V_o}{I_{L\min}} = \frac{10}{6 \times 10^{-3}} = 1.667\text{ k}\Omega$$

$$\text{and } I = I_{Z\min} + I_{L\max}$$

$$30\text{mA} = 5\text{mA} + I_{L\max}$$

$$I_{L\max} = 25\text{ mA}$$

$$\text{But } I_{L\max} = \frac{V_o}{R_{L\min}}$$

$$R_{L\min} = \frac{V_o}{I_{L\max}} = \frac{10}{25 \times 10^{-3}} = 400\text{ }\Omega$$

Hence, the range of I_L is 6mA to 25mA and that of R_L is 400 Ω to 1.667 k Ω .

Example 2.39 Determine the range of input voltage that maintains the output voltage of 10 V, for the regulator circuit shown.

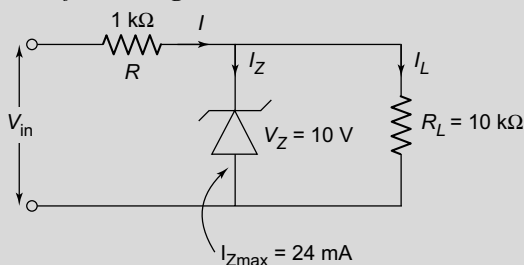


Fig. 2.22

[JNTU April 2003]

Solution As $V_o = 10\text{ V}$ constant and $R_L = 10\text{ k}\Omega$ constant, we have

$$I_L = \frac{V_o}{R_L} = \frac{10}{10 \times 10^3} = 1\text{ mA}$$

$$\text{When } V_{in} = V_{in(\max)}, I_Z = I_{Z\max}$$

$$\text{Now } I = I_Z + I_L$$

$$\text{Therefore, } I_{\max} = I_{Z\max} + I_L = 24\text{ mA} + 1\text{ mA} = 25\text{ mA.}$$

$$\frac{V_{in(\max)} - V_Z}{R} = 25\text{ mA}$$

Therefore, $V_{in(max)} - 10 = 1 \times 10^3 (25 \times 10^{-3})$

$$V_{in(max)} = 35 \text{ V}$$

When $V_{in} = V_{in(min)}$, $I_Z = I_{Zmin} = 5 \text{ mA}$

Therefore, $I_{min} = I_{Zmin} + I_L = 5 \text{ mA} + 1 \text{ mA} = 6 \text{ mA}$

$$\frac{V_{in(min)} - V_Z}{R} = 6 \times 10^{-3}$$

$$V_{in(min)} - 10 = 1 \times 10^3 (6 \times 10^{-3})$$

$$V_{in(min)} = 16 \text{ V}$$

Thus range of input voltage is 16V to 35V for which the output voltage will be of 10V.

2.7.2 Emitter-follower Type Regulator

In the Zener voltage regulator, the zener current varies over a wide range as the input voltage and load current vary. As a result, the output voltage which is equal to V_Z also changes by a small amount. This change in the output voltage can be minimised by reducing the change in the zener current with the help of a circuit called emitter-follower type regulator as shown in Fig. 2.23.

Here, the load resistance, R_L , is not connected across the zener directly as in the zener regulator, but is connected through an amplifier/buffer circuit.

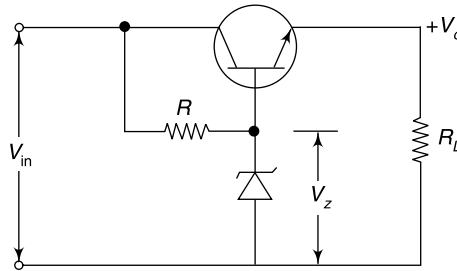


Fig. 2.23 Emitter Follower Type Regulator

Transistor is connected as an emitter-follower. As can be seen, the output voltage, $V_o = (V_Z - V_{BE})$.

However, the load current I_L is supplied by the transistor from the input voltage V_{in} , deriving its base current from the zener circuit. The base current I_B is equal to $\frac{I_L}{\beta}$, where β is the current-gain of the transistor. As far as the zener circuit is concerned, it is supplying only the base current. Any change in the load current is reduced by β times i.e. change in the zener current.

Overload Protection Figure 2.24 (a) shows overload protection circuit in which a small sensing resistance R_{SC} is added in series with the load resistance and two diodes are connected from the base of the transistor to the output.

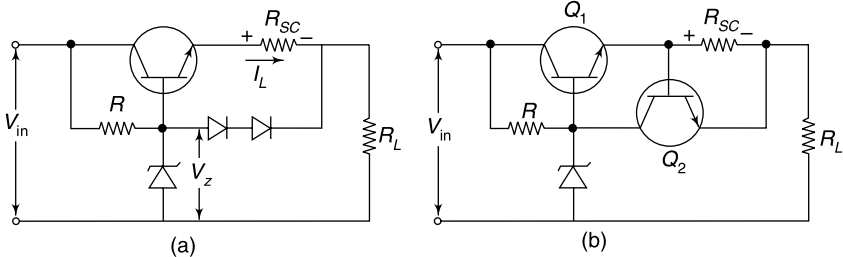


Fig. 2.24 (a) and (b) Overload Protection Circuit

The emitter voltage is equal to $(V_z - V_{BE})$. The voltage drop across the sensing resistance R_{SC} is equal to $(I_L \times R_{SC})$. As long as the voltage drop across R_{SC} is less than twice the cut-in voltage of the diode, the diodes are effectively as good as not connected in the circuit. If the voltage drop across R_{SC} increases suddenly due to over current in the load, then the diodes will be forward biased and will start conducting. This will divert a part of the base current, which will be directly led to the output, thus restricting the base current and hence, the transistor current. With a proper design, the transistor can be turned-off in the case of a short circuit.

The protective diodes can be replaced by another transistor Q_2 as shown in Fig. 2.24 (b). In this case, the voltage across R_{SC} is used in turning ON transistor Q_2 , giving the same effect as before.

2.7.3 Principle of Obtaining a Regulated Power Supply

If the control element of a regulator operates in its linear region, then the regulator is called a linear regulator. Linear regulators are generally of series mode type. The regulator circuit using Zener diode is vulnerable to the variations in supply voltage since the current through the Zener diode also changes correspondingly. Hence the linear regulator uses an op-amp as an error amplifier, and a pass transistor as a control element. The error output from the op-amp drives the control element, which allows current to the load accordingly and keeps the output voltage constant.

The basic circuit of a linear voltage regulator is shown in Fig. 2.25. The regulating circuit consists of a voltage reference (V_{ref}), a differential amplifier called *error amplifier* using op-amp and a series regulating element Q_1 connected as an emitter follower. The output voltage is sampled and fed back to the inverting input of the error amplifier through the potential divider $R_2 - R_3$. The error amplifier produces an output voltage that is proportional to the difference between the reference voltage and the sampled output voltage and it may be written as $V'_o = A[V_{ref} - \beta V_o]$, where A is the gain of the amplifier and β is the feedback factor which is equal to $R_3/(R_2 + R_3)$. Since the drop across the base-emitter junction of transistor Q_1 is small, the output V_o can be approximated to V'_o .

Thus
$$V'_o = V_o = A[V_{ref} - \beta V_o]$$

That is,
$$V_o = AV_{ref}/(1 + A\beta)$$

This equation implies that the output voltage is determined by the reference voltage and the feedback factor.

The output voltage thus obtained is kept at a constant level by the control of series element connected with the error amplifier. For instance, an increase in output voltage causes a corresponding decrease in the error amplifier output, which biases the series control transistor with reduced base current. This action causes an increase in collector-to-emitter voltage and thus the increase in the output is reduced.

On the other hand, when the output voltage reduces, the output of the differential amplifier increases. Then, the series transistor is biased heavily at its base and as a consequence, the collector-to-emitter voltage decreases. Thus the reduction in output is compensated and the output voltage is maintained constant.

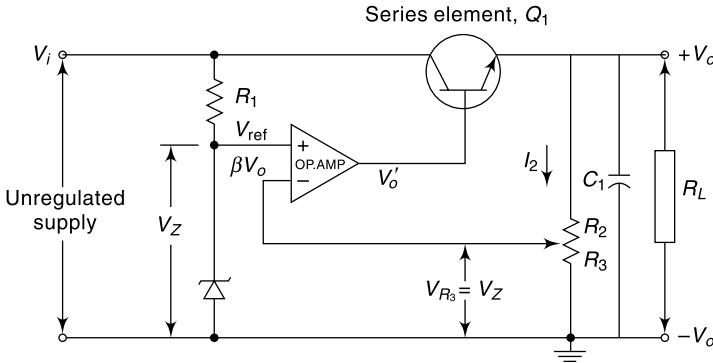


Fig. 2.25 Basic Circuit of a Linear Voltage Regulator

Example 2.40 Referring to Fig. 2.25, design a linear voltage regulator to produce an output of 15V with a maximum load current of 50 mA.

Solution Refer to Fig. 2.25. We know that

$$V_{i(\min)} = V_o + 3V = 15 + 3 = 18V$$

Assuming the ripple voltage $V_r = 2V$ (max), the input voltage is

$$V_i = V_{i(\min)} + \frac{V_r}{2} = 18 + 1 = 19V$$

Therefore, the input voltage, $V_i = 19V$ with a 2 V (max) ripple superimposed

$$\text{Then } V_z = \frac{V_i}{2} = \frac{19}{2} = 9.5V \quad (\text{use the Zener diode 1N758 for 10V})$$

$$\text{Therefore, } V_z = 10V$$

$$I_z \approx 20 \text{ mA}$$

$$R_1 = \frac{V_i - V_z}{I_z} = \frac{19 - 10}{20 \times 10^{-3}} = 450 \Omega$$

$$\text{Let } I_2 = I_{B(\max)} = 50 \mu A$$

$$R_2 = \frac{V_o - V_z}{I_2} = \frac{15 - 10}{50 \times 10^{-6}} = 100 \text{ k}\Omega$$

$$R_3 = \frac{V_z}{I_2} = \frac{105}{50 \times 10^{-6}} = 200 \text{ k}\Omega$$

Select $C_1 = 50 \mu\text{F}$.

Specification of transistor Q_1

$$V_{CE(\max)} = V_{i(\max)} = V_i + V_r/2 = 19 + 2/2 = 20 \text{ V}$$

$$I_E = I_L = 50 \text{ mA}$$

$$\begin{aligned} P &= V_{CE} \times I_L = (V_i - V_o) \times I_L \\ &= (16 - 15) \times 50 \times 10^{-3} = 200 \text{ mW} \end{aligned}$$

Use the transistor 2N718 for Q_1 .

2.7.4 Principle of Obtaining a Dual Tracking Voltage Regulator

The circuit diagram of a dual tracking voltage regulator using op-amps is shown in Fig. 2.26. The top half of the circuit is similar to the single polarity positive voltage regulator shown in Fig. 2.25. The bottom half of the circuit consisting of the components op-amp A_2 , PNP transistor Q_2 , resistors R_4 and R_5 , and capacitor C_2 constitutes a negative voltage regulator. The reference voltage for the negative voltage regulator is provided by the output of the positive voltage regulator circuit. The potential divider R_4 and R_5 is connected between the positive and negative output terminals. Any change in the negative voltage output is applied to the op-amp A_2 , which amplifies and inverts to correct the change accordingly.

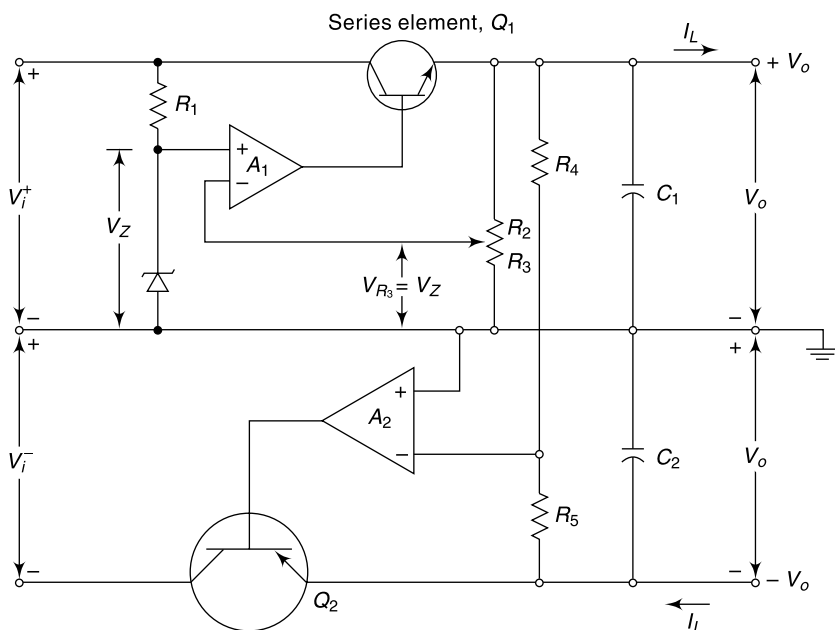


Fig. 2.26 Dual Tracking Voltage Regulator using Op-amps

When the resistors R_4 and R_5 are made equal, the output voltage between the positive and negative terminals is exactly twice the positive voltage. This gives a negative output that is equal to the positive output. This type of negative voltage regulator is called a *tracking regulator*, since the negative voltage output tracks the change in the positive output voltage.

The arrangement of this type of *plus-minus* power supply is possible only when there is no ground connection in the unregulated power supply.

2.7.5 Transistorised Shunt Regulator

In the transistorised shunt voltage regulator shown in Fig. 2.27, the output voltage is determined by the voltage drop across series resistor R_s . If I_L increases due to a load change, V_o will tend to decrease. However, the voltage across R_2 will also decrease, thereby reducing the forward bias on the transistor and driving it to cut-off. This results in less current flow through the transistor, thereby maintaining I_s almost constant, which keeps the voltage drop across R_s relatively unchanged. Thus, for a given input voltage, output voltage $V_o = V_i - I_s R_s$, remains substantially constant. The major drawback in this circuit is the large amount of power dissipated in R_s .

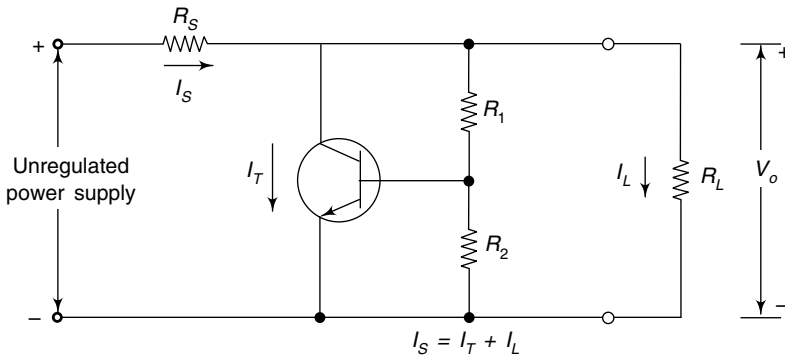


Fig. 2.27 Transistorised Shunt Regulator

2.7.6 Transistorised Series Regulator

If R_s is replaced by a transistor as shown in Fig. 2.28, a more efficient circuit results which is more sensitive to voltage changes and provides better regulation.

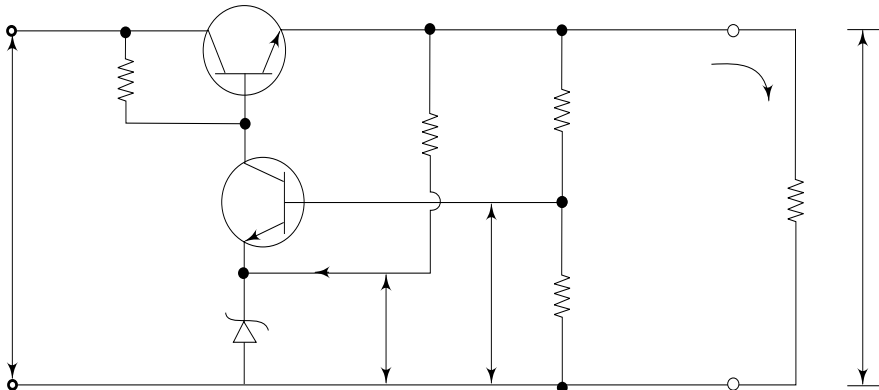


Fig. 2.28 Transistorised Series Regulator

Transistor Q_2 actually serves as a differential amplifier in which the fraction of the output voltage βV_o is compared with reference voltage V_z . The difference ($\beta V_o - V_z$) is amplified by Q_2 and appears at the base of Q_1 . This in turn determines the voltage drop that will occur across Q_1 . Because of the gain of Q_2 , it requires only a small change in V_o to have a large effect on Q_1 . Further, the output voltage may be varied over wide range using R_2 . The zener diode and transistor Q_2 can be chosen so that the temperature coefficients practically cancel.

If R_2 is adjusted for a lower output voltage, a greater voltage drop occurs across Q_1 . Maximum dissipation in Q_1 thus takes place at high load currents and low output voltage in variable regulated power supplies employing a series regulator.

Drawback of Transistorised Series Regulator The transistorised series voltage regulator has the limitation that the output voltage available is restricted by the V_{CEO} of the series transistor used. The power rating of the transistor used, as a series loser, depends on the voltage difference between the input and output voltages. This difficulty can be minimised to a great extent by using thyristors. Thyristors have the ability to control large power with minimal control power, and this control power does not have to remain continuous as in the case of the base current of a transistor. Transistors for a relatively high voltage, high power operations are rarely available.



Example 2.41 Design a series voltage regulator with the following specifications:

$$V_o = 20 \text{ V}; V_{in} = (22 - 30) \text{ V}; I_{L(\max)} = 50 \text{ mA}$$

Refer to Fig. 2.28.

Solution Selection of Zener diode

$$R_{L \min} = \frac{V_o}{I_{L(\max)}} = \frac{20}{50 \times 10^{-3}} = 400 \Omega$$

$$V_z \approx \frac{V_o}{2} = \frac{20}{2} = 10 \text{ V}$$

Hence, the zener diode 0.5Z10 is chosen.

$$\text{Since } I_{R1} > I_{B2}, I_{R1} > \frac{I_{C2}}{\beta}, I_{R1} > \frac{10 \times 10^{-3}}{150}$$

$$I_{R1} > 66.7 \mu\text{A}$$

$$\text{Let } I_{R1} \approx I_{R2} \approx I_{R3} = 10 \text{ mA (neglecting } I_{B2})$$

$$\text{Let } I_{C2} \approx I_{E2} = 10 \text{ mA}$$

So, the current flowing through the Zener,

$$I_z = I_{E2} + I_{R1} = 20 \text{ mA}$$

$$P_z = V_z I_z = 10 \times 20 \times 10^{-3} \\ = 0.2 \text{ W} < 0.5 \text{ W}$$

Hence selection of 0.5Z10 Zener diode is confirmed.

Selection of transistor Q_1

$$\begin{aligned} I_{E1} &= I_{R1} + I_{R2} + I_L \\ &= (10 + 10 + 50) \text{ mA} = 70 \text{ mA} \end{aligned}$$

$$V_{i(\max)} - V_0 = 30 - 20 = 10 \text{ V}$$

For transistor SL100, the ratings are

$$\begin{aligned} I_{C(\max)} &= 500 \text{ mA} \\ V_{CE(\max)} &= 50 \text{ V} \\ h_{FE} &= 50 - 280 \end{aligned}$$

Hence, SL100 can be chosen for Q_1 .

Selection of transistor Q_2

$$\text{From the figure, } V_{CE2(\max)} + V_z = (V_o + V_{BE1})$$

$$\text{Therefore, } V_{CE2(\max)} = (V_o + V_{BE1}) - V_z = 20.6 - 10 = 10.6 \text{ V}$$

For transistor BC107, the ratings are

$$\begin{aligned} V_{CEO(\max)} &= 45 \text{ V} \\ I_{C(\max)} &= 200 \text{ mA} \\ h_{FE} &= 125 - 300 \end{aligned}$$

Hence, transistor BC107 is selected for Q_2 .

Selection of resistors R_1 , R_2 and R_3

$$V_{R1} = V_o - V_z = 20 - 10 = 10 \text{ V}$$

$$R_1 = \frac{V_{R1}}{I_{R1}} = \frac{10}{10 \times 10^{-3}} = 1 \text{ k}\Omega$$

$$V_{R2} = V_o - V_{R3} = 20 - 10.6 = 9.4 \text{ V}$$

$$R_2 = \frac{V_{R2}}{I_{R2}} = \frac{9.4}{10 \times 10^{-3}} = 940 \Omega$$

$$\begin{aligned} V_{R3} &= V_z + V_{BE2(\text{sat})} \\ &= 10 + 0.6 = 10.6 \text{ V} \end{aligned}$$

$$R_3 = \frac{V_{R3}}{I_{R3}} = \frac{10.6}{10 \times 10^{-3}} = 1060 \Omega$$

Selection of resistor R_4

$$V_{B1} = V_{C2} = V_o + V_{BE1} = 20 + 0.6 = 20.6 \text{ V}$$

$$I_{B1} = \frac{I_{C1}}{\beta} = \frac{70 \times 10^{-3}}{50} = 1.4 \text{ mA}$$

$$I_{R4} = I_{B1} + I_{C2} = 11.4 \text{ mA}$$

$$R_{4(\max)} = \frac{V_{R4(\max)}}{I_{R4}} = \frac{V_{i(\max)} - V_{B1}}{I_{R4}}$$

$$\begin{aligned}
 &= \frac{30 - 20.6}{11.4 \times 10^{-3}} = 825 \, \Omega \\
 R_{4(\min)} &= \frac{V_{R4(\min)}}{I_{R4}} = \frac{V_{i(\min)} - V_{B1}}{I_{R4}} \\
 &= \frac{22 - 20.6}{11.4 \times 10^{-3}} = 123 \, \Omega \\
 R_4 &= \frac{R_{4(\max)} + R_{4(\min)}}{2} = 474 \, \Omega
 \end{aligned}$$

2.7.7 Adjustable Voltage Regulators

LM 723C is the general purpose adjustable voltage regulator. The output voltage is adjustable from 2 to 37 V. It will supply output currents up to 150 mA without external pass transistor and output currents excess of 10 A by adding external transistors. It can be used as either a linear or a switching regulator. Also, it can be used as a negative voltage regulator.

Regulated function of LM 723C The regulating function of this chip will be best understood by considering its internal circuit which is shown in Fig. 2.29.

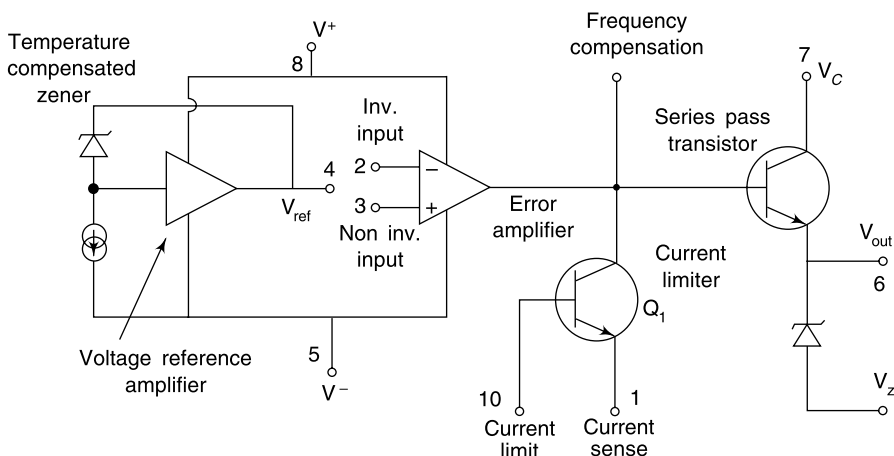


Fig. 2.29 Internal Circuit of LM 723C

A reference voltage is developed across the zener diode which is temperature compensated. The reference amplifier acts as a buffer and so the constant reference voltage is available at its output. The control amplifier has two inputs, one inverting and the other non-inverting. This amplifier compares the reference with a fixed part of the feedback output voltage and the resultant error is amplified. This error voltage controls the series transistor such that the output voltage remains at a constant level. Transistor Q_1 is used for current limiting.

The load current can be limited by providing a small resistance R_{sc} between the current limit and current sense terminals. The voltage across R_{sc} is used to bias the current limiting transistor, present inside the chip.

Hence, $V_{BE} = I_{\text{limit}} R_{sc}$, where V_{BE} is the bias voltage and I_{limit} is the limiting current

$$R_{sc} = \frac{V_{BE}}{I_{\text{limit}}}$$

Short Circuit Protection When the output terminals of a power supply are short circuited, the load current becomes too high and the power supply unit may be damaged. To guard against such an occurrence, current limiters have to be included in the regulator circuit. Current limiting can be provided by means of a fuse or a resistor (R_{sc}), rated for desired value of current. Figure 2.30 shows the regulator with current limiting. The resistance R_{sc} has to be chosen for the desired current limit.

$$R_{sc} = \frac{0.7}{I_{\text{limit}}}$$

By doing this, it is ensured that the power supply is being operated safely.

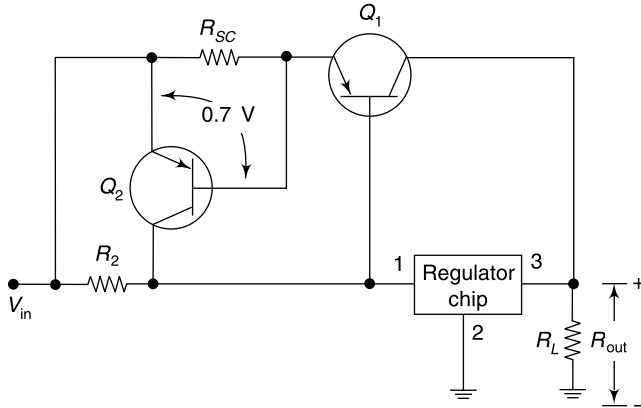


Fig. 2.30 Regular with Current Limiting

Over Voltage Protection The over voltage problems are divided into two types, namely, (i) *spikes* and (ii) *surges*. Spikes are high voltage transients which last for short duration of few μs . Surges are high voltage transients which will last for longer duration and will stretch for many ms.

Spike and surge protectors are designed to prevent over voltages from reaching the system. They absorb excess voltages before they enter the system. Surge protectors are connected between the system and the power line.

The most common over voltage protection devices are the metal oxide varistors. The varistors can chop and shunt away all voltages above a certain

level. These devices are designed to accept voltages as high as 6000 V and divert any voltage above 200 V to ground.

The excess energy does not disappear but turns into heat possible destroying the varistors. The most important characteristic of over voltage protection devices are how fast they work and how much energy they can dissipate. Other than varistors, semiconductors, ionised spark-gaps and ferro-resonant transformers are also used as surge protectors.

REVIEW QUESTIONS

1. What are the different types of power supplies?
2. What is the main difference between LMPS and SMPS?
3. Briefly explain the operation of linear power supply.
4. What are the requirements of the LMPS?
5. What is a rectifier?
6. Show that a PN diode works as rectifier.
7. Define the following terms.
(a) ripple factor, (b) peak inverse voltage, (c) efficiency, (d) transformer utilisation factor, (e) form factor, and (f) peak factor.
8. Draw the circuit diagram of an half wave rectifier, and explain its operation.
9. Derive expressions for rectification efficiency, ripple factor, transformer utilisation factor, form factor and peak factor of an half wave rectifier with resistive load.
10. A half wave rectifier has a load of $3.5\text{ k}\Omega$. If the diode resistance and secondary coil resistance together have a resistance of $800\ \Omega$ and the input voltage has a signal voltage of peak value 240 V , calculate
 - (i) Peak, average and rms value of current flowing
 - (ii) dc power output
 - (iii) ac power input
 - (iv) Efficiency of the rectifier

[Ans: (i) 58.81 mA , 17.78 mA and 27.9 mA
(ii) 1.1 W (iii) 3.35 W (iv) 32.9%]
11. Explain the action of a full-wave rectifier and give waveforms of input and output voltages.
12. Derive expressions of dc or average value of voltage and rms value of voltage of a full-wave rectifier with resistive load.
13. Derive an expression for a ripple factor in a full-wave rectifier with resistive load.
14. Determine the value of ripple factor in the full-wave rectifier operating at 50 Hz with a $100\ \mu\text{F}$ capacitor filter and $100\ \Omega$ load. [Ans: 29%]
15. Show that a full-wave rectifier is twice as efficient as a half-wave rectifier.
16. Describe the action of a full-wave bridge rectifier.
17. What are the advantages of a bridge rectifier?
18. Compare half wave, full-wave and bridge rectifiers.
19. What is the need for filters in power supplies?
20. Explain the various types of filters used in power supplies.

21. Obtain the ripple factor of a full-wave rectifier with shunt capacitor filter.
22. Derive an expression for the ripple factor in a full-wave rectifier using inductor filter.
23. Compare the performance of inductive, L -section and π -section filters.
24. An L - C filter is to be used to provide a dc output with 1% ripple from a full-wave rectifier operating at 50 Hz. Assuming $L/C = 0.01$, determine the required values of L and C .
[Ans: 1.093 H, 109.27 μ F]
25. In a full-wave rectifier using an L - C filter, it is known that $L = 10$ H, $C = 100$ μ F and $R_L = 500$ Ω . Calculate I_{dc} , V_{dc} , I_{ac} , V_{ac} , if $V_m = 30$ V and $f = 50$ Hz.
[Ans: 38.2 mA, 19.1 V, 1.43 mA, 22.7 mV]
26. List three reasons why an unregulated supply is not good enough for some applications.
27. Define line regulation and load regulation in a voltage regulator.
28. How does a zener diode maintain constant output voltage?
29. Design a zener shunt regulation with the following specifications:
 $V_o = 15$ V, $V_{in} = 20$ – 25 V, $I_L = 25$ – 50 mA, $I_Z = 20$ – 45 mA
30. Describe the operation of emitter-follower type voltage regulator.
31. Draw the short-circuit, overload protection circuit and explain its operation.
32. How is short-circuit current protection provided for an IC voltage regulator?
33. Mention the advantages of IC voltage regulators.
34. Compare linear and switched mode power supplies.
35. The turns ratio of the transformer used in a half-wave rectifier is 2:1 and the primary is connected to 230 V, 50 Hz power mains. Assuming the diodes to be ideal, determine (a) dc voltage across the load, (b) PIV of each diode, and (c) medium and average values of power delivered to the load having a resistance of 200 Ω . Also find the efficiency of the rectifier and output ripple frequency.
[Ans: 51.7 V, 162.2 V, 132.2 W, 13.5 W, 10.21%, 50 Hz]
36. In a full-wave rectifier, the voltage applied to each diode is $240 \sin 377t$, the load resistance is $R_L = 2000$ Ω and each diode has a forward resistance of 400 Ω . Determine the (a) peak value of current, (b) dc value of current, (c) rms value of current, (d) rectifier efficiency, (e) ripple factor, and (f) output ripple frequency.
[Ans: 100 mA, 63.8 mA, 70.7 mA, 67.6%, 0.482, 120 Hz]
37. In a bridge rectifier, the transformer is connected to 220 V, 60 Hz mains and the turns ratio of the step down transformer is 11:1. Assuming the diodes to be ideal, find (a) the voltage across the load, (b) I_{dc} and (c) PIV.
[Ans: 18 V, 18 mA, 28.28 V]
38. In a full-wave rectifier, the transformer rms secondary voltage from center tap to each end of secondary is 50 V. The load resistance is 900 Ω . If the diode resistance and transformer secondary winding resistance together has a resistance of 100 Ω . Determine the average load current and rms value of load current.
[Ans: 45 mA, 50 mA]
39. A zener diode shunt regulator circuit is to be designed to maintain a constant load current of 400 mA and voltage of 40 V. The input voltage is 90 ± 5 V.

- The zener diode voltage is 40 V and its dynamic resistance is $2.5\ \Omega$. Find the following quantities for the regulator: (a) the series dropping resistance, (b) zener power dissipation, and (c) load resistance. Assume the zener current to be 10% of load current. [Ans: $112.5\ \Omega$, $3.6\ \text{W}$, $100\ \Omega$]
40. In a half-wave rectifier, an ac voltage of peak value 24 V is connected in series with a silicon diode and load resistance of $480\ \Omega$. If the forward resistance of the diode is $20\ \Omega$, find the peak current flowing through the diode. [Ans: $46.6\ \text{mA}$]
41. A zener diode voltage regulator shown in Fig. 2.15 has the following specifications:
 $V_z = 15\ \text{V}$, $I_{z(\min)} = 2\ \mu\text{A}$, $P_z = 120\ \text{W}$, $R_z = 40\ \Omega$, $R_L = 5\ \text{k}\Omega$ and $V_{in} = 18\text{--}24\ \text{V}$. Determine the minimum and maximum value of series dropping resistance R . [Ans: $158\ \Omega$, $1\ \text{k}\Omega$]
42. Design a Zener regulator for the following specifications:
 output voltage, $V_o = 5\ \text{V}$, Load current, $I_L = 20\ \text{mA}$,
 Input voltage, $V_i = 12\ \text{V} \pm 3\ \text{V}$, Zener wattage $P_z = 500\ \text{mW}$.

OBJECTIVE TYPE QUESTIONS

1. A rectifier is used to
 - (a) convert a.c. voltage to d.c. voltage
 - (b) convert d.c. voltage to a.c. voltage
 - (c) both (a) and (b)
 - (d) convert voltage to current
2. The ripple factor of a half-wave rectifier is

(a) 1.21	(b) 0.482	(c) 0.406	(d) 0.121
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3. The peak inverse voltage of half-wave rectifier is

(a) V_m	(b) $2V_m$	(c) $V_m/2$	(d) $3V_m$
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4. The efficiency of a half-wave rectifier is

(a) 40.6%	(b) 81.2%	(c) 1.12%	(d) 48.2%
-----------	-----------	-----------	-----------
5. The ripple factor of a full-wave rectifier is

(a) 1.21	(b) 0.482	(c) 0.406	(d) 0.121
----------	-----------	-----------	-----------
6. The peak inverse voltage of a full-wave rectifier is

(a) V_m	(b) $2V_m$	(c) $V_m/2$	(d) $3V_m$
-----------	------------	-------------	------------
7. The efficiency of a full-wave rectifier is

(a) 40.6%	(b) 81.2%	(c) 1.12%	(d) 48.2%
-----------	-----------	-----------	-----------
8. The peak inverse voltage of a bridge rectifier is

(a) V_m	(b) $2V_m$	(c) $V_m/2$	(d) $3V_m$
-----------	------------	-------------	------------
9. The bridge rectifier requires

(a) 2 diodes	(b) 3 diodes	(c) 4 diodes	(d) 8 diodes
--------------	--------------	--------------	--------------
10. In a rectifier, the larger the value of the shunt capacitor filter, the
 - (a) longer time that the current pulse flows through the diode
 - (b) smaller the d.c. voltage across the load
 - (c) larger the d.c. voltage across the load
 - (d) larger the peak current in the rectifying diode

11. The primary function of a rectifier filter is to
 - (a) suppress old harmonics
 - (b) remove ripples
 - (c) stabilize the output d.c. level
 - (d) minimize the input a.c. variations
12. The major advantages of a bridge rectifier is that
 - (a) no centre-tap transformer is required
 - (b) the required peak inverse voltage of each diode is double of that for a full-wave rectifier
 - (c) the peak inverse voltage of each diode is half of that for a full-wave rectifier
 - (d) the output is more smooth
13. Which of the following circuits cannot be operated directly from the mains?
 - (a) Half-wave rectifier
 - (b) Full-wave rectifier
 - (c) Voltage doubler
 - (d) Center-tapped full-wave rectifier
14. The theoretical maximum efficiency of a full-wave circuit is
 - (a) the same as that of a half-wave circuit
 - (b) double as that of a half-wave circuit
 - (c) definitely more but less than double that of a half-wave circuit
 - (d) none of these
15. In a full-wave rectifier, the diodes conduct for
 - (a) one half-cycle
 - (b) full cycle
 - (c) alternate half-cycle
 - (d) none of these
16. The ripple frequency for a full-wave rectifier is
 - (a) equal to the supply frequency
 - (b) twice the supply frequency
 - (c) thrice the supply frequency
 - (d) none of these
17. In a full-wave rectifier circuit, the output voltage under load condition is
 - (a) the same as that under no load condition
 - (b) less than that under no load condition
 - (c) more than that under no load condition
 - (d) none of these
18. In a full-wave rectifier circuit, the peak inverse voltage per diode is
 - (a) the same as that in a half-wave circuit
 - (b) half the value in a half-wave circuit
 - (c) double the value in a half-wave circuit
 - (d) none of these
19. In a half-wave rectifier, the load current flows
 - (a) only for the positive half-cycle of the input signal
 - (b) for less than half-cycle of the input signal
 - (c) for more than half-cycle of the input signal
 - (d) for whole cycle of the input signal
20. In a full-wave rectifier, the current in each diode flows for
 - (a) whole cycle of the input signal
 - (b) half-cycle of the input signal
 - (c) more than half-cycle of the input signal
 - (d) None of the above
21. In a centre-tapped full-wave rectifier, if V_m is the peak voltage between the centre-tap and one end of the secondary, the maximum voltage coming across the reverse-biased diode is

- (a) V_m (b) $2V_m$ (c) $\frac{V_m}{2}$ (d) $\frac{V_m}{\sqrt{2}}$
22. In a full-wave bridge rectifier, if V_m is the peak voltage across the secondary of the transformer, the maximum voltage coming across each reverse-biased diode is
 (a) V_m (b) $2V_m$ (c) $\frac{V_m}{2}$ (d) $\frac{V_m}{\sqrt{2}}$
23. The ripple factor decreases with
 (a) decrease in C (b) increase in C
 (c) increase in frequency (d) decrease in frequency
24. The diode used in voltage regulator is
 (a) PN-junction diode (b) varactor diode
 (c) Zener diode (d) GUNN diode
25. The ripple factor of a inductor filter is
 (a) $\frac{R_L \omega}{\sqrt{2} \sqrt{3} L}$ (b) $\frac{R_L}{\sqrt{2} \sqrt{3} L}$ (c) $\frac{R_L}{3 \sqrt{2} \omega L}$ (d) $\frac{R_L}{\sqrt{3} 2 \omega L}$
26. A bleed resistor is used in a d.c. power supply because it
 (a) keeps the supply OFF (b) keeps the supply ON
 (c) improves filtering action (d) improve voltage regulation
27. The function of series pass transistor in a regulator is to
 (a) allow the current to pass to the load
 (b) adequately increase the level of output current to drive the load
 (c) compare the reference voltage
 (d) compensate any change with respect to temperature
28. Voltage regulator protection circuit includes
 (a) thermal shut down (b) current limiting
 (c) current boost (d) both (a) and (b)
29. The Zener-breakdown is observed in Zener diodes having V_z less than
 (a) 5 V (b) 10 V (c) 50 V (d) 100 V
30. Avalanche breakdown is observed in Zener diodes having V_z more than
 (a) 5 V (b) 8 V (c) 50 V (d) 100 V
31. For a Zener diode having maximum Zener current of 50 mA and $V_z = 10$ V, the maximum power dissipation is
 (a) 1 W (b) 5 mW (c) 50 mW (d) 0.5 W

State whether the following statements are true (T) or false (F)

32. The Zener diode is used as a constant voltage source in a regulator.
 33. The ripple factor of a half-wave rectifier is 2.31.
 34. The peak factor of a full-wave rectifier is $\sqrt{2}$.
 35. The ripple factor of the half-wave rectifier is 1.21.
 36. The transformer utilization factor for the half-wave rectifier is 0.287.
 37. The ripple factor of the full-wave rectifier is 0.482.
 38. The transformer utilization factor for the half-wave rectifier is 1.11.

39. For short-circuit protection of a power supply, the resistance $R_{SC} = \frac{0.7}{I_{\text{limit}}}$ has to be chosen for the desired current limit.
40. Spikes are high voltage transients which are lost for a short duration for a few microseconds.
41. The surges are high voltage transients which are lost for a longer duration for many milliseconds.
42. LM 723C is the general purpose adjustable voltage regulator.
43. Load regulation = $\frac{V_{\text{no load}} - V_{\text{full load}}}{V_{\text{no load}}}$
44. A Zener diode can be used in a voltage regulator circuit.
45. In a bridge rectifier, two diodes are used.
46. In a centre-tap full-wave rectifier, V_m is the peak voltage between the centre-tap and one end of the secondary. The PIV of the non-conducting diode is $2V_m$ when the filter is not connected.
47. The dc output voltage from a power supply increases with higher values of filter capacitance but decreases with more load current.
48. Zener diode is used as a rectifier.
49. Zener diode can be used in a voltage regulator circuit.
50. Zener diode operates in the reverse biased region.
51. The d.c. output voltage from a power supply increases with higher values of filter capacitance but decreases with more load current.

ANSWERS

- | | | | | |
|---------|---------|---------|---------|---------|
| 1. (a) | 2. (a) | 3. (a) | 4. (a) | 5. (b) |
| 6. (b) | 7. (b) | 8. (a) | 9. (c) | 10. (d) |
| 11. (b) | 12. (c) | 13. (d) | 14. (b) | 15. (c) |
| 16. (b) | 17. (b) | 18. (c) | 19. (a) | 20. (b) |
| 21. (b) | 22. (a) | 23. (b) | 24. (c) | 25. (c) |
| 26. (d) | 27. (b) | 28. (d) | 29. (a) | 30. (b) |
| 31. (d) | 32. (T) | 33. (F) | 34. (T) | 35. (T) |
| 36. (T) | 37. (T) | 38. (F) | 39. (T) | 40. (T) |
| 41. (T) | 42. (T) | 43. (T) | 44. (T) | 45. (T) |
| 46. (T) | 47. (F) | 48. (F) | 49. (T) | 50. (T) |
| 51. (F) | | | | |

3

BIPOLAR JUNCTION TRANSISTOR AND UJT

3.1 INTRODUCTION

A Bipolar Junction Transistor (BJT) is a three terminal semiconductor device in which the operation depends on the interaction of both majority and minority carriers and hence the name Bipolar. The BJT is analogous to a vacuum triode and is comparatively smaller in size. It is used in amplifier and oscillator circuits, and as a switch in digital circuits. It has wide applications in computers, satellites and other modern communication systems.

A single PN junction three terminal semiconductor switching device, commonly called Unijunction transistor (UJT), has three leads. Due to its negative resistance property, UJT can be employed in a variety of applications such as sawtooth waveform generator, pulse generator, switching, timing and phase control circuits. This chapter deals with the construction, operation, characteristics and applications of BJT and UJT.

3.2 CONSTRUCTION OF BJT AND BJT SYMBOLS

The BJT consists of a silicon (or germanium) crystal in which a thin layer of N-type Silicon is sandwiched between two layers of P-type silicon. This transistor is referred to as PNP. Alternatively, in a NPN transistor, a layer of P-type material is sandwiched between two layers of N-type material. The two types of the BJT are represented in Fig. 3.1.

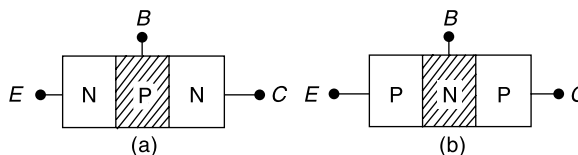


Fig. 3.1 Transistor (a) NPN and (b) PNP

The symbolic representation of the two types of the BJT is shown in Fig. 3.2. The three portions of the transistor are Emitter, Base and Collector, shown as E ,

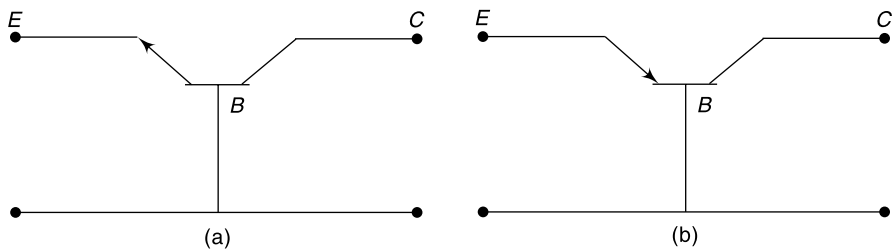


Fig. 3.2 Circuit Symbol (a) NPN Transistor and (b) PNP Transistor

B and *C*, respectively. The arrow on the emitter specifies the direction of current flow when the *EB* junction is forward biased.

Emitter is heavily doped so that it can inject a large number of charge carriers into the base. Base is lightly doped and very thin. It passes most of the injected charge carriers from the emitter into the collector. Collector is moderately doped.

3.3 TRANSISTOR BIASING

As shown in Fig. 3.3, usually the emitter-base junction is forward biased and collector-base junction is reverse biased. Due to the forward bias on the emitter-base junction an emitter current flows through the base into the collector. Though the,

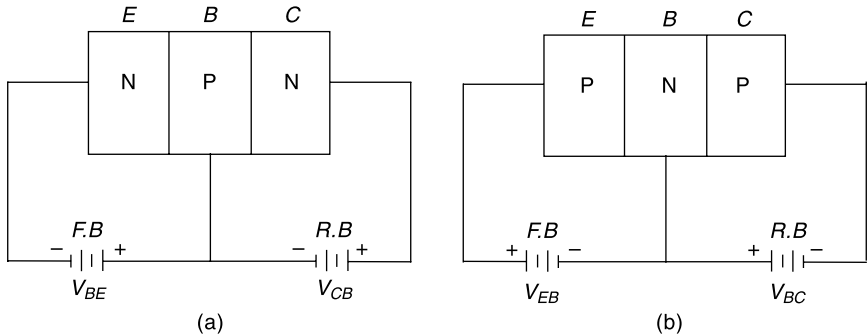


Fig. 3.3 Transistor Biasing (a) NPN Transistor and (b) PNP Transistor

collector-base junction is reverse biased, almost the entire emitter current flows through the collector circuit.

3.4 BJT OPERATION AND TRANSISTOR CURRENT COMPONENTS

3.4.1 Operation and Current Components of NPN Transistor

As shown in Fig. 3.4, the forward bias applied to the emitter base junction of an NPN transistor causes a lot of electrons from the emitter region to crossover to the

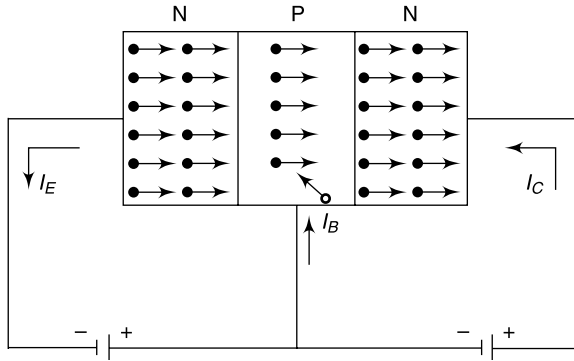


Fig. 3.4 Current in NPN Transistor

base region. As the base is lightly doped with P-type impurity, the number of holes in the base region is very small and hence the number of electrons that combine with holes in the P-type base region is also very small. Hence a few electrons combine with holes to constitute a base current I_B . The remaining electrons (more than 95%) crossover into the collector region to constitute a collector current I_C . Thus the base and collector current summed up gives the emitter current, i.e. $I_E = -(I_C + I_B)$.

In the external circuit of the NPN bipolar junction transistor, the magnitudes of the emitter current I_E , the base current I_B and the collector current I_C are related by $I_E = I_C + I_B$.

3.4.2 Operation and Current Components of PNP Transistor

As shown in Fig. 3.5, the forward bias applied to the emitter-base junction of a PNP transistor causes a lot of holes from the emitter region to crossover to the base region. As the base is lightly doped with N-type impurity, the number of electrons in the base region is very small and hence the number of holes combined with electrons in the N-type base region is also very small. Hence a few holes combined with electrons to constitute a base current I_B . The remaining holes (more than 95%) crossover into the collector region to constitute a collector current I_C . Thus the collector and base current when summed up gives the emitter current, i.e. $I_E = -(I_C + I_B)$.

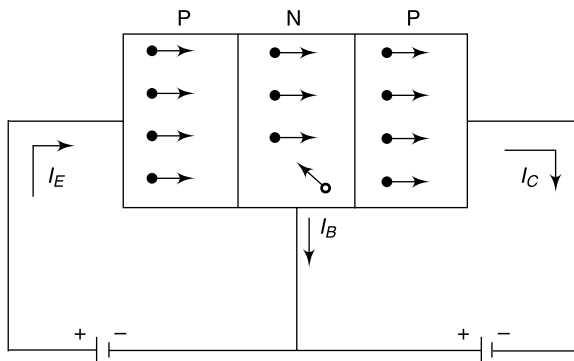


Fig. 3.5 Current in PNP Transistor

In the external circuit of the PNP bipolar junction transistor, the magnitudes of the emitter current I_E , the base current I_B and the collector current I_C are related by

$$I_E = I_C + I_B \quad (3.1)$$

This equation gives the fundamental relationship between the currents in a bipolar transistor circuit. Also, this fundamental equation shows that there are current amplification factors α and β in common base transistor configuration and common emitter transistor configuration respectively for the static (dc) currents, and for small changes in the currents.

Large-signal Current Gain(α) The large signal current gain of a common base transistor is defined as the ratio of the negative of the collector-current increment to the emitter-current change from cutoff ($I_E = 0$) to I_E , i.e.

$$\alpha = -\frac{(I_C - I_{CBO})}{I_E - 0} \quad (3.2)$$

where I_{CBO} (or I_{CO}) is the reverse saturation current flowing through the reverse biased collector-base junction, i.e. the collector to base leakage current with emitter open. As the magnitude of I_{CBO} is negligible when compared to I_E , the above expression can be written as

$$\alpha = \frac{I_C}{I_E} \quad (3.3)$$

Since I_C and I_E are flowing in opposite directions, α is always positive. Typical value of α ranges from 0.90 to 0.995. Also, α is not a constant but varies with emitter current I_E , collector voltage V_{CB} and temperature.

General Transistor Equation In the active region of the transistor, the emitter is forward biased and the collector is reverse biased. The generalised expression for collector current I_C for collector junction voltage V_C and emitter current I_E is given by

$$I_C = -\alpha I_E + I_{CBO} (1 - e^{V_C/V_T}) \quad (3.4)$$

If V_C is negative and $|V_C|$ is very large compared with V_T , then the above equation reduces to

$$I_C = -\alpha I_E + I_{CBO} \quad (3.5)$$

If V_C , i.e. V_{CB} , is few volts, I_C is independent of V_C . Hence the collector current I_C is determined only by the fraction α of the current I_E flowing in the emitter.

Relation among I_C , I_B and I_{CBO} From Eq. (3.5), We have

$$I_C = -\alpha I_E + I_{CBO}$$

Since I_C and I_E are flowing in opposite directions,

$$I_E = -(I_C + I_B)$$

Therefore,

$$\begin{aligned}
 I_C &= -\alpha [-(I_C + I_B)] + I_{CBO} \\
 I_C - \alpha I_C &= \alpha I_B + I_{CBO} \\
 I_C (1 - \alpha) &= \alpha I_B + I_{CBO} \\
 I_C &= \frac{\alpha}{1 - \alpha} I_B + \frac{I_{CBO}}{1 - \alpha}
 \end{aligned}$$

Since

$$\beta = \frac{\alpha}{1 - \alpha}, \quad (3.6)$$

the above expression becomes

$$I_C = (1 + \beta) I_{CBO} + \beta I_B \quad (3.7)$$

Relation Among I_C , I_B and I_{CEO} In the common-emitter (CE) transistor circuit, I_B is the input current and I_C is the output current. If the base circuit is open, i.e., $I_B = 0$, then a small collector current flows from the collector to emitter. This is denoted as I_{CEO} , the collector-emitter current with base open. This current I_{CEO} is also called the collector to emitter leakage current.

In this CE configuration of the transistor, the emitter-base junction is forward-biased and collector-base junction is reverse-biased and hence the collector current I_C is the sum of the part of the emitter current I_E that reaches the collector, and the collector-emitter leakage current I_{CEO} . Therefore, the part of I_E , which reaches collector is equal to $(I_C - I_{CEO})$.

Hence, the *large-signal current gain* (β) is defined as,

$$\beta = \frac{(I_C - I_{CEO})}{I_B} \quad (3.8)$$

From the equation, we have

$$I_C = \beta I_B + I_{CEO} \quad (3.9)$$

Relation Between I_{CBO} and I_{CEO} Comparing Eqs (3.7) and (3.9), we get the relationship between the leakage currents of transistor common-base (CB) and common-emitter (CE) configurations as

$$I_{CEO} = (1 + \beta) I_{CBO} \quad (3.10)$$

From this equation, it is evident that the collector-emitter leakage current (I_{CEO}) in CE configuration is $(1 + \beta)$ times larger than that in CB configuration. As I_{CBO} is temperature-dependent, I_{CEO} varies by large amount when temperature of the junctions changes.

Expression for Emitter Current The magnitude of emitter-current is

$$I_E = I_C + I_B$$

Substituting Eq. (3.7) in the above equation, we get

$$I_E = (1 + \beta) I_{CBO} + (1 + \beta) I_B \quad (3.11)$$

Substituting Eq. (3.6) into Eq. (3.11), we have

$$I_E = \frac{1}{1 - \alpha} I_{CBO} + \frac{1}{1 - \alpha} I_B \quad (3.12)$$

DC Current Gain (β_{dc} or h_{FE}) The dc current gain is defined as the ratio of the collector current I_C to the base current I_B . That is,

$$\beta_{dc} = h_{FE} = \frac{I_C}{I_B} \quad (3.13)$$

As I_C is large compared with I_{CEO} , the large signal current gain (β) and the dc current gain (h_{FE}) are approximately equal.

3.5 TYPES OF TRANSISTOR AMPLIFIER CONFIGURATION

When a transistor is to be connected in a circuit, one terminal is used as an input terminal, the other terminal is used as an output terminal and the third terminal is common to the input and output. Depending upon the input, output and common terminal, a transistor amplifier can be connected in three configurations. They are: (a) Common base (CB) configuration, (b) Common emitter (CE) configuration, and (c) Common collector (CC) configuration.

(a) CB configuration This is also called grounded base configuration. In this configuration, emitter is the input terminal, collector is the output terminal and base is the common terminal.

(b) CE configuration This is also called grounded emitter configuration. In this configuration, base is the input terminal, collector is the output terminal and emitter is the common terminal.

(c) CC configuration This is also called grounded collector configuration. In this configuration, base is the input terminal, emitter is the output terminal and collector is the common terminal.

The supply voltage connections for normal operation of an NPN transistor in the three configurations are shown in Fig. 3.6.

3.5.1 Common Base Configuration

The circuit diagram for determining the static characteristic curves of an NPN transistor in the common base configuration is shown in Fig. 3.7.

Input Characteristics To determine the input characteristics, the collector-base voltage V_{CB} is kept constant at zero Volt and the emitter current I_E is increased from zero in suitable equal steps by increasing V_{EB} . This is repeated for higher fixed values of V_{CB} . A curve is drawn between emitter current I_E and emitter-base voltage V_{EB} at constant collector-base voltage V_{CB} . The input characteristics thus obtained are shown in Fig. 3.8.

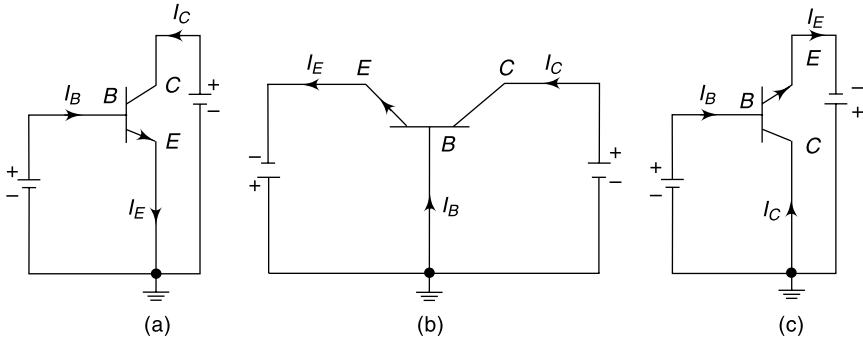


Fig. 3.6 Transistor Configuration: (a) Common Emitter (b) Common Base and (c) Common Collector

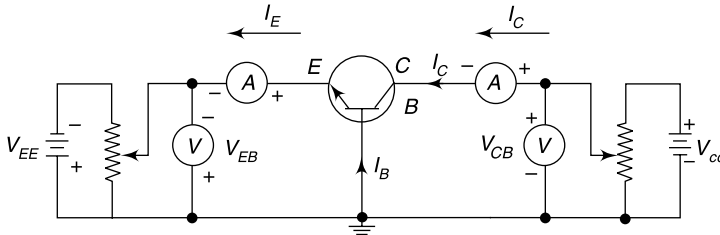


Fig. 3.7 Circuit to Determine CB Static Characteristics

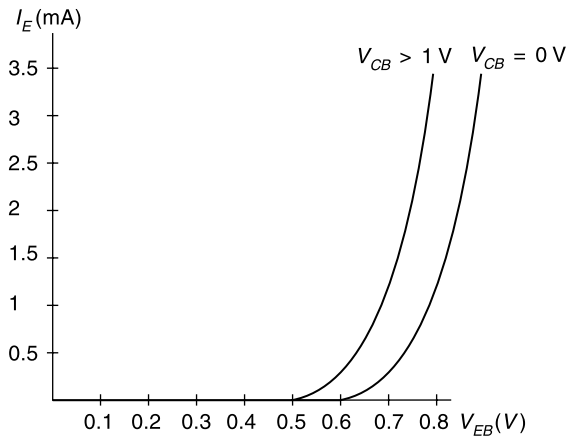


Fig. 3.8 CB Input Characteristics

When V_{CB} is equal to zero and the emitter-base junction is forward biased as shown in the characteristics, the junction behaves as a forward biased diode so that emitter current I_E increases rapidly with small increase in emitter-base voltage V_{EB} . When V_{CB} is increased keeping V_{EB} constant, the width of the base region will decrease. This effect results in an increase of I_E . Therefore, the curves shift towards the left as V_{CB} is increased.

Output Characteristics To determine the output characteristics, the emitter current I_E is kept constant at a suitable value by adjusting the emitter-base voltage V_{EB} . Then V_{CB} is increased in suitable equal steps and the collector current I_C is noted for each value of I_E . This is repeated for different fixed values of I_E . Now the curves of I_C versus V_{CB} are plotted for constant values of I_E and the output characteristics thus obtained is shown in Fig. 3.9.

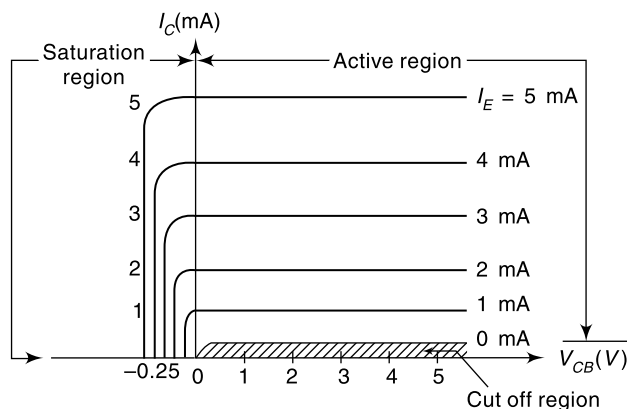


Fig. 3.9 CB Output Characteristics

From the characteristics, it is seen that for a constant value of I_E , I_C is independent of V_{CB} and the curves are parallel to the axis of V_{CB} . Further, I_C flows even when V_{CB} is equal to zero. As the emitter-base junction is forward biased, the majority carriers, i.e. electrons, from the emitter are injected into the base region. Due to the action of the internal potential barrier at the reverse biased collector-base junction, they flow to the collector region and give rise to I_C even when V_{CB} is equal to zero.

Early Effect or Base-width Modulation As the collector voltage V_{CC} is made to increase the reverse bias, the space charge width between collector and base tends to increase, with the result that the effective width of the base decreases. This dependency of base-width on collector-to-emitter voltage is known as the *Early effect*. This decrease in effective base-width has three consequences:

- There is less chance for recombination within the base region. Hence, α increases with increasing $|V_{CB}|$.
- The charge gradient is increased within the base, and consequently, the current of minority carriers injected across the emitter junction increases.
- For extremely large voltages, the effective base-width may be reduced to zero, causing voltage breakdown in the transistor. This phenomenon is called the *punch through*.

For higher values of V_{CB} , due to Early effect, the value of α increases. For example, α changes, say from 0.98 to 0.985. Hence, there is a very small positive slope in the CB output characteristics and hence the output resistance is not zero.

Transistor Parameters The slope of the CB characteristics will give the following four transistor parameters. Since these parameters have different dimensions, they are commonly known as common base *hybrid parameters* or *h-parameters*.

(a) **Input impedance (h_{ib})** It is defined as the ratio of the change in (input) emitter voltage to the change in (input) emitter current with the (output) collector voltage V_{CB} kept constant. Therefore,

$$h_{ib} = \frac{\Delta V_{EB}}{\Delta I_E}, V_{CB} \text{ constant.} \quad (3.14)$$

It is the slope of CB input characteristics I_E versus V_{EB} as shown in Fig. 3.8. The typical value of h_{ib} ranges from 20Ω to 50Ω .

(b) **Output admittance (h_{ob})** It is defined as the ratio of change in the (output) collector current to the corresponding change in the (output) collector voltage with the (input) emitter current I_E kept constant. Therefore,

$$h_{ob} = \frac{\Delta I_C}{\Delta V_{CB}}, I_E \text{ constant.} \quad (3.15)$$

It is the slope of CB output characteristics I_C versus V_{CB} as shown in Fig. 3.9. The typical value of this parameter is of the order of 0.1 to $10 \mu\text{mhos}$.

(c) **Forward current gain (h_{fb})** It is defined as a ratio of the change in the (output) collector current to the corresponding change in the (input) emitter current keeping the (output) collector voltage V_{CB} constant. Hence,

$$h_{fb} = \frac{\Delta I_C}{\Delta I_E}, V_{CB} \text{ constant.} \quad (3.16)$$

It is the slope of I_C versus I_E curve. Its typical value varies from 0.9 to 1.0 .

(d) **Reverse voltage gain (h_{rb})** It is defined as the ratio of the change in the (input) emitter voltage and the corresponding change in (output) collector voltage with constant (input) emitter current, I_E . Hence,

$$h_{rb} = \frac{\Delta V_{EB}}{\Delta V_{CB}}, I_E \text{ constant.} \quad (3.17)$$

It is the slope of V_{EB} versus V_{CB} curve. Its typical value is of the order of 10^{-5} to 10^{-4} .

3.5.2 Common Emitter Configuration

Input Characteristics To determine the input characteristics, the collector to emitter voltage is kept constant at zero Volt and base current is increased from zero in equal steps by increasing V_{BE} in the circuit shown in Fig. 3.10.

The value of V_{BE} is noted for each setting of I_B . This procedure is repeated for higher fixed values of V_{CE} , and the curves of I_B V_S V_{BE} are drawn. The input characteristics thus obtained are shown in Fig. 3.11.

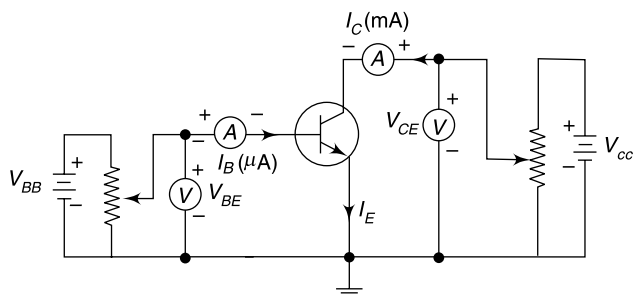


Fig. 3.10 Circuit to Determine CE Static Characteristics

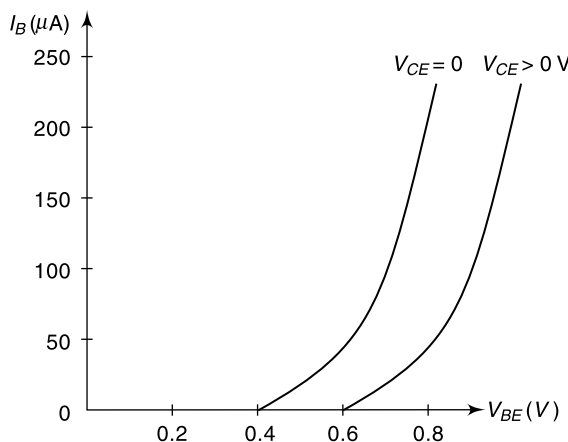


Fig. 3.11 CE Input Characteristics

When $V_{CE} = 0$, the emitter-base junction is forward biased and the junction behaves as a forward biased diode. Hence the input characteristic for $V_{CE} = 0$ is similar to that of a forward-biased diode. When V_{CE} is increased, the width of the depletion region at the reverse biased collector-base junction will increase. Hence the effective width of the base will decrease. This effect causes a decrease in the base current I_B . Hence, to get the same value of I_B as that for $V_{CE} = 0$, V_{BE} should be increased. Therefore, the curve shifts to the right as V_{CE} increases.

Output Characteristics To determine the output characteristics, the base current I_B is kept constant at a suitable value by adjusting base-emitter voltage, V_{BE} . The magnitude of collector-emitter voltage V_{CE} is increased in suitable equal steps from zero and the collector current I_C is noted for each setting of V_{CE} . Now the curves of I_C versus V_{CE} are plotted for different constant values of I_B . The output characteristics thus obtained are shown in Fig. 3.12.

From Eqs (3.6) and (3.7), we have

$$\beta = \frac{\alpha}{1 - \alpha} \quad \text{and} \quad I_C = (1 + \beta) I_{CBO} + \beta I_B$$

For larger values of V_{CE} , due to Early effect, a very small change in α is reflected in a very large change in β . For example, when $\alpha = 0.98$, $\beta = \frac{0.98}{1 - 0.98} = 49$. If α increases

to 0.985, then $\beta = \frac{0.985}{1-0.985} = 66$. Here, a slight increase in α by about 0.5% results in an increase in β by about 34%. Hence, the output characteristics of CE configuration show a larger slope when compared with CB configuration.

The output characteristics have three regions, namely, saturation region, cutoff region and active region. The region of curves to the left of the line OA is called the *saturation region* (hatched), and the line OA is called the saturation line. In this region, both junctions are forward

biased and an increase in the base current does not cause a corresponding large change in I_C . The ratio of $V_{CE(sat)}$ to I_C in this region is called saturation resistance.

The region below the curve for $I_B = 0$ is called the *cut-off region* (hatched). In this region, both junctions are reverse biased. When the operating point for the transistor enters the cut-off region, the transistor is OFF. Hence, the collector current becomes almost zero and the collector voltage almost equals V_{CC} , the collector supply voltage. The transistor is virtually an open circuit between collector and emitter.

The central region where the curves are uniform in spacing and slope is called the *active region* (unhatched). In this region, emitter-base junction is forward biased and the collector-base junction is reverse biased. If the transistor is to be used as a linear amplifier, it should be operated in the active region.

If the base current is subsequently driven large and positive, the transistor switches into the saturation region via the active region, which is traversed at a rate that is dependent on factors such as gain and frequency response. In this ON condition, large collector current flows and collector voltage falls to a very low value, called $V_{CE(sat)}$, typically around 0.2 V for a silicon transistor. The transistor is virtually a short circuit in this state.

High speed switching circuits are designed in such a way that transistors are not allowed to saturate, thus reducing switching times between ON and OFF times.

Transistor Parameters The slope of the CE characteristics will give the following four transistor parameters. Since these parameters have different dimensions, they are commonly known as common emitter *hybrid parameters* or *h-parameters*.

(a) **Input Impedance (h_{ie})** It is defined as the ratio of the change in (input) base voltage to the change in (input) base current with the (output) collector voltage V_{CE} kept constant. Therefore,

$$h_{ie} = \frac{\Delta V_{BE}}{\Delta I_B}, V_{CE} \text{ constant} \quad (3.18)$$

It is the slope of CE input characteristics I_B versus V_{BE} as shown in Fig. 3.11. The typical value of h_{ie} ranges from 500 to 2000 Ω .

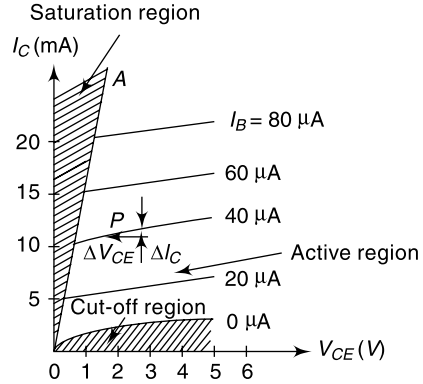


Fig. 3.12 CE Output Characteristics

(b) **Output Admittance (h_{oe})** It is defined as the ratio of change in the (output) collector current to the corresponding change in the (output) collector voltage with the (input) base current I_B kept constant. Therefore,

$$h_{oe} = \frac{\Delta I_C}{\Delta V_{CE}}, I_B \text{ constant.} \quad (3.19)$$

It is the slope of CE output characteristic I_C versus V_{CE} as shown in Fig. 3.12. The typical value of this parameter is of the order of 0.1 to 10 μmhos .

(c) **Forward Current Gain (h_{fe})** It is defined as a ratio of the change in the (output) collector current to the corresponding change in the (input) base current keeping the (output) collector voltage V_{CE} constant. Hence,

$$h_{fe} = \frac{\Delta I_C}{\Delta I_B}, V_{CE} \text{ constant.} \quad (3.20)$$

It is the slope of I_C versus I_B curve. Its typical value varies from 20 to 200.

(d) **Reverse Voltage Gain (h_{re})** It is defined as the ratio of the change in the (input) base voltage and the corresponding change in (output) collector voltage with constant (input) base current, I_B . Hence,

$$h_{re} = \frac{\Delta V_{BE}}{\Delta V_{CE}}, I_B \text{ constant.} \quad (3.21)$$

It is the slope of V_{BE} versus V_{CE} curve. Its typical value is of the order of 10^{-5} to 10^{-4} .

3.5.3 Common Collector Configuration

The circuit diagram for determining the static characteristics of an NPN transistor in the common collector configuration is shown in Fig. 3.13.

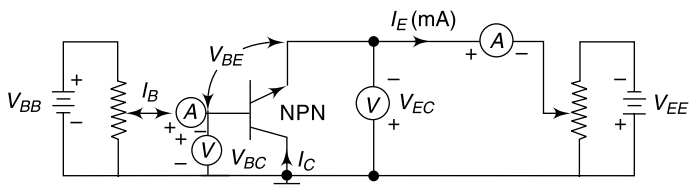


Fig. 3.13 Circuit to Determine CC Static Characteristics

Input Characteristics To determine the input characteristics, V_{EC} is kept at a suitable fixed value. The base-collector voltage V_{BC} is increased in equal steps and the corresponding increase in I_B is noted. This is repeated for different fixed values of V_{EC} . Plots of V_{BC} versus I_B for different values of V_{EC} shown in Fig. 3.14 are the input characteristics.

Output Characteristics The output characteristics shown in Fig. 3.15 are the same as those of the common emitter configuration.

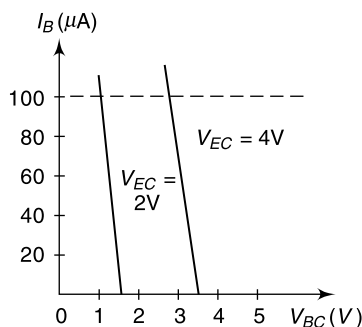


Fig. 3.14 CC Input Characteristics

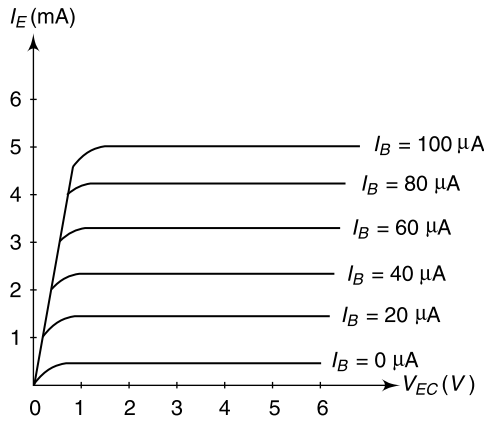


Fig. 3.15 CC Output Characteristics

3.5.4 Comparison

Table 3.1 A comparison of CB, CE and CC configurations

Property	CB	CE	CC
Input resistance	Low (about 100Ω)	Moderate (about 750Ω)	High (about $750 \text{ k}\Omega$)
Output resistance	High (about $450 \text{ k}\Omega$)	Moderate (about $45 \text{ k}\Omega$)	Low (about 25Ω)
Current gain	1	High	High
Voltage gain	About 150	About 500	Less than 1
Phase shift between input & output voltages	0 or 360°	180°	0 or 360°
Applications	For high frequency circuits	For audio frequency circuits	For impedance matching

3.5.5 Current Amplification Factor

In a transistor amplifier with ac input signal, the ratio of change in output current to the change in input current is known as the current amplification factor.

In the CB configuration the current amplification factor, $\alpha = \frac{\Delta I_C}{\Delta I_E}$ (3.22)

In the CE configuration the current amplification factor, $\beta = \frac{\Delta I_C}{\Delta I_B}$ (3.23)

In the CC configuration the current amplification factor, $\gamma = \frac{\Delta I_E}{\Delta I_B}$ (3.24)

Relationship Between α and β We know that $\Delta I_E = \Delta I_C + \Delta I_B$

By definition, $\Delta I_C = \alpha \Delta I_E$

Therefore, $\Delta I_E = \alpha \Delta I_E + \Delta I_B$

i.e $\Delta I_B = \Delta I_E (1 - \alpha)$

Dividing both sides by ΔI_C , we get

$$\frac{\Delta I_B}{\Delta I_C} = \frac{\Delta I_E}{\Delta I_C} (1 - \alpha)$$

Therefore,

$$\frac{1}{\beta} = \frac{1}{\alpha} (1 - \alpha)$$

$$\beta = \frac{\alpha}{(1 - \alpha)}$$

Rearranging, we also get

$$\alpha = \frac{\beta}{(1 + \beta)}, \text{ or } \frac{1}{\alpha} - \frac{1}{\beta} = 1 \quad (3.25)$$

From this relationship, it is clear that as α approaches unity, β approaches infinity. The CE configuration is used for almost all transistor applications because of its high current gain, β .

Relation Among α , β and γ In the CC transistor amplifier circuit, I_B is the input current and I_E is the output current.

From Eq. (3.22),

$$\gamma = \frac{\Delta I_E}{\Delta I_B}$$

Substituting

$$\Delta I_B = \Delta I_E - \Delta I_C,$$

we get

$$\gamma = \frac{\Delta I_E}{\Delta I_E - \Delta I_C}$$

Dividing the numerator and denominator on RHS by ΔI_E , we get

$$\gamma = \frac{\frac{\Delta I_E}{\Delta I_E}}{\frac{\Delta I_E}{\Delta I_E} - \frac{\Delta I_C}{\Delta I_E}} = \frac{1}{1 - \alpha}$$

Therefore,

$$\gamma = \frac{1}{1 - \alpha} = (\beta + 1) \quad (3.26)$$

3.6 TRANSISTOR AS AN AMPLIFIER

A load resistor R_L is connected in series with the collector supply voltage V_{CC} of CB transistor configuration as shown in Fig. 3.16.

A small change in the input voltage between emitter and base, say ΔV_{ib} , causes a relatively larger change in emitter current, say ΔI_E . A fraction of this change in current is collected and passed through R_L and is

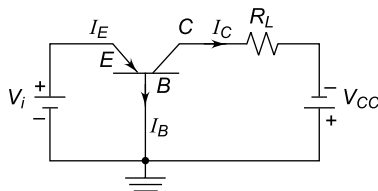


Fig. 3.16 Common Base Transistor Configuration

denoted by symbol α' . Therefore the corresponding change in voltage across the load resistor R_L due to this current is $\Delta V_o = \alpha' R_L \Delta I_E$.

Here, the voltage amplification $A_v = \frac{\Delta V_o}{\Delta V_i}$ is greater than unity and thus the transistor acts as an amplifier.

3.7

LARGE SIGNAL, DC, AND SMALL SIGNAL CE VALUES OF CURRENT GAIN

We know from the characteristics of CE configuration, the current amplification factor is $\beta \equiv \frac{\alpha}{1 - \alpha}$.

and

$$I_C = (1 + \beta) I_{CBO} + \beta I_B$$

The above equation can be expressed as

$$I_C - I_{CBO} = \beta I_{CBO} + \beta I_B$$

Therefore,

$$\beta = \frac{I_C - I_{CBO}}{I_B - (-I_{CBO})}$$

The output characteristics of CE configuration show that in the *cut-off* region, the values $I_E = 0$, $I_C = I_{CBO}$ and $I_B = -I_{CBO}$. Therefore, the above equation gives the ratio of the collector-current increment to the base-current change from cut off to I_B , and hence β is called the *large-signal current gain of common-emitter transistor*.

The dc current gain of the transistor is given by

$$\beta_{dc} \equiv h_{FE} \equiv \frac{I_C}{I_B}$$

Based on this h_{FE} value, we can determine whether the transistor is in saturation or not. For any transistor, in general, I_B is large compared to I_{CBO} . Under this condition, the value of $h_{FE} \approx \beta$.

The small-signal CE forward short-circuit gain β' is defined as the ratio of a collector-current increment ΔI_C for a small base-current change ΔI_B , at a fixed collector-to-emitter voltage V_{CE} .

i.e.

$$\beta' \equiv \left. \frac{\partial I_C}{\partial I_B} \right|_{V_{CE}}$$

If β is independent of currents, then $\beta' = \beta \approx h_{FE}$. However, β is a function of current, then $\beta' = \beta + (I_{CBO} + I_B) \frac{\partial \beta}{\partial I_B}$. By using $\beta' = h_{fe}$ and $\beta \approx h_{FE}$. Therefore, the above equation becomes

$$h_{fe} = \frac{h_{FE}}{1 - (I_{CBO} + I_B) \frac{\partial h_{FE}}{\partial I_C}}$$

In Fig. 3.17, the h_{FE} versus I_C shows a maximum and hence $h_{fe} > h_{FE}$ for smaller currents, and $h_{fe} < h_{FE}$ for larger currents. Therefore, the above equation is valid only for the active region.

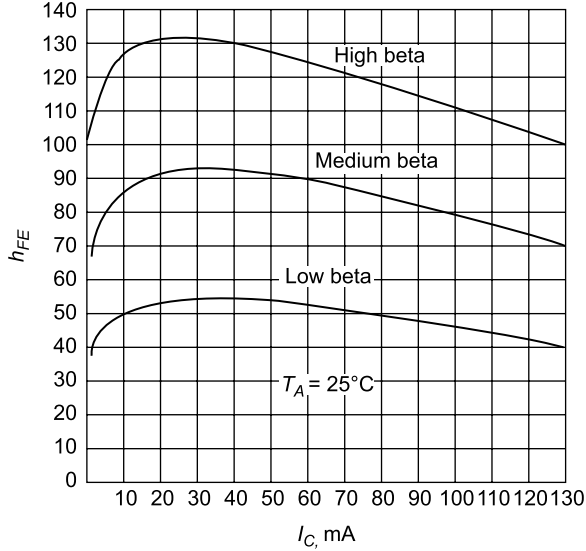


Fig. 3.17 Characteristic Curves of dc Current Gain h_{FE} (at $V_{CE} = -0.25$ V) Versus Collector Current for Low, Medium and High Beta Values

3.8 LIMITS OF OPERATION (BREAKDOWN IN TRANSISTORS)

There is a possibility of voltage breakdown in the transistor at high voltages even though the rated dissipation of the transistor is not exceeded. Therefore, there is an upper limit to the maximum allowable collector junction voltage. There are two types of breakdown, namely, **avalanche multiplication** or **avalanche breakdown** and **reach-through** or **punch-through**.

3.8.1 Avalanche Breakdown and Multiplication

When a diode is reverse biased, there is a limit on the voltage that can be applied which is the avalanche voltage. Similarly, in the transistor, the maximum reverse biasing voltage which may be applied before breakdown between the collector and base terminals with the emitter open is called breakdown voltage BV_{CBO} . Therefore, an upper limit is set on the collector voltage V_{CB} by avalanche breakdown in the reverse biased collector-base junction. Fields of order 10^6 V/m are required in the depletion layer for breakdown to occur, which usually limits V_{CB} to a maximum of several tens of volts.

Breakdown may occur because of Avalanche multiplication of the current I_{CO} that crosses the collector junction. As a result of this multiplication, the current becomes MI_{CO} , where M is the avalanche multiplication factor. At the breakdown voltage BV_{CBO} , multiplication factor M becomes infinite and the current rises

abruptly in the breakdown region, as shown in Fig. 3.18, there will be large changes in current with small changes in applied voltage.

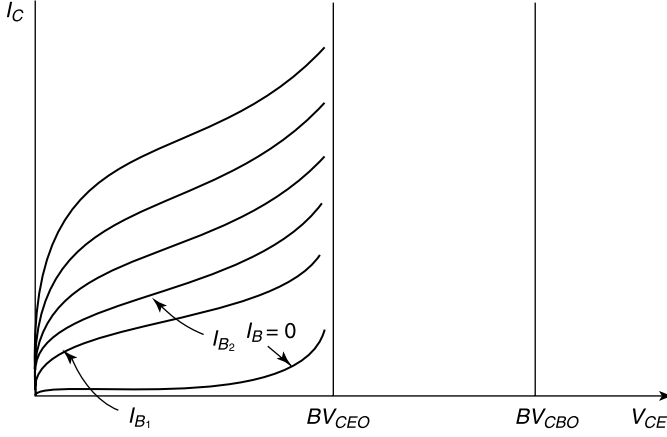


Fig. 3.18 CE Characteristics in the Breakdown Region

The Avalanche multiplication factor depends on the voltage V_{CB} between collector and base, which has been found to be given empirically by

$$M = \frac{1}{1 - \left(\frac{V_{CB}}{BV_{CBO}} \right)^n} \quad (3.27)$$

where the empirical constant, n , depends on the lattice material and the carrier type, which is usually in the range of about 2 to 10; for N-type silicon $n \approx 4$ and for P-type $n \approx 2$, and controls the sharpness of the onset breakdown. Taking Avalanche multiplication into account, I_C has the magnitude $M \alpha I_E$, where α is the common base current gain. As a result, in the presence of Avalanche multiplication, the current gain of CB transistor has become $M\mu$. As the effective current gain exceeds unity, the emitter circuit may then display negative resistance effects, which can lead to undesirable instabilities.

For the CE configuration, the collector to emitter breakdown voltage BV_{CEO} with base open is

$$BV_{CEO} = BV_{CBO} \sqrt[n]{\frac{1}{h_{FE}}} \quad (3.28)$$

In general, BV_{CEO} is 40 to 50% of BV_{CBO} . This is the upper limit of V_{CE} that can be placed across the transistor without damaging it.

3.8.2 Reach-Through or Punch-Through

According to Early effect, the width of the collector-junction transition region increases with increased collector-junction voltage. As the voltage applied across the junction V_{CB} increases the transition region penetrates deeper into the base and will have spread completely across the base to reach the emitter junction, as the

base is very thin. Thus, the collector voltage has reached through the base region. This effect, known as reach-through, also affects the output characteristics of a transistor since I_C Vs V_{CB} curves are no longer horizontal but take on a positive slope indicating that the device has a finite output impedance that is voltage dependent. Since the input characteristics are also affected, the input impedance is also influenced by V_{CB} .

It is possible to raise the punch-through voltage by increasing the doping concentration in the base, but this automatically reduces the emitter efficiency.

Punch-through takes place at a fixed voltage between collector and base and is not dependent on circuit configuration, whereas Avalanche multiplication takes place at different voltages depending upon the circuit configuration. Therefore, the voltage limit of a particular transistor is determined by either of the two types of breakdown, whichever occurs at lower voltage.



Example 3.1 In a common-base transistor circuit, the emitter current I_E is 10 mA and the collector current I_C is 9.8 mA. Find the value of the base current I_B .

Solution $I_E = 10$ mA and $I_C = 9.8$ mA

We know that emitter current (I_E), $I_E = I_B + I_C$

$$\text{i.e.,} \quad 10 \times 10^{-3} = I_B + 9.8 \times 10^{-3}$$

$$\text{Therefore,} \quad I_B = (10 - 9.8) \times 10^{-3} = 0.2 \text{ mA}$$



Example 3.2 In a common-base connection, the emitter current I_E is 6.28 mA and the collector current I_C is 6.20 mA. Determine the common-base dc current gain.

Solution Given $I_E = 6.28$ mA and $I_C = 6.20$ mA

We know that common-base dc current gain,

$$\alpha = \frac{I_C}{I_E} = \frac{6.20 \times 10^{-3}}{6.28 \times 10^{-3}} = 0.987$$



Example 3.3 The common-base dc current gain of a transistor is 0.967. If the emitter current is 10 mA, what is the value of base current?

Solution Given $\alpha = 0.967$ and $I_E = 10$ mA

The common-base dc current gain (α) is

$$\alpha = 0.967 = \frac{I_C}{I_E} = \frac{I_C}{10 \times 10^{-3}}$$

$$\text{Therefore,} \quad I_C = 0.967 \times 10 \times 10^{-3} = 9.67 \text{ mA}$$

The emitter current

$$\text{i.e.,} \quad 10 \times 10^{-3} = I_B + 9.67 \times 10^{-3}$$

$$\text{Therefore,} \quad I_B = (10 - 9.67) \text{ mA} = 0.33 \text{ mA}$$

Example 3.4 The transistor has $I_E = 10 \text{ mA}$ and $\alpha = 0.98$. Determine the value of I_C and I_B .

Solution Given $I_E = 10 \text{ mA}$ and $\alpha = 0.98$

The common-base dc current gain, $\alpha = \frac{I_C}{I_E}$

$$\text{i.e.,} \quad 0.98 = \frac{I_C}{10 \times 10^{-3}}$$

Therefore, $I_C = 0.98 \times 10^{-2} = 9.8 \text{ mA}$

The emitter current $I_E = I_B + I_C$

$$\text{i.e.,} \quad 10 \times 10^{-3} = I_B + 9.8 \times 10^{-3}$$

Therefore, $I_B = (10 - 9.8) \text{ mA} = 0.2 \text{ mA}$

Example 3.5 If a transistor has α of 0.97, find the value of β . If $\beta = 200$, find the value of α .

Solution If $\alpha = 0.97$, $\beta = \frac{\alpha}{1-\alpha} = \frac{0.97}{1-0.97} = 32.33$

$$\text{If} \quad \beta = 200, \alpha = \frac{\beta}{\beta+1} = \frac{200}{200+1} = 0.995$$

Example 3.6 A transistor has $\beta = 100$. If the collector current is 40 mA, find the value of emitter current.

Solution Given $\beta = 100$ and $I_C = 40 \text{ mA}$

$$\beta = 100 = \frac{I_C}{I_B} = \frac{40 \times 10^{-3}}{I_B}$$

$$\text{Therefore,} \quad I_B = \frac{40 \times 10^{-3}}{100} = 0.4 \text{ mA and}$$

$$I_E = I_B + I_C = (0.4 + 40) \times 10^{-3} = 40.4 \text{ mA}$$

Example 3.7 A transistor has $\beta = 150$. Find the collector and base currents, if $I_E = 10 \text{ mA}$.

Solution Given $\beta = 150$ and $I_E = 10 \text{ mA}$

$$\text{The common-base current gain,} \quad \alpha = \frac{\beta}{\beta+1} = \frac{150}{150+1} = 0.993$$

$$\text{Also,} \quad \alpha = \frac{I_C}{I_E}$$

$$\text{i.e.,} \quad 0.993 = \frac{I_C}{I_E}$$

Therefore, $I_C = 0.993 \times 10^{-2} = 9.93 \text{ mA}$
 The emitter current $I_E = I_B + I_C$
 i.e., $10 \times 10^{-3} = I_B + 9.93 \times 10^{-3}$
 Therefore, $I_B = (10 - 9.93) \times 10^{-3} = 0.07 \text{ mA}$

Example 3.8 Determine the value of I_B and I_E for the transistor circuit if $I_C = 80 \text{ mA}$ and $\beta = 170$.

Solution Given $\beta = 170$ and $I_C = 80 \text{ mA}$

We know that β ,
$$\beta = 170 = \frac{I_C}{I_B} = \frac{80 \times 10^{-3}}{I_B}$$

Therefore,
$$I_B = \frac{80 \times 10^{-3}}{170} = 0.47 \text{ mA} \quad \text{and}$$

$$I_E = I_B + I_C = (0.47 + 80) \text{ mA} = 80.47 \text{ mA}$$

Example 3.9 Determine the value of I_C and I_E for the transistor circuit of $\beta = 200$ and $I_B = 0.125 \text{ mA}$.

Solution Given $I_B = 0.125 \text{ mA}$ and $\beta = 200$

Therefore,
$$\beta = 200 = \frac{I_C}{I_B} = \frac{I_C}{0.125 \times 10^{-3}}$$

Therefore,
$$I_C = 200 \times 0.125 \times 10^{-3} = 25 \text{ mA}$$

and
$$I_E = I_B + I_C = (0.125 + 25) \times 10^{-3} = 25.125 \text{ mA}$$

Example 3.10 Determine the value of I_C and I_B for the transistor circuit if $I_E = 12 \text{ mA}$ and $\beta = 100$.

Solution Given $I_E = 12 \text{ mA}$ and $\beta = 100$

We know that base current,

$$I_B = \frac{I_E}{1 + \beta} = \frac{12 \times 10^{-3}}{1 + 100} = 0.1188 \text{ mA}$$

and collector current,

$$I_C = I_E - I_B = (12 - 0.1188) \times 10^{-3} = 11.8812 \text{ mA}$$

Example 3.11 A transistor has $I_B = 100 \text{ mA}$ and $I_C = 2 \text{ mA}$. Find (a) β of the transistor, (b) α of the transistor, (c) emitter current I_E , (d) if I_B changes by $+25 \text{ } \mu\text{A}$ and I_C changes by $+0.6 \text{ mA}$, find the new value of β .

Solution Given $I_B = 100 \text{ } \mu\text{A} = 100 \times 10^{-6} \text{ A}$ and $I_C = 2 \text{ mA} = 2 \times 10^{-3} \text{ A}$.

(a) To find β of the transistor

$$\beta = \frac{I_C}{I_B} = \frac{2 \times 10^{-3}}{100 \times 10^{-6}} = 20$$

(b) To find α of the transistor

$$\alpha = \frac{\beta}{\beta + 1} = \frac{20}{1 + 20} = 0.952$$

(c) To find Emitter current, I_E

$$\begin{aligned} I_E &= I_B + I_C = 100 \times 10^{-6} + 2 \times 10^{-3} \text{ A} \\ &= (0.01 + 2) \times 10^{-3} = 2.01 \times 10^{-3} \text{ A} = 2.01 \text{ mA} \end{aligned}$$

(d) To find new value of β when $\Delta I_B = 25 \mu\text{A}$ and $\Delta I_C = 0.6 \text{ mA}$

$$\text{Therefore, } I_B = (100 + 25) \text{ mA} = 125 \text{ mA}$$

$$I_C = (2 + 0.6) \text{ mA} = 2.6 \text{ mA}$$

New value of β of the transistor,

$$\beta = \frac{I_C}{I_B} = \frac{2.6 \times 10^{-3}}{125 \times 10^{-6}} = 20.8$$

Example 3.12 Calculate the collector current and emitter for a transistor with $\alpha_{dc} = 0.99$ and $I_{CBO} = 50 \mu\text{A}$ when the base current is $20 \mu\text{A}$.
[JNTU April/May 2007]

Solution

$$\begin{aligned} I_C &= \frac{\alpha_{dc} \cdot I_B}{1 - \alpha_{dc}} + \frac{I_{CBO}}{1 - \alpha_{dc}} \\ &= \frac{0.99 \times 50 \times 10^{-6}}{1 - 0.99} + \frac{50 \times 10^{-6}}{1 - 0.99} = 9950 \times 10^{-6} \text{ A} \end{aligned}$$

Example 3.13 For a transistor circuit having $\alpha = 0.98$, $I_{CBO} = I_{CO} = 5 \mu\text{A}$ and $I_B = 100 \mu\text{A}$, find I_C and I_E .

Solution Given $\alpha = 0.98$, $I_{CBO} = I_{CO} = 5 \mu\text{A}$ and $I_B = 100 \mu\text{A}$

The collector current is

$$I_C = \frac{\alpha \cdot I_B}{1 - \alpha} + \frac{I_{CO}}{1 - \alpha} = \frac{0.98 \times 100 \times 10^{-6}}{1 - 0.98} + \frac{5 \times 10^{-6}}{1 - 0.98} = 5.15 \text{ mA}$$

The emitter current is

$$I_E = I_B + I_C = 100 \times 10^{-6} + 5.15 \times 10^{-3} = 5.25 \text{ mA}$$

Example 3.14 Given an NPN transistor for which $\alpha = 0.98$, $I_{CO} = 2 \mu\text{A}$ and $I_{EO} = 1.6 \mu\text{A}$. A common emitter connection is used and $V_{CC} = 12 \text{ V}$ and $R_L = 4 \text{ k}\Omega$. What is the minimum base current required in order that transistor enter in to saturation region.

Solution Given $\alpha = 0.98$, $I_{CO} = 2 \mu A$, $I_{EO} = 1.6 \mu A$ and $R_L = 4 k\Omega$

We know that, $V_{CC} - I_C R_L - V_{CE(sat)} = 0$

$$12 - I_C \times 4 \times 10^3 - 0.3 = 0$$

$$I_C = \frac{11.7}{4 \times 10^3} = 2.925 \text{ mA}$$

We know that,

$$I_C = \frac{\alpha}{1-\alpha} I_B + \frac{I_{CO}}{1-\alpha}$$

$$2.925 \times 10^{-3} = \frac{0.98}{1-0.98} I_B + \frac{2 \times 10^{-6}}{1-0.98}$$

$$= 49 I_B + \frac{2 \times 10^{-6}}{0.02}$$

$$= 49 I_B + 0.1 \times 10^{-3}$$

Therefore,

$$I_B = \frac{2.825 \times 10^{-3}}{49} = 57.65 \mu A$$

Example 3.15 Calculate the values of I_C and I_E for a transistor with $\alpha_{dc} = 0.99$ and $I_{CBO} = 5 \mu A$. I_B is measured as $20 \mu A$.

[JNTU May 2003, Nov 2004]

Solution Given $\alpha_{dc} = 0.99$, $I_{CBO} = 5 \mu A$ and $I_B = 20 \mu A$

$$I_C = \frac{\alpha_{dc} I_B}{1-\alpha_{dc}} + \frac{I_{CBO}}{1-\alpha_{dc}} = \frac{0.99 \times 20 \times 10^{-6}}{1-0.99} + \frac{5 \times 10^{-6}}{1-0.99} = 2.48 \text{ mA}$$

Therefore, $I_E = I_B + I_C = 20 \times 10^{-6} + 2.48 \times 10^{-3} = 2.5 \text{ mA}$

Example 3.16 If $\alpha_{dc} = 0.99$ and $I_{CBO} = 50 \mu A$, find emitter current.

Solution Given $\alpha_{dc} = 0.99$ and $I_{CBO} = 50 \mu A$,

Assume that $I_B = 1 \text{ mA}$

$$\begin{aligned} I_C &= \frac{\alpha_{dc} I_B}{1-\alpha_{dc}} + \frac{I_{CBO}}{1-\alpha_{dc}} = \frac{0.99(1 \times 10^{-3})}{1-0.99} + \frac{50 \times 10^{-6}}{1-0.99} \\ &= \frac{0.99 \times 10^{-3}}{0.01} + \frac{50 \times 10^{-6}}{0.01} = 99 \text{ mA} + 5 \text{ mA} = 104 \text{ mA} \end{aligned}$$

$$I_E = I_C + I_B = 104 \text{ mA} + 1 \text{ mA} = 105 \text{ mA}$$

Example 3.17 A germanium transistor used in a complementary symmetry amplifier has $I_{CBO} = 10 \mu A$ at $27^\circ C$ and $h_{FE} = 50$. (a) Find I_C when $I_B = 0.25 \text{ mA}$ and (b) assuming h_{FE} does not increase with temperature, find the value of new collector current, if the transistor's temperature rises to $50^\circ C$.

Solution Given $I_{CBO} = 10 \mu\text{A}$ and $h_{FE} (= \beta) = 50$

(a) To find the value of collector current when $I_B = 0.25 \text{ mA}$

$$\begin{aligned} I_C &= \beta I_B + (1 + \beta) I_{CBO} \\ &= 50 \times (0.25 \times 10^{-3}) + (1 + 50) \times (10 \times 10^{-6}) \text{ A} \\ &= 13.01 \text{ mA} \end{aligned}$$

(b) To find the value of new collector current if temperature rises to 50°C

We know that I_{CBO} doubles for every 10°C rise in temperature. Therefore,

$$\begin{aligned} I'_{CBO} (\beta = 50) &= I_{CBO} \times 2^{(T_2 - T_1)/10} = 10 \times 2^{(50 - 27)/10} \mu\text{A} \\ &= 10 \times 2^{2.3} \mu\text{A} = 49.2 \mu\text{A} \end{aligned}$$

Therefore, the collector current at 50°C is

$$\begin{aligned} I_C &= \beta I_B + (1 + \beta) I'_{CBO} \\ &= 50 \times (0.25 \times 10^{-3}) + (1 + 50) \times 49.2 \times 10^{-6} \text{ A} \\ &= 15.01 \text{ mA} \end{aligned}$$

Example 3.18 When the emitter current of a transistor is changed by 1 mA , there is a change in collector current by 0.99 mA . Find the current gain of the transistor.

Solution The current gain of the transistor is

$$\alpha = \frac{\Delta I_C}{\Delta I_E} = \frac{0.99 \times 10^{-3}}{1 \times 10^{-3}} = 0.99$$

Example 3.19 The dc current gain of a transistor in CE mode is 100. Determine its dc current gain in CB mode.

Solution The dc current gain of the transistor in CB mode is

$$\alpha_{dc} = \frac{\beta_{dc}}{1 + \beta_{dc}} = \frac{100}{1 + 100} = 0.99$$

Example 3.20 When I_E of a transistor is changed by 1 mA , its I_C changes by 0.995 mA . Find its common base current gain α , and common-emitter current gain β .

Solution Common-base current gain is

$$\alpha = \frac{\beta_{dc}}{1 + \beta_{dc}} = \frac{100}{1 + 100} = 0.995$$

Common-emitter current gain is

$$\beta = \frac{\alpha}{1 - \alpha} = \frac{0.995}{1 - 0.995} = 199$$

Example 3.21 The current gain of a transistor in CE mode is 49. Calculate its common-base current gain. Find the base current when the emitter current is 3 mA. [JNTU Jan 2010]

Solution Given $\beta = 49$

We know that $\alpha = \frac{\beta}{1 + \beta}$

Therefore, the common base current gain is

$$\alpha = \frac{49}{1 + 49} = 0.98$$

We also know that

$$\alpha = \frac{I_C}{I_E}$$

Therefore,

$$I_C = \alpha I_E = 0.98 \times 3 \times 10^{-3} = 2.94 \text{ mA}$$

Example 3.22 Determine I_C , I_E and α for a transistor circuit having $I_B = 15 \mu\text{A}$ and $\beta = 150$.

Solution The collector current,

$$I_C = \beta I_B = 150 \times 15 \times 10^{-6} = 2.25 \text{ mA}$$

The emitter current,

$$I_E = I_C + I_B = 2.25 \times 10^{-3} + 15 \times 10^{-6} = 2.265 \text{ mA}$$

Common-base current gain,

$$\alpha = \frac{\beta}{1 + \beta} = \frac{150}{151} = 0.9934$$

Example 3.23 Determine the base, collector and emitter currents and V_{CE} for a CE circuit shown in Fig. 3.19. For $V_{CC} = 10 \text{ V}$, $V_{BB} = 4 \text{ V}$, $R_B = 200 \text{ k}\Omega$, $R_C = 2 \text{ k}\Omega$, $V_{BE} (\text{on}) = 0.7 \text{ V}$, $\beta = 200$.

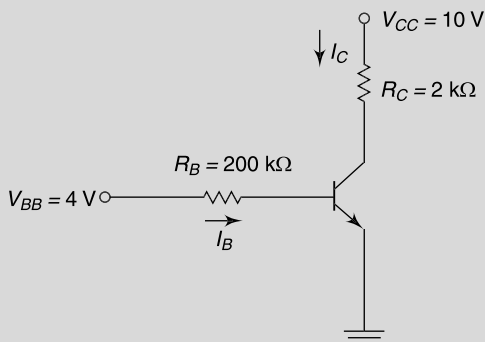


Fig. 3.19

Solution Referring to Fig. 3.19, the base current is

$$I_B = \frac{V_{BB} - V_{BE} (\text{on})}{R_B} = \frac{4 - 0.7}{200 \times 10^3} = 16.5 \mu\text{A}$$

The collector current is

$$I_C = \beta I_B = 200 \times 16.5 \times 10^{-6} = 3.3 \text{ mA}$$

The emitter current is

$$I_E = I_C + I_B = 3.3 \times 10^{-3} + 16.5 \times 10^{-6} = 3.3165 \text{ mA}$$

Therefore,

$$V_{CE} = V_{CC} - I_C R_C = 10 - 3.3 \times 10^{-3} \times 2 \times 10^3 = 3.4 \text{ V}$$

Example 3.24 For the circuit shown in Fig. 3.20, determine I_E , V_C and V_{CE} . Assume $V_{BE} = 0.7 \text{ V}$. [JNTU April/May 2007]

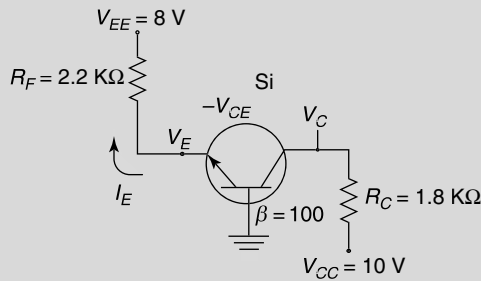


Fig. 3.20

Solution Given $V_{EE} = 8 \text{ V}$, $V_{CC} = 10 \text{ V}$, $V_{BE} = 0.7 \text{ V}$, $\beta = 100$, $R_E = 2.2 \text{ k}\Omega$ and $R_C = 1.8 \text{ k}\Omega$

To find I_E

From the circuit,

$$-8 = I_E R_E + 0.7$$

$$-8.7 = I_E R_E$$

$$I_E = \frac{-8.7}{2.2 \times 10^3} = -3.955 \text{ mA}$$

To find V_C

$$I_B = \frac{I_C}{\beta}$$

$$I_E = I_C + I_B = I_C + \frac{I_C}{\beta} = I_C \left(1 + \frac{1}{\beta} \right)$$

$$\text{Therefore, } -3.955 \text{ mA} = I_C \left(\frac{101}{100} \right)$$

Therefore, $I_C = -3.9154 \text{ mA}$

$$\begin{aligned} V_C &= 10 - I_C R_C \\ &= 10 - 3.9154 \times 10^{-3} \times 1.8 \times 10^3 = 2.952 \text{ V} \end{aligned}$$

To find V_{CE}

From the circuit,

$$\begin{aligned} 10 - I_C R_C - V_{CE} - I_E(2.2 \times 10^3) + 8 &= 0 \\ 10 - 7.048 - V_{CE} - 8.7 + 8 &= 0 \\ V_{CE} &= 2.252 \text{ V} \end{aligned}$$

Example 3.25 The reverse leakage current of the transistor when connected in CB configuration is $0.2 \mu\text{A}$ and it is $18 \mu\text{A}$ when the same transistor is connected in CE configuration. Calculate α_{dc} and β_{dc} of the transistor.

[JNTU May 2003]

Solution The leakage current $I_{CBO} = 0.2 \mu\text{A}$

$$I_{CEO} = 18 \mu\text{A}$$

Assume that

$$I_B = 30 \mu\text{A}$$

$$I_E = I_B + I_C$$

$$I_C = I_E - I_B = \beta I_B + (1 + \beta) I_{CBO}$$

We know that

$$I_{CEO} = \frac{I_{CBO}}{1 - \alpha} = (1 + \beta) I_{CBO}$$

$$\beta = \frac{I_{CEO}}{I_{CBO}} - 1 = \frac{18}{0.2} - 1 = 89$$

$$\begin{aligned} I_C &= \beta I_B + (1 + \beta) I_{CBO} \\ &= 89 (30 \times 10^{-6}) + (1 + 89) (0.2 \times 10^{-6}) \\ &= 2688 \times 10^{-6} = 2.668 \times 10^{-3} \text{ A} \end{aligned}$$

$$\begin{aligned} \alpha_{dc} &= 1 - \frac{I_{CBO}}{I_{CEO}} \\ &= 1 - \frac{0.2 \times 10^{-6}}{18 \times 10^{-6}} = 0.988 \end{aligned}$$

$$\begin{aligned} \beta_{dc} &= \frac{I_C - I_{CBO}}{I_B - (-I_{CBO})} \\ &= \frac{2.688 \times 10^{-3} - 0.2 \times 10^{-6}}{30 \times 10^{-6} + 0.2 \times 10^{-6}} = 89 \end{aligned}$$

Example 3.26 For the CE amplifier circuit shown below, find the percentage change in collector current if the transistor with $h_{fe} = 50$ is replaced by another transistor with $h_{fe} = 150$. Assume $V_{BE} = 0.6 \text{ V}$ (Fig. 3.21).

[JNTU May/June 2006]

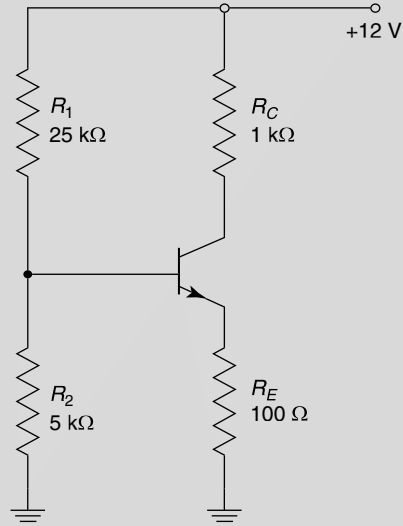


Fig. 3.21

Solution
$$V_B = \frac{R_2}{R_1 + R_2} \times V_{CC} = \frac{5 \times 10^3}{5 \times 10^3 + 25 \times 10^3} \times 12 = 2 \text{ V}$$

$$V_E = V_B - V_{BE} = 2 - 0.6 = 1.4 \text{ V}$$

Here,
$$I_E = \frac{V_E}{R_E} = \frac{1.4}{100} = 14 \text{ mA}$$

For $\beta = 50$,
$$I_B = \frac{I_E}{1 + \beta} = \frac{14 \times 10^{-3}}{51} = 274.5 \text{ } \mu\text{A}$$

Therefore, $I_{C1} = \beta I_B = 50 \times 274.5 \times 10^{-6} = 13.725 \text{ mA}$

For $\beta = 150$,
$$I_B = \frac{I_E}{1 + \beta} = \frac{14 \times 10^{-3}}{151} = 92.715 \text{ } \mu\text{A}$$

Therefore, $I_{C2} = \beta I_B = 150 \times 92.715 \times 10^{-6} = 13.907 \text{ mA}$

Hence, the percentage change in collector current is calculated as

$$\frac{I_{C2} - I_{C1}}{I_{C1}} \times 100 = \frac{13.907 \times 10^{-3} - 13.725 \times 10^{-3}}{13.725 \times 10^{-3}} \times 100 = 1.326\%$$

Example 3.27 Given an NPN transistor for which $\alpha = 0.98$, $I_{CO} = 2 \text{ } \mu\text{A}$ and $I_{CEO} = 16 \text{ } \mu\text{A}$. A common emitter connection is used and $V_{CC} = 12 \text{ V}$ and $R_L = 4 \text{ k}\Omega$. What is the minimum base current required in order that transistor enter into saturation region.

[JNTU Nov/Dec 2005]

Solution Given $\alpha = 0.98$, $I_{CO} = 2 \mu\text{A}$, $I_{CEO} = 1.6 \mu\text{A}$, $V_{CC} = 12 \text{ V}$ and $R_L = 4 \text{ k}\Omega$

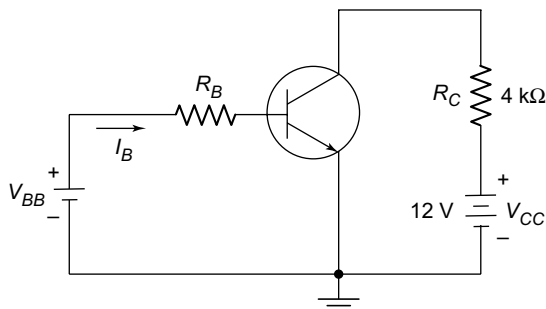


Fig. 3.22

$$I_{C(\text{sat})} = \frac{V_{CC}}{R_L} = \frac{12}{4 \times 10^3} = 3 \text{ mA}$$

$$\beta = \frac{\alpha}{1 - \alpha} = \frac{0.98}{1 - 0.98} = 49$$

$$I_{B(\text{min})} = \frac{I_{C(\text{sat})}}{\beta} = \frac{3 \times 10^{-3}}{49} = 61.224 \times 10^{-6} = 61.224 \mu\text{A}$$

Example 3.28 A transistor operating in CB configuration has $I_C = 2.98 \text{ mA}$, $I_E = 3 \text{ mA}$ and $I_{CO} = 0.01 \text{ mA}$. What current will flow in the collector circuit of this transistor when connected in CE configuration with a base current of $30 \mu\text{A}$? [JNTU May/June 2008]

Solution Given $I_C = 2.98 \text{ mA}$, $I_E = 3 \text{ mA}$, $I_{CO} = 0.01 \text{ mA}$ and $I_B = 30 \mu\text{A}$.
For CB configuration,

$$I_C = \alpha I_E + I_{CO}$$

$$\text{Therefore, } \alpha = \frac{I_C - I_{CO}}{I_E} = \frac{(2.98 - 0.01) \times 10^{-3}}{3 \times 10^{-3}} = 0.99$$

$$\beta = \frac{\alpha}{1 - \alpha} = \frac{0.99}{1 - 0.99} = 99$$

For CE configuration,

$$I_C = \beta I_B + (1 + \beta) I_{CO}$$

$$= 99 \times 30 \times 10^{-6} + (1 + 99) \times 0.01 \times 10^{-3} = 3.97 \text{ mA}$$

Example 3.29 The reverse saturation current in a transistor is $8 \mu\text{A}$. If the transistor common base current gain is 0.979, calculate the collector and emitter currents for $40 \mu\text{A}$ base current. [JNTU May/June 2008]

Solution

Given $I_{CO} = 8 \mu\text{A}$, $\alpha = 0.979$ and $I_E = 40 \mu\text{A}$

$$I_C = \alpha I_E + I_{CO} = 0.979 \times 40 \times 10^{-6} + 8 \times 10^{-6} = 47.16 \mu\text{A}$$

3.9 TWO-PORT DEVICES AND NETWORK PARAMETERS

A transistor can be treated as a two-port network. The terminal behaviour of any two port network can be specified by the terminal voltages v_1 and v_2 at ports 1 and 2, respectively, and currents i_1 and i_2 , entering ports 1 and 2, respectively, as shown in Fig. 3.23. Of these four variables v_1 , v_2 , i_1 and i_2 , two can be selected as independent variables and the remaining two can be expressed in terms of these independent variables. This leads to various two-port parameters out of which the following three are more important.

- (i) Z-parameters or Impedance parameters
- (ii) Y-parameters or Admittance parameters
- (iii) h-parameters or Hybrid parameters.

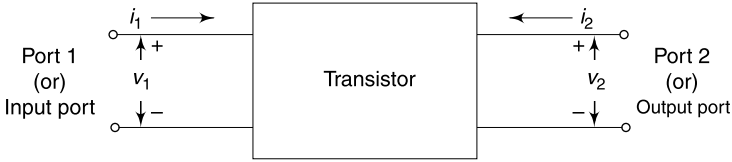


Fig. 3.23 Two Port Network

3.9.1 Z-Parameters or Impedance Parameters

Here, i_1 and i_2 are taken as independent variables. The voltages v_1 and v_2 are given by the equations

$$v_1 = Z_{11}i_1 + Z_{12}i_2 \quad (3.29)$$

$$v_2 = Z_{21}i_1 + Z_{22}i_2 \quad (3.30)$$

These four impedance parameters, Z_{11} , Z_{22} , Z_{12} and Z_{21} are defined as follows:

$$Z_{11} = \left[\frac{v_1}{i_1} \right] \text{ with } i_2 = 0$$

= input impedance with output port open circuited

$$Z_{22} = \left[\frac{v_2}{i_2} \right] \text{ with } i_1 = 0$$

= output impedance with input port open circuited

$$Z_{12} = \left[\frac{v_1}{i_2} \right] \text{ with } i_1 = 0$$

= reverse transfer impedance with port 1 open circuited

$$Z_{21} = \left[\frac{v_2}{i_1} \right] \text{ with } i_2 = 0$$

= forward transfer impedance with port 2 open circuited.

3.9.2 Y-Parameters or Admittance Parameters

Here, v_1 and v_2 are taken as independent variables. The currents i_1 and i_2 are given by the equations

$$i_1 = y_{11}v_1 + y_{12}v_2 \quad (3.31)$$

$$i_2 = y_{21}v_1 + y_{22}v_2 \quad (3.32)$$

y_{11} , y_{12} , y_{21} and y_{22} represent short-circuit admittance parameters or simply admittance parameters or y -parameters. They are defined as follows:

$$y_{11} = \left[\frac{i_1}{v_1} \right] \text{ with } v_2 = 0$$

= input admittance with port 2 short circuited

$$y_{22} = \left[\frac{i_2}{v_2} \right] \text{ with } v_1 = 0$$

= output admittance with port 1 short circuited

$$y_{12} = \left[\frac{i_1}{v_2} \right] \text{ with } v_1 = 0$$

= reverse transfer admittance with port 1 short circuited

$$y_{21} = \left[\frac{i_2}{v_1} \right] \text{ with } v_2 = 0$$

= forward transfer admittance with port 2 short circuited.

3.9.3 Hybrid Parameters or h -Parameters

If the input current i_1 and the output voltage v_2 are taken as independent variables, the input voltage v_1 and output current i_2 can be written as

$$v_1 = h_{11}i_1 + h_{12}v_2 \quad (3.33)$$

$$i_2 = h_{21}i_1 + h_{22}v_2 \quad (3.34)$$

The four hybrid parameters h_{11} , h_{12} , h_{21} and h_{22} are defined as follows:

$$h_{11} = \left[\frac{v_1}{i_1} \right] \text{ with } v_2 = 0$$

= input impedance with output port short circuited

$$h_{22} = \left[\frac{i_1}{v_2} \right] \text{ with } i_1 = 0$$

= output admittance with input port open circuited

$$h_{12} = \left[\frac{v_1}{v_2} \right] \text{ with } i_1 = 0$$

= reverse voltage transfer ratio with input port open circuited

$$h_{21} = \left[\frac{i_2}{i_1} \right] \text{ with } v_2 = 0$$

= forward current gain with output port short circuited.

The dimensions of h -parameters are as follows:

$$h_{11} - \Omega$$

$$h_{22} - \text{mhos}$$

$$h_{12}, h_{21} - \text{dimensionless}$$

As the dimensions are not alike, i.e. they are hybrid in nature, these parameters are called as hybrid parameters.

An alternative subscript notation recommended by IEEE is commonly used:

$$i = 11 = \text{input}; o = 22 = \text{output}$$

$$f = 21 = \text{forward transfer}; r = 12 = \text{reverse transfer}$$

3.9.4 Notations Used in Transistor Circuits

When h -parameters are applied to transistors, it is a common practice to add a second subscript to designate the type of configuration considered — e for common emitter, b for common base and c for common collector. Thus, for a common emitter (CE) configuration,

$$h_{ie} = h_{11e} = \text{short circuit input impedance}$$

$$h_{oe} = h_{22e} = \text{open circuit output admittance}$$

$$h_{re} = h_{12e} = \text{open circuit reverse voltage transfer ratio}$$

$$h_{fe} = h_{21e} = \text{short circuit forward current gain}$$

3.10 THE HYBRID MODEL FOR TWO-PORT NETWORK

Based on the definition of hybrid parameters the mathematical model for two-port networks known as h -parameter model can be developed. Equations (3.33) and (3.34) can be written as

$$v_1 = h_i i_1 + h_r v_2 \quad (3.35)$$

$$i_2 = h_f i_1 + h_o v_2 \quad (3.36)$$

The proposed model shown in Fig. 3.24 should satisfy these two equations and it can be readily verified by writing Kirchhoff's voltage law equation in the input loop and Kirchhoff's current law equation for the output node. It is to be noted that the input circuit have a dependent voltage generator and the output circuit contains a dependent current generator.

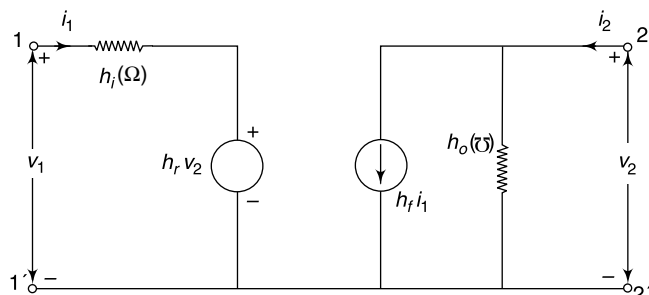


Fig. 3.24 Hybrid Model for a Two-port Network

3.10.1 Determination of h -parameters from Transistor Characteristics

On extending the hybrid model for two-port network to a transistor it is assumed that the signal excursion about the Q point is small so that the transistor parameters may be considered constant over the signal excursion. Use of h -parameters to describe a transistor have the following advantages:

- (i) h -parameters are real numbers upto radio frequencies
- (ii) they are easy to measure
- (iii) they can be determined from the transistor static characteristics curves
- (iv) they are convenient to use in circuit analysis and design
- (v) easily convertible from one configuration to other
- (vi) readily supplied by manufacturers.

In order to derive a hybrid model for transistor consider the CE circuit of Fig. 3.25. The variables are i_B , i_C , v_B ($= v_{BE}$) and v_C ($= v_{CE}$). i_B and v_C are considered as independent variables.

$$\text{Then, } v_B = f_1(i_B, v_C) \quad (3.37)$$

$$i_C = f_2(i_B, v_C) \quad (3.38)$$

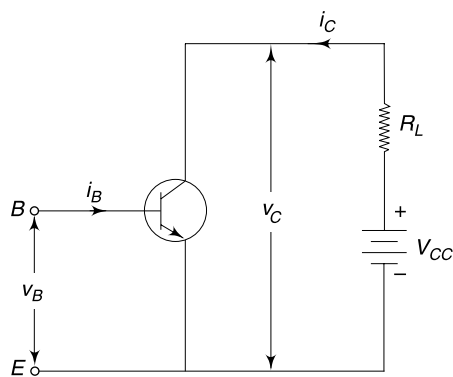


Fig. 3.25 CE Transistor Circuit

Making a Taylor's series expansion around the quiescent point I_B , V_C and neglecting higher order terms, the following two equations are obtained:

$$\Delta v_B = \left(\frac{\partial f_1}{\partial i_B} \right)_{V_C} \Delta i_B + \left(\frac{\partial f_1}{\partial v_C} \right)_{i_B} \Delta v_C \quad (3.39)$$

$$\Delta i_C = \left(\frac{\partial f_2}{\partial i_B} \right)_{V_C} \Delta i_B + \left(\frac{\partial f_2}{\partial v_C} \right)_{I_B} \Delta v_C \quad (3.40)$$

The partial derivatives are taken keeping the collector voltage or base current constant as indicated by the subscript attached to the derivative. Δv_B , Δv_C , Δi_B and Δi_C represent the small-signal (incremental) base and collector voltages and currents. They are represented by symbols v_b , v_c , i_b and i_c , respectively. Hence, Eqs (3.39) and (3.40) may be written as

$$v_b = h_{ie} i_b + h_{re} v_c$$

$$i_c = h_{fe} i_b + h_{oe} v_c$$

where

$$h_{ie} = \left(\frac{\partial f_1}{\partial i_B} \right)_{V_C} = \left(\frac{\partial v_B}{\partial i_B} \right)_{V_C} \approx \left(\frac{\Delta v_B}{\Delta i_B} \right)_{V_C} = \left(\frac{v_b}{i_b} \right)_{V_C} \quad (3.41)$$

$$h_{re} = \left(\frac{\partial f_1}{\partial v_C} \right)_{I_B} = \left(\frac{\partial v_B}{\partial v_C} \right)_{I_B} \approx \left(\frac{\Delta v_B}{\Delta v_C} \right)_{I_B} = \left(\frac{v_b}{v_c} \right)_{I_B} \quad (3.42)$$

$$h_{fe} = \left(\frac{\partial f_2}{\partial i_B} \right)_{V_C} = \left(\frac{\partial i_C}{\partial i_B} \right)_{V_C} \approx \left(\frac{\Delta i_C}{\Delta i_B} \right)_{V_C} = \left(\frac{i_c}{i_b} \right)_{V_C} \quad (3.43)$$

$$h_{oe} = \left(\frac{\partial f_2}{\partial v_C} \right)_{I_B} = \left(\frac{\partial i_C}{\partial v_C} \right)_{I_B} \approx \left(\frac{\Delta i_C}{\Delta v_C} \right)_{I_B} = \left(\frac{i_c}{v_c} \right)_{I_B} \quad (3.44)$$

The above equations define the h -parameters of the transistor in CE configuration. The same theory can extended to transistors in other configurations.

The hybrid models and equations given in Table 3.2 are valid for NPN as well as PNP transistors and hold good for all types of loads and methods of biasing Table 3.3 gives the typical h -parameter values for a transistor and Table 3.4 gives the conversion formulae to find the h -parameters for CC and CB configurations given the h -parameters for CE configuration.

Table 3.2 Hybrid Model for the Transistor in Three Different Configurations

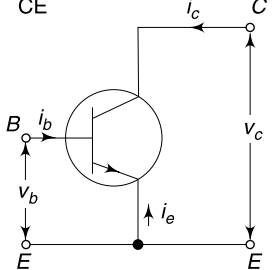
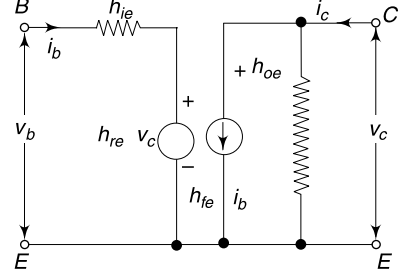
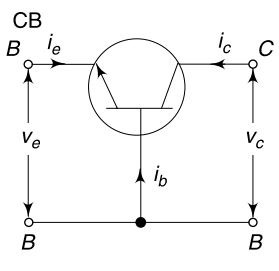
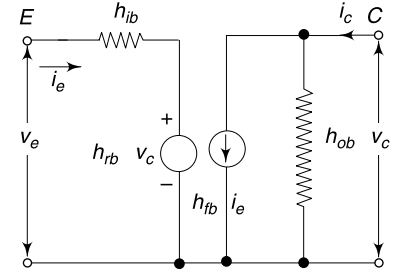
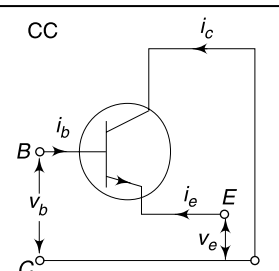
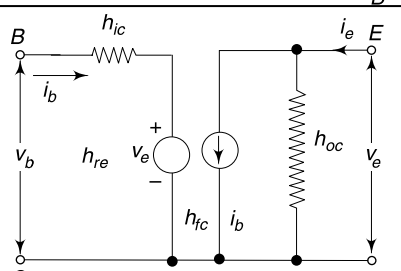
CE 		$v_b = h_{ie} i_b + h_{re} v_c$ $i_c = h_{fe} i_b + h_{oe} v_c$
CB 		$v_e = h_{ib} i_e + h_{rb} v_c$ $i_e = h_{fb} i_e + h_{ob} v_c$
CC 		$v_b = h_{ic} i_b + h_{rc} v_e$ $i_e = h_{fc} i_b + h_{oc} v_e$

Table 3.3 Typical *h*-Parameter Values for a Transistor

Parameter	CE	CC	CB
h_i	1,100 Ω	1,100 Ω	22 Ω
h_r	2.5×10^{-4}	1	3×10^{-4}
h_f	50	-51	-0.98
h_o	25 $\mu A/V$	25 $\mu A/V$	0.49 $\mu A/V$

Table 3.4 Conversion Formulae for Hybrid Parameters

CC	CB
$h_{ic} = h_{ie}$	$h_{ib} = \frac{h_{ie}}{1 + h_{fe}}$
$h_{rc} = 1$	$h_{rb} = \frac{h_{ie} h_{oe}}{1 + h_{fe}} - h_{re}$
$h_{fc} = -(1 + h_{fe})$	$h_{fb} = \frac{-h_{fe}}{1 + h_{fe}}$
$h_{oc} = h_{oe}$	$h_{ob} = \frac{h_{oe}}{1 + h_{fe}}$

3.11 COMPARISON OF CB, CE AND CC TRANSISTOR AMPLIFIER CONFIGURATIONS

The characteristics of three configurations are summarised in Table 3.5. Here the quantities A_i , A_v , R_i , R_o and A_p are calculated for a typical transistor whose h -parameters are given in Table 3.3. The values of R_L and R_S are taken as 3 k Ω .

Table 3.5 Performance Schedule of Three Transistor Configurations

Quantity	CB	CC	CE
A_i	0.98	47.5	-46.5
A_v	131	0.989	-131
A_p	128.38	46.98	6091.5
R_i	22.6 Ω	144 k Ω	1065 Ω
R_o	1.72 M Ω	80.5 Ω	45.5 k Ω

The values of current gain, voltage gain, input impedance and output impedance calculated as a function of load and source impedances can be shown graphically as in Fig. 3.26.

From Table 3.5 and Fig. 3.26 the performance of the CB, CC and CE amplifiers can be summarised as follows:

Characteristics of Common Base Amplifier

- Current gain is less than unity and its magnitude decreases with the increase of load resistance R_L ,
- Voltage gain A_v is high for normal values of R_L ,
- The input resistance R_i is the lowest of all the three configurations, and
- The output resistance R_o is the highest of all the three configurations.

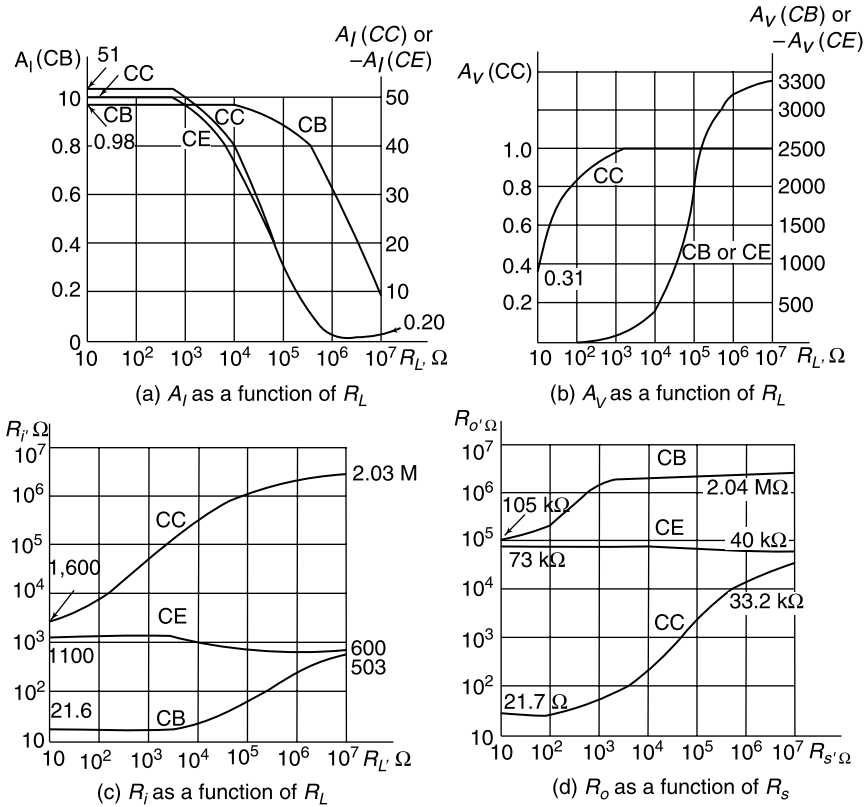


Fig. 3.26 Comparison of Transistor Amplifier Configurations (a) Current Gain as a Function of R_L , (b) Voltage Gain as a Function of R_L , (c) Input Impedance as a Function of R_L , (d) Output Impedance as a Function of R_s

Applications The CB amplifier is not commonly used for amplification purpose. It is used for

- (i) matching a very low impedance source
- (ii) as a non-inverting amplifier with voltage gain exceeding unity
- (iii) for driving a high impedance load
- (iv) as a constant current source

Characteristics of Common Collector Amplifier

- (i) For low value of R_L ($< 10 \text{ k}\Omega$), the current gain A_I is high and almost equal to that of a CE amplifier,
- (ii) The voltage gain A_V is less than unity,
- (iii) The input resistance is the highest of all the three configurations, and
- (iv) The output resistance is the lowest of all the three configurations.

Applications The CC amplifier is widely used as a buffer stage between a high impedance source and a low impedance load. The CC amplifier is called the emitter follower.

Characteristics of Common Emitter Amplifier

- (i) The current gain A_I is high for $R_I < 10 \text{ k}\Omega$,
- (ii) The voltage gain is high for normal values of load resistance R_L ,
- (iii) The input resistance R_i is medium, and
- (iv) The output resistance R_o is moderately high.

Applications Of the three configurations CE amplifier alone is capable of providing both voltage gain and current gain. Further the input resistance R_i and the output resistance R_o are moderately high. Hence CE amplifier is widely used for amplification purpose.

3.12 UJT (UNIJUNCTION TRANSISTOR)

UJT is a three terminal semiconductor switching device. As it has only one PN junction and three leads, it is commonly called as Unijunction transistor.

The basic structure of UJT is shown in Fig. 3.27(a). It consists of a lightly doped N-type Silicon bar with a heavily doped P-type material alloyed to its one side closer to B_2 for producing single PN junction. The circuit symbol of UJT is shown in Fig. 3.27(b). Here the emitter leg is drawn at an angle to the vertical and the arrow indicates the direction of the conventional current.

Characteristics of UJT Referring to Fig. 3.27(c), the interbase resistance between B_2 and B_1 of the silicon bar is $R_{BB} = R_{B1} + R_{B2}$. With emitter terminal open, if voltage V_{BB} is applied between the two bases, a voltage gradient is established along the N-type bar. The voltage drop across R_{B1} is given by $V_1 = \eta V_{BB}$, where the *intrinsic stand-off ratio* $\eta = R_{B1}/(R_{B1} + R_{B2})$. The typical value of η ranges from 0.56 to 0.75. This voltage V_1 reverse biases the PN junction and emitter current is cut-off. But a small leakage current flows from B_2 to emitter due to minority carriers. If a positive voltage V_E is applied to the emitter, the

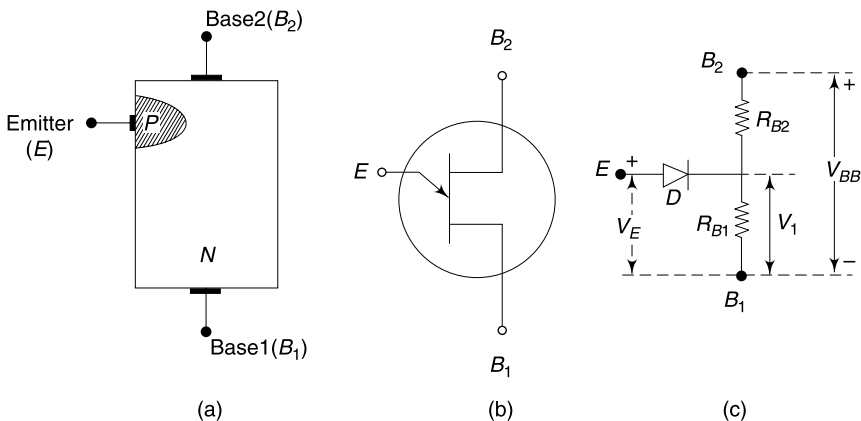


Fig. 3.27 UJT (a) Basic Structure, (b) Circuit Symbol, and (c) Equivalent Circuit

PN junction will remain reverse biased so long as V_E is less than V_1 . If V_E exceeds V_1 by the cutin voltage V_γ , the diode becomes forward biased. Under this condition, holes are injected into N-type bar. These holes are repelled by the terminal B_2 and are attracted by the terminal B_1 . Accumulation of holes in E to B_1 region reduces the resistance in this section and hence emitter current I_E is increased and is limited by V_E . The device is now in the 'ON' state.

If a negative voltage is applied to the emitter, PN junction remains reverse biased and the emitter current is cut off. The device is now in the 'OFF' state.

Figure 3.28 shows a family of input characteristics of UJT. Here, up to the peak point P , the diode is reverse biased and hence, the region to the left of the peak point is called *cut-off region*. The UJT has a stable firing voltage V_P which depends linearly on V_{BB} and a small firing current I_P ($\approx 25\mu A$). At P , the peak voltage $V_P = \eta V_{BB} + V_\gamma$, the diode starts conducting and holes are injected into N-layer. Hence, resistance decreases thereby decreasing V_E for the increase in I_E . So, there is a *negative resistance region* from peak point P to valley point V . After the valley point, the device is driven into saturation and behaves like a conventional forward biased PN junction diode. The region to the right of the valley point is called saturation region. In the valley point, the resistance changes from negative to positive. The resistance remains positive in the saturation region. For very large I_E , the characteristic asymptotically approaches the curve for $I_{B2} = 0$.

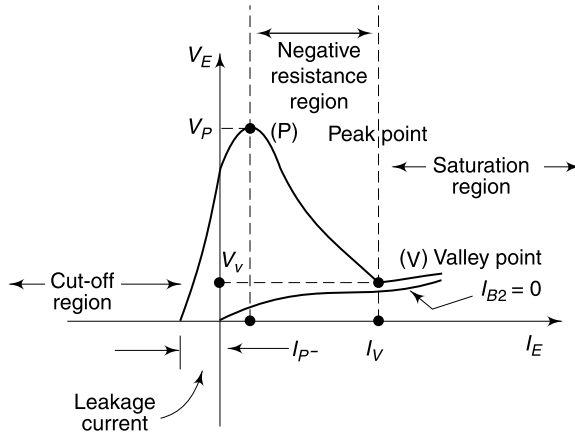


Fig. 3.28 Input Characteristics of UJT

A unique characteristic of UJT is, when it is triggered, the emitter current increases regeneratively until it is limited by emitter power supply. Due to this negative resistance property, a UJT can be employed in a variety of applications, viz. sawtooth wave generator, pulse generator, switching, timing and phase control circuits.



Example 3.30 If $\eta = 0.8$, $V_{BB} = 15\text{ V}$, and $V_D = 0.7\text{ V}$, find the value of V_P .

Solution We know that $V_P = \eta V_{BB}$

Therefore, $V_P = 0.8 \times 15 = 12\text{ V}$

UJT Relaxation Oscillator The relaxation oscillator using UJT which is meant for generating sawtooth waveform is shown in Fig. 3.29. It consists of a UJT and a capacitor C_E which is charged through R_E as the supply voltage V_{BB} is switched ON.

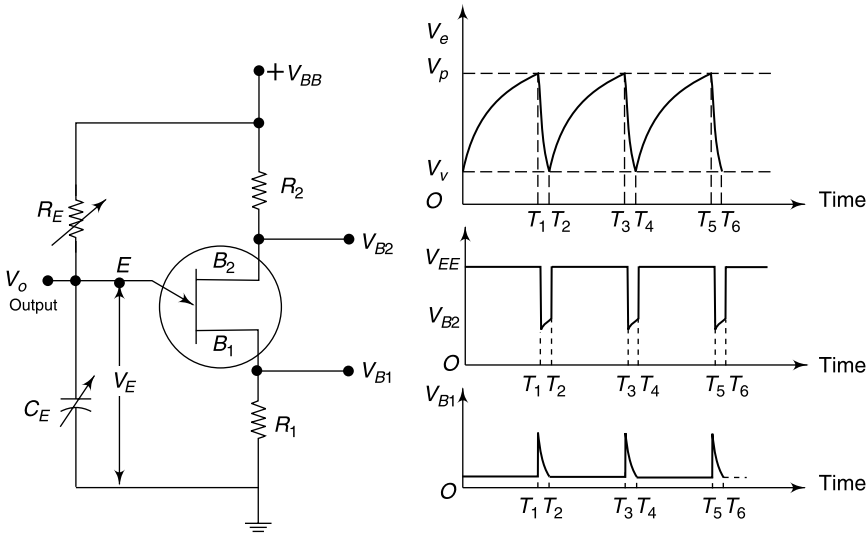


Fig. 3.29 UJT Relaxation Oscillator

The voltage across the capacitor increases exponentially and when the capacitor voltage reaches the peak point voltage V_P , the UJT starts conducting and the capacitor voltage is discharged rapidly through EB_1 and R_1 . After the peak point voltage of UJT is reached, it provides negative resistance to the discharge path which is useful in the working of the relaxation oscillator. As the capacitor voltage reaches zero, the device then cuts off and capacitor C_E starts to charge again. This cycle is repeated continuously generating a *sawtooth waveform* across C_E .

The inclusion of external resistors R_2 and R_1 in series with B_2 and B_1 provides spike waveforms. When the UJT fires, the sudden surge of current through B_1 causes drop across R_1 , which provides positive going spikes. Also, at the time of firing, fall of V_{EB1} causes I_2 to increase rapidly which generates negative going spikes across R_2 .

By changing the values of capacitance C_E or resistance R_E , frequency of the output waveform can be changed as desired, since these values control the time constant $R_E C_E$ of the capacitor charging circuit.

Frequency of Oscillation The time period and hence the frequency of the sawtooth wave can be calculated as follows. Assuming that the capacitor is initially uncharged, the voltage V_C across the capacitor prior to breakdown is given by

$$V_c = V_{BB} (1 - e^{-t/R_E C_E})$$

where $R_E C_E$ = charging time constant of resistor-capacitor circuit, and t = time from the commencement of the waveform.

The discharge of the capacitor occurs when V_C is equal to the peak-point voltage V_P , i.e.

$$V_P = \eta V_{BB} = V_{BB} (1 - e^{-t/R_E C_E})$$

$$\eta = 1 - e^{-t/R_E C_E}$$

$$e^{-t/R_E C_E} = (1 - \eta)$$

Therefore,

$$\begin{aligned} t &= R_E C_E \log_e \frac{1}{(1 - \eta)} \\ &= 2.303 R_E C_E \log_{10} \frac{1}{(1 - \eta)} \end{aligned}$$

If the discharge time of the capacitor is neglected, then $t = T$, the period of the wave.

Therefore, frequency of oscillation of sawtooth wave,

$$f = \frac{1}{T} = \frac{1}{2.303 R_E C_E \log_{10} \frac{1}{(1 - \eta)}}$$

Example 3.31 A UJT has a firing potential of 20 V. It is connected across the capacitor of a series RC circuit with $R = 100 \text{ k}\Omega$ and $C = 1000 \text{ pF}$ supplied by a source of 40 V dc. Calculate the time period of the saw tooth waveform generated. [JNTU Dec 2005]

Solution Given $R_E = 100 \text{ k}\Omega$ and $C_E = 1000 \text{ pF}$

$$V_{BB} = 40 \text{ V}, V_P = 20 \text{ V}$$

$$V_P = V_{BB} \left(1 - e^{\frac{-t}{R_E C_E}} \right)$$

$$20 = 40 \left(1 - e^{\frac{-t}{R_E C_E}} \right)$$

$$e^{-t/R_E C_E} = \frac{1}{2}$$

$$\frac{-t}{R_E C_E} = \ln \left(\frac{1}{2} \right)$$

$$\frac{t}{R_E C_E} = \ln(2)$$

$$t = \ln(2) \times R_E C_E$$

$$\text{Therefore, } t = 0.693 \times 100 \times 10^3 \times 1000 \times 10^{-12} = 0.693 \times 10^{-4} = 69.3 \mu\text{s}$$

Example 3.32 Design a UJT relaxation oscillator to generate a sawtooth waveform at a frequency of 500 Hz. Assume the supply voltage $V_{BB} = 20 \text{ V}$, $V_P = 2.9 \text{ V}$, $V_V = 1.118 \text{ V}$, $I_P = 1.6 \text{ mA}$ and $I_V = 3.5 \text{ mA}$.

Solution We know that

$$f = \frac{1}{2.303 R_E C_E \log_{10} \frac{1}{(1-\eta)}}$$

We know that $\eta_{\min} = 0.56$

For determining R_E , we have

$$R_E < \frac{V_{BB} - V_P}{I_V}, \quad \text{i.e.} \quad R_E < \frac{20 - 2.9}{1.6 \times 10^{-3}} = 10.7 \text{ k}\Omega$$

$$R_E > \frac{V_{BB} - V_V}{I_V}, \quad \text{i.e.} \quad R_E > \frac{20 - 1.118}{3.5 \times 10^{-3}} = 5.36 \text{ k}\Omega$$

Therefore, R_E is selected as 10 k Ω .

$$\frac{1}{500} = 2.303 \times 10 \times 10^3 C_E \log_{10} \left(\frac{1}{(1-0.56)} \right)$$

$$\text{Therefore,} \quad C_E = \frac{1}{500 \times 2.303 \times 10^4 \times 0.36} = 0.24 \mu\text{F}$$

So, C_E is selected as 0.22 μF .

Let the required pulse voltage at $B_1 = 5 \text{ V}$

Let the peak pulse current, $I_E = 250 \text{ mA}$.

$$\text{Therefore,} \quad R_1 = \frac{V_{R1}}{I_E} = \frac{5}{250 \times 10^{-3}} = 20 \Omega$$

So, R_1 is selected to be 22 Ω .

We select the voltage characteristics for $V_{B1B2} = 4 \text{ V}$.

$$\text{Therefore,} \quad V_{R2} = 20 - (4 + 5) = 11 \text{ V}$$

$$R_2 = \frac{11}{250} \times 10^3 = 44 \Omega$$

So, R_2 is selected as 100 Ω .

3.13 SPECIFICATIONS OF BJT AND UJT

The specifications of some of the commonly used BJT-NPN small signal transistor, BJT-PNP small signal transistor and UJT are given in Tables A.6(a) and (b) and Table A.9 respectively in Appendix-A.

REVIEW QUESTIONS

1. What is a Bipolar junction transistor? How are its terminals named?
2. Explain the operation of NPN and PNP transistors.
3. What are the different configurations of BJT?
4. Explain the input and output characteristics of a transistor in CB configuration.
5. Explain the early effect and its consequences.
6. Derive the relationship between α and β .
7. Why does the CE configuration provide large current amplification while the CB configuration does not?
8. Draw the circuit diagram of an NPN junction transistor CE configuration and describe the static input and output characteristics. Also, define active, saturation and cutoff regions, and saturation resistance of a CE transistor.
9. How will you determine h -parameters from the characteristics of CE configuration?
10. Determine the h -parameters from the characteristics of CB configuration?
11. What is the relation between I_B , I_E and I_C in CB configuration?
12. Explain the laboratory setup for obtaining the CC characteristics.
13. Compare the performance of a transistor in different configurations.
14. Define α , β and γ of a transistor. Show how they are related to each other?
15. Describe the two types of breakdown in transistors.
16. In a transistor, 99.6% of the carriers injected into the base cross the collector-base junction. If the leakage current is $5\mu\text{A}$ and the collector current is 20 mA, calculate (a) α_{dc} and (b) I_E .
(Ans: (a) $\alpha_{dc} = 0.996$, (b) $I_E = 20.07\text{ mA}$)
17. Calculate the values of collector current and base current for a transistor with $\alpha_{dc} = 0.99$, $I_{CBO} = 10\mu\text{A}$ and $I_E = 8\text{ mA}$.
(Ans: $I_C = 7.93\text{ mA}$ and $I_B = 70\mu\text{A}$)
18. If $\beta = 100$, $I_{CBO} = 10\mu\text{A}$ and $I_B = 80\mu\text{A}$, find I_E . (Ans: $I_E = 9.09\text{ mA}$)
19. If $\alpha = 0.98$, $I_{CBO} = 10\mu\text{A}$ and $I_B = 100\mu\text{A}$, find I_E . (Ans: $I_E = 505\text{ mA}$)
20. (a) Find α if $\beta = 50$ (b) Find β if $\alpha = 0.995$.
(Ans: (a) $\alpha = 0.9804$ (b) $\beta = 199$)
21. If the base current in a transistor is $20\mu\text{A}$ and the emitter current is 6.4 mA, calculate α , β and I_C . (Ans: $\alpha = 0.9968$, $\beta = 319$ and $I_C = 6.38\text{ mA}$)
22. The reverse leakage current of the transistor is $0.1\mu\text{A}$ when connected in common-base configuration. If it is $16\mu\text{A}$ when the same transistor is connected in common-emitter configuration, calculate α and β .
(Ans: $\alpha = 0.993$, $\beta = 159$)
23. Calculate the base current if the emitter current is 8 mA and $\alpha = 0.99$?
(Ans: $I_B = 80\mu\text{A}$)
24. If the base current is $30\mu\text{A}$ and the emitter current is 7.2 mA, calculate α , β and I_C .
(Ans: $\alpha = 0.995$, $\beta = 239$, $I_C = 7.17\text{ mA}$)

25. If the emitter current is 10 mA which is 1.04 times larger than the collector current, find the base current. (Ans: $I_B = 384 \mu\text{A}$).
26. A transistor has $I_B = 18 \text{ mA}$ and $I_C = 1.6 \text{ mA}$. Find (a) α (b) β (c) I_E and (d) if I_B changes by +25 mA and I_C changes by +.05 mA, find new β .
27. A germanium transistor used in a complementary symmetry amplifier has $I_{CBO} = 10 \mu\text{A}$ at 27°C and $h_{FE} = 100$. Find (a) I_C when $I_B = 0.25 \text{ mA}$ and assuming h_{FE} does not increase with temperature, find the value of new collector current, if the transistor's temperature rises to 50°C .
28. When the emitter current of a transistor is changed by 1 mA, there is a change in collector current by 0.98 mA. Find the current gain of the transistor.
29. Derive the network parameters for two-port devices.
30. Why hybrid parameters are called so? Define them.
31. What are the salient features of hybrid parameters?
32. Derive the equations for voltage gain, current gain, input impedance and output admittance for a BJT using low frequency h -parameter model for (a) CE configuration (b) CB configuration and (c) CC configuration.
33. Compare the performance of a BJT as an amplifier in CE, CB & CC configurations.
34. Justify the validity of approximate hybrid model applicable in low frequency region.
35. Draw the equivalent circuit of UJT and explain its operation with the help of emitter characteristics.
36. Explain the $V-I$ characteristics of UJT.
37. Define intrinsic stand-off ratio of UJT.
38. Explain the terms (a) peak point voltage V_P , and (b) valley point voltage V_V of a UJT
39. Mention some of the applications of UJT.
40. Explain the difference between UJT and a conventional bipolar transistor.
41. Explain with the help of a circuit diagram the working of a UJT relaxation oscillator.
42. A silicon UJT has an interbase resistance $R_{BB} = 10 \text{ k}\Omega$ and $R_{B1} = 6 \text{ k}\Omega$ with $I_E = 0$. If $V_{BB} = 20 \text{ V}$ and $V_E < V_P$, find UJT current, η and V_P . [Ans. 2 mA, 0.6, 12.7 V]
43. An UJT has $R_{BB} = 10 \text{ k}\Omega$ and $R_{B2} = 3.5 \text{ k}\Omega$. Find its intrinsic stand-off ratio [Ans. 0.65]
44. Design an UJT relaxation oscillator to generate a sawtooth waveform at a frequency of 600 Hz. Assume the supply voltage $V_{BB} = 18 \text{ V}$, $V_P = 2.9 \text{ V}$ and $V_V = 1.118 \text{ V}$.
45. The base one of an UJT has resistance of $4.7 \text{ k}\Omega$ and the value of intrinsic stand-off ratio of the device is 0.58. If an inter-base voltage of 10 V is applied across the two base s, calculate the value of I_B . [Ans. 1.23 mA]

OBJECTIVE TYPE QUESTIONS

1. A collector collects
 - (a) electrons from the base in case PNP transistors
 - (b) electrons from the emitter in case of PNP transistors
 - (c) holes from the base in case of NPN transistors
 - (d) holes from the base in case of PNP transistors
2. In a PNP transistor with normal bias,
 - (a) the collector junction has negligible resistance
 - (b) only holes cross the collector junction
 - (c) the collector-base junction is reverse biased and the emitter base junction is forward biased
 - (d) only majority carriers cross the collector junction
3. A PNP transistor is made of
 - (a) silicon
 - (b) germanium
 - (c) either silicon or germanium
 - (d) none of the above
4. In most transistors, the collector region is made physically larger than the emitter region
 - (a) for dissipating heat
 - (b) to distinguish it from other regions
 - (c) as it is sensitive to ultraviolet rays
 - (d) to reduce resistance in the path of flow of electrons
5. The three terminals of a bipolar junction transistor are called
 - (a) PNP
 - (b) NPN
 - (c) anode, cathode and gate
 - (d) emitter, base and collector
6. For a NPN transistor, the N regions are
 - (a) emitter and base
 - (b) base and collector
 - (c) emitter and collector
 - (d) None
7. For operation of PNP amplifier, the base of the amplifier must be
 - (a) 0 V
 - (b) positive with respect to collector
 - (c) negative with respect to collector
 - (d) greater than the collector current
8. In a transistor, the region that is very lightly doped and very thin is the
 - (a) emitter
 - (b) base
 - (c) collector
 - (d) none of the above
9. In an NPN transistor, the emitter
 - (a) emits or injects holes into the collector
 - (b) emits or injects electrons into the collector
 - (c) emits or injects electrons into the base
 - (d) emits or injects holes into the base
10. In a PNP transistor with normal bias, the emitter junction
 - (a) is always reverse biased
 - (b) offers very high resistance
 - (c) offers a low resistance
 - (d) remains open

11. In a NPN transistor, when the emitter junction is forward biased and the collector junction is reverse biased, the transistor will operate in the
 - (a) active region
 - (b) saturation region
 - (c) cut-off region
 - (d) inverted region
12. In a PNP transistor, electrons flow
 - (a) into the transistor at the collector only
 - (b) into the transistor at the base and the collector leads
 - (c) out of the transistor at the base and the collector leads
 - (d) out of the transistor at the base collector as well as emitter leads
13. The arrow head on a transistor symbol indicates
 - (a) direction of electron current in the emitter
 - (b) direction of hole current in the emitter
 - (c) diffusion current in the emitter
 - (d) drift current in the emitter
14. Power transistors are invariably provided with
 - (a) soldered connections
 - (b) heat sink
 - (c) metallic casing
 - (d) none of the above
15. The largest current flow of a bipolar transistor occurs
 - (a) in emitter
 - (b) in base
 - (c) in collector
 - (d) through emitter-collector
16. Conventional biasing of a bipolar transistor has
 - (a) EB forward biased and CB forward biased
 - (b) EB reversed biased and CB forward biased
 - (c) EB forward biased and CB reverse biased
 - (d) EB reversed biased and CB reverse biased
17. The common emitter transistor circuit has
 - (a) high gain
 - (b) low gain
 - (c) negligible gain
 - (d) zero gain
18. In an NPN transistor, if both the emitter junction and collector junction are reverse biased, the transistor will operate in
 - (a) active region
 - (b) saturation region
 - (c) cut-off region
 - (d) inverted region
19. In a normally biased NPN transistor, the main current crossing the collector junction is
 - (a) a drift current
 - (b) a hole current
 - (c) a diffusion current
 - (d) same as the base current
20. In a PNP transistor, the electrons flow into the transistor at the
 - (a) collector only
 - (b) emitter only
 - (c) emitter and base
 - (d) collector and base
21. The forward current gain, h_{fe} , is defined as
 - (a) $\frac{\Delta V_{BE}}{\Delta I_B}, V_{CE} \text{ constant}$
 - (b) $\frac{\Delta I_B}{\Delta I_C}, V_{CE} \text{ constant}$
 - (c) $\frac{\Delta I_C}{\Delta I_B}, V_{CE} \text{ constant}$
 - (d) $\frac{\Delta I_C}{\Delta I_B}, V_{BE} \text{ constant}$

22. The alpha (α) and beta (β) of a transistor are related to each other as

$$(a) \alpha = \frac{\beta}{\beta + 1} \quad (b) \beta = \frac{\alpha}{1 + \alpha} \quad (c) \beta = \frac{1 + \alpha}{\alpha} \quad (d) \alpha = \frac{1 + \beta}{\beta}$$

23. The transistor configuration which provides highest output impedance is

- (a) Common Base (b) Common Emitter
(c) Common Collector (d) None of the above

24. A semiconductor transistor operates in the active region only when

- (a) the emitter-base junction is forward-biased and the collector-base junction is reverse biased
(b) both emitter-base and collector-base junctions are forward-biased
(c) both emitter-base and collector-base junctions are reverse-biased
(d) the collector-base junction is forward-biased and emitter-base junction is reverse biased

25. When a healthy NPN transistor is connected to an ohmmeter such that the base terminal is connected to the red lead of the meter and the emitter terminal to the black lead then the meter shown

- (a) an open circuit (b) some medium resistance
(c) a very small resistance (d) none of the above

26. A conducting bipolar transistor dissipates least power when operating in the

- (a) saturation region (b) cut-off region
(c) active region (d) reverse-active region

27. A bipolar transistor in sinusoidal oscillator configuration is operating in the

- (a) active region (b) saturation region
(c) cut-off region (d) reverse-active region

28. The maximum reverse collector to emitter breakdown voltage with base open is referred to as

- (a) V_{CEO} (b) V_C (c) V_{CBO} (d) V_{EBO}

29. When a transistor is used as a switch, the base current required to switch on the transistor for a given collector current is calculated from

- (a) h_{fe} (b) α (c) γ (d) σ

30. For $\alpha = 0.99$, the value of β is

- (a) 9.9 (b) 49 (c) 99 (d) 100

31. With increase in the collector-base reverse voltage

- (a) the base width increases (b) the base width decreases
(c) the base width is not affected (d) the base width can increase or decrease

32. Which of the transistor configurations is capable of providing both voltage and current gains?

- (a) Common base (b) Common collector
(c) Common emitter (d) Both common emitter and common base

33. If the emitter-base junction is open, the value of collector voltage is

- (a) 0 V (b) 0.2 V (c) Floating (d) V_{CC}

34. For operation of a bipolar junction transistor as an amplifier, the transistor operates in

- (a) cut off region (b) saturation region
(c) active region (d) deep in saturation

35. What is the phase difference between input and output signal in a CE amplifier?
(a) 0° (b) 45° (c) 180° (d) 90°
36. The current gain of a CB amplifier is
(a) less than 1 (b) greater than 1
(c) approximately equal (d) none of the above
37. The voltage gain of CC amplifier is
(a) less than 1 (b) greater than 1
(c) equal to 1 (d) none of the above
38. The input and output impedance of a CC amplifier is
(a) large and small (b) small and large
(c) small and small (d) large and large
39. The power gain of a CC amplifier is approximately equal to
(a) current gain (b) voltage gain
(c) none of the above
40. The power gain of a CB amplifier is approximately equal to
(a) current gain (b) voltage gain
(c) none of the above
41. A Cascode amplifier comprises
(a) CB and CC amplifier (b) CB and CE amplifier
(c) CE and CB amplifier (d) CC and CE amplifier
42. Which of the following amplifiers is known as an emitter follower?
(a) CE amplifier (b) CC amplifier
(c) CB amplifier (d) Cascode amplifier
43. Which of the following amplifiers is used in impedance matching?
(a) CB amplifier (b) CC amplifier
(c) CE amplifier (d) Cascode amplifier
44. Which of the following power amplifiers operates with least distortion?
(a) Class A (b) Class B (c) Class C (d) Class D
45. In an RC coupled amplifier for improving the low frequency response,
(a) lower R_L is provided (b) more bias is used
(c) less gain is provided (d) higher C_C is used
46. A unijunction transistor has
(a) anode, cathode and a gate (b) two bases and one emitter
(c) two anodes and one gate (d) anode, cathode and two gates
47. A UJT has $R_{BB} = 10 \text{ k}\Omega$ and $R_{B2} = 4 \text{ k}\Omega$. Its intrinsic stand-off ratio is
(a) 0.6 (b) 0.4 (c) 2.5 (d) $5/3$

State whether the following statements are true (T) or false (F)

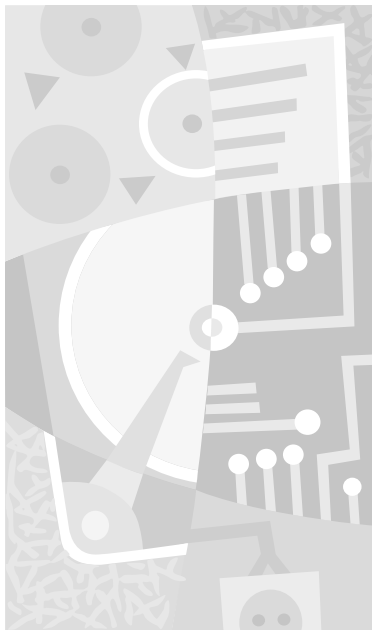
48. Forward bias is applied to an NPN transistor by positive voltage to the emitter and negative voltage to the base.
49. A PNP transistor has a wafer of an N-type semiconductor forming two junctions with two P-type semiconductors.
50. Base region is lightly doped and thin.
51. I_{CO} in a diode and I_{CBO} in a transistor consist of majority carriers.
52. The collector circuit in a transistor amplifier always has a reverse bias, while the base emitter circuit has a forward bias.

53. The common emitter amplifier circuit is used most often because it has the best combination of voltage and current gain.
54. An ordinary transistor is called “Bipolar Junction Transistor” because it has two poles-one positive and the other negative.
55. Silicon transistors have much less leakage current as compared to germanium transistors.
56. PNP transistors are used as amplifiers while NPN transistors are used as rectifiers only.
57. In the schematic symbol for transistor, the arrowhead on the emitter indicates the direction of hole current into the base for P-emitter or out from the base for an N-emitter.
58. In the NPN transistor, the collector current is hole charges supplied by the emitter.
59. The positive voltage applied to the collector of an NPN transistor is the polarity for reverse voltage.
60. In an NPN transistor, the positive voltage at the base with respect to the emitter provides a forward bias.
61. The collector region is made physically larger than the emitter region in order to dissipate more heat.
62. The β -characteristics of transistors compare collector current and the base current.
63. Typical values of β for transistors are 0.98 to 0.99.
64. An NPN transistor has a P-type semiconductor at the centre forming two junctions with two N-type semiconductors.
65. In CE mode of transistors, the leakage current is low.
66. In an NPN transistor amplifier circuit, the negative voltage is required at the collector.
67. In the CE circuit, the input signal is applied to the base and the output is taken from the collector.
68. Reverse voltage gain, h_{re} , of a CE mode transistor is $\frac{\Delta V_{BE}}{\Delta V_{CE}}$, I_B , constant.
69. The output admittance is obtained from input characteristics of a transistor.
70. The input impedance is obtained from the input characteristics of a transistor.
71. In the CC configuration, the current amplification factor, $\gamma = \frac{\Delta I_E}{\Delta I_B}$
72. $\alpha = (1 + \beta)$
73. $\alpha = \frac{\beta}{1 + \beta}$
74. $\gamma = (1 + \beta)$
75. $\beta = \frac{\alpha}{1 + \alpha}$
76. $\beta = \frac{\alpha}{1 - \alpha}$
77. The α and β of a transistor are 0.99 and 99 respectively. If its I_{CBO} is 0.1A, then its I_{CEO} will be 10A.

78. An ordinary transistor is called 'Bipolar Junction Transistor' because it has two poles-one positive and the other negative.
79. I_{CO} in a transistor consists of majority carriers.
80. A CE amplifier has a large current gain.
81. Voltage gain of a CC amplifier is approximately one unit.
82. The power gain of a CB amplifier is equal to the current gain.
83. A CE amplifier provides both voltage gain and current gain.
84. A CE amplifier is widely used for amplification purposes.
85. A CC amplifier gives almost unity voltage gain.
86. In a CB amplifier, the current gain increases with increase in load resistance.
87. The emitter bypass capacitor C_E determines the lower cut-off frequency.

ANSWERS

- | | | | | |
|---------|---------|---------|---------|---------|
| 1. (d) | 2. (c) | 3. (c) | 4. (a) | 5. (d) |
| 6. (c) | 7. (b) | 8. (b) | 9. (b) | 10. (c) |
| 11. (a) | 12. (b) | 13. (a) | 14. (b) | 15. (a) |
| 16. (c) | 17. (a) | 18. (c) | 19. (a) | 20. (d) |
| 21. (c) | 22. (a) | 23. (a) | 24. (a) | 25. (c) |
| 26. (a) | 27. (a) | 28. (a) | 29. (a) | 30. (c) |
| 31. (b) | 32. (c) | 33. (d) | 34. (c) | 35. (c) |
| 36. (c) | 37. (c) | 38. (a) | 39. (a) | 40. (b) |
| 41. (c) | 42. (b) | 43. (b) | 44. (a) | 45. (d) |
| 46. (b) | 47. (a) | 48. (F) | 49. (T) | 50. (T) |
| 51. (F) | 52. (T) | 53. (T) | 54. (F) | 55. (T) |
| 56. (F) | 57. (T) | 58. (F) | 59. (T) | 60. (T) |
| 61. (T) | 62. (T) | 63. (F) | 64. (T) | 65. (T) |
| 66. (F) | 67. (T) | 68. (T) | 69. (F) | 70. (T) |
| 71. (T) | 72. (F) | 73. (T) | 74. (T) | 75. (F) |
| 76. (T) | 77. (T) | 78. (F) | 79. (F) | 80. (T) |
| 81. (T) | 82. (F) | 83. (T) | 84. (T) | 85. (T) |
| 86. (F) | 87. (T) | | | |



4

TRANSISTOR BIASING AND STABILIZATION

4.1 INTRODUCTION

The quiescent operating point of a transistor amplifier should be established in the active region of its characteristics. Since the transistor parameters such as β , I_{CO} and V_{BE} are functions of temperature, the operating point shifts with changes in temperature. The stability of different methods of biasing transistor (BJT, FET and MOSFET) circuits and compensation techniques for stabilizing the operating point are discussed in this chapter.

4.2 BIAS STABILITY

4.2.1 Operating Point, Need for Biasing and Bias Stabilization Against Variations in V_{BE} and β

In order to produce distortion-free output in amplifier circuits, the supply voltages and resistances in the circuit must be suitably chosen. These voltages and resistances establish a set of dc voltage V_{CEQ} and current I_{CQ} to operate the transistor in the active region. These voltages and currents are called *quiescent values* which determine the *operating point* or Q-point for the transistor. The process of giving proper supply voltages and resistances for obtaining the desired Q-point is called *biasing*. The circuits used for getting the desired and proper operating point are known as *biasing circuits*.

The collector current for common-emitter amplifier is expressed by

$$I_C = \beta I_B + I_{CEO} = (1 + \beta)I_{CO}$$

Here the three variables h_{FE} i.e. β , I_B and I_{CO} are found to increase with temperature. For every 10°C rise in temperature, I_{CO} doubles itself. When I_{CO} increases, I_C increases significantly. This causes power dissipation to increase and hence to make I_{CO} increase. This will cause I_C to increase further and the process becomes cumulative which will lead to thermal runaway that will destroy the transistor. In addition,

the quiescent operating point can shift due to temperature changes and the transistor can be driven into the region of saturation. The effect of β on the Q-point is shown in Fig. 4.1. One more source of bias instability is to be considered due to the variation of V_{BE} with temperature. V_{BE} is about 0.6V for a silicon transistor and 0.2V for a germanium transistor at room temperature. As the temperature increases, $|V_{BE}|$ decreases at the rate of 2.5 mV/°C for both silicon and germanium transistors. The transfer characteristic curve shifts to the left at the rate of 2.5 mV/°C (at constant I_C) for increasing temperature and hence the operating point shifts accordingly. To establish the operating point in the active region, compensation techniques are needed.

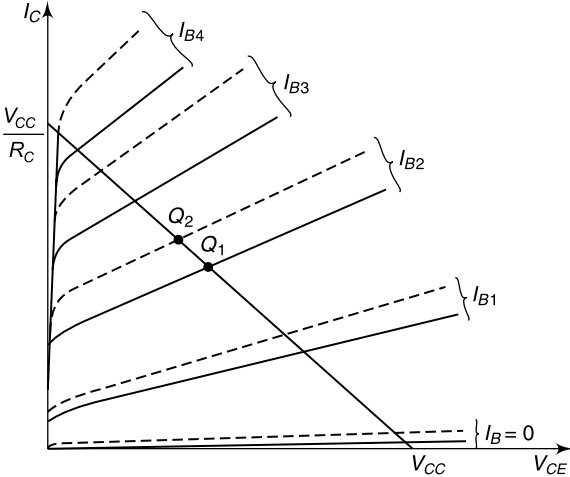


Fig. 4.1

4.2.2 DC Load Line

Referring to the biasing circuit of Fig. 4.2(a), the values of V_{CC} and R_C are fixed and I_C and V_{CE} are dependent on R_B .

Applying Kirchhoff's voltage law to the collector circuit in Fig. 4.2(a), we get $V_{CC} = I_C R_C + V_{CE}$.

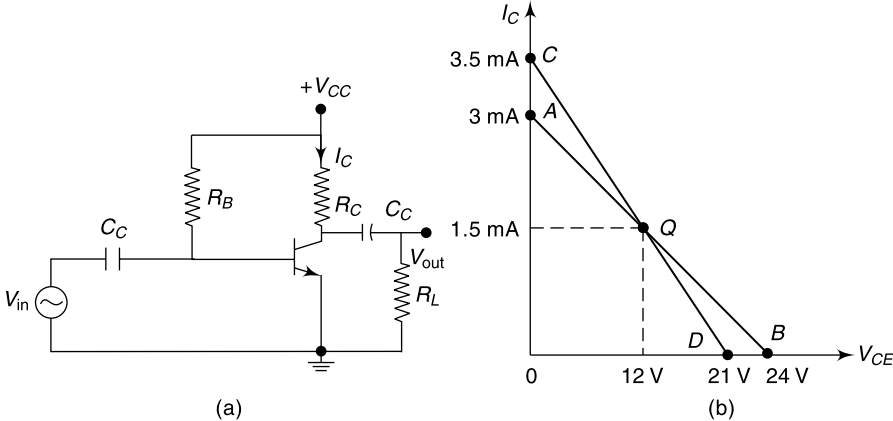


Fig. 4.2 (a) Biasing Circuit (b) CE Output Characteristics and Load Line

The straight line represented by AB in Fig. 4.2(b) is called the dc *load line*. The coordinates of the end point A are obtained by substituting $V_{CE} = 0$ in the above equation. Then $I_C = \frac{V_{CC}}{R_C}$. Therefore, the coordinates of A are $V_{CE} = 0$ and $I_C = \frac{V_{CC}}{R_C}$.

The coordinates of B are obtained by substituting $I_C = 0$ in the above equation. Then $V_{CE} = V_{CC}$. Therefore the coordinates of B are $V_{CE} = V_{CC}$ and $I_C = 0$. Thus, the dc load line AB can be drawn if the values of R_C and V_{CC} are known.

As shown in Fig. 4.2(b), the optimum Q-point is located at the midpoint of the dc load line AB between the saturation and cutoff regions, i.e. Q is exactly midway between A and B . In order to get faithful amplification, the Q-point must be well within the active region of the transistor.

Even though the Q-point is fixed properly, it is very important to ensure that the operating point remains stable where it is originally fixed. If the Q-point shifts nearer to either A or B , the output voltage and current get clipped, thereby output signal is distorted.

In practice, the Q-point tends to shift its position due to any or all of the following three main factors.

- (a) Reverse saturation current, I_{CO} , which doubles for every 10°C increase in temperature
- (b) Base-emitter voltage, V_{BE} , which decreases by $2.5 \text{ mV per } ^\circ\text{C}$
- (c) Transistor current gain, β i.e. h_{FE} which increases with temperature

Referring to Fig. 4.2(a), the base current I_B is kept constant since I_B is approximately equal to V_{CC}/R_B . If the transistor is replaced by another one of the same type, one cannot ensure that the new transistor will have identical parameters as that of the first one. Parameters such as β vary over a range. This results in the variation of collector current I_C for a given I_B . Hence, in the output characteristics, the spacing between the curves might increase or decrease which leads to the shifting of the Q-point to a location which might be completely unsatisfactory.

4.2.3 AC Load Line

After drawing the dc load line, the operating point Q is properly located at the center of the dc load line. This operating point is chosen under zero input signal condition of the circuit. Hence, the ac load line should also pass through the operating point Q. The effective ac load resistance, R_{ac} , is the combination of R_C parallel to R_L , i.e. $R_{ac} = R_C \parallel R_L$. So the slope of the ac load line CQD will be $\left(-\frac{1}{R_{ac}}\right)$.

To draw an ac load line, two end points, viz. maximum V_{CE} and maximum I_C when the signal is applied are required.

Maximum $V_{CE} = V_{CEQ} + I_{CQ}R_{ac}$, which locates the point D (OD) on the V_{CE} axis.

Maximum $I_C = I_{CQ} + \frac{V_{CEQ}}{R_{ac}}$, which locates the point C (OC) on the I_C axis.

By joining points C and D , ac load line CD is constructed. As $R_C > R_{ac}$, the dc load line is less steep than the ac load line.

When the signal is zero, we have the exact dc conditions. From the Fig. 4.2(b), it is clear that the intersection of dc and ac load lines is the operating point Q .

Example 4.1 In the transistor amplifier shown in Fig. 4.2(a), $R_C = 8\text{ k}\Omega$, $R_L = 24\text{ k}\Omega$ and $V_{CC} = 24\text{ V}$. Draw the dc load line and determine the optimum operating point. Also draw the ac load line.

Solution (a) *DC load line:*

Referring to Fig. 4.2(a), we have $V_{CC} = V_{CE} + I_C R_C$.

For drawing dc load line, the two end points, viz. maximum V_{CE} point (at $I_C = 0$) and maximum I_C point (at $V_{CE} = 0$) are required.

$$\text{Maximum } V_{CE} = V_{CC} = 24\text{ V}$$

$$\text{Maximum } I_C = \frac{V_{CC}}{R_C} = \frac{24}{8 \times 10^3} = 3\text{ mA}$$

Therefore, the dc load line AB is drawn with the point B ($OB = 24\text{ V}$) on the V_{CE} axis and the point A ($OA = 3\text{ mA}$) on the I_C axis, as shown in Fig. 4.2(b).

- (b) For fixing the optimum operating point Q , mark the middle of the dc load line AB and the corresponding V_{CE} and I_C values can be found.

$$\text{Here, } V_{CEQ} = \frac{V_{CC}}{2} = 12\text{ V} \quad \text{and} \quad I_{CQ} = 1.5\text{ mA}$$

(c) *AC load line:*

To draw an ac load line, two end points viz. maximum V_{CE} and maximum I_C when the signal is applied are required.

$$\text{The ac load, } R_{ac} = R_C \parallel R_L = \frac{8 \times 10^3 \times 24 \times 10^3}{(8 + 24) \times 10^3} = 6\text{ k}\Omega$$

$$\begin{aligned} \text{Maximum } V_{CE} &= V_{CEQ} + I_{CQ} R_{ac} \\ &= 12 + 1.5 \times 10^{-3} \times 6 \times 10^3 = 21\text{ V} \end{aligned}$$

This locates the point D ($OD = 21\text{ V}$) on the V_{CE} axis.

$$\begin{aligned} \text{Maximum collector current, } I_C &= I_{CQ} + \frac{V_{CEQ}}{R_{ac}} \\ &= 1.5 \times 10^{-3} + \frac{12}{6 \times 10^3} = 3.5\text{ mA} \end{aligned}$$

This locates the point C ($OC = 3.5\text{ mA}$) on the I_C axis. By joining points C and D , ac load line CD is constructed.

Example 4.2 For the transistor amplifier shown in Fig. 4.3(a), $V_{CC} = 12\text{ V}$, $R_1 = 8\text{ k}\Omega$, $R_2 = 4\text{ k}\Omega$, $R_C = 1\text{ k}\Omega$, $R_E = 1\text{ k}\Omega$ and $R_L = 1.5\text{ k}\Omega$. Assume $V_{BE} = 0.7\text{ V}$. (a) draw the dc load line, (b) determine the operating point, and (c) draw the ac load line.

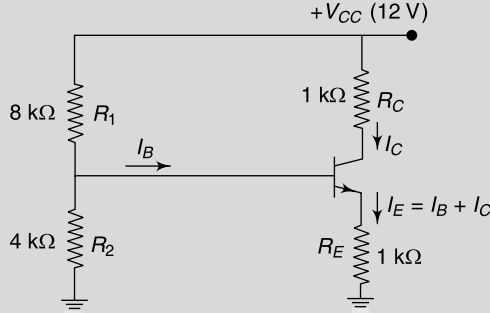


Fig. 4.3(a)

Solution (a) *DC load line:*

Referring to Fig. 4.3(a), we have
 $V_{CC} = V_{CE} + I_C(R_C + R_E)$.

To draw the dc load line, we need two end points, viz. maximum V_{CE} point (at $I_C = 0$) and maximum I_C point (at $V_{CE} = 0$).

Maximum $V_{CE} = V_{CC} = 12\text{ V}$, which locates the point B ($OB = 12\text{ V}$) of the dc load line.

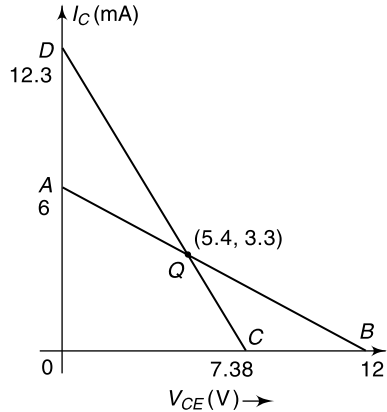


Fig. 4.3(b)

$$\text{Maximum } I_C = \frac{V_{CC}}{R_C + R_E} = \frac{12}{(1+1) \times 10^3} = 6\text{ mA}$$

This locates the point A ($OA = 6\text{ mA}$) of the dc load line. Figure 4.3(b) shows the dc load line AB , with $(12\text{ V}, 6\text{ mA})$.

(b) *Operating point Q:*

The voltage across R_2 is
$$V_2 = \frac{R_2}{R_1 + R_2} V_{CC}$$

Therefore,
$$V_2 = \frac{4 \times 10^3}{12 \times 10^3} \times 12 = 4\text{ V}$$

$$V_2 = V_{BE} + I_E R_E$$

Therefore,
$$I_E = \frac{V_2 - V_{BE}}{R_E} = \frac{4 - 0.7}{1 \times 10^3} = 3.3 \text{ mA}$$

$$I_C \approx I_E = 3.3 \text{ mA}$$

$$\begin{aligned} V_{CE} &= V_{CC} - I_C(R_C + R_E) \\ &= 12 - 3.3 \times 10^{-3} \times 2 \times 10^3 = 5.4 \text{ V} \end{aligned}$$

Therefore, the operating point Q is at 5.4 V and 3.3 mA, which is shown on the dc load line.

(c) *AC load line:*

To draw the ac load line, we need two end points, viz. maximum V_{CE} and maximum I_C when signal is applied.

AC load,
$$R_{ac} = R_C \parallel R_L = \frac{1 \times 1.5}{2.5} = 0.6 \text{ k}\Omega$$

Therefore, maximum
$$\begin{aligned} V_{CE} &= V_{CEQ} + I_{CQ} R_{ac} \\ &= 5.4 + 3.3 \times 10^{-3} \times 0.6 \times 10^3 = 7.38 \text{ V} \end{aligned}$$

This locates the point C ($OC = 6.24 \text{ V}$) on the V_{CE} axis.

Maximum
$$\begin{aligned} I_C &= I_{CQ} + \frac{V_{CEQ}}{R_{ac}} \\ &= 3.3 \times 10^{-3} + \frac{5.4}{0.6 \times 10^3} = 12.3 \text{ mA} \end{aligned}$$

This locates the point D ($OD = 12.3 \text{ mA}$) on the I_C axis. By joining points C and D, ac load line CD is constructed.

Example 4.3 Design the circuit shown in Fig. 4.4, given Q-point values are to be $I_{CQ} = 1 \text{ mA}$ and $V_{CEQ} = 6 \text{ V}$. Assume that $V_{CC} = 10 \text{ V}$, $\beta = 100$ and $V_{BE(on)} = 0.7 \text{ V}$.

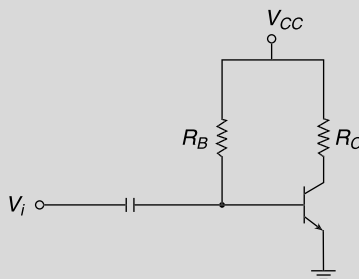


Fig. 4.4

Solution The collector resistance is

$$R_C = \frac{V_{CC} - V_{CEQ}}{I_{CQ}} = \frac{10 - 6}{1 \times 10^{-3}} = 4 \text{ k}\Omega$$

The base current is
$$I_{BQ} = \frac{I_{CQ}}{\beta} = \frac{1 \times 10^{-3}}{100} = 10 \mu\text{A}$$

The base resistance is
$$R_B = \frac{V_{CC} - V_{BE(on)}}{I_{BQ}} = \frac{10 - 0.7}{10 \times 10^{-6}} = 0.93 \text{ M}\Omega$$

Example 4.4 Determine the characteristics of a circuit shown in Fig. 4.5. Assume that $\beta = 100$ and $V_{BE}(on) = 0.7$ V.

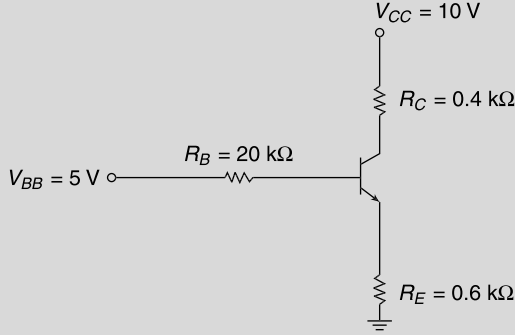


Fig. 4.5

Solution Referring to Fig. 4.5, Kirchoff's voltage law equation is

$$V_{BB} = I_B R_B + V_{BE}(on) + I_E R_E$$

We know that

$$I_E = I_B + I_C = I_B + \beta I_B = (1 + \beta) I_B$$

$$\text{The base current } I_B = \frac{V_{BB} - V_{BE}(on)}{R_B + (1 + \beta) R_E} = \frac{5 - 0.7}{20 \times 10^3 + 101 \times 600} = 53.34 \mu\text{A}$$

$$\text{Therefore, } I_C = \beta I_B = 100 \times 53.34 \times 10^{-6} = 5.334 \text{ mA}$$

$$I_E = I_C + I_B = 5.334 \times 10^{-3} + 53.34 \times 10^{-6} = 5.38734 \text{ mA}$$

$$V_{CE} = V_{CC} - I_C R_C - I_E R_E = 10 - 5.334 \times 10^{-3} \times 400 \\ - 5.38734 \times 10^{-3} \times 600 = 4.634 \text{ V}$$

The Q point is at $V_{CEQ} = 4.634$ V and $I_{CQ} = 5.334$ mA

4.2.4 Stability Factor (S)

The extent to which the collector current I_C is stabilised with varying I_{CO} is measured by a stability factor S . It is defined as the rate of change of collector current I_C with respect to the collector base leakage current I_{CO} , keeping both the current I_B and the current gain β constant

$$S = \frac{\partial I_C}{\partial I_{CO}} \approx \frac{dI_C}{dI_{CO}} \approx \frac{\Delta I_C}{\Delta I_{CO}}, \beta \text{ and } I_B \text{ constant} \quad (4.1)$$

The collector current for a CE amplifier is given by

$$I_C = \beta I_B + (\beta + 1) I_{CO} \quad (4.2)$$

Differentiating the above equation with respect to I_C , we get

$$1 = \beta \frac{dI_B}{dI_C} + (\beta + 1) \frac{dI_{CO}}{dI_C}$$

Therefore,

$$\left(1 - \beta \frac{dI_B}{dI_C}\right) = \frac{(\beta + 1)}{S}$$

$$S = \frac{1 + \beta}{1 - \beta \left(\frac{dI_B}{dI_C}\right)} \quad (4.3)$$

From this equation, it is clear that this factor S should be as small as possible to have better thermal stability.

Stability factors S' and S'' The stability factor S' is defined as the rate of change of I_C with V_{BE} , keeping I_{CO} and β constant

$$S' = \frac{\partial I_C}{\partial V_{BE}} \approx \frac{\Delta I_C}{\Delta V_{BE}}$$

The stability factor S'' is defined as the rate of change of I_C with respect to β , keeping I_{CO} and V_{BE} constant

$$S'' = \frac{\partial I_C}{\partial \beta} \approx \frac{\Delta I_C}{\Delta \beta}$$

4.3 METHODS OF TRANSISTOR BIASING

The stability factors for some commonly used biasing circuits are discussed below.

4.3.1 Fixed Bias or Base Resistor Method

A common emitter amplifier using fixed bias circuit is shown in Fig. 4.6. The dc analysis of the circuit yields the following equation.

$$V_{CC} = I_B R_B + V_{BE} \quad (4.4)$$

Therefore,

$$I_B = \frac{V_{CC} - V_{BE}}{R_B}$$

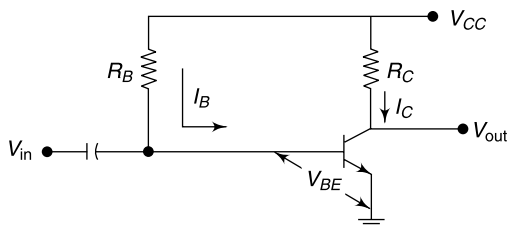


Fig. 4.6 Fixed Bias Circuit

Since this equation is independent of current I_C , $dI_B/dI_C = 0$ and the stability factor given in Eq. (4.3) reduces to

$$S = 1 + \beta$$

Since β is a large quantity, this is a very poor bias stable circuit. Therefore, in practice, this circuit is not used for biasing the base.

The advantage of this method are: (a) simplicity, (b) small number of components required, and (c) if the supply voltage is very large as compared to V_{BE} of the transistor, then the base current becomes largely independent of the voltage V_{BE} .

Example 4.5 In the fixed bias compensation method (refer to Fig. 4.6), a silicon transistor with $\beta = 100$ is used. $V_{CC} = 6\text{ V}$, $R_C = 3\text{ k}\Omega$, $R_B = 530\text{ k}\Omega$. Draw the dc load line and determine the operating point. What is the stability factor?

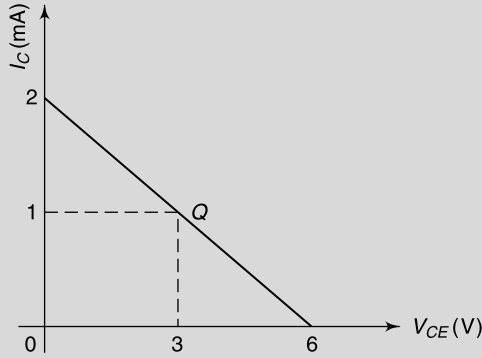


Fig. 4.7 DC Load Line

Solution (a) *DC load line:*

$$V_{CE} = V_{CC} - I_C R_C$$

When

$$I_C = 0, V_{CE} = V_{CC} = 6\text{ V}$$

When

$$V_{CE} = 0, I_C = \frac{V_{CC}}{R_C} = \frac{6}{3 \times 10^3} = 2\text{ mA}$$

(b) *Operating point Q:*

For silicon transistor,

$$V_{BE} = 0.7\text{ V}$$

$$V_{CC} = I_B R_B + V_{BE}$$

Therefore,

$$I_B = \frac{V_{CC} - V_{BE}}{R_B} = \frac{6 - 0.7}{530 \times 10^3} = 10\text{ }\mu\text{A}$$

Therefore,

$$I_C = \beta I_B = 100 \times 10 \times 10^{-6} = 1\text{ mA}$$

$$V_{CE} = V_{CC} - I_C R_C = 6 - 1 \times 10^{-3} \times 3 \times 10^3 = 3\text{ V}$$

Therefore operating point is $V_{CEQ} = 3\text{ V}$ and $I_{CQ} = 1\text{ mA}$.

(c) *Stability factor:*

$$S = 1 + \beta = 1 + 100 = 101$$

Example 4.6 Find the collector current and collector-to-emitter voltage for the given circuit as shown in Fig. 4.8.

[JNTU June 2009]

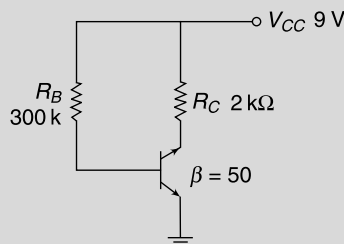


Fig. 4.8

Solution For a silicon transistor $V_{BE} = 0.7 \text{ V}$

$$\text{Base current } I_B = \frac{V_{CC} - V_{BE}}{R_B} = \frac{9 - 0.7}{300 \times 10^3} = 27.67 \mu\text{A}$$

$$\text{Collector current } I_C = \beta I_B = 50 \times 27.67 \times 10^{-6} = 1.38 \text{ mA}$$

$$\text{Collector-to-emitter voltage } V_{CE} = V_{CC} - I_C R_C = 9 - 1.38 \times 10^{-3} \times 2 \times 10^3 = 6.24 \text{ V}$$

Example 4.7 A Germanium transistor having $\beta = 100$ and $V_{BE} = 0.2 \text{ V}$ is used in a fixed bias amplifier circuit where $V_{CC} = 16 \text{ V}$, $R_C = 5 \text{ k}\Omega$ and $R_B = 790 \text{ k}\Omega$. Determine its operating point.

[JNTU Aug/Sep 2008]

Solution For a germanium transistor, $V_{BE} = 0.2 \text{ V}$

Applying KVL to the base circuit, we have

$$V_{CC} - I_B R_B - V_{BE} = 0$$

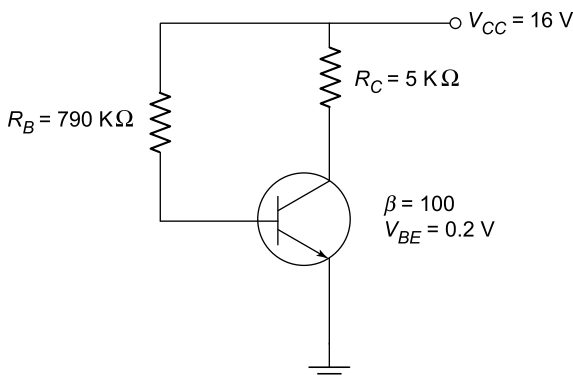


Fig. 4.9

$$\text{Therefore, } I_B = \frac{V_{CC} - V_{BE}}{R_B} = \frac{16 - 0.2}{790 \times 10^3} = 20 \mu\text{A}$$

$$I_C = \beta I_B = 100 \times 20 \mu\text{A} = 2 \text{ mA}$$

Applying KVL to the collector circuit, we have

$$V_{CC} - I_C R_C - V_{CE} = 0$$

$$V_{CE} = V_{CC} - I_C R_C = 16 - 2 \times 10^{-3} \times 5 \times 10^3 = 6 \text{ V}$$

Hence, the operating point is $I_C = 2 \text{ mA}$ and $V_{CE} = 6 \text{ V}$.

Example 4.8 The circuit as shown in Fig. 4.10 has fixed bias using NPN transistor. Determine the value of base current, collector current and collector to emitter voltage. [JNTU Aug/Sep 2008]

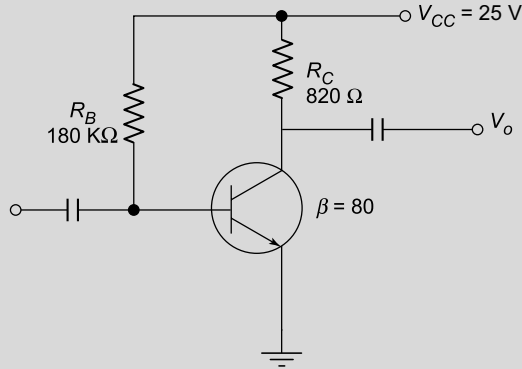


Fig. 4.10

Solution Applying KVL to the base circuit, we have

$$V_{CC} - I_B R_B - V_{BE} = 0$$

$$\text{Therefore, } I_B = \frac{V_{CC} - V_{BE}}{R_B} = \frac{25 - 0.7}{180 \times 10^3} = 135 \mu\text{A}$$

$$I_C = \beta I_B = 80 \times 135 \times 10^{-6} = 10.8 \mu\text{A}$$

Applying KVL to the collector circuit, we have

$$V_{CC} - I_C R_C - V_{CE} = 0$$

$$\text{Therefore, } V_{CE} = V_{CC} - I_C R_C = 25 - 10.8 \times 10^{-6} \times 820 = 16.144 \text{ V}$$

Example 4.9 For a fixed bias configuration, determine I_C , R_C , R_B and V_{CE} using the following specifications: $V_{CC} = 12 \text{ V}$, $V_C = 6 \text{ V}$, $\beta = 80$, $I_B = 40 \mu\text{A}$. [JNTU May/June 2008]

Solution Assume $V_{BE} = 0.7 \text{ V}$ for a Silicon transistor.

$$I_C = \beta I_B = 80 \times 40 \mu\text{A} = 3.2 \text{ mA}$$

$$R_C = \frac{V_{CC} - V_C}{I_C} = \frac{12 - 6}{3.2 \times 10^{-3}} = 1.875 \text{ k}\Omega$$

$$R_B = \frac{V_{CC} - V_{BE}}{I_B} = \frac{12 - 0.7}{40 \times 10^{-6}} = 282.5 \text{ k}\Omega$$

Since emitter is grounded, $V_E = 0$.

$$V_{CE} = V_C = 6 \text{ V}$$

4.3.2 Emitter-Feedback Bias

The emitter-feedback bias network shown in Fig. 4.11 contains an emitter resistor for improving the stability level over that of the fixed-bias configuration. The analysis will be performed by first examining the base-emitter loop and then using the results to investigate the collector-emitter loop.

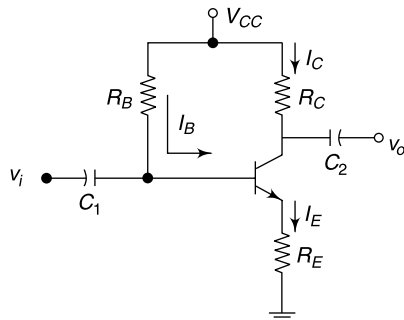


Fig. 4.11 Emitter-Feedback Bias Circuit

Base-Emitter Loop Applying Kirchoff's voltage law for the base-feedback emitter loop, we get

$$V_{CC} - I_B R_B - V_{BE} - I_E R_E = 0 \quad (4.5)$$

$$V_{CC} - I_B R_B - V_{BE} - (I_B + I_C) R_E = 0$$

$$V_{CC} - I_B (R_B + R_E) - V_{BE} - I_C R_E = 0$$

$$V_{CC} - V_{BE} = I_B (R_B + R_E) + I_C R_E$$

Therefore,
$$I_B = \frac{V_{CC} - V_{BE}}{R_E + R_B} - \left(\frac{R_E}{R_E + R_B} \right) I_C \quad (4.6)$$

Here V_{BE} is independent of I_C .

Hence,
$$\frac{dI_B}{dI_C} = - \left(\frac{R_E}{R_E + R_B} \right) \quad (4.7)$$

Substituting Eq. (4.7) in Eq. (4.3), we get the stability factor as

$$S = \frac{1 + \beta}{1 + \beta \frac{R_E}{R_E + R_B}} \quad (4.8)$$

Since $1 + \frac{\beta R_E}{(R_E + R_B)} > 1$, $S < (1 + \beta)$. Note that the value of the stability factor S

is always lower in emitter-feedback bias circuit than that of the fixed bias circuit. Hence it is clear that a better thermal stability can be achieved in emitter-feedback bias circuit than the fixed-bias circuit.

Collector-Emitter Loop Applying Kirchoff's voltage law for the collector-emitter loop, we get

$$I_E R_E + V_{CE} + I_C R_C - V_{CC} = 0$$

Substituting $I_E = I_C$, we have

$$V_{CE} - V_{CC} + I_C (R_C + R_E) = 0$$

and

$$V_{CE} = V_{CC} - I_C (R_C + R_E) \quad (4.9)$$

V_E is the voltage from emitter to ground and is determined by

$$V_E = I_E R_E \quad (4.10)$$

The voltage from collector to ground can be determined from

$$V_{CE} = V_C - V_E$$

and

$$V_C = V_{CE} + V_E \quad (4.11)$$

or

$$V_C = V_{CC} - I_C R_C \quad (4.12)$$

The voltage at the base with respect to ground can be determined from

$$V_B = V_{CC} - I_B R_E \quad (4.13)$$

or

$$V_B = V_{BE} + V_E \quad (4.14)$$

Example 4.10 For the emitter-feedback bias circuit, $V_{CC} = +10V$, $R_C = 1.5 k\Omega$, $R_B = 270 k\Omega$ and $R_E = 1 k\Omega$. Assuming $\beta = 50$, determine (a) Stability factor, S (b) I_B (c) I_C (d) V_{CE} (e) V_C (f) V_E (g) V_B and (h) V_{BC} .

Solution (a) The stability factor is

$$\begin{aligned} S &= \frac{1 + \beta}{1 + \frac{\beta R_E}{(R_E + R_B)}} = \frac{1 + 50}{1 + \frac{(50 \times 1 \times 10^3)}{1 \times 10^3 + 270 \times 10^3}} \\ &= \frac{51}{1 + 0.185} = \frac{51}{1.185} = 43.04 \end{aligned}$$

$$(b) I_B = \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1) R_E} = \frac{10 - 0.7}{27 \times 10^3 + (51)(1 \times 10^3)} = \frac{9.3}{321} = 28.97 \mu A$$

$$(c) I_C = \beta I_B = (50)(28.97 \times 10^{-6}) = 1.45 \text{ mA}$$

$$\begin{aligned} (d) V_{CE} &= V_{CC} - I_C (R_C + R_E) \\ &= 10 - 1.45 \times 10^{-3} (1.5 \times 10^3 + 1 \times 10^3) = 10 - 3.62 = 6.38 \text{ V} \end{aligned}$$

$$(e) V_C = V_{CC} - I_C R_C = 10 - 1.45 \times 10^{-3} (1.5 \times 10^3) = 7.825 \text{ V}$$

$$(f) V_E = V_C - V_{CE} = 7.825 - 6.38 = 1.445 \text{ V or}$$

$$V_E = I_E R_E \cong I_C R_E = 1.45 \times 10^{-3} \times 1 \times 10^3 = 1.45 \text{ V}$$

$$(g) V_B = V_{BE} + V_E = 0.7 + 1.45 = 2.15 \text{ V}$$

$$(h) V_{BC} = V_B - V_C = 2.15 - 7.825 = -5.675 \text{ V} \quad (\text{reverse bias as required})$$

Example 4.11 Calculate dc bias voltage and currents in the circuit in Fig. 4.12 [Neglect V_{BE} of transistor].
[JNTU 2008 and June 2009]

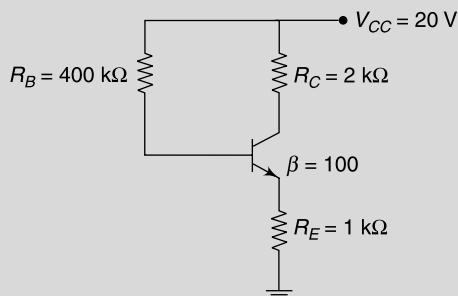


Fig. 4.12

Solution Given $V_{CC} = 20 \text{ V}$; $R_B = 400 \text{ k}\Omega$, $\beta = 100$, $R_E = 1 \text{ k}\Omega$; $R_C = 2 \text{ k}\Omega$

$$I_B R_B + V_{BE} + I_E R_E = V_{CC}$$

$$\frac{I_C}{\beta} R_B + 0 + (I_C + I_B) R_E = 20$$

$$I_C \left[\frac{R_B}{\beta} + R_E + \frac{R_E}{\beta} \right] = 20$$

$$\text{Therefore, } I_C = \frac{20}{\left[\frac{400 \times 10^3}{100} + 1 \times 10^3 + 10 \right]} = 4 \text{ mA}$$

$$I_B = \frac{I_C}{\beta} = \frac{4 \times 10^{-3}}{100} = 0.4 \mu\text{A}$$

$$V_B = V_{BE} + I_E R_E$$

$$= 0 + 4 \times 10^{-3} \times 1 \times 10^3 = 4 \text{ V, since } I_C \approx I_E$$

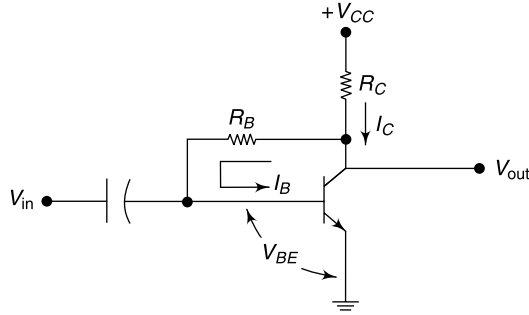
4.3.3 Collector-to-Base Bias or Collector-Feedback Bias

A common emitter amplifier using collector-to-base bias circuit is shown in Fig. 4.13. This circuit is the simplest way to provide some degree of stabilization to the amplifier operating point.

If the collector current I_C tends to increase due to either increase in temperature or the transistor has been replaced by the one with a higher β , the voltage drop across R_C increases, thereby reducing the value of V_{CE} . Therefore, I_B decreases which, in turn, compensates the increase in I_C . Thus, greater stability is obtained.

The loop equation for this circuit is

$$V_{CC} = (I_B + I_C)R_C + I_B R_B + V_{BE} \quad (4.15)$$

**Fig. 4.13** Collector-to-Base Bias Circuit

$$\text{i.e.} \quad I_B = \frac{V_{CC} - V_{BE} - I_C R_C}{R_C + R_B} \quad (4.16)$$

$$\text{Therefore,} \quad \frac{dI_B}{dI_C} = \frac{R_C}{R_C + R_B} \quad (4.17)$$

Substituting Eq. (4.17) into Eq. (4.3), we get

$$S = \frac{1 + \beta}{1 + \beta \left(\frac{R_C}{R_C + R_B} \right)} \quad (4.18)$$

As can be seen, this value of the stability factor is smaller than the value obtained by fixed bias circuit. Also, S can be made small and the stability can be improved by making R_B small or R_C large.

If R_C is very small, then $S = (\beta + 1)$, i.e. stability is very poor. Hence, the value of R_C must be quite large for good stabilization. Thus, collector-to-base bias arrangement is not satisfactory for the amplifier circuits like transformer coupled amplifier where the dc load resistance in collector circuit is very small. For such amplifiers, emitter bias or self bias will be the most satisfactory transistor biasing for stabilization.

Example 4.12 In the biasing with feedback resistor method, a silicon transistor with feedback resistor is used. The operating point is at 7 V, 1 mA and $V_{CC} = 12$ V. Assume $\beta = 100$. Determine (a) the value of R_B , (b) stability factor, and (c) what will be the new operating point if $\beta = 50$ with all other circuit values are same.

Solution Refer to Fig. 4.6. We know that for a silicon transistor, $V_{BE} = 0.7$ V.

(a) To determine R_B :

The operating point is at $V_{CE} = 7$ V and $I_C = 1$ mA

Here,

$$R_C = \frac{V_{CC} - V_{CE}}{I_C} = \frac{12 - 7}{1 \times 10^{-3}} = 5 \text{ k}\Omega$$

$$I_B = \frac{I_C}{\beta} = \frac{1 \times 10^{-3}}{100} = 10 \text{ }\mu\text{A}$$

Using the relation,

$$R_B = \frac{V_{CC} - V_{BE} - I_C R_C}{I_B} = \frac{12 - 0.7 - 1 \times 10^{-3} \times 5 \times 10^3}{10 \times 10^{-6}} = 630 \text{ k}\Omega$$

(b) *Stability factor*,
$$S = \frac{1 + \beta}{1 + \beta \left[\frac{R_C}{R_C + R_B} \right]}$$

$$= \frac{1 + 100}{1 + 100 \left[\frac{5 \times 10^3}{(5 + 630) \times 10^3} \right]} = 56.5$$

(c) *To determine new operating point when $\beta = 50$*

$$\begin{aligned} V_{CC} &= \beta I_B R_C + I_B R_B + V_{BE} \\ &= I_B (\beta R_C + R_B) + V_{BE} \\ \text{i.e. } 12 &= I_B (50 \times 5 \times 10^3 + 630 \times 10^3) + 0.7 \end{aligned}$$

$$I_B = \frac{11.3}{880 \times 10^3} = 12.84 \text{ }\mu\text{A}$$

Therefore, $I_C = \beta I_B = 50 \times 12.84 \times 10^{-6} = 0.642 \text{ mA}$

$$V_{CE} = V_{CC} - I_C R_C = 12 - 0.642 \times 10^{-3} \times 5 \times 10^3 = 8.79 \text{ V}$$

Therefore the coordinates of the new operating point are $V_{CEQ} = 8.79 \text{ V}$ and $I_{CQ} = 0.642 \text{ mA}$.

Example 4.13 An NPN transistor if $\beta = 50$ is used in common emitter circuit with $V_{CC} = 10 \text{ V}$ and $R_C = 2 \text{ k}\Omega$. The bias is obtained by connecting $100 \text{ k}\Omega$ resistor from collector to base. Find the quiescent point and stability factor. [JNTU May 2003, 2004 & 2005, May/June 2006, Nov. 2010]

Solution Given, $V_{CC} = 10 \text{ V}$, $R_C = 2 \text{ k}\Omega$,

$\beta = 50$ and collector to base resistor $R_B = 100 \text{ k}\Omega$

To determine the quiescent point: We know that for the collector to base bias transistor circuit

$$V_{CC} = \beta I_B R_C + I_B R_B + V_{BE}$$

Therefore,
$$I_B = \frac{V_{CC} - V_{BE}}{R_B + \beta \cdot R_C}$$

$$= \frac{10 - 0.7}{100 \times 10^3 + 50 \times 2 \times 10^3} = 46.5 \text{ }\mu\text{A}$$

Hence, $I_C = \beta \cdot I_B = 50 \times 46.5 \times 10^{-6} = 2.325 \text{ mA}$

$$V_{CE} = V_{CC} - I_C R_C = 10 - 2.325 \times 10^{-3} \times 2 \times 10^3 = 5.35 \text{ V}$$

Therefore, the co-ordinates of the new operating point are

$$V_{CEQ} = 5.35 \text{ V and } I_{CQ} = 2.325 \text{ mA}$$

To find the stability factors S :

$$\begin{aligned} S &= \frac{1 + \beta}{1 + \beta \left[\frac{R_C}{R_C + R_B} \right]} \\ &= \frac{1 + 50}{1 + 50 \left[\frac{2 \times 10^3}{2 \times 10^3 + 100 \times 10^3} \right]} = 25.75 \end{aligned}$$

Example 4.14 In a collector to base CE amplifier circuit of Fig. 4.6 having $V_{CC} = 12 \text{ V}$, $R_C = 250 \text{ k}\Omega$, $I_B = 0.25 \text{ mA}$, $\beta = 100$ and $V_{CEQ} = 8 \text{ V}$, calculate R_B and stability factor.

Solution $R_B = \frac{V_{CEQ}}{I_B} = \frac{8}{0.25 \times 10^{-3}} = 32 \text{ k}\Omega$

Stability factor, $S = \frac{1 + \beta}{1 + \beta \left(\frac{R_C}{R_C + R_B} \right)} = \frac{101}{1 + 100 \left(\frac{250}{32 + 250} \right)} = 56.9$

Example 4.15 Calculate the quiescent current and voltage of collector to base bias arrangement using the following data:

$V_{CC} = 10 \text{ V}$, $R_B = 100 \text{ k}\Omega$, $R_C = 2 \text{ k}\Omega$, $\beta = 50$ and also specify a value of R_B so that $V_{CE} = 7 \text{ V}$. [JNTU May/June 2008]

Solution

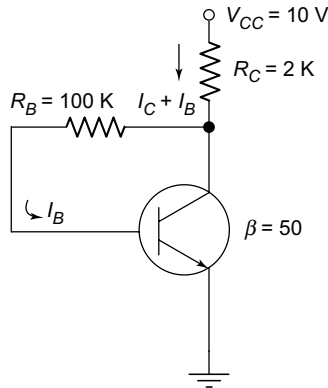


Fig. 4.14

(a) Applying KVL to the base circuit, we have

$$V_{CC} - I_B(1 + \beta)R_C - I_B R_B - V_{BE} = 0$$

$$\text{Therefore, } I_B = \frac{V_{CC} - V_{BE}}{R_B + (1 + \beta)R_C} = \frac{10 - 0.7}{100 \times 10^3 + (1 + 50) \times 2 \times 10^3} = 46 \mu\text{A}$$

$$I_C = \beta I_B = 50 \times 46 \mu\text{A} = 2.3 \text{ mA}$$

Applying KVL to the collector circuit, we have

$$V_{CC} - (I_B + I_C)R_C - V_{CE} = 0$$

$$\begin{aligned} \text{Therefore, } V_{CE} &= V_{CC} - (I_B + I_C)R_C \\ &= 10 - (46 \times 10^{-6} + 2.3 \times 10^{-3}) \times 2 \times 10^3 = 5.308 \text{ V} \end{aligned}$$

Quiescent current, $I_{CQ} = 2.3 \text{ mA}$ and

Quiescent voltage, $V_{CEQ} = 5.308 \text{ V}$

(b) Given $V_{CE} = 7 \text{ V}$

$$(I_B + I_C)R_C = V_{CC} - V_{CE}$$

$$(1 + \beta)I_B R_C = V_{CC} - V_{CE}$$

$$I_B = \frac{V_{CC} - V_{CE}}{(1 + \beta)R_C} = \frac{10 - 7}{(1 + 50) \times 2 \times 10^3} = 29.41 \mu\text{A}$$

We have,

$$V_{CC} = I_B R_B + V_{BE}$$

$$R_B = \frac{V_{CC} - V_{BE}}{I_B} = \frac{7 - 0.7}{29.41 \times 10^{-6}} = 214.2 \text{ k}\Omega$$

4.3.4 Collector-Emitter Feedback Bias

Figure 4.15 shows the collector-emitter feedback bias circuit that can be obtained by applying both the collector-feedback and emitter-feedback. Here collector-feedback is provided by connecting a resistance R_B from the collector to the base and emitter-feedback is provided by connection an emitter resistance R_E from the emitter to ground. Both the feedbacks are used to control the collector current I_C and the base current I_B in the opposite direction to increase the stability as compared to the previous biasing circuits.

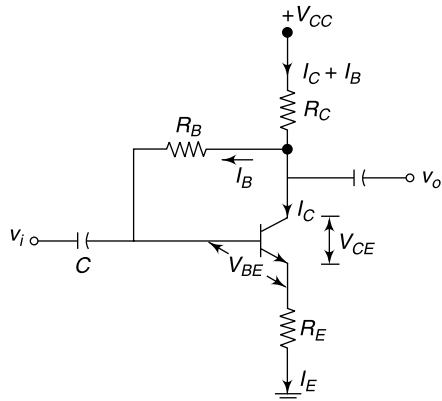


Fig. 4.15 Collector-Emitter Feedback Circuit

Applying Kirchoff's voltage law to the current, we get

$$(I_B + I_C)R_E + V_{BE} + I_B R_B + (I_B + I_C) R_C - V_{CC} = 0$$

Therefore,

$$I_B = \frac{V_{CC} - V_{BE}}{R_E + R_C + R_B} - \left(\frac{R_E + R_C}{R_E + R_C + R_B} \right) I_C$$

Since V_{BE} is independent of I_C ,

$$\frac{dI_B}{dI_C} = - \left(\frac{R_E + R_C}{R_E + R_C + R_B} \right)$$

Substituting the above equation in Eq. (4.3), we get

$$S = \frac{1 + \beta}{1 + \frac{\beta(R_E + R_C)}{R_E + R_C + R_B}} \quad (4.19)$$

From this, it is clear that the stability of the collector-emitter feedback bias circuit is always better than that of the collector-feedback and emitter feedback circuits.

4.3.5 Voltage Divider Bias, Self Bias, or Emitter Bias

A simple circuit used to establish a stable operating point is the self-biasing configuration. The self bias, also called as emitter bias, or emitter resistor and potential divider circuit, that can be used for low collector resistance, is shown in Fig. 4.16. The current in the emitter resistor R_E causes a voltage drop which is in the direction to reverse bias the emitter junction. For the transistor to remain in the active region, the base-emitter junction has to be forward biased. The required base bias is obtained from the power supply through the potential divider network of the resistances R_1 and R_2 .

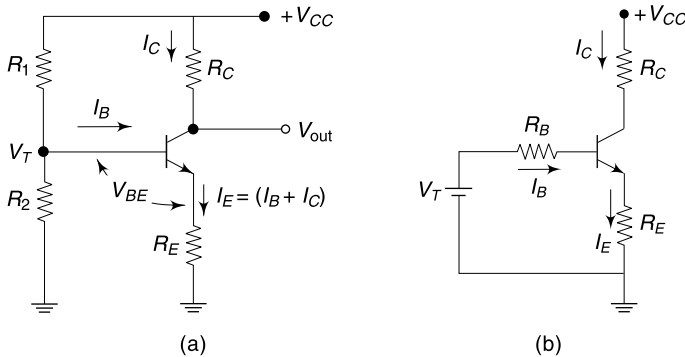


Fig. 4.16 (a) Self Bias (b) Thevenin's Equivalent Circuit

Use of Self-Bias Circuit as a Constant Current Circuit If I_C tends to increase, say, due to increase in I_{CO} with temperature, the current in R_E increases. Hence, the voltage drop across R_E increases thereby decreasing the base current. As a result, I_C is maintained almost constant.

4.4 STABILIZATION FACTORS

To Determine Stability Factor, S Applying Thevenin's Theorem to the circuit of Fig. 4.16, for finding the base current, we have,

$$V_T = \frac{R_2 V_{CC}}{R_1 + R_2} \quad \text{and} \quad R_B = \frac{R_1 R_2}{R_1 + R_2}$$

The loop equation around the base circuit can be written as

$$V_T = I_B R_B + V_{BE} + (I_B + I_C) R_E$$

Differentiating this equation with respect to I_C , we get,

$$\frac{dI_B}{dI_C} = -\frac{R_E}{R_E + R_B}$$

Substituting this equation in Eq. (4.3), we get

$$S = \frac{1 + \beta}{1 + \beta \left(\frac{R_E}{R_E + R_B} \right)}$$

Therefore,

$$S = (1 + \beta) \frac{1 + \frac{R_B}{R_E}}{1 + \beta + \frac{R_B}{R_E}} \quad (4.20)$$

As can be seen, the value of S is equal to one if the ratio R_B/R_E is very small as compared to 1. As this ratio becomes comparable to unity, and beyond towards infinity, the value of the stability factor goes on increasing till $S = 1 + \beta$.

This improvement in the stability up to a factor equal to 1 is achieved at the cost of power dissipation. To improve the stability, the equivalent resistance R_B must be decreased, forcing more current in the voltage divider network of R_1 and R_2 .

Often, to prevent the loss of gain due to the negative feedback, R_E is shunted by a capacitor C_E . The capacitive reactance X_{CE} must be equal to about one-tenth of the value of the resistance R_E at the lowest operating frequency.

To Determine the Stability Factor S' The stability factor S' is defined as the rate of change of I_C with V_{BE} , keeping I_{CO} and β constant.

$$S' = \frac{\partial i_C}{\partial V_{BE}} = \frac{\Delta I_C}{\Delta V_{BE}}$$

From Fig. 4.16(b),

$$\begin{aligned} V_T &= I_B R_B + V_{BE} + I_E R_E \\ &= I_B [R_B + R_E] + I_C R_E + V_{BE} \text{ since } [I_E = I_B + I_C] \end{aligned} \quad (4.21)$$

From Eq. (4.2), we have

$$I_B = \frac{I_C - (1 + \beta) I_{CO}}{\beta} \quad (4.22)$$

Substituting Eq. (4.22) in Eq. (4.21), we get

$$V_T = \frac{I_C}{\beta} (R_B + R_E) + V_{BE} + I_C R_E + \frac{I_{CO}}{\beta} (1 + \beta) \{ R_B + R_E \} \quad (4.23)$$

Differentiating the above eq. w.r.t. V_{BE} we get

$$\begin{aligned} 0 &= \frac{dI_C}{dV_{BE}} \left(\frac{R_B + R_E}{\beta} \right) + 1 + R_E \frac{dI_C}{dV_{BE}} + 0 \\ -1 &= \frac{dI_C}{dV_{BE}} \left[R_E + \frac{R_B + R_E}{\beta} \right] \\ -1 &= \frac{dI_C}{dV_{BE}} \left[\frac{R_B + (1 + \beta) R_E}{\beta} \right] \end{aligned}$$

Therefore,

$$S' = \frac{dI_C}{dV_{BE}} = \frac{-\beta}{R_B + (1 + \beta) R_E} \quad (4.24)$$

To Determine the Stability of S'' The stability factor S'' is defined as the rate of change of I_C w.r.t. to β , keeping I_{CO} and V_{BE} constant.

Rearranging Eq. (4.23), we have

$$I_C = \frac{\beta (V_T - V_{BE})}{R_B + (1 + \beta) R_E} + \frac{\beta \left(\frac{1 + \beta}{\beta} \right) I_{CO} (R_B + R_E)}{R_B + (1 + \beta) R_E} \quad (4.25)$$

Since $\beta \gg 1$, the numerator of the second term can be written as

$$(R_B + R_E) \left(\frac{1 + \beta}{\beta} \right) I_{CO} \approx (R_B + R_E) I_{CO} \quad (4.26)$$

Substituting Eq. (4.26) in Eq. (4.25), we have

$$I_C = \frac{\beta(V_T - V_{BE})}{R_B + (1 + \beta)R_E} + \frac{\beta(R_B + R_E) I_{CO}}{R_B + (1 + \beta)R_E}$$

Therefore,
$$I_C = \frac{\beta[V_T - V_{BE} + (R_B + R_E) I_{CO}]}{R_B + (1 + \beta)R_E}$$

Let,
$$V' = (R_B + R_E) I_{CO},$$

Therefore,
$$I_C = \frac{\beta[V_T - V_{BE} + V']}{R_B + (1 + \beta)R_E} \quad (4.27)$$

Differentiating the above equation w.r.t. β and simplifying, we obtain

$$S'' = \frac{dI_C}{d\beta} = \frac{I_C}{\beta \left[1 + \beta \left(\frac{R_E}{R_E + R_B} \right) \right]} = \frac{SI_C}{\beta(1 + \beta)} \quad (4.28)$$

Example 4.16 In a CE germanium transistor amplifier circuit, the bias is provided by self bias, i.e. emitter resistor and potential divider arrangement (refer to Fig. 4.7). The various parameters are: $V_{CC} = 16 \text{ V}$, $R_C = 3 \text{ k}\Omega$, $R_E = 2 \text{ k}\Omega$, $R_1 = 56 \text{ k}\Omega$, $R_2 = 20 \text{ k}\Omega$ and $\alpha = 0.985$. Determine (a) the coordinates of the operating point, and (b) the stability factor S .

Solution For a germanium transistor, $V_{BE} = 0.3 \text{ V}$. As $\alpha = 0.985$,

$$\beta = \frac{\alpha}{1 - \alpha} = \frac{0.985}{1 - 0.985} = 66$$

(a) To find the coordinates of the operating point

Referring to Fig. 4.16, we have

$$\begin{aligned} \text{Thevenin's voltage, } V_T &= \frac{R_2}{R_1 + R_2} V_{CC} \\ &= \frac{20 \times 10^3}{76 \times 10^3} \times 16 = 4.21 \text{ V} \end{aligned}$$

$$\begin{aligned} \text{Thevenin's resistance, } R_B &= \frac{R_1 R_2}{R_1 + R_2} = \frac{20 \times 10^3 \times 56 \times 10^3}{76 \times 10^3} \\ &= 14.737 \text{ k}\Omega \end{aligned}$$

The loop equation around the base circuit is

$$V_T = I_B R_B + V_{BE} + (I_B + I_C) R_E$$

$$\begin{aligned}
 &= \frac{I_C}{\beta} R_B + V_{BE} + \left(\frac{I_C}{\beta} + I_C \right) R_E \\
 4.21 &= \frac{I_C}{66} \times 14.737 \times 10^3 + 0.3 + I_C \left(\frac{1}{66} + 1 \right) \times 2 \times 10^3 \\
 3.91 &= I_C [0.223 + 2.03] \times 10^3
 \end{aligned}$$

Therefore,
$$I_C = \frac{3.91}{2.253 \times 10^3} = 1.73 \text{ mA}$$

Since I_B is very small, $I_C \approx I_E = 1.73 \text{ mA}$

Therefore,
$$\begin{aligned}
 V_{CE} &= V_{CC} - I_C R_C - I_E R_E \\
 &= V_{CC} - I_C [R_C + R_E] = 16 - 1.73 \times 10^{-3} \times 5 \times 10^3 \\
 &= 7.35 \text{ V}
 \end{aligned}$$

Therefore, the coordinates of the operating point are $I_C = 1.73 \text{ mA}$ and $V_{CE} = 7.35 \text{ V}$.

(b) To find the stability factor S ,

$$\begin{aligned}
 S &= (1 + \beta) \frac{1 + \frac{R_B}{R_E}}{1 + \beta + \frac{R_B}{R_E}} \\
 &= (1 + 66) \frac{1 + \frac{14.737}{2}}{1 + 66 + \frac{14.737}{2}} \\
 &= 67 \times \frac{8.3685}{74.3685} = 7.537
 \end{aligned}$$

Example 4.17 Consider the self bias circuit where $V_{CC} = 22.5 \text{ Volts}$, $R_C = 5.6 \text{ k}\Omega$, $R_2 = 10 \text{ k}\Omega$ and $R_1 = 90 \text{ k}\Omega$, $h_{fe} = 55$, $V_{BE} = 0.6 \text{ V}$. The transistor operates in active region. Determine (a) operating point and (b) stability factor. [JNTU June 2009]

Solution For the given circuit,

$$V_{BE} = 0.6 \text{ V}, h_{fe} = 55$$

(a) To determine the operating point:

Thevenin's voltage,
$$V_T = \frac{R_2}{R_1 + R_2} V_{CC} = \frac{10 \times 10^3}{100 \times 10^3} \times 22.5 = 2.25 \text{ V}$$

Thevenin's resistance, $R_B = \frac{R_1 R_2}{R_1 + R_2} = \frac{10 \times 10^3 \times 90 \times 10^3}{100 \times 10^3} = 9 \text{ k}\Omega$

The loop equation around the base circuit is

$$\begin{aligned} V_B &= I_B R_B + V_{BE} + (I_B + I_C) R_E \\ &= \frac{I_C}{h_{fe}} R_B + V_{BE} + \left(\frac{I_C}{h_{fe}} + I_C \right) R_E \\ 2.25 &= \frac{I_C}{55} \times 9 \times 10^3 + 0.6 + \left(\frac{1}{55} + 1 \right) I_C \times 1 \times 10^3 \\ 2.25 &= I_C \times 0.16 \times 10^3 + 0.6 + 1.01 \times I_C \times 10^3 \\ 2.25 &= I_C \times 1.17 \times 10^3 + 0.6 \end{aligned}$$

Therefore, $I_C = \frac{2.25 - 0.6}{1.17 \times 10^3} = 1.41 \text{ mA}$

Since I_B is very small, $I_C \approx I_E = 1.41 \text{ mA}$

Therefore, $V_{CE} = V_{CC} - I_C R_C - I_E R_E = V_{CC} - I_C (R_C + R_E)$

$$= 22.5 - 1.41 \times 10^{-3} \times 6.6 \times 10^3 = 13.19 \text{ V}$$

Operating point coordinates are $V_{CE} = 13.19 \text{ V}$ and $I_C = 1.41 \text{ mA}$

(b) To find the stability factor, S

$$S = (1 + \beta) \frac{1 + \frac{R_B}{R_E}}{1 + \beta + \frac{R_B}{R_E}} = (1 + 55) \frac{1 + \frac{9 \times 10^3}{1 \times 10^3}}{1 + 55 + \frac{9 \times 10^3}{1 \times 10^3}} = \frac{56 \times 10}{65} = \frac{560}{65} = 8.6$$

Example 4.18 The Fig. 4.17 shows that dc bias circuit of a common emitter transistor amplifier. Find the percentage change in collector current, if the transistor with $h_{fe} = 50$ is replaced by another transistor with $h_{fe} = 150$. It is given that the base emitter drop $V_{BE} = 0.6 \text{ V}$.

[JNTU June 2009]

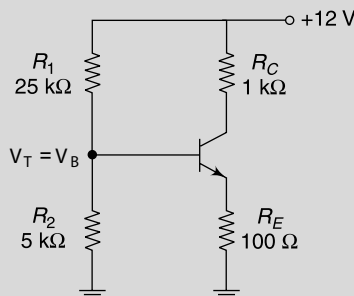


Fig. 4.17

Solution (a) For the given circuit, $V_{BE} = 0.6 \text{ V}$, $h_{fe} = 50$

$$\text{Thevenin's voltage, } V_T = \frac{R_2}{R_1 + R_2} V_{CC} = \frac{5 \times 10^3}{30 \times 10^3} \times 12 = 2 \text{ V}$$

$$\text{Thevenin's resistance, } R_B = \frac{R_1 R_2}{R_1 + R_2} = \frac{25 \times 10^3 \times 5 \times 10^3}{30 \times 10^3} = 4.16 \text{ k}\Omega$$

The loop equation around the base circuit is

$$\begin{aligned} V_T &= V_B = I_B R_B + V_{BE} + (I_B + I_C) R_E \\ &= \frac{I_C}{h_{fe}} R_B + V_{BE} + \left(\frac{I_C}{h_{fe}} + I_C \right) R_E \\ 2 &= \frac{I_C}{50} \times 4.6 \times 10^3 + 0.6 + \left(\frac{1}{50} + 1 \right) \times I_C \times 0.1 \times 10^3 \\ 2 - 0.6 &= I_C \times (0.08 + 0.102) \times 10^3 \end{aligned}$$

$$\text{Therefore, } I_C = \frac{1.4}{0.182 \times 10^3} = 7.69 \text{ mA}$$

(b) For the given circuit, $V_{BE} = 0.6 \text{ V}$, $h_{fe} = 150$

The loop equation around the base circuit is

$$\begin{aligned} V_B &= I_B R_B + V_{BE} + (I_B + I_C) R_E \\ &= \frac{I_C}{h_{fe}} R_B + V_{BE} + \left(\frac{I_C}{h_{fe}} + I_C \right) R_E \\ 2 &= \frac{I_C}{50} \times 4.6 \times 10^3 + 0.6 + \left(\frac{1}{50} + 1 \right) \times I_C \times 0.1 \times 10^3 \\ 2 - 0.6 &= I_C \times (0.028 + 0.1) \times 10^3 \end{aligned}$$

$$\text{Therefore, } I_C = \frac{1.4}{0.128 \times 10^3} = 10.93 \text{ mA}$$

$$\text{Change in collector current} = \frac{10.93 - 7.69}{7.69} = 0.42. \text{ i.e. } 42\%$$

There is 42% change in I_C when h_{fe} changes from 50 to 150.

Example 4.19 If the various parameters of a CE amplifier which uses the self bias method are $V_{CC} = 12 \text{ V}$, $R_1 = 10 \text{ k}\Omega$, $R_2 = 5 \text{ k}\Omega$, $R_C = 1 \text{ k}\Omega$, $R_E = 2 \text{ k}\Omega$ and $\beta = 100$, find

(a) the coordinates of the operating point and

(b) the stability factor, assuming the transistor to be silicon. [JNTU 2008]

Solution (a) *To find the coordinates of the operating point*

Refer to Fig. 4.16.

$$\text{Thevenin's voltage, } V_T = \frac{R_2}{R_1 + R_2} V_{CC} = \frac{5 \times 10^3}{15 \times 10^3} \times 12 = 4 \text{ V}$$

$$\text{Thevenin's resistance, } R_B = \frac{R_1 R_2}{(R_1 + R_2)} = \frac{5 \times 10^3 \times 10 \times 10^3}{15 \times 10^3} = 3.33 \text{ k}\Omega$$

The loop equation around the basic circuit is

$$\begin{aligned} V_T &= I_B R_B + V_{BE} + (I_B + I_C) R_E \\ &= \frac{I_C}{\beta} R_B + V_{BE} + \left(\frac{I_C}{\beta} + I_C \right) R_E \\ 4 &= \frac{I_C}{100} \times 3.33 \times 10^3 + 0.7 + I_C \left(\frac{1}{100} + 1 \right) \times 2 \times 10^3 \\ 3.3 &= (33.3 + 2020) I_C \\ I_C &= \frac{3.3}{2053.3} = 1.61 \text{ mA} \end{aligned}$$

Since I_B is very small, $I_C \approx I_E = 1.61 \text{ mA}$

Therefore, $V_{CE} = V_{CC} - I_C R_C - I_E R_E$

$$\begin{aligned} &= V_{CC} - I_C [R_C + R_E] = 12 - 1.61 \times 10^{-3} \times 3 \times 10^3 \\ &= 7.17 \text{ V} \end{aligned}$$

Therefore, the coordinates of the operating point are $I_C = 1.61 \text{ mA}$ and $V_{CE} = 7.17 \text{ V}$.

(b) *To find the stability factor S ,*

$$S = (1 + \beta) \frac{1 + \frac{R_B}{R_E}}{1 + \beta + \frac{R_B}{R_E}} = (1 + 100) \frac{1 + \frac{3.33 \times 10^3}{2 \times 10^3}}{1 + 100 + \frac{3.33 \times 10^3}{2 \times 10^3}} = 2.6$$

Example 4.20 Determine the quiescent current and collector to emitter voltage for a germanium transistor with $\beta = 50$ in self biasing arrangement. Draw the circuit with a given component value with $V_{CC} = 20 \text{ V}$, $R_C = 2 \text{ k}\Omega$, $R_E = 100 \Omega$, $R_1 = 100 \text{ k}\Omega$ and $R_2 = 5 \text{ k}\Omega$. Also find the stability factor.

Solution For a germanium transistor, $V_{BE} = 0.3 \text{ V}$ and $\beta = 50$

To find the coordinates of the operating point:

$$\begin{aligned}\text{Thevenin's voltage, } V_T &= \frac{R_2}{R_1 + R_2} V_{CC} \\ &= \frac{5 \times 10^3}{105 \times 10^3} \times 20 = 0.95 \text{ V}\end{aligned}$$

$$\text{Thevenin's resistance, } R_B = \frac{R_1 R_2}{R_1 + R_2} = \frac{100 \times 10^3 \times 5 \times 10^3}{105 \times 10^3} = 4.76 \text{ k}\Omega$$

The loop equation around the base circuit is

$$\begin{aligned}V_T &= I_B R_B + V_{BE} + (I_B + I_C) R_E \\ &= \frac{I_C}{\beta} R_B + V_{BE} + \left(\frac{I_C}{\beta} + I_C \right) R_E \\ 0.95 &= \frac{I_C}{50} \times 4.76 \times 10^3 + 0.3 + I_C \times \frac{51}{50} \times 100\end{aligned}$$

$$0.65 = 197.2 I_C$$

$$\text{Therefore, } I_C = \frac{0.65}{197.2} = 3.296 \text{ mA}$$

Since I_B is very small, $I_C \approx I_E = 3.296 \text{ mA}$

$$\begin{aligned}\text{Therefore, } V_{CE} &= V_{CC} - I_C R_C - I_E R_E \\ &= V_{CC} - I_C (R_C + R_E) \\ &= 20 - 3.296 \times 10^{-3} \times 2.01 \times 10^3 = 13.375 \text{ V}\end{aligned}$$

Therefore, the coordinates of the operating point are $I_C = 3.296 \text{ mA}$ and $V_{CE} = 13.375 \text{ V}$.

To find the stability factor S :

$$S = (1 + \beta) \frac{1 + \frac{R_B}{R_E}}{1 + \beta + \frac{R_B}{R_E}} = (1 + 50) \frac{1 + \frac{4.76 \times 10^3}{100}}{1 + 50 + \frac{4.76 \times 10^3}{100}} = 25.18$$

Example 4.21 A germanium transistor is used in a self biasing circuit configuration as shown below with $V_{CC} = 16\text{ V}$, $R_C = 1.5\text{ k}\Omega$ and $\beta = 50$. The operating point desired is $V_{CE} = 8\text{ V}$ and $I_C = 4\text{ mA}$. If a stability factor $S = 10$ is desired, calculate the values of R_1 , and R_2 and R_E of the circuit (Fig. 4.18).

[JNTU May/June 2006]

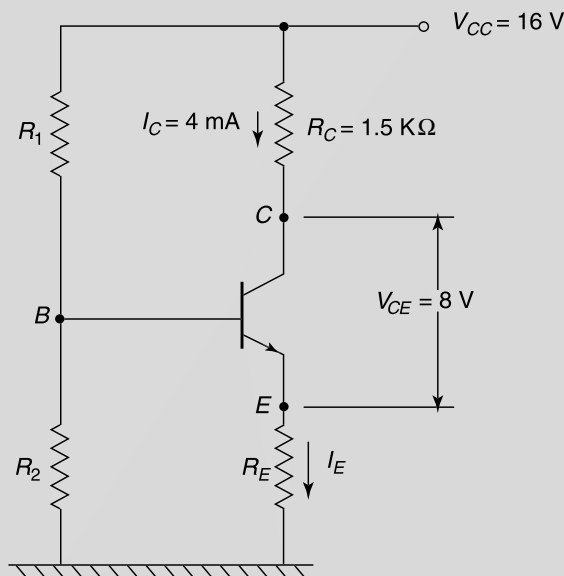


Fig. 4.18

Solution To determine R_E

We know that, $V_{CC} = V_{CE} + I_C(R_C + R_E)$

$$16 = 8 + 4 \times 10^{-3} (1.5 \times 10^3 + R_E)$$

Therefore, $R_E = 500\ \Omega$

To determine R_{TH}

Given $S = 10$

$$\text{Stability factor } S = \frac{1 + \beta}{1 + \beta \frac{R_E}{R_{TH} + R_E}} = \frac{1 + 50}{1 + 50 \left(\frac{500}{R_{TH} + 500} \right)}$$

Upon solving, we get $R_{TH} = 5.58\text{ k}\Omega$

To determine R_2

$$R_2 = 0.1 \beta R_E = 27.98 \text{ k}\Omega$$

To determine R_1

We know that,

$$R_{TH} = R_1 \parallel R_2 = \frac{R_1 R_2}{R_1 + R_2}$$

$$5.58 \times 10^3 = \frac{R_1 \times 27.98 \times 10^3}{R_1 + 27.98 \times 10^3}$$

Therefore,

$$R_1 = 6.97 \text{ k}\Omega$$

Example 4.22 A CE transistor amplifier with voltage divider bias circuit of Fig. 4.16 is designed to establish the quiescent point at $V_{CE} = 12 \text{ V}$, $I_C = 2 \text{ mA}$ and stability factor ≤ 5.1 . If $V_{CC} = 24 \text{ V}$, $V_{BE} = 0.7 \text{ V}$, $\beta = 50$ and $R_C = 4.7 \text{ k}\Omega$, determine the values of resistors R_E , R_1 and R_2 .

Solution (a) To determine R_E

$$\begin{aligned} V_{CE} &= V_{CC} - I_C R_C - I_E R_E \\ &= V_{CC} - I_C [R_C + R_E], \text{ since } I_C \approx I_E \\ 12 &= 24 - 2 \times 10^{-3} [4.7 \times 10^3 + R_E] \end{aligned}$$

Therefore,

$$R_E = 1.3 \text{ k}\Omega$$

(b) To determine R_1 and R_2

Stability factor,

$$S = \frac{1 + \beta}{1 + \beta \left(\frac{R_E}{R_E + R_B} \right)}, \text{ where } R_B = \frac{R_1 R_2}{(R_1 + R_2)}$$

$$5.1 = \frac{51}{1 + 50 \left(\frac{1.3 \times 10^3}{1.3 \times 10^3 + R_B} \right)}$$

i.e.

$$1 + 50 \left(\frac{1.3 \times 10^3}{1.3 \times 10^3 + R_B} \right) = \frac{51}{5.1} = 10$$

Therefore,

$$\left(\frac{50 \times 1.3 \times 10^3}{1.3 \times 10^3 + R_B} \right) = 9$$

$$1.3 \times 10^3 + R_B = \frac{50 \times 1.3 \times 10^3}{9} = 7.2 \text{ k}\Omega$$

$$R_B = 5.9 \text{ k}\Omega$$

Also, we know that for a good voltage divider, the value of resistor $R_2 = 0.1 \beta R_E$

Therefore, $R_2 = 0.1 \times 50 \times 1.3 \times 10^3 = 6.5 \text{ k}\Omega$

$$R_B = \frac{R_1 R_2}{R_1 + R_2}$$

$$5.9 \times 10^3 = \frac{R_1 \times 6.5 \times 10^3}{R_1 + 6.5 \times 10^3}$$

Simplifying, we get $R_1 = 64 \text{ k}\Omega$

Example 4.23 In the circuit shown, if $I_C = 2 \text{ mA}$ and $V_{CE} = 3 \text{ V}$, calculate R_1 and R_3 (Fig. 4.19).

[JNTU April/May 2007]

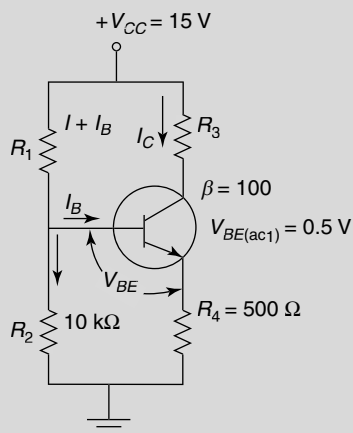


Fig. 4.19

Solution Given $\beta = 100$, $I_C = 2 \text{ mA}$, $V_{CE} = 3 \text{ V}$, $V_{BE} = 0.6 \text{ V}$, $R_2 = 10 \text{ k}\Omega$ and $R_4 = 500 \Omega$

We know that $\beta = \frac{I_C}{I_B}$

Hence, $I_B = \frac{I_C}{\beta} = \frac{2 \times 10^{-3}}{100} = 20 \mu\text{A}$

$$V_{CC} = I_C R_3 + V_{CE} + I_E R_4$$

$$I_E = I_C + I_B = 20 \times 10^{-6} + 2 \times 10^{-5} = 2.02 \text{ mA}$$

Substituting the values, we get

$$15 = 2 \times 10^{-3} \times R_3 + 3 + 2.02 \times 10^{-3} \times 500$$

Therefore, $R_3 = 5.495 \text{ k}\Omega$

$$V_B = V_{BE} + I_E R_4 = 0.6 + 2.02 \times 10^{-3} \times 500 = 1.61$$

From the circuit, $V_B = \frac{R_2 V_{CC}}{R_1 + R_2}$

$$1.61 = \frac{10 \times 10^3 \times 15}{R_1 + 10 \times 10^3}$$

Therefore, $R_1 = 83.17 \text{ k}\Omega$

Example 4.24 Given a voltage divider bias circuit shown in Fig. 4.20, determine the Q point. Let $R_1 = 56 \text{ k}\Omega$, $R_2 = 12.2 \text{ k}\Omega$, $R_C = 2 \text{ k}\Omega$, $R_E = 400 \Omega$, $V_{CC} = 10 \text{ V}$, $V_{BE}(\text{on}) = 0.7 \text{ V}$ and $\beta = 150$.

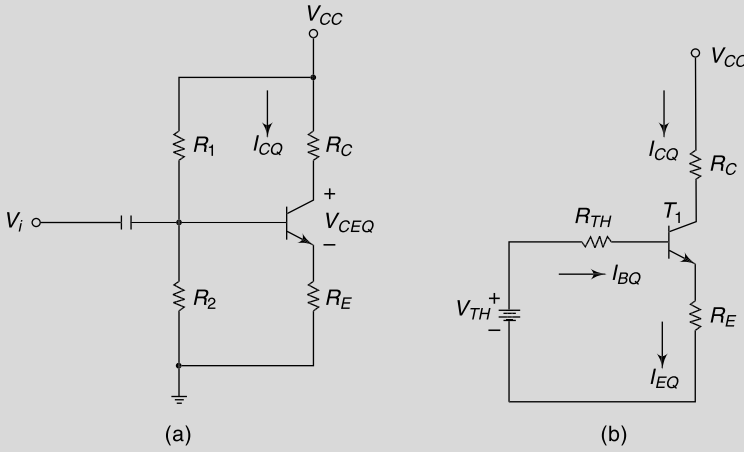


Fig. 4.20

Solution From the Thevenin equivalent circuit shown in Fig. 4.20(b), we get

$$R_{TH} = R_1 \parallel R_2 = 56 \text{ k}\Omega \parallel 12.2 \text{ k}\Omega = 10 \text{ k}\Omega$$

$$V_{TH} = \left(\frac{R_2}{R_1 + R_2} \right) V_{CC} = \frac{12.2}{56 + 12.2} \times 10 = 1.79 \text{ V}$$

With the help of Kirchhoff's voltage law equation, we get

$$I_{BQ} = \frac{V_{TH} - V_{BE}(\text{on})}{R_{TH} + (1 + \beta) R_E} = \frac{1.79 - 0.7}{10 + 151 \times 0.4} = 15.5 \mu\text{A}$$

Therefore, $I_{CQ} = \beta I_{BQ} = 150 \times 15.5 \times 10^{-6} = 2.32 \text{ mA}$

$$I_{EQ} = I_{BQ} + I_{CQ} = 15.5 \mu\text{A} + 2.32 \text{ mA} = 2.336 \text{ mA}$$

$$\begin{aligned}
 V_{CEQ} &= V_{CC} - I_{CQ}R_C - I_{EQ}R_E \\
 &= 10 - 2.32 \times 10^{-3} \times 2 \times 10^3 - 2.336 \times 10^{-3} \times 400 \\
 &= 4.426 \text{ V}
 \end{aligned}$$

The quiescent point is at $V_{CEQ} = 4.426 \text{ V}$ and $I_{CQ} = 2.32 \text{ mA}$

Example 4.25 For a circuit shown in Fig. 4.21 $V_{CC} = 22 \text{ V}$, $R_C = 2 \text{ k}\Omega$, $\beta = 60$, $V_{BE \text{ active}} = 0.6 \text{ V}$, $R_1 = 100 \text{ k}\Omega$. Calculate I_B , V_{CE} , I_C and stability factor S .

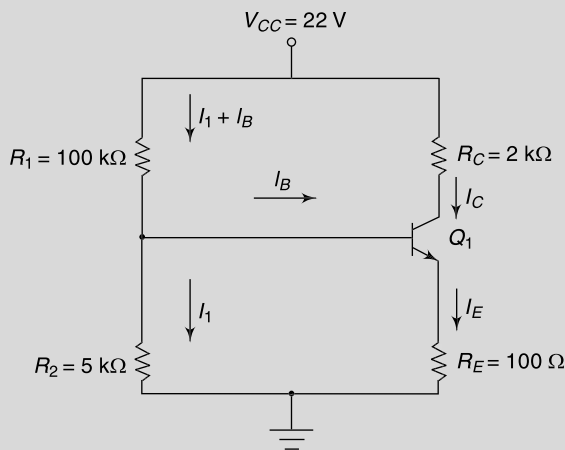


Fig. 4.21

Solution For the given circuit

$$\begin{aligned}
 V_{CC} &= R_1[I_1 + I_B] + I_1 R_2 \\
 I_1 &= \frac{V_{CC} - I_B R_1}{R_1 + R_2} \quad (1)
 \end{aligned}$$

Further $V_{CC} = R_1[I_1 + I_B] + V_{BE} + I_E R_E$

As $I_E = I_C + I_B$
 $= \beta I_B + I_B = (1 + \beta) I_B$

Hence $V_{CC} = R_1[I_1 + I_B] + V_{BE} + (1 + \beta) I_B R_E$

Substituting for I_1 from Eq. (1), we get

$$\begin{aligned}
 V_{CC} &= R_1 \left[\frac{V_{CC} - I_B R_1}{R_1 + R_2} + I_B \right] + V_{BE} + (1 + \beta) I_B R_E \\
 V_{CC} &= R_1 \left[\frac{V_{CC} + I_B R_2}{R_1 + R_2} \right] + V_{BE} + (1 + \beta) I_B R_E
 \end{aligned}$$

Substituting for V_{CC} , R_1 , R_2 , V_{BE} , β and R_E

$$22 = 100 \times 10^3 \left[\frac{22 + I_B \times 5 \times 10^3}{(100 + 5) \times 10^3} \right] + 0.6 + (1 + 60) I_B \times 100$$

$$22 = 0.952[22 + I_B \times 5 \times 10^3] + 0.6 + 6100 I_B$$

$$22 = 20.944 + 11,100 I_B + 0.6$$

$$I_B = 41.08 \mu\text{A}$$

$$I_C = \beta I_B = 60 \times 41.08 \times 10^{-6} = 246 \text{ mA}$$

Applying KVL to collector circuit, we get

$$V_{CC} = I_C R_C + V_{CE} + I_E R_E = I_C R_C + V_{CE} + (1 + \beta) I_B R_E$$

Hence

$$\begin{aligned} V_{CE} &= V_{CC} - I_C R_C - (1 + \beta) I_B R_E \\ &= 22 - (246) \times 10^{-3} \times 2 \times 10^3 - (61)(41.08 \times 10^{-6})(100) \\ &= 22 - 4.92 - 0.25 = 16.83 \text{ V} \end{aligned}$$

To find stability factor, (S): Stability factor for voltage divider bias is

$$\begin{aligned} S &= \frac{1 + \beta}{1 + \beta \left(\frac{R_E}{R_E + R_B} \right)}, \text{ where } R_B = R_1 \parallel R_2 \\ S &= \frac{1 + 60}{1 + 60 \left(\frac{100}{100 + \left(\frac{100 \times 10^3 \times 5 \times 10^3}{(100 + 5) \times 10^3} \right)} \right)} \\ &= \frac{61}{1 + 60 \left(\frac{100}{100 + 4761.9} \right)} \\ &= \frac{61}{2.234} = 27.3 \end{aligned}$$

Example 4.26 An NPN transistor if $\beta = 50$ is used in common emitter circuit with $V_{CC} = 10 \text{ V}$ and $R_C = 2 \text{ k}\Omega$. The bias is obtained by connecting $100 \text{ k}\Omega$ resistor from collector to base. Find the quiescent point and stability factor.

Solution Given $V_{CC} = 10 \text{ V}$, $R_C = 2 \text{ k}\Omega$,

$\beta = 50$ and collector to base resistor $R_B = 100 \text{ k}\Omega$

To determine the quiescent point:

We know that for the collector to base bias transistor circuit

$$V_{CC} = \beta I_B R_C + I_B R_B + V_{BE}$$

Therefore,

$$I_B = \frac{V_{CC} - V_{BE}}{R_B + \beta \cdot R_C} = \frac{10 - 0.7}{100 \times 10^3 + 50 \times 2 \times 10^3} = 46.5 \mu\text{A}$$

Hence,

$$I_C = \beta I_B = 50 \times 46.5 \times 10^{-6} = 2.325 \text{ mA}$$

$$V_{CE} = V_{CC} - I_C R_C = 10 - 2.325 \times 10^{-3} \times 2 \times 10^3 = 5.35 \text{ V}$$

Therefore, the co-ordinates of the new operating point are

$$V_{CEQ} = 5.35 \text{ V and } I_{CQ} = 2.325 \text{ mA}$$

To find the stability factor S :

$$S = \frac{1 + \beta}{1 + \beta \left[\frac{R_C}{R_C + R_B} \right]} = \frac{1 + 50}{1 + 50 \left[\frac{2 \times 10^3}{2 \times 10^3 + 100 \times 10^3} \right]} = 25.75$$

Example 4.27 Design a voltage divider bias network using a supply of 24V, $\beta = 110$ and $I_{CQ} = 4 \text{ mA}$ $V_{CEQ} = 8 \text{ V}$. Choose $V_E = V_{CC} / 8$.
[JNTU May/June 2008]

Solution Given $I_{CQ} = 4 \text{ mA}$, $V_{CEQ} = 8 \text{ V}$, $V_E = V_{CC} / 8$, $V_{CC} = 24 \text{ V}$, $\beta = 110$

(a) To determine I_B , I_E and V_E

$$I_B = \frac{I_{CQ}}{\beta} = \frac{4 \times 10^{-3}}{110} = 36.36 \mu\text{A}$$

$$I_E = I_B + I_C = 36.36 \times 10^{-6} + 4 \times 10^{-3} = 4.03636 \text{ mA}$$

$$V_E = \frac{V_{CC}}{8} = \frac{24}{8} = 3 \text{ V}$$

(b) To determine R_E and R_2

$$R_E = \frac{V_E}{I_E} = \frac{0.3}{4.03636 \times 10^{-3}} = 743.244 \Omega$$

Applying KVL to the collector circuit,

$$V_{CC} - I_C R_C - V_{CE} - V_E = 0$$

$$\text{Therefore, } R_C = \frac{V_{CC} - V_{CE} - V_E}{I_C} = \frac{24 - 8 - 3}{4 \times 10^{-3}} = 3.25 \text{ k}\Omega$$

(c) To determine R_1 and R_2

$$V_B = V_E + V_{BE} = 3 + 0.7 = 3.7 \text{ V}$$

Referring to Fig. 4.16, consider the current through R_1 be $I + I_B$ and that through R_2 be I . Resistors R_1 and R_2 forms the potential divider. For proper operation of potential divider, current I should be atleast ten times the I_B . i.e. $I \geq 10I_B$. Therefore,

$$I = 10I_B = 10 \times 36.36 \times 10^{-6} = 363.6 \mu\text{A}$$

$$R_2 = \frac{V_B}{I} = \frac{3.7}{363.6 \times 10^{-6}} = 10.176 \text{ k}\Omega$$

$$R_1 = \frac{V_{CC} - V_B}{I + I_B} = \frac{24 - 3.7}{(363.6 + 36.36) \times 10^{-6}} = 50.755 \text{ k}\Omega$$

Example 4.28 Determine the stability factor for the circuit shown in Fig. 4.22. [JNTU June 2009]

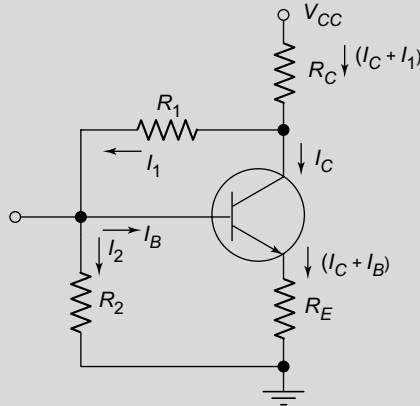


Fig. 4.22

$$\begin{aligned} \text{Solution } I_2 &= \frac{V_{BE} + (I_C + I_B)R_E}{R_2} \\ I_1 &= I_B + I_2 \\ I_1 &= I_B + \frac{V_{BE} + (I_C + I_B)R_E}{R_2} \\ &= \frac{I_B R_2 + V_{BE} + (I_C + I_B)R_E}{R_2} \end{aligned} \quad (1)$$

Applying KVL to the collector, base-emitter loop, we have

$$\begin{aligned} V_{CC} - (I_C + I_1)R_C - I_1R_1 - V_{BE} - (I_C + I_B)R_E \\ V_{CC} = (I_C + I_1)R_C + I_1R_1 + V_{BE} + (I_C + I_B)R_E \\ = I_C R_C + I_1 R_C + I_1 R_1 + V_{BE} + I_C R_E + I_B R_E \\ = I_C (R_C + R_E) + I_1 (R_C + R_1) + V_{BE} + I_B R_E \end{aligned}$$

Substituting the value of I_1 from Eqn.(1), we get

$$\begin{aligned} V_{CC} = I_C (R_C + R_E) + \frac{I_B R_2 + V_{BE} + (I_C + I_B)R_E}{R_2} (R_C + R_1) + V_{BE} + I_B R_E \\ = I_C \left[R_C + R_E + \frac{R_E (R_C + R_1)}{R_2} \right] + I_B \left[R_E + \frac{(R_E + R_2)(R_C + R_1)}{R_2} \right] \\ + \left[1 + \frac{(R_C + R_1)}{R_2} \right] V_{BE} \end{aligned}$$

We know that $I_C = \beta I_B + (1 + \beta) I_{CO}$

Therefore,
$$I_B = \frac{I_C - (1 + \beta) I_{CO}}{\beta}$$

Substituting the value of I_B , we get

$$\begin{aligned} V_{CC} = I_C \left[R_C + R_E + \frac{R_E (R_C + R_1)}{R_2} \right] + \frac{I_C - (1 + \beta) I_{CO}}{\beta} \\ \left[R_E + \frac{(R_E + R_2)(R_C + R_1)}{R_2} \right] + \left[1 + \frac{(R_C + R_1)}{R_2} \right] V_{BE} \end{aligned}$$

We know that $S = \frac{dI_C}{dI_{CO}}$. Hence, differentiating the above equation and assuming V_{BE} constant, we get

$$\begin{aligned} 0 = \frac{\partial I_C}{\partial I_{CO}} \left[R_C + R_E + \frac{R_E (R_C + R_1)}{R_2} \right] + \frac{\partial I_C}{\partial I_{CO}} \times \frac{1}{\beta} \left[R_E + \frac{(R_E + R_2)(R_C + R_1)}{R_2} \right] \\ - \frac{(1 + \beta)}{\beta} \left[R_E + \frac{(R_E + R_2)(R_C + R_1)}{R_2} \right] \\ \frac{1 + \beta}{\beta} \left[\frac{R_2 R_E + R_E R_C + R_E R_1 + R_2 R_C + R_1 R_2}{R_2} \right] = \end{aligned}$$

$$\begin{aligned}
& \frac{\partial I_C}{\partial I_{CO}} \left[\frac{R_2 R_C + R_2 R_E + R_E R_C + R_E R_1}{R_2} \right] + \frac{\partial I_C}{\partial I_{CO}} \times \frac{1}{\beta} \\
& \quad \left[\frac{R_2 R_E + R_E R_C + R_E R_1 + R_2 R_C + R_1 R_2}{R_2} \right] \\
& \frac{1 + \beta}{\beta} \left[\frac{R_2 (R_C + R_1) + R_E (R_1 + R_2 + R_C)}{R_2} \right] = \\
& \frac{\partial I_C}{\partial I_{CO}} \left[\frac{R_2 R_C + R_E (R_1 + R_2 + R_C)}{R_2} \right] + \frac{\partial I_C}{\partial I_{CO}} \times \frac{1}{\beta} \\
& \quad \left[\frac{R_2 (R_C + R_1) + R_E (R_1 + R_2 + R_C)}{R_2} \right] \\
& \frac{1 + \beta}{\beta} \left[\frac{R_2 (R_C + R_1) + R_E (R_1 + R_2 + R_C)}{R_2} \right] = \\
& \frac{\partial I_C}{\partial I_{CO}} \left[\frac{R_2 R_C + R_E (R_1 + R_2 + R_C)}{R_2} \right] + \left[\frac{R_2 (R_C + R_1) + R_E (R_1 + R_2 + R_C)}{\beta R_2} \right] \\
& \frac{\partial I_C}{\partial I_{CO}} = \frac{\frac{1 + \beta}{\beta} [R_2 (R_C + R_1) + R_E (R_1 + R_2 + R_C)]}{R_2 R_C + R_E (R_1 + R_2 + R_C) + \frac{R_2 (R_C + R_1) + R_E (R_1 + R_2 + R_C)}{\beta}} \\
& = \frac{\frac{1 + \beta}{\beta} [R_2 (R_C + R_1) + R_E (R_1 + R_2 + R_C)]}{\frac{\beta (R_2 R_C + R_E (R_1 + R_2 + R_C)) + R_2 (R_C + R_1) + R_E (R_1 + R_2 + R_C)}{\beta}} \\
& S = \frac{\partial I_C}{\partial I_{CO}} = \frac{(1 + \beta) [R_2 (R_C + R_1) + R_E (R_1 + R_2 + R_C)]}{R_1 R_2 + (\beta + 1) [R_2 R_C + R_E (R_1 + R_2 + R_C)]}
\end{aligned}$$

4.4.1 Common Base Stability

In a common base amplifier circuit, the equation for collector current I_C is given by

$$I_C = \alpha I_E + I_{CO}$$

$$S \approx dI/dI_{CO} = 1$$

Since this is highly stable, the common base amplifier circuit is not in need of bias stabilization.

4.4.2 Advantage of Self-Bias (Voltage Divider Bias) Over Other Types of Biasing

In fixed bias method discussed in Section 4.3.1, the stability factor is given by

$$S = 1 + \beta$$

Since β , is normally a large quantity, this circuit provides very poor stability. Therefore the fixed biasing technique is not preferred for biasing the base.

In collector-to-base bias method discussed in Section 4.3.3, when R_C is very small, $S \approx 1 + \beta$, which is equal to that of fixed bias. Hence, collector-to-base bias method is also not preferable. In self bias method discussed in Section 4.3.5, when R_B/R_E is very small, $S \approx 1$, which provides good stability. Hence, self bias method is the best one over other types of “biasing”.

4.5 BIAS COMPENSATION USING DIODE AND TRANSISTOR

The various biasing circuits considered in the previous sections used some types of negative feedback to stabilise the operation point. Also, diodes, thermistors and sensistors can be used to compensate for variations in current.

4.5.1 Diode Compensation

Figure 4.23 shows a transistor amplifier with a diode D connected across the base-emitter junction for compensation of change in collector saturation current I_{CO} . The diode is of the same material as the transistor and it is reverse biased by the base-emitter junction voltage V_{BE} , allowing the diode reverse saturation current I_O to flow through diode D . The base current $I_B = I - I_O$.

As long as temperature is constant, diode D operates as a resistor. As the temperature increases, I_{CO} of the transistor increases. Hence, to compensate for this, the base current I_B should be decreased.

The increase in temperature will also cause the leakage current I_O through D to increase and thereby decreasing the base current I_B . This is the required action to keep I_C constant.

This method of bias compensation does not need a change in I_C to effect the change in I_B , as both I_O and I_{CO} can track almost equally according to the change in temperature.

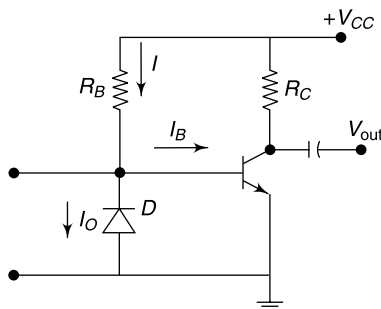


Fig. 4.23 Diode Bias Compensation

4.5.2 Thermistor Compensation

In Fig. 4.24, a thermistor, R_T , having a negative temperature coefficient is connected in parallel with R_2 . The resistance of thermistor decreases exponentially

with increase of temperature. An increase in temperature will decrease the base voltage V_{BE} , reducing I_B and I_C . Bias stabilization is also provided by R_E and C_E .

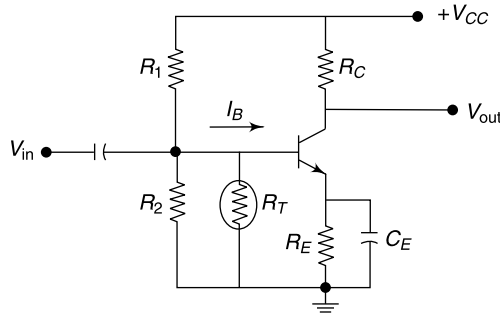


Fig. 4.24 Thermistor Bias Compensation

4.5.3 Sensistor Compensation

In Fig. 4.25, a sensistor, R_S , having a positive temperature coefficient is connected across R_1 (or R_E). R_S increases with temperature. As temperature increases, the equivalent resistance of the parallel combination of R_1 and R_S also increases and hence the base voltage V_{BE} decreases, reducing I_B and I_C . This reduced I_C compensates for the increased I_C caused by the increase in I_{CO} , V_{BE} and β due to temperature rise.

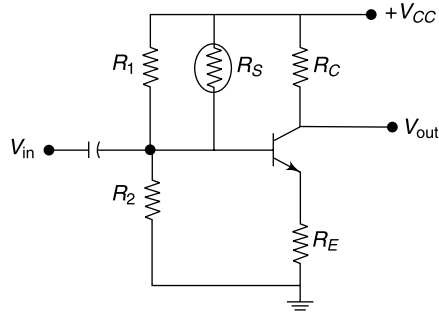


Fig. 4.25 Sensistor Bias Compensation

4.6 THERMAL RUNAWAY

The collector current for the CE circuit of Fig. 4.2 is given by $I_C = \beta I_B + (1 + \beta) I_{CO}$. The three variables in the equation, β , I_B and I_{CO} increase with rise in temperature. In particular, the reverse saturation current or leakage current I_{CO} changes greatly with temperature. Specifically, it doubles for every 10°C rise in temperature. The collector current I_C causes the collector-base junction temperature to rise which, in turn, increase I_{CO} , as a result I_C will increase still further, which will further rise the temperature at the collector-base junction. This process will become cumulative leading to “thermal runaway.” Consequently, the ratings of the transistor are exceeded which may destroy the transistor itself.

The collector is normally made larger in size than the emitter in order to help dissipate the heat developed at the collector junction.

However, if the circuit is designed such that the base current I_B is made to decrease automatically with rise in temperature, then the decrease in βI_B will compensate for the increase in $(1 + \beta) I_{CO}$, keeping I_C almost constant.

In power transistors, the heat developed at the collector junction may be removed by the use of heat sink, which is a metal sheet fitted to the collector and whose surface radiates heat quickly.

4.7 THERMAL RESISTANCE

Consider a transistor used in a circuit where the ambient temperature of the air around the transistor is T_A °C and the temperature of the collector-base junction of the transistor is T_J °C. Due to heating within the transistor T_J is higher than T_A . As the temperature difference $T_J - T_A$ is greater, the power dissipated in the transistor, P_D will be greater, i.e. $T_J - T_A \propto P_D$.

This equation can be written as $T_J - T_A = \Theta P_D$, where Θ is the constant of proportionality and is called the Thermal resistance. Rearranging the above equation $\Theta = T_J - T_A / P_D$. Hence Θ is measured in °C/W which may be as small as 0.2°C/W for a high power transistor that has an efficient heat sink or up to 1000°C/W for small signal, low power transistors which have no cooling provision.

As Θ represents total thermal resistance from a transistor junction to the ambient temperature, it is commonly referred to as Θ_{J-A} . However, for power transistors, thermal resistance is given from junction to case, Θ_{J-C} .

The amount of power that may be safely dissipated in the transistor is given by

$$P_D = (T_J - T_A) / \Theta_{J-A}$$

or

$$P_D = (T_J - T_C) / \Theta_{J-C}$$

The thermal resistance from junction to ambience is considered to consist of two parts.

$$\Theta_{J-A} = \Theta_{J-C} + \Theta_{C-A}$$

which indicates that heat dissipated in the junction must make its way to the surrounding air through two series paths from junction to case and from case to air. Hence, the power dissipated

$$\begin{aligned} P_D &= (T_J - T_A) / \Theta_{J-A} \\ &= (T_J - T_A) / (\Theta_{J-C} + \Theta_{C-A}) \end{aligned}$$

Θ_{J-C} is determined by the type of manufacture of the transistor and how it is located in the case, but Θ_{C-A} is determined by the surface area of the case or flange and its contact with air. If the effective surface area of the transistor case could be increased, the resistance to heat flow, or Θ_{C-A} , could be decreased. This can be achieved by the use of a heat sink.

The heat sink is a relatively large, finned, usually black metallic heat conducting device placed in close contact with the transistor case or flange. Many versions of heat sink exist depending upon the shape and size of the transistor.

Larger the heat sink, smaller will be its thermal resistance, Θ_{HS-A} . This thermal resistance is not added to Θ_{C-A} in series, but is instead in parallel with it and if Θ_{HS-A} is much less than Θ_{C-A} , then Θ_{C-A} will be reduced significantly, thereby improving the dissipation capability of the transistor.

Thus,

$$\Theta_{J-A} = \Theta_{J-C} + \Theta_{C-A} \parallel \Theta_{HS-A}$$

Example 4.29 For a given transistor, the thermal resistance is 8°C/W and the ambient temperature T_A is 27°C . If the transistor dissipates 3 W of power, calculate the junction temperature T_J .

Solution We know that, $T_J = T_A + \Theta P_D$

$$\begin{aligned}\text{Therefore, } T_J &= 27^\circ\text{C} + (8^\circ\text{C/W}) \times 3\text{ W} \\ &= 27^\circ\text{C} + 24^\circ\text{C} = 51^\circ\text{C}\end{aligned}$$

Example 4.30 For a transistor, $T_J = 160^\circ\text{C}$, $T_A = 40^\circ\text{C}$ and $\Theta_{J-A} = 80^\circ\text{C/W}$. Calculate the power that the transistor can safely dissipate in free air.

$$\text{Solution } P_D = \frac{T_J - T_A}{\Theta_{J-A}} = \frac{160 - 40}{80} = \frac{120}{80} = 1.5\text{ W}$$

Example 4.31 Determine the power dissipation capability of a transistor which has been mounted with a heat sink having thermal resistance $\Theta_{HS-A} = 8^\circ\text{C/W}$, $T_A = 40^\circ\text{C}$, $T_J = 160^\circ\text{C}$, $\Theta_{J-C} = 5^\circ\text{C/W}$ and $\Theta_{C-A} = 85^\circ\text{C/W}$.

Solution We know that $\Theta_{J-A} = \Theta_{J-C} + \Theta_{C-A} \parallel \Theta_{HS-A}$

$$\begin{aligned}&= 5 + 85 \parallel 8 \\ &= 5 + \frac{85 \times 8}{85 + 8} = 5 + 7.31 = 12.31^\circ\text{C/W} \\ P_D &= \frac{T_J - T_A}{\Theta_{J-A}} = \frac{160 - 40}{12.31} = \frac{120}{12.31} = 9.75\text{ W}\end{aligned}$$

4.8 CONDITION FOR THERMAL STABILITY

For preventing thermal runaway, the required condition is that the rate at which the heat is released at the collector junction should not exceed the rate at which the heat can be dissipated under steady state condition. Hence, the condition to be satisfied to avoid thermal runaway is given by

$$\frac{\partial P_C}{\partial T_J} < \frac{1}{\Theta} \quad (4.29)$$

If the circuit is properly designed, then the transistor cannot runaway below a specified ambient temperature or even under any conditions.

In the self bias circuit, the transistor is biased in the active region. The power generated at the collector junction without any signal is

$$P_C = I_C V_{CB} \approx I_C V_{CE}$$

Let us assume that the quiescent collector and emitter currents are equal. Then,

$$P_C = I_C V_{CC} - I_C^2 (R_E + R_C) \quad (4.30)$$

The condition to prevent thermal runaway can be rewritten as

$$\frac{\partial P_C}{\partial I_C} \frac{\partial I_C}{\partial T_J} < \frac{1}{\Theta} \quad (4.31)$$

As Θ and $\frac{\partial I_C}{\partial T_J}$ are positive, $\frac{\partial P_C}{\partial I_C}$ should be negative in order to satisfy the above condition.

Differentiating Eq. (4.30) with respect to I_C , we get

$$\frac{\partial P_C}{\partial I_C} = V_{CC} - 2 I_C (R_E + R_C) \quad (4.32)$$

Hence to avoid thermal runaway, it is necessary that

$$I_C > \frac{V_{CC}}{2(R_E + R_C)} \quad (4.33)$$

Since $V_{CE} = V_{CC} - I_C(R_E + R_C)$, then Eq. (4.33) implies that $V_{CE} < V_{CC}/2$. If the inequality of Eq. (4.33) is not satisfied and $V_{CE} < V_{CC}/2$, then from Eq. (4.32), $\frac{\partial P_C}{\partial I_C}$ is positive, and corresponding Eq. (4.31) should be satisfied; otherwise, thermal runaway will occur.

When the effect of I_{CO} dominates, $\Delta I_C = S \Delta I_{CO}$. Therefore, Eq. (4.31) becomes

$$\frac{\partial P_C}{\partial I_C} \left(S \frac{\partial I_{CO}}{\partial T_J} \right) < \frac{1}{\Theta} \quad (4.34)$$

Since the reverse saturation current for either silicon or germanium increases about 7 percent/ $^{\circ}\text{C}$.

$$\frac{\partial I_{CO}}{\partial T_J} = 0.07 I_{CO} \quad (4.35)$$

Substituting Eqs (4.32) and (4.35) in Eq. (4.34), we get

$$[V_{CC} - 2I_C(R_E + R_C)](S)(0.07I_{CO}) < \frac{1}{\Theta} \quad (4.36)$$

The amplifier circuits operating at low current and lower value of stability factor i.e. $S < 10$ are not susceptible to thermal runaway. But in power amplifiers operating at high power levels, the value of R_E is low for power efficiency and hence the stability factor S becomes high. As a result, thermal runaway occurs. Hence to avoid thermal runaway, a heat sink may be attached to the collector of the power transistor.

4.9 TYPES OF HEAT SINKS

Low Power Transistor Type The small signal low power transistors can be mounted directly on the metal chassis to increase the sufficient heat dissipation capability. Care should be taken while doing this because very often the collector of the transistor is connected to the transistor case to increase heat-dissipation capabilities. Hence, some provision for insulating the case from the chassis, which is usually at ground potential, must be provided unless a common collector is being employed.

One method of achieving this is to use a beryllium oxide insulating washer which has a good thermal conductivity, as shown in Fig. 4.26(a). By using a zinc oxide film Silicon compound between the washer and the chassis, heat transfer from the transistor case to the chassis may be improved. An insulated clamp over the top of the transistor may be used to help improve thermal dissipation and increase pressure.

When the transistor is mounted in Teflon (PTFE Poly Tere Fluoro Ethylene) sockets, it does not provide thermal conduction from transistor case to chassis. Therefore, a press-on fin type of a black anodized heat sink may be used, as shown in Fig. 4.26(b), for mounting transistors that are encased in a metal TO-5 package.

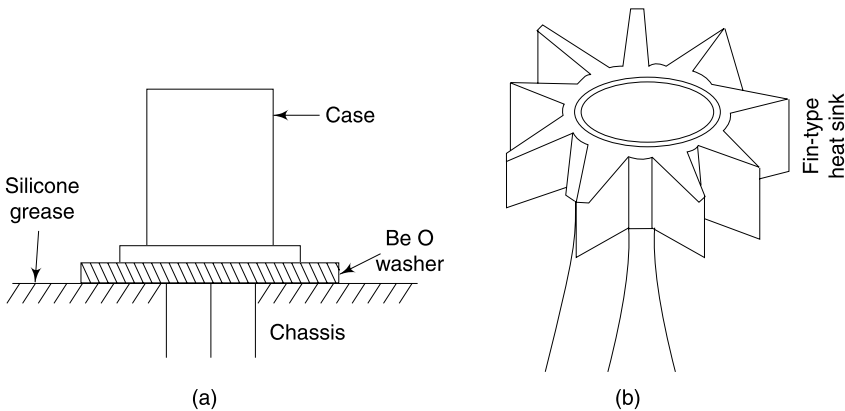


Fig. 4.26 (a) Mounting the Transistor Case Close to the Chassis using a Beryllium Oxide Insulating Washer (b) Using a Separate Heat Sink Pressed onto the Transistor

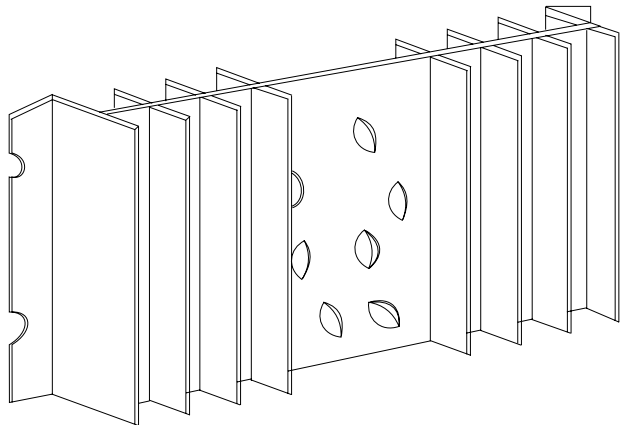


Fig. 4.27 Power Transistor Heat Sink

Power Transistor Heat Sinks The diamond shaped TO-3 and TO-66 types are the popular mounting packages used for the power transistors which have dissipation in the order of 100 W. These have two leads for emitter and base but the case, or the mounting flange of the case, is the collector terminal. So, it is necessary to insulate the case from the heat sink by the use of an insulating washer. Figure 4.27 shows a typical heat sink that can accommodate a TO-3 power transistor package that provides cooling by conduction, convection and radiation. Although measuring only 11.5 cm by 7.8 cm, it has a thermal dissipation equal to that of a flat aluminium sheet 25 cm $\times$ 20 cm $\times$ 0.32 cm. The thermal resistance of this heat sink is 3°C/W.

4.10 ANALYSIS OF A TRANSISTOR AMPLIFIER CIRCUIT USING *h*-PARAMETERS

A transistor amplifier can be constructed by connecting an external load and signal source as indicated in Fig. 4.28 and biasing the transistor properly.

The two port active network of Fig. 4.29 represents a transistor in any one of its configuration. The hybrid equivalent circuit is valid for any type of load whether it is pure resistance or impedance or another transistor. It is assumed that *h*-parameters remain constant over the operating range. Further, the input is sinusoidal and I_1 , V_1 , I_2 and V_2 are phasor quantities.

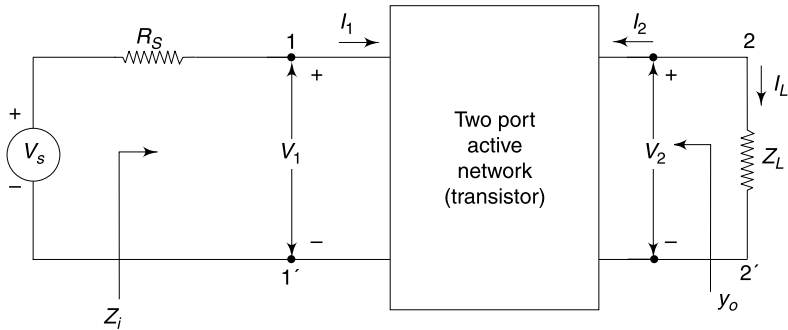


Fig. 4.28 Basic Amplifier Circuit

4.10.1 Current Gain or Current Amplification, A_i

For a transistor amplifier the current gain A_i is defined as the ratio of output current to input current, i.e.,

$$A_i = \frac{I_L}{I_1} = \frac{-I_2}{I_1} \quad (4.37)$$

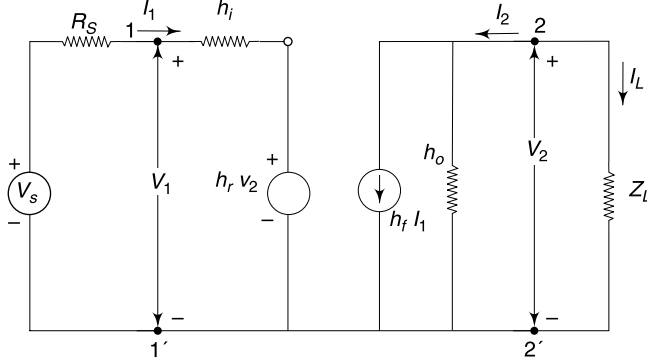


Fig. 4.29 Circuit of Fig. 4.28 with Transistor Replaced by its Hybrid Model

From the circuit of Fig. 4.29,

$$I_2 = h_f I_1 + h_o V_2 \quad (4.38)$$

Substituting $V_2 = I_L Z_L = -I_2 Z_L$,

$$I_2 = h_f I_1 - I_2 Z_L h_o$$

$$I_2 + I_2 Z_L h_o = h_f I_1$$

$$I_2 (1 + Z_L h_o) = h_f I_1$$

$$A_i = \frac{-I_2}{I_1} = \frac{-h_f}{1 + h_o Z_L} \quad (4.39)$$

Therefore,

$$A_i = \frac{-h_f}{1 + h_o Z_L}$$

4.10.2 Input Impedance, Z_i

In the circuit of Fig. 4.29, R_S is the signal source resistance. The impedance seen when looking into the amplifier terminals (1,1') is the amplifier input impedance Z_i , i.e.,

$$Z_i = \frac{V_1}{I_1} \quad (4.40)$$

From the input circuit of Fig. 4.29, $V_1 = h_i I_1 + h_r V_2$

Hence,

$$\begin{aligned} Z_i &= \frac{h_i I_1 + h_r V_2}{I_1} \\ &= h_i + h_r \frac{V_2}{I_1} \end{aligned}$$

Substituting

$$\begin{aligned} V_2 &= -I_2 Z_L = A_I I_1 Z_L \\ Z_i &= h_i + h_r \frac{A_I I_1 Z_L}{I_1} \end{aligned}$$

resulting in

$$Z_i = h_i + h_r A_I Z_L \quad (4.41)$$

Substituting for A_I ,

$$\begin{aligned} Z_i &= h_i - \frac{h_f}{1 + h_o Z_L} h_r Z_L \\ &= h_i - \frac{h_f h_r}{Z_L \left(\frac{1}{Z_L} + h_o \right)} Z_L \end{aligned}$$

Taking the load admittance as $Y_L = \frac{1}{Z_L}$

$$Z_i = h_i - \frac{h_f h_r}{Y_L + h_o} \quad (4.42)$$

Note that the input impedance is a function of load impedance.

4.10.3 Voltage Gain or Voltage Amplification Factor, A_V

The ratio of output voltage v_2 to input voltage v_1 gives the voltage gain of the transistor, i.e.

$$A_V = \frac{V_2}{V_1} \quad (4.43)$$

Substituting

$$\begin{aligned} V_2 &= -I_2 Z_L = A_I I_1 Z_L \\ A_V &= \frac{A_I I_1 Z_L}{V_1} = \frac{A_I Z_L}{Z_i} \end{aligned} \quad (4.44)$$

4.10.4 Output Admittance, Y_o

By definition, Y_o is obtained by setting V_S to zero, Z_L to infinity and by driving the output terminals from a generator V_2 . If the current drawn from V_2 is I_2 , then

$$Y_o \equiv \frac{I_2}{V_2} \text{ with } V_S = 0 \text{ and } R_L = \infty$$

From the circuit of Fig. 4.29,

$$I_2 = h_f I_1 + h_o V_2$$

Dividing by V_2 ,

$$\frac{I_2}{V_2} = h_f \frac{I_1}{V_2} + h_o \quad (4.45)$$

With $V_S = 0$, by KVL in input circuit,

$$R_S I_1 + h_i I_1 + h_r V_2 = 0 \quad (4.46)$$

$$I_1 (R_S + h_i) + h_r V_2 = 0$$

Hence,

$$\frac{I_1}{V_2} = \frac{-h_r}{R_S + h_i}$$

Substituting Eq. (4.46) in Eq. (4.45),

$$\frac{I_2}{V_2} = h_f \left(\frac{-h_r}{R_S + h_i} \right) + h_o$$

$$Y_o = h_o - \frac{h_f h_r}{h_i + R_s} \quad (4.47)$$

From Eq. (4.47), the output admittance is a function of source resistance. If the source impedance is resistive then Y_o is real.

4.10.5 Voltage Amplification (A_{VS}) Taking into Account the Resistance (R_S) of the Source

This overall voltage gain A_{VS} is given by

$$A_{VS} = \frac{V_2}{V_S} = \frac{V_2}{V_1} \frac{V_1}{V_S} = A_V \frac{V_1}{V_S} \quad (4.48)$$

From the equivalent input circuit using Thevenin's equivalent for the source shown in Fig. 4.30,

$$V_1 = \frac{V_S Z_i}{Z_i + R_S} \quad (4.49)$$

$$\frac{V_1}{V_S} = \frac{Z_i}{Z_i + R_S}$$

Then, $A_{VS} = \frac{A_V Z_i}{Z_i + R_S} \quad (4.50)$

Substituting $A_V = \frac{A_I Z_L}{Z_i}$

$$A_{VS} = \frac{A_I Z_L}{Z_i + R_S} \quad (4.51)$$

Note that if $R_S = 0$, then $A_{VS} = \frac{A_I Z_L}{Z_i} = A_V$. Hence, A_V is the voltage gain with an ideal voltage source (with $R_S = 0$). In practice, A_{VS} is more meaningful than A_V because source resistance has an appreciable effect on the overall amplification.

4.10.6 Current Amplification (A_{IS}) Taking into Account the Source Resistance (R_S)

The modified input circuit using Norton's equivalent circuit for the source for the calculation of A_{IS} is shown in Fig. 4.31.

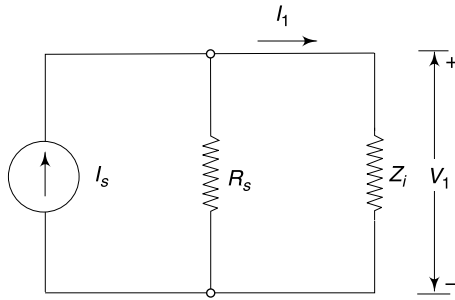


Fig. 4.31 Modified Input Equivalent Circuit

Overall current gain, $A_{IS} = \frac{-I_2}{I_S} = \frac{-I_2}{I_1} = \frac{I_1}{I_S} = A_I \frac{I_1}{I_S} \quad (4.52)$

From Fig. 4.31, $I_1 = I_S \frac{R_S}{R_S + Z_i} \quad (4.53)$

$$\frac{I_1}{I_S} = \frac{R_S}{R_S + Z_i}$$

and hence,
$$A_{IS} = A_I \frac{R_S}{R_S + Z_i} \quad (4.54)$$

If $R_S = \infty$, then $A_{IS} = A_I$. Hence, A_I is the current gain with an ideal current source (one with infinite source resistance).

From Eq. (4.51),

$$A_{VS} = \frac{A_I Z_L}{Z_i + R_S} \frac{R_S}{R_S}$$

Then,
$$A_{VS} = \frac{A_{IS} Z_L}{R_S} \quad (4.55)$$

4.10.7 Operating Power Gain, A_p

From Fig. 4.29 average power delivered to the load is $P_2 = |V_2|I_L| \cos\theta$, where θ is the phase angle between V_2 and I_L . Assume that Z_L is resistive i.e. $Z_L = R_L$. Since h -parameters are real at low frequencies, the power delivered to the load is $P_2 = V_2 I_L = -V_2 I_2$. Since the input power $P_1 = V_1 I_1$, the operating power gain A_p of the transistor is defined as

$$A_p = \frac{P_2}{P_1} = \frac{-V_2 I_2}{V_1 I_1} = A_V A_I = A_I A_I \frac{R_L}{R_i}$$

$$A_p = A_I^2 \left(\frac{R_L}{R_i} \right) \quad (4.56)$$

The important relations derived above are summarised in Table 4.1.

Table 4.1 Small Signal Analysis of a Transistor Amplifier

$A_I = \frac{-h_f}{1 + h_o Z_L}$	$A_V = \frac{A_I Z_L}{Z_i}$
$Z_i = h_i + h_r A_I Z_L = h_i - \frac{h_f h_r}{Y_L + h_o}$	$A_{VS} = \frac{A_V Z_i}{Z_i + R_S} = \frac{A_I Z_L}{Z_i + R_S} = A_{IS} \frac{Z_L}{R_S}$
$Y_o = h_o - \frac{h_f h_r}{h_i + R_S} = \frac{1}{Z_o}$	$A_{IS} = \frac{A_I R_S}{Z_i + R_S} = A_{VS} \frac{R_S}{Z_L}$

4.11 SIMPLIFIED CE HYBRID MODEL

As the h -parameters themselves vary widely for the same type of transistor, it is justified to make approximations and simplify the expressions for A_I , A_V , A_P , R_i and R_o . In addition, a better understanding of the behaviour of the transistor circuit can be obtained by using the simplified hybrid model. Since CE configuration is more useful and general, it is taken for consideration.

The h -parameter equivalent circuit of the transistor in the CE configuration is shown in Fig. 4.32. Here, $\frac{1}{h_{oe}}$ is in parallel with R_L . The parallel combination of two unequal impedances, i.e. $\frac{1}{h_{oe}}$ and R_L is approximately equal to the lower value, i.e. R_L . Hence, if $\frac{1}{h_{oe}} \gg R_L$, then the term h_{oe} may be neglected provided that $h_{oe}R_L \ll 1$. Further, if h_{oe} is omitted, the collector current I_C is given by $I_C = h_{fe}I_b$.

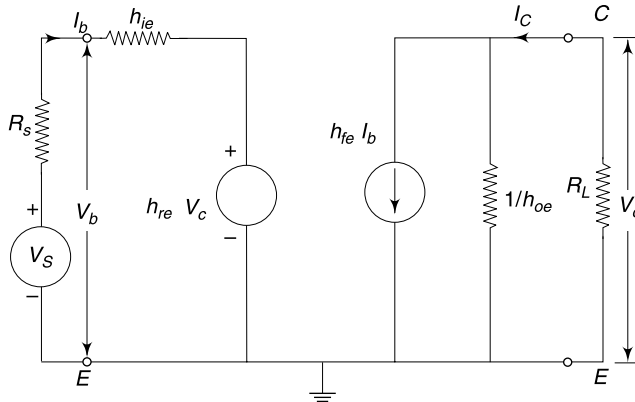


Fig. 4.32 Exact CE Hybrid Model

Under this condition the magnitude of voltage of generator in the emitter circuit is

$$h_{re}|V_C| = h_{re}I_C R_L = h_{re}h_{fe}I_b R_L$$

Since $h_{re}h_{fe} \approx 0.01$, this voltage may be neglected in comparison with the voltage drop across $h_{ie} = h_{ie}I_b$ provided that R_L is not too large.

To conclude, if the load resistance R_L is small it is possible to neglect the parameter h_{re} and h_{oe} and obtain the approximate equivalent circuit as shown in Fig. 4.33. It can be shown that if $h_{oe}R_L \leq 0.1$ the error in calculating A_I , A_V , R_i and R_o for CE configuration is less than 10%.

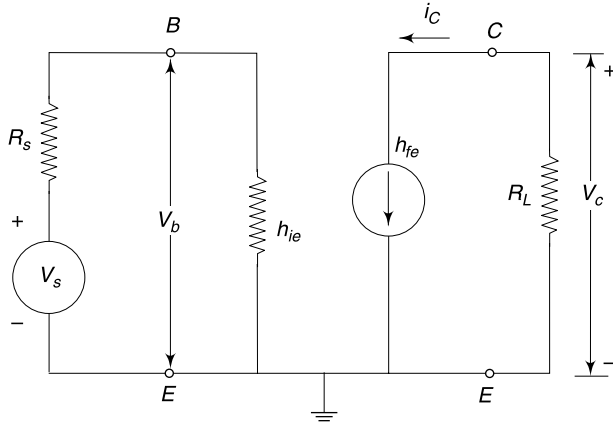


Fig. 4.33 Approximate CE Hybrid Model

4.11.1 Generalised Approximate Model

Figure 4.34 shows the simplified hybrid circuit that can be used for any configuration by simply grounding the appropriate terminal. The signal is connected between input and ground and the load is connected between output and ground. The errors introduced in calculating the various parameters using the approximate hybrid model are to be found out now. This is done for CE configuration.

Current gain From Eq. (4.39), the current gain for CE configuration

$$A_I = \frac{-h_f}{1 + h_{oe}R_L}$$

If

$$h_{oe}R_L < 0.1$$

$$A_I \approx -h_{fe} \quad (4.57)$$

This new A_I overestimates the magnitude of current gain by less than 10% provided $h_{oe}R_L < 0.1$.

Input Impedance From Eq. (4.41), the input impedance

$$R_i = h_{ie} + h_{re}A_I R_L$$

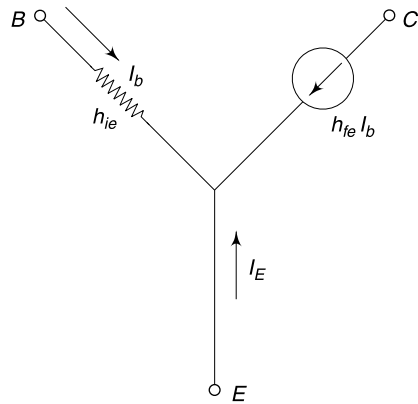


Fig. 4.34 Approximate Hybrid Model Valid for All Configurations

It may be put in the form

$$R_i = h_{ie} \left(1 - \frac{h_{re} h_{fe} |A_I| h_{oe} R_L}{h_{ie} h_{oe} h_{fe}} \right)$$

Using the typical values for the h -parameters,

$$\frac{h_{re} h_{fe}}{h_{ie} h_{oe}} \approx 0.5$$

Further, $|A_I| = \frac{h_{fe}}{1 + h_{oe} R_L} \approx h_{fe}$

Hence, the equation approximates to

$$R_i \approx h_{ie} \left(1 - \frac{0.5 h_{fe} h_{oe} R_L}{h_{fe}} \right)$$

If $h_{oe} R_L < 0.1$

$$R_i \approx h_{ie} \frac{V_b}{I_b} \quad (4.58)$$

This overestimates the input resistance by less than 5%.

Voltage Gain From Eq. (4.44),

Voltage gain, $A_V = A_I \frac{R_L}{R_i} = - \frac{h_{fe} R_L}{h_{ie}} \quad (4.59)$

By taking the logarithm of this equation and then the differential,

$$\frac{dA_V}{A_V} = \frac{dA_I}{A_I} - \frac{dR_i}{R_i}$$

with $h_{oe} R_L < 0.1$, $\frac{dA_I}{A_I} = +0.1$ and $\frac{dR_i}{R_i} = +0.05$

Therefore, $\frac{dA_V}{A_V} = 0.1 - 0.05 = 0.05$

$$\frac{dA_V}{A_V} \% = 5\%$$

Hence, the maximum error in voltage gain is 5% and the magnitude of A_V is over estimated by this amount.

Output Impedance It is the ratio of V_C to I_C with $V_S = 0$ and R_L excluded. The simplified circuit has infinite output impedance because with $V_S = 0$ and external voltage source applied at the output, it is found that $I_b = 0$ and hence, $I_C = 0$.

However the actual value of output impedance depends on the source resistance R_S and lies between 40 k Ω and 80 k Ω . With the load resistance R_L included, the output resistance R_o calculated using the approximate model increases, but not more than 10%.

Example 4.32 A CE amplifier is drawn by a voltage source of internal resistance $r_s = 800 \Omega$ and the load impedance is a resistance $R_L = 1000 \Omega$. The h -parameters are $h_{ie} = 1 \text{ k}\Omega$, $h_{re} = 2 \times 10^{-4}$, $h_{fe} = 50$ and $h_{oe} = 25 \mu\text{A/V}$. Compute the current gain A_I , input resistance R_i , voltage gain A_v and output resistance R_o using exact analysis and using approximate analysis.

Solution Exact analysis

$$\begin{aligned} \text{Current gain, } A_I &= \frac{-h_{fe}}{1 + h_{oe} R_L} \\ &= \frac{-50}{1 + 25 \times 10^{-6} \times 10^3} = -48.78 \end{aligned}$$

$$\begin{aligned} \text{Input resistance, } R_i &= h_{ie} - \frac{h_{fe} h_{re}}{h_{oe} + \frac{1}{R_L}} \\ &= 1000 - \frac{50 \times 2 \times 10^{-4}}{25 \times 10^{-6} + \frac{1}{1000}} = 990.24 \Omega \end{aligned}$$

$$\begin{aligned} \text{Voltage gain, } A_V &= A_I \frac{R_L}{R_i} \\ &= (-48.78) \times \frac{1000}{990.24} = -49.26 \end{aligned}$$

Output resistance, R_o

$$\begin{aligned} Y_o &= h_{oe} - \frac{h_{fe} h_{re}}{h_{ie} + R_S} \\ &= 25 \times 10^{-6} - \frac{50 \times 2 \times 10^{-4}}{1000 + 800} \\ &= 1.94 \times 10^{-5} \text{ mho} \\ R_o &= \frac{1}{Y_o} = 51.42 \text{ k}\Omega \end{aligned}$$

Approximate analysis

$$\begin{aligned} A_I &= -h_{fe} = -50 \\ R_i &= h_{ie} = 1 \text{ k}\Omega \\ A_V &= -\frac{h_{fe} R_L}{h_{ie}} = -\frac{50 \times 1000}{1000} = -50 \\ R_o &= \infty \end{aligned}$$

Example 4.33 Determine A_v , A_i , R_i , R_o for a CE amplifier using NPN transistor with $h_{ie} = 1200 \Omega$, $h_{re} = 0$, $h_{fe} = 36$ and $h_{oe} = 2 \times 10^{-6} \text{ Mhos}$. $R_L = 2.5 \text{ k}\Omega$, $R_s = 500 \Omega$ (neglect the effect of biasing circuit).

Solution To find current gain (A_i):

$$A_i = \frac{-h_{fe}}{1 + h_{oe} R_L} = \frac{-36}{1 + 2 \times 10^{-6} \times 2.5 \times 10^3} = -35.82$$

To find input resistance (R_i):

$$R_i = h_{ie} - \frac{h_{fe} h_{re}}{h_{oe} + \frac{1}{R_L}} = 1200$$

To find voltage gain (A_v):

$$A_v = A_i \frac{R_L}{R_i} = -35.82 \times \frac{2.5 \times 10^3}{1200} = -74.625$$

To find output resistance (R_o):

$$\begin{aligned} Y_o &= h_{oe} - \frac{h_{fe} h_{re}}{h_{ie} + R_s} \\ &= 2 \times 10^{-6} - \frac{36 \times 0}{1200 + 500} = 2 \times 10^{-6} \text{ mhos} \end{aligned}$$

Therefore,
$$R_o = \frac{1}{Y_o} = \frac{1}{2 \times 10^{-6}} = 500 \text{ k}\Omega$$

Example 4.34 The h -parameters of a transistor in the CE amplifier mode are $h_{ie} = 1100 \Omega$, $h_{re} = 2.5 \times 10^{-4}$, $h_{fe} = 50$ and $h_{oe} = 25 \mu\text{mho}$. Determine the current gain and input resistance of the amplifier for a load resistance $R_L = 1 \text{ k}\Omega$.

Solution Given $h_{ie} = 1100 \Omega$, $h_{re} = 2.5 \times 10^{-4}$, $h_{fe} = 50$, $h_{oe} = 25 \mu\text{mho}$ and $R_L = 1 \text{ k}\Omega$.

(a) The current gain of the amplifier is

$$A_i = \frac{-h_{fe}}{1 + h_{oe} R_L} = \frac{-50}{1 + 25 \times 10^{-6} \times 10^3} = -48.78$$

(b) The input resistance of the amplifier is

$$Z_i = h_{ie} - \frac{h_{re} h_{fe}}{h_{oe} + \frac{1}{R_L}} = 1100 - \frac{2.5 \times 10^{-4} \times 50}{25 \times 10^{-6} + 10^{-3}} = 1087.85 \Omega$$

Example 4.35 For a CE amplifier, if $R_L = R_S = 1000 \Omega$, $h_{ie} = 1100 \Omega$, $h_{re} = 2.5 \times 10^{-4}$, $h_{fe} = 50$ and $h_{oe} = 25 \mu A/V$, find A_I, R_i, A_V, A_{VS} and A_{IS} .

Solution Given $R_L = R_S = 1000 \Omega$, $h_{ie} = 1100 \Omega$, $h_{re} = 2.5 \times 10^{-4}$, $h_{fe} = 50$, and $h_{oe} = 25 \mu A/V$

$$(a) A_I = \frac{-h_{fe}}{1 + h_{oe} R_L} = \frac{-50}{1 + 25 \times 10^{-6} \times 10^3} \\ = \frac{-50}{1 + 25 \times 10^{-3}} = \frac{-50}{1.025} = -48.78$$

$$(b) R_i = h_{ie} - \frac{h_{fe} h_{re}}{h_{oe} + \frac{1}{R_L}} = 1100 - \frac{50 \times 2.5 \times 10^{-4}}{25 \times 10^{-6} + 10^{-3}} = 1100 - 12.19 = 1088 \Omega$$

$$(c) A_V = \frac{A_I R_L}{R_i} = -\frac{48.78 \times 1000}{1088} = -44.83$$

$$(d) A_{VS} = \frac{A_V R_i}{R_i + R_S} = \frac{-44.83 \times 1088}{2088} = -23.36$$

$$(e) A_{IS} = \frac{A_I R_S}{R_i + R_S} = -\frac{48.78 \times 1000}{2088} = -23.36$$

Example 4.36 The h -parameters of a transistor used in a CE circuit are $h_{ie} = 1 k\Omega$, $h_{re} = 10 \times 10^{-4}$, $h_{fe} = 50$ and $h_{oe} = 100 \mu A/V$. The load resistance for the transistor is $1 k\Omega$. Determine R_p, R_o, A_V, A_I in the amplifier stage. Assume $R_S = 1000 \Omega$. [JNTU 2008]

Solution (a) Current gain,

$$A_I = \frac{-h_{fe}}{1 + h_{oe} R_L} = \frac{-50}{1 + 100 \times 10^{-6} \times 10^3} = -45.45$$

$$(b) \text{ Input resistance, } R_i = h_{ie} - \frac{h_{fe} h_{re}}{h_{oe} + \frac{1}{R_L}} = 1 \times 10^3 - \frac{50 \times 10 \times 10^{-4}}{100 \times 10^{-6} + \frac{1}{10^3}} \\ = 954.55 \Omega$$

$$(c) \text{ Voltage gain, } A_V = A_I \frac{R_L}{R_i} = -45.45 \times \frac{10^3}{954.55} = -47.61$$

(d) Output resistance, R_o :

$$Y_o = h_{oe} - \frac{h_{fe} h_{re}}{h_{ie} + R_S} = 100 \times 10^{-6} - \frac{50 \times 10 \times 10^{-4}}{10^3 + 10^3} = 5 \times 10^{-5} \text{ mho}$$

$$\text{Therefore, } R_o = \frac{1}{Y_o} = \frac{1}{5 \times 10^{-5}} = 20 k\Omega$$

$$(e) R_{ot} = \frac{R_o \times R_L}{R_o + R_L} = \frac{20 \times 10^3 \times 1 \times 10^3}{20 \times 10^3 + 1 \times 10^3} = 952.4 \Omega$$

Approximate Analysis:

$$(a) A_I = -h_{fe} = -50$$

$$(b) R_i = h_{ie} = 1 \text{ k}\Omega$$

$$(c) A_V = -\frac{h_{fe} R_L}{h_{ie}} = -\frac{50 \times 10^3}{1 \times 10^3} = -50$$

$$(d) R_o = \infty$$

$$(e) R_{ot} = \infty \parallel R_L = R_L = 1 \text{ k}\Omega$$

4.11.2 Common Emitter Amplifier with Emitter Resistor

A simple and effective way to provide voltage gain stabilisation in a CE amplifier is to add an emitter resistor R_E which provides feedback as shown in Fig. 4.35. An approximate solution for this arrangement can be obtained by considering the simplified hybrid model equivalent circuit.

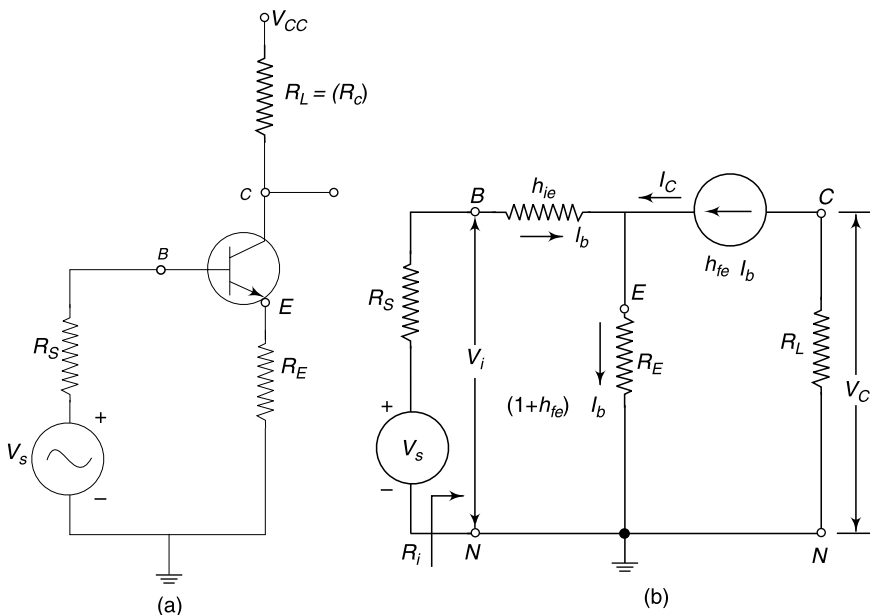


Fig. 4.35 (a) CE Amplifier with Emitter Resistor, (b) Approximate Small-signal Equivalent Circuit

Current Gain, A_I

$$A_I = -\frac{I_c}{I_b} = \frac{-h_{fe} I_b}{I_b} = -h_{fe}$$

Thus, the current gain is equal to the short circuit current gain with $R_E = 0$ and is unaffected by R_E .

Input Resistance, R_i

$$R_i = \frac{V_i}{I_b} = \frac{[h_{ie} + (1 + h_{fe}) R_E] I_b}{I_b} \quad (4.60)$$

$$R_i = h_{ie} + (1 + h_{fe}) R_E$$

Comparing with Eq. (4.58), the input resistance is augmented by $(1 + h_{fe}) R_E$ and may be very much larger than h_{ie} . For example, with $R_E = 1 \text{ k}\Omega$, $h_{fe} = 60$, $(1 + h_{fe}) R_E = 61 \text{ k}\Omega \gg h_{ie} \approx 1 \text{ k}\Omega$.

Hence, R_E greatly increases the input resistance of the amplifier.

Voltage Gain, A_V From Eq. (4.44),

$$A_V = A_I \frac{R_L}{R_i} = - \frac{h_{fe} R_L}{h_{ie} + (1 + h_{fe}) R_E} \quad (4.61)$$

Thus, the addition of emitter resistor R_E greatly reduces the voltage amplification as R_i has increased from h_{ie} to $h_{ie} + (1 + h_{fe}) R_E$. This reduction in gain is compensated by stability improvement.

$$\text{If } (1 + h_{fe}) R_E \gg h_{ie} \gg 1$$

$$\text{Then } A_V \approx - \frac{h_{fe} R_L}{(1 + h_{fe}) R_E} \approx - \frac{R_L}{R_E} \quad (4.62)$$

Hence, under these approximations A_V is completely stable and independent of all transistor parameters provided stable resistances are used for R_L and R_E .

Output Resistance, R_o Output resistance R_o with R_L excluded is infinite and with R_L included, it is equal to R_L and is independent of R_E .

Example 4.37 A CE amplifier uses load resistor $R_C = 2 \text{ k}\Omega$ in the collector circuit and is given by the voltage source V_S of internal resistance 1000Ω . The h -parameters of the transistor are $h_{ie} = 1300 \Omega$, $h_{re} = 2 \times 10^{-4}$, $h_{fe} = 55$ and $h_{oe} = 22 \mu\text{Mhos}$. Neglecting the biasing resistors supply, compute the current gain A_I , input resistance R_i , voltage gain A_V , output resistance R_o and output terminal resistance R_{oT} for the following values of emitter resistor R_E inserted in the emitter circuit: (a) 200Ω (b) 400Ω and (c) 1000Ω . Use the approximate model for the transistor if permissible.

Solution (a) For $R_E = 200 \Omega$,

$$h_{oe} \times (R_E + R_C) = (2 \times 10^3 + 200) \times (22 \times 10^{-6}) = 0.0484$$

Since $h_{oe} \times (R_E + R_C) < 0.1$, the approximate model is permissible.

$$A_I = - h_{fe} = - 55$$

$$R_i = h_{ie} + (1 + h_{fe}) R_E = 12.5 \text{ k}\Omega$$

$$\begin{aligned}
 A_V &= A_I \frac{R_C}{R_i} \\
 &= -55 \times \frac{2000}{12,500} = -8.8
 \end{aligned}$$

Output resistance, $R_o = \infty$

Output terminal resistance, $R_{OT} = R_o \parallel R_C = 2 \text{ k}\Omega$

(b) For $R_E = 400 \Omega$

$$h_{oe} \times (R_E + R_C) = (2 \times 10^3 + 400) \times (22 \times 10^{-6}) = 0.0528$$

Since $h_{oe} \times (R_E + R_C) < 0.1$, approximate model is permissible.

$$\begin{aligned}
 A_I &= -h_{fe} = -55 \\
 R_i &= h_{ie} + (1 + h_{fe})R_E = 23.7 \text{ k}\Omega \\
 A_V &= A_I \frac{R_C}{R_i} \\
 &= -55 \times \frac{2000}{23,700} = -4.64
 \end{aligned}$$

Output resistance, $R_o = \infty$

Output terminal resistance, $R_{OT} = R_o \parallel R_C = 2 \text{ k}\Omega$

(c) For $R_E = 1000 \Omega$

Since $h_{oe} \times (R_E + R_C) < 0.1$, approximate model is permissible.

$$\begin{aligned}
 A_I &= -h_{fe} = -55 \\
 R_i &= h_{ie} + (1 + h_{fe})R_E = 57.3 \text{ k}\Omega \\
 A_V &= A_I \frac{R_C}{R_i} \\
 &= -55 \times \frac{2000}{57,300} = -1.92
 \end{aligned}$$

Output resistance, $R_o = \infty$

Output terminal resistance, $R_{OT} = R_o \parallel R_C = 2 \text{ k}\Omega$

4.12 ANALYSIS OF CC AMPLIFIER USING THE APPROXIMATE MODEL

Figure 4.36 shows the equivalent circuit of CC amplifier using the approximate model, with the collector grounded, input signal applied between base and ground and load connected between emitter and ground.

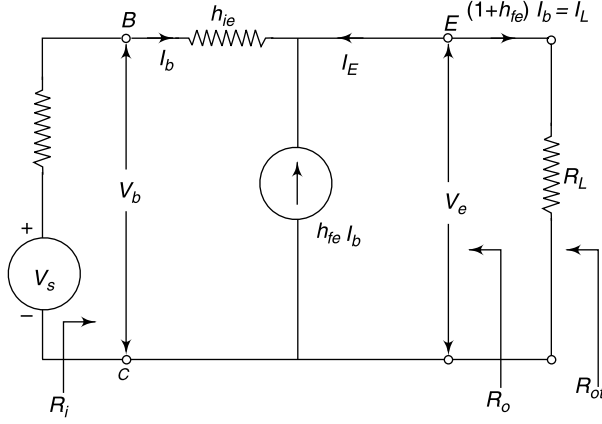


Fig. 4.36 Simplified Hybrid Model for the CC Circuit

Current Gain

Current gain,
$$A_I = \frac{I_L}{I_b} = 1 + h_{fe} \quad (4.63)$$

Input Resistance

From the circuit of Fig. 4.36, $V_b = I_b h_{ie} + (1 + h_{fe}) I_b R_L$

Input resistance,
$$R_i = \frac{V_b}{I_b} = h_{ie} + (1 + h_{fe}) R_L \quad (4.64)$$

Voltage Gain

$$\begin{aligned} A_V &= \frac{V_e}{V_b} = \frac{(1 + h_{fe}) I_b R_L}{[h_{ie} I_b + (1 + h_{fe}) I_b R_L]} \\ &= \frac{(1 + h_{fe}) R_L}{[h_{ie} + (1 + h_{fe}) R_L]} \end{aligned} \quad (4.65)$$

$$\begin{aligned} &= \frac{h_{ie} + (1 + h_{fe}) R_L h_{ie}}{[h_{ie} + (1 + h_{fe}) R_L]} \\ &= 1 - \frac{h_{ie}}{h_{ie} + (1 + h_{fe}) R_L} \end{aligned} \quad (4.66)$$

Therefore,
$$A_V = 1 - \frac{h_{ie}}{R_i} \quad (4.67)$$

Output Impedance

$$\text{Output admittance } (Y_o) = \frac{\text{short circuit current in output terminals}}{\text{open circuit voltage between output terminals}}$$

$$\text{Short circuit current in output terminals} = 1 + (h_{fe}) I_b = \frac{(1 + h_{fe}) V_s}{h_{ie} + R_s}$$

$$\text{Open circuit voltage between output terminals} = V_s$$

$$Y_o = \frac{1 + h_{fe}}{h_{ie} + R_s} \quad (4.68)$$

$$\text{and} \quad Z_o = \frac{1}{Y_o} = \frac{h_{ie} + R_s}{1 + h_{fe}} \quad (4.69)$$

Output impedance including R_L , i.e. $R'_o = R_o \parallel R_L$.

Example 4.38 A voltage source of internal resistance $R_s = 900 \Omega$ drives a CC amplifier using load resistance $R_L = 2000 \Omega$. The CE h -parameters are $h_{ie} = 1200 \Omega$, $h_{re} = 2 \times 10^{-4}$, $h_{fe} = 60$ and $h_{oe} = 25 \mu A/V$. Compute the current gain A_I the input impedance R_i , voltage gain A_V and output resistance R_o using approximate analysis and exact analysis.

Solution Conversion formulae: $h_{ic} = h_{ie} = 1200 \Omega$, $h_{fc} = -(1 + h_{fe}) = -61$

$$h_{rc} = 1, h_{oc} = h_{oe} = 25 \mu A/V$$

Exact analysis

$$\begin{aligned} \text{Current gain, } A_I &= \frac{-h_{fc}}{1 + h_{oc} R_L} \\ &= \left[\frac{-(-61)}{1 + 25 \times 10^{-6} \times 2 \times 10^3} \right] = 58.095 \end{aligned}$$

$$\begin{aligned} \text{Input impedance, } R_i &= h_{ic} + h_{rc} A_I R_L \\ &= 1200 + (1) (58.095) (2000) = 117.39 \text{ k}\Omega \end{aligned}$$

$$\text{Voltage gain, } A_V = \frac{A_I R_L}{R_i} = \frac{(58.095)(2 \times 10^3)}{117.39 \times 10^3} = 0.9897$$

Output resistance, R_o :

$$Y_o = \frac{1}{R_o} = h_{oc} - \frac{h_{fc} h_{rc}}{h_{ic} + R_s}$$

$$= 25 \times 10^{-6} - \frac{(-61)(1)}{(1200) + (900)}$$

$$R_o = 34.396 \, \Omega$$

Approximate analysis

Current gain, $A_I = 1 + h_{fe} = 1 + 60 = 61$

Input impedance, $R_i = h_{ie} + (1 + h_{fe}) R_L = 1200 + (61)(2000) = 123.2 \, \text{k}\Omega$

Voltage gain, $A_V = 1 - \frac{h_{ie}}{R_i} = 1 - \frac{1200}{123.2 \times 10^3} = 0.9903$

Output resistance, R_o :

$$Y_o = \frac{1}{R_o} = \frac{1 + h_{fe}}{h_{ie} + R_S} = \frac{1 + 60}{1200 + 900} = 0.029 \, \text{mho}$$

$$R_o = 34.43 \, \Omega$$

4.13

ANALYSIS OF CB AMPLIFIER USING THE APPROXIMATE MODEL

Figure 4.37 shows the equivalent circuit of CB amplifier using the approximate model, with the base grounded, input signal applied between emitter and base and load connected between collector and base.

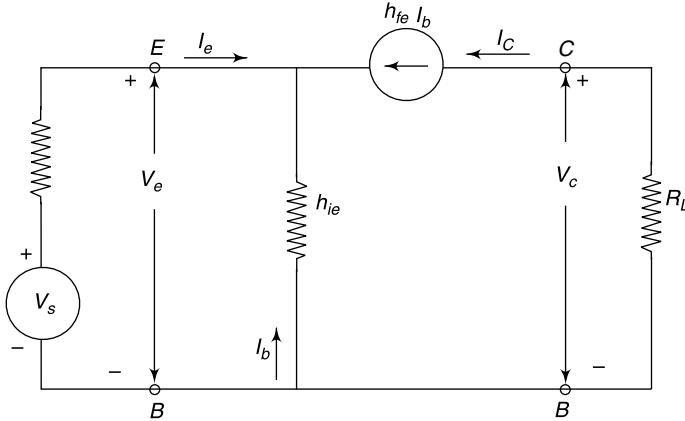


Fig. 4.37 Simplified Hybrid Model for the CB Circuit

Current Gain

$$\begin{aligned} A_I &= \frac{-I_c}{I_e} = \frac{-h_{fe} I_b}{I_e} \\ &= \frac{-h_{fe} I_b}{-(h_{fe} I_b + I_b)} = \frac{h_{fe}}{1 + h_{fe}} = -h_{fb}, \text{ from Table 3.4} \end{aligned}$$

Hence, current gain, $A_I = \frac{h_{fe}}{1+h_{fe}} = -h_{fb}$ (4.70)

Input Resistance $R_i = \frac{V_e}{I_e}$

$$= \frac{-I_b h_{ie}}{-(1+h_{fe})I_b} = \frac{h_{ie}}{1+h_{fe}}$$

$$= h_{ib} \quad (4.71)$$

Voltage Gain

$$A_V = \frac{V_c}{V_e}$$

$$= \frac{-h_{fe} I_b R_L}{-I_b h_{ie}}$$

$$= \frac{h_{fe} R_L}{h_{ie}} \quad (4.72)$$

A_I , A_V and R_i do not differ from exact values by more than 10%.

Output Impedance

$$R_o = \frac{V_e}{I_c} \text{ with } V_S = 0, R_L = \infty$$

With $V_S = 0, I_e = 0$ and $I_b = 0$

Hence, $I_c = 0$

Therefore, $R_o = \infty$ using the approximate model.

Example 4.39 For a CB transistor amplifier driven by a voltage source of internal resistance $R_s = 1200 \Omega$, the load impedance is a resistor $R_L = 1000 \Omega$. The h -parameters are $h_{ib} = 22 \Omega$, $h_{rb} = 3 \times 10^{-4}$, $h_{fb} = -0.98$ and $h_{ob} = 0.5 \mu A/V$. Compute the current gain A_I , the input impedance R_i , voltage gain A_V , overall voltage gain A_{VS} , overall current gain A_{IS} , output impedance Z_o , and power gain A_P using exact analysis and approximate analysis.

Solution Exact analysis

Current gain, $A_I = \frac{-h_{fb}}{1+h_{ob}R_L} = \frac{-(-0.98)}{1+0.5 \times 10^{-6} \times 1000}$

$$= 0.98$$

Input impedance, $R_i = h_{ib} + h_{rb} A_I R_L$

$$= 22 + (3 \times 10^{-4}) \times 0.98 \times 1000 = 22.3 \Omega$$

Voltage gain, $A_V = \frac{A_I R_L}{R_i} = \frac{0.98 \times 1000}{22.3} = 43.946$

$$\text{Overall voltage gain, } A_{VS} = \frac{A_V R_i}{R_i + R_s} = \frac{43.946 \times 22.3}{22.3 + 1200} = 0.802$$

$$\text{Overall current gain, } A_{IS} = \frac{A_I R_s}{R_i + R_s} = \frac{0.98 \times 1200}{22.3 + 1200} = 0.962$$

$$\begin{aligned} \text{Output admittance, } Y_o &= h_{ob} - \frac{h_{fb} h_{rb}}{h_{ib} + R_s} = 0.5 \times 10^{-6} - \frac{(-0.98)(3 \times 10^{-4})}{22 + 1200} \\ &= 0.7405 \times 10^{-6} \text{ mhos} \end{aligned}$$

$$R_o = \frac{1}{Y_o} = 1.35027 \text{ M}\Omega$$

$$\text{Power gain, } A_P = A_V A_I = 0.98 \times 43.946 = 43.06$$

Approximate analysis

$$\text{Current gain, } A_I = -h_{fb} = 0.98$$

$$\text{Input impedance, } R_i = h_{ib} = 22 \Omega$$

$$\text{Voltage gain, } A_V = \frac{h_{fe} R_L}{h_{ie} - h_{fe}}$$

$$\text{From Table 3.4, } h_{fb} = \frac{-h_{fe}}{1 + h_{fe}}$$

$$\text{Rearranging this equation, } h_{fe} = \frac{-h_{fb}}{1 + h_{fb}}$$

$$\text{From the given data, } h_{fe} = \frac{-(-0.98)}{1 - 0.98} = 49$$

$$\text{From Table 3.4, } h_{ib} = \frac{h_{ie}}{1 + h_{fe}}$$

$$\begin{aligned} h_{ie} &= h_{ib}(1 + h_{fe}) \\ &= 22(1 + 49) = 1100 \Omega \end{aligned}$$

$$A_V = \frac{49 \times 1000}{1100} = 44.54$$

$$\text{Output impedance, } R_o = \infty$$

$$\text{Overall voltage gain, } A_{VS} = \frac{A_V R_i}{R_i + R_s} = \frac{44.54 \times 22}{22 + 1200} = 0.802$$

$$\text{Overall current gain, } A_{IS} = \frac{A_I R_s}{R_i + R_s} = \frac{0.98 \times 1200}{22 + 1200} = 0.962$$

$$\text{Power gain, } A_P = A_V A_I = 44.54 \times 0.98 = 43.65$$

REVIEW QUESTIONS

1. What is meant by Q -point?
2. What is the need for biasing a transistor?
3. What factors are to be considered for selecting the operating point Q for an amplifier?
4. Distinguish between dc and ac load lines with suitable diagrams.
5. Briefly explain the reasons for keeping the operating point of a transistor as fixed.
6. What is thermal runaway? How can it be avoided?
7. What three factors contribute to thermal instability?
8. Define 'stability factor.' Why would it seem more reasonable to call this an instability factor?
9. Draw a fixed bias circuit and derive an expression for the stability factor.
10. If the coordinates of the operating point of a CE amplifier using fixed bias or base resistor method of biasing are $V_{CE} = 6\text{ V}$ and $I_C = 1\text{ mA}$, determine the value of R_C and R_B .

[Ans: $R_C = 3\text{ k}\Omega$, $R_B = 300\text{ k}\Omega$]
11. Consider a common emitter NPN transistor with fixed bias as shown in Fig. 4.6. If $\beta = 80$, $R_B = 390\text{ k}\Omega$, $R_C = 1.5\text{ k}\Omega$ and $V_{CC} = 30\text{ V}$, find the coordinates of the Q -point.

[Ans: 21 V , 6 mA]
12. A germanium transistor having $\beta = 100$ and $V_{BE} = 0.2\text{ V}$ is used in a fixed bias amplifier circuit where $V_{CC} = 16\text{ V}$, $R_C = 5\text{ k}\Omega$ and $R_B = 790\text{ k}\Omega$. Determine its operating point.
13. Derive an expression for the stability factor of an emitter-feedback bias circuit.
14. Derive an expression for the stability factor of a collector-emitter feedback bias circuit.
15. Derive an expression for the stability factor of a collector-to-base bias circuit.
16. Mention the disadvantages of collector-to-base bias. Can they be overcome?
17. In a germanium transistor CE amplifier biased by feedback resistor method, $V_{CC} = 20\text{ V}$, $V_{BE} = 0.2\text{ V}$, $\beta = 100$ and the operating point is chosen such that $V_{CE} = 10.4\text{ V}$ and $I_C = 9.9\text{ mA}$. Determine the value of R_B and R_C .

[Ans: $100\text{ k}\Omega$, $1\text{ k}\Omega$]
18. Draw a circuit diagram of CE transistor amplifier using emitter biasing. Describe qualitatively the stability action of the circuit.
19. Draw a voltage divider bias circuit and derive an expression for its stability factor.
20. Why does potential divider method of biasing become universal?
21. If the various parameters of a CE amplifier which uses the self bias method are $V_{CC} = 12\text{ V}$, $R_1 = 10\text{ k}\Omega$, $R_2 = 5\text{ k}\Omega$, $R_C = 1\text{ k}\Omega$, $R_E = 2\text{ k}\Omega$ and $\beta = 100$, find (i) the coordinates of the operating point, and (ii) the stability factor, assuming the transistor to be of silicon.

[Ans: $V_{CE} = 7.05\text{ V}$, $I_C = 1.65\text{ mA}$, $S = 2.62$]

22. In a CE germanium transistor amplifier using self bias circuit, $R_C = 2.2 \text{ k}\Omega$, $\beta = 50$, $V_{CC} = 9 \text{ V}$ and the operating point is required to be set at $I_C = 2 \text{ mA}$ and $V_{CE} = 3 \text{ V}$. Determine the values of R_1 , R_2 and R_E .
 [Ans: $R_1 = 17.75 \text{ k}\Omega$, $R_2 = 4.75 \text{ k}\Omega$, $R_E = 800 \Omega$]
23. Determine the operating point for the circuit of a potential divider bias arrangement with $R_2 = R_C = 5 \text{ k}\Omega$, $R_E = 1 \text{ k}\Omega$ and $R_1 = 40 \text{ k}\Omega$.
 [Ans: $V_{CE} = 6 \text{ V}$, $I_C = 1 \text{ mA}$]
24. Calculate the values of R_1 and R_C in the voltage divider bias circuit so that Q-point is at $V_{CE} = 6 \text{ V}$ and $I_C = 2 \text{ mA}$. Assume the transistor parameters are: $\alpha = 0.985$, $I_{CBO} = 4 \mu\text{A}$ and $V_{BE} = 0.2 \text{ V}$.
 [Ans: $R_C = 3 \text{ k}\Omega$, $R_1 = 5.54 \text{ k}\Omega$]
25. Determine the stability factor for a CB amplifier circuit.
26. Draw a circuit which uses a diode to compensate for changes in I_{CO} . Explain how stabilization is achieved in the circuit.
27. How will you provide temperature compensation for the variations of V_{BE} and stabilization of the operating point?
28. What is the principle of providing thermal stabilization by means of different methods of transistor biasing? How does this differ from the compensation techniques using a diode or thermistor or sensistor?
29. What is a heat sink? How does it contribute to increase in power dissipation?
30. Why are power transistors provided with heat sinks?
31. Briefly explain the commonly available heat sinks.
32. What is the condition for thermal stability? Explain.
33. Show that thermal runaway cannot take place if the quiescent point is located at $V_{CE} < V_{CC}/2$.
34. Derive the equations for voltage gain, current gain, input impedance and output impedance for a BJT using the approximate h -parameter model for (a) CE configuration (b) CB configuration and (c) CC configuration.
35. A CE amplifier is drawn by a voltage source of internal resistance $r_s = 1000 \Omega$ and the load impedance is a resistance $R_L = 1200 \Omega$. The h -parameters are $h_{ie} = 1.2 \text{ k}\Omega$, $h_{re} = 2 \times 10^{-4}$, $h_{fe} = 60$ and $h_{oe} = 25 \mu\text{A/V}$. Compute the current gain A_I , input resistance R_i , voltage gain A_V and output resistance R_o using exact analysis and using approximate analysis.
 [Ans: Exact analysis: $A_I = -58.25$, $R_i = 1.186 \text{ k}\Omega$, $A_V = -58.937$, $R_o = 51.162 \text{ k}\Omega$, Approximate Analysis: $A_i = -60$, $R_i = 1.2 \text{ k}\Omega$, $A_v = -60$, $Z_o = \infty$.]
36. A CE amplifier uses load resistor $R_C = 2.5 \text{ k}\Omega$ in the collector circuit and is given by the voltage source V_S of internal resistance 600Ω . The h -parameters of the transistor are $h_{ie} = 1300 \Omega$, $h_{re} = 2 \times 10^{-4}$, $h_{fe} = 55$ and $h_{oe} = 22 \mu\text{Mhos}$. Neglecting the biasing resistors across the V_{CC} supply, compute the current gain A_I , input resistance R_i , voltage gain A_V , output resistance R_o and output terminal resistance R_{oT} for the following values of emitter resistor R_E inserted in the emitter circuit. (i) 200Ω (ii) 400Ω and (iii) 1000Ω . Use the approximate model for the transistor if permissible.

[Ans: (i) $A_I = -55$, $R_i = 12.5 \text{ k}\Omega$,
 $A_v = -11$, $R_o = \infty$, $R_{oT} = 2.5 \text{ k}\Omega$]

- (ii) $A_I = -55$, $R_i = 23.7 \text{ k}\Omega$,
 $A_v = -5.8$, $R_o = \infty$, $R_{oT} = 2.5 \text{ k}\Omega$
- (iii) $A_I = -55$, $R_i = 57.3 \text{ k}\Omega$,
 $A_v = -2.399$, $R_o = \infty$, $R_{oT} = 2.5 \text{ k}\Omega$]
37. For a CB transistor amplifier driven by a voltage source of internal resistance $R_s = 600 \Omega$, the load impedance is a resistor $R_L = 1200 \Omega$. The h -parameters are $h_{ib} = 22 \Omega$, $h_{rb} = 4 \times 10^{-4}$, $h_{fb} = -0.98$ and $h_{ob} = 0.25 \mu\text{A/V}$. Compute the current gain A_I , the input impedance R_i , voltage gain A_v , overall voltage gain A_{VS} , overall current gain A_{IS} , output impedance Z_o and power gain A_P .
 [Ans: Exact analysis: $A_I = 0.9797$, $R_i = 22.47 \Omega$, $A_v = 52.32$, $A_{VS} = 1.8886$, $A_{IS} = 0.9443$, $R_o = 1.13607 \text{ M}\Omega$, $A_P = 51.257$. Approximate analysis $A_I = 0.98$, $R_i = 22 \Omega$, $A_v = 53.4545$, $A_{VS} = 1.8906$, $A_{IS} = 0.9453$, $R_o = \infty$, $A_P = 52.381$]
38. A voltage source of internal resistance $R_s = 600 \Omega$ drives a CC amplifier using load resistance $R_L = 1000 \Omega$. The CE h -parameters are $h_{ie} = 1200 \Omega$, $h_{re} = 2 \times 10^{-4}$, $h_{fe} = 60$ and $h_{oe} = 25 \mu\text{A/V}$. Compute the current gain A_I , the input impedance R_i , voltage gain A_v and output resistance R_o using approximate analysis and exact analysis.
 [Ans: Exact analysis: $A_I = 59.512$, $R_i = 60.712 \text{ k}\Omega$, $A_v = 0.9802$, $R_o = 29.486 \Omega$. Approximate analysis $A_I = 61$, $R_i = 62.2 \text{ k}\Omega$, $A_v = 0.9807$, $R_o = 29.508 \Omega$]

OBJECTIVE TYPE QUESTIONS

1. The operating point variation is due to
 (a) I_{CO} (b) V_{BE} (c) β (d) all the above
2. In cut-off, the value of V_{CE} is
 (a) 0 V (b) V_{CC} (c) minimum (d) $V_{BE} = 0.6 \text{ V}$
3. The saturation value of V_{CE} is
 (a) equal to V_{CE} (b) 0 V (c) 0.7 V (d) $\frac{V_{CC}}{2}$
4. Which of the following conditions ensures that the transistor does not undergo thermal run-away?
 (a) $V_{CE} = V_{CC}/2$ (b) $V_{CE} < V_{CC}/2$ (c) $V_{CE} < V_{CC}$ (d) $V_{CE} > V_{CC}/2$
5. The stability factor of a fixed bias is
 (a) $1 + \beta$ (b) $1 - \beta$ (c) $\beta/1 - \beta$ (d) $1/1 - \beta$
6. The leakage current in CE configuration may be around
 (a) few nanoamperes (b) few microamperes
 (c) few hundred microamperes (d) few milliamperes
7. The quiescent point of a transistor biasing circuit implies
 (a) zero bias (b) no output
 (c) no distortion (d) no input signal
8. For normal amplification, the Q point should be established in the
 (a) active region (b) saturation region
 (c) cut-off region (d) none of the above

9. The following biasing technique gives good stability:
(a) Collector-to-base bias (b) Fixed bias
(c) Self bias (d) none of the above
10. Stability factor 'S' is approximately unity for
(a) Collector-to-base bias (b) Fixed bias
(c) Self bias (d) none of the above
11. The following transistor parameters(s) are functions of temperature:
(a) β alone (b) $0 I_{CO}$ alone (c) V_{BE} alone (d) All of the above
12. The self bias arrangement gives an improved Q-point stability when
(a) R_E is low (b) β is small, but R_E is large
(c) both β and R_E are large (d) None
13. The biasing method which is considered independent of transistor β_{dc} is
(a) fixed biasing (b) collector feedback bias
(c) voltage divider bias (d) base bias with collector feedback
14. The biasing configuration that offers least stability is
(a) fixed-bias configuration
(b) collector-to-base-bias configuration
(c) voltage-divider-bias configuration
(d) none of the above
15. Which one of the following statements is correct?
(a) Both I_{CO} and V_{BE} increase with temperature.
(b) Both I_{CO} and V_{BE} decrease with temperature.
(c) I_{CO} increases with temperature and V_{BE} decreases with temperature.
(d) I_{CO} decreases with temperature and V_{BE} increases with temperature.
16. The collector current for the CE circuit is given $I_C = \beta I_B + (1 + \beta) I_{CO}$. The three variables β , I_B and I_{CO}
(a) increase with rise in temperature
(b) increase with fall in temperature
(c) decrease with rise in temperature
(d) decrease with fall in temperature
17. The resistance of thermistor decreases exponentially with
(a) increase of temperature (b) decrease of temperature
(c) none of the above
18. Which of the following has a negative temperature co-efficient?
(a) Thermistor (b) Sensistor (c) Resistor (d) Both (a) and (b)
19. Which of the following has a positive temperature co-efficient?
(a) Thermistor (b) Sensistor (c) Both (a) and (b) (d) Resistor
20. The resistance of sensistor increases exponentially with
(a) increase of temperature (b) decrease of temperature
(c) none of the above
21. In thermistor compensation technique, thermistor is connected parallel with
(a) collector resistance (b) emitter resistance
(c) base to ground (d) base to V_{CC}
22. In the fixed-bias circuit if the base resistor is shorted then
(a) the transistor may get damaged
(b) the base voltage will be zero

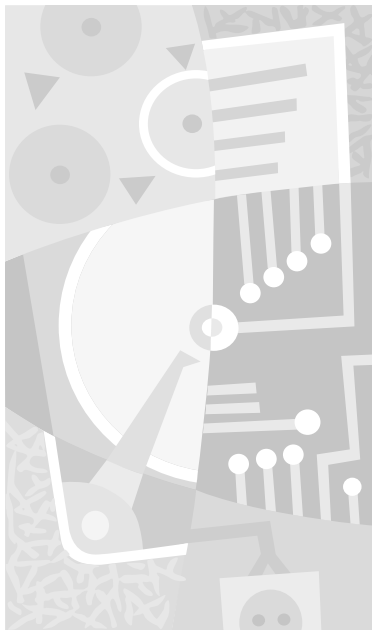
- (c) the collector voltage will be equal to the supply voltage
(d) the collector current is zero
23. As temperature is increased, the h-parameters h_{ie} , h_{re} , h_{fe} and h_{oe} of a transistor
(a) decreases (b) increases (c) remains constant (d) None
24. In power transistors, the heat developed at the collector junction may be removed by the use of
(a) heat sink (b) transistor with high β
(c) transistor with low b (d) none of these
25. The condition to prevent thermal runaway is given by
(a) $\frac{\partial P_C}{\partial T_j} < \Theta$ (b) $\frac{\partial T_j}{\partial P_C} < \frac{1}{\Theta}$ (c) $\frac{\partial P_C}{\partial T_j} < \frac{1}{\Theta}$ (d) none of these
26. When a transistor is connected in a common-emitter mode, it will have
(a) negligible input resistance and high output resistance
(b) high input resistance and low output resistance
(c) medium input resistance and high output resistance
(d) low input resistance as well as output resistance
27. The following is a non-inverting amplifier with voltage gain exceeding unity.
(a) CE amplifier (b) CB amplifier
(c) CC amplifier (d) none of these
28. The voltage gain of CB amplifier has the same magnitude as that of
(a) CE amplifier (b) CC amplifier
(c) both CE and CC amplifiers (d) none of these
29. The BJT amplifier configuration with the lowest output resistance is
(a) CE amplifier (b) CB amplifier
(c) CC amplifier (d) none of these
30. The BJT amplifier configuration with the highest output resistance is
(a) CE amplifier (b) CB amplifier
(c) CC amplifier (d) none of these
31. The BJT amplifier configuration with the lowest input resistance is
(a) CE amplifier (b) CB amplifier
(c) CC amplifier (d) none of these
32. The BJT amplifier having the highest output resistance is
(a) CE amplifier (b) CB amplifier
(c) CC amplifier (d) none of these
33. For an amplifier, the input resistance is high, the current gain is high and voltage gain is near unity. It is a
(a) common base (b) common emitter
(c) common collector (d) Darlington amplifier
33. An amplifier has good voltage, current and power gains and the input resistance is low. It is a
(a) common base (b) common emitter
(c) Common collector amplifier (d) None
34. The maximum current gain of an amplifier is unity. It is a
(a) common base (b) common collector
(c) Common emitter amplifier (d) None

State whether the following statements are true (T) or false (F)

36. The practical range of stability factor is from 1 to 10.
37. D.C. and A.C. load lines intersect at the Q point.
38. Stability factor should be as high as possible to have better thermal stability
39. Collector-to-base bias gives better stability than fixed bias.
40. The quiescent point is otherwise called operating point.
41. In a transistor, the I_{CO} the doubles for every 10°C increase in temperature.
42. Bias stabilization prevents thermal run away.
43. Thermal runaway enhances thermal stability of a transistor.
44. The larger the heat sink, the smaller will be its thermal resistance.
45. The amplifier circuits operating at low current and lower value of stability factor are not susceptible to thermal energy.
46. To avoid thermal runaway, a heat sink may be attached to the collector of the power transmitter.
47. An amplifier in voltage divider bias configuration is the most widely used one.
48. As the temperature difference between the temperature of the collector-base junction of the transistor and the ambient temperature of the air around the transistor is greater, the power dissipated in the transistor will be greater.
49. A thermistor has a positive temperature coefficient.

ANSWERS

- | | | | | |
|---------|---------|---------|---------|---------|
| 1. (d) | 2. (b) | 3. (c) | 4. (b) | 5. (a) |
| 6. (b) | 7. (c) | 8. (a) | 9. (c) | 10. (c) |
| 11. (d) | 12. (a) | 13. (c) | 14. (a) | 15. (c) |
| 16. (a) | 17. (a) | 18. (a) | 19. (b) | 20. (a) |
| 21. (c) | 22. (a) | 23. (b) | 24. (a) | 25. (c) |
| 26. (c) | 27. (b) | 28. (a) | 29. (c) | 30. (c) |
| 31. (b) | 32. (b) | 33. (c) | 34. (b) | 35. (a) |
| 36. (T) | 37. (T) | 38. (F) | 39. (T) | 40. (T) |
| 41. (T) | 42. (T) | 43. (F) | 44. (T) | 45. (T) |
| 46. (T) | 47. (T) | 48. (T) | 49. (F) | |



5

FIELD EFFECT TRANSISTORS AND FET AMPLIFIERS

5.1 INTRODUCTION

The FET is a device in which the flow of current through the conducting region is controlled by an electric field. Hence the name Field Effect Transistor (FET). As current conduction is only by majority carriers, FET is said to be a unipolar device.

Based on the construction, the FET can be classified into two types as *Junction FET (JFET)* and *Metal Oxide Semiconductor FET (MOSFET)* or *Insulated Gate FET (IGFET)* or *Metal Oxide Silicon Transistor (MOST)*.

Depending upon the majority carriers, JFET has been classified into two types, namely, (1) *N-channel JFET* with electrons as the majority carriers, and (2) *P-Channel JFET* with holes as the majority carriers.

5.2 CONSTRUCTION OF N-CHANNEL JFET

It consists of a N-type bar which is made of silicon. Ohmic contacts (terminals), made at the two ends of the bar, are called Source and Drain.

Source (S) This terminal is connected to the negative pole of the battery. Electrons which are the majority carriers in the N-type bar enter the bar through this terminal.

Drain (D) This terminal is connected to the positive pole of the battery. The majority carriers leave the bar through this terminal.

Gate (G) Heavily doped P-type silicon is diffused on both sides of the N-type silicon bar by which PN junctions are formed. These layers are joined together and called Gate G .

Channel The region BC of the N-type bar between the depletion region is called the channel. Majority carriers move from the source to drain when a potential difference V_{DS} is applied between the source and drain.

5.3

PRINCIPLE OF OPERATION, PINCH-OFF VOLTAGE, VOLT-AMPERE CHARACTERISTICS AND SYMBOLS OF JFET

When $V_{GS} = 0$ and $V_{DS} = 0$ When no voltage is applied between drain and source, and gate and source, the thickness of the depletion regions round the PN junction is uniform as shown in Fig. 5.1.

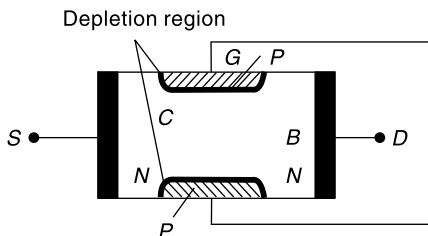


Fig. 5.1 JFET Construction

When $V_{DS} = 0$ and V_{GS} is Decreased from Zero In this case, the PN

junctions are reverse biased and hence the thickness of the depletion region increases. As V_{GS} is decreased from zero, the reverse bias voltage across the PN junction is increased and hence, the thickness of the depletion region in the channel increases until the two depletion regions make contact with each other. In this condition, the channel is said to be cut-off. The value of V_{GS} which is required to cut-off the channel is called the cut-off voltage V_C .

When $V_{GS} = 0$ and V_{DS} is Increased from Zero Drain is positive with respect to the source with $V_{GS} = 0$. Now the majority carriers (electrons) flow through the N-channel from source to drain. Therefore the conventional current I_D flows from drain to source. The magnitude of the current will depend upon the following factors:

1. The number of majority carriers (electrons) available in the channel, i.e. the conductivity of the channel.
2. The length L of the channel.
3. The cross-sectional area A of the channel at B .
4. The magnitude of the applied voltage V_{DS} . Thus the channel acts as a resistor of resistance R given by

$$R = \frac{\rho L}{A} \quad (5.1)$$

$$I_D = \frac{V_{DS}}{R} = \frac{AV_{DS}}{\rho L} \quad (5.2)$$

where ρ is the resistivity of the channel. Because of the resistance of the channel and the applied voltage V_{DS} , there is a gradual increase of positive potential along the channel from source to drain. Thus the reverse voltage across the PN junctions increases and hence the thickness of the depletion regions also increases. Therefore, the channel is wedge shaped as shown in Fig. 5.2.

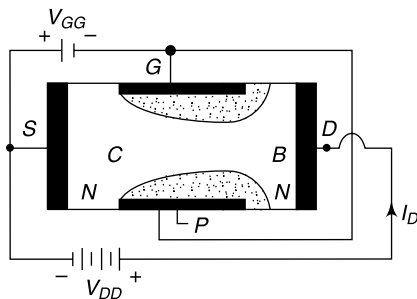


Fig. 5.2 JFET Under Applied Bias

As V_{DS} is increased, the cross-sectional area of the channel will be reduced. At a certain value V_P of V_{DS} , the cross-sectional area at B becomes minimum. At this voltage, the channel is said to be pinched off and the drain voltage V_P is called the pinch-off voltage.

As a result of the decreasing cross-section of the channel with the increase of V_{DS} , the following results are obtained.

(a) As V_{DS} is increased from zero, I_D increases along OP , and the rate of increase of I_D with V_{DS} decreases as shown in Fig. 5.3. The region from $V_{DS} = 0$ V to $V_{DS} = V_P$ is called the ohmic region. In the channel ohmic region the drain to source resistance $\frac{V_{DS}}{I_D}$ is related to the gate voltage V_{GS} , in an almost linear manner. This is useful as a voltage variable resistor (VVR) or voltage dependent resistor (VDR).

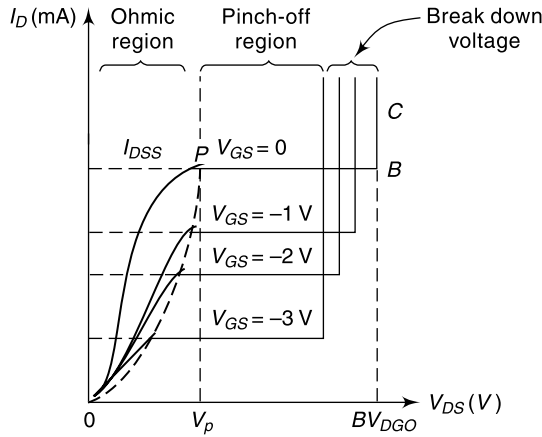


Fig. 5.3 Drain Characteristics

(b) When $V_{DS} = V_P$, I_D becomes maximum. When V_{DS} is increased beyond V_P , the length of the pinch-off or saturation region increases. Hence, there is no further increase of I_D .

(c) At a certain voltage corresponding to the point B , I_D suddenly increases. This effect is due to the Avalanche multiplication of electrons caused by breaking of covalent bonds of silicon atoms in the depletion region between the gate and the drain. The drain voltage at which the breakdown occurs is denoted by BV_{DGO} . The variation of I_D with V_{DS} when $V_{GS} = 0$ is shown in Fig. 5.3 by the curve $OPBC$.

When V_{GS} is Negative and V_{DS} is Increased When the gate is maintained at a negative voltage less than the negative cut-off voltage, the reverse voltage across the junction is further increased. Hence for a negative value of V_{GS} , the curve of I_D versus V_{DS} is similar to that for $V_{GS} = 0$, but the values of V_P and BV_{DGO} are lower, as shown in Fig. 5.3.

From the curves, it is seen that above the pinch-off voltage, at a constant value of V_{DS} , I_D increases with an increase of V_{GS} . Hence, a JFET is suitable for use as a voltage amplifier, similar to a transistor amplifier.

It can be seen from the curve that for voltage $V_{DS} = V_P$, the drain current is not reduced to zero. If the drain current is to be reduced to zero, the ohmic voltage drop along the channel should also be reduced to zero. Further, the reverse biasing to the gate-source PN junction essential for pinching off the channel would also be absent.

The drain current I_D is controlled by the electric field that extends into the channel due to reverse biased voltage applied to the gate; hence, this device has been given the name *Field Effect Transistor*.

In a bar of P-type semiconductor, the gate is formed due to N-type semiconductor. The working of the P-channel JFET will be similar to that of N-channel JFET with proper alterations in the biasing circuits; in this case holes will be the current carriers instead of electrons. The circuit symbols for N-channel and P-channel JFETs are shown in Fig. 5.4. It should be noted that the direction of the arrow points in the direction of conventional current which would flow into the gate if the PN junction was forward biased.

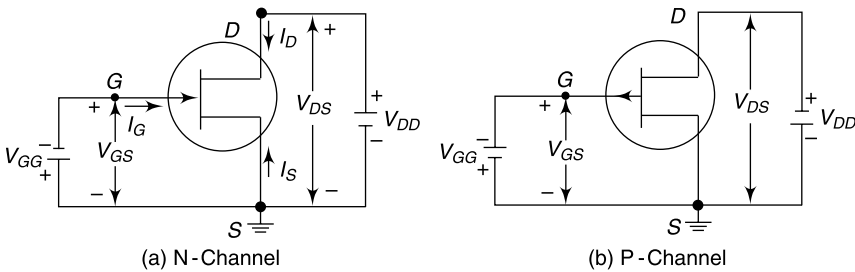


Fig. 5.4 Circuit Symbols for N- and P-channel JFET

5.4 CHARACTERISTIC PARAMETERS OF THE JFET

In a JFET, the drain current I_D depends upon the drain voltage V_{DS} and the gate voltage V_{GS} . Any one of these variables may be fixed and the relation between the other two are determined. These relations are determined by the three parameters which are defined below.

Mutual Conductance or Transconductance, g_m It is the slope of the transfer characteristic curves, and is defined by

$$g_m = \left(\frac{\partial I_D}{\partial V_{GS}} \right)_{V_{DS}} = \frac{\Delta I_D}{\Delta V_{GS}}, \quad V_{DS} \text{ held constant.}$$

It is the ratio of a small change in the drain current to the corresponding small change in the gate voltage at a constant drain voltage. The change in I_D and V_{GS} should be taken on the straight part of the transfer characteristics. It has the unit of conductance in mho.

Drain Resistance, r_d It is the reciprocal of the slope of the drain characteristics and is defined by

$$r_d = \left(\frac{\partial V_{DS}}{\partial I_D} \right)_{V_{GS}} \\ = \frac{\Delta V_{DS}}{\Delta I_D}, \quad V_{GS} \text{ held constant.}$$

It is the ratio of a small change in the drain voltage to the corresponding small change in the drain current at a constant gate voltage. It has the unit of resistance in ohms.

The drain resistance at $V_{GS} = 0$ V, i.e. when the depletion regions of the channel are absent, is called as *drain-source ON resistance*, represented as R_{DS} or $R_{DS(ON)}$.

The reciprocal of r_d is called the drain conductance. It is denoted by g_d or g_{os} .

Amplification Factor, μ It is defined by

$$\mu = - \left(\frac{\partial I_{DS}}{\partial V_{GS}} \right) I_D = - \frac{\Delta V_{DS}}{\Delta V_{GS}}, \quad I_D \text{ held constant.}$$

It is the ratio of a small change in the drain voltage to the corresponding small change in the gate voltage at a constant drain current. Here, the negative sign shows that when V_{GS} is increased, V_{DS} must be decreased for I_D to remain constant.

Relationship Among FET Parameters As I_D depends on V_{DS} and V_{GS} , the functional equation can be expressed as

$$I_D = f(V_{DS}, V_{GS})$$

If the drain voltage is changed by a small amount from V_{DS} to $(V_{DS} + \Delta V_{DS})$ and the gate voltage is changed by a small amount from V_{GS} to $(V_{GS} + \Delta V_{GS})$, then the corresponding small change in I_D may be obtained by applying Taylor's theorem with neglecting higher order terms. Thus the small change ΔI_D is given by

$$\Delta I_D = \left(\frac{\partial I_D}{\partial V_{DS}} \right)_{V_{GS}} \Delta V_{DS} + \left(\frac{\partial I_D}{\partial V_{GS}} \right)_{V_{DS}} \Delta V_{GS}$$

Dividing both the sides of this equation by ΔV_{GS} , we obtain

$$\frac{\Delta I_D}{\Delta V_{GS}} = \left(\frac{\partial I_D}{\partial V_{DS}} \right)_{V_{GS}} \left(\frac{\Delta V_{DS}}{\Delta V_{GS}} \right) + \left(\frac{\partial I_D}{\partial V_{GS}} \right)_{V_{DS}}$$

If I_D is constant, then $\frac{\Delta I_D}{\Delta V_{GS}} = 0$

Therefore, we have

$$0 = \left(\frac{\partial I_D}{\partial V_{DS}} \right)_{V_{GS}} \left(\frac{\Delta V_{DS}}{\Delta V_{GS}} \right)_{I_D} + \left(\frac{\partial I_D}{\partial V_{GS}} \right)_{V_{DS}}$$

Substituting the values of the partial differential coefficients, we get

$$0 = \left(\frac{1}{r_d} \right) (-\mu) + g_m$$

Hence, $\mu = r_d g_m$

Therefore, amplification factor (μ) is the product of drain resistance (r_d) and transconductance (g_m).

Power dissipation, P_D The FET's continuous power dissipation, P_D , is the product of I_D and V_{DS} .

Pinch-off Voltage V_p A single-ended-geometry junction FET is shown in Fig. 5.5 in which the diffusion is done from one side only. The substrate is of P-type material which is epitaxially grown on an N-type channel. A P-type gate is then diffused into the N-type channel. The substrate functions as a second gate which is of relatively low resistivity material. The diffused gate is also of very low resistivity material, allowing the depletion region to spread mostly into the N-type channel. A slab of N-type semiconductor is sandwiched between two layers of P-type material forming two PN junctions in this device.

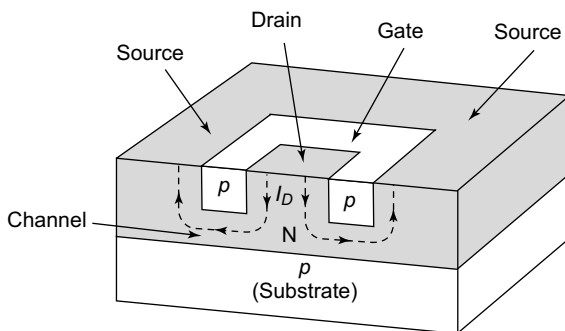


Fig. 5.5 Single-ended-geometry junction FET

The gate reverse voltage that removes all the free charge from the channel is called the pinch-off voltage V_p . We consider that the P-type region is doped with N_A acceptor atoms, the N-type region is doped with N_D donor atoms and the junction formed is abrupt.

Moreover, if the acceptor impurity density is assumed to be much larger than the donor density, then the depletion region width in the P-region will be much smaller than the depletion width of the N-region. i.e., $N_A \gg N_D$, then $W_p \ll W_n$ and $W_n \approx W$. Referring to Fig. 1.18, the relationship between potential and charge density is given by

$$\frac{d^2V}{dx^2} = \frac{-qN_D}{\epsilon}$$

Integrating the above equation subject to boundary conditions, we get

$$\frac{dV}{dx} = \frac{-qN_D}{\epsilon}(x - W)$$

Integrating again, we get

$$V = \frac{-qN_D}{2\epsilon}(x^2 - 2Wx)$$

At $x = W$, $V = V_B$ which is the junction or barrier potential. Thus,

$$V_B = \frac{qN_D W^2}{2\epsilon}$$

As the barrier potential represents a reverse voltage, it is lowered by an applied forward voltage $V(x)$ at x and it is expressed as $V_B = V_o - V(x)$. Now, the space-charge width, $W_n(x) = W(x)$ at a distance x along the channel in Fig. 5.6 becomes

$$W(x) = a - b(x) = \left\{ \frac{2\epsilon}{qN_D} [V_o - V(x)] \right\}^{\frac{1}{2}}$$

where ϵ is the dielectric constant of channel material, q is the magnitude of electronic charge, V_o is the junction contact potential at x , $V(x)$ is the applied potential across space-charge region at x and is a negative value for an applied reverse bias, a is the metallurgical channel thickness between the substrate and P^+ gate region, $a - b(x)$ is the penetration $W(x)$ of depletion region into channel at a point x along channel as shown in Fig. 5.6 and W_{PO} is the depletion region width at pinch-off.

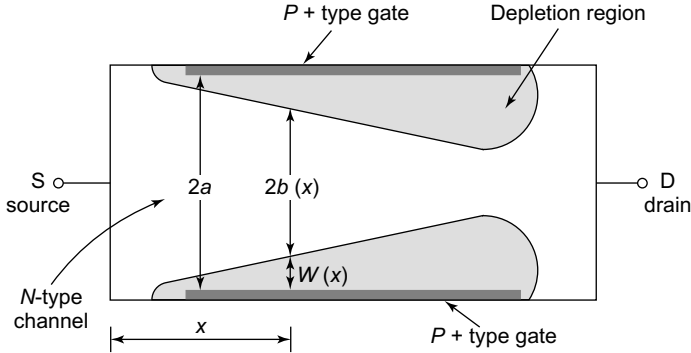


Fig. 5.6 Detailed Structure of N-channel JFET

If the drain current is zero, $b(x)$ and $V(x)$ are independent of x and hence $b(x) = b$. If we substitute $b(x) = b = 0$ in the previous equation and solve for V with the assumption that $|V_o| \ll |V|$, the pinch-off voltage V_P can be obtained as

$$|V_P| = \frac{qN_D}{2\epsilon} a^2$$

Hence, at pinch-off, the depletion width $W_{PO} = a$ when $|V_P| = |V_{GS}|$ and hence

$$W_{PO} = \left(\frac{2\epsilon}{qN_D} V_{GS} \right)^{1/2}$$

To study the effect of V_{GS} , given $V_{GS} = V_o - V(x)$ in the space charge width equation, we get

$$a - b = \left\{ \frac{2\epsilon}{qN_D} V_{GS} \right\}^{1/2}$$

Upon solving, we get

$$a - b = \left[\frac{a^2}{V_P} V_{GS} \right]^{1/2}$$

Further simplifying,

$$\frac{a - b}{a} = \left(\frac{V_{GS}}{V_P} \right)^{1/2}$$

Therefore,
$$\left(1 - \frac{b}{a} \right)^2 = \frac{V_{GS}}{V_P}$$

Hence, the gate source voltage is

$$V_{GS} = \left(1 - \frac{b}{a} \right)^2 V_P$$

For a reverse biased P^+N junction, V_{GS} must be a negative voltage across the gate junction and is independent of distance along the channel if $I_D = 0$.

Example 5.1 When a reverse gate voltage of 12 V is applied to JFET, the gate current is 1 nA. Determine the resistance between gate and source.

Solution $V_{GS} = 12 \text{ V}, I_G = 10^{-9} \text{ A}$.

Therefore, gate-to-source resistance = $\frac{V_{GS}}{I_G} = \frac{12}{10^{-9}} = 12,000 \text{ M}\Omega$

Example 5.2 When the reverse gate voltage of JFET changes from 4.0 to 3.9 V, the drain current changes from 1.3 to 1.6 mA. Find the value of transconductance.

Solution $\Delta V_{GS} = 4.0 - 3.9 = 0.1 \text{ V}$

$$\Delta I_D = 1.6 - 1.3 = 0.3 \text{ mA}$$

Therefore, transconductance, $g_m = \frac{\Delta I_D}{\Delta V_{GS}} = \frac{0.3 \times 10^{-3}}{0.1} = 3 \text{ m mhos}$.

Example 5.3 A FET has a driven current of 4 mA. If $I_{DSS} = 8 \text{ mA}$ and $V_{GS(off)} = -6 \text{ V}$. Find the values of V_{GS} and V_P .

Solution

$$I_D = I_{DSS} \left[1 - \frac{V_{GS}}{V_{GS(off)}} \right]^2$$

$$4 \times 10^{-3} = 8 \times 10^{-3} \left[1 + \frac{V_{GS}}{6} \right]^2$$

$$1 + \frac{V_{GS}}{6} = \sqrt{\frac{4 \times 10^{-3}}{8 \times 10^{-3}}} = \frac{1}{\sqrt{2}} = 0.707$$

Therefore,

$$V_{GS} = -1.76 \text{ V}$$

$$V_P = |V_{GS(off)}| = 6 \text{ V}$$

5.5 EXPRESSION FOR SATURATION DRAIN CURRENT

For the transfer characteristics, V_{DS} is maintained constant at a suitable value greater than the pinch-off voltage V_P . The gate voltage V_{GS} is decreased from zero till I_D is reduced to zero. The transfer characteristics I_D versus V_{GS} is shown in Fig. 5.7. The shape of the transfer characteristic is very nearly a parabola. It is found that the characteristic is approximately represented by the parabola,

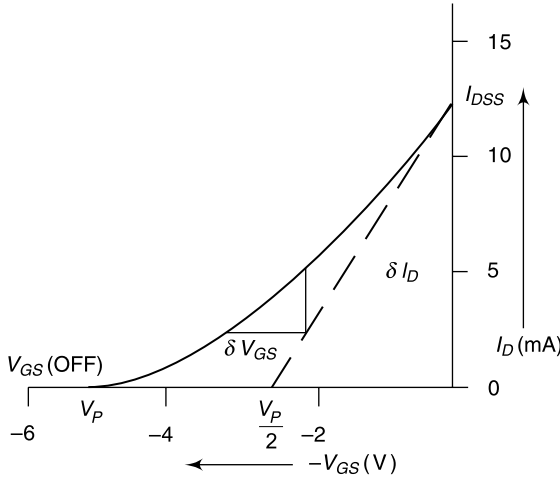


Fig. 5.7 Transfer Characteristic of JFET

$$I_{DS} = I_{DSS} \left(1 - \frac{V_{GS}}{V_P} \right)^2 \quad (5.3)$$

where I_{DS} is the saturation drain current, I_{DSS} is the value of I_{DS} when $V_{GS} = 0$, and V_P is the pinch-off voltage.

Differentiating Eq. (5.3) with respect to V_{GS} , we can obtain an expression for g_m .

$$\frac{\partial I_{DS}}{\partial V_{GS}} = I_{DSS} \times 2 \left(1 - \frac{V_{GS}}{V_P} \right) \left(-\frac{1}{V_P} \right)$$

We know that, $g_m = \frac{\delta I_{DS}}{\delta V_{GS}}$, V_{DS} is constant.

$$\text{Therefore, } g_m = \frac{-2 I_{DSS}}{V_P} \left(1 - \frac{V_{GS}}{V_P} \right) \quad (5.4)$$

Now from Eq. (5.3), we have

$$\left(1 - \frac{V_{GS}}{V_P} \right) = \sqrt{\frac{I_{DS}}{I_{DSS}}} \quad (5.5)$$

Substituting this value in Eq. (5.4), we get

$$g_m = \frac{2 \sqrt{I_{DS} I_{DSS}}}{V_P}$$

Suppose $g_m = g_{mo}$, when $V_{GS} = 0$, then from Eq. (5.4)

$$g_{mo} = -\frac{2 I_{DSS}}{V_P} \quad (5.6)$$

Therefore, from Eqs (5.4) and (5.6), we get

$$g_m = g_{mo} \left(1 - \frac{V_{GS}}{V_P} \right) \quad (5.7)$$

Equation (5.5) shows that g_m varies as the square root of the saturation drain current I_{DS} , and Eq. (5.7) shows that g_m decreases linearly with increase of V_{GS} .

5.6 SLOPE OF THE TRANSFER CHARACTERISTIC AT I_{DSS}

From Eq. (5.5), we have

$$g_m = \frac{2 \sqrt{I_{DS} I_{DSS}}}{V_P}$$

or

$$\frac{\partial I_{DS}}{\partial V_{GS}} = \frac{2 \sqrt{I_{DS} I_{DSS}}}{V_P}$$

Substituting $I_{DS} = I_{DSS}$,

$$\frac{\partial I_{DS}}{\partial V_{GS}} = -\frac{2 I_{DSS}}{V_P} = \frac{I_{DSS}}{\frac{-V_P}{2}}$$

This equation shows that the tangent to the curve at $I_{DS} = I_{DSS}$, $V_{GS} = 0$, will have an intercept at $\frac{-V_P}{2}$ on the axis of V_{GS} as shown in Fig. 5.4. Therefore, the value of V_P can be found by drawing the tangent at $I_{DS} = I_{DSS}$, $V_{GS} = 0$.

The gate-source cut off voltage, $V_{GS(off)}$, on the transfer characteristic is equal to the pinch-off voltage, V_P , on the drain characteristics, i.e. $V_P = |V_{GS(off)}|$.

$$\text{Therefore, } I_D = I_{DSS} \left[1 - \frac{V_{GS}}{V_{GS(off)}} \right]^2$$

Example 5.4 An N-channel JFET has $I_{DSS} = 8 \text{ mA}$ and $V_P = -5 \text{ V}$. Determine the minimum value of V_{DS} for pinch-off region and the drain current I_{DS} , for $V_{GS} = -2 \text{ V}$ in the pinch-off region.

Solution The minimum value of V_{DS} for pinch-off to occur for $V_{GS} = -2 \text{ V}$ is

$$V_{DSmin} = V_{GS} - V_P = -2 - (-5) = 3 \text{ V}$$

$$I_{DS} = I_{DSS} \left[1 - \frac{V_{GS}}{V_P} \right]^2$$

$$= 8 \times 10^{-3} [1 - (-2)/(-5)]^2 = 2.88 \text{ mA}$$

Example 5.5 For an N-Channel silicon FET with $a = 3 \times 10^{-4} \text{ cm}$ and $N_D = 10^{15} \text{ electrons/cm}^3$, find the pinch-off voltage. [JNTU June 2009]

Solution We know that the pinch-off voltage is

$$|V_P| = \frac{q N_D}{2 \epsilon} a^2$$

where $\epsilon = \text{dielectric constant of channel material (Si)} = \epsilon_r \epsilon_0 = 12 \epsilon_0$,

$q = \text{magnitude of electronic charge} = 1.602 \times 10^{-19} \text{ C}$

a is in metres and N_D is in electrons/ m^3

$$\text{Therefore, } |V_P| = \frac{1.602 \times 10^{-19} \times 10^{21}}{2 \times 12 \times 8.854 \times 10^{-12}} \times (3 \times 10^{-6})^2 = 5.8 \text{ V}$$

Example 5.6 Determine the pinch-off voltage for an N-channel silicon FET with a channel width of $5.6 \times 10^{-4} \text{ cm}$ and a donor concentration of $10^{15}/\text{cm}^3$. Given that dielectric constant of Si is 12.

Solution $N_D = 10^{15}/\text{cm}^3 = 10^{21}/\text{m}^3$, $\epsilon = \epsilon_r \epsilon_0 = 12 \epsilon_0$

$$a = 5.6 \times 10^{-4} \text{ cm} = 5.6 \times 10^{-6} \text{ m}$$

The pinch-off voltage, $|V_p| = \frac{q N_D}{2\epsilon} a^2$

where ϵ = dielectric constant of channel material (Si) = $\epsilon_r \epsilon_0 = 12\epsilon_0$
 q = magnitude of electronic charge
 a is in metres and N_D is in electrons/ m^3 .

$$|V_p| = \frac{1.602 \times 10^{-19} \times 10^{21}}{2 \times 12 \times 8.854 \times 10^{-12}} \times (5.6 \times 10^{-6})^2 = 23.6 \text{ V.}$$

Example 5.7 For a P -channel silicon FET, with an effective channel width, $a = 2 \times 10^{-4} \text{ cm}$ and channel resistivity $\rho = 20 \Omega - \text{cm}$, find the pinch-off voltage.

Solution We know that the pinch-off voltage for P -channel FET,

$$V_p = \frac{q N_A}{2\epsilon} \times a^2$$

For silicon, $\epsilon = 12\epsilon_0$ and $\mu_p = 500 \text{ cm}^2 / \text{V} - \text{sec}$

$$\sigma = \frac{1}{\rho} = p\mu_p \times q = N_A \mu_p \times q$$

i.e.
$$q N_A = \frac{1}{\rho \mu_p} = \frac{1}{20 \times 500}$$

$$q N_A = 1 \times 10^{-4}$$

Therefore,
$$V_p = \frac{q N_A}{2\epsilon} \times a^2 = \frac{1 \times 10^{-4}}{2 \times 12 \times \left(\frac{1}{36\pi \times 10^{11}} \right)} \times (2 \times 10^{-4})^2 = 1.89 \text{ V}$$

5.7 THE JFET SMALL-SIGNAL MODEL

From the drain and transfer characteristics of the Field Effect Transistor, (JFET and MOSFET), the drain current of an FET is a function of Drain to Source voltage (v_{DS}) and Gate to Source Voltage (v_{GS}). The linear small signal equivalent circuit for FET can be drawn analogous to the BJT.

Assuming varying currents and voltages for an FET,

$$i_D = f(v_{GS}, v_{DS}) \quad (5.8)$$

If both gate and drain voltages are varied, the change in drain current is given approximately by first two terms in the Taylor's series expansion of Eq. (5.8).

$$\Delta i_D = \left(\frac{\partial i_D}{\partial v_{GS}} \right)_{V_{DS}} \Delta v_{GS} + \left(\frac{\partial i_D}{\partial v_{DS}} \right)_{V_{GS}} \Delta v_{DS} \quad (5.9)$$

In small signal notation as for BJT,

$$\Delta i_D = i_d, \Delta v_{GS} = v_{gs} \text{ and } \Delta v_{DS} = v_{ds}$$

so that Eq. (5.9) can be written as

$$i_d = g_m v_{gs} + \frac{1}{r_d} v_{ds} \quad (5.10)$$

where mutual conductance or transconductance of the FET is defined as

$$g_m = \left(\frac{\partial i_D}{\partial v_{GS}} \right)_{V_{DS}} \approx \left(\frac{\Delta i_D}{\Delta v_{GS}} \right)_{V_{DS}} = \left(\frac{i_d}{v_{gs}} \right)_{V_{DS}} \quad (5.11)$$

and drain (or output) resistance of the FET is defined as

$$r_d = \left(\frac{\partial v_{DS}}{\partial i_D} \right)_{V_{GS}} \approx \left(\frac{\Delta v_{DS}}{\Delta i_D} \right)_{V_{GS}} = \left(\frac{v_{ds}}{i_d} \right)_{V_{GS}} \quad (5.12)$$

The reciprocal of r_d is the drain conductance g_d .

An amplification factor μ for an FET may be defined as

$$\mu = \left(-\frac{\partial v_{DS}}{\partial v_{GS}} \right)_{I_D} \approx \left(-\frac{\Delta v_{DS}}{\Delta v_{GS}} \right)_{I_D} = \left(-\frac{v_{ds}}{v_{gs}} \right)_{i_d=0} \quad (5.13)$$

From Eq. (5.10) by setting $i_d = 0$, it can be verified that μ , r_d and g_m are related by

$$\mu = r_d g_m \quad (5.14)$$

A small-signal model for FET in common source configuration can be drawn satisfying Eq. (5.10) as shown in Fig. 5.8(a).

This low frequency model for FET has a Norton's output circuit with a dependent current generator whose magnitude is proportional to the gate-to-source voltage. The proportionality factor is the transconductance ' g_m '. The output resistance is r_d . The input resistance between the gate and source is infinite, since it is assumed that the reverse biased gate draws no current. For the same reason, the resistance between gate and drain is assumed to be infinite.

The small signal models for the common source FET can be used for analyzing the three basic FET amplifier configurations: (i) Common source (CS), (ii) Common drain (CD) or *source-follower*, and (iii) Common gate (CG). The CS amplifier which provides good voltage amplification is most frequently used. The CD amplifier with high input impedance and near-unity voltage gain is used as a buffer amplifier and the CG amplifier is used as high frequency amplifier. The small signal current-source model for the FET in CS configuration is shown in Fig. 5.8(a) and the voltage-source model shown in Fig. 5.8(b) can be derived by finding the Thevenin's equivalent for the output part of Fig. 5.8(a). μ , r_d and g_m are the amplification factor, drain resistance and mutual conductance of the FET.

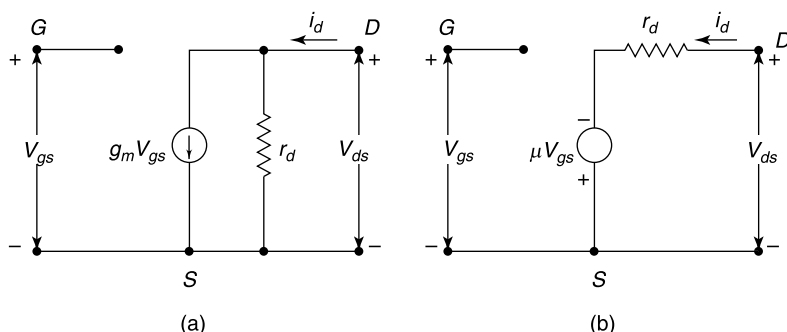


Fig. 5.8 (a) Small Signal Current-source Model for FET in CS Configuration
(b) Voltage-source Model for FET in CS Configuration

5.8 APPLICATIONS OF JFET

1. FET is used as a buffer in measuring instruments, receivers since it has high input impedance and low output impedance.
2. FETs are used in RF amplifiers in FM tuners and communication equipment for the low noise level.
3. Since the input capacitance is low, FETs are used in cascade amplifiers in measuring and test equipments.
4. Since the device is voltage controlled, it is used as a voltage variable resistor in operational amplifiers and tone controls.
5. FETs are used in mixer circuits in FM and TV receivers, and communication equipment because inter modulation distortion is low.
6. It is used in oscillator circuits because frequency drift is low.
7. As the coupling capacitor is small, FETs are used in low frequency amplifiers in hearing aids and inductive transducers.
8. FETs are used in digital circuits in computers, LSD and memory circuits because of its small size.

5.9 METAL OXIDE SEMICONDUCTOR FIELD EFFECT TRANSISTOR (MOSFET)

MOSFET is the common term for the Insulated Gate Field Effect Transistor (IGFET). There are two basic forms of MOSFET: (i) Enhancement MOSFET and (ii) Depletion MOSFET.

Principle By applying a transverse electric field across an insulator, deposited on the semiconducting material, the thickness and hence the resistance of a conducting channel of a semiconducting material can be controlled.

In a depletion MOSFET, the controlling electric field reduces the number of majority carriers available for conduction, whereas in the enhancement MOSFET, application of electric field causes an increase in the majority carrier density in the conducting regions of the transistor.

5.10

CONSTRUCTION, PRINCIPLE OF OPERATION AND SYMBOLS OF ENHANCEMENT MOSFET

Construction The construction of an N-channel enhancement MOSFET is shown in Fig. 5.9(a) and the circuit symbols for an N-channel and a P-channel enhancement MOSFET are shown in Figs 5.9(b) and (c), respectively. As there is no continuous channel in an enhancement MOSFET, this condition is represented by the broken line in the symbols.

Two highly doped N^+ regions are diffused in a lightly doped substrate of P-type silicon. One N^+ region is called the source S and the other one is called the drain D . They are separated by 1 mil (10^{-3} inch). A thin insulating layer of SiO_2 is grown over the surface of the structure and holes are cut into the oxide layer, allowing contact with source and drain. Then a thin layer of metal aluminium is formed over the layer of SiO_2 . This metal layer covers the entire channel region and it forms the gate G .

The metal area of the gate, in conjunction with the insulating oxide layer of SiO_2 and the semiconductor channel forms a parallel plate capacitor. This device is called the insulated gate FET because of the insulating layer of SiO_2 . This layer gives an extremely high input impedance for the MOSFET.

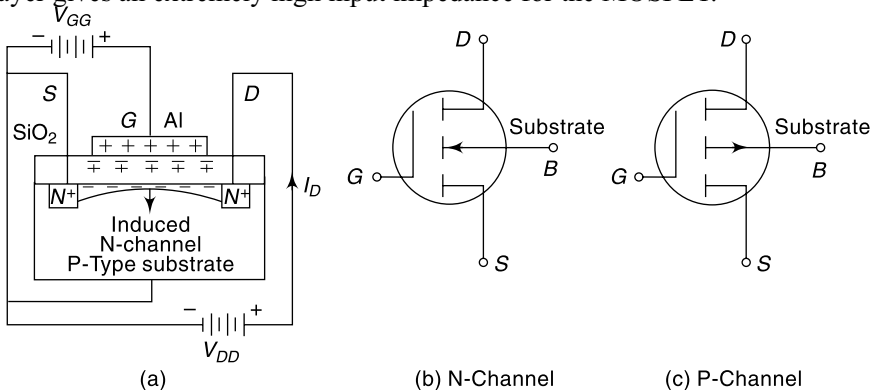


Fig. 5.9 (a) N-channel Enhancement MOSFET, (b) and (c) Circuit Symbols for Enhancement MOSFET

Operation If the substrate is grounded and a positive voltage is applied at the gate, the positive charge on G induces an equal negative charge on the substrate side between the source and drain regions. Thus, an electric field is produced between the source and drain regions. The direction of the electric field is perpendicular to the plates of the capacitor through the oxide. The negative charge of electrons which are minority carriers in the P-type substrate forms an inversion layer. As the positive voltage on the gate increases, the induced negative charge in the semiconductor increases. Hence, the conductivity increases and current flows from source to drain through the induced channel. Thus the drain current is enhanced by the positive gate voltage as shown in Fig. 5.10.

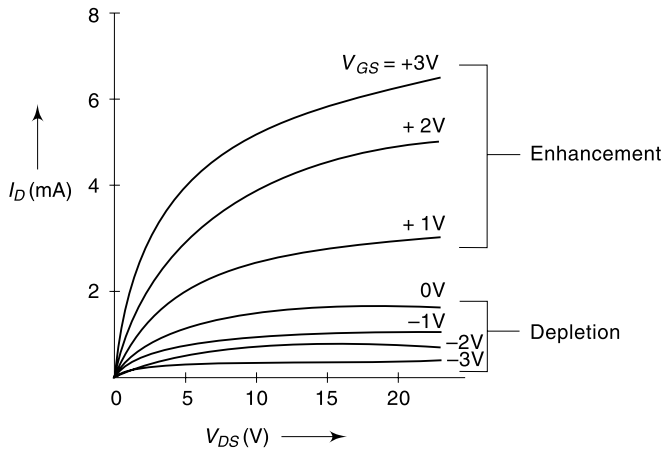


Fig. 5.10 Volt-ampere Characteristics of MOSFET

5.11 CONSTRUCTION, PRINCIPLE OF OPERATION AND SYMBOLS OF DEPLETION MOSFET

The construction of an N-channel depletion MOSFET is shown in Fig. 5.11(a) where an N-channel is diffused between the source and drain to the basic structure of MOSFET. The circuit symbols for an N-channel and a P-channel depletion MOSFET are shown in Figs 5.11(b) and (c), respectively.

With $V_{GS} = 0$ and the drain D at a positive potential with respect to the source, the electrons (majority carriers) flow through the N-channel from S to D . Therefore, the conventional current I_D flows through the channel D to S . If the gate voltage is made negative, positive charge consisting of holes is induced in the channel through SiO_2 of the gate-channel capacitor. The introduction of the positive charge causes depletion of mobile electrons in the channel. Thus a depletion region is produced in the channel. The shape of the depletion region depends on V_{GS} and V_{DS} . Hence the channel will be wedge shaped as shown in Fig. 5.11. When V_{DS} is increased, I_D increases and it becomes practically constant at a certain value of V_{DS} , called the pinch-off voltage. The drain current I_D almost gets saturated beyond the pinch-off voltage.

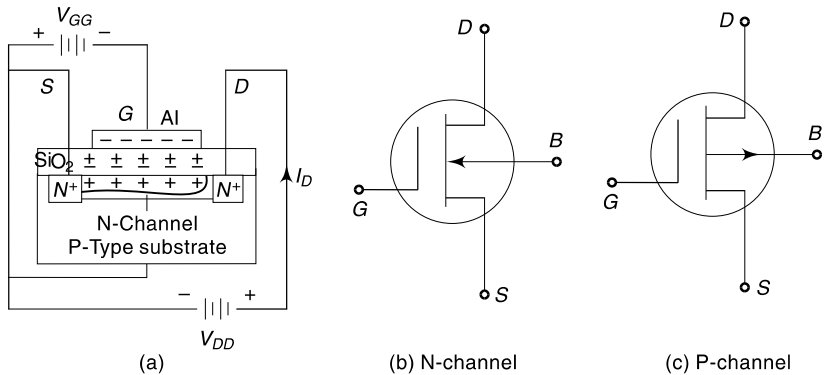


Fig. 5.11 (a) N-channel Depletion MOSFET, (b) and (c) Circuit Symbols for Depletion MOSFETs

Since the current in an FET is due to majority carriers (electrons for an N-type material), the induced positive charges make the channel less conductive, and I_D drops as V_{GS} is made negative.

The depletion MOSFET may also be operated in an enhancement mode. It is only necessary to apply a positive gate voltage so that negative charges are induced into the N-type channel. Hence, the conductivity of the channel increases and I_D increases. As the depletion MOSFET can be operated with bipolar input signals irrespective of doping of the channel, it is also called as *dual mode MOSFET*. The volt-ampere characteristics are indicated in Fig. 5.10.

The curve of I_D versus V_{GS} for constant V_{DS} is called the transfer characteristics of MOSFET and is shown in Fig. 5.12.

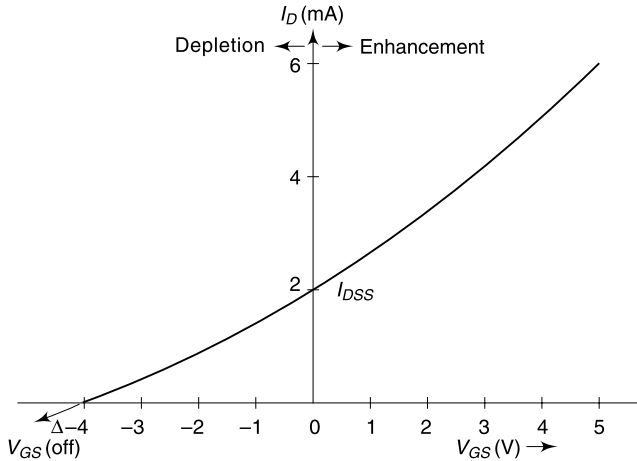


Fig. 5.12 Transfer Characteristics of MOSFET

Example 5.8 The data sheet for an enhanced MOSFET gives $I_D = 4.5 \text{ mA}$ at $V_{GS} = 12 \text{ V}$ and $V_{GS(th)} = 6 \text{ V}$. Determine the value of I_D for $V_{GS} = 10 \text{ V}$. [JNTU May/June 2006]

Solution Given $I_D = 4.5 \text{ mA}$ at $V_{GS} = 12 \text{ V}$ and $V_{GS(th)} = 6 \text{ V}$

$$I_{DSS} = \frac{I_D}{\left(1 - \frac{V_{GS}}{V_{GS(th)}}\right)} = \frac{4.5 \times 10^{-3}}{\left(1 - \frac{12}{6}\right)^2} = 4.5 \text{ mA}$$

$$\text{For } V_{GS} = 10 \text{ V, } I_D = 4.5 \times 10^{-3} \left(1 - \frac{12}{6}\right)^2 = 2 \text{ mA.}$$

5.12 COMPARISON OF MOSFET WITH JFET

1. In enhancement and depletion types of MOSFET, the transverse electric field induced across an insulating layer deposited on the semiconductor material controls the conductivity of the channel. In the JFET the transverse electric field across the reverse biased PN junction controls the conductivity of the channel.

2. The gate leakage current in a MOSFET is of the order of 10^{-12} A. Hence the input resistance of a MOSFET is very high in the order of 10^{10} to $10^{15} \Omega$. The gate leakage current of a JFET is of the order of 10^{-9} A and its input resistance is of the order of $10^8 \Omega$.
3. The output characteristics of the JFET are flatter than those of the MOSFET and hence, the drain resistance of a JFET (0.1 to 1 M Ω) is much higher than that of a MOSFET (1 to 50 k Ω).
4. JFETs are operated only in the depletion mode. The depletion type MOSFET may be operated in both depletion and enhancement mode.
5. Comparing to JFET, MOSFETs are easier to fabricate.
6. MOSFET is very susceptible to overload voltage and needs special handling during installation. It gets damaged easily if it is not properly handled.
7. MOSFET has zero offset voltage. As it is a symmetrical device, the source and drain can be interchanged. These two properties are very useful in analog signal switching.
8. Special digital CMOS circuits are available which involve near-zero power dissipation and very low voltage and current requirements. This makes them most suitable for portable systems.

MOSFETs are widely used in digital VLSI circuits than JFETs because of their advantages.

5.13 HANDLING PRECAUTIONS FOR MOSFET

The MOSFET has the drawback of being very susceptible to overload voltage and may require special handling during installation. The MOSFET gets damaged easily if it is not properly handled. A very thin layer of SiO_2 between the gate and channel is damaged due to high voltage and even by static electricity. The static electricity may result from the sliding of a device in a plastic bag. If a person picks up the transistor by its case and brushes the gate against some grounded object, a large electrostatic discharge may result. In a relatively dry atmosphere, a static potential of 300 V is not uncommon on a person who has high resistance soles on his footwear.

MOSFETs are protected by a shorting ring that is wrapped around all four terminals during shipping and must remain in place until after the device is soldered into position. Prior to soldering, the technician should use a shorting strap to discharge his static electricity and make sure that the tip of the soldering iron is grounded. Once in circuit, there are usually low resistances present to prevent any excessive accumulation of electrostatic charge. However, the MOSFET should never be inserted into or removed from a circuit with the power ON. JFET is not subject to these restrictions, and even some MOSFETs have a built in gate protection known as “integral gate protection,” a system built into the device to get around the problem of high voltage on the gate causing a puncturing of the oxide layer. The manner in which this is done is shown in the cross sectional view of Fig. 5.13. The symbol clearly shows that between each and the source is placed a back-to-back (or front-to-front) pair of diodes, which are built right into the P-type substrate.

These diodes are designed so that if either gate exceeds $+10\text{ V}$ typically, with respect to the source, the upper diode will conduct and the lower diode will breakdown by Zener effect, providing a shunt path for excessive charge from gate to source. Similarly, if the gate voltage exceeds -10 V , the lower diode conducts and the upper diode breaks down. The breakdown voltage from either gate to source is in a range such that normal application of signal will not be affected by the gate protection diodes. In addition to providing protection against preinstallation high static voltages, these diodes also guard against in-circuit transients.

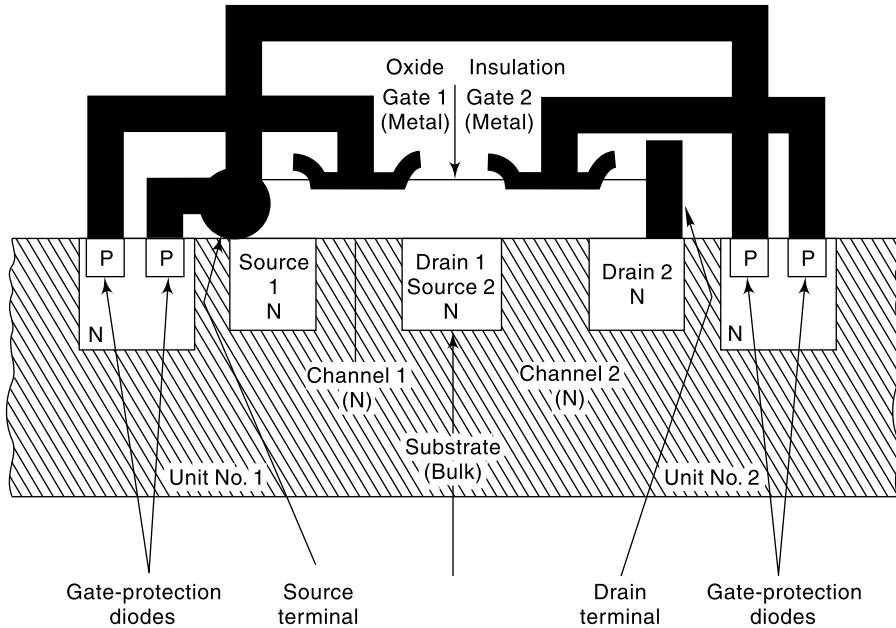


Fig. 5.13 Cross Sectional View of the Dual Gate Protected N-channel Depletion Type MOSFET

5.14 COMPARISON OF N- WITH P-CHANNEL MOSFETS

1. The P-channel enhancement MOSFET is very popular because it is much easier and cheaper to produce than the N-channel device.
2. The hole mobility is nearly 2.5 times lower than the electron mobility. Thus, a P-channel MOSFET occupies a larger area than an N-channel MOSFET having the same I_D rating.
3. The drain resistance of P-channel MOSFET is three times higher than that for an identical N-channel MOSFET.
4. The N-channel MOSFET has the higher packing density which makes it faster in switching applications due to the smaller junction areas and lower inherent capacitances.
5. The N-channel MOSFET is smaller for the same complexity than P-channel device.
6. Due to the positively charged contaminants, the N-channel MOSFET may turn ON prematurely, whereas the P-channel device will not be affected.

5.15 COMPARISON OF N- WITH P-CHANNEL FETS

1. In an N-channel JFET the current carriers are electrons, whereas the current carriers are holes in a P-channel JFET.
2. Mobility of electrons is large in N-channel JFET; mobility of holes is poor in P-channel JFET.
3. The input noise is less in N-channel JFET than that of P-channel JFET.
4. The transconductance is larger in N-channel JFET than that of P-channel JFET.

5.16 ADVANTAGES OF BJT OVER MOSFET

BJTs have some advantages over MOSFETs for at least two digital applications. Firstly, in high speed switching, they do not have the *larger* capacitance factor imposed by the gate, which when multiplied by the resistance of the channel gives the intrinsic time constant of the process. The intrinsic time constant places a limit on the speed performance at which a MOSFET can operate, since the higher frequency signals may be filtered out. Widening the channel reduces the resistance of the channel. However, it increases the capacitance by the same amount. On the other hand, reducing the width of the channel increases the resistance, but reduces the capacitance by the same amount. For example, if $R \times C = T_{c1}$, the value of $0.5R \times 2C = T_{c1}$ and $2R \times 0.5C = T_{c1}$. There is no other way to minimize the intrinsic time constant for a certain process. The various processes using different channel lengths, channel heights, gate thicknesses and materials will also have varying intrinsic time constants. This problem is mostly avoided while using a BJT, since it does not use a gate.

The second application where the BJTs prove more advantageous over the MOSFETs stems from the first reasoning. While driving several fan-out gates, the resistance of the MOSFET is in series with the gate capacitances of the other FETs, creating a secondary time constant. Delay circuits use this fact to create a fixed signal delay by using a small CMOS device to send a signal to many other, larger CMOS fan-out devices. The secondary time constant can be minimized by increasing the driving FET's channel width to decrease its resistance and decreasing the channel widths of the FETs being driven, for reducing the fan-out or driven capacitance. However, its drawback is that, this increases the capacitance of the driving FET and increases the resistance of the FETs being driven. Usually this drawback is a negligible factor, while comparing with the timing problem. The BJTs are better able to drive the other gates, since they can provide more output current than the MOSFETs. This allows the FETs being driven to get charged faster. Therefore, many integrated circuits employ the MOSFET driven inputs and BiCMOS outputs.

5.17 INTRODUCTION

Field Effect Transistor (FET) amplifiers provide an excellent voltage gain with the added feature of high input impedance. They have low power consumption with a good frequency range, and minimal size and weight. The noise output level is low. This feature makes them very useful in the amplifier circuits meant for very small signal amplifications. JFETs, depletion MOSFETs and enhancement MOSFETs are used in the design of amplifiers having comparable voltage gains. However, the depletion MOSFET circuit realizes much higher input impedance than the equivalent JFET configuration. Because of the high input impedance characteristic of FETs, the ac equivalent model is somewhat simpler than that employed for BJTs. The common source configuration is the most popular one, providing an inverted and amplified signal. However, one also finds the common drain (source follower) circuits providing unity gain with no inversion and common gate circuits providing gain with no inversion. Due to very high input impedance, the input current is generally assumed to be negligible, and it is of the order of few microamperes and the current gain is an undefined quantity. Output impedance values are comparable for both the BJT and FET circuits.

5.18 COMMON SOURCE (CS) AMPLIFIER

A simple common-source amplifier is shown in Fig. 5.14(a), and the associated small signal equivalent circuit using the voltage-source model of FET is shown in Fig. 5.14(b).

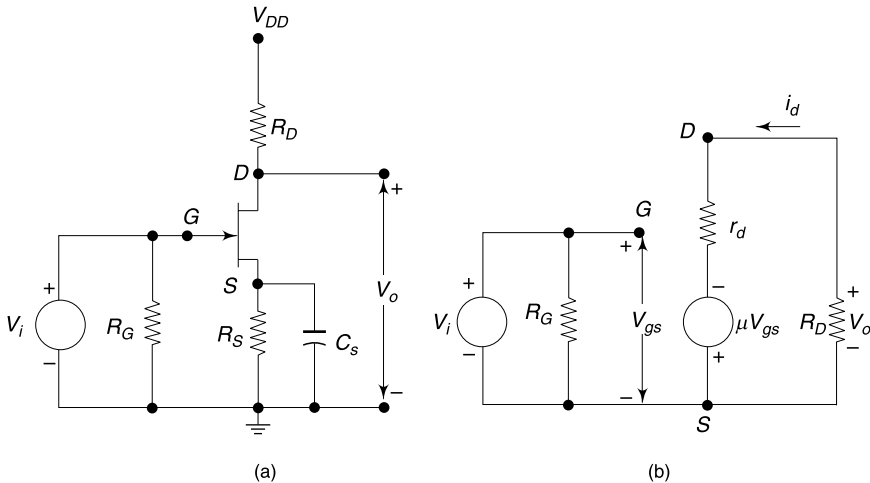


Fig. 5.14 (a) Common Source Amplifier (b) Small Signal Equivalent Circuit of CS Amplifier

Voltage Gain Source resistor (R_s) is used to set the Q-point but is bypassed by C_s for mid-frequency operation. From the small signal equivalent circuit, the output voltage,

$$V_o = \frac{-R_D}{R_D + r_d} \mu V_{gs} \quad (5.15)$$

where $V_{gs} = V_i$, the input voltage.

Hence, the voltage gain,

$$A_v = \frac{V_o}{V_i} = \frac{-\mu R_D}{R_D + r_d} \quad (5.16)$$

Input Impedance From Fig. 5.14(b), Input Impedance is

$$Z_i = R_G$$

For voltage divider bias as in CE Amplifiers of BJT

$$R_G = R_1 \parallel R_2$$

Output Impedance Output Impedance is the impedance measured at the output terminals with the input voltage $V_i = 0$

From Fig. 5.14(b) when $V_i = 0$, $V_{gs} = 0$ and hence

$$\mu V_{gs} = 0$$

Then the equivalent circuit for calculating Output Impedance is given in Fig. 5.14(c).

Output impedance $Z_o = r_d \parallel R_D$

Normally r_d will be far greater than R_D [Typical values are $r_d = 35 \text{ k}\Omega$ and $R_D = 5 \text{ k}\Omega$]

Hence $Z_o \approx R_D$

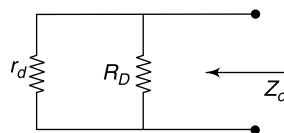


Fig. 5.14(c) Calculation of Output Impedance

Example 5.9 In the CS amplifier of Fig. 5.14(b), let $R_D = 5 \text{ k}\Omega$, $R_G = 10 \text{ M}\Omega$, $\mu = 50$, and $r_d = 35 \text{ k}\Omega$. Evaluate the voltage gain A_v , input impedance Z_i , and output impedance Z_o .

Solution The voltage gain,

$$\begin{aligned} A_v &= \frac{V_o}{V_i} = \frac{-\mu R_D}{R_D + r_d} \\ &= \frac{-50 \times 5 \times 10^3}{5 \times 10^3 + 35 \times 10^3} \\ &= \frac{-250 \times 10^3}{40 \times 10^3} = -6.25 \end{aligned}$$

The minus sign indicates a 180° phase shift between V_i and V_o .

$$\begin{aligned}\text{input impedance} \quad Z_i &= R_G \\ &= 10 \text{ M}\Omega\end{aligned}$$

$$\begin{aligned}\text{output impedance} \quad Z_o &\approx R_D \\ &= 5 \text{ k}\Omega\end{aligned}$$

Example 5.10 A FET amplifier in the common-source configuration uses a load resistance of $500 \text{ k}\Omega$. The ac drain resistance of the device is $100 \text{ k}\Omega$ and the transconductance is 0.8 mA/V . Calculate the voltage gain of

Solution Given load resistance, $R_L = R_D = 500 \text{ k}\Omega$, $r_d = 100 \text{ k}\Omega$ and $g_m = 0.8 \text{ mA/V}$

The transconductance, $\mu = g_m r_d = 0.8 \times 10^{-3} \times 100 \times 10^3 = 80$

$$\begin{aligned}\text{The voltage gain,} \quad A_v &= -\frac{g_m R_D}{R_D + r_d} \\ &= \frac{-80 \times 500 \times 10^3}{500 \times 10^3 + 100 \times 10^3} \\ &= -\frac{40 \times 10^6}{600 \times 10^3} \\ &= -66.67\end{aligned}$$

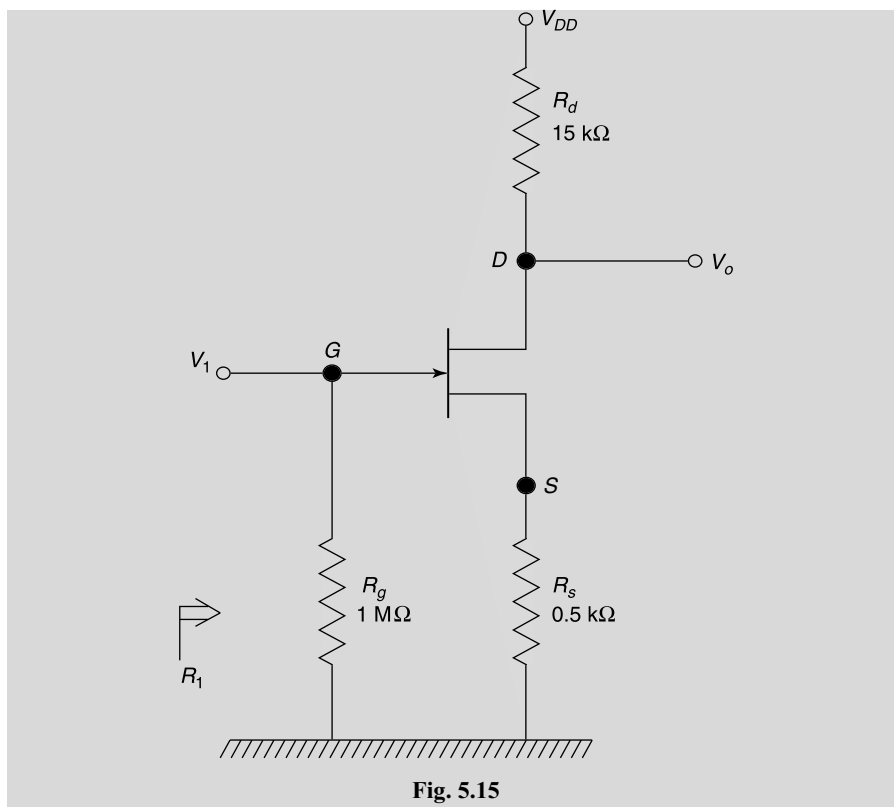
Example 5.11 A common source FET amplifier circuit shown in Fig. 5.15 with unbypassed R_S has the following circuit parameters: $R_d = 15 \text{ k}\Omega$, $R_g = 1 \text{ M}\Omega$, $r_d = 5 \text{ k}\Omega$, $g_m = 15 \text{ mho}$ and $V_{DD} = 20 \text{ V}$, calculate A_v and R_o .
[JNTU May/June 2006]

Solution Given $g_m = 5 \times 10^{-3} \text{ S}$, $r_d = 5 \text{ k}\Omega$ and $R_d = 15 \text{ k}\Omega$

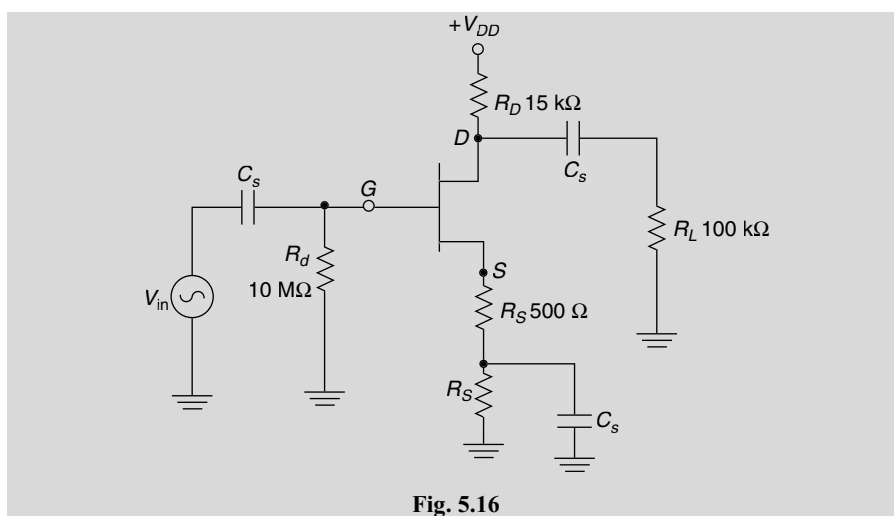
Hence, $\mu = r_d g_m = 5 \times 10^3 \times 5 \times 10^{-3} = 25$

$$\begin{aligned}\text{Therefore, voltage gain,} \quad A_v &= \frac{-\mu R_d}{R_d + r_d} \\ &= \frac{-25 \times 15 \times 10^3}{15 \times 10^3 + 5 \times 10^3} \\ &= -18.75\end{aligned}$$

The output resistance, $R_o \cong R_d = 15 \text{ k}\Omega$



Example 5.12 The figure (Fig. 5.16) is a swamped FET amplifier. Determine the voltage gain when $R_L = 100 \text{ k}\Omega$. Neglect the FET output resistance (r_d). Take $g_m = 4 \text{ mS}$. [JNTU April/May 2007]



Solution $V_{in} - V_{gs} - V_{rs} = 0$

$$V_{gs} = V_{in} - V_{rs} = V_{in} - I_D \cdot r_s$$

Here,

$$I_D = g_m V_{gs}$$

Output voltage

$$V_O = -I_D \cdot r_L = -g_m V_{gs} \cdot r_L$$

$$V_O = -g_m (V_{in} - I_D \cdot r_s) r_L$$

$$I_D = g_m V_{gs} = g_m (V_{in} - I_D r_s)$$

Therefore,

$$I_D = \frac{g_m V_{in}}{1 + g_m \cdot r_s}$$

Output voltage,

$$V_O = -I_D \cdot r_L = -\frac{g_m V_{in} \cdot r_L}{1 + g_m \cdot r_s}$$

Voltage gain,

$$A_v = \frac{V_O}{V_{in}} = \frac{-g_m \cdot r_L}{1 + g_m \cdot r_s} = \frac{4 \times 10^{-3} \times 13.04 \times 10^3}{1 + (4 \times 10^{-3} \times 500)} = -17.3$$

5.19 COMMON DRAIN (CD) AMPLIFIER

A simple common-drain amplifier is shown in Fig. 5.17(a) and the associated small signal equivalent circuit using the voltage-source model of FET is shown in Fig. 5.17(b). Since voltage V_{gd} is more easily determined than V_{gs} , the voltage source in the output circuit is expressed in terms of V_{gd} using Thevenin's theorem. The output voltage,

$$V_o = \frac{R_S}{R_S + \frac{r_d}{\mu + 1}} \times \frac{\mu}{\mu + 1} V_{gd} = \frac{\mu R_S V_{gd}}{(\mu + 1) R_S + r_d} \quad (5.17)$$

where $V_{gd} = V_i$, the input voltage.

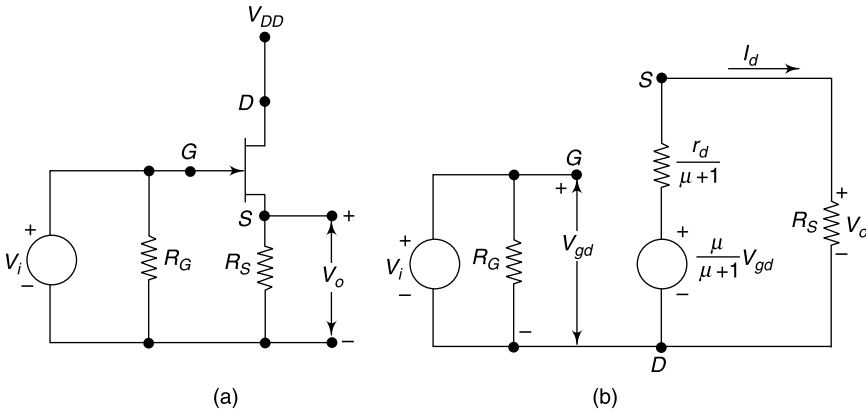


Fig. 5.17 (a) Common Drain Amplifier, (b) Small Signal Equivalent Circuit of CD Amplifier

Hence, the voltage gain,

$$A_v = \frac{V_o}{V_i} = \frac{\mu R_S}{(\mu + 1) R_S + r_d} \quad (5.18)$$

Input Impedance From Fig. 5.15(b), Input Impedance $Z_i = R_G$

Output Impedance From Fig. 5.15(b), Output Impedance measured at the output terminals with input voltage $V_i = 0$ can be simply calculated from the following equivalent circuit.

$$\text{As } V_i = 0; \quad V_{gd} = 0; \quad \frac{\mu}{\mu + 1} V_{gd} = 0$$

Output Impedance

$$Z_o = \frac{r_d}{\mu + 1} \parallel R_S$$

when $\mu \gg 1$ (typical value of $\mu = 50$)

$$Z_o = \frac{r_d}{\mu} \parallel R_S = \frac{1}{g_m} \parallel R_S$$

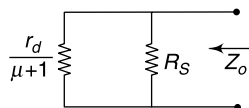


Fig. 5.17(c) Calculation of Output Impedance

Example 5.13 In the CD amplifier of Fig. 5.17(b), let $R_S = 4 \text{ k}\Omega$, $R_G = 10 \text{ M}\Omega$, $\mu = 50$, and $r_d = 35 \text{ k}\Omega$. Evaluate the voltage gain A_v , Input Impedance Z_i and Output Impedance Z_o .

Solution The voltage gain,

$$\begin{aligned} A_v &= \frac{V_o}{V_i} = \frac{\mu R_S}{(\mu + 1) R_S + r_d} \\ &= \frac{50 \times 4 \times 10^3}{(50 + 1) \times 4 \times 10^3 + 35 \times 10^3} = 0.836 \end{aligned}$$

The positive value indicates that V_o and V_i are in-phase and further note that $A_v < 1$ for CD amplifier.

Input impedance $Z_i = R_G = 10 \text{ M}\Omega$

$$\begin{aligned} \text{Output impedance } Z_o &= \frac{1}{g_m} \parallel R_S \\ &= \left(\frac{r_d}{\mu} \right) \parallel R_S \end{aligned}$$

$$Z_o = \frac{35 \times 10^3}{50} \parallel 4 \times 10^3 = 595.7 \Omega$$

5.20 COMMON GATE (CG) AMPLIFIER

A simple common-gate amplifier is shown in Fig. 5.18(a) and the associated small signal equivalent circuit using the current-source model of FET is shown in Fig. 5.18(b).

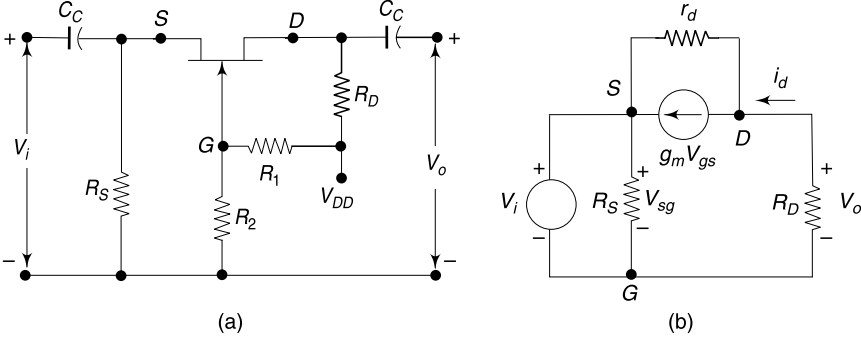


Fig. 5.18 (a) Common Gate Amplifier, (b) Small Signal Equivalent Circuit of CG Amplifier

Voltage Gain From the small signal equivalent circuit by applying KCL, $i_r = i_d - g_m V_{gs}$. Applying KVL around the outer loop gives

$$V_o = (i_d - g_m V_{gs}) r_d + V_{sg}$$

But

$$V_{sg} = -V_{gs} = V_i \quad \text{and} \quad i_d = \frac{-V_o}{R_D}$$

Thus,

$$V_o = \left(\frac{-V_o}{R_D} + g_m V_i \right) r_d + V_i$$

Hence, the voltage gain,

$$A_v = \frac{V_o}{V_i} = \frac{(g_m r_d + 1) R_D}{R_D + r_d} \quad (5.19)$$

Input Impedance Figure 5.18(b) is modified for calculation of Input Impedance as shown in Fig. 5.18(c).

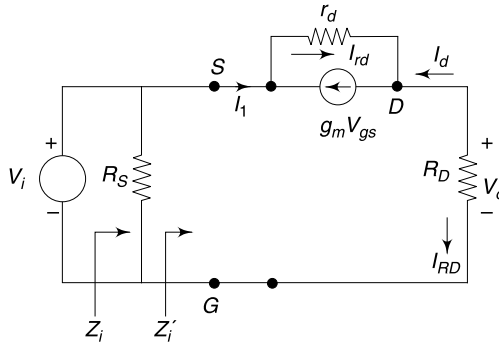


Fig. 5.18(c) Modified Equivalent Circuit

Current through r_d is

$$I_{rd} = -I_r = I_1 + g_m V_{gs}$$

$$I_1 = I_{rd} - g_m V_{gs}$$

where

$$\begin{aligned} I_{rd} &= \frac{V_i - V_o}{r_d} \\ &= \frac{V_i - I_{RD} R_D}{r_d} \end{aligned}$$

Hence

$$I_1 = \frac{V_i - I_{RD} R_D}{r_d} - g_m V_{gs}$$

From Fig. 5.18(c),

$$\begin{aligned} V_i &= -V_{gs} \\ I_1 &= \frac{V_i - I_{RD} R_D}{r_d} + g_m V_i \\ &= \frac{V_i}{r_d} - \frac{I_{RD} R_D}{r_d} + g_m V_i \\ I_1 + \frac{I_{RD} R_D}{r_d} &= \frac{V_i}{r_d} + g_m V_i \end{aligned}$$

From Fig. 5.18(c),

$$I_1 = I_{RD}$$

Therefore

$$\begin{aligned} I_1 \left[\frac{r_d + R_D}{r_d} \right] &= V_i \left[\frac{1}{r_d} + g_m \right] \\ \frac{V_i}{I_1} &= \frac{r_d + R_D}{1 + g_m r_d} = Z' \end{aligned}$$

From Fig. 5.18(c),

$$\begin{aligned} Z_i &= R_S + Z'_i \\ &= R_S + \frac{r_d + R_D}{1 + g_m r_d} \end{aligned}$$

In practice, $r_d \gg R_D$ and $g_m r_d \gg 1$.

Therefore,

$$\begin{aligned} Z_i &= R_S + \frac{r_d}{g_m r_d} \\ &= R_S + \frac{1}{g_m} \end{aligned}$$

Output Impedance It is the impedance seen from the output, terminals with input short circuited.

From Fig. 5.18(c) when $V_i = 0$, $V_{sg} = 0$, the resultant equivalent circuit is shown in Fig. 5.18(d).

$$Z_o = r_d \parallel R_D$$

as

$$r_d \gg R_D$$

$$Z_o \approx R_D$$

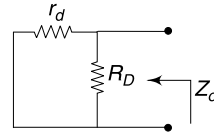


Fig. 5.18(d) Equivalent Circuit for Output Impedance

Example 5.14 The figure shows the circuit of a common gate JFET amplifier. The JFET has $g_m = 2500 \mu S$. Determine the voltage gain and input resistance. [JNTU April/May 2007]

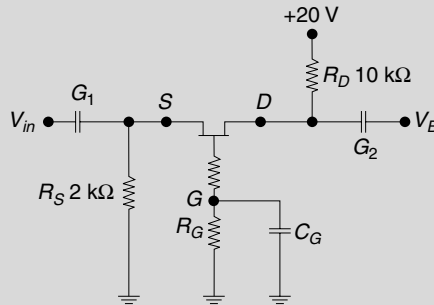


Fig. 5.19

Solution Voltage gain, $A_v = g_m R_D = 2500 \times 10^{-6} \times 10 \times 10^3 = 25$

Input resistance, $R_i = R_S \parallel \frac{1}{g_m} = 2 \times 10^3 \parallel \frac{1}{2500 \times 10^{-6}} = 333.33 \Omega$

Example 5.15 In the CG amplifier of Fig. 5.19(b), let $R_D = 2 k\Omega$, $R_S = 1 k\Omega$, $g_m = 1.43 \times 10^{-3} mho$, and $r_d = 35 k\Omega$. Evaluate the voltage gain A_v . Input impedance Z_i , and output impedance Z_o .

Solution The voltage gain,

$$\begin{aligned} A_v &= \frac{V_o}{V_i} = \frac{(g_m r_d + 1) R_D}{R_D + r_d} \\ &= \frac{(1.43 \times 10^{-3} \times 35 \times 10^3 + 1) 2 \times 10^3}{2 \times 10^3 + 35 \times 10^3} = 2.75 \end{aligned}$$

Input impedance, $Z_i = R_S \parallel \frac{1}{g_m}$

$$= 1 \times 10^3 \parallel \frac{10^3}{1.4}$$

$$= 0.41 k\Omega$$

Output impedance, $Z_o \approx R_D = 2 k\Omega$

5.21 A GENERALIZED FET AMPLIFIER

A generalized FET amplifier shown in Fig. 5.17 can be used to analyse the CS, CG and CD amplifier circuits at low frequencies. It consists of three independent signal sources, v_i in series with the gate, v_s in series with the source and v_a in series with the drain. For the CS amplifier, $v_s = v_a = 0$, and the output is v_{o1} taken at the drain. For the CG amplifier, $v_i = v_a = 0$, the signal is v_s with a source resistance R_s , and the output is v_{o1} . For the CD circuit i.e. source follower, $R_D = 0$, $v_s = v_a = 0$, the signal voltage is v_i and the output is v_{o2} taken at the source. The signal-source resistance is negligible small when it is in series with the gate that draws negligible current.

5.21.1 Output from the Drain

The Thevenin's equivalent circuits from drain to ground, and from source to ground of the generalized FET amplifier of Fig. 5.20 are shown in Fig. 5.21(a) and (b) respectively. Looking into the drain of the FET of Fig. 5.21(a), for small-signal operation, an equivalent circuit is found consisting of two generators in series, one is $-\mu$ times the gate-signal voltage v_i and the second is $(\mu + 1)$ times the source-signal voltage v_s and the resistance is $r_d + (\mu + 1)R_s$. Here the source voltage v_s and the resistance in the source lead are both multiplied by the factor $(\mu + 1)$.

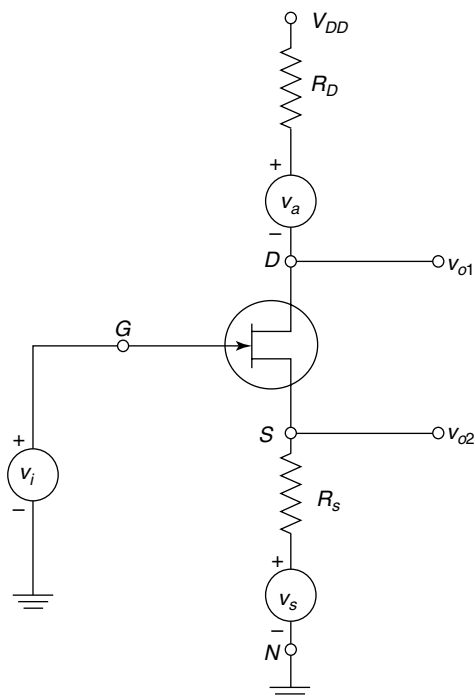


Fig. 5.20 A Generalized FET Amplifier

5.21.2 For CS Amplifier

From Fig. 5.21(a), with $v_s = v_a = 0$, the voltage gain is

$$A_V = \frac{v_{o1}}{v_i} = \frac{-\mu R_D}{r_d + (\mu + 1)R_s + R_s} = \frac{-g_m R_D}{1 + g_m R_s + g_d (R_s + R_D)}$$

where the minus sign indicates a 180° phase shift between input and output.

The resistance R_o , looking into the drain, is increased by $(\mu + 1)R_s$ from its value r_d for $R_s = 0$. Considering R_D , the net output resistance R_o is

$$R_o' = (r_d + (\mu + 1)R_s) \parallel R_D$$

The addition of R_s will reduce the voltage gain and increase the output impedance. Since the gate junction is reverse-biased, the input impedance is over 100 MΩ.

5.21.3 For CG Amplifier

From Fig. 5.21(a), with $v_i = v_a = 0$, the voltage gain is

$$A_V = \frac{v_{o1}}{v_s} = \frac{(\mu + 1)R_D}{r_d + (\mu + 1)R_s + R_D} = \frac{(g_m + g_d)R_D}{1 + g_m R_s + g_d (R_s + R_D)}$$

There is no phase shift between input and output. Also, since $g_m \gg g_d$, the magnitude of the amplification is approximately equal to that of CS amplifier, with $R_s \approx 0$.

The output resistance R_o' is $(r_d + (\mu + 1)R_s) \parallel R_D$. Unless R_s is very small, R_o' will be larger than $r_d \parallel R_D$. Referring to Fig. 5.21(b), the input impedance R_i' between source and ground is obtained as

$$R_i' = \left(\frac{r_d + R_D}{\mu + 1} \right) \parallel R_s.$$

The CG amplifier has low input resistance and high output resistance.

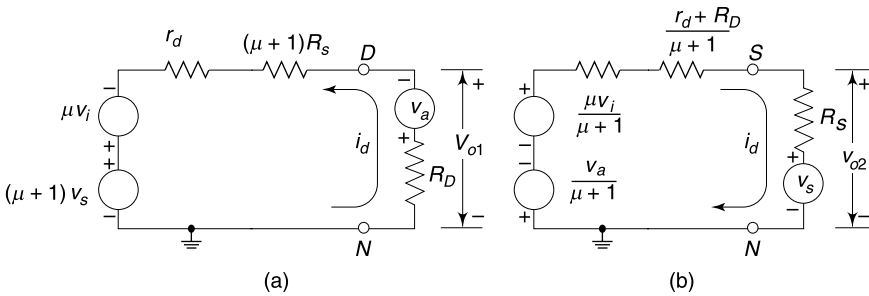


Fig. 5.21 Thevenin's Equivalent Circuits for the Generalized Amplifier of Fig. 5.20 Looking into (a) the Drain and (b) the Source, $\mu = g_m r_d$

5.21.4 Output from the Source

Looking into the source of the FET of Fig. 5.21(b), for small-signal operation, an equivalent circuit is found consisting of two generators in series, one is $\mu(\mu + 1)$ times the gate-signal voltage v_i and the second is $1/(\mu + 1)$ times the drain-signal voltage v_a and the resistance is $(r_d + R_D)/(\mu + 1)$. Here the voltage v_a and the resistance in the drain circuit are both divided by the factor $(\mu + 1)$.

5.21.5 For CD Amplifier

The voltage gain A_V of the source follower is obtained from Fig. 5.21(b) with $v_s = v_a = 0$ and $R_D = 0$.

$$A_V = \frac{V_{o2}}{v_i} = \frac{\mu R / (\mu + 1)}{r_d / (\mu + 1) + R_s} = \frac{g_m R_s}{1 + (g_m + g_d) R_s}$$

From Fig. 5.21(b), the output impedance R_o of the source follower at lower frequencies, with $R_D = 0$ and with R_s considered external to the amplifier is

$$R_o = \frac{r_d}{\mu + 1} = \frac{1}{g_m + g_d}$$

The output impedance R'_o considering the source resistance R_s is $R'_o = R_o \parallel R_s$.

5.22 BIASING THE FET

For the proper functioning of a linear FET amplifier, it is necessary to maintain the operating point Q stable in the central portion of the pinch off region. The Q-point should be independent of device parameter variations and ambient temperature changes. This can be achieved by suitably selecting the gate to source voltage (V_{GS}) and drain current (I_D) which is referred to as biasing.

5.22.1 Fixing the Q-point

The Q-point, the quiescent point or operating point for a self-biased JFET is established by determining the value of drain current I_D for a desired value of gate-to-source voltage, V_{GS} , or vice-versa. However, if the data sheet of JFET includes a transfer characteristics curve, then the Q-point may be determined by using the procedure given below.

- (i) Select a convenient value of drain current whose value is generally taken half of the maximum possible value of drain current, I_{DSS} , Then find the voltage drop across source resistor, R_S , by

$$V_S = I_D R_S$$

and the gate-to-source voltage from the equation

$$V_{GS} = -V_S$$

- (ii) Plot the assumed value of drain current, I_D , and the corresponding gate-to-source voltage, V_{GS} , on the transfer characteristics curve.
- (iii) Draw a line through the plotted point and the origin. The point of intersection of the line and the curve gives the desired Q-point. Then, read the co-ordinates of Q-point.

It is necessary to fix the Q-point near the mid-point of the transfer characteristic curve of a JFET. The mid-point bias allows a maximum amount of drain current swing between the values of I_{DSS} and the origin.

The following analytical method or graphical method can be used for the design of self bias circuit.

Analytical Method

The values of the maximum drain current, I_{DSS} , and the gate-to-source cut-off voltage, $V_{GS(off)}$ are noted down from the data sheets of JFET.

The value of the drain current is determined by

$$I_D = \left[1 - \frac{V_{GS}}{V_{GS(off)}} \right]^2$$

For example, if we select the gate-to-source voltage, $V_{GS} = V_{GS(off)} / 4$, then the value of the drain current will be

$$I_D = I_{DSS} [1 - 0.25]^2 = I_{DSS} (0.75)^2 = 0.56 I_{DSS}$$

Here, the drain current is slightly more than one-half of I_{DSS} . But it will bias the JFET close to the mid-point of the curve. The value of the drain resistor, R_D , is selected in such a way that the drain voltage, V_D , is equal to half the drain supply voltage, R_D . The value of gate resistor, R_G , is chosen arbitrarily large, so that it prevents loading on the driving stages.

Graphical Method

A self-bias line is drawn such that it intersects the transfer characteristic curve near its mid-point gives the required Q-point. Then the co-ordinates of the Q-point are obtained. The value of source resistance, R_S , is expressed by the ratio of gate-to-source voltage, V_{GS} , to the drain current, I_D .

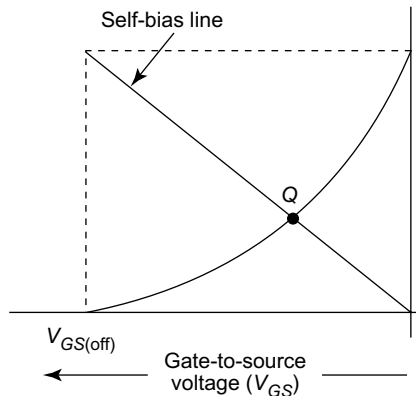


Fig. 5.22

Therefore, the source resistance is given by

$$R_S = \frac{V_{GS}}{I_D}$$

However, a more accurate method is to draw a self-bias line through the coordinates of I_{DSS} and $V_{GS(off)}$ as shown in Fig. 5.22. Then the point of intersection of self-bias line and the transfer characteristic curve locates the Q-point. The value of the source resistor is expressed by the relation

$$R_S = \frac{V_{GS(off)}}{I_{DSS}}$$

The value of drain resistor, R_D , and the gate resistor, R_G , are selected in the same way as discussed above for the analytical method.

An FET may have a combination of self bias and fixed bias to provide stability of the quiescent drain current against device and temperature variations.

5.22.2 Self-bias

Figure 5.23 shows the self-bias circuit for an N-channel FET. When the drain voltage V_{DD} is applied, a drain current I_D flows even in the absence of gate voltage (V_G). The voltage drop across resistor R_S produced by the drain current is given by $V_S = I_D R_S$. This voltage drop reduces the gate to source reverse voltage required for FET operation. The feedback resistor R_S prevents any variation in FET drain current.

The drain voltage, $V_D = V_{DD} - I_D R_D$

The drain-to-source voltage,

$$\begin{aligned} V_{DS} &= V_D - V_S = (V_{DD} - I_D R_D) - I_D R_S \\ &= V_{DD} - I_D (R_D + R_S) \end{aligned}$$

The gate-to-source voltage

$$V_{GS} = V_{GG} - V_S = 0 - I_D R_S = -I_D R_S$$

When drain current increases, the voltage drop across R_S increases. The increased voltage drop increases the reverse gate to source voltage, which decreases the effective width of the channel and hence, reduces the drain current. Now the reduced drain current decreases the gate to source voltage which, in turn, increases the effective width of channel thereby increasing the value of drain current.

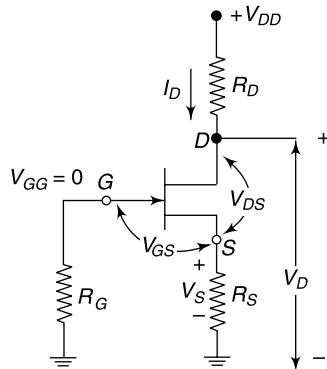


Fig. 5.23 Self-Bias Circuit for an N-Channel JFET

5.22.3 Voltage Divider Bias

Figure 5.24(a) shows the voltage divider bias circuit and its Thevenin's equivalent is shown in Fig. 5.24(b). Resistors R_1 and R_2 connected on the gate side forms a voltage divider. The gate voltage,

$$V_{GG} = \left(\frac{R_2}{R_1 + R_2} \right) V_{DD} \quad \text{and} \quad R_G = \frac{R_1 R_2}{R_1 + R_2}$$

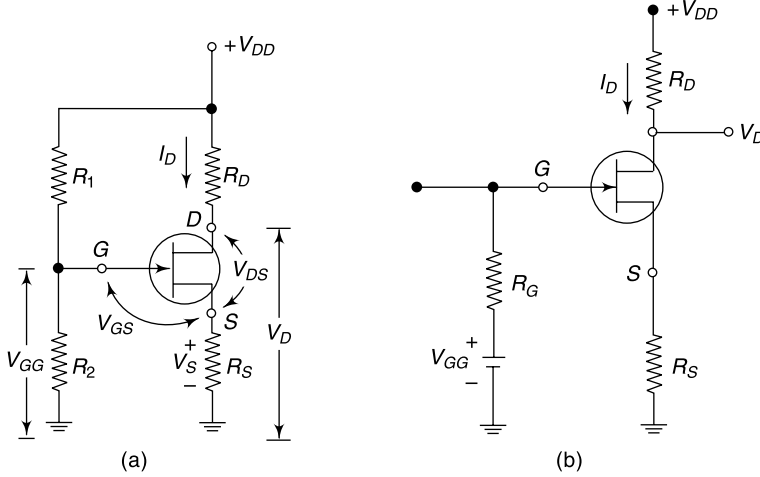


Fig. 5.24 (a) Voltage Divider Bias Circuit, and (b) Thevenin's Equivalent Circuit

The bias line satisfies the equation $V_{GS} = V_{GG} - I_D R_S$.

The drain to ground voltage, $V_D = V_{DD} - I_D R_D$. If the gate voltage V_{GG} is very large as compared to gate to source V_{GS} , the drain current is approximately constant. In practice, the voltage divider bias is less effective with JFET than BJT.

This is because, in BJT, $V_{BE} \approx 0.7$ V (silicon) with only minor variations from one transistor to another. But in a JFET, the V_{GS} can vary several volts from one JFET to another.

5.22.4 Fixed Bias

The FET device needs DC bias for setting the gate to source voltage V_{GS} to give desired drain-current I_D . For a JFET, the drain current is limited by I_{DSS} . Since the FET has a high input impedance, it does not allow the gate current to flow and the dc voltage of the gate set by a voltage divider or a fixed battery is not affected or loaded by the FET.

The fixed bias circuit for an N-Channel JFET shown in Fig. 5.25 is obtained by using a supply V_{GG} . This supply ensures that the gate is always negative with respect to source and no current flows through resistor R_G and gate terminal i.e. $I_G = 0$. The V_{GG} supply provides a voltage V_{GS} to bias the N-channel JFET,

but no resulting current is drawn from the battery V_{GG} . Resistor R_G is included to allow any ac signal applied through capacitor C to develop across R_G . While any ac signal will develop across R_G , the dc voltage drop across R_G is equal to $I_G R_G$ which is equal to zero Volt.

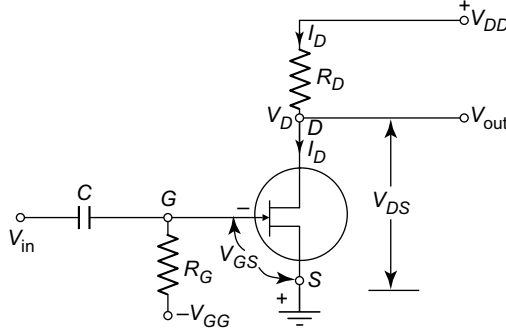


Fig. 5.25 Fixed bias circuit for an N-Channel JFET

Then, the gate to source voltage V_{GS} is

$$V_{GS} = V_G - V_s = -V_{GG} - 0 = -V_{GG}$$

The drain-source current I_D is then fixed by the gate-source voltage. This current will cause a voltage drop across the drain resistor R_D and is given as

$$V_{DD} = I_D \cdot R_D + V_{DS}$$

$$I_D = \frac{V_{DD} - V_{DS}}{R_D}$$

5.23 BIASING THE MOSFET

5.23.1 Biasing of Enhancement MOSFET

Figure 5.26 shows the drain-to-gate bias circuit for enhancement mode MOSFET. Here, the gate bias voltage is

$$V_{GS} = \left[\frac{R_1}{R_1 + R_f} \right] V_{DS}$$

the dc stabilisation through the feedback resistor R_f . However, the input resistance is reduced because of Miller effect.

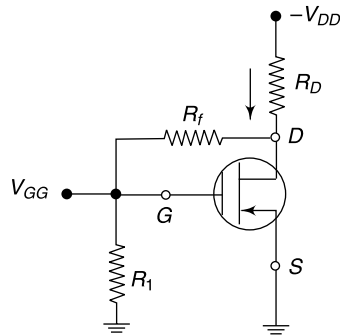


Fig. 5.26 Drain-to-Gate Bias Circuit for Enhancement MOSFET

Also, the voltage divider biasing technique given for JFET can be used for the enhancement MOSFET. Here, the dc stability is accomplished by the dc feedback through R_S .

But the self-bias technique given for JFET cannot be used for establishing an operating point for the enhancement MOSFET because the voltage drop across R_S is in a direction to reverse-bias the gate and it actually needs forward gate bias.

Figure 5.27 shows an N-channel enhancement mode MOSFET common source circuit with source resistor. The gate voltage is

$$V_G = V_{GS} = \left(\frac{R_2}{R_1 + R_2} \right) (V_{DD})$$

and the gate-to-source voltage is

$$V_{GS} = V_{DD} - V_G$$

Assuming that $V_{GS} > V_{TN}$ and the MOSFET is biased in the saturation region, the drain current is

$$I_D = K_N (V_{GS} - V_{TN})^2$$

Here the threshold voltage V_{TN} and conduction parameter K_N are functions of temperature

The drain-to-source voltage is

$$V_{DS} = V_{DD} - I_D R_D$$

If $V_{DS} > V_{DS}(sat) = V_{GS} - V_{TN}$, then the MOSFET is biased in the saturation region. If $V_{DS} < V_{DS}(sat) = V_{GS} - V_{TN}$, then the MOSFET is biased in the non-saturation region, and the drain current is given by

$$I_D = K_N [2(V_{GS} - V_{TN}) V_{DS} - V_{DS}^2]$$

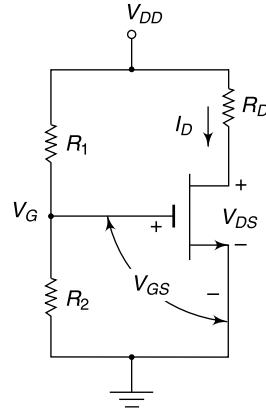


Fig. 5.27 N-Channel Enhancement Mode MOSFET Common Source Circuit with Source resistor

5.23.2 Biasing of Depletion MOSFET

Both the self-bias technique and voltage divider bias circuit given for JFET can be used to establish an operating point for the depletion mode MOSFET.

Example 5.16 Calculate the operating point of the self biased JFET having the supply voltage $V_{DD} = 20V$, maximum value of drain current $I_{DSS} = 10 \text{ mA}$ and $V_{GS} = -3 \text{ V}$ at $I_D = 4 \text{ mA}$. Also, determine the values of resistors R_D and R_S to obtain this bias condition.

Solution We know that the value of drain current at Q-point,

$$I_{DQ} = \frac{I_{DSS}}{2} = \frac{10 \times 10^{-3}}{2} = 5 \text{ mA}$$

and the value of drain-to-source voltage at Q-point,

$$V_{DSQ} = \frac{V_{DD}}{2} = \frac{20}{2} = 10 \text{ V}$$

Therefore, the operating point is at $V_{DS} = 10 \text{ V}$ and $I_D = 5 \text{ mA}$.

Also, we know that the drain-to-source voltage,

$$\begin{aligned} V_{DS} &= V_{DD} - I_D R_D \\ 10 &= 20 - (4 \times 10^{-3}) R_D \end{aligned}$$

Therefore,
$$R_D = \frac{20 - 10}{4 \times 10^{-3}} = 2.5 \text{ k}\Omega$$

The source voltage or voltage across the source resistor R_S is

$$V_S = -V_{GS} = -3 \text{ V}$$

Also,
$$V_S = I_D R_S \text{ i.e. } 3 = (4 \times 10^{-3}) R_S$$

Therefore,
$$R_S = \frac{3}{4 \times 10^{-3}} = 750 \text{ }\Omega$$

 **Example 5.17** Calculate the values of R_S required to self bias and N-channel JFET with $I_{DSS} = 40 \text{ mA}$, $V_P = -10 \text{ V}$ and $V_{GSQ} = -5 \text{ V}$.

Solution We know that
$$I_D = I_{DSS} \left[1 - \frac{V_{GS}}{V_P} \right]^2$$

Substituting the given values, we get

$$I_D = 40 \times 10^{-3} \left[1 - \frac{(-5)}{(-10)} \right]^2 = 10 \text{ mA}$$

Therefore,
$$R_S = \left| \frac{V_{GSQ}}{I_D} \right| = \frac{5}{10 \times 10^{-3}} = 500 \text{ }\Omega$$

 **Example 5.18** A JFET amplifier with a voltage divider biasing circuit, shown in Fig. 5.24, has the following parameters: $V_P = -2 \text{ V}$, $I_{DSS} = 4 \text{ mA}$, $R_D = 910 \text{ }\Omega$, $R_S = 3 \text{ k}\Omega$, $R_1 = 12 \text{ M}\Omega$, $R_2 = 8.57 \text{ M}\Omega$ and $V_{DD} = 24 \text{ V}$. Find the value of the drain current I_D at the operating point. Verify whether the FET will operate in the pinch-off region.

Solution From Fig. 5.24, we obtain

$$V_{GG} = V_{DD} \frac{R_2}{R_1 + R_2} = 24 \frac{8.57 \times 10^6}{(12 + 8.57) \times 10^6} = 10 \text{ V}$$

We know that

$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P} \right)^2$$

$$= I_{DSS} \left(1 - \frac{V_{GG} - I_D R_S}{V_P} \right), \text{ where } V_{GS} = V_{GG} - I_D R_S$$

Expressing I_D and I_{DSS} in mA, we have

$$I_D = 4 \times \left(1 - \frac{10 - I_D \times 3}{-2} \right)^2$$

$$\text{i.e.,} \quad 9I_D^2 - 73I_D + 144 = 0$$

Therefore, $I_D = 3.39 \text{ mA}$ or 4.72 mA

As $I_D = 4.72 \text{ mA} > 4 \text{ mA} = I_{DSS}$, this value is inappropriate. So, $I_{DQ} = 3.39 \text{ mA}$ is selected.

Therefore,

$$V_{GSQ} = V_{GG} - I_{DQ} R_S = 10 - (3.39 \times 10^{-3} \times 3 \times 10^3) = -0.17 \text{ V}$$

$$\text{and } V_{DSQ} = V_{DD} - I_{DQ} (R_D + R_S) = 24 - 3.39 \times 10^{-3} (0.91 + 3) \times 10^3$$

$$= 10.745 \text{ V}$$

$$\text{Then } V_{DGQ} = V_{DSQ} - I_{GQS} = 10.745 + 0.17 = 10.915 \text{ V}$$

which is greater than $|V_P| = 2 \text{ V}$. Hence, the FET is in the pinch-off region.

Example 5.19 A voltage divider bias is provided to an N-channel JFET circuit as shown in Fig. 5.28. To establish $I_{DSS} = 10 \text{ mA}$, $V_P = -3.5 \text{ V}$, $R_1 + R_2 = 120 \text{ k}\Omega$, $I_D = 5 \text{ mA}$ and $V_{DS} = 5 \text{ V}$, determine the values of R_1 , R_2 and R_D .

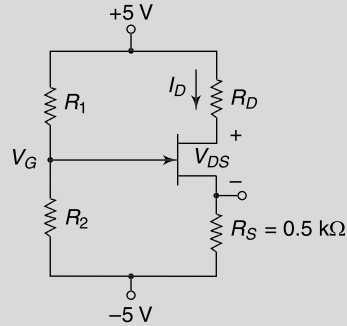


Fig. 5.28

Solution Let us assume that the JFET is biased in the saturation region. Then the dc drain current is given by

$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P} \right)^2$$

Therefore,

$$5 = 10 \left(1 - \frac{V_{GS}}{(-3.5)} \right)^2$$

By solving, we get $V_{GS} = -1.008$ V

The voltage at the source terminal is

$$V_S = I_D R_S - 5 = (5)(0.5) - 5 = -2.5$$

The gate voltage is

$$V_G = V_{GS} + V_S = -1.008 - 2.5 = -3.508$$

The gate voltage can be written as

$$V_G = \left(\frac{R_2}{R_1 + R_2} \right) (10) - 5$$

Therefore,

$$-3.508 = \frac{R_2}{120} (10) - 5$$

i.e.,

$$R_2 = 2.984 \text{ k}\Omega$$

and

$$R_1 = 17.016 \text{ k}\Omega$$

The drain-to-source voltage is

$$V_{DS} = 5 - I_D R_D - I_D R_S - (-5)$$

Substituting the specified values, we get

$$R_D = \frac{10 - V_{DS} - I_D R_S}{I_D} = \frac{10 - 5 - (5)(0.5)}{5} = 0.5 \text{ k}\Omega$$

$$V_{GS} - V_P = -1.24 - (-3.5) = 2.26 \text{ V}$$

Here, since $V_{DS} > (V_{GS} - V_P)$, the JFET is biased in the saturation region, which satisfies the initial assumption.

Example 5.20 For the circuit shown Fig. 5.29, find the values of V_{DS} and V_{GS} . Given, $I_D = 5 \text{ mA}$, $V_{DD} = 10 \text{ V}$, $R_D = 1 \text{ k}\Omega$ and $R_S = 500 \Omega$. [JNTU April/May 2007]

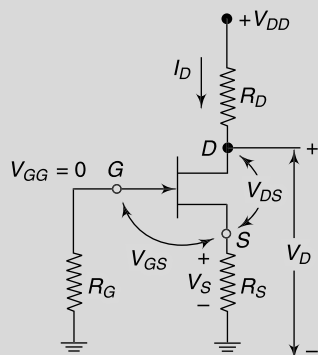


Fig. 5.29

Solution

$$V_{GG} = V_{GS} + I_D R_S$$

Since $V_{GG} = 0$, $V_{GS} = -I_D R_S = -5 \times 10^{-3} \times 500 = -2.5 \text{ V}$

We know that $V_{DD} = I_D(R_D + R_S) + V_{DS}$

Therefore, $V_{DS} = V_{DD} - I_D(R_D + R_S) = 10 - 5 \times 10^{-3}(1500) = 2.5 \text{ V}$

Example 5.21 For the common-source N-channel MOSFET circuit shown in Fig. 5.30 with the threshold voltage $V_{TN} = 1.5 \text{ V}$, conduction parameter $K_N = 1 \text{ mA/V}^2$, the channel-length modulation parameter $\lambda = 0.01 \text{ V}^{-1}$, $R_i = R_1 \parallel R_2 = 100 \text{ k}\Omega$ and the current at the transition point $I_{Dt} = 4 \text{ mA}$. Design the MOSFET circuit with voltage divider bias such that $I_{DQ} = 1.5 \text{ mA}$ and Q-point is in the middle of the saturation region.

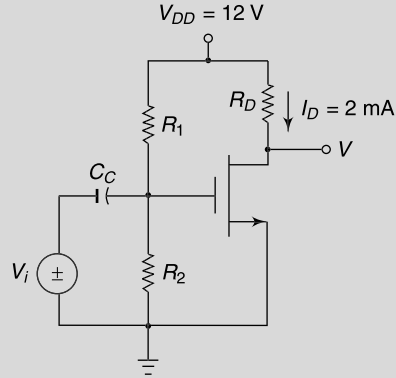


Fig. 5.30 Common-Source N-Channel MOSFET Circuit

Solution To determine V_{DSI} :

We know that

$$\begin{aligned} I_{Dt} = 4 &= K_N (V_{GSt} - V_{TN})^2 \\ &= 1 (V_{GSt} - 1.5)^2 \end{aligned}$$

where the subscript t indicates transition point values.

Solving, we get

$$V_{GSt} = 3.5 \text{ V}$$

Therefore,

$$V_{DSI} = V_{GSt} - V_{TN} = 3.5 - 1.5 = 2 \text{ V}$$

If the Q-point is in the middle of the saturation region, then $V_{DSQ} = 7 \text{ V}$, which gives 10 V peak-to-peak symmetrical output voltage.

From Fig. 5.29,

$$V_{DSQ} = V_{DD} - I_{DQ} R_D$$

Therefore,
$$R_D = \frac{V_{DD} - V_{DSQ}}{I_{DQ}} = \frac{12 - 7}{1.5} = 3.33 \text{ k}\Omega$$

Then

$$\begin{aligned} I_{DQ} = 1.5 &= K_N (V_{GSQ} - V_{TN})^2 \\ &= (1) (V_{GSQ} - 1.5)^2 \end{aligned}$$

Therefore, $V_{GSQ} = 2.73 \text{ V}$

Then,
$$V_{GSQ} = 2.73 = \left(\frac{R_2}{R_1 + R_2} \right) (V_{DD}) = \left(\frac{1}{R_1} \right) \left(\frac{R_1 R_2}{R_1 + R_2} \right) (V_{DD})$$

$$= \frac{R_i}{R_1} (V_{DD}) = \frac{(100)(12)}{R_1}$$

By solving, we get

$$R_1 = 439.6 \text{ k}\Omega \text{ and } R_2 = 129.45 \text{ k}\Omega$$

Example 5.22 For the N-channel depletion mode MOSFET circuit shown in Fig. 5.31, $V_{TS} = -2 \text{ V}$ and $K_N = 0.1 \text{ mA/V}^2$. Assume that $V_{DD} = 5 \text{ V}$ and $R_S = 5 \text{ k}\Omega$. Determine I_D and V_{DS} .

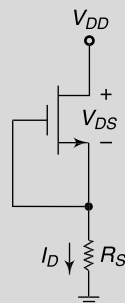


Fig. 5.31 N-Channel Depletion Mode MOSFET Circuit

Solution Let us assume that the MOSFET is biased in the saturation region. Then the dc drain current is

$$\begin{aligned} I_D &= K_N (V_{GS} - V_{TN})^2 = K_N (-V_{TN})^2 \\ &= (0.1) (-(-2))^2 = 0.4 \text{ mA} \end{aligned}$$

The dc drain-to-source voltage is

$$V_{DS} = V_{DD} - I_D R_S = 5 - (0.4)(5) = 3 \text{ V}$$

Then,

$$V_{DS(sat)} = V_{GS} - V_{TN} = 0 - (-2) = 2 \text{ V}$$

Since $V_{DS} > V_{DS(sat)}$, the MOSFET is biased in the saturation region.

Example 5.23 Determine the following for the network shown in Fig. 5.32

- (i) V_{GSQ}
- (iii) V_D
- (v) V_S

- (ii) V_{DS}
- (iv) V_G

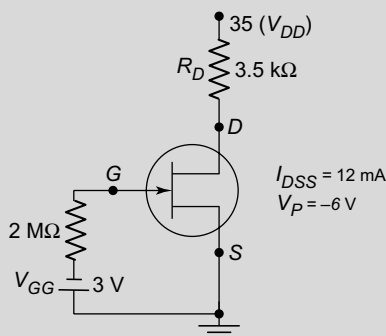


Fig. 5.32

Solution

$$(i) V_{GSQ} = -V_{GG} = -3 \text{ V}$$

$$(ii) I_{DQ} = I_{DSS} \left(1 - \frac{V_{GS}}{V_P} \right)^2 = 12 \times 10^{-3} \left(1 - \frac{-3}{-6} \right)^2 = 3 \text{ mA}$$

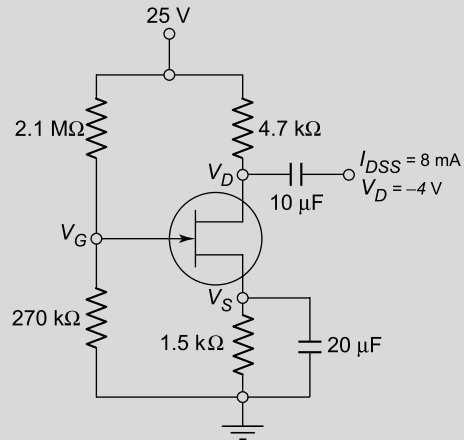
$$V_{DS} = V_{DSQ} = V_{DD} - I_{DQ} R_D \\ = 35 - 3 \times 10^{-3} \times 3.5 \times 10^3 = 24.5 \text{ V}$$

$$(iii) V_D = V_{DS} + V_S = 24.5 + 0 = 24.5 \text{ V}$$

$$(iv) V_G = -3 \text{ V}$$

$$(v) V_S = 0 \text{ V}$$

Example 5.24 Determine I_{DQ} , V_{GSQ} , V_D , V_S , V_{DS} and V_{DG} for the given network shown in Fig. 5.33.

**Fig. 5.33****Solution**

To find expression for V_{GS} :

$$V_{GS} = \frac{R_2}{R_1 + R_2} \times V_{DD} = \frac{270}{(2100 + 270)} \times 20 = 2.28 \text{ V}$$

$$V_S = 1.5 I_D$$

Therefore, $V_{GS} = V_G - V_S = (2.28 - 1.5 I_D)$

To find I_D :

$$I_D = I_{DSS} \left[1 - \frac{V_{GS}}{V_P} \right]^2 \text{ mA}$$

$$I_D = 8 \left[1 - \frac{(2.28 - 1.5 I_D)}{4} \right]^2 \text{ mA}$$

$$\text{Therefore, } I_D = \frac{8}{16} [4 + 2.28 - 1.5I_D]^2 = 0.5(6.28 - 1.5I_D)^2$$

$$2I_D = 39.44 - 18.84I_D + 2.25I_D^2$$

$$2.25I_D^2 - 20.84I_D + 39.44 = 0$$

$$\text{Therefore, } I_D = \frac{20.84 \pm \sqrt{(20.84)^2 - (4 \times 2.25 \times 39.44)}}{2 \times 2.25} = 6.6 \text{ mA or } 2.65 \text{ mA}$$

For $I_D = 6.6 \text{ mA}$, V_{DS} is negative and hence this value may be neglected. Let us choose $I_D = 2.65 \text{ mA}$.

Therefore, $I_{DQ} = 2.65 \text{ mA}$

To find V_{GSQ} :

$$V_{GSQ} = 2.28 - 1.5I_{DQ} = 2.28 - (1.5 \times 2.65) = -1.695 \text{ V}$$

To find V_{DSQ} :

$$V_{DSQ} = V_{DD} - I_{DQ}(R_D + R_S)$$

$$\text{Therefore, } V_{DSQ} = 20 - 2.65 \times 10^{-3} (4.7 + 1.5) \times 10^{-3} = 3.57 \text{ V}$$

To find V_D , V_S and V_{DG} :

$$V_S = I_D R_S = 2.65 \times 10^{-3} \times 1.5 \times 10^{-3} = 3.975 \text{ V}$$

$$V_D = V_S + V_{DS} = 3.975 + 3.57 = 7.545 \text{ V}$$

$$\text{Hence, } V_{DG} = V_D - V_G = 7.545 - 2.28 = 5.265 \text{ V}$$

Example 5.25 For the given measurement $V_S = 1.7 \text{ V}$ for the network as shown in Fig. 5.34, determine

- | | |
|------------------|----------------|
| (i) I_{DQ} | (ii) V_{GSQ} |
| (iii) I_{DSSS} | (iv) V_D |
| (v) V_{DS} | |

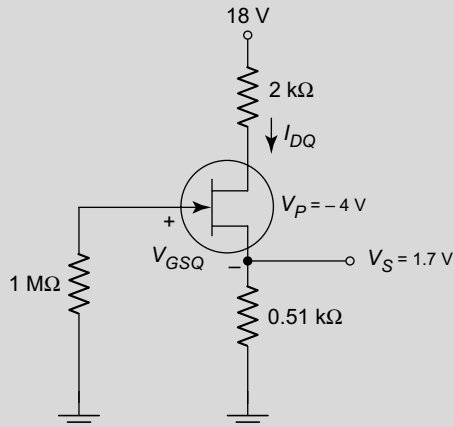


Fig. 5.34

Solution Given $V_S = 1.7 \text{ V}$

$$(i) V_S = I_D R_S$$

$$I_{DQ} = \frac{V_S}{R_S} = \frac{1.7}{510} = 3.33 \text{ mA}$$

$$(ii) V_{GSQ} = V_G - V_S = -V_S = -1.7 \text{ V}$$

$$(iii) I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P} \right)^2$$

$$I_{DSS} = \frac{I_D}{\left(1 - \frac{V_{GS}}{V_P} \right)^2} = \frac{3.33 \times 10^{-3}}{\left(1 - \frac{(-1.7)}{(-4)} \right)^2} = 10 \text{ mA}$$

$$(iv) V_D = V_{DD} - I_D R_D = 18 - 3.33 \times 10^{-3} \times 2 \times 10^3 = 11.34 \text{ V}$$

$$(v) V_{DS} = V_D - V_S = 11.34 - 1.7 = 9.64 \text{ V}$$

Example 5.26 For a circuit shown in Fig. 5.35, calculate V_o , Z_i and Z_o . Given input is $V_i = 0.2 \text{ V (rms)}$, $I_{DSS} = 9 \text{ mA}$ and $V_P = -4.5 \text{ V}$.

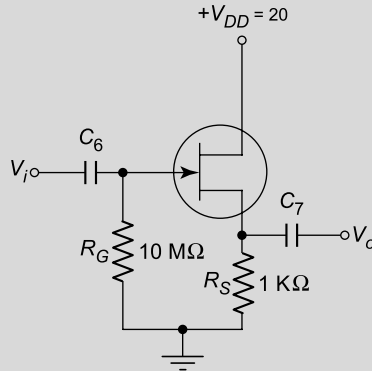


Fig. 5.35

Solution

$$Z_i = R_G = 10 \text{ M}\Omega$$

$$\begin{aligned} I_D &= I_{DSS} \left(1 - \frac{V_{GS}}{V_P} \right)^2 \\ &= 9 \times 10^{-3} \left(1 - \frac{-I_D R_S}{V_P} \right)^2 \\ &= 9 \times 10^{-3} \left(1 - \frac{(-1000 I_D)}{-4.5} \right)^2 \\ &= 9 \times 10^{-3} (1 - 222.22 I_D)^2 \end{aligned}$$

$$= 9 \times 10^{-3} (1 - 444.44 I_D + 49383 I_D^2)$$

$$I_D = 9 \times 10^{-3} - 4 I_D + 444.45 I_D^2$$

Therefore, $444.45 I_D^2 - 5 I_D + 9 \times 10^{-3} = 0$

Solving the quadratic equation, we get

$$I_D = 2.25 \text{ mA or } 9 \text{ mA}$$

Since $I_D < I_{DSS}$, we take $I_D = 2.25 \text{ mA}$. Therefore,

$$g_{mo} = \frac{2I_{DSS}}{|V_P|} = \frac{2 \times (9 \times 10^{-3})}{4.5} = 4 \text{ mS}$$

$$g_m = g_{mo} \left(1 - \frac{V_{GSQ}}{V_P} \right)$$

where $V_{GSQ} = -I_D R_S = -2.25 \times 10^{-3} \times 1000 = -2.25 \text{ V}$

$$g_m = 4 \times 10^{-3} \left(1 - \frac{(-2.25)}{(-4.5)} \right) = 2 \text{ mS}$$

$$Z_o = \frac{1}{g_m} \parallel R_S = \frac{1}{2 \times 10^{-3}} \parallel 1 \text{ k}\Omega = 333.33 \Omega$$

$$A_v = \frac{g_m (r_d \parallel R_S)}{1 + g_m (r_d \parallel R_S)} = \frac{g_m R_S}{1 + g_m R_S} = \frac{2 \times 10^{-3} \times 1 \times 10^3}{1 + (2 \times 10^{-3} \times 1 \times 10^3)} = 0.667$$

$$V_o = V_i \times A_v = 0.2 \times 0.667 = 0.133 \text{ V}$$

Example 5.27 An N-channel JFET having $V_P = -4\text{V}$ and $I_{DSS} = 10 \text{ mA}$ is used in the circuit of Fig. 5.36. The parameter values are $V_{DD} = 18\text{V}$, $R_S = 2\text{k}\Omega$, $R_1 = 450\text{k}\Omega$, and $R_2 = 90\text{k}\Omega$. Determine I_D and V_{DS} .

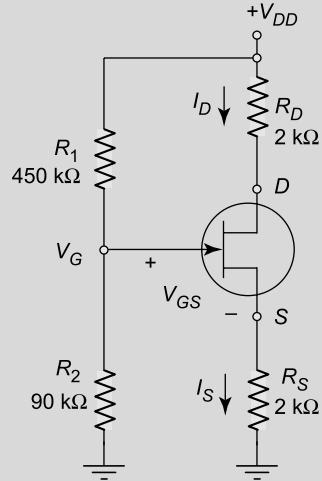


Fig. 5.36

Solution

To find V_{GS} :

$$V_{GS} = V_G - I_D R_S$$

$$V_G = \frac{R_2}{R_1 + R_2} \times V_{DD} = \frac{90 \times 10^3}{(450 + 90) \times 10^3} \times 18 = 3 \text{ V}$$

Therefore, $V_{GS} = (3 - 2 \times 10^3 I_D)$

To find I_D :

$$\begin{aligned} I_D &= I_{DSS} \left[1 - \frac{V_{GS}}{V_P} \right]^2 = 10 \times 10^{-3} \left[1 - \frac{(3 - 2 \times 10^3 I_D)}{-4} \right]^2 \\ &= \frac{10 \times 10^{-3}}{16} [4 + 3 - 2 \times 10^3 I_D]^2 \end{aligned}$$

$$\begin{aligned} \text{Therefore, } 1.6 I_D &= 10^{-3} [7 - 2 \times 10^3 I_D]^2 \\ &= 0.049 - 28 I_D + 4 \times 10^3 I_D^2 \\ 4 \times 10^3 I_D^2 - 29.6 I_D + 0.049 &= 0 \\ I_D &= \frac{29.6 \pm \sqrt{(29.6)^2 - 4 \times 4 \times 10^3 \times 0.049}}{2 \times 4 \times 10^3} \end{aligned}$$

Therefore, $I_D = 4.9 \text{ mA}$ or 2.5 mA

If $I_D = 4.9 \text{ mA}$, then V_{DS} would be negative and hence this is not acceptable.

Therefore, $I_{DQ} = 2.5 \text{ mA}$

To find V_{DS} :

$$V_{DS} = V_{DD} - I_{DQ} (R_D + R_S) = 18 - 2.5 \times 10^{-3} (2 + 2) \times 10^3 = 8 \text{ V}$$

Example 5.28 Determine V_{GS} , I_D , V_{DS} , V_D , and V_G for the circuit shown in Fig. 5.37.

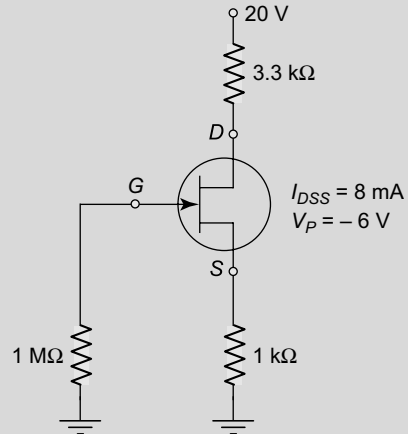


Fig. 5.37

Solution

To find V_{GSQ} :

For a self bias circuit, $V_{GSQ} = -I_D R_S = -I_D \times 10^3$

To find I_D :

$$\begin{aligned} I_D &= I_{DSS} \left[1 - \frac{V_{GS}}{V_P} \right]^2 = 8 \times 10^{-3} \left[1 + \frac{(1 \times 10^3 I_D)}{-6} \right]^2 \\ &= 8 \times 10^{-3} \left[1 - \frac{1000 I_D}{6} \right]^2 \end{aligned}$$

$$\begin{aligned} \text{Therefore, } 36 I_D &= 8 \times 10^{-3} [6 - 1000 I_D]^2 \\ &= 8 \times 10^{-3} [36 - 12 \times 10^3 I_D + 10^6 I_D^2] \\ &= 8 \times 10^3 I_D^2 - 132 I_D + 0.288 = 0 \end{aligned}$$

$$I_D = \frac{132 \pm \sqrt{(132)^2 - 4 \times 8 \times 10^3 \times 0.288}}{2 \times 8 \times 10^3}$$

Therefore, $I_D = 13.9 \text{ mA}$ or 2.5 mA

But I_D cannot be higher than I_{DSS} , Therefore, $I_D = 4.225 \text{ mA}$

To find V_{DS} :

$$V_{DS} = V_{DD} - I_D (R_D + R_S) = 20 - 2.5 \times 10^{-3} (3.3 + 1) \times 10^3 = 9.25 \text{ V}$$

To find V_{GS} , V_D and V_S :

$$V_{GS} = 1 \times 10^3 \times I_D = 1 \times 10^3 \times 2.5 \times 10^{-3} = 2.5 \text{ V}$$

$$V_S = I_D R_S = 2.5 \times 10^{-3} \times 1 \times 10^3 = 2.5 \text{ V}$$

$$V_D = V_S + V_{DS} = 2.5 + 9.25 = 11.75 \text{ V}$$

Example 5.29 Determine the following for the network shown in Fig. 5.38

$-V_{GSQ}$, V_{DQ} , V_D , V_G , V_S and V_{DS}

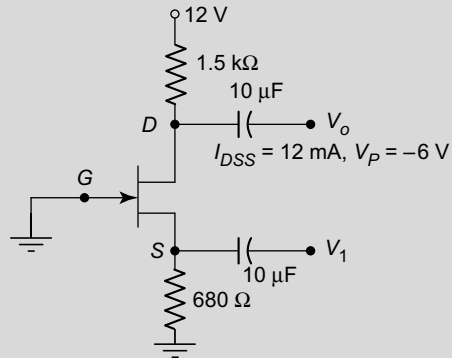


Fig. 5.38

Solution

$$V_G = 0, V_S = I_D R_S$$

$$V_{GS} = V_G - V_S = 0 - I_D R_S = -680 I_D$$

To find I_D :

$$\begin{aligned} \text{We know that } I_D &= I_{DSS} \left(1 - \frac{V_{GS}}{V_P} \right)^2 = 12 \times 10^{-3} \left(1 - \left(\frac{-680 I_D}{-6} \right) \right)^2 \\ &= 152.6 I_D^2 - 2.7 I_D + 0.01 \end{aligned}$$

$$\text{Therefore, } I_D = \frac{2.7 \pm \sqrt{(2.7)^2 - 4 \times 152.6 \times 0.01}}{2 \times 152.6} = 12.4 \text{ mA or } 5.28 \text{ mA}$$

Since I_D is less than I_{DSS} ,

$$I_{DQ} = 5.28 \text{ mA}$$

To find V_{GSQ} :

$$V_{GSQ} = -680 I_D = -680 \times 5.28 \times 10^{-3} = -3.6 \text{ V}$$

To find V_S :

$$\begin{aligned} V_S &= I_D R_S \\ &= 5.28 \times 10^{-3} \times 680 = 3.6 \text{ V} \end{aligned}$$

To find V_{DS} :

$$\begin{aligned} V_{DS} &= V_{DD} - I_D (R_D + R_S) \\ &= 12 - 5.28 \times 10^{-3} (1.5 \times 10^3 + 680) = 12 - 11.51 = 0.49 \text{ V} \end{aligned}$$

To find V_D :

$$V_D = V_S + V_{DS} = 3.6 + 0.49 = 4.09 \text{ V}$$

5.24 THE FET MODEL AT HIGH FREQUENCY

In the high frequency model of FET, the capacitances between nodes have to be added in the low frequency model. The resulting equivalent circuit is shown in Figure.

C_{gs} represents the barrier capacitance between gate and source. C_{gd} is the barrier capacitance between gate and drain. C_{ds} is the drain to source capacitance of the channel. These internal capacitances leads to feedback from output to input and the voltage amplification decreases at higher frequencies.

The parameter of FET shown in Fig. 5.39 will have their magnitudes as given in Table 5.1.

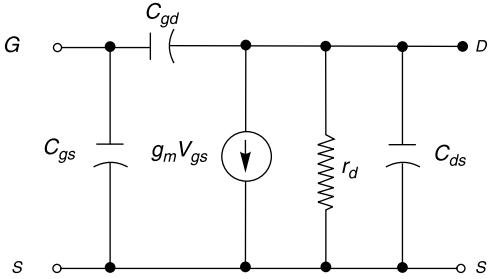


Fig. 5.39 The High Frequency Model of FET

Table 5.1 Parameter Values of FET (JFET and MOSFET)

DEVICE	PARAMETER	G_M	R_D	C_{DS}	C_{GS}, C_{GD}	R_{GS}	R_{GD}
JFET	Range	0.10–10 mA/V M Ω	0.1–1 pF	0.1–1 pF	1–10	$>10^8 \Omega$	$>10^8 \Omega$
MOSFET		0.1–20 mA/V	1–50 k Ω	0.1–1 pF	1–10 pF	$>10^{10} \Omega$	$>10^{14} \Omega$

5.24.1 The Common Source (CS) Amplifier at High Frequencies

The circuit of Fig. 5.40 shows the CS amplifier.

The equivalent circuit at high frequencies is shown in Fig. 5.41.

The Norton’s equivalent circuit between D & S can be obtained by finding the short circuit current from D to S and impedance Z seen from output point with independent voltage sources short circuited and independent current sources open circuited. With $V_i = 0$, current $g_m V_i = 0$ the circuit of Fig. 5.41 reduces to circuit of Fig. 5.42.

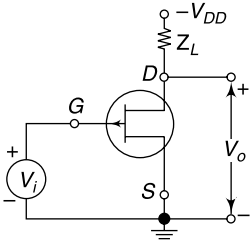


Fig. 5.40 CS Amplifier Circuit

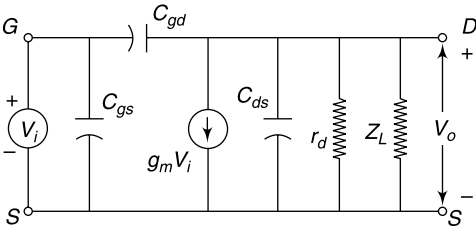


Fig. 5.41 Small Signal Equivalent Circuit of CS Amplifier at High Frequencies

Hence admittance at the output point

$$Y = \frac{1}{Z} = Y_L + g_d + Y_{ds} + Y_{gd} \quad (5.20)$$

where $Y_L = \frac{1}{Z_L}$ is admittance corresponding to Z_L

$g_d = \frac{1}{r_d}$ is conductance corresponding to r_d

$Y_{ds} = j\omega C_{ds}$ is admittance corresponding to C_{ds}

$Y_{gd} = j\omega C_{gd}$ is admittance corresponding to C_{gd} .

The equivalent circuit to find the short circuit current from D to S is shown in Fig. 5.43.

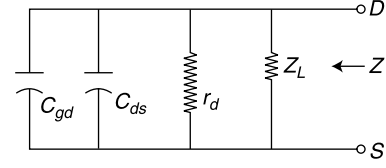


Fig. 5.42 Equivalent Circuit to Find Z

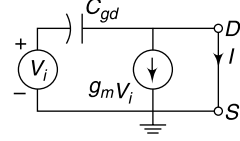


Fig. 5.43 Equivalent Circuit to Find I

$$\text{Hence current } I = -g_m V_i + V_i Y_{gd} \quad (5.21)$$

Voltage Gain A_V Voltage gain (amplification) A_V with load Z_L included is given by

$$A_V = \frac{V_o}{V_i} = \frac{IZ}{V_i} = \frac{I}{V_i Y}$$

From Equations (5.20) and (5.21),

$$A_V = \frac{-g_m + Y_{gd}}{Y_L + g_d + Y_{ds} + Y_{gd}} \quad (5.22)$$

At low frequencies FET capacitances can be neglected and hence

$$Y_{ds} = Y_{gd} = 0$$

Eq. (5.22) at low frequencies reduces to

$$\begin{aligned} A_V &= \frac{-g_m}{Y_L + g_d} = \frac{-g_m}{\frac{1}{Z_L} + \frac{1}{r_d}} \\ A_V &= \frac{-g_m r_d Z_L}{r_d + Z_L} = -g_m Z'_L \end{aligned} \quad (5.23)$$

where $Z'_L = Z_L \parallel r_d$

Input Admittance From Fig. 5.41, it is found that the Gate circuit is not isolated from drain circuit, but connected by C_{gd} .

According to Miller's theorem, an impedance Z' connected between two points (1) & (2) of a circuit can be replaced by $Z_1 = \frac{Z'}{1 - A_V}$ from (1) to Ground and $Z_2 = \frac{Z' A_V}{A_V - 1}$ from (2) to Ground, where A_V is the voltage gain V_2/V_1 .

Applying Miller's theorem to circuit of Fig. 5.41, the circuit of Fig. 5.44 is obtained, where capacitances are replaced by equivalent admittances.

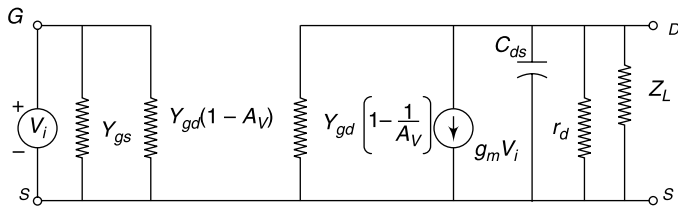


Fig. 5.44 CS Amplifier Equivalent Circuit after Applying Miller's Theorem

Hence the input admittance is given by

$$Y_i = Y_{gs} + (1 - A_v)Y_{gd} \quad (5.24)$$

As $Y_{gs} = j\omega C_{ds}$ and $Y_{gd} = j\omega C_{gd}$ for an FET to possess negligible input admittance over a wide range of frequencies, the gate-source and gate-drain capacitances must be negligible.

Input Capacitance (Miller Effect) From Eq. (5.23), the voltage gain $A_v = -g_m Z'_L$ where $Z'_L = Z_L \parallel r_d$. For an FET with drain-circuit resistance R_d , the voltage gain $A_v = -g_m R'_d$ where $R'_d = R_d \parallel r_d$.

From Eq. (5.24)

$$\begin{aligned} Y_i &= Y_{gs} + (1 + g_m R'_d) Y_{gd} \\ Y_i &= j\omega C_{gs} + (1 + g_m R_d) j\omega C_{gd} \\ \frac{Y_i}{j\omega} &= C_i = C_{gs} + (1 + g_m R'_d) C_{gd} \end{aligned} \quad (5.25)$$

The increase in input capacitance C_i over the capacitance from gate to source is the Miller effect.

In multistage (cascaded amplifiers), this input capacitance appears in shunt with output impedance of previous stage. As capacitive reactance decrease with increase in frequency, the resultant output impedance will be lower at higher frequencies, thereby reducing the gain.

Output Admittance The output impedance for the CS amplifier of Fig. 5.41 is obtained by setting input voltage $V_i = 0$ and looking from the output point. The resulting equivalent circuit is shown in Fig. 5.45.

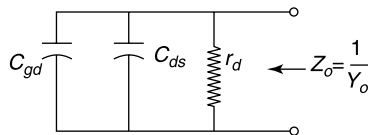


Fig. 5.45 Calculation of Output Impedance

The output admittance with Z_L considered external to CS amplifier circuit is given by

$$Y_o = g_d + Y_{ds} + Y_{gd} \quad (5.26)$$

5.24.2 The Common-Drain Amplifier at High Frequencies

The Common-Drain amplifier (Source-follower) circuit is shown in Fig. 5.46 and its high frequency equivalent circuit is shown in Fig. 5.46.

The small signal high frequency equivalent circuit of CD Amplifier is shown in Fig. 5.47.

Voltage Gain The output voltage V_o is the product of the short-circuit current and the impedance between terminals S and N. It is found to be, voltage gain

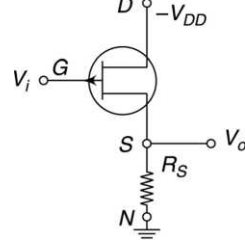


Fig. 5.46 Common-Drain Amplifier

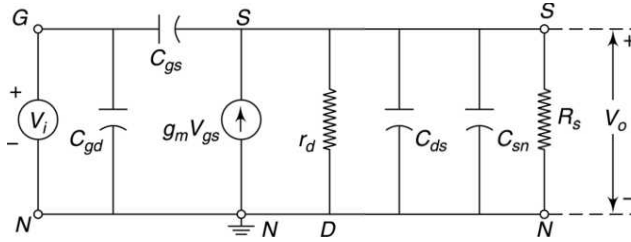


Fig. 5.47 Small Signal High Frequency Equivalent Circuit of Common-drain Amplifier

$$A_v = \frac{(g_m + j\omega C_{gs}) R_S}{1 + (g_m + g_d + j\omega C_T) R_S} \quad (5.27)$$

where $C_T \equiv C_{gs} + C_{ds} + C_{sn}$.

C_{gs} is the capacitance from gate to source, C_{ds} is the capacitance from drain to source and C_{sn} is the capacitance from source to ground.

At low frequencies, the voltage gain reduces to

$$A_v \approx \frac{g_m R_S}{1 + (g_m + g_d) R_S}$$

The amplification is positive and has a value less than unity. If $g_m R_S \gg 1$,

$$\text{then } A_v \approx \frac{g_m R_S}{g_m R_S + g_d R_S} = \frac{g_m}{g_m + g_d} = \frac{\mu}{\mu + 1}$$

Input Admittance The input admittance Y_i is obtained by applying Miller's theorem to C_{gs} .

$$Y_i = j\omega C_{gd} + j\omega C_{gs} (1 - A_v)$$

$$Y_i \approx j\omega C_{gd} \text{ as } A_v \approx 1.$$

The CD amplifier offers the important advantage of lower input capacitance than the CS amplifier.

Output Admittance The output admittance with input voltage set to zero is given by

$$Y_o = g_m + g_d + j\omega C_T.$$

where R_s is considered external to the amplifier. At low frequencies the output admittance

$$Y_o = g_m + g_d$$

and output resistance $R_o = \frac{1}{g_m + g_d} = \frac{1}{g_m}$, since $g_m \gg g_d$.

The CD amplifier (Source follower) is used for the same application as emitter follower, in applications requiring high input impedance and low output impedance.

Note: Small signal model and analysis of FET amplifiers are equally applicable for both JFET and MOSFET, with suitable changes in biasing.

5.25 FREQUENCY RESPONSE OF FET AMPLIFIER

In the case of FET amplifier, the high frequency characteristic of the amplifier is determined by the interelectrode and wiring capacitances. The capacitor C_{gs} and C_{gd} typically vary from 1–10pF, while the capacitance C_{ds} is usually quite a bit smaller, ranging from 0.1–1pF. At high frequencies C_i (miller capacitance) will approach a short-circuit equivalent and V_{gs} will drop in value and reduce the overall gain.

The cut-off frequencies defined by the input and output circuits can be obtained by first finding the Thevenin equivalent circuits for each section as shown in Fig. 5.48.

For the input circuit shown in Fig. 5.48(b),

$$f_{Hi} = 1/(2\pi R_{Thi} C_i) \text{ and}$$

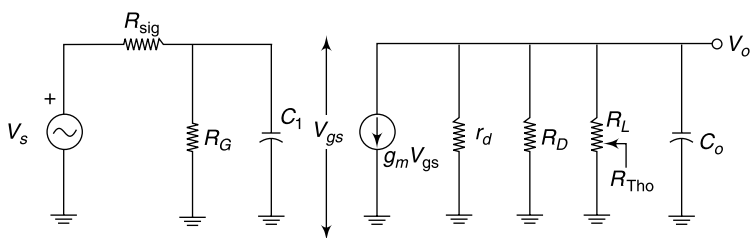


Fig. 5.48 (a) Modified High Frequency ac Equivalent Circuit (CS Amplifier)

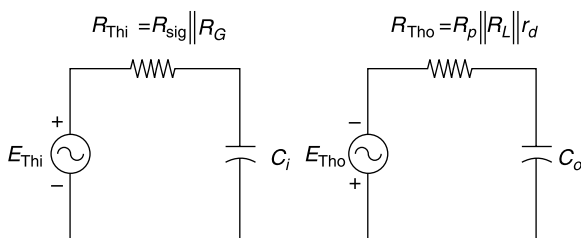


Fig. 5.48 (b) The Thevenin Equivalent Circuit for (i) Input Circuit (ii) Output Circuit

where

$$R_{Thi} = R_{Sig} || R_G$$

$$C_i = C_{gs} + (1 + g_m R_d) C_{gd}$$

and for the output circuit shown in Fig. 5.48(b),

$$f_{Ho} = 1/(2\pi R_{Tho} C_o)$$

where

$$R_{Tho} = R_D || R_L || r_d$$

5.26 COMPARISON OF JFET AND BJT

1. FET operation depends only on the flow of majority carriers—holes for P-channel FETs and electrons for N-channel FETs. Therefore, they are called Unipolar devices. Bipolar transistor (BJT) operation depends on both minority and majority current carriers.
2. As FET has no junctions and the conduction is through an N-type or P-type semiconductor material, FET is less noisy than BJT.
3. As the input circuit of FET is reverse biased, FET exhibits a much higher input impedance (in the order of 100 MΩ) and lower output impedance and there will be a high degree of isolation between input and output. So, FET can act as an excellent buffer amplifier but the BJT has low input impedance because its input circuit is forward biased.
4. FET is a voltage controlled device, i.e. voltage at the input terminal controls the output current, whereas BJT is a current controlled device, i.e. the input current controls the output current.
5. FETs are much easier to fabricate and are particularly suitable for ICs because they occupy less space than BJTs.
6. The performance of BJT is degraded by neutron radiation because of the reduction in minority-carrier lifetime, whereas FET can tolerate a much higher level of radiation since they do not rely on minority carriers for their operation.
7. The performance of FET is relatively unaffected by ambient temperature changes. As it has a negative temperature coefficient at high current levels, it prevents the FET from thermal breakdown. The BJT has a positive temperature coefficient at high current levels which leads to thermal breakdown.
8. Since FET does not suffer from minority carrier storage effects, it has higher switching speeds and cut-off frequencies. BJT suffers from minority carrier storage effects and therefore has lower switching speed and cut-off frequencies.
9. FET amplifiers have low gain bandwidth product due to the junction capacitive effects and produce more signal distortion except for small signal operation.
10. BJTs are cheaper to produce than FETs.

5.27 FET AS VOLTAGE-VARIABLE RESISTOR (VVR)

FET is operated in the constant-current portion of its output characteristics for the linear applications. In the region before pinch-off, where V_{DS} is small, the drain to source resistance r_d can be controlled by the bias voltage V_{GS} . The FET is useful as a *Voltage Variable Resistor (VVR)* or *Voltage Dependent Resistor (VDR)*.

In JFET, the drain-to-source conductance $g_d = \frac{I_D}{V_{DS}}$ for small values of V_{DS} , which may also be expressed as,

$$g_d = g_{do} \left[1 - \left(\frac{V_{GS}}{V_P} \right)^2 \right]$$

where g_{do} is the value of drain conductance when the bias voltage V_{GS} is zero.

The variation of the r_d with V_{GS} can be closely approximated by the empirical expression.

$$r_d = \frac{r_o}{1 - KV_{GS}}$$

where r_o = drain resistance at zero gate bias, and K = a constant, dependent upon FET type.

Thus, small signal FET drain resistance r_d varies with applied gate voltage V_{GS} and FET acts like a variable passive resistor.

FET finds wide applications where VVR property is useful. For example, the VVR can be used in Automatic Gain Control (AGC) circuit of a multistage amplifier.

Note: Salient features of different configuration of FET are discussed in Chapter-6.

5.28 SPECIFICATIONS OF JFET AND MOSFET

The specifications of some of the commonly used N-Channel JFET, P-Channel JFET, Power MOSFET, Small Signal MOSFET, are given in Tables A-7(a) and (b) and Tables A-8(a) respectively in Appendix-A.

REVIEW QUESTIONS

1. Why a Field Effect Transistor is called so?
2. Explain the construction of N channel JFET.
3. With the help of neat sketches and characteristic curves explain the operation of the junction FET.
4. How does the FET behave for small and large values of $|V_{DS}|$?
5. Define the pinch-off voltage V_P . Sketch the depletion region before and after pinch-off.

6. Explain the four distinct regions of the output characteristics of a JFET.
7. Define and explain the parameters transconductance g_m , drain resistance r_d and amplification factor μ of a JFET. Establish the relation between them.
8. Assuming that the saturation drain current I_{DS} is given by the parabolic relation

$$I_{DS} = I_{DSS} \left(1 - \frac{V_{GS}}{V_P} \right)^2$$

Prove that the transconductance g_m is given by

$$g_m = g_{m_0} \left(1 - \frac{V_{GS}}{V_P} \right) = -\frac{2}{V_P} \sqrt{I_{DSS} I_{DS}}$$

where g_{m_0} is the value of g_m for $V_{GS} = 0$.

9. Explain how the transconductance of a JFET varies with drain current and gate voltage.
10. What are the relative merits of an N-channel and a P-channel FET?
11. Explain why BJTs are called bipolar devices while FETs are called unipolar devices.
12. Explain why a low power FET is called as a square law device.
13. Briefly describe some applications of JFET.
14. A certain JFET operates in the linear region with a constant drain voltage of 1 V. When the gate voltage is 2 V, the drain current of 10 mA flows, but when the gate voltage is changed to 1 V, the drain current becomes 22.8 mA. Find (a) the pinch-off voltage of the device and (b) the channel resistance for zero gate voltage.
[Ans: 3 V, 18.6 Ω]
15. Show that if a JFET is operated at sufficiently low drain voltage, it behaves as a resistance R given approximately by

$$R = \frac{R_o}{\left[1 - \left(\frac{V_{GS}}{V_P} \right)^{\frac{1}{2}} \right]}$$

where R_o is the channel resistance for zero gate voltage.

16. What is a MOSFET? How many types of MOSFETs are there?
17. With the help of suitable diagrams explain the working of different types of MOSFET.
18. How does the constructional feature of a MOSFET differ from that of a JFET.
19. Explain qualitatively the shapes of the I_D Vs V_{DS} and I_D Vs V_{GS} characteristics for the three types of FETs.
20. Why are N-channel MOSFETs preferred over P-channel MOSFETs?
21. Draw the small signal model of FET for low frequency and high frequency regions and compare them with the BJT models.

22. Draw the small signal equivalent circuit of FET amplifier in CS connection and derive the equations for voltage gain, Input Impedance and Output Impedance.
23. Draw the small signal equivalent circuit of FET amplifier in CG connection and derive the equation for voltage gain, Input Impedance and Output Impedance.
24. Derive the expression for voltage gain, input impedance and output impedance of CD amplifier configuration under small signal low frequency conditions.
25. Draw a generalized FET amplifier circuit and its Thevenin's Equivalent circuits and find the voltage gain output resistance, input resistance of CS, CG and CD amplifier circuits.
26. Explain in detail about the biasing of FET.
27. Draw the source self bias and voltage divider bias circuits for FET.
28. Explain self bias circuit using N-channel JFET.
29. Draw two biasing circuits for a JFET or a depletion type MOSFET.
30. Determine the values of resistors R_D and R_S for a self - biased P-channel JFET having the following parameters: $V_P = 5$ V, $I_{DSS} = 12$ mA, $V_{DD} = 12$ V, $I_D = 5$ mA and $V_{DS} = 6$ V. [Ans. $R_D = 1.5$ k Ω ; $R_S = 525$ Ω]
31. Determine the value of R_S required to self-bias an N-channel JFET with $I_{DSS} = 50$ mA, $V_P = -10$ V and $V_{GSQ} = -5$ V. [Ans. $R_S = 400$ Ω]
32. In a self-bias N-channel JFET circuit, the operating point is to be set at $I_D = 1.5$ mA and $V_{DS} = 10$ V. The JFET parameters are $I_{DSS} = 5$ mA and $V_P = -2$ V. Find the values of R_S and R_D . Given that $V_{DD} = 20$ V. [Ans. $R_S = 0.6$ k Ω ; $R_D = 6$ k Ω .]
33. In an N-channel JFET biased by potential divider method, it is desired to set the operating point at $I_D = 2.5$ mA and $V_{DS} = 8$ V. If $V_{DD} = 30$ V, $R_1 = 1$ M Ω and $R_2 = 500$ k Ω . Find the value of R_S . The parameters of JFET are $I_{DSS} = 10$ mA and $V_P = -5$ V. [Ans. $R_S = 5$ k Ω .]
34. Draw two biasing circuits for an enhancement type MOSFET.
35. Explain in detail the biasing of MOSFET.
36. Draw the biasing circuits for the following:
N-Channel JFET, P-Channel JFET, N-Channel EMOSFET, P-Channel EMOSFET, N-Channel DEMOSFET and P-Channel DEMOSFET.
37. Explain how an FET is used as a voltage variable resistor.
38. Compare JFET with BJT.

OBJECTIVE TYPE QUESTIONS

1. The JFET is

(a) a bipolar device	(b) unipolar device
(c) voltage controlled device	(d) b and c
2. The channel of a JFET exists between

(a) gate and source	(b) drain and source
(c) gate and drain	(d) input and output
3. For low values of V_{DS} , the JFET behaves like a

(a) resistance	(b) constant voltage device
(c) constant current device	(d) negative resistor

4. In an N channel JFET,
 - (a) the current carriers are holes
 - (b) the current carriers are electrons
 - (c) V_{GS} is positive
 - (d) the input resistance is very low
5. In a P channel JFET
 - (a) the current carriers are electrons
 - (b) the current carriers are holes
 - (c) V_{GS} is negative
 - (d) the input resistance is very small
6. For an N channel JFET
 - (a) V_{GS} can vary between zero negatively to V_{GSO}
 - (b) V_{GS} can vary between zero positively to V_{GSO}
 - (c) pinch off occurs for positive V_{GS}
 - (d) V_{DD} is negative
7. An FET cannot operate at $V_{GS} = 0V$. The FET is
 - (a) JFET
 - (b) D-MOSFET
 - (c) E-MOSFET
 - (d) both (a) and (b)
8. The transconductance g_m of JFET is defined
 - (a) $\frac{\Delta I_D}{\Delta V_{GS}}$
 - (b) $\frac{\Delta I_D}{\Delta V_{DS}}$
 - (c) $\frac{\Delta V_{GS}}{\Delta I_D}$
 - (d) $\frac{\Delta I_{DSS}}{\Delta I_D}$
9. The amplification factor μ of JFET is given by
 - (a) $g_m \times r_d$
 - (b) $\frac{g_m}{r_d}$
 - (c) $g_m \cdot I_{DSS}$
 - (d) $g_m \left(1 - \frac{V_{GS}}{V_P}\right)$
10. The Shockley equation is given by
 - (a) $I_D = I_{DSS} \left[1 - \frac{V_{GS}}{V_P}\right]^2$
 - (b) $I_D = I_{DSS} \left[1 - \frac{V_{GS}}{V_P}\right]$
 - (c) $g_m = g_{mo} \left[1 - \frac{V_{GS}}{V_P}\right]$
 - (d) $I_D = C[V_{GS} - V_{GS}^{(th)}]^2$
11. Ideally, the equivalent circuit of an FET consists of
 - (a) a resistance between drain and source
 - (b) a current source between the gate and the source
 - (c) a current source between the drain and the source
 - (d) a current source between the gate and the drain
12. The magnitude of the current source in the ac equivalent circuit of an FET depends on
 - (a) the dc supply voltage
 - (b) V_{DS}
 - (c) externally drain resistance
 - (d) transconductance and gate to source voltage
13. Which one of the following has the highest input resistance?
 - (a) NPN transistor in CB configuration
 - (b) PNP transistor in CE configuration
 - (c) N type channel JFET
 - (d) P type channel MOSFET

14. The operation of a JFET involved:
 - (a) a flow of minority carriers (b) a flow of majority carriers
 - (c) Recombination (d) Negative resistance
15. When the positive voltage on the gate of a P-channel JFET is increased, its drain current:
 - (a) increases (b) decreases
 - (c) remains the same (d) None of the above
16. Which of the following statements is true?
 - (a) FET and BJT, both are unipolar
 - (b) FET and BJT, both are bipolar
 - (c) FET is bipolar and BJT is unipolar
 - (d) FET is unipolar and BJT is bipolar
17. An FET has a
 - (a) very high input resistance
 - (b) very low input resistance
 - (c) high connection emitter junction
 - (d) forward based PN junction
18. For small values of drain-to-source voltage, JFET behaves like a
 - (a) resistor (b) constant-current source
 - (c) constant-voltage-source (d) negative resistance
19. In a JFET, the primary control on drain current is exerted by
 - (a) channel resistance (b) size of depletion regions
 - (c) voltage drop across channel (d) gate reverse bias
20. After V_{DS} reaches pinch-off value V_P in a JFET, the drain current I_D becomes
 - (a) zero (b) low (c) saturated (d) reversed
21. In a JFET, as external bias applied to the gate is increased
 - (a) channel resistance is decreased
 - (b) drain current is increased
 - (c) pinch-off voltage is reached at lower values of I_D
 - (d) size of depletion regions is reduced
22. In a JFET, drain current is maximum when V_{GS} is
 - (a) zero (b) negative (c) positive (d) equal to V_P
23. A JFET has the disadvantage of
 - (a) being noisy
 - (b) having small gain bandwidth product
 - (c) possessing positive temperature coefficient
 - (d) having low input impedance
24. The drain source voltage at which the drain current becomes nearly constant is called,
 - (a) barrier voltage (b) breakdown voltage
 - (c) pick-off voltage (d) pinch-off voltage
25. The transconductance ' g_m ' of a JFET is equal to
 - (a) $-\frac{2I_{DSS}}{V_P}$ (b) $\frac{2}{|V_P|} \sqrt{I_{DSS} I_D}$
 - (c) $-\frac{2I_{DSS}}{V_P} \left(1 - \frac{V_{GS}}{V_P}\right)$ (d) $\frac{I_{DSS}}{V_P} \left(1 - \frac{V_{GS}}{V_P}\right)$

26. A depletion MOSFET differs from a JFET in the sense that it has no
 - (a) channel
 - (b) gate
 - (c) PN-junctions
 - (d) substrate
27. For the operation of enhancement N-channel MOSFET, the gate voltage will be
 - (a) high positive
 - (b) high negative
 - (c) low positive
 - (d) zero
28. The extremely high input impedance of a MOSFET is primarily due to the
 - (a) absence of its channel
 - (b) negative gate-source voltage
 - (c) depletion of current carriers
 - (d) extremely small leakage current of its gate capacitor
29. The main factor which makes a MOSFET likely to break down during normal handling is its
 - (a) very low gate capacitance
 - (b) high leakage current
 - (c) high input resistance
 - (d) both (a) and (b)
30. The main factor which differentiates a depletion MOSFET from an enhancement only MOSFET is the absence of
 - (a) insulated gate
 - (b) electrons
 - (c) channel
 - (d) PN-junction
31. The value of current source in ac equivalent circuit of FET depends on the
 - (a) transconductance and gate to source voltage
 - (b) dc supply voltage
 - (c) external drain resistance
 - (d) answers (b) and (c)
32. In a CS amplifier, $V_{DS} = 4.2$ V and $V_{GS} = 140$ mV. The voltage gain is
 - (a) 4.2
 - (b) 42
 - (c) 30
 - (d) 3
33. In a CS amplifier, $R_D = 2.2$ k Ω , $R_S = 330$ Ω , $V_{DD} = 10$ V and $g_m = 4500$ μ S. If the source resistor is completely bypassed, the voltage gain is
 - (a) 99
 - (b) 990
 - (c) 9.9
 - (d) 20.45
34. If the source bypass capacitor is removed in the CS amplifier,
 - (a) the voltage gain will increase
 - (b) the transconductance will increase
 - (c) the voltage gain will decrease
 - (d) the Q point will shift
35. If the external load resistance in CS amplifier is removed, the output voltage will
 - (a) remain constant
 - (b) decrease
 - (c) increase
 - (d) be zero
36. A CD amplifier with $R_S = 1.2$ k Ω has a g_m of 6000 μ S. The voltage gain is
 - (a) 1
 - (b) 0.878
 - (c) 0.5
 - (d) 7.2
37. The voltage gain of common drain amplifier is always
 - (a) equal to 1
 - (b) less than 1
 - (c) greater than 1
 - (d) zero
38. CG amplifier differs from both CS and CD configurations by having
 - (a) much higher voltage gain
 - (b) much lower voltage gain
 - (c) much higher input resistance
 - (d) much lower input resistance
39. The best location for setting a Q-point on dc load line of an FET amplifier is at
 - (a) saturation point
 - (b) cut-off point
 - (c) mid-point
 - (d) none of these

40. Which of the following bias methods provides a solid Q-point in JFET amplifiers?
(a) Gate bias (b) Self-bias
(c) Voltage divider bias (d) Current source bias
41. Which of the following technique is used for biasing the enhancement type MOSFET?
(a) Voltage divider bias (b) Collector feedback bias
(c) Current source bias (d) self-bias
42. The threshold voltage of an N-channel enhancement mode MOSFET is 0.5 V, when the device is biased at a gate voltage of 3 V, pinch-off would occur at a drain voltage of
(a) 1.5 V (b) 2.5 V (c) 3.5 V (d) 4.5 V
43. If properly biased, JFET with act as a
(a) current controlled current source
(b) voltage controlled voltage source
(c) voltage controlled current source
(d) current controlled voltage source
44. In a FET amplifier, the source follower is
(a) CS amplifier (b) CG amplifier
(c) CD amplifier (d) None of the above
45. The voltage gain of the CG amplifier if $R_D = 4\text{ k}\Omega$, $g_m = 2 \times 10^{-3}\text{ mho}$ and $r_d = 40\text{ k}\Omega$ is
(a) 7.36 (b) -7.36 (c) 2.6 (d) 1.66
46. The voltage gain of the CS amplifier if $R_D = 4\text{ k}\Omega$, $\mu = 40$ and $r_d = 40\text{ k}\Omega$ is
(a) -3.64 (b) 0.66 (c) -0.66 (d) 1.66
47. The voltage gain of a given common-source JFET amplifier does not depend on its
(a) input impedance (b) amplification factor
(c) dynamic drain resistance (d) drain load resistance
48. The voltage gain of the CD amplifier if $R_S = 2\text{ k}\Omega$, $\mu = 40$ and $r_d = 40\text{ k}\Omega$ is
(a) 1 (b) 0.66 (c) 0.5 (d) 1.66

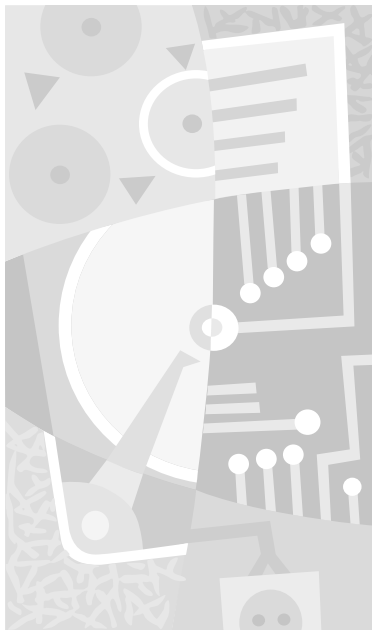
State whether the following statements are true (T) or false (F)

49. The FET has a very high input resistance.
50. The gate terminal of an FET corresponds to the collector of a BJT.
51. JFET has higher input impedance than MOSFET.
52. An FET is a unipolar transistor.
53. In a Field Effect Transistor (FET) the gate to source voltage that gives zero drain current is called pitch-off voltage.
54. A fixed bias configuration has a fixed dc voltage applied from gate to source to establish the operating point.
55. The input impedance for most FET configurations is low.
56. The output impedance of CS and CG amplifier configurations is determined primarily by R_D
57. For source follower (CD amplifier), the output impedance is determined by R_S and g_m .

58. There is no phase shift between input and output for the CD amplifier and CG amplifier configurations.
59. In CS JFET amplifier, there is a phase shift of 180° between the input and output sinusoidal voltages.
60. The output impedance of FETs is different in magnitude to that of conventional BJTs.
61. In BJTs and FETs, output voltage is the controlled variable.
62. The gate terminal of an FET corresponds to the collector of a BJT.
63. FET is used as a variable resistor.
64. For the proper functioning of a linear FET amplifier, it is necessary to maintain the operating point Q stable in the central portion of the pinch-off region.
65. Common source FET amplifier is also called source follower.
66. Generally the voltage gain of a JFET amplifier is lower than that of a BJT amplifier.
67. The voltage gain of a CD amplifier is less than 1.
68. The other name of CD amplifier is source follower.
69. The phase shift between input and output is zero for a CG amplifier.
70. The phase shift between input and output of a CS amplifier is 180° .
71. The voltage gain of CG amplifier is less than 1.
72. The gate to source voltage controls the drain to source current of FET.

ANSWERS

- | | | | | |
|---------|---------|---------|---------|---------|
| 1. (d) | 2. (a) | 3. (a) | 4. (d) | 5. (c) |
| 6. (c) | 7. (a) | 8. (d) | 9. (d) | 10. (c) |
| 11. (c) | 12. (a) | 13. (d) | 14. (d) | 15. (c) |
| 16. (d) | 17. (a) | 18. (a) | 19. (d) | 20. (c) |
| 21. (a) | 22. (c) | 23. (c) | 24. (c) | 25. (c) |
| 26. (b) | 27. (b) | 28. (d) | 29. (c) | 30. (d) |
| 31. (b) | 32. (c) | 33. (b) | 34. (c) | 35. (a) |
| 36. (a) | 37. (a) | 38. (b) | 39. (c) | 40. (a) |
| 41. (b) | 42. (d) | 43. (c) | 44. (c) | 45. (a) |
| 46. (d) | 47. (d) | 48. (c) | 49. (T) | 50. (F) |
| 51. (F) | 52. (T) | 53. (T) | 54. (T) | 55. (F) |
| 56. (T) | 57. (T) | 58. (T) | 59. (T) | 60. (F) |
| 61. (F) | 62. (F) | 63. (T) | 64. (T) | 65. (F) |
| 66. (T) | 67. (T) | 68. (T) | 69. (T) | 70. (T) |
| 71. (T) | 72. (T) | | | |



APPENDIX A

SPECIFICATIONS OF SEMICONDUCTOR DEVICES

INTRODUCTION

Data sheets published by the semiconductor device manufacturer specify the electrical characteristics of each type of device. The data sheet establishes the communication link between the manufacturer and the user. The information provided by the manufacturer must be recognized and correctly interpreted by the user. It can avoid device failures and give optimum performance.

Here, the specification sheets of some of the commonly used PN Junction Diode, Zener Diode, Varactor Diode, Tunnel Diode, Photo Diode, BJT–NPN Small Signal Transistor, BJT–PNP Small Signal Transistor, N-Channel JFET, P-Channel JFET, Power MOSFET, Small Signal MOSFET, UJT “and SCR ”are given in the following Tables.

Table A.1 PN Junction Diode

Parameters Type	V_R (max) (V)	I_R (max) (mA)	Continuous		Peak		Reverse recovery (ns)	Capacitance (10 V) (pF)	Class	Comments
			V_F (V)	I_F (mA)	V_F (V)	I_F (A)				
IN4002	100	50	0.9	1000	2.3	25	3500	15	1-amp rect	Indus std; 7-member family Silicon
IN4007	1000	50	0.9	1000	2.3	25	5000	10		Silicon
OA79	30	1.5 to 150	1.2 to 3.8	35	1.4 to 4.0	100	—	—	Detector Circuits	Germanium
BY126	650	10	—	—	1.5	5	—	—	Television receivers	Silicon
BY127	1250	10	—	—	1.5	5	—	—	Television receivers	Silicon
DR-25	25	200	—	—	1000 (mV)	250	—	—	Low power rectifier	Germanium
DS-10	10	500	—	—	22.5	20	—	—	Stabilizing element	Germanium
IN4148	75	0.025	1	10	1	200	4	4	Small Signal	Silicon
IN5404	400	5	—	—	1.1	3	—	—	GP rectifier	Silicon
IN5408	1000	5	—	3	1.1	3	—	—	GP rectifier	Silicon

Table A.2 Zener Diode

<i>Type</i>	<i>Parameter</i>	<i>Volt V_Z @ IZT (Volt)</i>	<i>Watts</i>	<i>Test Current IZT (mA)</i>
IN4728		3.3	1	—
IN4732		4.7	1	53
IN4734		5.6	1	45
IN4736		6.8	1	37
IN4740		10	1	25
IN4742		12	1	21
IN4744		15	1	17
IN4747		20	1	12.5
IN4749		24	1	10.5
IN4751		30	1	8.5
IN750		4.7	0.4	20
IN752		5.6	0.4	20
IN754		6.8	0.4	18.5
IN755		7.5	0.4	16.5
IN758		10	0.4	14.0
IN759		12	0.4	11.5
IN965		15	0.4	9.5
IN968		20	0.4	7.0
IN970		24	0.4	5.6
IN972		30	0.4	5.3

Table A.3 Varactor Diode

<i>Type</i>	<i>Parameter</i>	<i>Reverse breakdown voltage (min) (Volts)</i>	<i>Reverse leakage current (max) (nA)</i>	<i>Operating temperature range °C</i>	<i>Nominal capacitance pF</i>	<i>Minimum Q (figure of merit)</i>	<i>Tuning ratio</i>	
							<i>Min.</i>	<i>Max.</i>
ZC830B (SMT)		25	20	−55°C to +150°C	10	300	4.5	60
SOD 323 (SMT)		30	20 at $T_A = 25^\circ\text{C}$	−55°C to +125°C	—	—	—	—
SOD 123 (SMT)		30	10 at $T_A = 25^\circ\text{C}$	−55°C to +125°C	19.5	—	—	—
BB139		30	50	−55°C to +150°C	6.3	150	—	—

Table A.4 Tunnel Diode

Parameter	Maximum reverse voltage (Volts)	Maximum forward current (mA)	Maximum dissipation at 25°C (mW)	Maximum reverse leakage current at 10 V (nA)	Series inductance (nH)	Q at 2V dc	Applications
Type							
MV1403	12	250	400	10	6	200	AM broadcast band, general A.F.C. and tuning applications in lower RF frequencies AFC applications in Radio, TV and general electronic tuning.
MV1600	20	250	400	100	4	—	

Table A.5 Photo Diode

Parameter	Light Current		$V_{(BGR)}$ Volts (min)	Dark Current		Response Time	
	μA typ	@ V_R Volts		nA (max)	@ Volts		ns Typ
Type							
MRD 500	9.0	20	5.0	100	2.0	20	1.0
MRD 510	2.0	20	5.0	100	2.0	20	1.0
MRD 721	4.0	20	5.0	100	10	20	1.0

Table A.6(a) BJT–NPN Small Signal Transistor

Parameter Type	I_C (max) (mA)	P_D (max) (mW)	V_{CEO} (max) (Volts)	V_{CBO} (max) (Volts)	H_{fe} (min-max) @ I_C (mA)	V_{CE}	F_T @ MHz	Complement	Applications
BC107	100	300	45	50	110-450@2	5	300	BC177	Audio driver
BC108	100	300	20	30	110-800@2	5	300	BC178	General purpose
BC109	100	300	20	30	200-800@2	5	300	BC179	Low noise audio
BC547	100	625	45	50	110-800@2	5	300	BC557	Amplifier
BC548	100	625	30	30	110-800@2	5	300	BC558	Amplifier
BC549	100	625	30	30	110-800@2	5	250	BC559	Low noise audio
2N2369A	200	360	15	40	20(min)@100	1	500(min)	—	High speed switch
2N3904	200	350	40	60	100-300@10	1	300(min)	2N3906	Low level amp.
2N4401	600	350	40	60	100-300@150	2	250(max)	2N4403	General purpose
2N3053	700	5.0W	40	60	50-250@150	10	100	—	General purpose
2N2222A	800	500	40	75	100-300@150	10	300	—	High speed switch
BFY50	1000	2800	35	80	30(min)@150	10	60(min)	—	General Purpose
BUY82	10000	30W	60	150	40(min)@1.5A	5	60	—	High power switch
2N3055	15A	115W	60	70	20-70@4A	1.1	0.8	BDY 20	O/P - SW
2N916		360	25	45	50-200@10	0.5	300	2N2222A	

(Contd.)

Table A.6(a) (Contd.)

Parameter Type	I_C (max) (mA)	P_D (max) (mW)	V_{CEQ} (max) (Volts)	V_{CBO} (max) (Volts)	H_{fe} (min-max) @ I_C (mA)	V_{CE}	F_T @ MHz	Complement	Applications
2N2369		360		40	40-120@10	0.35	500	—	
2N3040		360	30	40	40-160@150	0.20	50	2N3040	For IF amplifier
BF184	30	145	20	30	75 to 750	10 V	300	—	Low noise AM and FM applications
BF185	30	145	20	30	67	10 V	220	—	
BF195	30	250	20	30	67	10 V	200	—	
SL100	500	800	50	60	50-280	7 V	—	SK100	AM for receivers Class B push pull stage
PT4	2 Amps	4W	20	32	80-320	10 V	3	PT6	Amplifier and Radio receivers
AC176	1 Amp	155 mW	32	32	83	—	Mc/S	AC128	Radio receivers, Tape recording
BF115	30	145 mW	30	50	45-165	—	230	—	Gen. Broadcast and television

Table A.6(b) BJT–PNP Small Signal Transistor

Parameter Type	V_{CE} (Volts)	V_{CB} (Volts)	I_C (mA)	$V_{ces}@I_C$	$H_{fe}@$ I_C (mA)	$F_T@$ I_{MHz}	P to t (mW)	Use	Comparable Types
BC157	–45	–50	–100	–0.25 10	75–260	2 150	10 300	S.S. Amp	BC177, BC212, BC307
BC158	–25	–30	–100	0.25 10	75–500	2 150	10 300	S.S. Amp	BC175, BC308
BC159	–20	–25	–100	–0.25 10	125–500	2 150	10 300	S.S. Amp	BC179 BC309
BC557	–45	–50	–100	—	110–300	2 150	— 500	4p. small sig.	BC157 DS557
BC558	–30	–30	–100	—	75	2 150	— 500	4p. small sig.	BC158 DS558
AF115	–32	–32	–10	—	150	75	75	RF. Amp, mixer oscillator in S.W. receivers	AF125 AF200
2N2904	–40	–60	–0.6 A	—	20–40	200	0.6 W	Switching & driving applications	—
AD162	–20	–32	–2 A	–1 v	50–300	1.5	6 W	Audio matched pair	—
BC177	–45	–50	–100	–5 v	175–500	200	300 mW	Audio driver	BC 107

Table A.7(a) N-Channel JFET (Absolute Maximum Rating at 25°C)

Parameter	Gate Source voltage V_{GS} (Volts)	Drain-Gate voltage V_{DG} (Volts)	Drain-Source voltage V_{DS} (Volts)	Forward Gate current I_{GF} (mA)	Device Dissipation P_D (mW)	Storage Temperature Range (°C) T_{sig}	Common uses	Features
Type								
BFW10	-30	-30	±30	10	300	-65° to 200°C	UHF amplifiers	Low Capacitance High Transconductor
BFW11	-30	-30	±30	10	300	-65° to 200°C	UHF amplifiers	Low Capacitance High Transconductor

Table A.7(b) P-Channel JFET (Absolute Maximum Rating at 25°C)

Parameter	Gate Source voltage V_{GS} (Volts)	Drain-Gate voltage V_{DG} (Volts)	Drain-Source voltage V_{DS} (Volts)	Forward Gate current I_{GF} (mA)	Device Dissipation P_D (mW)	Storage Temperature Range (°C) T_{sig}	Common uses	Features
Type								
E174	30	30	±30	10	350	-55°C to +150°C	Analog switches choppers commutators	Low cost Low on resistance Fast switching speed
2N5018	30	30	±30	10	300	-65°C to +200°C	Digital switch, sample and hold multiplexers	Low on resistance, High speed switch

Table A.8(a) Power MOSFET

Parameter Type	$V_{(BR)}$ Volts mini.	DSS max.	V_{DS} (ohms) max.	I_D (Amps)	I_D Continuous (Amps) max.	P_D $T_C = 25^\circ\text{C}$ watts
MTM6N55	550		1.2	3	6.0	150
2N6762	500		1.5	3	4.5	75
2N6760	400		1	3.5	5.5	75
2N6758	200		0.4	6	9	75
2N6757	150		0.6	5	8	75

Table A.8(b) Small Signal MOSFET

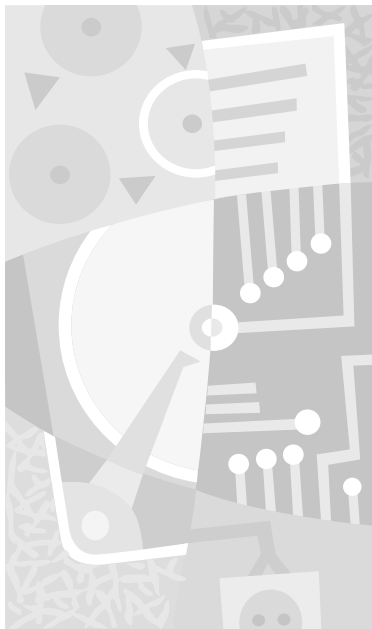
Parameter Type	$V_{(BR)}$ Volts mini.	DSS Volts mini.	V_{DS} (ohms) max.	I_D (Amps)	I_D Continuous (Amps) max.	P_D $T_C = 25^\circ\text{C}$ watts
BS107	200		14	0.2	0.25	0.6
BS170	200		5	0.2	0.50	0.6
2N6661	90		4	1	2	6.25

Table A.9 Uni Junction Transistor (Absolute Maximum Rating at 25°C)

Parameter Type	Emitter-Base two reverse voltage V_{EB2} (v)	Inter-base voltage V_{BB} (v)	Continuous emitter current I_E (mA)	Peak-emitter current I_P (A)	Continuous device dissipation at (or below) 25°C (mW)	Storage temperature range	Lead temperature 1/16 inch from the case for 10 seconds
2N2646	-30	35	50	2	300	-65 to 150°C	260°C
2N2647	-30	35	50	2	300	-65 to 150°C	260°C
MU10	30	35	50	1.0	300	-65 to 150°C	-

Table A.10 SCR

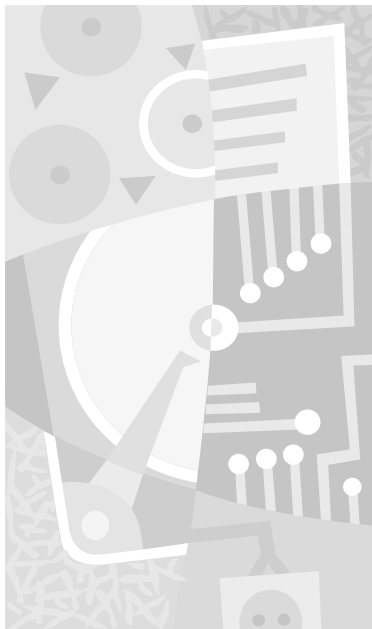
Parameter Type	Repetitive Reverse Voltage (Max.) V_{RRM} in Volts	Forward Current (RMS) I_T (RMS) in Amp.	Forward Current (Average) I_T (AV) in Amp.	Non-Repetitive Surge Current (Max.) I_{TSM} in Amp.	Gate Power (Max.) P_{GM} in Watts	Forward Gate Current (Max.) I_{GM} in Amp.	Reverse Gate Voltage (Max.) V_{RGM} in Volts	Operating Junction Temp. T_j in °C
MCR 106	30	4.0	2.55	25	0.5	0.2	6.0	-40 to +110
MCR 115	150	0.8	—	—	0.1	1.0	5.0	-40 to +110
C 106A	100	4.0	2.55	20	0.5	0.2	6.0	-40 to +110
TY6004	50	8.0	—	—	5.0	2.0	—	-40 to +125



APPENDIX B

PROBABLE VALUE OF GENERAL PHYSICAL CONSTANTS

<i>Constant</i>	<i>Symbol</i>	<i>Value</i>
Electronic charge	q	$1.602 \times 10^{-19} \text{ C}$
1 electron volt	eV	$1.602 \times 10^{-19} \text{ Joules}$
Electronic mass	m	$9.109 \times 10^{-31} \text{ kg}$
Ratio of charge to mass of an electron	q/m	$1.759 \times 10^{11} \text{ C/kg}$
Planck's constant	h	$6.626 \times 10^{-34} \text{ J-s}$
Boltzmann constant	k	$8.620 \times 10^{-5} \text{ eV/}^\circ\text{K}$
Velocity of light	c	$2.998 \times 10^8 \text{ m/s}$
Acceleration of gravity	g	9.807 m/s^2
Permeability of free space	μ_0	$1.257 \times 10^{-6} \text{ H/m}$
Permittivity of free space	ϵ_0	$8.854 \times 10^{-12} \text{ F/m}$
1 Joule	J	$6.25 \times 10^{18} \text{ eV}$



APPENDIX C

CONVERSION FACTORS AND PREFIXES

<i>Constant</i>	<i>Value</i>
1 ampere (A)	1 C/s
1 angstrom unit (Å)	$10^{-10} \text{ m} = 10^{-8} \text{ cm}$
1 coulomb (C)	1 A-s
1 farad (F)	1 C/V
1 henry (H)	1 V-s/A
1 hertz (Hz)	1 cycle/s
1 lumen	0.0016 W (at 0.55 μm)
1 mil	$10^{-3} \text{ inch} = 25 \mu\text{m}$
1 micron	$1 \mu\text{m} = 10^{-6} \text{ m}$
1 Newton (N)	1 kg-m/s ²
1 Volt (V)	1 W/A
1 Watt (W)	1 J/s
1 Weber (Wb)	1 V-s
1 Weber per square meter (Wb/m ²)	10 ⁴ Gauss
1 tesla (T)	1 Wb/m ²

Code No: A109210203

R09

Set No. 1

II B.Tech-I Sem Examinations, November-2010

Electronic Devices and Circuits

(Common to EEE, ECE, CSE, EIE, BME, IT, MCT, ECC, ETM, ICE)

Time: 3 hours

Maximum: 75 Marks

Answer any FIVE Questions

All Questions carry equal marks

1. (a) Derive an expression for total diode current starting from Boltzmann relationship in terms of the applied voltage.
(b) The reverse saturation current of a silicon $p-n$ junction diode at an operating temperature of 27°C is 50 nA . Compute the dynamic forward and reverse resistances of the diode for applied voltages of 0.8 V and -0.4 V respectively. [15]
2. (a) Define the following terms of a rectifier and filter:
(i) Ripple Factor
(ii) Regulation
(iii) Rectification Efficiency
(iv) Form Factor
(b) What is the ripple factor if a power supply of 220 V , 50 Hz is to be full wave rectified and filtered with a $220\text{ }\mu\text{F}$ capacitor before delivering to a resistive load of 120Ω ? Compute the value of the capacitor for the ripple factor to be less than 15% . [15]
3. (a) With the help of input and output characteristics, explain the operation of a BJT in Common Emitter Configuration.
(b) For an NPN transistor with $\alpha_N = 0.98$, $I_{eo} = 2\text{ }\mu\text{A}$ and $I_{EO} = 1.6\text{ }\mu\text{A}$ connected in Common Emitter Configuration, calculate the minimum base current for which the transistor enters into saturation region. V_{CC} and load resistance are given as 12 V and $4.0\text{ k}\Omega$ respectively [15]
4. (a) Explain how self biasing can be done in a BJT with relevant sketches and waveforms.
(b) Design a self bias circuit for the following specifications:
 $V_{CC} = 12\text{ V}$; $V_{CE} = 2\text{ V}$; $I_C = 4\text{ mA}$; $h_{fe} = 80$. Assume any other design parameters required. Draw the designed circuit. [15]
5. (a) Explain the concept of biasing for amplification and principle of amplification with a BJT.
(b) For a transistor amplifier, show that the input resistance R_i is given by

$$R_i = \frac{h_i}{(1 - h_r A_V)} \quad [15]$$

6. (a) Detail the construction of an n -channel MOSFET of depletion type. Draw and explain its characteristics.
- (b) A self biased p -channel JFET has a pinch-off voltage of $V_p = 5$ V and $I_{DSS} = 12$ mA. The supply voltage is 12 V. Determine the values of R_D and R_S so that $I_D = 5$ mA and $V_{DS} = 6$ V. [15]
7. (a) Draw the symbol and equivalent circuit of a UJT. Explain the operation of UJT with the help of its $V-I$ characteristics.
- (b) With the help of relevant schematic, explain the functioning of a common amplifier. [15]
8. (a) Explain how a variable capacitance can be built using a varactor diode.
- (b) Compute the zener and load currents, power dissipation in the zener diode for the voltage regulator drawn below. [15]

Solutions

1. (a) Expression for Diode Current Equation:

Let p_p = hole concentration in P -type at the edge of depletion region.

n_n = electron concentration in n -type at the edge of depletion region

p_n = hole concentration in n -region

n_p = electron concentration in p -region.

When a p - n junction diode is forward biased, holes move from P -region to n -region and electrons from n -region to p -region. The *Boltzmann Relation* gives the relationship between ' p_p ' and ' p_n '.

$$\text{i.e.,} \quad p_p = p_n \exp (V_B / V_T) \quad (1)$$

where V_B = barrier potential,

V_T = Volt-equivalent of temperature.

When a p - n junction diode is forward biased the no. of minority charges in either regions increase above their thermal equilibrium value. Equation (1) now transforms into

$$p_{p0} = p_n(0) \exp[(V_J - V)/V_T] \quad (2)$$

where p_{p0} = hole concentration in p -region

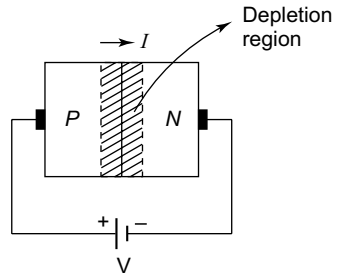
$p_n(0)$ = concentration of holes in n -region at the junction.

Under unbiased condition, $V = 0$

$$\therefore \quad p_{p0} = p_{n0} \exp [V_J / V_T] \quad (3)$$

where p_{n0} = concentration of holes in n -region at the junction.

As the concentration of holes in the entire p -region remains constant, (2) = (3)



$$\Rightarrow p_n(0) \exp [(V_J - V)/V_T] = p_{n0} \exp [V_J/V_T]$$

$$\Rightarrow p_n(0) = p_{n0} \exp [V/V_T] \quad (4)$$

Equation (4) is known as Law of Junction.

The difference between two hole concentrations at the junction under biased and unbiased condition is called injected/excess hole concentration and is denoted as $p_n(0)$

$$P_n(0) = p_n(0) - p_{n0}$$

Substituting (4) in (5), we get,

$$\begin{aligned} P_n(0) &= p_{n0} \exp \left(\frac{V}{V_T} \right) - p_{n0} \\ &= p_{n0} \left[\exp \left(\frac{V}{V_T} \right) - 1 \right] \end{aligned} \quad (6)$$

Hole current crossing the junction from p - to n -side is given by,

$$I_{pn}(0) = \frac{Aq D_p p_n(0)}{L_p} \quad (7)$$

where A = Area of cross-section of junction

D_p = Diffusion const. of holes

L_p = Diffusion length for holes

Similarly, electron current crossing the junction from n -side to p -side is given by,

$$I_{np}(0) = \frac{Aq D_n N_p(0)}{L_p} \quad (8)$$

Total current, $I = I_{pn}(0) + I_{np}(0)$

$$\Rightarrow I = \left[\frac{Aq D_p p_{n0}}{L_p} + \frac{Aq D_n n_{p0}}{L_p} \right] \exp \left(\frac{V}{V_T} - 1 \right)$$

$$\Rightarrow I = I_0 \exp \left(\frac{V}{V_T} - 1 \right) \quad (9)$$

where I_0 = reverse saturation current = $\frac{Aq D_p p_{n0}}{L_p} + \frac{Aq D_n n_{p0}}{L_p}$

For any diode the general diode current equation is given by,

$$I = I_0 \exp \left(\frac{V}{\eta V_T} - 1 \right) \quad (10)$$

where $\eta = \begin{cases} 1 & \text{for Ge} \\ 2 & \text{for Si} \end{cases}$

(b) Given data

$$T = (27 + 273)^\circ\text{K} = 300 \text{ K} \Rightarrow V_T = \frac{T}{11,600} = \frac{300}{11,600} = 26 \text{ mV}$$

$$I_0 = 50 \text{ nA}$$

$$V_f = 0.8 \text{ V}, V_R = -0.4 \text{ V}$$

$$\eta = 2 \text{ (}\therefore \text{ diode is made of Silicon)}$$

To find forward resistance (r_f)

We know that, $r_f \cong \frac{\eta V_T}{I_f}$

and $I_f = I_0 \left[\exp \left(\frac{V_f}{\eta V_T} \right) - 1 \right]$

$$I_f = 50 \times 10^{-9} \left[\exp \left(\frac{0.8}{2 \times 26 \times 10^{-3}} \right) - 1 \right]$$

$$I_f = 229 \text{ mA}$$

$$\therefore r_f = \frac{\eta V_T}{I_f} = \frac{2 \times 26 \times 10^{-3}}{229 \times 10^{-3}} = 0.227 \Omega$$

$$r_f = 0.227 \Omega$$

To find reverse resistance (r_r)

We know that $r_r = \frac{\eta V_T}{I_R + I_o} \approx \frac{\eta V_T}{I_o} = \frac{2 \times 26 \times 10^{-3}}{50 \times 10^{-9}}$

$$r_r = 1.04 \text{ M}\Omega$$

2. (a) (i) **Ripple Factor (γ):** It is defined as the ratio of *rms* value of ac component of output of rectifier to the average (or) DC value of the output.

$$\gamma = \frac{(I_{ac})_{rms}}{I_{DC}}$$

for any rectifier the expression for ripple factor is,

$$\gamma = \sqrt{\frac{I_{rms}^2}{I_{DC}^2} - 1}$$

For *HWR* and *FWR*, the values of ripple factors are 1.21 and 0.48 respectively.

- (ii) **Regulation:** It is defined as the ratio of difference between the no load o/p voltage and full load o/p voltage to the full load o/p voltage.

$$\% \text{ Regulation} = \frac{V_{NL} - V_{FL}}{V_{FL}} \times 100$$

where
$$V_{NL} = \begin{cases} V_m/\pi & \text{for HWR} \\ 2V_m/\pi & \text{for FWR} \end{cases}$$

and
$$V_{FL} = I_{DC} \cdot R_L$$

- (iii) **Rectification Efficiency:** It is defined as the ratio of *DC* power delivered to the load of a rectifier to the input ac power.

$$\% \eta = \frac{P_{DC}}{P_{AC}} \times 100$$

where
$$P_{DC} = I_{DC} \cdot R_L, P_{AC} = V_{rms} \cdot I_{rms}$$

$$\% \eta = \begin{cases} 40.6 & \text{for HWR} \\ 81.2 & \text{for FWR} \end{cases}$$

- (iv) **Form Factor:** It is defined as the ratio of *rms* value to average value of the o/p of a rectifier.

i.e.
$$FF = \frac{V_{rms}}{V_{DC}}$$

for a *HWR*,
$$FF = \frac{V_m/2}{V_m/\pi} = 1.57$$

for a *FWR*,
$$FF = \frac{V_m/\sqrt{2}}{2V_m/\pi} = \frac{V_m}{\sqrt{2}} \cdot \frac{\pi}{2V_m} = 1.11$$

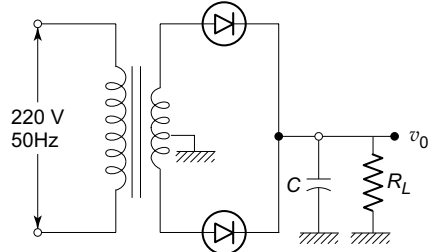
(b) Given data,

$$V_{p(rms)} = 220 \text{ V}$$

$$f = 50 \text{ Hz}$$

$$C = 220 \text{ } \mu\text{F}$$

$$R_L = 120 \text{ } \Omega.$$



For a full wave rectifier with a capacitor filter,

$$\text{Ripple Factor, } \gamma = \frac{1}{4\sqrt{3}fCR_L}$$

$$\Rightarrow \gamma = \frac{1}{4\sqrt{3}(50)(220 \times 10^{-6})(120)}$$

$$\Rightarrow \gamma = 0.1093$$

Given that $\gamma \leq 15\% \leq 0.15$

Taking $\gamma = 0.15$, $R_L = 120 \text{ } \Omega$, $f = 50 \text{ Hz}$

$$C = \frac{1}{4\sqrt{3}f\gamma R_L} = \frac{1}{4\sqrt{3}(50)(0.15)(120)}$$

$$\therefore C = 160.379 \text{ } \mu\text{F}$$

3. (a) Input Characteristics of BJT in CE Configuration

Fig. (a) shows the circuit diagram of BJT in CE configuration. In CE configuration, the input is applied between 'B' and 'E' and o/p is taken between 'C' and 'E'.

Input characteristics are the graphs plotted between input parameters (I_B and V_{BE}) for different values of output voltage (V_{CE}).

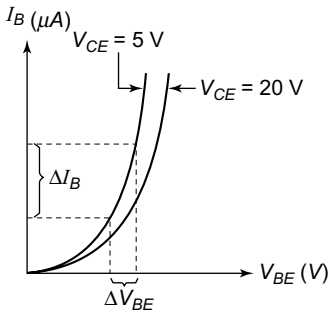


Fig. (b)

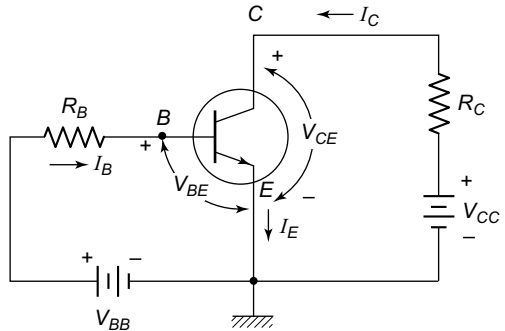


Fig. (a)

Input characteristic resembles forward characteristics of a diode because the o/p junction (J_E) is usually forward biased. After the cut-in voltage [V_{BE} (cut-in)] the output current (I_B) increases rapidly.

The input impedance of the BJT in CE configuration is very small since for a small change in input voltage there will be a large change in the input current.

$$r_i = \left. \frac{\Delta V_{BE}}{\Delta I_B} \right|_{V_{CE} = \text{const.}}$$

For a fixed value of V_{BE} , ' I_B ' decreases as V_{CE} is increased. Because larger V_{CE} results larger reverse bias across ' J_C ' and the effective base width decreases due to this the base current decreases. This is known as *base-width modulation*.

Output Characteristics

Output characteristics of BJT in CE configuration are graphs plotted between output parameters (I_C and V_{CE}) for different values of input current (I_B). For a fixed value of V_{CE} , if we take the ratio of small change in I_C to small change in I_B , we get ac beta,

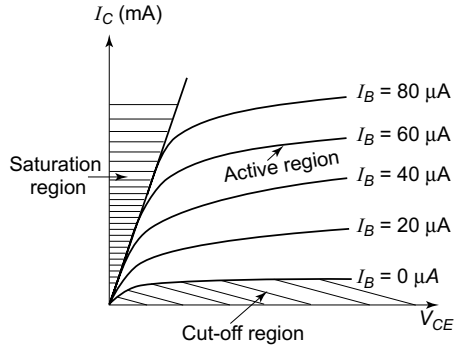
$$\beta_{ac} = \left. \frac{\Delta I_C}{\Delta I_B} \right|_{\Delta V_{CE} = 0}$$

There are three regions in the output characteristics. They are active region, saturation region and cut-off region.

Active Region: The region lying above $I_B = 0 \mu A$ and right of $V_{CE(SAT)}$ is known as active region. In this region emitter junction (J_E) is forward biased and the collector junction is reverse biased.

In this region the relationship between I_B and I_C is given as

$$I_C = \beta I_B + (\beta + 1) I_{CO}$$



Saturation Region: The region in which $V_{CE} < V_{CE(SAT)}$ and $I_B > 0 \mu A$ is known as saturation region. In this region both the junction (J_E and J_C) are forward biased.

The output current is maximum and S independent of input current.

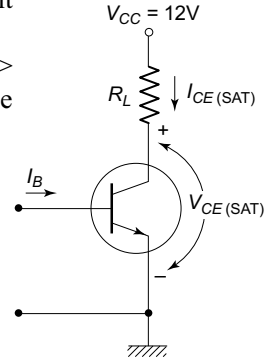
Cut-off Region: In this region $I_B < 0 \mu A$ and $V_{CE} > V_{CE(SAT)}$ and both the junctions are said to be reverse biased. ($I_C = I_{CEO}$)

(b) Given Data,

$$\alpha_N = 0.98, I_{CO} = 2 \mu A,$$

$$I_{EO} = 1.6 \mu A$$

$$V_{CC} = 12 V, R_L = 4 k\Omega$$



From figure,

$$I_{C(SAT)} = \frac{V_{CC} - V_{CE(SAT)}}{R_L} = \frac{12 - 0.2}{4 \times 10^3} = 2.95 \text{ mA}$$

$$\beta = \frac{\alpha}{1 - \alpha} = \frac{0.98}{1 - 0.98} = 49$$

For the transistor to be in saturation,

$$I_B \geq \frac{I_{C(SAT)}}{\beta}$$

$$\Rightarrow I_{B(\min)} = \frac{I_{C(SAT)}}{\beta} = \frac{2.95 \times 10^{-3}}{49} = 60.2 \mu A$$

$$\therefore I_{B(\min)} = 60.2 \mu A$$

4. (a) Refer to 4(a) of set-2

(b) Given data,

$$V_{CC} = 12V, V_{CE} = 2V, I_C = 4 \text{ mA}, h_{fe} = \beta = 80$$

Step 1: Obtain I_B , I_E and V_E

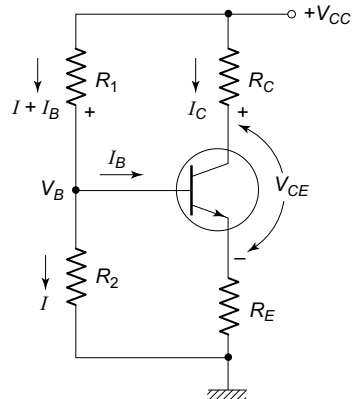
$$I_B = \frac{I_{CQ}}{\beta} = \frac{4 \times 10^{-3}}{80} = 50 \mu A$$

$$\text{wkt, } I_E = I_B + I_C = 50 \times 10^{-6} + 4 \times 10^{-3} = 4.05 \text{ mA}$$

Let us assume that

$$V_E = 0.1 V_{CC}$$

$$\therefore V_E = (0.1)(12) = 1.2 \text{ V}$$



Step 2: Obtain R_E and R_C

$$R_E = \frac{V_E}{I_E} = \frac{1.2}{4.05 \times 10^{-3}} = 296.296 \Omega$$

$$R_C = \frac{V_{CC} - V_{CE} - V_E}{I_C} = \frac{12 - 2 - 1.2}{4 \times 10^{-3}} = 2.2 \text{ k}\Omega$$

Step 3: Obtain R_1 and R_2

$$V_B = V_E + V_{BE} = 1.2 + 0.7 = 1.9 \text{ V}$$

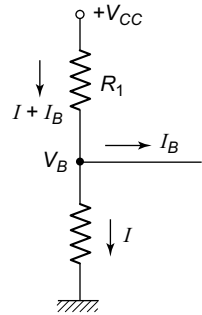
$$[\because V_{BE} = V_B - V_E]$$

For proper operation of self-bias circuit

$$I \geq 10 I_B$$

$$\Rightarrow I_{\min} = 10 I_B = (10) (50 \times 10^{-6}) = 500 \mu\text{A}$$

$$\text{From figure, } R_2 = \frac{V_B}{I} = \frac{1.9}{500 \times 10^{-6}} = 3.8 \text{ k}\Omega$$



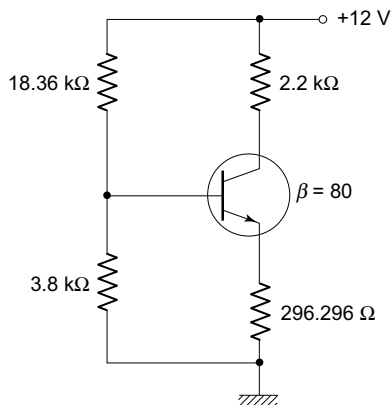
$$R_1 = \frac{V_{CC} - V_B}{I + I_B} = \frac{12 - 1.9}{500 \times 10^{-6} + 50 \times 10^{-6}} = \frac{10.1}{550 \times 10^{-6}} = 18.363 \text{ k}\Omega$$

Therefore the designed values of R_1 , R_2 , R_C and R_E are

$$R_1 = 18.363 \text{ k}\Omega, R_2 = 3.8 \text{ k}\Omega, R_C = 2.2 \text{ k}\Omega,$$

$$R_E = 296.296 \Omega$$

and the designed circuit is as shown below:



5. (a) **Biasing of BJT:**

In order to operate a *BJT* in the required region (active, saturation (or) cut-off) it is necessary to apply external *DC* voltages to the *BJT*. This application of external *DC* voltages is known as Biasing of a *BJT*.

Consider a *CE* circuit as shown below. The transistor (*BJT*) in the circuit is biased with a common supply (V_{CC}) so as to forward bias the emitter junction and reverse bias the collector junction. Thereby *BJT* operates in active region.

Applying *KVL* around the output loop of the ckt shown, we get,

$$\begin{aligned}
 -V_{CC} + I_C R_C + V_{CE} &= 0 \\
 \Rightarrow I_C &= \left(\frac{-1}{R_C} \right) V_{CE} + \left(\frac{V_{CC}}{R_C} \right) \quad (1)
 \end{aligned}$$

This equation is of the form $y = mx + c$. This can be plotted in the output characteristics of *BJT* in *CE* configuration as shown below:

The coordinates of points 'A' and 'B' can be obtained by putting $V_{CE} = 0$ and $I_C = 0$ respectively in (1).

This line is called the dc load line. The point of intersection of dc load line and output characteristic curve is known as operating/Quiescent point/*Q*-point. This point can lie in any of the three regions depending on values of V_{CC} , R_C and R_B .

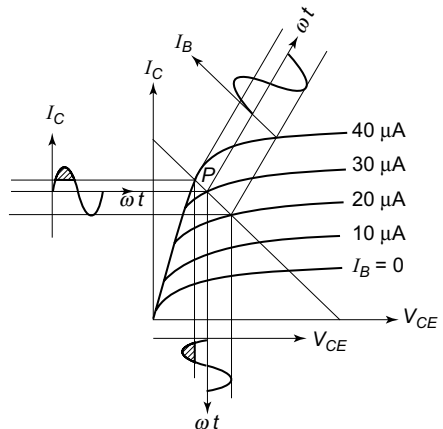
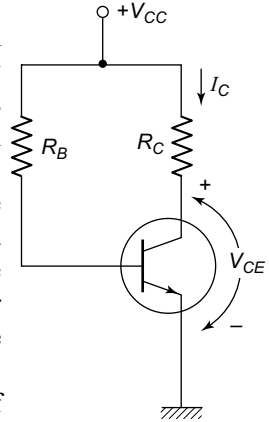
For the faithful reproduction of input ac signal of *CE* amplifier the *Q*-point has to be fixed in the active region.

Let us see the consequences of fixing the *Q*-point in all the three regions of operation of *BJT*.

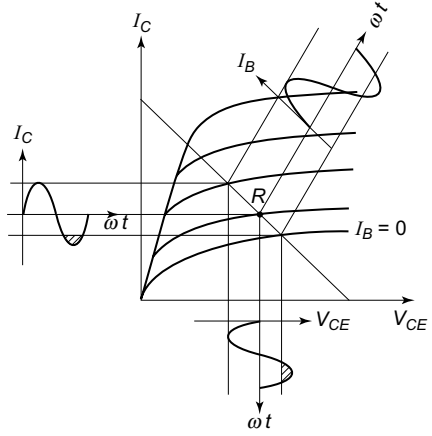
Case 1: Q-point in saturation region: (operating pt. at 'P')

When the *Q*-point is fixed in the saturation region it is observed from the above figure that a portion of the output signal is clipped and thereby there is no proper faithful reproduction of input signal.

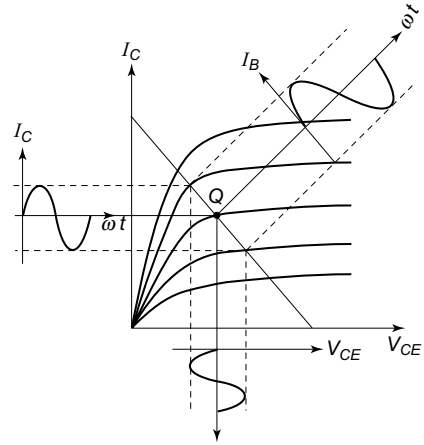
Operating pt. at 'P' is not the desired location for faithful reproduction of input signal.



Case 2: Q -point in cut-off region: (operating pt. at ' R ')
 When the Q -point is fixed in cut-off region, it is observed that the negative portion of o/p current signal gets clipped off and hence ' R ' is not the desired location of Q -point for having faithful reproductions of i/p signal.



Case 3: Q -point in the active region (operating point at ' Q ')
 When the Q -point is fixed in the active region no portion of the input signal is clipped and therefore ' Q ' is the most desired location for operating point to have faithful reproduction of input signal and good amplification of the input signal.



- (b) Consider an amplifier with 1 and 1' as input and 2 and 2' as o/p terminals as shown in the figure (a) below:

The fig. (b) shows the amplifier circuit with the BJT replaced with hybrid model.

For the amplifier circuit shown in the figure, the expressions for input impedance and voltage gain are

$$R_i = h_i + h_r A_I R_L \quad (1)$$

$$A_V = \frac{A_I R_L}{R_i} \quad (2)$$

$$\Rightarrow A_I = \frac{A_V R_i}{R_L} \quad (3)$$

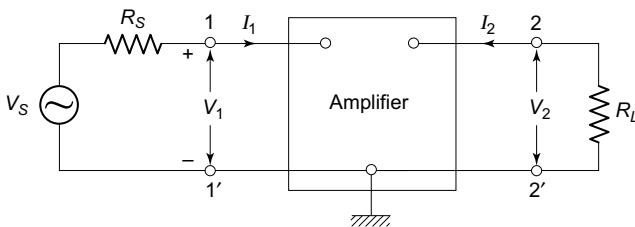


Fig. (a)

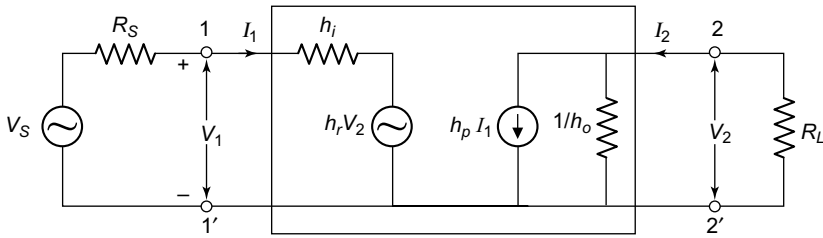


Fig. (b)

* Note: Take derivations of R_i and A_v from Q. No. 5(a) of set-2. Substituting (3) in (1), we get,

$$R_i = h_i + h_r \left(\frac{A_v R_i}{R_L} \right)$$

$$\Rightarrow R_i [1 - h_r A_v] = h_i$$

$$\Rightarrow R_i = \frac{h_i}{1 - h_r A_v} \quad (4)$$

Hence proved.

6. (a) **Construction of n -channel D-MOSFET:**

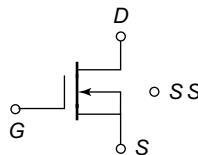
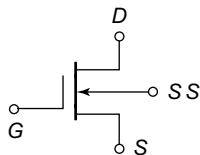
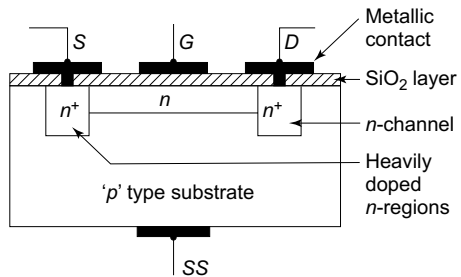
A lightly doped p -type substrate is taken and three n -type regions (n^+ , n -ch, and n^+) are introduced in to it through diffusion process.

Three layers of SiO_2 (insulating) layer are laid on the top surface of P -type substrate.

Three metallic contacts are made and the middle metallic contact acts like gate terminal of the transistor.

The two n -type regions on the two sides of the n -channel act as drain and source respectively.

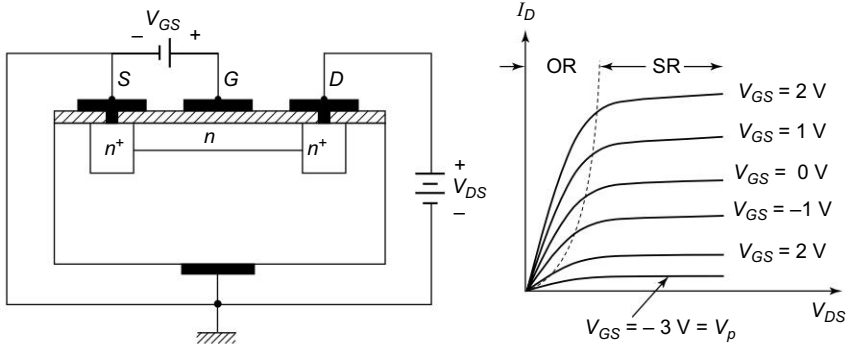
The figure below shows the circuit symbol of an n -channel D-MOSFET.



VI Characteristics of *n*-channel D-MOSFET

Drain characteristics:

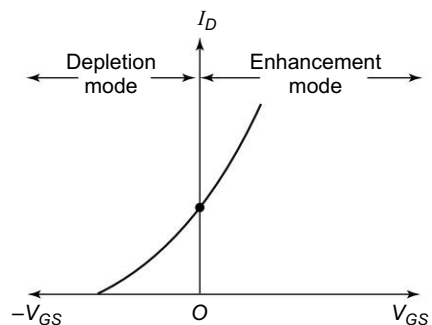
- These are the graphs plotted between I_D and V_{DS} for different values of V_{GS} .
- The figure below shows the drain characteristics of an *n*-channel D-MOSFET.



- The application of positive gate voltages enhances the level of free carriers in the channel compared to those at $V_{GS} = 0\text{ V}$.
- The region in which the drain current increase almost linearly is known as Ohmic region (OR).
- The region in which the drain current is saturated (remains constant) is known as saturation region (SR).
- When the V_{DS} is increased in the saturation region, the drain current is almost saturated because in saturation region the channel width becomes minimum and channel is said to be “pinched-off”.
- As the gate voltage increases in the negative direction the drain current reduces because negative gate voltage attracts holes in the channel that the result in the recombination. As a result the free elections in the channel are lost and hence the conduction of device falls down.

Transfer Characteristics

- These are the graphs drawn between ' I_D ' and ' V_{GS} ' by keeping V_{DS} constant.
- The transfer characteristic of an *n*-channel D-MOSFET has two parts: (i) enhancement mode and (ii) depletion mode as shown below.
- As V_{GS} increases in the positive direction its attracts electrons into the channel and enhances the conductivity of the channel and hence the mode of operation is enhancement mode.



- As V_{GS} increases in the negative direction, it attracts holes into the channel resulting in the recombinations in the channel and the drain current decreases and the mode of operation is depletion mode.
6. (b) Given data,

$$V_P = 5 \text{ V}, I_{DSS} = 12 \text{ mA}, I_D = 5 \text{ mA}, V_{DS} = 6 \text{ V}$$

for a p -channel JFET,

$$I_D = I_{DSS} \left[1 - \frac{V_{GS}}{V_P} \right]^2 \quad (1)$$

$$5 \times 10^{-3} = 12 \times 10^{-3} \left[1 - \frac{V_{GS}}{5} \right]^2$$

$$\text{Therefore, } V_{GS} = 1.7725 \text{ V} \quad (2)$$

KVL around output loop,

$$\begin{aligned} -V_{DD} + I_D R_D + V_{DS} + I_D R_S &= 0 \Rightarrow R_S + R_D \\ &= \frac{V_{DD} - V_{DS}}{I_D} \end{aligned}$$

$$\therefore R_S + R_D = \frac{12 - 6}{5 \times 10^{-3}} = 1.2 \text{ k}\Omega$$

$$R_S + R_D = 1200 \Omega$$

KVL around input loop,

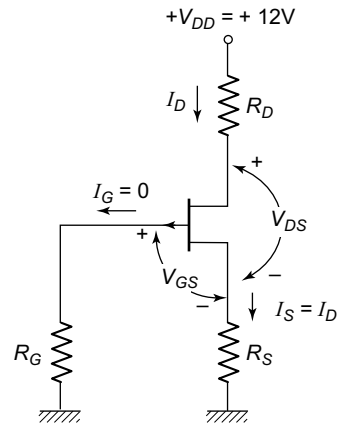
$$-I_G R_G + V_{GS} - I_D R_S = 0$$

$$\Rightarrow R_S = \frac{V_{GS}}{I_D} \quad [\because I_G = 0]$$

$$\Rightarrow R_S = \frac{1.7725}{5 \times 10^{-3}} = 354.6 \Omega \Rightarrow R_S = 354.6 \Omega$$

$$\therefore \text{From (3), } R_D = 846 \Omega$$

7. (a) **Uni Junction Transistor:** As the name implies this device has only one PN junction. The circuit symbol and equivalent ckt of UJT is as shown below:



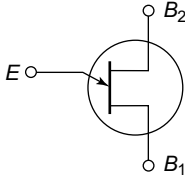


Fig. (a): ckt symbol

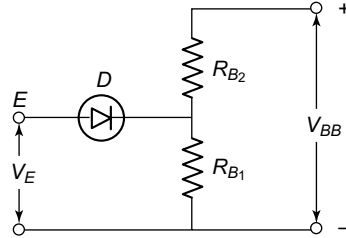


Fig. (b): Equivalent ckt

Operation:

V_{BB} is the base supply voltage applied between B_1 and B_2 .
 V_E is the emitter voltage applied between 'E' and GND.

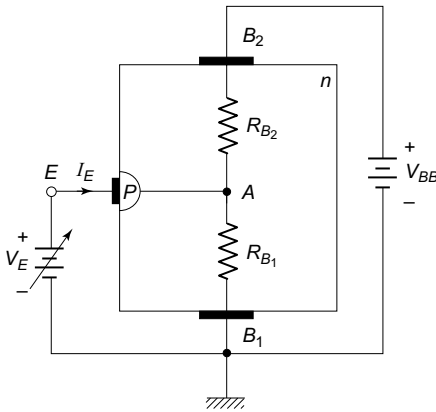


Fig. (c)

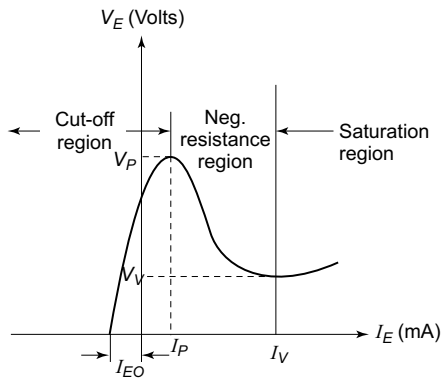


Fig. (d)

Case 1: $V_E < V_P$

As long as $V_E < V_P$, the pn junction in the UJT is reverse biased and hence the device is off state.

Case 2: $V_E \geq V_P$

When emitter voltage exceeds emitter firing potential (V_P) the pn junction becomes forward biased and I_E flows through the ckt and UJT is said to be ON. The emitter firing potential of UJT is given by

$$V_P = V_D + \eta V_{BB}$$

where V_D = diode drop

η = intrinsic stand-off ratio

Characteristics: The graph plotted between V_E and I_E is known as UJT firing characteristics.

The graph has three different regions.

- (i) *Cut-off region*: The region in which $V_E < V_P$ where the device is completely OFF is known as cut-off region. In this region small reverse leakage current ' I_{EO} ' flows through the device.
 - (ii) *Negative resistance region*: When V_E exceeds V_P the pn junction becomes forward biased and holes from P -region enter into lower base ' n ' region. This results in the decrease in the resistance R_B , of the base region and which makes the V_E decreasing. In this region ' I_E ' increases and V_E decreases.
 - (iii) *Saturation region*: Increase in I_E beyond ' I_V ' increases the emitter voltage. This region is known as saturation region.
8. (a) **Varactor Diode**

When the reverse voltage across a pn junction diode increases, the width of the depletion region increases and hence the no. of uncovered ion concentration in the depletion region increases. This increase in the no. of uncovered charge with the increase in reverse voltage applied is known as transition capacitance. (C_T)

As the transition capacitance of a diode varies with the applied voltage, it can be used as variable capacitance in many applications. In practice, special types of diodes that exhibit the property of variable capacitance are manufactured and are popularly known as *varactor diodes* (or) *varicap*.

The circuit symbol and equivalent ckt of varactor diode are as shown below:

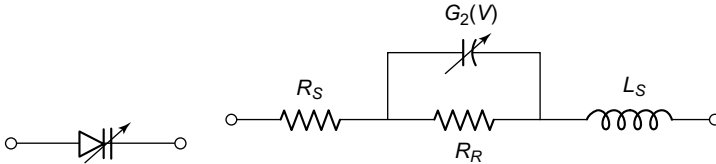


Fig. (a) Symbol

Fig. (b) Eqt. ckt

For a varactor diode, the transition capacitance is expressed as a function of applied reverse voltage as shown below:

$$C_T = \frac{k}{(V_J + V_R)^n}$$

where k = constant, V_J = Junction potential, V_R = Reverse voltage

$$\eta = \begin{cases} 1/2 & \text{for alloy function} \\ 1/3 & \text{for diffused function} \end{cases}$$

Code No: A109210203

R09

Set No. 2

II B.Tech-I Sem Examinations, November-2010

Electronic Devices and Circuits

(Common to EEE, ECE, CSE, EIE, BME, IT, MCT, ECC, ETM, ICE)

Time: 3 hours

Maximum: 75 Marks

Answer any FIVE Questions

All Questions carry equal marks

1. (a) Explain the operation of silicon $p-n$ junction diode and obtain the forward bias and reverse bias Volt-Ampere characteristics.
(b) Obtain the transition capacitance C_T of a junction diode at a reverse bias voltage of 12 V if C_T of the diode is given as 15 PF at a reverse bias of 8 V. Differential between transition and diffusion capacitances. [15]
2. (a) Derive expressions for ripple factor of a full wave rectifier with and without a capacitive filter.
(b) Compute the average and RMS load currents, TUF of an unfiltered centre tapped full wave rectifier specified below.
Input voltage to transformer = 220 V/50 Hz.
Step down ratio of centre tapped transformer = 4 : 1 (Primary to each section secondary).
Sum of transformer secondary winding in each secondary segment and diode forward resistance = 100 Ω .
Load resistance, $R_L = 200 \Omega$. [15]
3. (a) Compare the characteristics of a BJT in CB, CE and CC configurations.
(b) A Silicon BJT is connected in common Emitter configuration with collector-to-base bias. Calculate the base resistance R_b for the quiescent collector-to-emitter voltage, V_{CE} has to be 4 V. V_{CC} and R_C are given as 12 V and 1 k Ω respectively. Assume $\beta = 100$, V_{BE} to be zero volts. Also find the stability factor of the circuit. [15]
4. (a) Explain how biasing is provided to a transistor through potential divider bias. List the assumptions made. List the need of bias compensation methods.
(b) An NPN transistor with $\beta = 50$ is used in common Emitter configuration with $V_{CC} = 10V$ and $R_C = 2.2 k\Omega$. Biasing is done through a 100 k Ω resistance from collector-to-base. Assuming V_{BE} to be zero volts, Find
(i) The quiescent point
(ii) The stability Factor, 'S'. [15]

5. (a) Draw the hybrid equivalent circuit of an NPN-BJT in CE configuration. Derive the expressions for A_v , A_p , R_{in} and R_o .
- (b) Determine Z_i , Z_o and AV for the following network for the specifications listed below.
 $h_{fe} = 110$; $h_{ie} = 1.1 \text{ k}\Omega$; $h_{re} = 2 \times 10^{-4}$ and $h_{oe} = 20 \text{ }\mu\text{A/V}$ [15]
6. (a) Explain the significance of threshold voltage of a MOSFET. Discuss the methods to reduce threshold voltage, V_T .
- (b) A FET follows the relation $I_D = I_{DSS} \left[1 - \frac{V_{GS}}{V_P} \right]^2$. What are the value of I_D and g_m for $V_{GS} = -1.5 \text{ V}$ if I_{DSS} and V_P are given as 8.4 mA and -3 V respectively. [15]
7. (a) With a neat schematic, explain how amplification takes place in a common drain amplifier. [15]
- (b) Describe the application of a UJT as a relaxation oscillator.
8. Explain the principle of operation of the following devices:
 - (a) Schottky Barrier diode
 - (b) Tunnel diode through Energy band diagrams. [15]

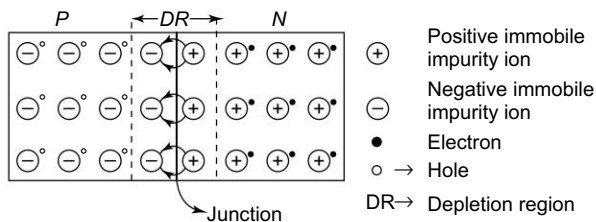
Solutions

1. (a) Operation of silicon p-n junction diode:

Case 1: Under no applied bias

As soon as a PN junction is just formed the following actions take place.

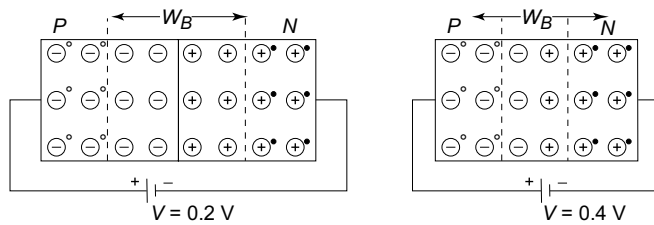
- (i) Due to concentration gradient across the junction holes from p -side and electrons from n -side move in either direction. (diffuse)
- (ii) Excess holes in n -region participate in recombination with electrons of n -region and due to this the covered ions in n -region become uncovered. Also excess electrons in P -region recombine with holes of P -region and hence the covered ions in P -region become uncovered.
- (iii) There exists an electric field between negative uncovered ion of P -region and positive uncovered ion of n -region, which stops the further diffusion of charges in both the directions.



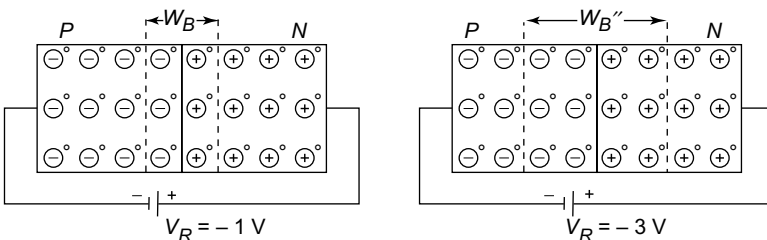
- (iv) This electric field thus acts like a barrier. The potential difference across this barrier is known as barrier potential (or) built-in voltage.
- (v) The region across the junction where the uncovered are present and which is depleted (lack of) of mobile charges (electrons and holes) is known as DEPLETION REGION.

Case 2: Under Forward bias

- (i) When the p -side and n -side of a pn junction are connected to positive and negative terminals of a battery respectively, the diode is said to be forward biased.
- (ii) Due to the applied forward bias holes of p -region and electrons of n -region experience repulsive force and they try to move towards the junction.
- (iii) Thereby these majority carriers try to cover the uncovered ions in the depletion region. Due to this the no. of uncovered ions in the depletion region reduces and hence the width of the depletion region decreases.
- (iv) When the applied forward bias equals the built-in potential the barrier (depletion region) disappears and from there onwards the majority carriers can move freely across the junction and this constitutes a current called forward current. The applied forward voltage at which the barrier across the junction disappears is known as cut-in voltage (or) threshold voltage (or) knee voltage.

**Case 3: Under Reverse Bias**

- (i) When the p -side and n -side of a pn junction are connected to negative and positive terminals of a battery respectively, the diode is said to be reverse biased.
- (ii) Due to the applied reverse bias the majority carriers of either regions experience attractive force and there by they try to move away from the junction.
- (iii) Thereby uncovering of immobile ions on both the sides take place and hence the width of the depletion region increases.
- (iv) The minority carriers in either regions experience repulsive forces due to the applied bias and they try to cross the junction which results in a small leakage current called reverse saturation current.



Voltage–Ampere Characteristics

Forward Characteristic: As the forward voltage across the PN junction diode increases, the width of the depletion region decreases and at cut-in voltage the barrier disappears and the flow of majority carriers across the junction constitutes forward current. This current increases with the increase in the applied forward bias exponentially as per the equation shown below:

$$I = I_0 \left[\exp\left(\frac{V}{\eta V_T}\right) - 1 \right] \quad (1)$$

where I_0 = reverse saturation current

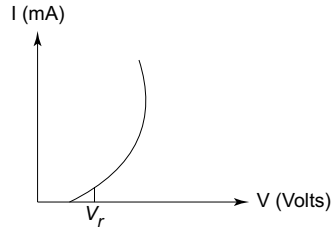
V = applied bias

$$\eta = \begin{cases} 1 & \text{for Ge} \\ 2 & \text{for Si} \end{cases}$$

$$V_T = \frac{T}{11,600} = \text{Volt-equivalent of temperature.}$$

The forward characteristic of a pn junction diode is as shown below:

' V_r ' is cut-in voltage above which the forward current through the diode increases rapidly with the increase in the voltage applied across the diode.



Reverse Characteristic: As the reverse voltage across the diode increases, the width of the depletion region increases. A small *reverse saturation current* of constant magnitude flows through the diode. This current is due to the flow of minority carriers.

When the reverse voltage across the pn junction diode reaches *breakdown voltage* (V_{BD}) breakdown occurs and the reverse current increases rapidly from there onwards as shown in the figure below:

The reverse saturation current gets doubled for every 10°C raise in temperature.

(b) Given data,

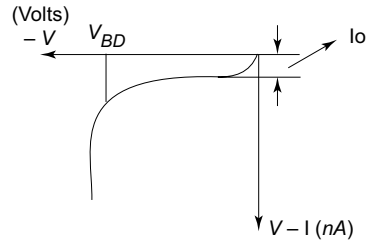
$$C_{T_2} = 15_p F, V_2 = -8 V$$

$$C_{T_1} = ?, V_1 = -12 V$$

We know that

$$C_T \propto \frac{1}{W} \text{ and } W \propto \sqrt{V_B} \text{ i.e. } W \propto \sqrt{V_J - V}$$

$$\therefore C_T \propto \frac{1}{\sqrt{V_J - V}}, \text{ where } V_J = \begin{cases} 0.6V & \text{for Si} \\ 0.2V & \text{for Ge} \end{cases}$$



$$\Rightarrow \frac{C_{T_1}}{C_{T_2}} = \sqrt{\frac{V_J - V_2}{V_J - V_1}}$$

$$\therefore \frac{C_{T_1}}{15P} = \sqrt{\frac{0.6 - (-8)}{0.6 - (-12)}}$$

$$\Rightarrow C_{T_1} = 12.39 \text{ PF}$$

Differences between Transition Capacitance and Diffusion Capacitance	
Diffusion Capacitance (C_D)	Transition Capacitance (C_T)
(1) It is, defined as the increase in the excess minority charges in either regions of pn junction diode due to the increase in the applied forward bias. $C_D = \left \frac{dQ}{dV} \right $ dQ = increase in excess minority charge dV = increase in applied forward bias	(1) It is defined as the increase in the uncovered immobile ion concentration in the depletion region due to the increase in the applied in the applied reverse bias. $C_T = \left \frac{dQ}{dV} \right $ dQ = increase in uncovered imurity non concentration dV = increase in applied reverse bias
(2) The expression for 'C_D' is given as $C_D = \frac{\tau I}{\eta V_T}$ where τ = mean lifetime of holes. I = forward current through diode	(2) The expression for 'C_T' is given as, $C_T = \frac{\epsilon A}{W}$ where ϵ = permittivity of material A = Area of cross-section of depletion region W = Width of depletion region.
(3) It is of the order of nano Farads to micro Farads.	(3) It is of the order of pico Farads.

2. (a) Expression for Ripple Factor of FWR without 'C' Filter

The ripple factor of a FWR is defined as

$$\gamma_{\text{FWR}} = \frac{\text{RMS value lof ac component of o/p}}{\text{Average/DC value of o/p}}$$

As the output of FWR is a pulsating DC output, the o/p current of FWR is composed of both ac and dc components.

As per the definition

$$\gamma_{\text{FWR}} = \frac{(I_{\text{ac}})_{\text{RMS}}}{I_{\text{DC}}} \quad (1)$$

$(I_{ac})_{RMS}$ = RMS value of ac component of o/p.

I_{DC} = DC component of o/p

I_{RMS} = RMS value of total o/p current

$$\Rightarrow I_{RMS} = \sqrt{I_{ac}^2 + I_{DC}^2}$$

$$\Rightarrow (I_{AC})_{RMS} = \sqrt{I_{RMS}^2 - I_{DC}^2} \quad (2)$$

Substituting (2) in (1), we get

$$\gamma_{FWR} = \sqrt{\left(\frac{I_{RMS}}{I_{DC}}\right)^2 - 1} \quad (3)$$

For a full wave rectifier,
RMS value of total o/p current is
given by,

$$I_{RMS} = \frac{I_m}{\sqrt{2}} \quad (4)$$

(where I_m = peak value of load current)

DC value of o/p current is given by,

$$I_{DC} = \frac{2I_m}{\pi} \quad (5)$$

Substituting (4) and (5) in (3), we get,

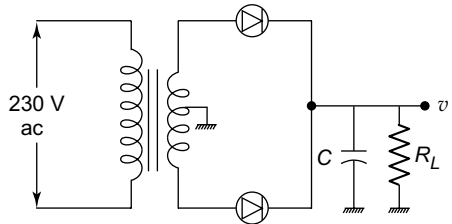
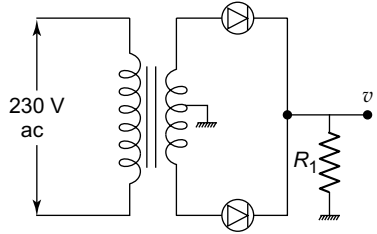
$$\begin{aligned} \gamma_{FWR} &= \sqrt{\frac{(I_m/\sqrt{2})^2}{(2I_m/\pi)^2} - 1} = \sqrt{\left(\frac{I_m^2}{2} \cdot \frac{\pi^2}{4I_m^2}\right) - 1} \\ &= 0.48 \end{aligned}$$

Expression for Ripple Factor of a FWR with Capacitor Filter

Let ' V_r ' be the peak-to-peak value of the ripple voltage.

Since the o/p of a FWR with 'c' filter resembles a triangular waveform, the relationship between rms value and peak-to-peak value of o/p is given by,

$$V_{rms} = \frac{V_r}{2\sqrt{3}} \quad (1)$$



During the time interval ' T_2 ', the capacitor discharges through the load resistance ' R_L '. Therefore the amount of charge lost during ' T_2 ' is given by,

$$Q_{\text{lost}} = I_{DC} \cdot T_2 \quad (2)$$

During the time interval ' T_1 ' the capacitor charges through the diode and it gains the charge equal to

$$Q_{\text{gained}} = C \cdot V_r \quad (3)$$

According to Law of conservation of charge,

Charge lost = Charge gained

$$I_{DC} \cdot T_2 = C \cdot V_r$$

$$V_r = \frac{I_{DC} \cdot T_2}{C}$$

$$V_r \cong \frac{I_{DC}}{2fC} \quad [\because T_2 \gg T_1 \text{ \& } T_1 + T_2 = T/2 = \frac{1}{2f}]$$

$$V_r = \frac{V_{DC}}{2fC R_L} \quad \left[\because I_{DC} = \frac{V_{DC}}{R_L} \right]$$

$$2\sqrt{3} V_{rms} = \frac{V_{DC}}{2fC R_L} \quad [\text{from (1)}]$$

$$\frac{V_{rms}}{V_{DC}} = \gamma = \frac{1}{4\sqrt{3} fC R_L}$$

$\therefore$ Ripple factor of a FWR with capacitor filter is given by

$$\gamma_{FWR} = \frac{1}{4\sqrt{3} fC R_L} \quad (4)$$

(b) Given data,

$$V_1 = 220 \text{ V}, f_1 = 50 \text{ Hz}$$

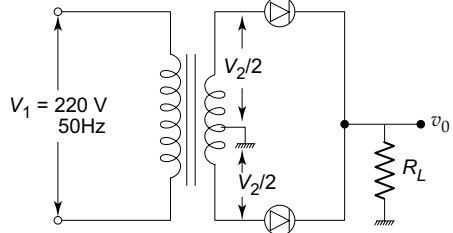
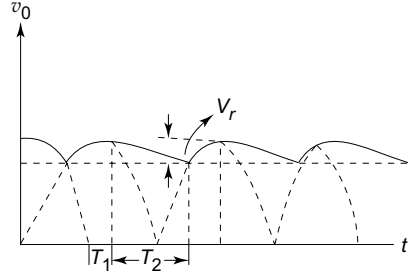
$$\frac{N_1}{N_2} = \frac{1}{4}$$

$$R_s + R_f = 100$$

$$R_L = 200 \text{ W}$$

For a transformer,

$$\frac{V_2}{V_1} = \frac{N_1}{N_2} \Rightarrow V_2 = \left(\frac{N_1}{N_2} \right) V_1 = \frac{1}{4} (220) = 55 \text{ V}$$



∴ RMS value of secondary voltage across 1st half

$$(V_2)_{rms} = \frac{V_2}{2} = \frac{55}{2} = 27.5 \text{ V}$$

Peak value of the secondary voltage,

$$V_m = (V_2)_{rms} \cdot \sqrt{2} = (27.5) \sqrt{2} = 38.89 \text{ V}$$

Peak value of the load current,

$$I_m = \frac{V_m}{R_s + R_t + R_L} = \frac{38.89}{100 + 220} = 0.1215 \text{ A}$$

Average value of load current,

$$I_{DC} = \frac{2I_m}{\pi} = 0.0773 \text{ A} = 77.34 \text{ mA}$$

RMS value of load current, $I_{rms} = \frac{I_m}{\sqrt{2}} = 85.91 \text{ mA}$

Secondary T.U.F. = $\frac{\text{DC power to the load}}{\text{AC power rating of secondary}}$

$$= \frac{\frac{I_{DC}^2 R_L}{\sqrt{2}} \cdot \frac{I_m}{\sqrt{2}}}{\left(\frac{I_m^2 R_L}{2} \right)} = \frac{\left(\frac{2I_m}{\pi} \right)^2 \cdot R_L}{\left(\frac{I_m^2 R_L}{2} \right)} = 0.812$$

Primary T.U.F. = $2 \times \text{T.U.F. of HWR}$

$$= 2 \times 0.287$$

$$= 0.574$$

$$\text{Avg. T.U.F.} = \frac{\text{Secondary T.U.F.} + \text{Primary T.U.F.}}{2}$$

$$\Rightarrow \text{Avg. T.U.F.} = \frac{0.182 + 0.574}{2} = 0.693$$

3. (a) Comparison of characteristics of BJT in CB, CE and CC configurations

S.No.	Characteristic	CB	CE	CC
1.	Input Impedance (R_i)	Very Low (20 Ω)	Low (1 k Ω)	High (500 k Ω)
2.	Output Impedance (R_o)	Very High (1 M Ω)	High (40 k Ω)	Low (50 Ω)
3.	Input Current	I_E	I_B	I_B
4.	Output current	I_C	I_C	I_E
5.	Input voltage applied between	Emitter and Base	Base and Emitter	Base and Collector

6.	Output voltage taken between	Collector and Base	Collector and Emitter	Emitter and Collector
7.	Current Amplification factor	$\alpha_{DC} = \frac{I_C}{I_E}$	$\beta_{DC} = \frac{I_C}{I_B}$	$\gamma_{dc} = \frac{I_E}{I_B}$
8.	Voltage Gain	Medium	Medium	Less than unity
9.	Current Gain	Less than unity	High (20 to few hundreds)	High (20 to few hundreds)
10.	Applications	As in i/p stage of multistage amplifier	For audio signal amplification	For impedance matching

(b) Given data

$$V_{CE} = 4V, V_{CC} = 12V, R_C = 1 \text{ k}\Omega$$

$$\beta = 100, V_{BE} = 0 \text{ V}$$

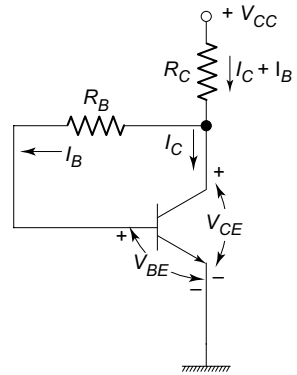
KVL around o/p loop

$$-V_{CC} + I_C R_C + V_{CE} + I_B R_C = 0$$

$$\Rightarrow I_{CRC} + \left(\frac{I_C}{\beta} \right) R_C = V_{CC} - V_{CE}$$

$$\Rightarrow I_C = \frac{V_{CC} - V_{CE}}{R_C \left(1 + \frac{1}{\beta} \right)} = 7.92 \text{ mA}$$

$$\Rightarrow I_B = I_C / \beta = \frac{7.92 \times 10^{-3}}{100} = 79.2 \mu\text{A}$$



From figure,

$$R_B = \frac{V_{CE} - V_{BE}}{I_B} = \frac{4 - 0}{79.2 \times 10^{-6}} = 50.5 \text{ k}\Omega$$

$$\text{Stability factor, } S = \frac{1 + \beta}{1 + \beta \left[\frac{R_C}{R_C + R_B} \right]} = \frac{1 + 100}{1 + 100 \left[\frac{1}{1 + 50.5} \right]}$$

$$\therefore R_B = 50.5 \text{ k}\Omega, S = 34.33$$

4. (a) Voltage Divider Bias

The above figures shows the voltage divider biasing arrangement.

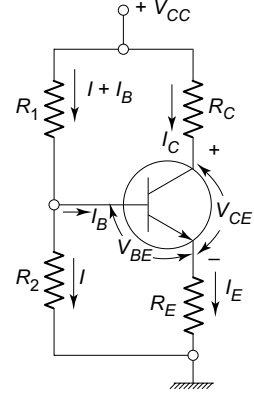
In this ckt, the biasing is provided by V_{CC} , R_1 , R_2 , R_C and R_E .

R_1 and R_2 act as a voltage divider giving a fixed voltage to the base of transistor.

If the collector current (I_C) increases due to temperature, the emitter current (I_E) also increases and the voltage drop across R_E increases, reducing the voltage difference between base and emitter ($V_{BE}^{\downarrow} = V_B - V_E^{\uparrow}$).

Due to the decrease in V_{BE} , base current (I_B) decreases and hence collector current also decreases ($I_C^\downarrow \approx \beta I_B^\downarrow$).

Therefore we can say that negative feedback exists in the emitter ckt. This reduction in ' I_C ' compensates for the original change in ' I_C ' due to change in temperature. Simplified ckt for voltage divider biasing ckt can be obtained by applying Thevenin's theorem between 'b' and GND. It is as shown below:



$$V_{TH} = V_{CC} \left(\frac{R_2}{R_1 + R_2} \right) \quad (1)$$

$$R_B = R_{Th} = R_1 \parallel R_2 \quad (2)$$

Application of KVL around input and o/p loops give the Quiscent point of the ckt.

KVL around i/p Loop

$$-V_{TH} + I_B R_B + V_{BE} + (\beta + 1) I_B R_E = 0$$

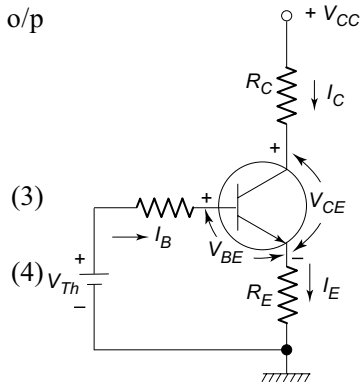
$$\Rightarrow I_B = \frac{V_{Th} - V_{BE}}{R_B + (\beta + 1) R_E}$$

$$\text{also } I_{CQ} = \beta I_B$$

KVL around o/p Loop

$$-V_{CC} + I_C R_C + V_{CE} + I_C R_E = 0$$

$$\Rightarrow V_{CEQ} = V_{CC} - I_C (R_C + R_E) \quad (5)$$



Eqns. (4) and (5) give the coordinates of Quiscent-point.

$$\therefore Q = (V_{CEQ}, I_{CQ}) \quad (6)$$

Stability factor ' S ' of the voltage divider ckt can be obtained as shown below:

$$S = \frac{\partial I_C}{\partial I_{C0}} \quad (7)$$

$$\text{We know that } S = \frac{1 + \beta}{1 - \beta(\partial I_B / \partial I_C)} \quad (8)$$

From KVL of i/p loop, we have

$$-V_{Th} + I_B R_B + V_{BE} + I_C R_E + I_B R_E = 0$$

$$\Rightarrow I_B = \frac{V_{Th} - V_{BE} - I_C R_E}{R_B + R_E}$$

$$\Rightarrow \frac{\partial I_B}{\partial I} = \frac{-R_E}{R_B + R_E} \quad (9)$$

Substituting (9) in (8), we get

$$S = \frac{1 + \beta}{1 + \left(\frac{\beta R_E}{R_E + R_B} \right)} \quad (10)$$

$$\Rightarrow S = (1 + \beta) \left[\frac{1 + (R_B/R_E)}{1 + \beta + (R_B/R_E)} \right]$$

If $R_B/R_E \ll 1$ then

$$S \approx 1 \quad (11)$$

Need for Compensating Method

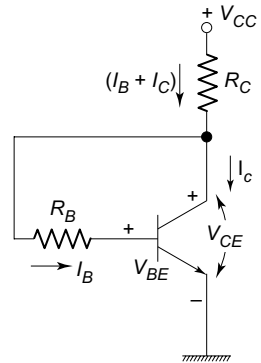
The voltage divider biasing technique introduces negative feedback for the stabilization of Q-point, which in turn reduces the amplification of the signal.

If the loss in signal is intolerable and extremely stable biasing conditions are required, then it is necessary to use compensation techniques.

(b) Given data

$$\beta = 50, V_{CC} = 10V, R_C = 2.2 \text{ k}\Omega$$

$$R_B = 100 \text{ k}\Omega, V_{BE} = 0 \text{ V.}$$



(i) To find Quiescent Point

KVL around i/p loop,

$$-V_{CC} + (I_C + I_B) R_C + I R_B + V_{BE} = 0$$

$$\Rightarrow I_C \left[1 + \frac{1}{\beta} \right] R_C + \frac{I_C}{\beta} R_B = V_{CC} \quad [\because V_{BE} = 0]$$

$$\Rightarrow I_C \left[\left(1 + \frac{1}{\beta} \right) + \frac{R_B}{\beta} \right] = V_{CC}$$

$$\Rightarrow I_C \left[2.2k \left(1 + \frac{1}{50} \right) + \frac{100 \times 10^3}{50} \right] = 10$$

$$I_C [4.244 \text{ K}] = 10$$

$$\Rightarrow I_C = 2.3562 \text{ mA} \Rightarrow I_B = \frac{I_C}{\beta} = \frac{2.3562 \times 10^{-3}}{50} = 0.04712 \text{ mA}$$

KVL Around o/p Loop

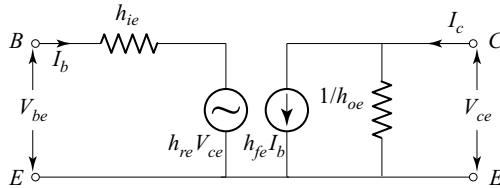
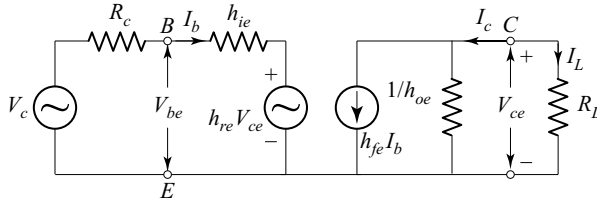
$$-V_{CC} + (I_C + I_B) R_C + V_{CE} = 0 \Rightarrow V_{CE} = V_{CC} - (\beta + 1) I_B R_C = 4.7126 \text{ V}$$

Therefore the Quiscent Point is

$$Q (4.7126 \text{ V}, 2.3562 \text{ mA})$$

(ii) To find the stability factor, 'S'

$$S = \frac{1 + \beta}{1 + \beta \left[\frac{R_C}{R_C + R_B} \right]} = \frac{1 + 50}{1 + 50 \left(\frac{2.2}{100 + 2.2} \right)} = 24.56$$

5. (a) Hybrid Equivalent circuit of NPN BJT in CE Configuration:**Derivation of Expressions for A_p , R_p , A_v and R_0 :****Current Gain**

' A_I ' is defined as ratio of output current to input current.

Mathematically,

$$A_I = \frac{I_L}{I_b} = \frac{-I_C}{I_b} \quad [\because I_L = -I_C] \quad (1)$$

From figure, KCL at node 'C' gives,

$$I_C = h_{fe} I_b + h_{oe} V_{ce} \quad (2)$$

also

$$V_{ce} = I_L R_L = -I_C R_L \quad (3)$$

Substituting (3) in (2), we get

$$I_C = h_{fe} I_b - (I_C R_L) h_{oe}$$

$$I_C [1 + h_{oe} R_L] = h_{fe} I_b$$

$$\frac{I_C}{I_b} = \frac{-h_{fe}}{1 + h_{oe} R_L}$$

$$\therefore \text{From (1), } A_I = \frac{-I_C}{I_b} = \frac{h_{fe}}{1 + h_{oe} R_L} \quad (4)$$

Input Impedance (R_{in})

It is the impedance looking into the amplifier input terminals (B and E).

$$\text{i.e., } R_{in} = \frac{V_{be}}{I_b} \quad (5)$$

KVL around i/p loop gives,

$$V_{be} = h_{ie} I_b + h_{re} V_{ce} \quad (6)$$

Dividing (6) by I_b , we get

$$\frac{V_{be}}{I_b} = R_{in} = h_{ie} + \frac{h_{re} V_{ce}}{I_b}$$

$$\Rightarrow R_{in} = h_{ie} + \frac{h_{re} (-I_C R_L)}{I_b} \quad [\text{from (3)}]$$

$$\Rightarrow R_{in} = h_{ie} + h_{re} A_I R_L \quad (7) \left[\because A_I = \frac{-I_C}{I_b} \right]$$

Voltage Gain (A_V)

It is defined as the ratio of o/p voltage (V_{ce}) to i/p voltage (V_{be}).

$$\text{i.e., } A_V = \frac{V_{ce}}{V_{be}} = \frac{-I_C R_L}{I_b R_{in}} \quad [\text{from (3) and (5)}]$$

$$\Rightarrow A_V = \frac{A_I R_L}{R_{in}} \quad (8) \left[\text{from (i), } A_I = \frac{-I_C}{I_b} \right]$$

Output Impedance (R_0)

It is the ratio of output voltage (V_{ce}) to the output current (I_C) when the source voltage (V_s) is zero.

$$\text{i.e., } R_0 = \frac{V_{ce}}{I_C} \quad V_s = 0 \quad (9)$$

From (2), we have

$$I_C = h_{fe} I_b + h_{oe} V_{ce}$$

dividing (2) by V_{ce} , we get

$$\frac{1}{R_0} = \frac{I_C}{V_{ce}} = \frac{h_{fe} I_b}{V_{ce}} + h_{oe}$$

$$\frac{1}{R_0} = h_{oe} + \frac{h_{fe} I_b}{V_{ce}} \quad (10)$$

When $V_S = 0$, KVL around the o/p loop

$$I_b R_s + I_b h_{ie} + h_{re} V_{ce} = 0$$

$$\Rightarrow I_b (R_s + h_{ie}) = -h_{re} V_{ce}$$

$$\Rightarrow \frac{I_b}{V_{ce}} = \frac{-h_{re}}{R_s + h_{ie}} \quad (11)$$

Substituting (11) in (10), we get,

$$\frac{1}{R_0} = h_{oe} - \frac{h_{fe} h_{re}}{R_s + h_{ie}}$$

$$\Rightarrow R_o = \frac{1}{h_{oe} - \frac{h_{fe} h_{re}}{R_s + h_{ie}}} \quad (12)$$

(b) Given data,

$$h_{fe} = 110, h_{ie} = 1.1 \text{ k}\Omega, h_{re} = 2 \times 10^{-4} \text{ and } h_{oe} = 20 \mu\text{A/V}$$

Assuming $R_L = 1 \text{ k}\Omega$ and $R_s = 500 \Omega$

For a CE amplifier,

$$\text{Current Gain, } A_I = \frac{-h_{re}}{1 + h_{oe} R_L} = \frac{-110}{1 + (20 \times 10^{-6})(10^3)} = -107.84$$

$$\begin{aligned} \text{Input Impedance, } Z_i &= h_{ie} + h_{re} A_I R_L = 1100 + (2 \times 10^{-4})(-107.843)(1000) \\ &= Z_i = 1078.4314 \Omega \end{aligned}$$

$$\text{Voltage Gain, } A_V = \frac{A_I R_L}{Z_i} = \frac{(-107.843)(1000)}{1078.4314}$$

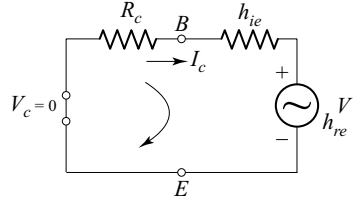
$$A_V = -99.998$$

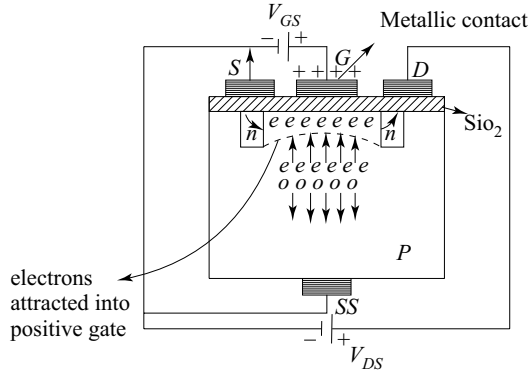
$$\text{Output Impedance, } Z_o = \frac{1}{h_{oe} - \frac{h_{fe} h_{re}}{h_{oe} + R_s}} = 160 \text{ k}\Omega$$

$$\therefore Z_0 = 160 \text{ k}\Omega$$

6. (a) In an n -channel E -MOSFET if we increase the magnitude of V_{GS} in the positive direction, the concentration of electrons near SiO_2 surface increases. At a particular value of V_{GS} an appreciable amount of current (I_D) flows between drain and source. This value of V_{GS} is called Threshold voltage (V_T). Thus in an n -channel E -MOSFET, a positive gate voltage above V_T induces a channel and hence the drain current flows through the enhanced channel.

For any gate voltage below V_T there is no channel and hence the device is said to be OFF.





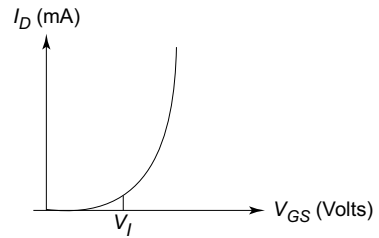
The below figures shows the transfer characteristic of an n -channel E -MOS-FET.

For $V_{GS} > V_T$, the relationship between ' I_D ' and ' V_{GS} ' is given by,

$$I_D = K (V_{GS} - V_T)^2 \quad (1)$$

' K ' is a constant depending on the construction of the device.

The value of V_T can be reduced by heavily doping the P -type substrate with n -type impurities.



(b) Given that

$$V_{GS} = -1.5 \text{ V}, I_{DSS} = 8.4 \text{ mA and } V_P = -3 \text{ V.}$$

also,

$$I_D = I_{DSS} \left[1 - \frac{V_{GS}}{V_P} \right]^2 \quad (1)$$

differentiating (1) wrt V_{GS} , we get,

$$g_m = \frac{dI_D}{dV_{GS}} = \left(\frac{-2I_{DSS}}{V_P} \right) \left[1 - \frac{V_{GS}}{V_P} \right]$$

$$\Rightarrow g_m = \left(\frac{2I_{DSS}}{V_P} \right) \left[\frac{V_{GS}}{V_P} - 1 \right] \quad (2)$$

From (1),

$$I_D = 8.4 \times 10^{-3} \left[1 - \frac{(-1.5)}{(-3)} \right]^2$$

$$\Rightarrow I_D = 2.1 \text{ mA}$$

From (2),

$$g_m = \frac{2(8.4 \times 10^{-3})}{(-3)} \left[\frac{(-1.5)}{(-3)} - 1 \right]$$

$$\Rightarrow g_m = 2.8 \Omega$$

7. (a) Common Drain Amplifier

In CD amplifier input is applied between gate (G) and source (S) and o/p is taken between source (S) and drain (D).

In a common drain amplifier circuit the source voltage is given as,

$$V_S = V_G + V_{GS} \quad (1)$$

When an ac signal is applied to JFET gate (G) via ' G ', V_G varies in accordance with the signal. As V_{GS} is fairly constant and from (1), V_S varies with V_i .

Because the o/p voltage at the source (V_S) follows the changes in the signal applied to the gate (G), this ckt is also called as SOURCE FOLLOWER.

(b) UJT as a relaxation oscillator:

Construction: R_1 and R_2 are biasing resistors which are so selected that they are lower than interbase resistances RB_1 and RB_2 respectively. The elements R_T and C_T are the timing elements and they decide the rate of oscillations. ' R_T ' is so selected that the operating point of UJT lies in its negative resistance region.

Operation: As soon as the circuit is switched ON, the capacitor ' C_T ' charges towards V_{BB} through ' R_T '. As long as the capacitor voltage is less than ' V_P ', UJT remains OFF.

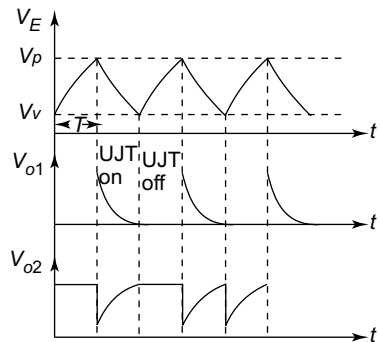
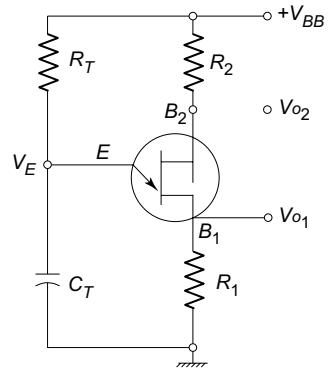
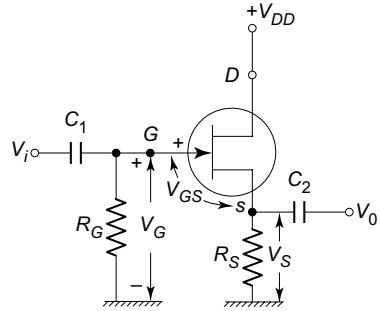
The emitter firing potential of UJT is given by

$$V_P = \eta V_{BB} + V_D \quad (1)$$

where η =intrinsic stand-off ratio

V_D = cut-in voltage of the pn junction.

When capacitor voltage exceeds ' V_P ', the UJT starts conducting and thereby ' C_T ' now discharges through UJT and ' R_1 '. When capacitor voltage (V_E)



reaches ' V_V ' UJT is turned OFF and capacitor ' C_T ' starts charging again. And the waveforms at ' E ', ' B ', and ' B_2 ' are as shown below:

Since the circuit relaxes back after every time period ' T ' the ckt is known as relaxation oscillator.

Analysis: The charging equation of capacitor C_1 is given by,

$$v_c(t) = V_V + V_{BB} [1 - e^{-t/R_T C_T}] \quad (2)$$

from figure at $t=T$, $v_c(t) = V_p$

therefore,

$$V_p = V_V + V_{BB} [1 - e^{-T/R_T C_T}]$$

$$\eta V_{BB} + V_D = V_V + V_{BB} [1 - e^{-T/R_T C_T}] \quad [\text{from (1)}]$$

$$\eta V_{BB} \cong V_{BB} [1 - e^{-T/R_T C_T}] \quad [\text{neglecting } V_D \text{ and } V_V]$$

$$T = R_T C_T \ln \left[\frac{1}{1 - \eta} \right]$$

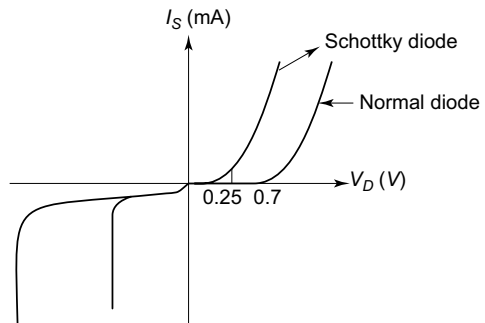
$$f = \frac{1}{R_T C_T \ln \left(\frac{1}{1 - \eta} \right)} \quad (3)$$

where

f = oscillating frequency

8. (a) Principle of Operation of Schottky Barrier Diode

The normal Pn junction diodes take certain amount of time to attain OFF state when they are switched from ON to OFF state. This time is called reverse recovery time. For frequencies upto 10 MHz it is very small but above 10 MHz its is large and puts a limit on its use at very high frequencies.



The diodes that are specially manufactured to solve this problem of fast switching are called Schottky Diodes.

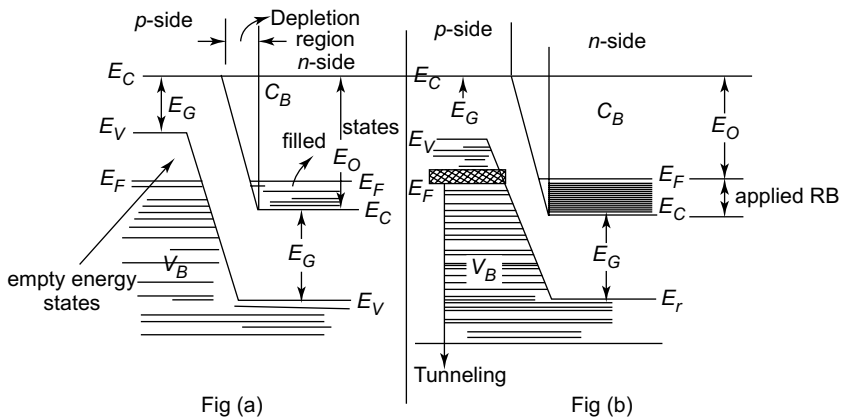
Construction: It consists of a metal to semiconductor junction as shown in the figure below.

The circuit symbol of a Schottky diode is also shown below.

Usually n -type Si is used as semiconducting material. Metals such as platinum, chrome (or) tungsten are used to get different set of characteristics such as increased frequency range, lower forward bias etc.

Operation: In both the metal and semiconductors electrons as a majority carriers. When a contact is made between the two the electrons from n -type material immediately flow into the adjoining metal. This is because the kinetic energy of majority carriers of n -type is higher than that of electrons in a metal. These injected carriers due to KE are called as hot carriers. The heavy flow of electrons into the metal creates a region near the junction surface, depleted of carriers in the Silicon material. This is similar to the depletion region of a conventional diode. The additional carriers in the metal establish a negative wall in the metal at the boundary between the materials which is called as surface barrier that prevents the further current.

Due to the lack of minority carriers in the metal, Schottky diode cannot store the charge and hence it can switch off very fast for frequencies above 300 MHz. The barrier potential and breakdown voltage of Schottky diode are less than those of conventional diode. These Schottky diodes find their applications in Schottky T/L logics, mixers, radar systems etc. Their VI characteristics are as shown below:



(b) Operation of Tunnel Diode using Energy Band Diagrams

The diodes in which the impurity atom concentration is greatly increased upto 1 in 103 that exhibit tunneling phenomenon are called as tunnel diodes. Due to this heavy doping the width of the depletion region is very small due to which electrons can easily penetrate (tunnel) through this region. This process is called as tunneling.

Let us see the process of tunneling using energy band diagrams of tunnel diode under different applied biases.

Figures (a) and (b) show the energy band diagrams of a tunnel diode under open circuited condition and reverse bias condition respectively. Under open circuited condition, the Fermi-level (E_F) in the p -side is at the same energy as that in the n -side. Since there are no filled energy states that are at a greater height than empty energy states, there is no movement of charges

and hence the device is OFF and there is no current through the device.

When reverse bias is applied the height of the barrier is increased above ' E_0 '. Hence the n -side energy levels shift downwards with respect to those of P -side. Due to the availability of energy states (shaded in black) in the valence band of p -side, the electrons in these energy states tunnel from p -side to n -side giving rise to reverse current. As the reverse voltage increases, the electrons in the shaded portion increases and hence the reverse current also increases.

When a tunnel diode is forward biased there exist occupied states in the conduction band of n -side (shown as shaded in Fig. (c)) which are at the same energy at that of allowed empty states of VR of P -side.

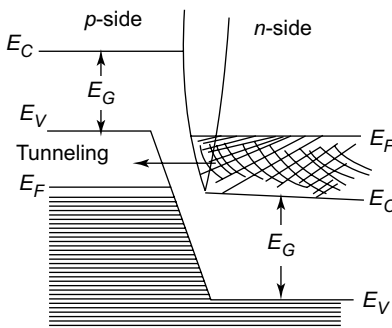


Fig (c)

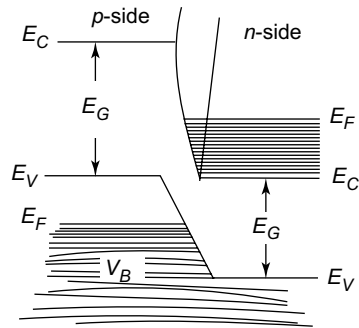
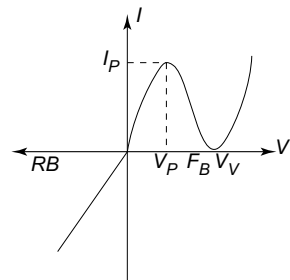


Fig (d)

Due to this electrons from n -side tunnel through the barrier and enter into the P -side. This movement causes the flow of forward current. When the forward bias is increased n -side level shifts further upwards and more no. of electrons try to tunnel, giving rise to IP (peak current). Further increase in forward bias reduces the height of the shade portion on n -side and hence forward current decreases as shown in the figure.

When the filled energy states in the shaded portion of n -side becomes zero tunneling stops f the device acts like a normal diode and the current increases with the increase in the applied bias.



Code No: A109210203

R09

Set No. 3

II B.Tech-I Sem Examinations, November-2010

Electronic Devices and Circuits

(Common to EEE, ECE, CSE, EIE, BME, IT, MCT, ECC, ETM, ICE)

Time: 3 hours

Maximum: 75 Marks

Answer any FIVE Questions

All Questions carry equal marks

1. Difference between
 - (i) Statics and dynamic resistances of a $p-n$ diode.
 - (ii) Transition and Diffusion capacitances of a $p-n$ diode.
 - (iii) Volt-Ampere characteristics of a single silicon $p-n$ diode and two identical silicon $p-n$ diodes connected in parallel.
 - (iv) Avalanche and Zener break down mechanisms. 15
2. (a) Define Ripple factor and form factor. Establish a relation between them.
(b) Explain the necessity of a bleeder resistor in an L -section filter used with a full wave filter.
(c) Compute ripple factor of an L – section choke input filter used at the output of a full wave rectifier. Inductor and capacitor values of the filter are given as $10H$ and $8.2 \mu F$ respectively. 15
3. (a) Describe the significance of the terms, ' α ' and ' β '. Establish a relation between them.
(b) A transistor is operated at a forward emitter current of 2 mA and with the collector open-circuited. Assuming $\alpha_N = 0.98$, $I_{E0} = 1.6 \mu A$ and $I_{C0} = 2 \mu A$, determine
 - (i) The junction voltages V_C and V_E
 - (ii) The collector to emitter voltage V_{CE}
 - (iii) The region of transistor operation (saturation/active/cut-off).

Assume any other values necessary. 15
4. (a) Justify statement "Potential divider bias is the most commonly used biasing method" for BJT circuits. Explain how bias compensation can be done in such biasing through diodes.
(b) An NPN transistor with $\beta = 100$ is used in common Emitter configuration with collector-to-Base bias. If $V_{CC} = 10V$, $R_C = 1 \text{ K}\Omega$ and $V_{BE} = 0V$, determine
 - (i) R_b such that quiescent collector-to-emitter Voltage is $4V$.
 - (ii) The stability factor, ' S '. 15
5. (a) Define all the four hybrid parameters of a BJT in CE configuration. Draw the circuit and its equivalent circuit.
(b) The source and load resistances connected to a BJT amplifier in CE configuration are 680Ω and $1K\Omega$ respectively. Calculate the voltage

gain A_v and the input resistance R_i if the h-parameters are listed as $h_{ie} = 1.1 \text{ k}\Omega$; $h_{re} = 2 \times 10^{-4}$; $h_{fe} = 50$ and $h_{oe} = 20 \text{ }\mu\text{mhas}$. Compute A_v and R_i using both approximate and exact analysis. 15

6. (a) Explain how a FET can be made to act as a switch.
- (b) Show that the transconductance, g_m and drain current, I_{DS} of FET are related through $g_m = \frac{2}{|V_P|} \sqrt{I_{DSS} I_{DS}}$
Define other terms of the equation.
- (c) List any four merits of MOSFET to show that they are more suitable than JFETs in Integrated circuits. 15
7. (a) With the help of a neat schematic, explain the functioning of a common source amplifier.
- (b) Bring out the differences between BJT and FET. Compare the three configurations of JFET amplifiers. 15
8. (a) What is Schottky effect? Elaborate Schottky effect for the functioning of a Schottky barrier diode.
- (b) Describe the construction, principle of operation and performance characteristics of a silicon controlled rectifier. 15

Solutions

1. (a) *Static Resistance:*

1. The ratio of voltage drop across the diode to the current flowing through the diode at any point in the VI characteristics of *pn* junction diode is known as static resistance

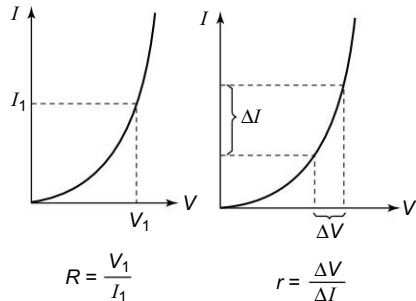
2. It is not a constant. Its value varies from point to point

Dynamic Resistance:

1. It is defined as the reciprocal of the slope of VI characteristic of a *pn* junction diode.

Mathematically it is defined as

$$r = \frac{\Delta V}{\Delta I}$$



- (b) Refer to 1(b) of Set No. 2

	Avalanche Breakdown		Zener Breakdown
1.	It is due to the kinetic energy acquired by the minority in either regions	1.	It is due to the strong electric field across the junction due to the heavy doping.
2.	This occurs for zener diodes with V_{BD} greater than 6V	2.	This occurs for zener diodes with V_{BD} less than 6V
3.	The temperature coefficient is positive	3.	It is negative
4.	Occurs for lightly doped diodes	4.	Occurs for heavily doped diodes.

2. (a) **Ripple Factor:** Refer to Q.No. 2(a) of Set No. 1

Form Factor: Refer to Q.No. 2(a) of Set No. 1.

For definitions

$$\text{Ripple Factor, } \gamma = \sqrt{\left(\frac{V_{rms}}{V_{DC}}\right)^2 - 1}$$

$$\text{Form Factor, } FF = \frac{V_{rms}}{V_{DC}}$$

$$\therefore \gamma = \sqrt{FF^2 - 1}$$

2. (b) **Necessity of Bleeder Resistor in L-section filter:**

For a full wave rectifier with an

LC filter,

$$I_{DC} = \frac{2}{\pi} \left[\frac{V_m}{R + R_x} \right] \quad (1)$$

where $R = R_B \parallel R_L$

R_x = internal resistance of 'L'

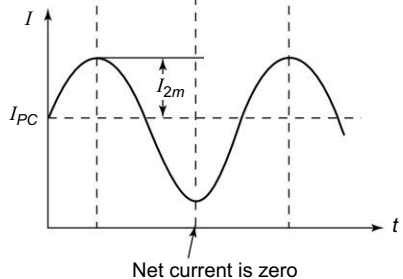
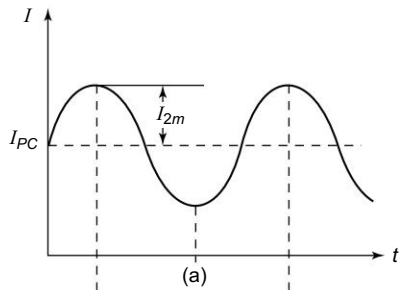
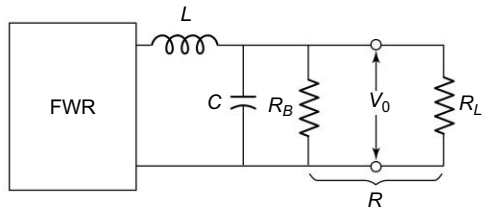
Also the magnitude of second harmonic of the O/P is,

$$I_{2m} = \frac{4}{3\pi} \left(\frac{V_m}{2\omega L} \right) \quad (2)$$

From (1) we can say that I_{DC} depend on R_L while I_{2m} does not. The second harmonic is superimposed over I_{DC} as shown in Fig. (a).

If R_L is increased, I_{DC} will decrease, but I_{2m} does not. If ' RL ' is further increased stage may come where I_{DC} becomes less than I_{2m} as shown in fig. (b) In such situation, the net current is zero at a particular instant. In order to overcome this problem a bleeder resistance ' R_B ' is connected in the ckt shown above.

The bleeder resistance is so selected that it draws, a minimum current through choke, the condition for overcoming the above mentioned problem is



(b)

$$I_{DC} = I_{2m}$$

$$\Rightarrow \frac{2}{\pi} \left[\frac{V_m}{R_x + R} \right] = \frac{4}{3\pi} \left(\frac{V_m}{2\omega L} \right)$$

$$\Rightarrow R + R_x = 3\omega L \quad (3)$$

$$\text{Usually } R_x \ll R \text{ then } R \cong 3\omega L \quad (4)$$

Considering the worst case that R_L is not connected than $R = R_B$

$$\therefore R \cong 3\omega L \quad (5)$$

2. (c) Given that

$$L = 10 \text{ H}$$

$$C = 8.2 \mu\text{F}$$

Ripple factor of an

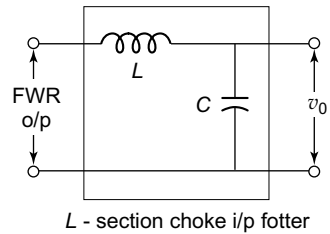
L-section choke i/p filter is given by,

$$\gamma_{LC} = \frac{1}{6\sqrt{2}\omega^2 LC}$$

Assuming $f = 50\text{Hz} \Rightarrow \omega = 2\pi f = 100\pi \text{ rad/s}$

$$\therefore \gamma_{LC} = \frac{1}{6\sqrt{2} (100\pi)^2 (10) (8.2) (10^{-6})}$$

$$\begin{aligned} \gamma_{LC} &= \frac{1}{6\sqrt{2} (\pi^2) (0.82)} \\ &= 0.01456 \end{aligned}$$



3. (a) **Relation between ' α_{DC} ' and ' β_{DC} ' :**

α_{DC} : defined as the ratio of collector current to emitter current

$$\text{i.e.} \quad \alpha_{DC} = \frac{I_C}{I_E} \quad (1)$$

β_{DC} : defined as the ratio of collector current to base current

$$\text{i.e.,} \quad \beta_{DC} = \frac{I_C}{I_B} \quad (2)$$

For any BJT, we know that

$$I_E = I_C + I_B$$

$$\Rightarrow I_B = I_E - I_C \quad (3)$$

Using (3) in (2), we get,

$$\beta_{DC} = \frac{I_C}{I_E - I_C}$$

Dividing numerator and denominator of RHS by ' I_E ',

$$\beta_{DC} = \frac{I_C/I_E}{I_E/I_E - I_C/I_E}$$

$$\beta_{DC} = \frac{\alpha_{DC}}{1 - \alpha_{DC}} \quad (4)[\text{from 1}]$$

Dividing both sides of 4 by $(1 + \beta_{DC})$,

$$\begin{aligned} \frac{\beta_{DC}}{(1 + \beta_{DC})} &= \left(\frac{\alpha_{DC}}{1 - \alpha_{DC}} \right) \left(\frac{1}{1 + \beta_{DC}} \right) \\ \frac{\beta_{DC}}{1 + \beta_{DC}} &= \left(\frac{\alpha_{DC}}{1 - \alpha_{DC}} \right) \left(\frac{1}{1 + \frac{\alpha_{DC}}{1 - \alpha_{DC}}} \right) \\ &= \frac{\alpha_{DC}}{1 - \alpha_{DC}} \left[\frac{1 - \alpha_{DC}}{1 - \alpha_{DC} + \alpha_{DC}} \right] \end{aligned}$$

$$\text{Therefore,} \quad \alpha_{DC} = \frac{\beta_{DC}}{1 + \beta_{DC}} \quad (5)$$

4. (a) Refer to 4(a) of set 1

4. (b) Given data

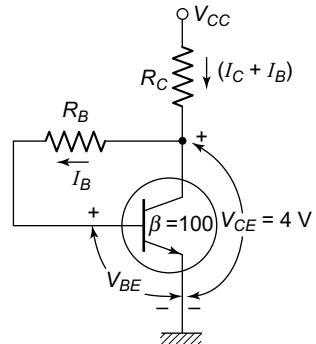
$$V_{CC} = 10V, \beta = 100, R_C = 1k\Omega, V_{BE} = 0V$$

Apply KVL around O/P loop,

$$-V_{CC} + (I_C + I_B) R_C + V_{CE} = 0$$

$$I_C + I_B = \frac{V_{CC} - V_{CE}}{R_C}$$

$$(\beta + 1) I_B = \frac{V_{CC} - V_{CE}}{R_C}$$



$$\Rightarrow I_B = \frac{V_{CC} - V_{CE}}{R_C (\beta + 1)} = \frac{10 - 4}{1 \times 10^3 (100 + 1)} = 59.4 \mu A$$

From figure,
$$R_B = \frac{V_{CE} - V_{BE}}{I_B} = \frac{4 - 0}{59.4 \times 10^{-6}} = 67.34 \text{ K}\Omega$$

(ii) Stability factor,
$$S = \frac{\beta + 1}{1 + \beta \left(\frac{R_C}{R_C + R_B} \right)}$$

$$S = \frac{1 + 100}{1 + 100 \left(\frac{1}{1 + 67.34} \right)} = 41.0023$$

Therefore, $S = 41.0023$ and $R_B = 67.34 \text{ K}\Omega$

5. (a) *Short circuit input impedance (h_{ie})*

Defined as the ratio of input voltage (V_{BE}) to the input current of BJT in C_E configurations when O/P is shorts circuited

$$h_{ie} = \left. \frac{V_{be}}{I_b} \right|_{V_{ce}} = \text{constant} \quad (1)$$

Reverse Voltage Transfer Ratio (h_{re})

Defined as the ratio of input voltage to output voltage when i/p is open circuited

i.e.,
$$h_{re} = \frac{V_{be}}{V_{ce}} \quad I_b = 0 \quad (2)$$

5. (b) Given data.

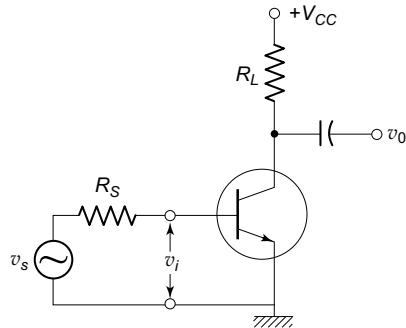
$$R_s = 680 \Omega, R_L = 1 \text{ k}\Omega,$$

$$h_{ie} = 1,100 \Omega, h_{re} = 2 \times 10^{-4},$$

$$h_{fe} = 50, h_{oe} = 20 \mu A/V$$

Using exact analysis,

$$A_I = \frac{-h_{fe}}{1 + h_{oe} R_L} = \frac{-50}{1 + (20 \times 10^{-6}) (1 \times 10^3)} = 49.0196$$



$$R_I = h_{ie} + h_{re} A_I R_L = 1,100 + (2 \times 10^{-4}) (-49.0196) (1000)$$

$$\Rightarrow R_I = 1090.196 \, \Omega,$$

$$A_v = \frac{A_I R_L}{R_I} = \frac{(-49.0196) (1000)}{1090.196}$$

$$\Rightarrow A_v = -44.964$$

Using approximate analysis

$$A_I \approx -h_{fe} = -50$$

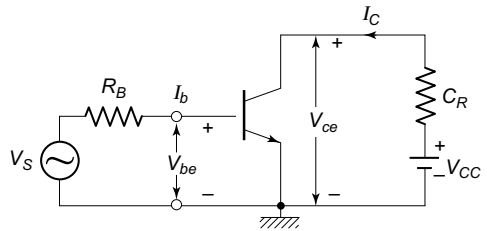
$$R_I \approx h_{oe} = 1,100 \, \Omega$$

$$A_v = \frac{A_I R_L}{R_I} = \frac{(-50) (1000)}{(1100)}$$

$$= -45.45$$

Short Circuit current Gain (h_{fe})

Defined as the ratio of output current (I_C) to the input current (I_B) when the output is short-circuited i.e.



$$\text{i.e. } h_{fe} = \left. \frac{I_C}{I_b} \right|_{V_{ce}=0} \quad (3)$$

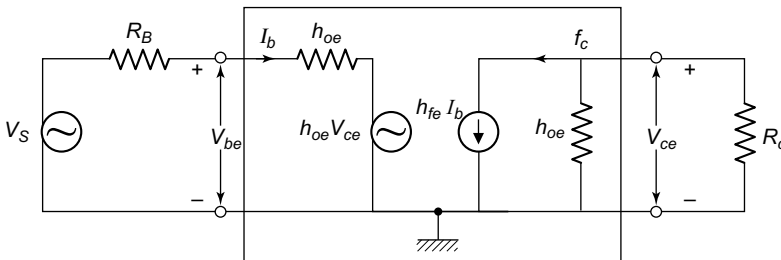
Output admittance (h_{oe})

Defined as the ratio of output current (I_C) to the output voltage (V_{ce}) when input is open circuited.

$$\text{i.e. } h_{oe} = \left. \frac{I_C}{V_{ce}} \right|_{I_b=0} \quad (4)$$

BJT in C_E configuration

Equivalent circuit



6. (a) **FET as a switch**

Consider the transfer characteristics of an n-channel JFET.

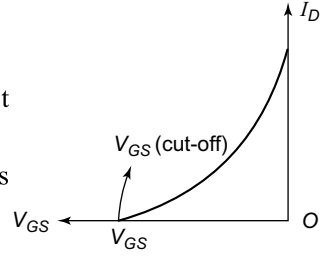
According to Shockley's equation,

$$I_D = I_{DSS} \left[1 - \frac{V_{GS}}{V_P} \right]^2 \quad (1)$$

As V_{GS} increases, I_D decreases.

When $V_{GS} = 0V$, $I_D = I_{DSS}$ and JFET is said to act like an ON switch.

When $V_{GS} = V_{GS}(\text{cut-off}) = V_P$, $I_D = 0$ and JFET is said to be Off.



6. (b) Shockley's equation is given by,

$$I_{DS} = I_{DSS} \left[1 - \frac{V_{GS}}{V_P} \right]^2 \quad (1)$$

where I_{DSS} = saturation drain current when $V_{GS} = 0$

V_P = pinch-off voltage.

Transconductance (g_m)

Defined as the ratio of slope of the transfer characteristic of a JFET.

$$\text{i.e.} \quad g_m = \frac{\Delta I_{DS}}{\Delta V_{GS}} \quad (2)$$

Differentiating (1) wrt V_{GS} , we get

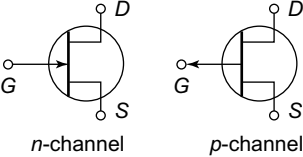
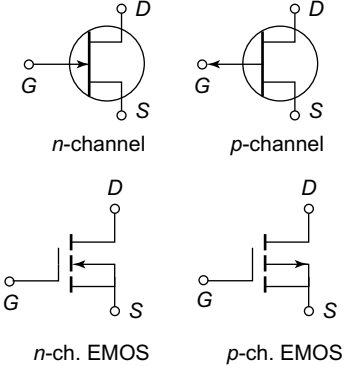
$$\frac{\partial I_{DS}}{\partial V_{GS}} = 2 I_{DSS} \left[1 - \frac{V_{GS}}{V_P} \right] \left[\frac{-1}{V_P} \right]$$

$$\Rightarrow g_m = \frac{-2 I_{DSS}}{V_P} \left[1 - \frac{V_{GS}}{V_P} \right]$$

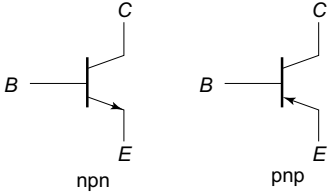
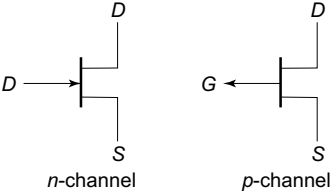
$$\Rightarrow g_m = \frac{-2 I_{DSS}}{V_P} \sqrt{\frac{I_{DS}}{I_{DSS}}}$$

$$\Rightarrow g_m = \frac{2 I_{DS}}{|V_P|} \sqrt{I_{DS} I_{DSS}} \quad \text{Hence proved.} \quad (3)$$

6. (c)

S. No.	Parameter	JFET	MOSFET
1.	Types	(a) <i>n</i> -channel (b) <i>p</i> -channel	(a) <i>n</i> -channel <i>D</i> -MOSFET (b) <i>P</i> -channel <i>D</i> -MOSFET (c) <i>n</i> -channel <i>E</i> -MOSFET (d) <i>P</i> -channel <i>E</i> -MOSFET
2.	Symbols		
3.	Operation Mode	Operated in depletion mode	Can be operated in both depletion and enhancement modes
4.	Input Impedance	High ($> 10\text{ M}\Omega$)	Very High ($> 10,000\text{ M}\Omega$)
5.	Channel	Exists permanently	Exists permanently in D MOS whereas it is enhanced in EMOSFE

7. (a) Differences between BJT and JFET

S. No.	Parameter	BJT	FET
1.	Control element	Current controlled current device	Voltage controlled current device
2.	Device type	Bipolar	Unipolar
3.	Types	npn and pnp	<i>n</i> -channel and <i>p</i> -channel
4.	Symbols		
5.	Size	Bigger	Smaller
6.	Input Impedance	Less	High

7.	Thermal Stability	Less	More
8.	Relation between input and output	Linear	Non-linear

7.(b) Comparison of JFET amplifier

S. No.	Characteristic	CG	CS	CD
1.	Input current	I_s	$I_G \approx 0$	$I_G \approx 0$
2.	Output current	I_D	I_D	I_s
3.	Input voltage	Source and Gate	Gate and Source	Gate and Drain
4.	Output voltage taken between	Drains and Gate	Drain and Source	Source and Drain
5.	Voltage Amplification Factor	$A_v = \frac{R_d \left(g_m + \frac{1}{r_d} \right)}{1 + g_m R_s + \frac{R_s + R_d}{r_d}}$ almost same as that of cs	$A_v = -g_m R_D$ (> 1)	$A_v = \frac{g_m R_s}{1 + g_m R_s}$ (always less than unity)
6.	Phase shift between input and output	0°	180°	0°
7.	Input Impedance	Finite $R_i = \frac{r_d + R_d}{1 + g_m R_d}$	∞	∞
8.	Output Impedance	$R_o [r_d + (1 + \beta) R_s] \parallel R_D$ (High)	$R_o = r_d \parallel R_D$ (Medium)	$R_o = \frac{1}{g_m} \parallel R_D$ Very Low

8. (a) Refer to Q.No. 8(a) of Set No. 2

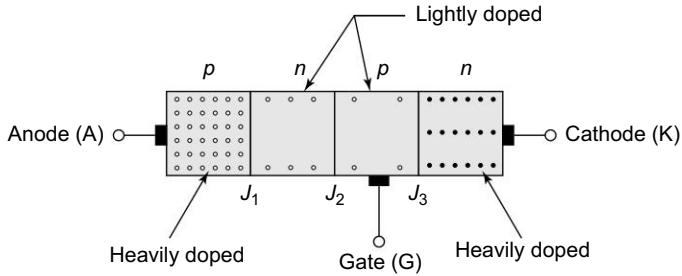
8. (b) **Construction of SCR:** It is Silicon controlled rectifier.

It is a four layer $p-n-p-n$ device when p and n are alternately arranged. The outer layers are heavily doped.

There are three pn junctions J_1 , J_2 , and J_3 .

The outer player acts as anode and outer n -layer as cathode.

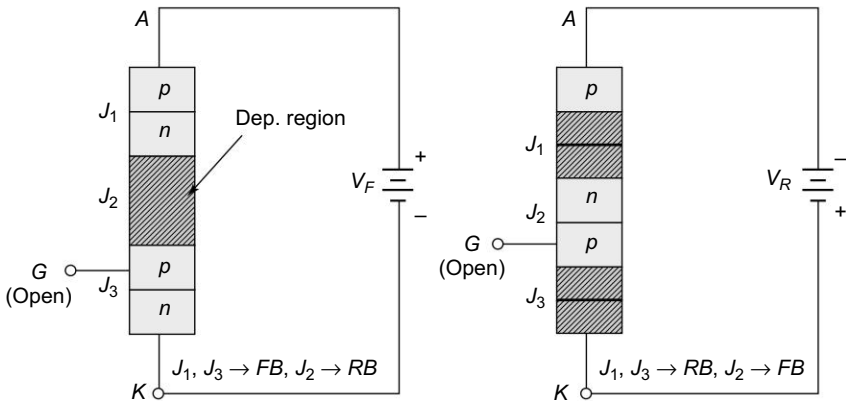
The middle p -layer is called as gate.



Operation of SCR:

Case 1: When gate (G) is open

When anode (A) is connected to +ve and cathode (K) is connected to -ve terminals of a battery then the junctions J_1 and J_3 become forward biased and J_2 is reverse biased.



Only leakage current flows through SCR and SCR is OFF and said to be operating in *forward blocking state*.

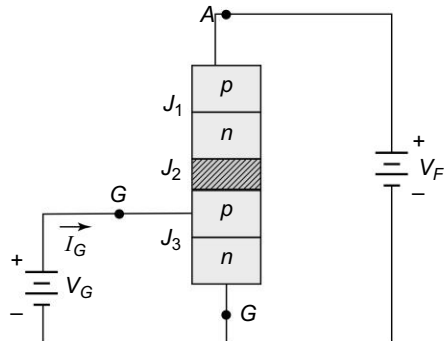
When the reverse voltage (V_R) is applied between anode (A) and cathode (K) with gate (G) open as shown in figure above, the junctions J_1, J_2 are Reverse biased and J_2 is forward biased. Here also small leakage current flows through SCR, SCR is OFF and said to be operating in *reverse blocking state*.

In forward blocking state if V_F is increased beyond a particular value, J_2 breakdown occurs and SCR starts conducting heavily. This voltage is forward breakover voltage (V_{BO}).

Case 2: When gate is closed

Consider that a voltage V_G is applied between 'G' and 'C' as shown in the figure below:

The electrons from n-type cathode which are majority carriers cross the junction J_3 to reach positive of the battery.



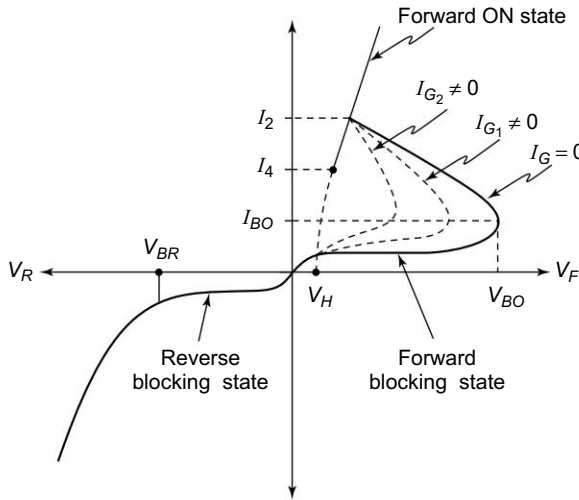
The holes from p-type move toward negative of the battery, this constitutes ' I_G '.

This current (I_G) increases the anode current as some of the electrons cross junction ' J_2 '. As anode current increases, more electrons cross J_2 and anode current further increases.

Due to regenerative action, ' J_2 ' breakdown occurs and SCR conducts heavily. *Characteristics of SCR:*

Forward Characteristics:

It shows a forward blocking region, when $I_G = 0$ It also shows when V_F increases upto V_{BO} , the SCR turns ON and high current results.



It also shows that, if gate bias (VG) is used then I_G increases, less voltage (V_F) is required to turn ON SCR.

Holding current (I_H): If forward current through SCR falls below ' I_H ', then depletion region begins to develop across ' J_2 ' and the device goes into forward blocking region.

Latching current (I_L): When SCR is turned from OFF state to ON state the resulting current is ' I_L '.

Reverse Characteristics:

For small reverse voltage the device remains in reverse blocking state as the junctions J_1 and J_3 are reverse biased.

As ' V_R ' is increased, at a particular value (V_{BR}) avalanche breakdown of J_1 and J_3 occurs and a large current flows through SCR. This breakdown voltage is known as Reverse Breakdown voltage (or) Reverse Breakover voltage.

Code No: A109210203

R09

Set No. 4

II B.Tech-I Sem Examinations, November-2010

Electronic Devices and Circuits

(Common to EEE, ECE, CSE, EIE, BME, IT, MCT, ECC, ETM, ICE)

Time: 3 hours

Maximum: 75 Marks

Answer any FIVE Questions

All Questions carry equal marks

- 1 (a) Define the following terms for a PN diode
 - (i) Dynamic resistance
 - (ii) Load line
 - (iii) Difference capacitance
 - (iv) Reverse saturation current
- (b) A reverse bias voltage of 90V is applied to Germanium diode through a resistance R. The reverse saturation current of the diode is $50\mu\text{A}$ at an operating temperature of 25°C . Compute the diode current and voltage for
 - (i) $R = 10\text{ M}\Omega$
 - (ii) $R = 100\text{ M}\Omega$
2. (a) List out the merits and demerits of bridge type full wave rectifiers over centre tapped type full wave rectifiers.
- (b) The secondary voltages of a centre tapped transformer are given as 60V-0V-60V the total resistance of secondary coil and forward diode resistance of each section of transformer secondary is 62Ω . Compute the following for a load resistance of $1\text{ K}\Omega$.
 - (i) Average load current
 - (ii) Percentage regulation
 - (iii) Rectification efficiency
 - (iv) Ripple factor for 240 V/50Hz supply to primary of transformer.
- (c) What is bleeder resistance in L-section filters ? 15
3. (a) Describe the functioning of a BJT in common base configuration.
- (b) Determine the collector current of a BJT with both of its junctions reverse biased.

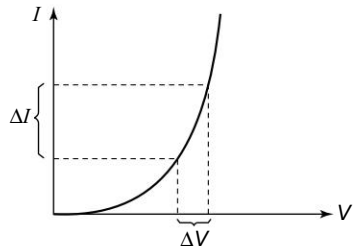
Assume $I_{CO} = 5\mu\text{A}$, $I_{EO} = 3.58\mu\text{A}$, $\alpha_N = 0.98$ and any other parameter values as required.
- (c) How do you identify the region of operation of a BJT to be saturation region from the values of various circuit currents ? 15
4. (a) Describe the significance of operating point, DC and AC load lines to ensure active region operation of a BJT in CE amplifier application.
- (b) Calculate the Q-point for the DC biased circuit shown below. 15

5. (a) With the help of a hybrid equivalent circuit of a BJT amplifier, derive expression for voltage gain and current gain when the source and load resistances of finite values are connected.
 (b) List out the typical values of h-parameters in the three BJT configurations (CE, CB and CC).
 (c) Describe how h_{ie} and h_{fe} can be determine from BJT characteristics. 15
6. (a) Differentiate between enhancement and depletion modes of a MOSFET with the help of its characteristics and construction.
 (b) Determine the pinch off voltage for an N-channel silicon. JEET if the thickness of its gate region is given as 3.2×10^{-4} cm and the donor density in n-type region is $1.2 \times 10^{15}/\text{cm}^3$.
 (c) Establish a relation between the three JEET parameters, μ , r_d and g_m . 15
7. (a) Describe how a FET can be used as a voltage variable resistance (VVR).
 (b) With the help of circuit diagram and its equivalent circuit of a source follower, derive an expression for the voltage gain possible. 15
8. Describe the following briefly:
 (a) Principle of operation of a photodiode.
 (b) Energy band structure and V-I characteristics of a tunnel diode. 15

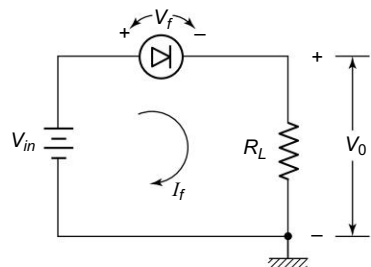
Solutions

- 1.(a) (i) **Dynamic Resistance (r):** It is the reciprocal of the slope of V - I characteristic (forward).

i.e.,
$$r = \frac{\Delta V}{\Delta I}$$



- (ii) **Load Line:** To obtain the DC load line of a diode a voltage V_{in} is applied so that the diode 'D' is forward biased. Output is taken across load resistance ' R_L '. Applying KVL around the loop in Fig.



$$-V_{in} + V_f + I_f R_L = 0$$

$\Rightarrow$

$$V_f = V_{in} - I_f R_L \quad (1)$$

This equation (1) is known as load line of diode.

- (iii) **Diffusion capacitance:**

When a diode is forward biased, the minority carrier concentration in either regions increases. This increase in the excess minority charges

due to the increase in applied forward bias is known as Diffusion Capacitance (C_D).

i.e.,
$$C_D = \left| \frac{dQ}{dv} \right|$$

where

dQ = increase in excess minority concentration

dV = increase in applied forward bias.

(iv) Reverse Saturation Current (I_0):

When a diode is reverse biased, the width of the depletion region increases and minority charges in either regions cross the junction. This movement constitutes a small leakage current of constant magnitude. This current is known as reverse saturation current.

2. (a) Merits of Bridge Rectifier over center-tapped FWR:

- (i) The weight of ordinary transformer is less than center-tapped one. Therefore Bridge Rectifier circuit is lighter in weight than center tapped FWR.
- (ii) The cost of bridge rectifier is very less when compared to center tapped FWR.
- (iii) The TUF of bridge rectifier is 0.812 and that of center-tapped one is 0.693 thereby the transformer utilization is move in bridge rectifier.
- (iv) PIV of each diode rectifier is half of that in center-tapped FWR.

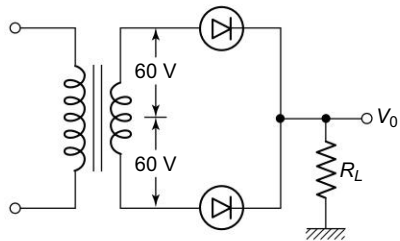
Demerits: The usage of four diodes causes additional voltage drop as indicated by the term $2R_f$ in the ' I_m ' expression. This reduces the o/p voltage.

2. (b) Given data,

$$(V_2)_{rms} = 60 \text{ V}$$

$$R_s + R_f = 62 \text{ } \Omega$$

$$R_L = 1 \text{ k } \Omega.$$



Peak value of secondary voltage,

$$V_m = \sqrt{2} (V_2)_{rms} = (\sqrt{2}) (60) = 84.8528 \text{ V}$$

Peak value of load current,

$$I_m = \frac{V_m}{R_s + R_f + R_L} = \frac{84.8528}{62 + 1000} = 79.9 \text{ mA}$$

(i) Average Load current, $I_{DC} = \frac{2I_m}{\pi} = \frac{2(79.9 \times 10^{-3})}{\pi}$

$$\Rightarrow \boxed{I_{DC} = 50.8659 \text{ mA}}$$

(ii) Percentage Load Regulation

$$\% LR = \frac{V_{NL} - V_{FL}}{V_{FL}} \times 100$$

$$V_{NL} = \frac{2}{\pi} V_m = \frac{2(84.8528)}{\pi} = 54.01897 \text{ V}$$

$$V_{FL} = I_{DC} \cdot R_L = (50.8659 \times 10^{-3}) (1 \times 10^3) = 50.8659 \text{ V}$$

$$\therefore \% LR = \frac{-50.8659 + 54.01897}{50.8659} \times 100$$

$$\therefore \boxed{\% LR = 6.1987\%}$$

(iii) Rectification efficiency $\% \eta = \frac{P_{DC}}{P_{AC}} \times 100$

$$P_{DC} = V_{DC} \cdot I_{DC} = \left(\frac{2V_m}{\pi} \right) \left(\frac{2V_m}{\pi} \right)$$

$$P_{AC} = V_{RMS} \cdot I_{RMS}$$

$$\therefore \% \eta = \frac{8}{\pi^2} \left[\frac{1}{1 + \left(\frac{R_f + R_s}{R_L} \right)} \right] 100$$

$$\Rightarrow \% \eta = \frac{8}{\pi^2} \left[\frac{100}{1 + \frac{62}{1000}} \right]$$

$$\Rightarrow \boxed{\% \eta = 76.32\%}$$

(iv) Ripple factor for 240 V/50 Hz supply

$$\gamma = \sqrt{\left(\frac{I_{RMS}}{I_{DC}} \right)^2 - 1}$$

$$I_{RMS} = \frac{I_m}{\sqrt{2}} = \frac{79.9 \text{ mA}}{\sqrt{2}} = 56.49 \text{ mA}$$

$$\therefore \gamma = \sqrt{\left(\frac{56.49}{50.86} \right)^2 - 1}$$

$$\boxed{\gamma = 0.4833}$$

(c) Refer to Q.No. 2(b) of Set-3.

3. (a) **Operation of npn BJT in CB configuration**

In common base configuration 'E' acts as input terminal and 'C' acts as output terminal.

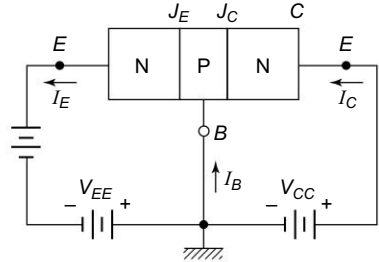
Due to V_{EE} and V_{CC} the junction J_E and J_C are forward biased and reverse biased respectively.

As ' J_E ' is forward biased, the width of depletion region across J_E decreases and as ' J_C ' is reverse biased the width of depletion region across ' J_C ' increases.

The forward biased J_E causes the electrons of n -type emitter region to flow into the p -type base region. This constitute ' I_E '.

A few of these electrons in p -type base participate in recombination with holes. The bulk recombination results in the base current I_B .

Rest of the electrons cross the reverse biased ' J_C ' and enter into collector region. This constitutes collector current ' I_C '.

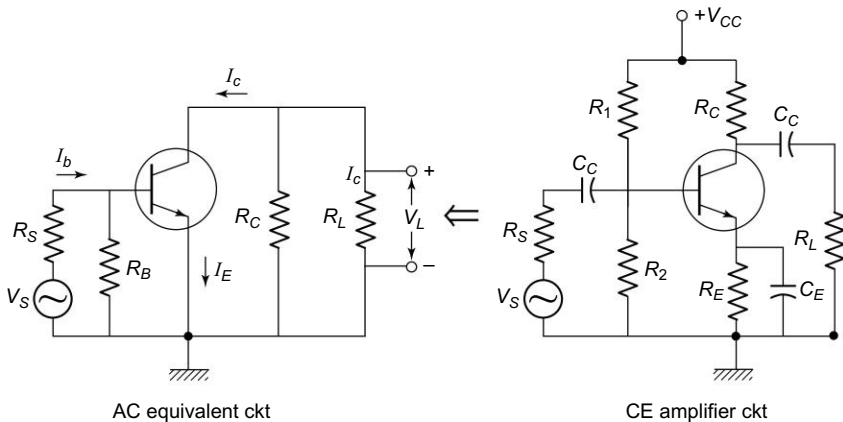


- (c) For a BJT to be in saturation region both the junctions J_E and J_C are to be forward biased. If the base current flowing through BJT follows the below stated condition BJT can be identified to be operating in saturation region:

$$I_B > \frac{I_C}{\beta_{DC}} \quad \text{and} \quad V_{CE} = V_{CE(SAT)}$$

4. (a) For operating point, DC load line refer to **Q.No. 5(a) of Set-1.**

AC load line: Consider a CE amplifier circuit as shown below:



Applying KVL to the collector circuit of ac equivalent ckt, we have

$$V_{ce} = I_c R_{ac}$$

(1)

where

V_{ce} = ac collector-to-emitter voltage

I_c = ac collector current.

Since $I_c = i_c - I_{CQ}$ and $V_{ce} = V_{CEQ} - V_{CE}$

$$(1) \text{ becomes, } \boxed{V_{CEQ} - v_{CE} = (i_c - I_{CQ})R_{ac}} \quad (2)$$

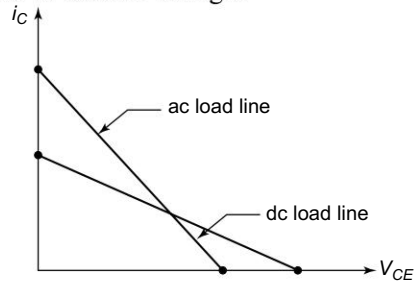
i_c = Total instantaneous collector current

V_{CE} = Total instantaneous collector to emitter voltage.

Rearranging (2), we have,

$$\boxed{i_c = \frac{V_{CEQ}}{R_{ac}} - \frac{V_{CE}}{R_{ac}} + I_{CQ}} \quad (3)$$

This equation is known as ac load line.



5. (a) Refer to Q.No. 5(a) of Set-2

(b) Typical values of h -parameters of BJT in CE, CB and CC config.

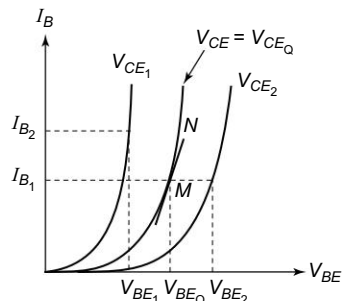
S.No.	Parameter	CE	CB	CC
1	h_i	1100 Ω	21.6 Ω	1100 Ω
2.	h_r	2.5×10^{-4}	2.9×10^{-4}	≈ 1
3.	h_f	50	-0.98	-51
4.	h_o	25 $\mu\text{A/V}$	0.49 $\mu\text{A/V}$	25 $\mu\text{A/V}$

(c) Determination of h_{ie} from input characteristic curves:

$$h_{ie} = \left. \frac{\Delta V_{BE}}{\Delta I_B} \right|_{V_{CE} = \text{constant}} = \frac{V_{BE_2} - V_{BE_1}}{I_{B_2} - I_{B_1}}$$

The parameter h_{ie} can be obtained as the change in the base voltage ($V_{BE_2} - V_{BE_1}$) divided by the change in the base current ($I_{B_2} - I_{B_1}$) for a constant collector voltage (V_{CE}) at a Quiescent point.

The slope of line MN, drawn tangent to the input characteristics curve at the point 'Q' gives h_{ie} .

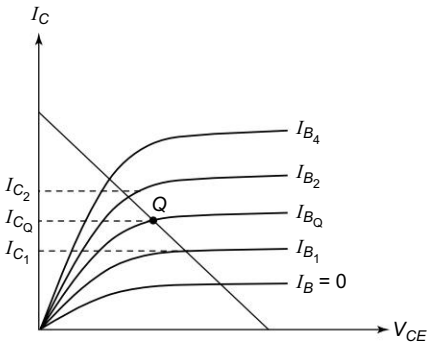


Determination of h_{fe} from output characteristics

The parameter ' h_{fe} ' of a *BJT* is defined as

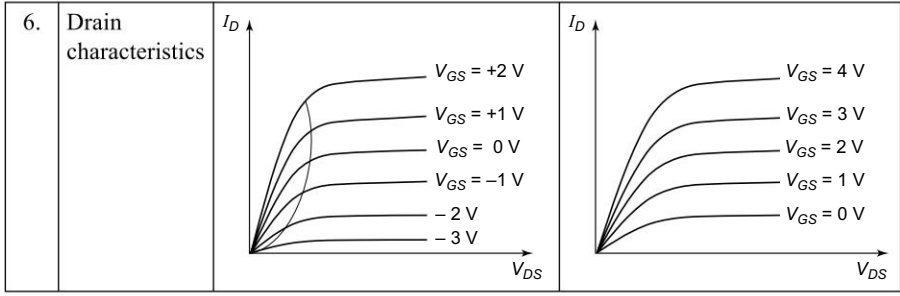
$$h_{fe} = \left. \frac{\Delta I_C}{\Delta I_B} \right|_{V_{CE} = \text{const.}} = \frac{I_{C_2} - I_{C_1}}{I_{B_2} - I_{B_1}}$$

It is the ratio of change in collector current taken around the *Q*-pt. to the corresponding change in base current, for a constant value of V_{CE} at *Q*-point.



6. (a) Comparison of E-MOSFET and D-MOSFET

S. No	Parameter	D-MOSFET	E-MOSFET
1.	Symbols		
2.	Channel	Exists permanently	Channel is physically absent. It is induced after application of +ve V_{GS} above the V_T for <i>n</i> -channel and -ve V_{GS} above V_T for <i>p</i> -channel.
3.	Operation	Can be operated in both enhancement and depletion mode	Can operate only in enhancement mode
4.	Current flow	Drain current flows on application of V_{DS} at $V_{GS} = 0$.	Practically no current flows on the application of V_{DS} at $V_{GS} = 0$.
5.	Transfer characteristics		



(b) Given data,

$$a = 3.2 \times 10^{-4} \text{ cm} = 3.2 \times 10^{-6} \text{ m}$$

$$N_D = 1.2 \times 10^5 / \text{cm}^3 = 1.2 \times 10^5 / (10^{-2})^3 \text{ m}^3$$

$$\Rightarrow N_D = 1.2 \times 10^5 \times 10^6 / \text{m}^3 = 1.2 \times 10^{11} / \text{m}^3$$

$$\text{Pinch-off voltage, } V_P = \frac{qN_D}{2\epsilon}(a^2)$$

$$\epsilon = 12\epsilon_0 = 12 \times 8.85 \times 10^{-12} = 106.2 \times 10^{-12} \text{ F/m.}$$

$$\therefore V_P = \frac{1.6 \times 10^{-19} \times 1.2 \times 10^{11} \times (3.2 \times 10^{-6})^2}{2(106.2 \times 10^{-12})}$$

$$= 0.0925 \times 10^{-8} \text{ V}$$

(c) Relationship between μ , r_d and g_m of a JFET

For a JFET,

Drain Resistance (r_d) is defined as the reciprocal of slope of drain characteristics of JFET.

$$\text{i.e., } r_d = \left. \frac{\Delta V_{DS}}{\Delta I_P} \right|_{V_{GS} = \text{constant}} \quad (1)$$

Transconductance (g_m) is defined as the slope of transfer characteristics of JFET

$$\text{i.e., } g_m = \left. \frac{\Delta I_D}{\Delta V_{GS}} \right|_{V_{DS} = \text{constant}} \quad (2)$$

Amplification factor (μ) is defined as the ratio of change in V_{PS} to the change in V_{GS} .

$$\text{i.e., } \mu = \left. \frac{\Delta V_{DS}}{\Delta V_{GS}} \right|_{I_D = \text{constant}} \quad (3)$$

$$(1) \times (2)$$

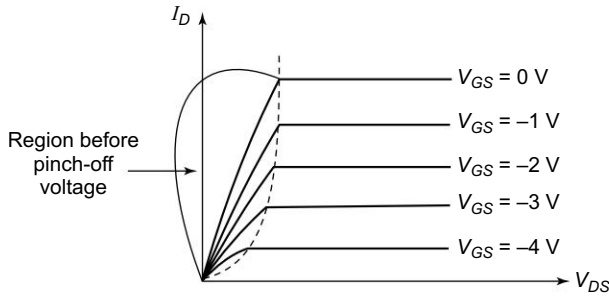
$$\Rightarrow g_m \cdot r_d = \left(\frac{\Delta I_D}{\Delta V_{GS}} \right) \left(\frac{\Delta V_{DS}}{\Delta I_D} \right) = \frac{\Delta V_{DS}}{\Delta V_{GS}}$$

$$\Rightarrow \boxed{\mu = g_m r_d} \quad (4)$$

This is the relationship between μ , g_m and r_d .

7. (a) **FET as a Voltage Variable Resistor (VVR)**

Let us consider the drain characteristics of *FET* as shown below: In this characteristics we can see a region before pinch-off voltage where drain characteristics is linear i.e., operation of *FET* is linear. In this region *FET* operates as voltage-controlled resistor i.e., the drain to source resistance is controlled by the voltage V_{GS} . Hence *FET* in this region acts as *VVR*.



- (b) **Source Follower:** The figure shows the circuit diagram of a common drain amplifier which is also called as source follower.

To obtain voltage gain of this amplifier circuit the *FET* in the ac equivalent *ckt* has to be replaced with its small signal model.

The *ac* equivalent circuit of the above circuit can be obtained by replacing all the capacitors with short circuits and replacing the *DC* voltage sources with ground. It is as shown below: Replacing the *FET* with its small signal equivalent, we get

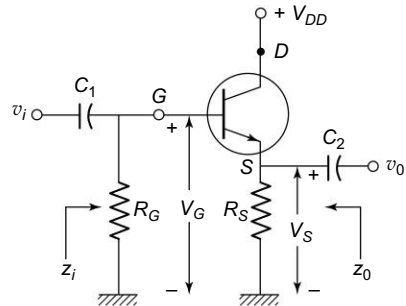
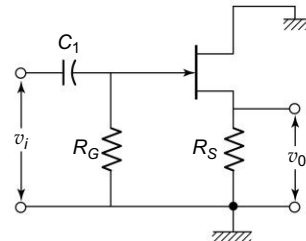
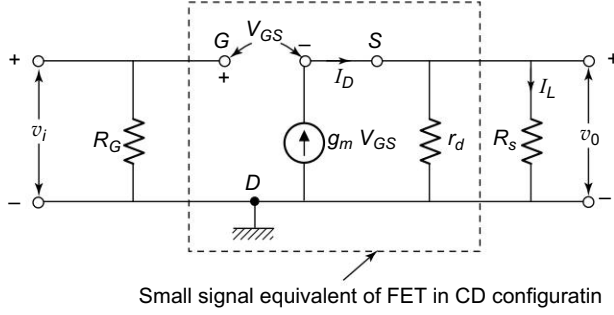


Fig. (a) Source Follower





Voltage gain (A_V) of this circuit is defined as the ratio of output voltage to the input voltage.

i.e.,
$$A_V = \frac{v_o}{v_i} \quad (1)$$

From figure,
$$v_o = (g_m V_{gs}) [r_d \parallel R_s] \quad (2)$$

Applying KVL around the outer loop, we get

$$\begin{aligned} -v_i + V_{gs} + v_o &= 0 \\ \Rightarrow v_i &= V_{gs} + v_o \\ \Rightarrow v_i &= V_{gs} + [g_m V_{gs}] [r_d \parallel R_s] \quad [\text{from (2)}] \\ \Rightarrow v_i &= V_{gs} [1 + g_m (r_d \parallel R_s)] \quad (3) \end{aligned}$$

Using (2) and (3) in (1), we get

$$\begin{aligned} A_V &= \frac{g_m V_{gs} [r_d \parallel R_s]}{V_{gs} [1 + g_m (r_d \parallel R_s)]} \\ \Rightarrow A_V &= \frac{g_m (r_d \parallel R_s)}{1 + g_m (r_d \parallel R_s)} \quad (4) \end{aligned}$$

If $r_d \gg R_s$ then $r_d \parallel R_s \approx R_s$

$$A_V \approx \frac{g_m R_s}{1 + g_m R_s} \quad (5)$$

This is the expression for voltage gain of a source follower.

8. (a) **Principle of operation of photodiode:**

The photodiode is a semiconductor $p-n$ junction device whose region of operation is limited to reverse biased region.

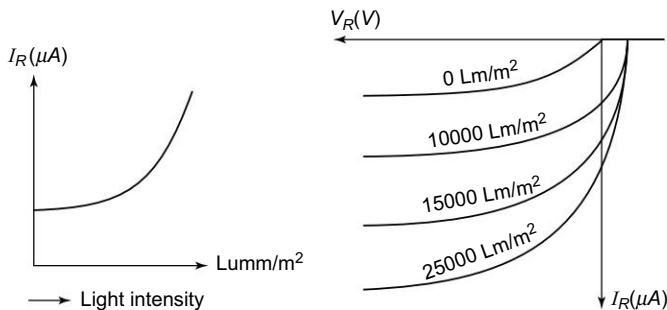
The figures (a) and (b) below show the symbol of photodiode and its principle of operation respectively.

The photodiode is connected in reverse bias condition. Due to the applied reverse bias the width of the depletion region increases.

Under normal conditions (absence of light), the diode carries small reverse leakage current due to the flow of minority carriers. This current is known as **dark current**.

When some light is incident through a glass window on the p-n junction, photons in the light collide with the p-n junction and some energy is imparted to the valence electrons of silicon, atoms. Due to this the valence electrons are dislodged from the covalent bonds and become free electrons. And also the dislodged electrons leave holes in the vacancies. Therefore electron-hole pairs are generated and the reverse current through the diode increases.

The characteristics of photo diode are as shown below:



(b) Refer to Q.No. 8(b) of Set-2.